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· Mushkarov · Nikolov

TOPICS

in

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Topics in Functional Equations

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*The worthwhile problems are the ones you can really solve
or help solve, the ones you can really contribute something to.
... No problem is too small or too trivial if we can really
do something about it.*

– Richard P. Feynman

Introduction

Functional equations have been of significant importance in mathematics for the last couple of centuries, as well as a constant presence at high-level mathematical competitions. Historically, in mathematical research, functional equations have been closely associated with analysis — some of the greatest mathematicians have dedicated themselves to solving them. In present-day research, certain differential equations are of central importance in analysis, differential geometry, and other various areas of theoretical physics. Because of their importance and popularity, differential equations are thoroughly studied, and there is an established theory of classical differential equations, taught in universities, as well as a multitude of resources which present more recent advances in the field.

However, this is not the case for functional equations that are given in high-school or even university contests. Due to the constraints of maintaining an elementary threshold, they have evolved along a quite different route, in that there is a scarcity of math Olympiad textbooks that implement a systematic approach to such equations. Few books accessible to high-school students move beyond discussing the ubiquitous Cauchy's Equations and their slight generalizations. If one were to inquire about a strong comprehensive mathematical Olympiad book about functional equations, Venkatachala's fine book [24] and Small's book [23] come to mind. There are several good articles and notes on this topic scattered throughout the literature, but we have not encountered any book that offers a comprehensive study of this subject.

We deem that functional equations are severely underestimated by competitive mathematical literature. This subject occupies a special position in

current mathematical contests. Functional equations are consistently featured in competitions, especially at the International Mathematical Olympiad, but it also appears that they are usually not considered as a distinct subject, but rather as a subset of algebra — a collection of problems and a collection of tricks to solve them.

We think that functional equations should be treated as a separate topic instead. An experienced competitor will say that every area of mathematics has its own distinct 'flavor', a different mindset and approach to handling its problems. This is why some are better at synthetic geometry than inequalities, or more skilled at number theory than combinatorics, despite equal time devoted to each of these topics. The authors, having been strongly involved in mathematical Olympiads for many years, both as competitors and as coaches, are certain that functional equations have a 'feel' of their own. To solve advanced functional equations, one needs to have a deep aptitude for algebraic manipulations, creativity for noticing clever substitutions, and a wide experience in all areas of mathematics from which functional equations' ideas stem. The way one searches for a solution for a functional equation can be quite different from the way one tries to solve a geometry problem or an inequality. Thus, functional equations belong to a field of their own.

It is known that mathematical competitions are a very efficient way to learn and hone problem-solving skills. These skills are later essential in several high-powered quantitative careers, including research in mathematics, physics, or computer science; or more applied work in engineering, finance, or computer programming. Functional equations, as part of problem solving, play their own important role in teaching the mind how to think. If one were to compare training the brain to training the body, functional equations would be like a set of exercises targeting certain muscle groups (but involving all other muscles to some extent as well).

This book is a systematic and comprehensive approach to functional equations as a whole. Unlike in other branches of competitive mathematics, there is very little theory – rather, the methods and techniques utilized in solving these equations play the most important part. For this reason, the book has taken a highly practical path, including lots of problems designed to teach students how to familiarize themselves with every strategy employed, as well

as to experiment in combining and manipulating different techniques.

This work contains all the important functional equations given at contests in recent years, classified by the way they are solved. It explains the reasoning behind each method, and offers advice on how to invent a meaningful solution. We are confident that talented and motivated students who work through this book will significantly expand their horizons and capabilities in solving functional equations given in any mathematical context. And, in the meantime, they will have acquired a certain way of thinking about problems, which will not only be useful in other areas of mathematics, but later in life as well.

The book begins with the most popular functional equation — Cauchy's equation. Several versions and extensions are presented. The chapter is also enhanced with a lot of examples and exercises designed to aid readers in fully absorbing this important equation, and learn how to apply it in other contexts. Two other chapters focus on key functional equations, but the rest of the chapters are centered around methods rather than specific equations. Some chapters touch on issues not mentioned anywhere else in this type of literature. For example, the chapter on approximating by linear functions or the subsection on polynomial recurrences and continuity demonstrate the uniqueness of this work. Because functional equations are so diverse, many of these chapters are independent from one another and could be read in a different order than as presented here. A reader with a specific goal in mind can proceed directly to a chapter they desire. Some chapters are shorter than others because there are much fewer problems illustrating the corresponding methods presented — we strove for efficiency rather than artificial homogeneity in chapter size. The second to last chapter of the book contains a list of problems labeled "miscellaneous"; these are questions that were hard to categorize, although they are still grouped in subsections of related problems. This is intended to be the final chapter one reads; a lot of problems that are very challenging. We advise the reader not to tackle these questions before they are comfortable with the methods employed in the rest of the book. The final chapter contains the solutions to all the problems in the book. Though it is recommended that each question may be attempted, we recognize that you can sometimes learn more from reading the solution to a hard problem rather than producing a solution to an easier problem by yourself.

We would like to thank our dear friend Gabriel Dospinescu for the problems suggested, as well as for his pertinent remarks.

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Chapter 1

Cauchy's Equations

1.1 Additive Cauchy Equation

Let D be a set of real numbers such that $x + y \in D$ for all $x, y \in D$.

A function $f : D \rightarrow \mathbb{R}$ is called additive if

$$f(x + y) = f(x) + f(y) \quad (1.1)$$

for all $x, y \in D$. The simplest examples of functions satisfying (1.1) are the linear functions of the form $f(x) = ax$, where $a \in \mathbb{R}$. For $D = \mathbb{R}$ the functional equation (1.1) has been first studied systematically by Cauchy [8] and it is now known as the additive Cauchy equation. It is natural to ask if the linear functions $f(x) = ax$ are the only solutions of the additive Cauchy equation. As we shall see later the answer to this question is *No* in general and *Yes* if we impose some additional assumptions on the functions under consideration.

The next proposition is a direct consequence of (1.1).

Proposition 1.1. Let D be a set of real numbers such that $0 \in D$, $x + y \in D$ and $\frac{x}{n} \in D$ for all $x, y \in D$ and all integers n . If $f : D \rightarrow \mathbb{R}$ is an additive function then

$$f\left(\sum_{k=1}^n r_k x_k\right) = \sum_{k=1}^n r_k f(x_k)$$

for arbitrary $x_k \in D$ and rational numbers r_k , $1 \leq k \leq n$.

Proof. Setting $y = 0$ and $y = -x$ in (1.1) gives $f(0) = 0$ and $f(-x) = -f(x)$, $x \in D$. Moreover, it follows by induction on n that

$$f\left(\sum_{k=1}^n x_k\right) = \sum_{k=1}^n f(x_k). \quad (1.2)$$

In particular, $f(nx) = nf(x)$ for any $x \in D$ and $n \in \mathbb{N}$.

Let m and n be positive integers, $r = \frac{m}{n}$ and $x \in D$. Then $mx \in D$ and $\frac{mx}{n} \in D$, i.e. $rx \in D$. Hence $nf(rx) = f(nrx) = f(mx) = mf(x)$ and therefore $f(rx) = rf(x)$. This identity also holds true for any non-positive rational number r since $f(0) = 0$ and $f(-y) = -f(y)$ for all $y \in D$. Now taking into account (1.2) we get that

$$f\left(\sum_{k=1}^n r_k x_k\right) = \sum_{k=1}^n f(r_k x_k) = \sum_{k=1}^n r_k f(x_k). \quad \square$$

Example 1.1. Each additive function $f : \mathbb{Q} \rightarrow \mathbb{R}$ has the form $f(x) = ax$, where $a \in \mathbb{R}$. Indeed, if $x \in \mathbb{Q}$ then $f(x) = xf(1) = ax$, where $a = f(1)$.

Example 1.2. Let $\mathbb{Q}(\sqrt{2}) = \{p + q\sqrt{2} \mid p, q \in \mathbb{Q}\}$. For any $x = p + q\sqrt{2}$ set $\bar{x} = p - q\sqrt{2}$. Then any additive function $f : \mathbb{Q}(\sqrt{2}) \rightarrow \mathbb{R}$ has the form $f(x) = ax + b\bar{x}$, where $a, b \in \mathbb{R}$. To see this let $x = p + q\sqrt{2}$. Then it follows from Proposition 1.1 that $f(x) = pf(1) + qf(\sqrt{2})$. On the other hand, $p = \frac{x + \bar{x}}{2}$ and $q = \frac{x - \bar{x}}{2\sqrt{2}}$. Hence

$$f(x) = \left(\frac{x + \bar{x}}{2}\right) f(1) + \left(\frac{x - \bar{x}}{2\sqrt{2}}\right) f(\sqrt{2})$$

and setting

$$a = \frac{f(1)}{2} + \frac{f(\sqrt{2})}{2\sqrt{2}}, \quad b = \frac{f(1)}{2} - \frac{f(\sqrt{2})}{2\sqrt{2}}$$

we get that $f(x) = ax + b\bar{x}$. Conversely, any function $f : \mathbb{Q}(\sqrt{2}) \rightarrow \mathbb{R}$ of this form is additive since

$$f(x + y) = a(x + y) + b(\overline{x + y}) = ax + b\bar{x} + ay + b\bar{y} = f(x) + f(y)$$

for all $x, y \in \mathbb{Q}(\sqrt{2})$.

In the next theorem we give some necessary and sufficient conditions for an additive function f to be of the form $f(x) = ax$.

Theorem 1.1. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be an additive function. Then the following four conditions are equivalent:

- (i) f has the form $f(x) = ax$ for some $a \in \mathbb{R}$;
- (ii) f is bounded above (below) on an interval;
- (iii) f is increasing (decreasing) on an interval;
- (iv) f is continuous at a point.

Proof. It is clear that the condition (i) implies the other three. We shall prove now that (ii) $\rightarrow$ (i), (iii) $\rightarrow$ (ii) and (iv) $\rightarrow$ (ii).

(ii) $\rightarrow$ (i). Set $g(x) = f(x) - f(1)x$. Then $g : \mathbb{R} \rightarrow \mathbb{R}$ is an additive function and it follows from Proposition 1.1 that $g(r) = 0$ for any $r \in \mathbb{Q}$. Since the function $f(x)$ is bounded above on an interval (p, q) , it follows that the function $g(x)$ is also bounded above on this interval. Let $g(x) < C$ for any $x \in (p, q)$ and let x_0 be an arbitrary real number. The interval $(p - x_0, q - x_0)$ contains a rational number r . Since $r + x_0 \in (p, q)$ we get

$$g(x_0) = g(r + x_0) - g(r) = g(r + x_0) < C.$$

Hence $g(x) < C$ for any $x \in \mathbb{R}$. Then

$$g(x_0) = \frac{1}{n}g(nx_0) \leq \frac{C}{n} \text{ and } g(x_0) = -\frac{1}{n}g(-nx_0) \geq -\frac{C}{n}$$

for any $n \in \mathbb{N}$. Thus

$$-\frac{C}{n} \leq g(x_0) \leq \frac{C}{n}$$

for any $n \in \mathbb{N}$ which clearly implies that $g(x_0) = 0$. Since $x_0 \in \mathbb{R}$ is arbitrary we get that $f(x) - f(1)x = g(x) = 0$, i.e. $f(x) = ax$, where $a = f(1)$.

The case when the function $f(x)$ is bounded below on an interval is considered analogously.

(iii) $\rightarrow$ (ii). If the function f is increasing (decreasing) on an interval then f is bounded above (below) on a smaller interval.

(iv) $\rightarrow$ (ii). Let the function $f(x)$ be continuous at a point $x_0 \in \mathbb{R}$. Then there is $\delta > 0$ such that if $|x - x_0| < \delta$ then $|f(x) - f(x_0)| < 1$. Hence $f(x) < 1 + f(x_0)$ ($f(x) > f(x_0) - 1$) for any $x \in (x_0 - \delta, x_0 + \delta)$, i.e. f is bounded above (below) on the interval $(x_0 - \delta, x_0 + \delta)$. $\square$

Corollary 1.1. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be an additive function which is also multiplicative, i.e. $f(x + y) = f(x) + f(y)$ and $f(xy) = f(x)f(y)$ for all $x, y \in \mathbb{R}$. Then either $f(x) \equiv 0$ or $f(x) \equiv x$.

Proof. Since $f(x^2) = f(x)^2 \geq 0$ it follows that f is an additive function which is bounded below on the interval $(0, +\infty)$. Hence it follows from Theorem 1.1 that $f(x) = ax$, where a is the constant. Then

$$axy = f(xy) = f(x)f(y) = a^2xy$$

for any $x, y \in \mathbb{R}$. Hence $a = a^2$ and we get that $a = 0$ or $a = 1$. $\square$

Theorem 1.1 shows that if an additive function $f : \mathbb{R} \rightarrow \mathbb{R}$ does not satisfy one of the conditions (i) – (iv) then it does not satisfy the other three. We shall show that such additive functions do exist and that they have very strange behaviour.

Theorem 1.2. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be an additive function which does not have the form $f(x) = ax$. Then any square (circle) in the plane contains a point from the graph of f .

Proof. We shall give two proofs of this theorem.

First proof. It is enough to prove the theorem for a square with sides parallel to the coordinate axes. Denote by $P_0(x_0, y_0)$ the center of such a square and let $2\varepsilon > 0$ be the length of its sides. Then for any point $M(x, y)$ inside the square we have $x_0 - \varepsilon < x < x_0 + \varepsilon$ and $y_0 - \varepsilon < y < y_0 + \varepsilon$. It follows from Theorem 1.1 that the function $f(x)$ is unbounded above and below on the interval $(x_0 - \varepsilon, x_0 + \varepsilon)$. Hence there exist points $x_1, x_2 \in (x_0 - \varepsilon, x_0 + \varepsilon)$ such that $y_1 = f(x_1) < y_0$ and $y_2 = f(x_2) > y_0$. Then the points $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$ lie in the strip determined by the vertical lines $x = x_0 - \varepsilon$ and $x = x_0 + \varepsilon$, and on different sides of the horizontal line $y = y_0$. If one of these

points lies in the square, then we are done. Hence we may assume that both of them lie outside the square. Consider the midpoint $x_3 = \frac{x_1 + x_2}{2}$ of the interval $[x_1, x_2]$. Since the function f is additive we have that

$$y_3 = f(x_3) = f\left(\frac{x_1 + x_2}{2}\right) = \frac{f(x_1) + f(x_2)}{2} = \frac{y_1 + y_2}{2}.$$

Hence y_3 is the midpoint of the interval $[y_1, y_2]$ and therefore

$$|y_3 - y_0| \leq \frac{y_2 - y_1}{2}.$$

If the point $P_3(x_3, y_3)$ lies inside the square the theorem is proved. So, we may assume that $P_3(x_3, y_3)$ lies outside the square. If $f(x_3) > y_0$ ($f(x_3) < y_0$), we set

$$x_4 = \frac{x_1 + x_3}{2} \quad \left(x_4 = \frac{x_2 + x_3}{2} \right).$$

In both cases we have that

$$|y_4 - y_0| \leq \frac{y_2 - y_1}{2^2},$$

where $y_4 = f(x_4)$. Continuing in the same way we construct a sequence of points $P_n(x_n, y_n)$ on the graph of f such that $x_n \in (x_0 - \varepsilon, x_0 + \varepsilon)$ and

$$|y_n - y_0| \leq \frac{y_2 - y_1}{2^n}$$

for any $n \in \mathbb{N}$. It follows that $\lim_{n \rightarrow \infty} y_n = y_0$ and therefore there exists $n \in \mathbb{N}$ such that $y_n \in (y_0 - \varepsilon, y_0 + \varepsilon)$. Hence the point $P_n(x_n, y_n)$ lies inside the square and the theorem is proved.

Second proof. We shall use the fact that for any real number there is a rational number which is arbitrarily close to it. If the function $f(x)$ does not have the form $f(x) = ax$ then there exist $x_1, x_2 \neq 0$ such that $\frac{f(x_1)}{x_1} \neq \frac{f(x_2)}{x_2}$. Hence the vectors $v_1 = (x_1, f(x_1))$ and $v_2 = (x_2, f(x_2))$ are linearly independent and for any vector v there exist real numbers ρ_1 and ρ_2 such that $v = \rho_1 v_1 + \rho_2 v_2$.

Therefore there exists a vector of the form $r_1v_1 + r_2v_2$ with rational r_1 and r_2 which is arbitrary close to v . Since

$$r_1v_1 + r_2v_2 = r_1(x_1, f(x_1)) + r_2(x_2, f(x_2)) = (r_1x_1 + r_2x_2, f(r_1x_1 + r_2x_2))$$

it follows that for any point inside the square there is a point on the graph of f which is arbitrary close to it. Hence this point also lies inside the square. $\square$

Using standard mathematical terminology the above theorem says that if the graph of an additive function is not a line then it is a dense subset of the plane which is unbounded around each of its points. The existence of such additive functions was proved by Hamel [11] in 1905 by using the so-called *Axiom of Choice* which says that for any family $\{F_\alpha\}$ of nonempty sets F_α there exists a set $F = \{x_\alpha\}$, where $x_\alpha \in F_\alpha$ for any α .

Note that the Axiom of Choice has some quite unexpected consequences that contradict normal thinking. That is why many mathematicians do not accept it and this led to the development of branches of mathematics where the Axiom of Choice is not valid.

Next we shall use the following theorem of Hamel [11] proved by means of the Axiom of Choice.

Theorem 1.3. There exists a set H of real numbers such that:

- (a) $1 \in H$;
- (b) if $h_1, h_2, \dots, h_n \in H$ and $r_1, r_2, \dots, r_n$ are rational numbers with

$$r_1h_1 + r_2h_2 + \dots + r_nh_n = 0,$$

then $r_1 = \dots = r_n = 0$;

- (c) for any real number x there exist rational numbers $r_1, \dots, r_n$ and real numbers $h_1, h_2, \dots, h_n \in H$ such that $x = r_1h_1 + \dots + r_nh_n$.

Any set H of real numbers having the above three properties is called a *Hamel basis*. Note that for any $x \in \mathbb{R}$, $x \neq 0$, the sum in (c) is finite but the number of the summands depends on x . Hence any $x \in \mathbb{R}$ can be written in the form

$$x = \sum r_\alpha h_\alpha, \tag{1.3}$$

where $\{h_\alpha\} = H$ and only finite of the rational numbers r_α are nonzero. Suppose that $x = \sum r_\alpha h_\alpha$ and $x = \sum s_\alpha h_\alpha$. Then

$$0 = x - x = \sum (r_\alpha - s_\alpha) h_\alpha$$

and it follows from (b) that $r_\alpha = s_\alpha$. Hence the rational numbers r_α in the representation (1.3) are uniquely determined by x and can be considered as coordinates of x with respect to the Hamel basis H .

The importance of the Hamel bases for description of additive functions is manifested by the following

Theorem 1.4. Let H be a Hamel basis and let $s : H \rightarrow \mathbb{R}$ be an arbitrary function. Then there exists a unique additive function $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(h) = s(h)$ for any $h \in H$.

Proof. To use the same notation as above, we set $H = \{h_\alpha\}$. Given a function $s : H \rightarrow \mathbb{R}$ we can define a function $f : \mathbb{R} \rightarrow \mathbb{R}$ in the following way. Let x be an arbitrary real number. Then $x = \sum r_\alpha h_\alpha$, where $r_\alpha \in \mathbb{Q}$ and only finite of the numbers r_α are nonzero. Set $f(x) = \sum r_\alpha s(h_\alpha)$. Then it is clear that $f(h_\alpha) = s(h_\alpha)$ for any $h_\alpha \in H$ and we have to prove that f is an additive function. Let $x = \sum r'_\alpha h_\alpha$ and $y = \sum r''_\alpha h_\alpha$, where $r'_\alpha, r''_\alpha \in \mathbb{Q}$. Then it follows from Proposition 1.1 that

$$\begin{aligned} f(x+y) &= f\left(\sum (r'_\alpha + r''_\alpha) h_\alpha\right) = \sum (r'_\alpha + r''_\alpha) f(h_\alpha) = \\ &= \sum (r'_\alpha + r''_\alpha) s(h_\alpha) = \sum r'_\alpha s(h_\alpha) + \sum r''_\alpha s(h_\alpha) = f(x) + f(y) \end{aligned}$$

and the theorem is proved. $\square$

The above theorem shows that there is an one-to-one correspondence between the set of all additive functions $f : \mathbb{R} \rightarrow \mathbb{R}$ and the set of all functions $s : H \rightarrow \mathbb{R}$. We should note that the additive functions not of the form $f(x) = ax$, $a \in \mathbb{R}$, correspond to the functions $s : H \rightarrow \mathbb{R}$ such that the function $\frac{s(h)}{h}$ is not constant (if s corresponds to the function $f(x) = ax$ then $s(h) = ah$ for any $h \in H$ and $\frac{s(h)}{h} = a$). Hence any such function determines an additive function which does not satisfy the conditions (i)–(iv) of Theorem 1.1.

Example 1.3. Let $H = \{h_\alpha\}$ be a Hamel basis. Consider the additive function $f : \mathbb{R} \rightarrow \mathbb{R}$ determined by the function $s : H \rightarrow \mathbb{R}$ defined by:

$$s(h_\alpha) = \begin{cases} h_\alpha & \text{if } h_\alpha \neq 1 \\ 2 & \text{if } h_\alpha = 1. \end{cases}$$

Then according to Proposition 1.1 we have that $f(x) = xf(1) = xs(1) = 2x$ for any $x \in \mathbb{Q}$. Note that the function $f : \mathbb{R} \rightarrow \mathbb{R}$ is not of the form $f(x) = ax$ since $\frac{s(1)}{1} = 2$ but $\frac{s(h_\alpha)}{h_\alpha} = 1$ for $h_\alpha \neq 1$.

One can show (see eg. [19]) that for any Hamel basis H there exists a bijection $\varphi : [0, 1] \rightarrow H$. If we set $\varphi(t) = h_t$, then $H = \{h_t | t \in [0, 1]\}$. This observation leads to constructions of additive functions having interesting properties.

Example 1.4. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the additive function determined by the function $s : H \rightarrow \mathbb{R}$ defined by:

$$s(h_t) = \begin{cases} h_t & \text{if } t \in \left[0, \frac{1}{2}\right] \\ -h_t & \text{if } t \in \left(\frac{1}{2}, 1\right]. \end{cases}$$

Then it is easy to check that $f(f(x)) = x$ for any $x \in \mathbb{R}$.

Another interesting example of a discontinuous additive function is obtained by considering the function $s : H \rightarrow \mathbb{R}$ defined by

$$s(h_t) = \begin{cases} h_t & \text{if } t \in \left[0, \frac{1}{2}\right] \\ 0 & \text{if } t \in \left(\frac{1}{2}, 1\right]. \end{cases}$$

Then $f(f(x)) = f(x)$ for all $x \in \mathbb{R}$.

Finally we shall show that any additive function on $[0, \infty)$ can be extended in a unique way to an additive function on $\mathbb{R}$.

Theorem 1.5. Let $g : [0, \infty) \rightarrow \mathbb{R}$ be an additive function. Then there exists a unique additive function $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x) = g(x)$ for all $x \in [0, \infty)$.

Proof. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be an additive function such that $f(x) = g(x)$ for all $x \in [0, \infty)$. Since $f(-x) = -f(x)$ for all $x \in \mathbb{R}$ we see that

$$f(x) = \begin{cases} g(x) & \text{if } x \geq 0 \\ -g(-x) & \text{if } x \leq 0 \end{cases} \quad (1.4)$$

This proves the uniqueness of the extension f of g . To prove the existence let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the function defined by (1.4). We shall prove that f is an additive function. If $x \geq 0$ and $y \geq 0$, then $x + y \geq 0$ and we get

$$f(x) + f(y) = g(x) + g(y) = g(x + y) = f(x + y).$$

If $x \leq 0$ and $y \leq 0$, then

$$f(x) + f(y) = -(g(-x) + g(-y)) = -g(-x - y) = f(x + y).$$

Let now $x \leq 0$ and $y \geq 0$. If $x + y \geq 0$ then

$$f(y) = g(y) = g((x + y) + (-x)) = g(x + y) + g(-x) = f(x + y) - f(x)$$

and therefore $f(x + y) = f(x) + f(y)$. Proceeding in the same way we see that the latter identity also holds true if $x + y \leq 0$. Hence $f : \mathbb{R} \rightarrow \mathbb{R}$ is an additive function and the theorem is proved. $\square$

In particular, the above theorem shows that there is a one-to-one correspondence between the class of additive functions on $[0, \infty)$ and the class of additive functions on $\mathbb{R}$.

The additive Cauchy equation (1.1) is related to the additive structure of the set of real numbers and as we saw in Theorem 1.1 it can be used to characterize the class of linear functions $f(x) = ax$, $a \in \mathbb{R}$.

Next we consider some analogous functional equations that are related to both the additive and the multiplicative structure of $\mathbb{R}$. As we shall see these equations can be used to characterize the logarithmic function $\lg x$, the exponential function a^x , $a > 0$, and the power function x^a , $a \in \mathbb{R}$.

1.2 Logarithmic Cauchy Equation

A well known property of the logarithmic function $\lg x$ is that

$$\lg xy = \lg x + \lg y \text{ for all } x, y \in (0, \infty).$$

This identity suggests considering the functional equation

$$f(xy) = f(x) + f(y), \quad x, y \in (0, \infty), \quad (1.5)$$

which is called the logarithmic Cauchy equation [2]. We shall show that its solutions are related to the solutions of the additive Cauchy equation (1.1) by means of the logarithmic function.

Theorem 1.6. Any solution $f : (0, \infty) \rightarrow \mathbb{R}$ of the logarithmic Cauchy equation has the form $f(x) = g(\lg x)$, where $g : \mathbb{R} \rightarrow \mathbb{R}$ is an additive function.

Proof. We shall use the fact that for any $x \in (0, \infty)$ there is a unique $u \in \mathbb{R}$ such that $x = 10^u$. In fact $u = \lg x$. For any $u \in \mathbb{R}$ set $g(u) = f(10^u)$. Then

$$g(u + v) = f(10^{u+v}) = f(10^u \cdot 10^v) = f(10^u) + f(10^v) = g(u) + g(v).$$

Hence $g : \mathbb{R} \rightarrow \mathbb{R}$ is an additive function such that

$$f(x) = g(\lg x), \quad x \in (0, \infty). \quad \square$$

An immediate consequence of Theorems 1.1 and 1.6 is the following characterization of the logarithmic function:

Corollary 1.2. Let $f : (0, \infty) \rightarrow \mathbb{R}$ be a solution of the logarithmic Cauchy equation having one of the following properties:

- (i) f is bounded above (below) on an interval;
- (ii) f is increasing (decreasing) on an interval;
- (iii) f is continuous at a point.

Then $f(x) = a \lg x$, where $a \in \mathbb{R}$.

We next consider the logarithmic Cauchy equation for functions $f : D \rightarrow \mathbb{R}$ whose domain D is different from $(0, \infty)$. Note first that if $D = [0, \infty)$ then $f(x) = 0$ for all $x \in [0, \infty)$. Indeed, setting $y = 0$ in (1.5) we get

$$f(0) = f(x) + f(0), \text{ i.e. } f(x) = 0.$$

The case $D = \mathbb{R} \setminus \{0\}$ is more interesting.

Theorem 1.7. Any solution $f : \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R} \setminus \{0\}$ of the logarithmic Cauchy equation has the form $f(x) = g(\lg |x|)$, where $g : \mathbb{R} \rightarrow \mathbb{R}$ is an additive function.

If f has one of the properties (i)–(iii) in Corollary 1.2 then $f(x) = a \lg |x|$, where $a \in \mathbb{R}$.

Proof. Setting $x = y = 1$ in (1.5) we get that $f(1) = 0$. Then setting $x = y = -1$ gives $f(-1) = 0$. Moreover, it follows from Theorem 1.6 that $f(x) = g(\lg x)$ for any $x \in \mathbb{R} \setminus \{0\}$, where $g : \mathbb{R} \rightarrow \mathbb{R}$ is an additive function. Now for any $x \in \mathbb{R} \setminus \{0\}$ we have that

$$f(x) = f(|x| \cdot \operatorname{sgn} x) = f(|x|) + f(\operatorname{sgn} x) = f(|x|) = g(\lg |x|). \quad \square$$

Another interesting case is $D = (0, 1]$.

Theorem 1.8. Any solution $f : (0, 1] \rightarrow \mathbb{R}$ of the logarithmic Cauchy equation has the form $f(x) = g(-\lg x)$, where $g : [0, \infty) \rightarrow \mathbb{R}$ is an additive function. If g satisfies one of the conditions (i)–(iii) of Corollary 1.2, then $g(x) = a \lg x$, where $a \in \mathbb{R}$.

Proof. For any $x \in (0, 1]$ there is a unique $u \in [0, \infty)$ such that $x = 10^{-u}$. Then setting $g(u) = f(10^{-u})$ for any $u \in [0, \infty)$ we see as in the proof of Theorem 1.6 that $f : [0, \infty) \rightarrow \mathbb{R}$ is an additive function such that

$$f(x) = g(-\lg x) \text{ for any } x \in (0, 1].$$

The last statement follows from Theorem 1.1. $\square$

1.3 Exponential Cauchy Equation

We now discuss the functional equation

$$f(x + y) = f(x) \cdot f(y) \quad (1.6)$$

for functions $f : D \rightarrow \mathbb{R}$, where $D = \mathbb{R}$, $(0, \infty)$ or $[0, \infty)$. It can be considered as a characteristic equation of the exponential functions $f(x) = a^x$, where $a > 0$.

Theorem 1.9. Let $f : D \rightarrow \mathbb{R}$ be a solution of the exponential Cauchy equation, where $D = \mathbb{R}$ or $(0, \infty)$. Then either $f \equiv 0$ or $f(x) = 10^{g(x)}$, where $g : D \rightarrow \mathbb{R}$ is an additive function.

Proof. Suppose first that $f(x_0) = 0$ for some $x_0 \in D$. Then it follows from (1.6) that $f(x_0 + y) = f(x_0)f(y) = 0$ for any $y \in D$. If $D = \mathbb{R}$ this implies that $f(x) = 0$ for all $x \in \mathbb{R}$. If $D = (0, \infty)$ then $f(x) = 0$ for all $x \geq x_0$. Let now $x \in (0, x_0)$. Then there exists a positive integer k such that $kx > x_0$. On the other hand, using (1.6) it follows by induction on n that $f(nx) = f(x)^n$ and therefore $f(x)^k = f(kx) = 0$, i.e. $f(x) = 0$.

Now assume that $f(x) \neq 0$ for all $x \in D$. Then it follows from (1.6) that $f(x) = \left(f\left(\frac{x}{2}\right)\right)^2 > 0$ for all $x \in \mathbb{R}$. Set $f(x) = 10^{g(x)}$. Then using again (1.6) we see that for all $x, y \in D$ we have

$$10^{g(x)+g(y)} = 10^{g(x)} \cdot 10^{g(y)} = f(x)f(y) = f(x+y) = 10^{g(x+y)},$$

i.e. $g(x+y) = g(x) + g(y)$.

Conversely, if $f \equiv 0$ or $f(x) = 10^{g(x)}$, where $g : D \rightarrow \mathbb{R}$ is an additive function, then f is a solution of (1.6). $\square$

The following result is an immediate consequence of Theorems 1.1 and 1.9.

Corollary 1.3. Let $f : D \rightarrow \mathbb{R}$ be a solution of the exponential Cauchy equation, where $D = \mathbb{R}$ or $(0, \infty)$.

- (i) If f is strictly increasing (decreasing) on an interval, then $f(x) = a^x$, where $a > 1$ ($a < 1$).

- (ii) If f is continuous at a point or bounded above on an interval, then either $f \equiv 0$ or $f(x) = a^x$, where $a > 0$.
- (iii) If on an interval f is bounded below by a positive constant, then $f(x) = a^x$, where $a > 0$.

We now consider the exponential Cauchy equation (1.6) for functions f defined on $[0, \infty)$. In this case a new solution arises and this is the function

$$\delta(x) = \begin{cases} 1 & \text{if } x = 0 \\ 0 & \text{if } x > 0 \end{cases} \quad (1.7)$$

Theorem 1.10. Let $f : [0, \infty) \rightarrow \mathbb{R}$ be a solution of the exponential Cauchy equation. Then either $f \equiv 0$, $f = \delta$ or $f(x) = 10^{g(x)}$, where $g : [0, \infty) \rightarrow \mathbb{R}$ is an additive function.

Proof. Setting $x = y = 0$ in (1.6) gives $f(0) = f(0)^2$. If $f(0) = 0$, then setting $y = 0$ in (1.6) we get that $f(x) = 0$ for all $x \in [0, \infty)$. Suppose now that $f(0) \neq 0$. Then $f(0) = 1$. Using the same reasoning as in the proof of Theorem 1.9 we see that if $f(x_0) = 0$ for some $x_0 > 0$, then $f(x) = 0$ for all $x > 0$. Hence in this case $f = \delta$, where δ is the function defined by (1.7). Assume now that $f(x) \neq 0$ for all $x \in [0, \infty)$. Then $f(x) = 10^{g(x)}$, where $g : [0, \infty) \rightarrow \mathbb{R}$ is an additive function. Note that for any such function we have $g(0) = 0$ and therefore the condition $f(0) = 1$ is satisfied. $\square$

We should remark that in the case $D = [0, \infty)$ Theorem 1.1 implies the same statement as Corollary 1.3 but now we have to add the solution $f = \delta$ in the cases (ii) and (iii).

1.4 Multiplicative Cauchy Equation

Let D be a set of real numbers such that $xy \in D$ for all $x, y \in D$.

A function $f : D \rightarrow \mathbb{R}$ is called multiplicative if

$$f(xy) = f(x)f(y) \quad (1.8)$$

for all $x, y \in D$. The functional equation (1.8) is the multiplicative analogue of the additive Cauchy equation (1.1) and its typical solutions are the power

functions $f(x) = |x|^a$, where $a \in \mathbb{R}$. We next discuss the multiplicative Cauchy equation (1.8) for functions defined on the sets $D = (0, \infty)$, $[0, \infty)$, $\mathbb{R}$ and $\mathbb{R} \setminus \{0\}$.

Theorem 1.11. Let $f : (0, \infty) \rightarrow \mathbb{R}$ be a multiplicative function. Then either $f \equiv 0$ or $f(x) = 10^{g(\lg x)}$, where $g : \mathbb{R} \rightarrow \mathbb{R}$ is an additive function. In particular, if f is continuous at a point, bounded above or monotone on an interval, then either $f = 0$ or $f(x) = x^a$, where $a \in \mathbb{R}$.

Proof. Let $f(x_0) = 0$ for some $x_0 \in (0, \infty)$. Then $f(x) = f\left(\frac{x}{x_0}\right)f(x_0) = 0$ for all $x \in (0, \infty)$. Suppose now that $f(x) \neq 0$ for every $x \in (0, \infty)$ and set $f(x) = 10^{g(\lg x)}$. Then for all $x, y \in (0, \infty)$ we have that

$$10^{g(\lg xy)} = f(xy) = f(x)f(y) = 10^{g(\lg x)} \cdot 10^{g(\lg y)} = 10^{g(\lg x) + g(\lg y)}.$$

Hence $g(\lg x + \lg y) = g(\lg xy) = g(\lg x) + g(\lg y)$ for all $x, y \in (0, \infty)$ and therefore $g : \mathbb{R} \rightarrow \mathbb{R}$ is an additive function.

The last part of the theorem follows from Theorem 1.1. $\square$

Theorem 1.12. Let $f : [0, \infty) \rightarrow \mathbb{R}$ be a multiplicative function. Then either

$$f \equiv 0, f \equiv 1 \text{ or } f(x) = \begin{cases} 0 & \text{if } x = 0 \\ 10^{g(\lg x)} & \text{if } x > 0 \end{cases},$$

where $g : \mathbb{R} \rightarrow \mathbb{R}$ is an additive function.

Proof. Setting $y = 0$ in (1.8) gives $f(0)(1 - f(x)) = 0$ for all $x \in [0, \infty)$. If $f(0) \neq 0$, then $f(x) = 1$ for all $x \in [0, \infty)$. If $f(0) = 0$, the Theorem follows from Theorem 1.11 since f is a multiplicative function on $(0, \infty)$. $\square$

Theorem 1.13. Let $f : \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}$ be a multiplicative function. Then either $f \equiv 0$, $f(x) = 10^{g(\lg |x|)}$ or $f(x) = \operatorname{sgn} x \cdot 10^{g(\lg |x|)}$, where $g : \mathbb{R} \rightarrow \mathbb{R}$ is an additive function.

Proof. Setting $y = 1$ in (1.8) we get $f(x)[1 - f(1)] = 0$. If $f(1) \neq 1$, then $f(x) = 0$ for all $x \in \mathbb{R} \setminus \{0\}$. Suppose now that $f(x) \neq 0$. Then $f(1) = 1$ and

setting $x = y = -1$ in (1.8) gives $1 = f(-1)^2$. Let $f(-1) = 1$. Then for any $x \in \mathbb{R} \setminus \{0\}$ we have that

$$f(x) = f(|x| \cdot \operatorname{sgn} x) = f(|x|)f(\operatorname{sgn} x) = f(|x|).$$

Hence it follows from Theorem 1.11 that $f(x) = 10^{g(\lg|x|)}$, where $g : \mathbb{R} \rightarrow \mathbb{R}$ is an additive function. In the case $f(-1) = -1$ we get as above that

$$f(x) = \operatorname{sgn} x 10^{g(\lg|x|)},$$

where $g : \mathbb{R} \rightarrow \mathbb{R}$ is an additive function. □

Finally, combining Theorems 1.12 and 1.13 we get the following

Theorem 1.14. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a multiplicative function. Then either

$$f \equiv 0, \quad f \equiv 1, \quad f(x) = \begin{cases} 0 & \text{if } x = 0 \\ 10^{g(\lg|x|)} & \text{if } x \neq 0 \end{cases}$$

or

$$f(x) = \begin{cases} 0 & \text{if } x = 0 \\ \operatorname{sgn} x \cdot 10^{g(\lg|x|)} & \text{if } x \neq 0 \end{cases},$$

where $g : \mathbb{R} \rightarrow \mathbb{R}$ is an additive function.

1.5 Practice Problems

Problem 1.1. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be an additive function such that $f(1) = 0$. Then for each non-empty interval $I \subset \mathbb{R}$ we have $f(I) = f(\mathbb{R})$.

Problem 1.2. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be an additive function. Prove that:

- (a) either $\operatorname{Ker} f = \{0\}$ or $\operatorname{Ker} f$ is a dense subset of $\mathbb{R}$;
- (b) either $\operatorname{Im} f = \{0\}$ or $\operatorname{Im} f$ is a dense subset of $\mathbb{R}$.

Problem 1.3. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a non-injective additive function. Prove that for any $a \in \mathbb{R}$ the set $F_a = \{x \in \mathbb{R} \mid f(x) = f(a)\}$ is dense in $\mathbb{R}$.

Problem 1.4. (Romania TST 2006) Given $r, s \in \mathbb{Q}$, determine all functions $f : \mathbb{Q} \rightarrow \mathbb{Q}$ such that

$$f(x + f(y)) = f(x + r) + y + s$$

for all $x, y \in \mathbb{Q}$.

Problem 1.5. Find all functions $f : \mathbb{Q} \rightarrow \mathbb{Q}$ such that

$$f(x + f(x) + 2y) = 2x + 2f(f(y))$$

for all $x, y \in \mathbb{Q}$.

Problem 1.6. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be an additive function such that

$$f\left(\frac{1}{x}\right) = \frac{f(x)}{x^2}$$

for all $x \neq 0$. Prove that $f(x) = cx$, $c \in \mathbb{R}$.

Problem 1.7. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be an additive function such that $f(1) = 1$ and

$$f(x)f\left(\frac{1}{x}\right) = 1$$

for all $x \neq 0$. Prove that $f(x) = x$.

Problem 1.8. Determine all continuous functions $f : (0, \infty) \rightarrow \mathbb{R}$ such that

$$f(xy) = xf(y) + yf(x)$$

for all $x, y \in (0, \infty)$.

Problem 1.9. Let $\mathbb{Z}(\sqrt{2}) = \{a + b\sqrt{2} | a, b \in \mathbb{Z}\}$. Find all increasing additive functions $f : \mathbb{Z}(\sqrt{2}) \rightarrow \mathbb{R}$.

Problem 1.10. Let $D \subset \mathbb{R}$ be a non-empty set such that $x + y \in D$ for all $x, y \in D$ and let $f : D \rightarrow \mathbb{R}$ be an increasing additive function. Prove that $f(x) = ax$, $x \in D$, where $a \geq 0$ is a constant.

Problem 1.11. Find all continuous functions $f, g : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$\begin{cases} f(x+y) = f(x)f(y) + g(x)g(y) \\ g(x+y) = f(x)g(y) + g(x)f(y) \end{cases}$$

for all $x, y \in \mathbb{R}$.

Problem 1.12. (Romania TST 1996) Let $a \in \mathbb{R}$ and $f_1, f_2, \dots, f_n : \mathbb{R} \rightarrow \mathbb{R}$ be additive functions such that

$$f_1(x)f_2(x) \dots f_n(x) = ax^n$$

for all $x \in \mathbb{R}$. Prove that:

- (a) if $a \neq 0$, then $f_1, f_2, \dots, f_n$ are linear functions;
- (b) if $a = 0$, then there exists i such that $f_i \equiv 0$.

Problem 1.13. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function such that

$$|f(x+y) - f(x) - f(y)| \leq 1$$

for all $x, y \in \mathbb{R}$. Prove that there exists an additive function $g : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$|f(x) - g(x)| \leq 1$$

for all $x \in \mathbb{R}$.

Chapter 2

Generalized Cauchy Equations

2.1 Jensen's Equation

In this chapter we consider some natural generalizations of Cauchy's equations discussed above. For the sake of simplicity in most cases we shall determine only their continuous solutions.

Let $f(x) = ax + b$, $a, b \in \mathbb{R}$, be a linear function. Then it is easy to check that

$$f\left(\frac{x+y}{2}\right) = \frac{f(x) + f(y)}{2} \quad (2.1)$$

for all $x, y \in \mathbb{R}$. Considering (2.1) as a functional equation we come to the so-called Jensen's equation [2].

Note that if $g : \mathbb{R} \rightarrow \mathbb{R}$ is an additive function and c is a real constant then it follows from Proposition 1.1 that the function $f(x) = g(x) + c$ is a solution of (2.1). Conversely, we shall prove that these are the only solutions of Jensen's equation.

Theorem 2.1. Let $I \subset \mathbb{R}$ be an arbitrary interval and $f : I \rightarrow \mathbb{R}$ a function satisfying (2.1) for all $x, y \in I$. Then there exists a unique additive function $g : \mathbb{R} \rightarrow \mathbb{R}$ and a constant $c \in \mathbb{R}$ such that $f(x) = g(x) + c$ for all $x \in I$.

Proof. We shall consider only the case when I is a finite open interval since the other cases can be treated in the same way. Suppose first that the interval I is symmetric with respect to 0, i.e. $I = (-d, d)$ for some $d > 0$. Consider the function $\tilde{f} : I \rightarrow \mathbb{R}$ defined by $\tilde{f}(x) = f(x) - f(0)$. Then $\tilde{f}(0) = 0$ and (2.1) implies that $\tilde{f}\left(\frac{x+y}{2}\right) = \frac{1}{2}(\tilde{f}(x) + \tilde{f}(y))$ for all $x, y \in I$. Setting $y = 0$ we get $\tilde{f}\left(\frac{x}{2}\right) = \frac{\tilde{f}(x)}{2}$ and it follows by induction on n that

$$\tilde{f}(x) = 2^n \tilde{f}\left(\frac{x}{2^n}\right) \quad (2.2)$$

for any $x \in I$ and any nonnegative integer n .

Now we shall define an additive function $g : \mathbb{R} \rightarrow \mathbb{R}$ such that $g(x) = \tilde{f}(x)$ for any $x \in I$. To do this note that for any $x \in \mathbb{R}$ there exists a nonnegative integer n (depending on x) such that $\frac{x}{2^n} \in (-d, d)$. Then we set $g(x) = 2^n \tilde{f}\left(\frac{x}{2^n}\right)$. The function $g : \mathbb{R} \rightarrow \mathbb{R}$ is defined correctly since if $\frac{x}{2^n}, \frac{x}{2^m} \in (-d, d)$ and $m > n$, then it follows from (2.2) that

$$2^n \tilde{f}\left(\frac{x}{2^n}\right) = 2^n \left[2^{m-n} \tilde{f}\left(\frac{1}{2^{m-n}} \left(\frac{x}{2^n}\right)\right) \right] = 2^m \tilde{f}\left(\frac{x}{2^m}\right).$$

Let now x, y be arbitrary real numbers and n be a nonnegative integer such that $\frac{x}{2^n}, \frac{y}{2^n} \in (-d, d)$. Then $\frac{x+y}{2^{n+1}} \in (-d, d)$ and we get

$$\begin{aligned} g(x+y) &= 2^{n+1} \tilde{f}\left(\frac{x+y}{2^{n+1}}\right) = 2^{n+1} \left[\frac{1}{2} \left(\tilde{f}\left(\frac{x}{2^n}\right) + \tilde{f}\left(\frac{y}{2^n}\right) \right) \right] = \\ &= 2^n \tilde{f}\left(\frac{x}{2^n}\right) + 2^n \tilde{f}\left(\frac{y}{2^n}\right) = g(x) + g(y). \end{aligned}$$

Hence $g : \mathbb{R} \rightarrow \mathbb{R}$ is an additive function such that $g(x) = \tilde{f}(x)$ for all $x \in I$. Setting $c = f(0)$ we see that $f(x) = g(x) + c$ for all $x \in I$.

Suppose now that there is another additive function $h : \mathbb{R} \rightarrow \mathbb{R}$ and a constant c_1 such that $f(x) = h(x) + c_1$ for all $x \in I$. Since $g(0) = h(0) = 0$ we get $c = c_1 = f(0)$. Let x be an arbitrary real number and let n be a

nonnegative integer such that $\frac{x}{2^n} \in I$. Then it follows from Proposition 1.1 that

$$g(x) = 2^n g\left(\frac{x}{2^n}\right) = 2^n \left[f\left(\frac{x}{2^n}\right) - c \right] = 2^n \left[f\left(\frac{x}{2^n}\right) - c_1 \right] = 2^n h\left(\frac{x}{2^n}\right) = h(x)$$

and the uniqueness of g is proved.

Let now $I = (a, b)$ be an arbitrary finite open interval. Set $d = \frac{b-a}{2}$ and consider the function $F : (-d, d) \rightarrow \mathbb{R}$ defined by

$$F(x) = f\left(x + \frac{a+b}{2}\right).$$

Then

$$\begin{aligned} F\left(\frac{x+y}{2}\right) &= f\left(\frac{x+y}{2} + \frac{a+b}{2}\right) = f\left(\frac{1}{2}\left(x + \frac{a+b}{2} + y + \frac{a+b}{2}\right)\right) = \\ &= \frac{1}{2} \left[f\left(x + \frac{a+b}{2}\right) + f\left(y + \frac{a+b}{2}\right) \right] = \frac{1}{2} (F(x) + F(y)) \end{aligned}$$

and the theorem follows by applying the same arguments as above. $\square$

Theorem 2.1 together with Theorem 1.1 shows that if a function $f : I \rightarrow \mathbb{R}$ satisfies Jensen's equation and is continuous at a point, monotone or bounded above (below) on a subinterval of I then f is a linear function.

2.2 Linear Cauchy Equation

This functional equation is an extension of both Cauchy's and Jensen's equation. It has the form

$$f(ax + by + c) = pf(x) + qf(y) + r, \quad (2.3)$$

where a, b, c, p, q and r are real constants and $ab \neq 0$.

Theorem 2.2. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a nonconstant continuous function satisfying (2.3). Then $f(x) = sx + t$ for some constants $s \neq 0$, $t \in \mathbb{R}$ and $p = a$, $q = b$, $r = sc + t(1 - a - b)$.

Proof. Setting

$$(x, y) = \left(-\frac{c}{a}, 0\right), \left(\frac{u-c}{a}, 0\right), \left(-\frac{c}{a}, \frac{v}{b}\right), \left(\frac{u-c}{a}, \frac{v}{b}\right)$$

in (2.3) gives respectively

$$\begin{aligned} f(0) &= pf\left(-\frac{c}{a}\right) + qf(0) + r, \\ f(u) &= pf\left(\frac{u-c}{a}\right) + qf(0) + r, \\ f(v) &= pf\left(-\frac{c}{a}\right) + qf\left(\frac{v}{b}\right) + r, \\ f(u+v) &= pf\left(\frac{u-c}{a}\right) + qf\left(\frac{v}{b}\right) + r. \end{aligned}$$

Hence $f(u+v) = f(u) + f(v) + f(0)$ for all $u, v \in \mathbb{R}$.

Thus $g(u) = f(u) - f(0)$ is a continuous additive function and we conclude from Theorem 1.1 that $g(x) = sx$ for some constant $s \in \mathbb{R}$, i.e. $f(x) = sx + t$, where $t = f(0)$ and $s \neq 0$. Now plugging $f(x)$ in (2.3) shows that $p = a$, $q = b$ and $r = sc + t(1 - a - b)$. $\square$

Example 2.1. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function such that

$$f(1 + 2x + 3y) = 1 + 2f(x) + 3f(y)$$

for all $x, y \in \mathbb{R}$. Then it follows from Theorem 2.2 that $f(x) = sx + \frac{s-1}{4}$, where s is a constant.

2.3 Pexider's Equations

A special feature of some functional equations is that they determine not only one but several unknown functions. Classical examples are the so-called Pexider's equations [18], [2] which generalize the Cauchy equations.

Example 2.2. Determine all continuous functions $f, g, h : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x + y) = g(x) + h(y) \quad (2.4)$$

for all $x, y \in \mathbb{R}$.

Solution. Setting respectively $x = 0$ and $y = 0$ in (2.4) we see that

$$h(y) = f(y) - g(0) \text{ and } g(x) = f(x) - h(0).$$

Hence

$$f(x + y) = f(x) + f(y) - h(0) - g(0).$$

Consider the function $k : \mathbb{R} \rightarrow \mathbb{R}$ defined by $k(x) = f(x) - h(0) - g(0)$. Then k is a continuous additive function and Theorem 1.1 implies that $k(x) = ax$, where a is a real constant. Thus the continuous solutions of (2.4) have the form

$$f(x) = ax + b + c, \quad g(x) = ax + b, \quad h(x) = ax + c,$$

where a, b, c are arbitrary real constants.

Example 2.3. Determine all continuous functions $f, g, h : (0, \infty) \rightarrow \mathbb{R}$ such that

$$f(xy) = g(x)h(y) \quad (2.5)$$

for all $x, y \in (0, \infty)$.

Solution. Setting $x = y = 1$ in (2.5) we get that $f(1) = g(1)h(1)$. Suppose first that $f(1) \neq 0$. Then setting respectively $x = 1$ and $y = 1$ in (2.5) gives

$$h(y) = \frac{f(y)}{g(1)} \text{ and } g(x) = \frac{f(x)}{h(1)} \text{ for all } x, y \in (0, \infty).$$

$$\text{Hence } f(xy) = \frac{f(x)f(y)}{h(1)g(1)} \text{ for all } x, y \in (0, \infty).$$

Consider the function $k : (0, \infty) \rightarrow \mathbb{R}$ defined by $k(x) = \frac{f(x)}{h(1)g(1)}$. Then the above identity implies that k is a continuous multiplicative function and using Theorem 1.1 we get that $k(x) = x^\alpha$, where α is a real constant. Thus the functions f, g and h have the form $f(x) = abx^\alpha$, $g(x) = ax^\alpha$, $h(x) = bx^\alpha$, $ab \neq 0$. Assume now that $f(1) = g(1)h(1) = 0$. If $g(1) = 0$ ($h(1) = 0$) then setting $x = 1$ ($y = 1$) in (2.5) we get that $f \equiv 0$ and (2.5) implies that either $g \equiv 0$ and h is an arbitrary function or $h \equiv 0$ and g is an arbitrary function.

Example 2.4. Determine all continuous functions $f, g, h : \mathbb{R} \rightarrow (0, \infty)$ such that

$$f(x+y) = g(x)h(y) \quad (2.6)$$

for all $x, y \in \mathbb{R}$.

Solution. Set $f(x) = 10^{F(x)}$, $g(x) = 10^{G(x)}$ and $h(x) = 10^{H(x)}$ for all $x \in \mathbb{R}$. Then (2.6) implies that

$$F(x+y) = G(x) + H(y)$$

for all $x, y \in \mathbb{R}$. Now it follows from Example 2.2 that $F(x) = Ax + B + C$, $G(x) = Ax + B$ and $H(x) = Ax + C$, where A, B, C are real constants. Hence the continuous solutions of (2.6) are given by the exponential functions of the form

$$f(x) = bca^x, \quad g(x) = ba^x, \quad h(x) = ca^x,$$

where $a, b, c > 0$.

Example 2.5. Determine all continuous functions $f, g, h : (0, \infty) \rightarrow \mathbb{R}$ such that

$$f(xy) = g(x) + h(y) \quad (2.7)$$

for all $x, y \in (0, \infty)$.

Solution. Consider the functions $F, G, H : \mathbb{R} \rightarrow \mathbb{R}$ defined by $F(x) = f(10^x)$, $G(x) = g(10^x)$ and $H(x) = h(10^x)$. Then (2.7) implies that

$$F(x+y) = f(10^{x+y}) = f(10^x \cdot 10^y) = g(10^x) + h(10^y) = G(x) + H(y)$$

for all $x, y \in \mathbb{R}$. Using Example 2.2 we get that

$$F(x) = ax + b + c, \quad G(x) = ax + b, \quad H(x) = ax + c.$$

Hence the continuous solutions of (2.4) are given by

$$f(x) = a \lg x + b + c, \quad g(x) = a \lg x + b, \quad h(x) = a \lg x + c,$$

where a, b, c are arbitrary real constants.

2.4 Vincze's Equations

We now consider the functional equation

$$f(x+y) = g(x)k(y) + h(y) \quad (2.8)$$

which generalizes both additive and multiplicative Pexider's equations. This equation has been first studied by Vincze [25] (c.f. also [11], [23]) and we shall determine all its continuous solutions. The case when f is a constant function is easy and we leave it as an exercise to the reader.

Theorem 2.3. Let $f, g, h, k : \mathbb{R} \rightarrow \mathbb{R}$ be functions satisfying (2.8) and f be a nonconstant continuous function. Then either

$$f(x) = ax + b, \quad g(x) = \frac{ax + b - c}{d}, \quad h(y) = ay + c, \quad k(y) = d$$

or

$$f(x) = at^x + b, \quad g(x) = \frac{at^x + b - c}{d}, \quad h(y) = (c - b)t^y + b, \quad k(y) = dt^y,$$

where $a \neq 0, b, c, d \neq 0$ and $t > 0$ are arbitrary constants.

Proof. Set $k(0) = d$ and $h(0) = c$. Then $d \neq 0$ since f is a nonconstant function and putting $y = 0$ in (2.8) gives

$$g(x) = \frac{f(x) - c}{d}.$$

Now setting

$$\alpha(y) = \frac{k(y)}{d}, \quad \beta(y) = h(y) - \frac{ck(y)}{d}$$

we get from (2.8) that

$$f(x+y) = f(x)\alpha(y) + \beta(y). \quad (2.9)$$

In particular $f(y) = f(0)\alpha(y) + \beta(y)$ and setting $\gamma(x) = f(x) - f(0)$ we rewrite (2.9) as

$$\gamma(x+y) = \gamma(x)\alpha(y) + \gamma(y). \quad (2.10)$$

Thus

$$\gamma(x)\alpha(y) + \gamma(y) = \gamma(x+y) = \gamma(y+x) = \gamma(y)\alpha(x) + \gamma(x)$$

and we obtain

$$(\alpha(y) - 1)\gamma(x) = (\alpha(x) - 1)\gamma(y) \quad (2.11)$$

for all $x, y \in \mathbb{R}$. To solve this equation we consider two cases.

Case 1. $\alpha(y) = 1$ for all $y \in \mathbb{R}$.

In this case it follows from (2.10) that γ is a continuous additive function and therefore $\gamma(x) = ax$ for some constant $a \in \mathbb{R}$. Setting $f(0) = b$ we get that

$$f(x) = ax + b, \quad g(x) = \frac{f(x) - c}{d} = \frac{ax + b - c}{d}$$

and $k(y) = d$ since $\alpha(y) = 1$. Then (2.8) implies that $h(y) = ay + c$.

Case 2. $\alpha(y_0) \neq 1$ for some $y_0 \in \mathbb{R}$.

Then setting $y = y_0$ in (2.11) we get

$$\gamma(x) = a(\alpha(x) - 1),$$

where $a = \frac{\gamma(y_0)}{\alpha(y_0) - 1}$. Note that $a \neq 0$ since otherwise $\gamma(x) = 0$ and $f(x)$ is a constant function. Hence (2.10) is equivalent to the equation

$$\alpha(x+y) = \alpha(x)\alpha(y)$$

and Theorem 1.9 implies that $\alpha(y) = t^y$ for some constant $t > 0$. Thus $\gamma(x) = a(t^x - 1)$ and therefore $f(x) = at^x + b$, where $b = f(0) - a$. We have also

$$k(y) = d\alpha(y) = dt^y, \quad g(x) = \frac{f(x) - c}{d} = \frac{at^x + b - c}{d}$$

and (2.8) gives $h(y) = (c - b)t^y + b$. □

Example 2.6. (Romania TST 1998) Find all functions $u : \mathbb{R} \rightarrow \mathbb{R}$ such that there exists a strictly increasing function $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x+y) = f(x)u(y) + f(y) \quad (2.12)$$

for all $x, y \in \mathbb{R}$.

Solution. The equation (2.12) is a particular case of Vincze's equation (2.8) with $f = g = h$. Since the function f is strictly increasing it follows from the proof of Theorem 2.3 and Theorems 1.1 and 1.9 that either $f(x) = ax$, $u(y) = 1$ or $f(x) = a(t^x - 1)$, $u(y) = t^y$, where $a > 0$ and $t > 0$ are constants.

Finally, let us note that one can consider also the multiplicative analog of Vincze's equation

$$f(xy) = g(x)k(y) + h(y) \quad (2.13)$$

Using the same arguments as above one can prove the following

Theorem 2.4. Let $f, g, h, k : (0, \infty) \rightarrow \mathbb{R}$ be continuous functions satisfying (2.13) and f be a nonconstant function. Then either

$$f(x) = a \log x + b, \quad g(x) = \frac{a \log x + b - c}{d}, \quad h(y) = a \log y + c, \quad k(y) = d$$

or

$$f(x) = ax^t + b, \quad g(x) = \frac{ax^t + b - c}{d}, \quad h(y) = (c - b)y^t + b, \quad k(y) = dy^t$$

where $a \neq 0$, $b, c, d \neq 0$ and $t \neq 0$ are arbitrary constants.

2.5 Functions Preserving Mean Values

Let n be a nonzero integer. Then the n -th mean value of two nonnegative real numbers x and y is defined by

$$M_n(x, y) = \left(\frac{x^n + y^n}{2} \right)^{\frac{1}{n}}.$$

It is also convenient to set $M_0(x, y) = \sqrt{xy}$. Recall that $M_1(x, y)$, $M_0(x, y)$, $M_2(x, y)$ and $M_{-1}(x, y)$ are called respectively the Arithmetic mean, the Geometric mean, the Root mean square and the Harmonic mean of the numbers $x, y > 0$.

Note that Jensen's equation (2.1) can be written in the form

$$f(M_1(x, y)) = M_1(f(x), f(y)),$$

i.e. its solutions are the functions preserving the Arithmetic mean of two numbers. Next we shall consider some analogous functional equations which have the form

$$f(M_k(x, y)) = M_n(f(x), f(y)),$$

where k and n are integers. For the sake of brevity we shall determine only their continuous solutions.

Example 2.7. Determine all continuous functions $f : \mathbb{R} \rightarrow [0, \infty)$ such that

$$f\left(\frac{x+y}{2}\right) = \sqrt{f(x)f(y)} \quad (2.14)$$

for all $x, y \in \mathbb{R}$.

Solution. Suppose that $f(x_0) = 0$ for some $x_0 \in \mathbb{R}$ and let t be an arbitrary real number. Then it follows from (2.14) that

$$f\left(\frac{x_0+t}{2}\right) = \sqrt{f(x_0)f(t)} = 0.$$

Since the function $t \rightarrow \frac{x_0+t}{2}$ is surjective (its inverse is $y \rightarrow 2y - x_0$) this shows that f is identically zero.

Assume now that $f(x) > 0$ for all $x \in \mathbb{R}$ and set $f(x) = 10^{g(x)}$. Then it follows from (2.14) that the function $g : \mathbb{R} \rightarrow \mathbb{R}$ satisfies the Jensen equation (2.1). Since g is a continuous function, we conclude that $g(x) = px + q$ for some real constants p and q . Setting $A = 10^q > 0$ and $a = 10^p > 0$ we see that $f(x) = Aa^x$.

Thus any continuous solution of the equation (2.14) has the form $f(x) = Aa^x$, where $A \geq 0$ and $a > 0$.

Note that the functional equation (2.14) has been first considered by Lobachevski in his study of non-Euclidean geometry.

Example 2.8. Determine all continuous functions $f : \mathbb{R} \rightarrow [0, \infty)$ such that

$$f\left(\sqrt{\frac{x^2+y^2}{2}}\right) = \sqrt{f(x)f(y)} \quad (2.15)$$

for all $x, y \in \mathbb{R}$.

Solution. Consider the function $g : [0, \infty) \rightarrow [0, \infty)$ defined by $g(x) = f(\sqrt{x})$. Then (2.15) implies that for any $x, y \geq 0$ we have

$$g\left(\frac{x+y}{2}\right) = f\left(\sqrt{\frac{x+y}{2}}\right) = \sqrt{f(\sqrt{x})f(\sqrt{y})} = \sqrt{g(x)g(y)}.$$

Hence it follows from the solution of Example 2.7 that there are constants $A \geq 0$ and $a > 0$ such that $g(x) = Aa^x$, i.e. $f(x) = Aa^{x^2}$ for $x \geq 0$. On the other hand, we get from (2.15) that

$$f(-x) = \sqrt{f(-x)f(-x)} = f\left(\sqrt{\frac{x^2+x^2}{2}}\right) = \sqrt{f(x)f(x)} = f(x).$$

Hence all continuous solutions of (2.15) have the form $f(x) = Aa^{x^2}$, where $A \geq 0$ and $a > 0$.

Example 2.9. Determine all continuous functions $f : (0, \infty) \rightarrow [0, \infty)$ such that

$$f(\sqrt{xy}) = \sqrt{f(x)f(y)} \quad (2.16)$$

for all $x, y \in (0, \infty)$.

Solution. Consider the function $f : \mathbb{R} \rightarrow [0, \infty)$ defined by $g(x) = f(10^x)$. Then using (2.16) we get that

$$g\left(\frac{x+y}{2}\right) = f\left(10^{\frac{x+y}{2}}\right) = f(\sqrt{10^x \cdot 10^y}) = \sqrt{f(10^x) \cdot f(10^y)} = \sqrt{g(x)g(y)}$$

for all $x, y \in \mathbb{R}$. Hence it follows from Example 2.7 that there are constants $A \geq 0$ and $a > 0$ such that $g(x) = Aa^x$ for all $x \in \mathbb{R}$. Then for any $x > 0$ we have that

$$f(x) = g(\lg x) = Aa^{\lg x} = Ax^{\lg a} = Ax^\alpha.$$

Conversely, for any $A \geq 0$ and $\alpha \in \mathbb{R}$ the function $f(x) = Ax^\alpha$ is a solution of (2.16).

Example 2.10. Determine all continuous functions $f : (0, \infty) \rightarrow \mathbb{R} \setminus \{0\}$ such that

$$f\left(\frac{2}{\frac{1}{x} + \frac{1}{y}}\right) = \frac{2}{\frac{1}{f(x)} + \frac{1}{f(y)}} \quad (2.17)$$

for all $x, y \in (0, \infty)$.

Solution. Consider the function $g : (0, \infty) \rightarrow \mathbb{R} \setminus \{0\}$ defined by

$$g(x) = \frac{1}{f\left(\frac{1}{x}\right)}.$$

Then it follows from (2.17) that

$$g\left(\frac{x+y}{2}\right) = \frac{1}{f\left(\frac{2}{\frac{1}{x} + \frac{1}{y}}\right)} = \frac{\frac{1}{f\left(\frac{1}{x}\right)} + \frac{1}{f\left(\frac{1}{y}\right)}}{2} = \frac{g(x) + g(y)}{2}$$

for all $x, y \in (0, \infty)$. Using Theorem 1.14, we get that $g(x) = ax + b$. Since $g(x) \neq 0$ for all $x \in (0, \infty)$ it follows that $ab \geq 0$, $a^2 + b^2 \neq 0$. Hence all continuous functions $f : (0, \infty) \rightarrow \mathbb{R} \setminus \{0\}$ satisfying the equation (2.17) have the form $f(x) = \frac{1}{g\left(\frac{1}{x}\right)} = \frac{x}{a + bx}$, where $ab \geq 0$, $a^2 + b^2 \neq 0$.

2.6 Practice Problems

Problem 2.1. Find all functions $f : (1, 4) \rightarrow \mathbb{R}$ such that

$$f(x+y) = f(x) + f(y)$$

for all $x, y \in (1, 2)$.

Problem 2.2. Let $f : (1, 6) \rightarrow \mathbb{R}$ be a function such that $f(x+y) = f(x) + f(y)$ for all $x, y \in (1, 3)$. Prove that there is an additive function $g : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x) = g(x)$ for all $x \in (1, 6)$.

Problem 2.3. Find all increasing functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f\left(\frac{x+y}{3}\right) = \frac{f(x) + f(y)}{2}$$

for all $x, y \in \mathbb{R}$.

Problem 2.4. Find all pairs of continuous functions $f, g : \mathbb{R} \rightarrow \mathbb{R}$ that satisfy

$$f(x) + f(y) = g(x+y)$$

for all $x, y \in \mathbb{R}$.

Problem 2.5. Find all continuous functions $f, g, k : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x+y) = g(x) \cdot k(y) + x + 2y$$

for all $x, y \in \mathbb{R}$.

Problem 2.6. Find all continuous functions $f, g, k : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(xy) = g(x)k(y) + y^2$$

for all $x, y \in \mathbb{R}$.

Problem 2.7. Determine all continuous solutions of the following functional equations:

$$(i) \ f(\sqrt{xy}) = \frac{f(x) + f(y)}{2}, \ x, y \in (0, \infty);$$

$$(ii) \ f(\sqrt{xy}) = \sqrt{\frac{f^2(x) + f^2(y)}{2}}, \ x, y \in (0, \infty);$$

$$(iii) \ f(\sqrt{xy}) = \frac{2}{\frac{1}{f(x)} + \frac{1}{f(y)}}, \ x, y \in (0, \infty);$$

$$(iv) \ f \left(\sqrt{\frac{x^2 + y^2}{2}} \right) = \frac{f(x) + f(y)}{2}, \ x, y \in \mathbb{R};$$

$$(v) \ f \left(\sqrt{\frac{x^2 + y^2}{2}} \right) = \sqrt{\frac{f^2(x) + f^2(y)}{2}}, \ x, y \in \mathbb{R};$$

$$(vi) \ f \left(\sqrt{\frac{x^2 + y^2}{2}} \right) = \frac{2}{\frac{1}{f(x)} + \frac{1}{f(y)}}, \ x, y \in \mathbb{R};$$

$$(vii) \ f \left(\frac{2}{\frac{1}{x} + \frac{1}{y}} \right) = \frac{f(x) + f(y)}{2}, \ x, y \in (0, \infty);$$

$$(viii) \ f \left(\frac{2}{\frac{1}{x} + \frac{1}{y}} \right) = \sqrt{f(x)f(y)}, \ x, y \in (0, \infty);$$

$$(ix) \ f \left(\frac{2}{\frac{1}{x} + \frac{1}{y}} \right) = \sqrt{\frac{f^2(x) + f^2(y)}{2}}, \ x, y \in (0, \infty).$$

Problem 2.8. For a real number r , define the function $f_r: (0, \infty) \rightarrow (0, \infty)$ by

$$f_r(x) = \begin{cases} x^r & r \neq 0 \\ \ln(x) & r = 0 \end{cases}$$

(in the case $r = 0$, the range is $\mathbb{R}$) and let g_r be its inverse, i.e.

$$g_r(x) = \begin{cases} x^{\frac{1}{r}} & r \neq 0 \\ e^x & r = 0 \end{cases}$$

Show that the continuous solutions $f: (0, \infty)$ to the equation

$$f(M_k(x, y)) = M_n(f(x), f(y))$$

have the form $g_n(af_k(x) + b)$.

Chapter 3

Reducing to Cauchy

3.1 Examples

In the previous chapters we considered various types of Cauchy's equations. We have dedicated so much space to them because they represent in some sense the cornerstone of functional equations: one cannot claim to master functional equations before they fully understand and know how to apply Cauchy's equations. The multitude of examples is the best proof of it.

Example 3.1. Find all continuous functions $f: \mathbb{R} \rightarrow \mathbb{R}$ that satisfy

$$f(f(x+y)) = f(x) + f(y)$$

for all $x, y \in \mathbb{R}$.

First Solution. We have $f(x+y) + f(0) = f(f(x+y)) = f(x) + f(y)$. So the function $f(x) - f(0)$ is additive and hence $f(x) = ax + b$. Then $f(f(x+y)) = f(x) + f(y)$ means $a^2(x+y) + ab + b = a(x+y) + 2b$, so $a^2 = a$ and $ab = b$. Hence either $a = b = 0$ or $a = 1$ giving $f \equiv 0$ or $f(x) = x + b$, $b \in \mathbb{R}$.

Second Solution. We can apply Pexider's equation (see Chapter 2) to the functions $f \circ f, f, f$ and we get $(f \circ f)(x) = ax + b + c$, $f(x) = ax + b$, $f(x) = ax + c$. Hence $b = c$ and then $(f \circ f)(x) = a(ax + b) + b = a^2x + ab + b$. Since

$(f \circ f)(x) = ax + b + c = ax + 2b$, we obtain $a^2 = a, ab = b$. Thus $a = b = 0$ or $a = 1$ and the solutions are $f \equiv 0$ and $f(x) = x + b, b \in \mathbb{R}$.

Example 3.2. Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ that satisfy

$$f(f(x) + yz) = x + f(y)f(z)$$

for all $x, y, z \in \mathbb{R}$.

Solution. Note that f is injective as setting $x = x_1, x_2$ for $f(x_1) = f(x_2)$ would immediately imply $x_1 = x_2$. Note also that f is surjective because if we fix y, z , the right-hand side spans all $\mathbb{R}$, hence so does the left-hand side. Now pick z_1 with $f(z_1) = 1$ and set $x = 0, z = z_1$ to get $f(f(0) + yz_1) = f(y)$. Then the injectivity implies $f(0) + yz_1 = y$ for any y , possible only for $f(0) = 0, z_1 = 1$. So $f(0) = 0, f(1) = 1$. Next set $y = 0$ to get $f(f(x)) = x$. Now set $z = 1, x = f(u), y = v$ to get $f(u + v) = f(u) + f(v)$. Hence f is additive and if we set $x = 0$ we get $f(yz) = f(y)f(z)$. It follows from Corollary 1.1 that f is the identity function since $f(1) = 1$.

Example 3.3. Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(f(x)^2 + y) = x^2 + f(y)$$

for all $x, y \in \mathbb{R}$.

Solution. If $f(x_1) = f(x_2)$, then setting $x = x_1, x_2$ we get $x_1^2 = x_2^2$ so $x_2 = \pm x_1$. Now consider the function $h: (0, \infty) \rightarrow (0, \infty)$ defined by $h(x) = f(\sqrt{x})^2$. We can rewrite the condition as $f(h(x) + y) = x + f(y)$ for $x > 0$. Then $f(h(u) + h(v) + y) = u + f(h(v) + y) = u + v + f(y) = f(h(u + v) + y)$. Therefore $h(u) + h(v) + y = \pm(h(u + v) + y)$. For any fixed $u, v \in (0, \infty)$ the identity $h(u) + h(v) + y = -(h(u + v) + y)$ cannot hold for all y , so for at least one y we have $h(u) + h(v) + y = h(u + v) + y$. Thus h is additive. As h is non-negative by definition, $h(x) = cx$ where $c \geq 0$. Using the same reasoning as above, we deduce that $f(x) = \pm\sqrt{cx}$ for all $x \in \mathbb{R}$. Hence $f(cx^2 + y) = x^2 + f(y)$. If $f(y) = -\sqrt{cy}$ for some $y \neq 0$, then $f(cx^2 + y) = x^2 - \sqrt{cy}$. So $f(cx^2 + y)^2 = (x^2 - \sqrt{cy})^2 \neq c(cx^2 + y)^2$ for at least one x , because $(x^2 - \sqrt{cy})^2 = c(cx^2 + y)^2$ is equivalent to the not identically zero polynomial

equation $(c^3 - 1)x^4 + 2(c^2 + \sqrt{c})yx^2 = 0$. So $f(y) = \sqrt{c}y$. In this case we get analogously $(x^2 + \sqrt{c}y)^2 = c(cx^2 + y)^2$ which implies $c = 1$. Hence $f(x) = x$ for all x and this function satisfies our requirement.

Example 3.4. Find all functions $f: (0, \infty) \rightarrow (0, \infty)$ such that

$$f(xf(x) + f(y)) = f(x)^2 + y$$

for all $x, y \in (0, \infty)$.

Solution. Note that we shall meet the same equation in the upcoming Problem 4.17. This problem differs from Problem 4.17 only by the fact that the set $\mathbb{R}$ is replaced by $(0, \infty)$ but this increases its difficulty considerably. For example we can not use the value $f(0)$ (as we shall do in the solution of Problem 4.17) as well as to compute directly the value of f at some particular positive number. That is why we shall first reduce the given equation to the additive Cauchy equation.

To this end we set $f(1) = a$. Then

$$f(f(y) + a) = a^2 + y \tag{1}$$

and

$$f(xf(x) + a) = f(x)^2 + 1. \tag{2}$$

It follows from (1) that

$$f(y) + a + a^2 = f(f(f(y) + a) + a) = f(y + a + a^2).$$

Now induction on n gives

$$f(y + n(a + a^2)) = f(y) + n(a + a^2) \tag{3}$$

for all $n \in \mathbb{N}$. On the other hand (1) and (2) imply that

$$xf(x) + a + a^2 = f(f(xf(x) + a) + a) = f(f(x)^2 + 1 + a).$$

This together with the given equation implies that

$$f(f(f(x)^2 + 1 + a) + f(y)) = f(xf(x) + a + a^2 + f(y)) = f(x)^2 + y + a + a^2. \tag{4}$$

It follows from (1) that the function $f(x)$ attains any value greater than a^2 and therefore the function $f(x)^2 + 1 + a$ attains any value greater than $a^4 + a + 1$. Hence replacing $f(x)^2 + 1 + a$ by x in (4) shows that

$$f(f(x) + f(y)) = x + y + a^2 - 1 \quad (5)$$

for all $x > a^4 + a + 1, y > 0$. Now it follows from (3) that (5) is fulfilled for all $x, y > 0$. Indeed, given $x, y > 0$ choose $n \in \mathbb{N}$ such that $x + n(a + a^2) > a^4 + a + 1$. Then

$$\begin{aligned} f(f(x) + f(y)) &= f(f(x + n(a + a^2)) + f(y)) - n(a + a^2) \\ &= x + n(a + a^2) + y + a^2 - 1 - n(a + a^2) = x + y + a^2 - 1. \end{aligned}$$

Replacing x and y respectively by $f(x) + a$ and $f(y) + a$ in (5) and using (1), we get

$$f(x + y + 2a^2) = f(x) + f(y) + a^2 + 2a - 1. \quad (6)$$

Hence

$$f(x + 2a^2) + f(y - 2a^2) = f(x + y + 2a^2) + 1 - 2a - a^2 = f(x) + f(y).$$

This shows that $f(x + 2a^2) - f(x) = b$ for all $x \in (0, \infty)$, where b is a constant. Set $g(x) = f(x) + c$, where $c = a^2 + 2a - 1 - b$. Then (6) can be rewritten as $g(x + y) = g(x) + g(y)$ for all $x, y \in (0, \infty)$. Taking into account that $f(x) > 0$ we see that $g(x) > c$ for all $x \in (0, \infty)$. Hence it follows from Theorems 1.1 and 1.5 that $g(x) = g(1)x$, i.e. $f(x) = g(1)x - c$ for all $x \in (0, \infty)$. Now it is easy to check that this function satisfies the given conditions if and only if $f(x) = x$ for all $x \in (0, \infty)$.

Example 3.5. (Bulgaria 2004) Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$(f(x) - f(y))f\left(\frac{x+y}{x-y}\right) = f(x) + f(y)$$

for all $x \neq y$.

Solution. It follows that if $x \neq y$ and $f(x) = f(y)$, then

$$f(x) = f(y) = 0.$$

Assume now that $f(a) = 0$ for some $a \neq 0$. Then either $f(x) = 0$, or

$$f\left(\frac{x+a}{x-a}\right) = 1.$$

Because the function $\frac{x+a}{x-a}$ is injective on $\mathbb{R} \setminus \{a\}$, $f\left(\frac{x+a}{x-a}\right) = 1$ can only happen for at most one value of x . Thus $f(x) = 0$ for all x except maybe some x_0 . Choose some y such that $\frac{x_0+y}{x_0-y} \neq x_0$ and then the condition directly implies $f(x_0) = 0$. Thus, either $f \equiv 0$, or $f(x) \neq 0$ for $x \neq 0$.

Let now $f(x) \neq 0$ for $x \neq 0$. Then f is an injection. For $y = 0$ we get that $f(x)(f(1) - 1) = f(0)(f(1) + 1)$. Since f is not constant, then $f(1) = 1$ and $f(0) = 0$. Replacing y by xy , we obtain that

$$f\left(\frac{1+y}{1-y}\right) = \frac{f(x) + f(xy)}{f(x) - f(xy)}.$$

In particular,

$$f\left(\frac{1+y}{1-y}\right) = \frac{f(1) + f(y)}{f(1) - f(y)}.$$

Since $f(1) = 1$, it follows that

$$\frac{f(x) + f(xy)}{f(x) - f(xy)} = \frac{1 + f(y)}{1 - f(y)}.$$

It is easy to see that this equality is equivalent to

$$f(xy) = f(x)f(y),$$

i.e. f is a multiplicative function. In particular $f(x) > 0$ for $x > 0$ and the given identity implies that $f(x) > f(y)$ for $x > y > 0$. Hence by Theorem 1.11 $f(x) = x^\alpha$ for $x > 0$. Substituting this function in the given identity shows that $\alpha = 1$.

So the solutions are $f(x) = 0$ and $f(x) = x$ for any x .

Example 3.6. Given an integer $n \geq 2$ find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x^n + f(y)) = f(x)^n + y$$

for all $x, y \in \mathbb{R}$.

Solution. Set $f(0) = a$. Then

$$f(f(y)) = y + a^n. \quad (1)$$

Applying this identity twice we get

$$f(f(x^n + f(f(y)))) = x^n + f(f(y)) + a^n = x^n + y + 2a^n.$$

On the other hand using the given identity and (1) gives

$$f(f(x^n + f(f(y)))) = f(f(x)^n + f(y)) = f(f(x))^n + y = (x + a^n)^n + y.$$

Hence $x^n + 2a^n = (x + a^n)^n$ for all $x \in \mathbb{R}$. Comparing the coefficients of x^{n-1} on both sides we conclude that $a = 0$. Thus (1) takes the form $f(f(y)) = y$ and the given condition for $y = 0$ gives

$$f(x^n) = f(x)^n. \quad (2)$$

Then for all $x \in [0, \infty)$ and $y \in \mathbb{R}$ we get

$$f(x + y) = f((\sqrt[n]{x})^n + f(f(y))) = f(\sqrt[n]{x})^n + f(y) = f(x) + f(y),$$

i.e. $f(x+y) = f(x) + f(y)$. Now it follows easily that the function f is additive on the whole $\mathbb{R}$.

Next we will show that either $f(x) \geq 0$ for all $x \in (0, \infty)$ or $f(x) \leq 0$ for all $x \in (0, \infty)$. Set $f(1) = b$. Then for every $x \in \mathbb{R}$ and $r \in \mathbb{Q}_+$ we have

$$\begin{aligned} \sum_{k=0}^n \binom{n}{k} f(x^k) r^{n-k} &= f((x+r)^n) = f(x+r)^n \\ &= (f(x) + br)^n = \sum_{k=0}^n \binom{n}{k} f(x)^k (br)^{n-k}. \end{aligned}$$

Comparing the coefficients of r^{n-2} on both sides gives $f(x^2) = b^{n-2}f(x)^2$. This shows that for all $x \in (0, \infty)$ the sign of $f(x)$ is that of b^{n-2} .

Thus the additive function f is bounded below or above on $(0, \infty)$ and therefore (see Theorem 1.1) $f(x) = bx$ for all $x \in \mathbb{R}$. Going back to the given

condition we get $b(x^n + by) = b^n x^n + y$ for all $x, y \in \mathbb{R}$. Hence $b = b^n$ and $b^2 = 1$ which shows that $b = 1$ for n even and $b = \pm 1$ for n odd. Thus the solutions of the problem are the following functions: $f(x) = x$ if n is even; $f(x) = x$ and $f(x) = -x$ if n is odd.

Remarks. a) For $n = 2$ this is Problem 2 from IMO 1992.

b) If $n = 1$ the above equation is equivalent to $f(x + y) = f(x) + f(y)$ and $f(f(x)) = x$ for all $x, y \in \mathbb{R}$. We should note that there exist functions which satisfy these conditions but are unbounded on any interval (c.f. Example 1.4).

Example 3.7. Find all continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x + y - f(y)) = f(x) + f(y - f(y))$$

for all $x, y \in \mathbb{R}$.

Solution. We shall prove that $f(x) = x$. Setting $x = 0$ in the given equation we get that $f(0) = 0$. Assume now that $f(x) \neq x$. Then, by continuity, $x - f(x)$ spans an interval containing 0. We may assume that it contains an interval $\Delta = [0, a], a > 0$ (the case $a < 0$ is similar). It follows by induction that $f(x + nz) = f(x) + nf(z)$ for all $x \in \mathbb{R}, z \in \Delta$ and $n \in \mathbb{N}$. For a fixed $t > 0$ choose $n \in \mathbb{N}$ such that $\frac{t}{n} \in \Delta$. Then

$$f(x + t) = f(x) + nf\left(\frac{t}{n}\right) = f(x) + f(t).$$

Setting $x = -t$ in the above equation gives $f(-t) = -f(t)$ and hence $f : \mathbb{R} \rightarrow \mathbb{R}$ is an additive function. Since it is continuous we conclude that $f(x) \equiv x$, a contradiction.

3.2 Practice Problems

Problem 3.1. (Bulgaria 1994) Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$xf(x) - yf(y) = (x - y)f(x + y)$$

for all $x, y \in \mathbb{R}$.

Problem 3.2. Find all functions $f : \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}$ such that for all nonzero numbers x, y with $x \neq y$ we have

$$f(y) - f(x) = f(y)f\left(\frac{x}{x-y}\right).$$

Problem 3.3. (AMM 2001, Bulgaria 2003) Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x^2 + y + f(y)) = 2y + f(x)^2$$

for all $x, y \in \mathbb{R}$.

Problem 3.4. (Vietnam TST 2004) Find all real values of a for which there exists exactly one function $f : \mathbb{R} \mapsto \mathbb{R}$ satisfying the equation

$$f(x^2 + y + f(y)) = f(x)^2 + ay$$

for all $x, y \in \mathbb{R}$.

Problem 3.5. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ that satisfy

$$f(x+y) + f(xy) = f(x)f(y) + 1$$

for all $x, y \in \mathbb{R}$.

Problem 3.6. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x+y) + f(x)f(y) = f(xy) + f(x) + f(y)$$

for all $x, y \in \mathbb{R}$.

Problem 3.7. (IMO Shortlist 2007) Find all functions $f : (0, \infty) \rightarrow (0, \infty)$ such that

$$f(x + f(y)) = f(x+y) + f(y)$$

for all $x, y \in (0, \infty)$.

Problem 3.8. (Romania 2009) Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x^3 + y^3) = xf(x^2) + yf(y^2)$$

for all $x, y \in \mathbb{R}$.

Problem 3.9. Let n be a positive integer. Find all monotone functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x + f(y)) = f(x) + y^n$$

for all $x, y \in \mathbb{R}$.

Problem 3.10. Denote by T the set of real numbers greater than 1. Given a positive integer n find all functions $f: T \rightarrow \mathbb{R}$ such that

$$f(x^{n+1} + y^{n+1}) = x^n f(x) + y^n f(y)$$

for all $x, y \in T$.

Problem 3.11. (Russia 1993) Find all functions $f: (0, \infty) \rightarrow (0, \infty)$ such that

$$f(x^y) = f(x)^{f(y)}$$

for all $x, y \in (0, \infty)$.

Problem 3.12. (generalization of Example 15.3) Find all functions $f: (0, \infty) \rightarrow (0, \infty)$ which are bounded from above on an interval and such that

$$f(xf(y)) = yf(x)$$

for all $x, y \in (0, \infty)$.

Problem 3.13. (generalization of Example 15.4) Let S be the set of all real numbers greater than -1 . Find all functions $f: S \rightarrow S$ which are bounded from above on an interval and such that

$$f(x + f(y) + xf(y)) = y + f(x) + yf(x)$$

for all $x, y \in S$.

Problem 3.14. (IMO 2002) Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$(f(x) + f(z))(f(y) + f(t)) = f(xy - zt) + f(xt + yz)$$

for all $x, y, z, t \in \mathbb{R}$.

Problem 3.15. (Romania TST 2007) Find all functions $f: \mathbb{R} \rightarrow [0, \infty)$ that satisfy

$$f(x^2 + y^2) = f(x^2 - y^2) + f(2xy)$$

for all $x, y \in \mathbb{R}$.

Problem 3.16. Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ that satisfy

$$f(y + zf(x)) = f(y) + xf(z)$$

for all $x, y, z \in \mathbb{R}$.

Problem 3.17. Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x)f(yf(x) - 1) = x^2f(y) - f(x)$$

for all real numbers x and y .

Problem 3.18. (IMO Shortlist 2003) Find all nondecreasing functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that $f(0) = 0$, $f(1) = 1$, and

$$f(x) + f(y) = f(x)f(y) + f(x + y - xy)$$

for all $x < 1$ and $y > 1$.

Problem 3.19. Let n be an integer greater than 1. Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x^n + 2f(y)) = f(x)^n + y + f(y)$$

for all $x, y \in \mathbb{R}$.

Problem 3.20. (China TST 2009) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function such that for any pairwise distinct real numbers a, b, c, d satisfying the identity

$$\frac{a-b}{b-c} + \frac{a-d}{d-c} = 0,$$

the values $f(a), f(b), f(c), f(d)$ are pairwise distinct and

$$\frac{f(a) - f(b)}{f(b) - f(c)} + \frac{f(a) - f(d)}{f(d) - f(c)} = 0.$$

Prove that f is a linear function.

Chapter 4

Substitutions

4.1 Theory and Examples

Almost every functional equation involves some kind of substitution. A *substitution* means replacing one of the variables in an identity or inequality by some suitable expression. In this section we have gathered problems for which substitutions represent the key idea to solution, i.e. the main part is finding a clever substitution, or maybe an entire string of substitutions.

There are several underlying ideas for substitutions that are useful in many problems. Some of them - like symmetrization and additional variables - will be discussed in other sections and not here.

Here is a brief checklist of quick substitutions to look for while solving a functional equation:

- Set one of the variables equal to 0, 1 or some other constant.
- If there are more variables involved, express them as functions of less variables. For example set $x = z, y = -z$.
- Express variables as functions of *more* variables. Even though this sounds counterintuitive, it can simplify the problem via exploiting symmetry and cancellation. This idea will be treated in the following section and not here.
- Replace a variable with the value of the function ($x = f(y)$).
- In case something about the image of the function is known (often when f is surjective), reverse the above problem (find an x such that $f(x) = y$ and

plug it in).

- Express the given variables in terms of new variables in a way that would make the condition simple or more intuitive. For example, if one has the term $f(f(f(x)) + y)$ in the equation, it may make sense to set $z = f(f(x)) + y$ to transform that term into $f(z)$. Another common idea is the substitution $x = a + b, y = a - b$ or vice versa.

- Instead of modifying the variables, modify the function: introduce a new function $g(x)$, expressible in terms of the original $f(x)$ that would be easier to work with.

- Make a term in the functional equation equal to a desired constant or a desired variable. Usually this is possible after knowing something about the image of f or an intermediate identity.

Let us take off with easy examples.

Example 4.1. (IMO 2008) Find all functions $f : (0, \infty) \rightarrow (0, \infty)$ such that

$$\frac{f(x)^2 + f(y)^2}{f(z^2) + f(t^2)} = \frac{x^2 + y^2}{z^2 + t^2}$$

for all $x, y, z, t \in (0, \infty)$ with $xy = zt$.

Solution. Setting $x = y = z = t = 1$ yields $f(1)^2 = f(1)$ and hence $f(1) = 1$. Now set $y = 1$ and $z = t = \sqrt{x}$ to obtain

$$\frac{f(x)^2 + 1}{2f(x)} = \frac{x^2 + 1}{2x}$$

which is equivalent to

$$(f(x) - x)(xf(x) - 1) = 0.$$

Thus, for every $x > 0$, either $f(x) = x$ or $f(x) = \frac{1}{x}$. Obviously, the functions $f(x) = x$ and $f(x) = \frac{1}{x}$ satisfy the given condition. We shall show that these two functions are the only solutions.

Assume the contrary. Then $f(x) = x$ and $f(y) = \frac{1}{y}$ for some $x, y \neq 1$. Setting $z = t = \sqrt{xy}$ leads to

$$f(xy) = \frac{xy(x^2 + y^{-2})}{x^2 + y^2}.$$

On the other hand, we know that either $f(xy) = xy$ or $f(xy) = \frac{1}{xy}$. This implies that either $y = 1$ or $x = 1$, a contradiction.

Example 4.2. (Belarus 1995) Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(f(x+y)) = f(x+y) + f(x)f(y) - xy$$

for all $x, y \in \mathbb{R}$.

Solution. Set $y = 0$ to get $f(f(x)) = f(x)(1 + f(0))$. So $f(x) = (1 + f(0))x$ on $\text{Im}(f)$. Hence $f(f(x+y)) = (1 + f(0))f(x+y)$ and substituting in the given equation we get $f(x)f(y) - xy = f(x+y)f(0)$. Set $a = f(0)$, $x = -a$ and $y = a$ to get $f(a)f(-a) + a^2 = a^2$ so $f(a)f(-a) = 0$. Therefore $0 \in \text{Im}(f)$ hence $f(0) = (1 + a)0 = 0$ so $a = 0$. Hence we get $f(x)f(y) = xy$. In particular, $f(1)^2 = 1$, i.e. $f(1) = \pm 1$ and setting $y = 1$ we see that either $f(x) = x$ or $f(x) = -x$ for all $x \in \mathbb{R}$. Now an easy check shows that the given condition is satisfied by the function $f(x) = x$ only.

Example 4.3. Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ that obey

$$(x-y)f(x+y) - (x+y)f(x-y) = 4xy(x^2 - y^2)$$

for all $x, y \in \mathbb{R}$.

Solution. The substitution $v = x + y, u = x - y$ is immediately seen and it simplifies our condition to $uf(v) - vf(u) = uv(u^2 - v^2)$. Now if we set $v = 0$ we get $uf(0) = 0$ for all u so $f(0) = 0$. If $uv \neq 0$ we divide by uv to get $\frac{f(v)}{v} - \frac{f(u)}{u} = u^2 - v^2$, hence $\frac{f(v)}{v} + v^2 = \frac{f(u)}{u} + u^2$ for all $u, v \neq 0$. Therefore $\frac{f(x)}{x} + x^2 = c$ for some fixed c and all $x \neq 0$. Then $f(x) = cx - x^3$ and this relation holds also for $x = 0$. We substitute to have $u(cv - v^3) - v(cu - u^3) = u^3v - v^3u$ which is true. All solutions are therefore of form $f(x) = cx - x^3$, where c is a constant.

Example 4.4. (IMO 1986) Find all functions $f: [0, \infty) \rightarrow [0, \infty)$ that satisfy the following conditions:

- i) $f(2) = 0$;
- ii) $f(xf(y))f(y) = f(x+y)$ for $x, y \in [0, \infty)$;
- iii) $f(x) > 0$ for $x \in [0; 2)$.

Solution. Setting $y = 2$ into ii) we get $f(x+2) = 0$ and so f is identically zero on $[2; \infty)$. Thus $f(x) = 0$ if and only if $x \geq 2$. Now consider a fixed $y < 2$. From condition ii) we deduce that $f(xf(y)) = 0$ if and only if $f(x+y) = 0$, or $xf(y) \geq 2$ if and only if $x+y \geq 2$. Hence if $x < 2-y$ then $xf(y) < 2$ so $f(y) < \frac{2}{x}$. Taking $x \uparrow 2-y$ we deduce $f(y) \leq \frac{2}{2-y}$. But setting $x = 2-y$ we get $x+y = 2$. So $xf(y) \geq 2$ and $f(y) \geq \frac{2}{x} = \frac{2}{2-y}$. We conclude that $f(y) = \frac{2}{2-y}$. Indeed, the function f with $f(x) = 0$ for $x \geq 2$ and $f(x) = \frac{2}{2-x}$ otherwise satisfies the conditions. If $y \geq 2$, this is trivial and if $y < 2$, we need to check $f\left(\frac{2x}{2-y}\right) \frac{2}{2-y} = f(x+y)$. If $x \geq 2-y$, then $\frac{2x}{2-y} \geq 2$ and $x+y \geq 2$ and the both sides are 0. If $x < 2-y$ then $\frac{2x}{2-y} < 2$ so $f\left(\frac{2x}{2-y}\right) = \frac{2-y}{2-x-y}$ and $x+y < 2$. So $f(x+y) = \frac{2}{2-x-y}$ and we get the identity $\frac{2}{2-y} \cdot \frac{2-y}{2-x-y} = \frac{2}{2-x-y}$.

Example 4.5. Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ that satisfy

$$f(f(x-y)) = f(x) - f(y) + f(x)f(y) - xy$$

for all $x, y \in \mathbb{R}$.

Solution. Setting $y = x$ and $d = f(f(0))$, we get $d = f(x)^2 - x^2$. Thus

$$f(x) = \pm \sqrt{x^2 + d}$$

for all $x \in \mathbb{R}$. Setting $y = 0$ and $c = f(0)$, we obtain

$$a\sqrt{x^2 + 2d} = b(1+c)\sqrt{x^2 + d} - c,$$

where a, b depend on x , namely $a = \operatorname{sgn} f(f(x)), b = \operatorname{sgn} f(x)$. Letting $x \rightarrow \infty$, it follows that $c = d = 0$ and $a = b = \pm 1$. So $f(x) = \pm x$ for all x .

Now we prove $f(x) = x$ for all $x \in \mathbb{R}$. Indeed assume $f(x_0) = -x_0, x_0 \neq 0$. Then

$$\begin{aligned} \pm(x - x_0) &= f(f(x - x_0)) = f(x) - f(x_0) + f(x)f(x_0) - xx_0 \\ &= (f(x) - 1)(f(x_0) + 1) - xx_0 + 1 = (x - 1)(-x_0 + 1) - xx_0 + 1 = x + x_0 - 2x_0x \\ \text{or } (-x - 1)(-x_0 + 1) - xx_0 + 1 &= x_0 - x. \text{ Now the equalities} \end{aligned}$$

$$x - x_0 = x + x_0 - 2x_0x, \quad x - x_0 = x_0 - x, \quad x_0 - x = x + x_0 - 2x_0x$$

are equivalent respectively to $x = 1, x = x_0, x(x_0 - 1) = 0$ which have at most one solution each for $x_0 \neq 1$. Therefore for $x_0 \neq 1$ we have

$$f(f(x - x_0)) = x_0 - x, \quad f(x) = -x$$

for all x except three values. However this is impossible: if we take x such that neither x nor $x_0 - x$ is among these three values, then

$$f(f(x - x_0)) = f(x_0 - x) = x - x_0$$

while we have $f(f(x - x_0)) = x_0 - x$. So $f(x) = x$ for all $x \neq 1$. Finally if we set $x = 3, y = 1$, we get

$$2 = f(2) = f(3) - f(1) + f(3)f(1) - 3 = 3 - f(1) + 3f(1) - 3 = 2f(1)$$

and hence $f(1) = 1$. Thus f is the identity function. This function satisfies the condition.

Example 4.6. Find all functions $f: \mathbb{Q} \rightarrow \mathbb{Q}$ such that

$$f\left(\frac{x+y}{3}\right) = \frac{f(x) + f(y)}{2}$$

for all $x, y \in \mathbb{Q} \setminus \{0\}$.

Solution. If we set $y = 2x$, we get $f(x) = \frac{f(x) + f(2x)}{2}$ for any x therefore $f(2x) = f(x)$. This suggests that f is a constant function. Indeed, let S be the set of all integers k for which $f(kx) = f(x)$ for all x . Then we know that $1, 2 \in S$. Also if $a, b \in S$ then $3a - b \in S$ (just set $x \rightarrow (3a - b)x, y \rightarrow bx$), and clearly if $a, b \in S$ then $ab \in S$. Now we can prove by induction on $|k|$ that $k \in S$ for all $k \in \mathbb{Z} \setminus \{0\}$. Indeed, $4 = 2 \cdot 2 \in S$, $-1 = 3 \cdot 1 - 4 \in S$, $-2 = 3 \cdot (-1) - (-1) \in S$ and the basis is proven. To prove the induction step, we have to show that every k with $|k| \geq 3$ can be written as $3a - b$ where $|a|, |b| < k$. Indeed, if we denote $a = \left\lfloor \frac{k}{3} \right\rfloor, b = -3 \left\{ \frac{k}{3} \right\}$ then $k = 3a - b$ and $|a| \leq \left\lfloor \frac{k}{3} \right\rfloor < k, |b| \leq 2 < k$. Thus $\mathbb{Z}^* \subset S$. Therefore if $a, b \in \mathbb{Q} \setminus \{0\}$ and $\frac{a}{b} = \frac{p}{q}, p, q \in \mathbb{Z}$, then $qa = pb$ thus $f(a) = f(qa) = f(pb) = f(b)$ and so f is constant on $\mathbb{Q} \setminus \{0\}$. Also if $x \neq 0$, then setting $y = 0$ we get

$$f\left(\frac{x}{3}\right) = \frac{f(x) + f(0)}{2}$$

which as $f\left(\frac{x}{3}\right) = f(x)$ implies that $f(0) = f(x)$. So f is a constant on $\mathbb{Q}$ and all constant functions satisfy the requirements.

Example 4.7. (IMO 1999) Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x - f(y)) = f(f(y)) + xf(y) + f(x) - 1$$

for all $x, y \in \mathbb{R}$.

Solution. Setting $x = f(y)$ in the given equation gives

$$f(x) = \frac{c + 1 - x^2}{2} \quad (1)$$

for all $x \in \text{Im} f$, where $c = f(0)$. On the other hand if $y = 0$ we get

$$f(x - c) - f(x) = f(c) + cx - 1.$$

Hence $f(-c) - c = f(c) - 1$ which shows that $c \neq 0$. Thus for every $x \in \mathbb{R}$ there is $t \in \mathbb{R}$ such that $x = y_1 - y_2$, where $y_1 = f(t - c)$, $y_2 = f(t)$. Now using the given equation we get

$$\begin{aligned} f(x) &= f(y_1 - y_2) = f(y_2) + y_1 y_2 + f(y_1) - 1 \\ &= \frac{c + 1 - y_2^2}{2} + y_1 y_2 + \frac{c + 1 - y_1^2}{2} - 1 \\ &= c - \frac{(y_1 - y_2)^2}{2} = c - \frac{x^2}{2}. \end{aligned}$$

This together with (1) gives $c = 1$ and $f(x) = 1 - \frac{x^2}{2}$. Conversely, it is easy to check that this function satisfies the given equation.

Remark. During the solution of the above problem we have encountered the conclusion $\text{Im}(f) - \text{Im}(f) = \mathbb{R}$, i.e. that every number can be written as the difference of two values of f . This idea occurs seldom, but when it does, it is usually the key to the solution of the problem. See the next one.

Example 4.8. (BMO 2007) Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(f(x) + y) = f(f(x) - y) + 4f(x)y \quad \text{for all } x, y \in \mathbb{R}.$$

Solution. It is clear that the function $f \equiv 0$ is a solution.

Let $f \not\equiv 0$. Choosing $f(x_0) \neq 0$, then $4f(x_0)y$ spans $\mathbb{R}$ as y ranges over $\mathbb{R}$. Hence the given condition implies that

$$\text{Im}(f) - \text{Im}(f) = \mathbb{R}. \tag{1}$$

On the other hand, replacing y by $f(x) - y$, we get that

$$f(2f(x) - y) = f(y) + 4f(x)(f(x) - y),$$

i.e.

$$f(y) - y^2 = f(2f(x) - y) - (2(f(x) - y))^2. \tag{2}$$

Let now $y_1, y_2 \in \mathbb{R}$. By (1), we may choose x_1, x_2 such that

$$f(x_1) - f(x_2) = \frac{y_1 - y_2}{2}. \tag{3}$$

Then (2) implies that

$$f(y_1) - y_1^2 = f(2f(x_1) - y_1) - (2(f(x_1) - y_1))^2$$

and

$$f(y_2) - y_2^2 = f(2f(x_2) - y_2) - (2(f(x_2) - y_2))^2.$$

Using (3), we get that $f(y_1) - y_1^2 = f(y_2) - y_2^2$. This means that $f(x) - x^2$ is constant, i.e. $f(x^2) = x^2 + c$, $c \in \mathbb{R}$. It is easy to check that these functions satisfy the given identity.

Example 4.9. (Putnam 2000) Determine all continuous functions $f : [-1, 1] \rightarrow \mathbb{R}$ such that $f(2x^2 - 1) = 2xf(x)$ for all $x \in [-1, 1]$.

Solution. Note first that $f(1) = f(-1) = 0$. Moreover $f(x)$ is odd since $2xf(x) = f(2x^2 - 1) = 2(-x)f(-x)$ and for $x \neq 0$ we get $f(x) = -f(-x)$; by continuity, the same holds for $x = 0$ so $f(0) = 0$. Now for any $t \in \mathbb{R}$ that is not an integer multiple of π define $g(t) = f(\cos t)/\sin t$. Then it is easy to check that $g(t + \pi) = g(t)$. Moreover the given condition for f implies that

$$g(2t) = f(2\cos^2 t - 1)/\sin 2t = \frac{2\cos t f(\cos t)}{2\sin t \cos t} = g(t).$$

Hence $g(2^k t) = g(t)$ for all $k \in \mathbb{Z}$. In particular,

$$g(1 + n\pi/2^k) = g(2^k + n\pi) = g(2^k) = g(1)$$

for all $k, n \in \mathbb{Z}$. Note also that g is continuous on its domain of definition. For every $t \in \mathbb{R}$ there is a sequence $(\delta_m)_{m \geq 1}$ of binary fractions (i.e. fractions of the form $\frac{n}{2^k}$, $n, k \in \mathbb{Z}$) such that $\lim_{m \rightarrow \infty} \delta_m = \frac{t-1}{\pi}$, i.e. $t = \lim_{m \rightarrow \infty} (1 + \pi\delta_m)$. Hence

$$\begin{aligned} g(t) &= g(\lim_{m \rightarrow \infty} (1 + \pi\delta_m)) = \lim_{m \rightarrow \infty} g(1 + \pi\delta_m) \\ &= \lim_{m \rightarrow \infty} g(1) = g(1). \end{aligned}$$

Thus g is constant and since g is odd we see that $g = 0$ and therefore $f = 0$.

4.2 Practice Problems

Problem 4.1. Find all continuous functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x+y) - f(x) - f(y) = xy(x+y)$$

for all $x, y \in \mathbb{R}$.

Problem 4.2. Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(xy + 2x + 2y) = f(x)f(y) + 2x + 2y$$

for all $x, y \in \mathbb{R}$.

Problem 4.3. (USAMO 2002) Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x^2 - y^2) = xf(x) - yf(y)$$

for all $x, y \in \mathbb{R}$.

Problem 4.4. (IMO Shortlist 2005) Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x+y) + f(x)f(y) = f(xy) + 2xy + 1$$

for all $x, y \in \mathbb{R}$.

Problem 4.5. Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(f(x) - y^2) = f(x)^2 - 2f(x)y^2 + f(f(y))$$

for all $x, y \in \mathbb{R}$.

Problem 4.6. Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x - f(y)) = f(x) + xf(y) + f(f(y))$$

for all $x, y \in \mathbb{R}$.

Problem 4.7. Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x+y) - f(x-y) = 2f(xy+1) - f(x)f(y) - 4$$

for all $x, y \in \mathbb{R}$.

Problem 4.8. (IMO Shortlist 2000) Find all pairs of functions $f, g: \mathbb{R} \rightarrow \mathbb{R}$ that satisfy

$$f(x + g(y)) = xf(y) - yf(x) + g(x)$$

for all $x, y \in \mathbb{R}$.

Problem 4.9. (Bulgaria 2008) Find all continuous functions $f: \mathbb{R} \rightarrow \mathbb{R}$ that satisfy

$$(1 - f(x)f(y))f(x + y) = f(x) + f(y) + 2f(x)f(y)$$

for all $x, y \in \mathbb{R}$.

Problem 4.10. Find all continuous functions $f: (-1, 1) \rightarrow \mathbb{R}$ such that

$$f\left(\frac{x+y}{1-xy}\right) = f(x) + f(y)$$

for all $x, y \in (-1, 1)$.

Problem 4.11. Find all continuous functions $f: (a, b) \rightarrow \mathbb{R}$ that satisfy

$$f(xyz) = f(x) + f(y) + f(z)$$

whenever $xyz, x, y, z \in (a, b)$, where $1 < a^3 < b$.

Problem 4.12. Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ that obey

$$f(f(x) + y) = f(x^2 - y) + 4f(x)y$$

for all $x, y \in \mathbb{R}$.

Problem 4.13. (China 1991) Find all functions $f: [0, 1] \times [0, 1] \rightarrow \mathbb{R}$ such that:

i) $f(x, 1) = x$;

ii) $f(1, y) = y$;

iii) $f(zx, zy) = z^k f(x, y)$;

iv) $f(f(x, y), z) = f(x, f(y, z))$

for all $x, y, z \in [0, 1]$ and some positive constant k .

Problem 4.14. Find all continuous functions $f: \mathbb{R} \rightarrow \mathbb{R}$ that satisfy

$$3f(2x+1) = f(x) + 5x$$

for all $x \in \mathbb{R}$.

Problem 4.15. (IMO Shortlist 2002) Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(f(x) + y) = 2x + f(f(y) - x)$$

for all $x, y \in \mathbb{R}$.

Problem 4.16. (Bulgaria 1996) Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(f(x) + xf(y)) = xf(y+1)$$

for all $x, y \in \mathbb{R}$.

Problem 4.17. (BMO 1997, BMO 2000) Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(xf(x) + f(y)) = f(x)^2 + y$$

for all $x, y \in \mathbb{R}$.

Problem 4.18. (Ukraine 2003) Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(xf(x) + f(y)) = x^2 + y$$

for all $x, y \in \mathbb{R}$.

Problem 4.19. Find all functions $f: (0, \infty) \rightarrow \mathbb{R}$ such that

$$f(x+y) = f(x^2 + y^2)$$

for all $x, y \in (0, \infty)$.

Problem 4.20. (IMO Shortlist 2004) Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfying the equation

$$f(x^2 + y^2 + 2f(xy)) = f(x+y)^2$$

for all $x, y \in \mathbb{R}$.

Chapter 5

Symmetrization and Additional Variables

5.1 Examples

Sometimes we have functional equations such that one side of them is symmetric in x, y or we can obtain such a condition by an appropriate substitution. Then swapping x with y we get a new condition, which might prove helpful. In other cases we might need to add an additional variable to get one side of the equation symmetric and then to symmetrize the other side in order to get a new condition. In this chapter we consider functional equations that can be solved by using the above approach.

Example 5.1. Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ that satisfy

$$f(x)f(y - f(x)) = 4xy - 2f(x^2)$$

for all $x, y \in \mathbb{R}$.

Solution. It is easy to make the left hand side symmetric in two variables: just replace y by $y + f(x)$. We then get

$$f(x)f(y) = 4x(y + f(x)) - 2f(x^2) = 4xy + 4xf(x) - 2f(x^2).$$

By swapping x with y we get

$$f(x)f(y) = 4xy + 4yf(y) - 2f(y^2)$$

thus by comparing the two obtained identities we obtain

$$4xf(x) - 2f(x^2) = 4yf(y) - 2f(y^2);$$

that is, the expression $4xf(x) - 2f(x^2)$ is independent of x and takes some constant value a .

In particular $f(x)f(y) = 4xy + 4f(x) - 2f(x^2) = 4xy + a$. This readily implies that $a = 0$. There are many ways to see this. One route would be to note that if $a \neq 0$ then $f(x)f\left(-\frac{a}{4x}\right) = 0$ so if $f(x) \neq 0$ then $f\left(-\frac{a}{4x}\right) = 0$. In particular $f(y_0) = 0$ for some value y_0 , which is impossible since then $f(x)f(y_0) = 4xy_0 + a$ would be 0 for all x but that is impossible if $a \neq 0$. Another way is to see that $f(x)^2 = 4x^2 + a$ on one hand and $f(x)f(1) = 4x + a$ on the other, so $f(x)^2 f(1)^2$ equals simultaneously $f(1)^2(4x + a)^2$ and $4x^2 + a$ but $f(1)^2(4x + a)^2 = 4x^2 + a$ can hold identically for all x only for $a = 0$.

For $a = 0$ we get $f(x)^2 = 4x^2$ so $f(x) = \pm 2x$. The identity $f(x)f(y) = 4xy$ immediately implies that either $f(x) = 2x$ for all x or $f(x) = -2x$ for all x . Both are solutions.

Example 5.2. Find all continuous functions $f: \mathbb{R} \rightarrow \mathbb{R}$ that satisfy

$$f(xy + x + y) = f(xy) + f(x + y)$$

for all $x, y \in \mathbb{R}$.

Solution. This is an equation of the type $f(a + b) = f(a) + f(b)$ but we cannot take a, b to be arbitrary because $x + y$ and xy are constrained by the AM-GM inequality $(x + y)^2 \geq 4xy$. In this case, it is a good idea to add an extra variable because solving a system of two equations in three variables allows for more leverage and we can even hope to bypass any restrictions.

More precisely, let's replace y by $y + z$ to obtain

$$f(xy + xz + x + y + z) = f(xy + xz) + f(x + y + z),$$

which can be rewritten as

$$f(xy + xz + x + y + z) - f(xy + xz) = f(x + y + z).$$

Because the right-hand side is symmetric in x, y, z we can swap them and obtain that

$$f(xy + xz + x + y + z) - f(xy + xz) = f(xy + yz + x + y + z) - f(xy + yz).$$

This can be rewritten as

$$f(xy + xz + x + y + z) - f(xy + yz + x + y + z) = f(xy + xz) - f(xy + yz).$$

If we set $xy + xz = a$, $xy + yz = b$ and $x + y + z = s$ we get

$$f(a + s) - f(b + s) = f(a) - f(b).$$

We only need $s^2 \geq 4a$, $s^2 \geq 4b$ to solve the two equations which yield x, y, z . Thus $f(a) - f(b)$ equals $f(a + s) - f(b + s)$ for all sufficiently big s . Of course, this tells that $f(a) - f(b)$ depends only on $a - b$. In particular

$$f(a) - f(b) = f(a - b) - f(0)$$

which implies $f(x) - f(0)$ is additive so f is linear. By substituting into the original condition we see that the free term is zero so $f(x) = cx$.

Example 5.3. Find all continuous functions $f: \mathbb{R} \rightarrow (0, \infty)$ that satisfy

$$f(x + y)f(x^2 + y^2) = f((x + y)(x^2 + y^2))$$

for all $x, y \in \mathbb{R}$.

Solution. This is similar to the previous problem in the sense that it is the familiar multiplicative Cauchy equation that is verified on a restrained set of values - namely we have $f(a)f(b) = f(ab)$ but we need a, b to be a solution of the system $x + y = a$, $x^2 + y^2 = b$ and not all pairs are like that.

Again, we bypass the inconvenience by adding a new variable and symmetrizing. Namely $y \rightarrow y + z$ yields

$$f(x + y + z)f(x^2 + (y + z)^2) = f((x + y + z)(x^2 + (y + z)^2))$$

or

$$f(x+y+z) = \frac{f((x+y+z)(x^2+(y+z)^2))}{f(x^2+(y+z)^2)}.$$

The left-hand side is symmetric so we obtain

$$\frac{f((x+y+z)(x^2+(y+z)^2))}{f(x^2+(y+z)^2)} = \frac{f((x+y+z)(y^2+(x+z)^2))}{f(y^2+(x+z)^2)}$$

or

$$\frac{f(x^2+(y+z)^2)}{f(y^2+(x+z)^2)} = \frac{f((x+y+z)(x^2+(y+z)^2))}{f((x+y+z)(y^2+(x+z)^2))}.$$

By letting $a = x^2 + (y+z)^2$, $b = y^2 + (x+z)^2$, $s = x+y+z$ we obtain

$$\frac{f(a)}{f(b)} = \frac{f(as)}{f(bs)}.$$

Now note that if s is small enough (more precisely if $s^2 \leq 4a, 4b$) then there are x, y, z that yield these values of s : we solve the quadratic system

$$x^2 + (y+z)^2 = a, \quad x+y+z = s$$

to get $x = \frac{s - \sqrt{4a - s^2}}{2}$, similarly $y = \frac{s - \sqrt{4b - s^2}}{2}$ and then z will be

$$s - x - y = \frac{\sqrt{4a - s^2} + \sqrt{4b - s^2}}{2}.$$

It follows that $\frac{f(a)}{f(b)} = \frac{f(as)}{f(bs)}$ for all sufficiently small s (if a, b are both positive). This implies that $\frac{f(a)}{f(b)}$ only depends on $\frac{a}{b}$; in particular,

$$\frac{f(a)}{f(b)} = \frac{f\left(\frac{a}{b}\right)}{f(1)},$$

i.e. the function $g(x) = \frac{f(x)}{f(1)}$ is multiplicative on $(0, \infty)$, thus $f(x) = f(1)x^a$ and direct verification yields $f(1) = 1$, so $f(x) = x^a$ on $(0, \infty)$. But note that

for $s < 0$ we get in the above identity $\frac{f(a)}{f(b)} = \frac{f(-a)}{f(-b)}$ and this immediately implies that $f(-x) = cf(x)$ for some $c > 0$. It remains to observe that c must be 1 by continuity at 0 where f must take the value 1.

The answer is $f(x) = |x|^a$.

Example 5.4. Find all continuous functions $f, g, h: \mathbb{R} \rightarrow \mathbb{R}$ that satisfy

$$f(x+y) + g(xy) = h(x) + h(y)$$

for all $x, y \in \mathbb{R}$.

Solution. Set $y = 0$ to get $f(x) = h(x) + h(0) - g(0)$. So the condition rewrites as $h(x+y) - h(x) - h(y) = g(xy)$ where replace g by $g - g(0) - h(0)$ for simplicity. Thus

$$\begin{aligned} h(x+y+z) &= h(x) + h(y+z) + g(xy+xz) \\ &= h(x) + h(y) + h(z) + g(yz) + g(xy+xz). \end{aligned}$$

Symmetrizing this we conclude that

$$g(yz) + g(xy+xz) = g(xz) + g(xy+yz) = g(xy) + g(xz+yz).$$

As for $a, b, c > 0$ we can find x, y, z with $yz = a, xz = b, xy = c$ we get

$$g(a) + g(b+c) = g(b) + g(a+c) + g(c) + g(a+b)$$

and taking $c \rightarrow 0^+$ we get

$$g(a+b) + g(0) = g(a) + g(b).$$

Next if we take $a > 0, b < 0, c < 0$ we can also find x, y, z with $yz = a, xz = b, xy = c$ so

$$g(a) + g(b+c) = g(b) + g(a+c) + g(c) + g(a+b).$$

Now taking $c \rightarrow 0^-$ we get

$$g(a) + g(b) = g(0) + g(a+b).$$

Finally if we take $a < 0, b < 0, c > 0$ and take $c \rightarrow 0^+$, we get

$$g(a) + g(b) = g(a + b) + g(0)$$

in this case too. So

$$g(a + b) + g(0) = g(a) + g(b)$$

holds for all non-zero a, b by continuity and then $g(x) = ax + b$ is linear. So

$$h(x + y) - h(x) - h(y) = axy + b.$$

If we consider $H(x) = h(x) - \frac{a}{2}x^2 + b$ then we see that

$$H(x) + H(y) = H(x + y)$$

so $H(x) = cx$. It means that h is a quadratic function, and since

$$f(x) = h(x) + h(0) - g(0)$$

we find that

$$f(x) = ux^2 + vx + t, \quad h(x) = ux^2 + vx + w$$

which leads to $g(x) = 2ux + 2w - t$; and such triples satisfy the condition.

Example 5.5. Determine all continuous functions $f, g, h : (0, \infty) \rightarrow \mathbb{R}$ such that

$$f(x + y) = h(x) + h(y) + g(xy)$$

for all $x, y \in (0, \infty)$.

Solution. Setting $y = 1$ we get that $g(x) = f(x + 1) - h(x) - h(1)$. Hence the given condition takes the form

$$h(xy) = h(x) + h(y) + f(xy + 1) - f(x + y) - h(1) \quad (1)$$

for all $x, y \in (0, \infty)$. Now we use the fact that the multiplication of real numbers is associative, i.e. $x(yz) = (xy)z$. Using the identity (1) twice we get that

$$h(xyz) = h(x(yz)) = h(x) + h(yz) + f(xyz + 1) - f(x + y + z) - h(1)$$

$$= h(x) + h(y) + h(z) + f(yz + 1) - f(y + z) - h(1) + f(xyz + 1) - f(x + y + z).$$

Proceeding in the same way we also get that

$$\begin{aligned} h(xyz) &= h((xy)z) = h(x) + h(y) + h(z) + f(xy + 1) - f(xyz) - \\ &\quad h(1) + f(xyz + 1) - f(xy + z) - h(1). \end{aligned}$$

Comparing both expressions for $h(xyz)$ we see that

$$f(xy + 1) - f(x + y) - f(xy + z) = f(yz + 1) - f(y + z) - f(x + yz) \quad (2)$$

for all $x, y, z \in (0, \infty)$. Now fix $x, y \in (0, \infty)$ and let $z \rightarrow 0$. Then (2) implies that

$$f(xy + 1) - f(xy) - f(1) = f(x + y) - f(x) - f(y). \quad (3)$$

Set $F(x, y) = f(x + y) - f(x) - f(y)$. Then it is easy to check that

$$F(x + y, z) + F(x, y) = F(x, y + z) + F(y, z). \quad (4)$$

On the other hand, (3) can be written as $F(x, y) = F(xy, 1)$ and it follows from (4) that

$$F((x + y)z, 1) + F(xy, 1) = F(x(y + z), 1) + F(yz, 1). \quad (5)$$

Set $k(x) = F(x, 1)$. Then (5) can be written as

$$k(xz + yz) + k(xy) = k(xy + xz) + k(yz). \quad (6)$$

For any $u, v \in (0, \infty)$ setting $x = \sqrt{uv}$, $y = \frac{v}{\sqrt{uv}}$, $z = \frac{\sqrt{uv}}{v}$ in (6) gives

$$k(u + 1) + k(v) = k(u + v) + k(1). \quad (7)$$

In particular, $k(u + 1) + k(v) = k(v + 1) + k(u)$ and setting $v = 1$ we get that $k(u + 1) = k(u) + k(2) - k(1)$. Hence (7) takes the form

$$k(u) + k(v) = k(u + v) + b,$$

where $b = 2k(1) - k(2)$. Consider the function $t : (0, \infty) \rightarrow \mathbb{R}$ defined by $t(u) = k(u) - b$. Then $t(u) + t(v) = t(u + v)$ and since the function t is continuous it follows from Theorem 1.1 that $t(x) = ax$, i.e. $k(x) = ax + b$. Hence

$$f(x + y) - f(x) - f(y) = F(x, y) = F(xy, 1) = k(xy) = axy + b \quad (8)$$

for all $x, y \in (0, \infty)$.

Observe that the function $f_0(x) = \frac{ax^2}{2} - b$ satisfies the identity (8). Then setting $f(x) = f_0(x) + f_1(x)$ and using (8) we see that $f_1 : (0, \infty) \rightarrow \mathbb{R}$ is a continuous additive function and Theorem 1.1 implies that $f_1(x) = cx$. Hence $f(x) = \frac{ax^2}{2} + cx - b$. Plugging this expression of $f(x)$ into (1) we get that

$$h(xy) - h(x) - h(y) = f(xy) - f(x) - f(y) + d, \quad (9)$$

where $d = \frac{a}{2} - b + c - h(1)$. Consider the function $l(x) = h(x) - f(x) - d$. Then (9) implies that $l(xy) = l(x) + l(y)$ for all $x, y \in (0, \infty)$ and it follows from Corollary 1.2 that $l(x) = A \log x$. Hence all continuous solutions $f, g, h : (0, \infty) \rightarrow \mathbb{R}$ of (1) have the form

$$f(x) = ax^2 + bx + c, \quad g(x) = -A \log x - 2ax + c - 2d$$

and $h(x) = A \log x + ax^2 + bx + d$.

Example 5.6. (India 2003) Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x + y) + f(x)f(y) = f(x) + f(y) + f(xy)$$

for all $x, y \in \mathbb{R}$.

Solution. Using the given identity several times, we obtain that

$$\begin{aligned} f(x + y + z) &= f(x) + f(y + z) + f(xy + xz) - f(x)f(y + z) \\ &= f(x) + (1 - f(x))(f(y) + f(z) + f(yz) - f(y)f(z)) \\ &\quad + f(xy) + f(xz) + f(x^2yz) - f(xy)f(xz) \\ &= f(x) + f(y) + f(z) + f(xy) + f(yz) + f(zx) + f(x)f(y)f(z) \\ &\quad - f(x)f(y) - f(y)f(z) - f(z)f(x) \\ &\quad + f(x^2yz) - f(xy)f(xz) - f(x)f(yz). \end{aligned}$$

Hence the term in the last line is a symmetric function of x , y and z , which implies that

$$f(x^2yz) - f(xy)f(xz) - f(x)f(yz) = f(xy^2z) - f(xy)f(yz) - f(y)f(xz).$$

For $y = 1$ we get that

$$f(x^2z) = (a - 1)f(xz) + f(x)f(xz),$$

where $a = 2 - f(1)$. On the other hand, the given identity shows that

$$f(x^2z) = f(x + xz) + f(x)f(xz) - f(x) - f(xz).$$

Therefore,

$$f(x + xz) = af(xz) + f(x).$$

For $z = 0$ we obtain that $af(0) = 0$. If $a = 0$, then $f(1 + z) = f(1)$, that is, $f \equiv 2$. Let $f(0) = 0$. Then $f(x) = -af(-x) = a^2f(x)$. Suppose that f is not identically 0. Then $a^2 = 1$. If $a = -1$, then $f(x + xz) = -f(xz) + f(x)$ and setting $z = 1$ gives $f(2x) = 0$ for all $x \in \mathbb{R}$, a contradiction. So $a = 1$. Setting $z = \frac{y}{x}$ leads to $f(x + y) = f(x) + f(y)$ for all $x \neq 0$ and y . The same remains true if $x = 0$. It follows now by the given identity that $f(xy) = f(x)f(y)$ and Corollary 1.1 implies that $f(x) = x$.

So, the solutions of the problem are the functions $f \equiv 2$, $f \equiv 0$ and $f(x) = x$.

5.2 Practice Problems

Problem 5.1. Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ that satisfy

$$f(xy) = f(x)y^2 + f(y) - 1$$

for all $x, y \in \mathbb{R}$.

Problem 5.2. (MR 2012) Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfying

$$f(y + f(x)) = f(x)f(y) + f(f(x)) + f(y) - xy$$

for all $x, y \in \mathbb{R}$.

Problem 5.3. Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$xf(x) - yf(y) = (x - y)f(x + y)$$

for all $x, y \in \mathbb{R}$.

Problem 5.4. Find all pairs of functions $f, g: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x) - f(y) = (x - y)(g(x) + g(y))$$

for all $x, y \in \mathbb{R}$.

Problem 5.5. Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x + y) = f(x)f(y)f(xy)$$

for all $x, y \in \mathbb{R}$.

Problem 5.6. Find all functions $f: \mathbb{Q} \rightarrow \mathbb{Q}$ such that

$$f(xy) = f(x)f(y) - f(x + y) + 1$$

for all $x, y \in \mathbb{Q}$.

Problem 5.7. Find all continuous functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that

a) $f(x + y) + f(xy - 1) = f(x) + f(y) + f(xy);$

b) $f(x + y) + f(xy) = f(x) + f(y) + f(xy + 1)$

for all $x, y \in \mathbb{R}$.

Problem 5.8. (Hosszu's functional equation) Show that a function $f: \mathbb{R} \rightarrow \mathbb{R}$ which satisfies the identity

$$f(x + y - xy) + f(xy) = f(x) + f(y)$$

is an additive function plus some constant.

Chapter 6

Iterations and Recurrence Relations

6.1 Theory and Examples

It is often the case that functional equations are related to recurrence relations. For example, the solution of Cauchy's equation $f(x+y) = f(x) + f(y)$ was essentially based on the simple recurrence $a_{n+1} = a_n + a_1$ for $a_n = f(nx)$ to establish that $f(x) = f(1)x$ for all $x \in \mathbb{Q}$. Another idea which leads to recurrence relations is to iterate a function and then set the elements of the sequence be these iterates. For example, if we have $f(x) + f(f(x)) = x$ we might set $a_n = f(f(\dots f(x)))$ (f is taken n times) and have the recurrence relation $a_{n+1} + a_{n+2} = a_n$. This chapter is concerned with solving functional equations with this method by constructing sequences and exploiting the recurrence relations between them.

For a function $f : D \rightarrow D$, $D \subseteq \mathbb{R}$, we denote by $f^{(n)}$ the n -th iterate of f . It is defined inductively by $f^{(0)}(x) = x$, $f^{(n+1)}(x) = f(f^{(n)}(x))$, $n \geq 0$. If the function f is one-to-one, then we can also define $f^{(n)}$ for negative integers n . Namely, $f^{(-n)}(x)$ is the n -th iterate of its inverse function $f^{-1}(x)$. It is easy to check that

$$f^{(m+n)}(x) = f^{(m)}(f^{(n)}(x)), \quad m, n \in \mathbb{Z}.$$

For a fixed $x \in D$ the sequence

$$O(x) = \{f^{(0)}(x), f^{(1)}(x), f^{(2)}(x), \dots\}$$

is called the orbit of x .

Example 6.1. Find the n -th iterate of the function:

$$(a) f(x) = \begin{cases} 1 & \text{if } |x| \leq 1 \\ 0 & \text{if } |x| > 1. \end{cases}$$

$$(b) f(x) = ax + b;$$

$$(c) f(x) = x^2 + 2x;$$

$$(d) f(x) = \frac{x-3}{x+1}, x \neq \pm 1;$$

$$(e) f(x) = \frac{x}{a+bx};$$

$$(f) f(x) = \sqrt{2+x}.$$

Solution. (a) For every $x \in [-1, 1]$ we have $f(x) = 1$ and it follows by induction that $f^{(n)}(x) = 1$ for $n \geq 2$. If $|x| > 1$, then

$$f(x) = 0, f^{(2)}(x) = f(0) = 1$$

and it follows by induction that $f^{(n)}(x) = 1$ for $n \geq 2$. Thus $f^{(n)}(x) = 1$ for every $x \in \mathbb{R}$ and $n \geq 2$.

(b) It follows by induction on n that

$$f^{(n)}(x) = a^n x + (a^{n-1} + \dots + a + 1)b.$$

(c) Note that $f(x) = (x+1)^2 - 1$. Hence $f^{(n)}(x) = (x+1)^{2^n} - 1$.

(d) It is easy to check that $f^{(2)}(x) = \frac{-x-3}{x+1}$ and $f^{(3)}(x) = x$. Hence

$$f^{(n)}(x) = \begin{cases} x & \text{for } n = 3k \\ \frac{x-3}{x+1} & \text{for } n = 3k+1 \\ \frac{-x-3}{x+1} & \text{for } n = 3k+2 \end{cases}$$

(e) Prove by induction on n that

$$f^{(n)}(x) = \frac{x}{a^n + (1 + a + \cdots + a^{n-1})bx}.$$

(f) Let $|x| \leq 2$. Set $x = 2 \cos \theta, \theta \in [0, \pi]$. Then

$$f(x) = \sqrt{2 + 2 \cos \theta} = 2 \cos \frac{\theta}{2}.$$

Hence

$$f^{(2)}(x) = \sqrt{2 + f(x)} = \sqrt{2 + 2 \cos \frac{\theta}{2}} = 2 \cos \frac{\theta}{2^2}$$

and we get by induction on n that

$$f^{(n)}(x) = 2 \cos \frac{\theta}{2^n} = 2 \cos \frac{\arccos \frac{x}{2}}{2^n}, |x| \leq 2.$$

Let now $x > 2$. Then $x = t + \frac{1}{t}$ for some $t > 1$ and

$$f(x) = \sqrt{2 + x} = t^{\frac{1}{2}} + \frac{1}{t^{\frac{1}{2}}}.$$

Now it follows by induction on n that

$$f^{(n)}(x) = t^{\frac{1}{2^n}} + \frac{1}{t^{\frac{1}{2^n}}}.$$

Solving the equation $x = t + \frac{1}{t}, t > 1$ we get

$$t = \frac{x + \sqrt{x^2 - 4}}{2}, \frac{1}{t} = \frac{x - \sqrt{x^2 - 4}}{2}.$$

Thus

$$f^{(n)}(x) = \left(\frac{x + \sqrt{x^2 - 4}}{2} \right)^{\frac{1}{2^n}} + \left(\frac{x - \sqrt{x^2 - 4}}{2} \right)^{\frac{1}{2^n}}, \quad x > 2.$$

Two functions $f, g : D \rightarrow D$ are called conjugate if there is a bijection $\varphi : D \rightarrow D$ such that $f(x) = \varphi^{-1}(g(\varphi(x)))$. In this case we write $f \sim_{\varphi} g$ and it is easy to check that $f^{(n)} \sim_{\varphi} g^{(n)}$ for every $n \in \mathbb{N}$. We shall use this property to find the n -th iterate of some functions.

Example 6.2. Find the n^{th} iterate of the function:

- (a) $f(x) = \frac{x}{1+ax}$;
- (b) $f(x) = \frac{x}{\sqrt{1+ax^2}}, x \geq 0$;
- (c) $f(x) = 2x^2 - 1, x \in [-1, 1]$.

Solution. (a) It is clear that $f^{(n)}(0) = 0$ for every $n \in \mathbb{N}$. If $x \neq 0$ set $g(x) = x + a$ and $\varphi(x) = \frac{1}{x}$. Then $\varphi^{-1}(x) = \frac{1}{x}$ and we have

$$\varphi^{-1}(g(\varphi(x))) = \frac{1}{\frac{1}{x} + a} = \frac{x}{1+ax} = f(x).$$

Thus $f \sim_{\varphi} g$ and therefore

$$f^{(n)}(x) = \varphi^{-1}(g^{(n)}(\varphi(x))) = \frac{1}{\frac{1}{x} + na} = \frac{x}{1+na x}.$$

(b) Set $g(x) = \frac{x}{1+ax}$ and $\varphi(x) = x^2$. Then $\varphi^{-1}(x) = \sqrt{x}$ and it is easy to check that $f \sim_{\varphi} g$. Hence $f^{(n)} \sim_{\varphi} g^{(n)}$ and using (a) we get

$$f^{(n)}(x) = \varphi^{-1}(g^{(n)}(\varphi(x))) = \frac{x}{\sqrt{1+na x^2}}.$$

(c) Set $\varphi(x) = \arccos x$. Then $\varphi^{-1}(x) = \cos x$ and we get

$$g(x) = \varphi(f(\varphi^{-1}(x))) = \arccos(2 \cos^2 x - 1) = \arccos(\cos 2x) = 2x.$$

Hence $f \sim_{\varphi} g$ which implies $f^{(n)} \sim_{\varphi} g^{(n)}$. But $g^{(n)}(x) = 2^n x$ and we get that $f^{(n)}(x) = \cos(2^n \arccos x)$.

Next we recall how one can solve the recurrence relations of first and second degree. Those of first degree have the form $x_{n+1} = ax_n + b$ and can be solved by making an easy substitution: if $a \neq 1$, then by letting $y_n = x_n + \frac{b}{a-1}$ we get $y_{n+1} = ay_n$. Hence $y_n = a^n y_0$ thus $x_n = a^n \left(x_0 + \frac{b}{a-1} \right) - \frac{b}{a-1}$. If $a = 1$, then we get $x_n = x_0 + nb$. We may also encounter recurrence relations

of second degree, i.e. relations that can be written as $x_{n+2} + ax_{n+1} + bx_n = 0$. The way to solve them is to look at the characteristic equation $x^2 + ax + b = 0$. If it has two roots r_1, r_2 then all solutions to the recurrence relation can be written as $x_n = c_1 r_1^n + c_2 r_2^n$, where c_1, c_2 are found using the linear system $x_0 = c_1 + c_2$, $x_1 = c_1 r_1 + c_2 r_2$. If $x^2 + ax + b$ has a double root r , then all solutions to the recurrence relation may be written as $x_n = (un + v)r^n$, where u, v is the solution of the linear system $v = x_0, (u + v)r = x_1$.

The examples follow.

Example 6.3. (IMO Longlist 1982) Determine all functions $f : \mathbb{Z} \rightarrow \mathbb{R}$ such that

$$f(n)f(m) = f(n+m) + f(n-m)$$

for all $n, m \in \mathbb{Z}$ if:

$$(a) \quad f(1) = \frac{5}{2}; \quad (b) \quad f(1) = \sqrt{3}.$$

Solution. Setting $m = 0, n = 1$ we get $f(1)f(0) = 2f(1)$. Hence $f(0) = 2$ since $f(1) \neq 0$. Now setting $n = 0$ gives $f(m) = f(-m)$ and it is enough to determine $f(n)$ for $n \in \mathbb{N}$. To do this we set $n = 1$ in the given equation and obtain the recurrence relation

$$a_{n+1} - f(1)a_n + a_{n-1} = 0, \tag{1}$$

where $a_n = f(n), n \geq 0$.

(a) In this case $f(1) = \frac{5}{2}$ and the roots of the characteristic equation $x^2 - \frac{5}{2}x + 1 = 0$ of (1) are $x_1 = 2, x_2 = \frac{1}{2}$. Hence

$$a_n = A \cdot 2^n + B \cdot 2^{-n}$$

for some constants A and B . Since $a_0 = f(0) = 2$ and $a_1 = f(1) = \frac{5}{2}$ we get that $A = B = 1$. Thus $f(n) = 2^n + 2^{-n}$ for $n \geq 0$. Since $f(n) = f(-n)$ we conclude that $f(n) = 2^n + 2^{-n}$ for all $n \in \mathbb{Z}$.

It is easy to check that this function satisfies the given equation.

(b) In this case $f(1) = \sqrt{3}$ and the roots of the characteristic equation $x^2 - \sqrt{3}x + 1 = 0$ of (1) are $x_{1,2} = \cos \frac{\pi}{6} \pm i \sin \frac{\pi}{6}$. Hence

$$f(n) = a_n = A \cos \frac{n\pi}{6} + B \sin \frac{n\pi}{6}$$

for $n \geq 0$. Since $f(0) = 2$, $f(1) = \sqrt{3}$ we get $A = 2$, $B = 0$. Thus

$$f(n) = 2 \cos \frac{n\pi}{6}$$

for all $n \in \mathbb{Z}$ and it is easy to check that this function satisfies the given equation.

Remark. The solution from a), b) adapts easily to the general case - if

$$f(1) = t + \frac{1}{t}, \quad t \neq \pm 1$$

then we can prove by induction that $f(n) = t^n + \frac{1}{t^n}$. This can be done either by induction, using the identity

$$\left(t^n + \frac{1}{t^n}\right) \left(t^m + \frac{1}{t^m}\right) = \left(t^{n+m} + \frac{1}{t^{n+m}}\right) + \left(t^{n-m} + \frac{1}{t^{n-m}}\right),$$

or the recurrence $a_{n+1} - f(1)a_n + a_{n-1} = 0$ whose characteristic polynomial has two distinct roots $t, \frac{1}{t}$.

Example 6.4. Let a be a positive integer. Determine all functions $f : \mathbb{N}_0 \rightarrow \mathbb{R}$ such that

$$f(n+m) + f(n-m) = f(an)$$

for all nonnegative integers $n \geq m$.

Solution. Setting $m = 0$ gives $2f(n) = f(an)$. In particular $f(0) = 0$. Now setting $m = 1$ we get the recurrence relation

$$a_{n+1} - 2a_n + a_{n-1} = 0,$$

where $a_n = f(n)$, $n \geq 0$. The characteristic equation $x^2 - 2x + 1 = 0$ has a double root $x_1 = x_2 = 1$ and it follows that $f(n) = A + Bn$. Since $0 = f(0) = A$ we see that $f(n) = Bn$ for all $n \in \mathbb{N}_0$. Hence $B(n+m) + B(n-m) = Ban$ for all $n \geq m$ and we get $B(a-2) = 0$. Thus for $a \neq 2$ the only solution is the zero function whereas for $a = 2$ the solutions are the functions $f(n) = Bn$, where B is a constant.

Example 6.5. Determine all functions $f : [0, \infty) \rightarrow \mathbb{R}$ such that $f(0) = 0$ and

$$f(x) = 1 + 5f\left(\left\lfloor \frac{x}{2} \right\rfloor\right) - 6f\left(\left\lfloor \frac{x}{4} \right\rfloor\right)$$

for all $x > 0$.

Solution. If $0 \leq x < 2$, then $\left\lfloor \frac{x}{2} \right\rfloor = \left\lfloor \frac{x}{4} \right\rfloor = 0$ and

$$f(x) = 1 + 5f(0) - 6f(0) = 1.$$

In particular $f(1) = 1$. If $2 \leq x < 4$, then

$$\left\lfloor \frac{x}{2} \right\rfloor = 1, \quad \left\lfloor \frac{x}{4} \right\rfloor = 0$$

and $f(x) = 1 + 5f(1) - 6f(0) = 6$. Now it follows easily by induction on n that $f(x) = a_n$ for $x \in [2^n, 2^{n+1})$, $n \geq 0$, where $\{a_n\}_{n=0}^{\infty}$ is the sequence defined by $a_0 = 1$, $a_1 = 6$ and $a_n = 1 + 5a_{n-1} - 6a_{n-2}$ for $n \geq 2$. Set $a_n = \frac{1}{2} + b_n$. Then $b_0 = \frac{1}{2}$, $b_1 = \frac{11}{2}$ and $b_n - 5b_{n-1} + 6b_{n-2} = 0$ for $n \geq 2$. The characteristic equation $x^2 - 5x + 6 = 0$ of this recurrence relation has roots $x_1 = 2$, $x_2 = 3$ and we get that $b_n = \frac{1}{2}(3^{n+2} - 2^{n+3})$ and $a_n = \frac{1}{2}(1 + 3^{n+2} - 2^{n+3})$ for $n \geq 0$. Hence the equation has a unique solution $f(x)$ given by

$$f(x) = \begin{cases} 0 & \text{if } x = 0 \\ 1 & \text{if } x \in (0, 2) \\ \frac{1}{2}(1 + 3^{n+2} - 2^{n+3}) & \text{if } x \in [2^n, 2^{n+1}), n \geq 1. \end{cases}$$

Example 6.6. Find all functions $f : \mathbb{Z} \rightarrow \mathbb{Z}$ that obey

$$f(m+n) + f(mn-1) = f(m)f(n)$$

for all $m, n \in \mathbb{Z}$.

Solution. If $f = c$ is a constant, we have $2c = c^2$ so $c = 0, 2$. If f is not constant setting $m = 0$ gives us $f(n)(1 - f(0)) = -f(-1)$ possible only for

$f(-1) = 0, f(0) = 1$. Then set $m = -1$ to get $f(n-1) + f(-n-1) = 0$. Now set $m = 1$ to get $f(n+1) + f(n-1) = f(1)f(n)$. This is a quadratic recurrence relation in $f(n)$ with characteristic equation $x^2 - f(1)x + 1 = 0$. If $f(1) = 0$, we get $f(n-1) + f(n+1) = 0$ which implies $f(n+2) = -f(n)$ so $f(2k) = (-1)^k f(0) = (-1)^k$, $f(2k+1) = (-1)^k f(1) = 0$. This function does satisfy the equation as one easily checks. If $f(1) = -1$ then we get $f(n) = (n-1) \pmod 3 - 1$ for all n by induction on n . It also satisfies the condition as we can check by looking at m, n modulo 3. If $f(1) = 2$, then $f(n+1) - 2f(n) + f(n-1) = 0$ and $f(n) = n+1$ by induction on $|n|$. It also satisfies the condition as $(m+n+1) + mn = (m+1)(n+1)$. If $f(1) = 1$, then we have $f(n+1) + f(n-1) = f(n)$. Hence $f(n+2) + f(n) = f(n+1)$ and we get that $f(n+2) = -f(n-1)$. In particular, $f(8) = -f(5) = f(2) = -f(-1) = 0$ and analogously $f(15) = -1$, and $f(4) = -1$. Now plugging $m = n = 4$ in the given identity we obtain $-1 = f(8) + f(15) = f^2(4) = 1$, a contradiction. If $f(1) = -2$, then $f(n+1) + f(n-1) + f(n) = 0$ and the characteristic equation $x^2 + 2x + 1$ has a double root $x_1 = x_2 = -1$. Hence $f(n) = (an+b)(-1)^n$ and the conditions $f(-1) = 0, f(0) = 1$ imply that $f(n) = (-1)^n(n+1)$ which is not a solution.

Finally, if $f(1) \neq 0, 1, -1, 2, -2$, then the equation $x^2 - f(1)x + 1 = 0$ has two roots $\frac{f(1) \pm \sqrt{f^2(1)-4}}{2}$, one of which is greater than 1 in absolute value and the other one is smaller. If we solve the recurrence equation, we find that $f(n) = cr^n + ds^n$, where $c, d \neq 0$ and without loss of generality $|r| > 1, |s| < 1$. In this case we have $f(n) \sim cr^n$ for $n \rightarrow \infty$. Then $f(m+n) + f(mn-1) = f(m)f(n)$ cannot hold because for $m = n$ the left-hand side is asymptotically equivalent to cr^{n^2-1} for $n \rightarrow \infty$ while the right-hand side is asymptotically equivalent to c^2r^{2n} and $n^2 - 1$ is much larger than $2n$.

Example 6.7. Find all functions $f: \mathbb{N} \rightarrow \mathbb{N}$ that satisfy

$$f(n) + f(n+1) = f(n+2)f(n+3) - k$$

for all $n \in \mathbb{N}$, where $k+1$ is a prime.

Solution. This is mainly a sequence problem. Set $a_n = f(n)$ to get

$$a_n + a_{n+1} = a_{n+2}a_{n+3} - k.$$

Writing this condition for $n-1$ we get $a_n + a_{n-1} = a_{n+2}a_{n+1} - k$. Subtracting them we get

$$(a_{n+1} - a_{n-1}) = a_{n+2}(a_{n+3} - a_{n+1}).$$

So if we set $b_m = a_{m+2} - a_m$ we get $b_{n-1} = a_{n+2}b_{n+1}$ and so

$$|b_{n+1}| = \frac{|b_{n-1}|}{|a_{n+2}|}.$$

Thus $|b_{n+1}| \leq |b_n|$ and if $b_{n-1} \neq 0, a_{n+2} \neq 1$, then $|b_{n+1}| < |b_{n-1}|$ and $b_{n+1} \neq 0$. Then again if $a_{n+4} \neq 1$, we will get $|b_{n+3}| < |b_{n+1}|, b_{n+3} \neq 0$ and so on. This would produce an infinite sequence of decreasing positive integers which is impossible. Therefore either $b_{n-1} = 0$ or the sequence $a_{n+2}, a_{n+4}, \dots$, becomes eventually 1. Set $n = 3$ to see that either $a_2 = a_4$ or $a_{2m+1} = 1$ for all $m \geq m_0$. Likewise if we set $n = 2$ we see that either $a_3 = a_1$ or $a_{2m} = 1$ for all $m \geq m_0$.

a) Assume that $a_2 = a_4$. Then $b_2 = 0$ and hence by induction $b_{2m} = 0$. So $a_{2m+2} = a_{2m}$ hence $a_{2m} = a_2$. If $a_2 \neq 1$, then we must have $a_1 = a_3$ and hence by induction $a_{2m+1} = a_1$. Thus the condition written for $n = 1$ becomes $a_2 + a_1 = a_2a_1 - k$ or $(a_2 - 1)(a_1 - 1) = k + 1$. As $k + 1$ is prime, one of $a_2 - 1$ and $a_1 - 1$ is $k + 1$ and the other is 1. So either $f(n) = k + 2$ for even n and $f(n) = 2$ for odd n or viceversa: $f(n) = 2$ for odd n and $f(n) = k + 2$ for even n . Both functions satisfy the condition. Now if $a_2 = 1$, then $a_{2m} = 1$ and we have $1 + f(2m + 1) = f(2m + 3) - k$ so $f(2m + 3) = f(2m + 1) + k + 1$. We conclude that $f(2m) = 1, f(2m + 3) = m(k + 1) + a$, where $a = f(1)$. This function also satisfies the condition.

b) Assume that $a_{2m+1} = 1$ for all $m \geq m_0$. We know that either $a_{2p} = 1$ for all $p \geq p_0$ or $a_3 = a_1$. If a_{2p} becomes eventually 1, set $n \geq 2m_0 + 1, 2p_0$ to get $1 + 1 = 1 - k$, a contradiction. Hence $a_3 = a_1$ and like in a) we conclude $a_{2m+1} = a_1$. As a_{2m+1} is eventually 1 we have $a_1 = 1$. Then like in a) we conclude that $f(2m) = (m - 1)(k - 1) + a$ where $a = f(2)$. It also satisfies the condition.

Example 6.8. Find all functions $f: \mathbb{N} \rightarrow \mathbb{N}$ such that

$$f(f(f(n))) + f(f(n)) + n = 3f(n)$$

for all $n \in \mathbb{N}$.

Solution. Fix an $n \in \mathbb{N}$ and set $a_0 = n$, $a_{k+1} = f(a_k)$, $k \geq 0$. We get the recurrence relation

$$a_{k+3} + a_{k+2} - 3a_{k+1} + a_k = 0$$

with characteristic equation $x^3 + x^2 - 3x + 1 = 0$. Its roots are equal to 1 and $-1 \pm \sqrt{2}$, i.e.

$$a_k = c_0 + c_1(-1 + \sqrt{2})^k + c_2(-1 - \sqrt{2})^k$$

for all $k \geq 0$. Note that $|-1 - \sqrt{2}| > 1 > |-1 + \sqrt{2}|$. If $c_2 > 0$, then $a_{2k+1} \rightarrow -\infty$ which is a contradiction. Similarly, the case $c_2 < 0$ is impossible. Thus $c_2 = 0$. Then using that $a_0, a_1, a_2 \in \mathbb{Q}$ we easily get that $c_1 = 0$. (Prove it!). Hence $a_1 = a_0$, i.e. $f(n) = n$ for all $n \in \mathbb{N}$.

Example 6.9. (BMO 2002) Find all functions $f: \mathbb{N} \rightarrow \mathbb{N}$ such that

$$2n + 2001 \leq f(f(n)) + f(n) \leq 2n + 2002$$

for all $n \in \mathbb{N}$.

Solution. Fix an n and set

$$a_0 = n, a_{k+1} = f(a_k), c_k = a_{k+1} - a_k - 667, \quad k \geq 0.$$

Then

$$2a_k + 2001 \leq a_{k+2} + a_{k+1} \leq 2a_k + 2002,$$

$$0 \leq c_{k+1} + 2c_k \leq 1, \quad k \geq 0.$$

We shall prove that $c_0 = 0$. Assume the contrary. Then we may assume that $c_0 \geq 1$ since otherwise $c_1 \geq -2c_0 \geq 2$ and we consider the sequence $c_1, c_2, \dots$. We have

$$c_{2k+2} \geq -2c_{2k+1} \geq 4c_{2k} - 2 \geq 2c_{2k}$$

and it follows by induction that $c_{2k} \geq 2^k$, $k \geq 0$. Hence

$$a_{2k+2} = a_{2k} + c_{2k} + c_{2k+1} + 1334 \leq a_{2k} + 1335 - c_{2k} \leq$$

$$\leq a_{2k} + 1335 - 2^k, \quad k \geq 0.$$

Summing up these inequalities gives

$$a_{2k} \leq a_0 + 1335k - 2^k, \quad k \geq 0.$$

This inequality shows that $a_{2k} \leq 0$ for all sufficiently large k , a contradiction. Thus $c_0 = 0$ and $f(n) = n + 667$ for all n . It is easy to check that this function satisfies the given conditions.

Remark. The above problem is considered also in Chapter 14 (Problem 14.5), where it is solved by using the extremal element method.

Example 6.10. (Putnam 2001) Let $a, b \in (0, 1/2)$ and $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function such that

$$f(f(x)) = af(x) + bx$$

for all $x \in \mathbb{R}$. Prove that there exists a real constant c such that $f(x) = cx$ for all $x \in \mathbb{R}$.

Solution. Note first that the function f is injective and its continuity implies that f is strictly monotonic. Moreover, f is unbounded since the function bx is unbounded and therefore f is onto. For arbitrary $x_0 \in \mathbb{R}$ define $x_{n+1} = f(x_n)$ for $n > 0$ and $x_{n-1} = f^{-1}(x_n)$ for $n \leq 0$. Then it follows from the given condition for f that $x_{n+2} = ax_{n+1} + bx_n$ for all $n \in \mathbb{Z}$. Let t_1 and t_2 be the roots of the characteristic equation $x^2 - ax - b = 0$. Then $t_1 > 0 > t_2$, $1 > |t_1| > |t_2|$ and there exist $c_1, c_2 \in \mathbb{R}$ such that $x_n = c_1 t_1^n + c_2 t_2^n$ for all $n \in \mathbb{Z}$. Assume that f is increasing. If $c_2 > 0$, then $0 < x_n < x_{n+2}$ and $0 < x_{n+3} < x_{n+1}$ for odd $n < 0$ small enough, a contradiction since we get $f(x_n) > f(x_{n+2})$ but $x_n < x_{n+2}$. Similarly, the case $c_2 < 0$ is impossible and we conclude that $c_2 = 0$. So, $x_0 = c_1$ and $x_1 = c_1 t_1 = t_1 x_0$. Thus $f(x) = t_1 x$ for all $x \in \mathbb{R}$. The same arguments show that if f is decreasing, then $t_1 = 0$ and $f(x) = t_2 x$ for all $x \in \mathbb{R}$.

6.2 Practice Problems

Problem 6.1. Find the n -th iterate of the function:

- (a) $f(x) = 4x(1 - x), x \in [0, 1]$;
- (b) $f(x) = ax^2 + bx + c, a \neq 0, 4ac = b^2 - 2b$.

Problem 6.2. Find all functions $f: \mathbb{Q}^+ \rightarrow \mathbb{R}$ that satisfy

$$f(x+y) + f(x-y) = (2^y + 2^{-y})f(x)$$

for all $x, y \in \mathbb{Q}^+$.

Problem 6.3. (Bulgaria 1996) Find all functions $f: \mathbb{Z} \rightarrow \mathbb{Z}$ such that

$$3f(n) - 2f(f(n)) = n$$

for all $n \in \mathbb{Z}$.

Problem 6.4. (MOSP 2001) Find all functions $f: \mathbb{N} \rightarrow \mathbb{N}$ such that

$$f(f(f(n))) + 6f(n) = 3f(f(n)) + 4n + 2001$$

for all $n \in \mathbb{N}$.

Problem 6.5. (Putnam 1988) Prove that there exists a unique function $f: (0, \infty) \rightarrow (0, \infty)$ such that

$$f(f(x)) + f(x) = 6x$$

for all $x \in (0, \infty)$.

Problem 6.6. Find all functions $f: (0, \infty) \rightarrow (0, \infty)$ such that

$$f(f(f(x))) + f(f(x)) = 2x + 5$$

for all $x \in (0, \infty)$.

Problem 6.7. Find all continuous functions $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfying

$$f(2x+1) = f(x)$$

for all $x \in \mathbb{R}$.

Problem 6.8. Find all continuous functions $f: \mathbb{R} \rightarrow \mathbb{R}$ that satisfy

$$f(f(x)) = f(x) + 2x$$

for all $x \in \mathbb{R}$.

Problem 6.9. (IMO Longlist 1979) Find all monotonic bijections $f: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x) + f^{-1}(x) = 2x$$

for all $x \in \mathbb{R}$.

Problem 6.10. Find all functions $f: \mathbb{Z} \rightarrow \mathbb{Z}$ that satisfy

$$f(m+n) + f(mn-1) = f(m)f(n) + k$$

for all $m, n \in \mathbb{Z}$ and a given $k \in \mathbb{Z}$.

Problem 6.11. (Belarus 1998) Prove that:

a) if $a \leq 1$ then there is no function $f: (0, \infty) \rightarrow (0, \infty)$ such that

$$f\left(f(x) + \frac{1}{f(x)}\right) = x + a \tag{1}$$

for all $x \in (0, \infty)$;

b) if $a > 1$, then there are infinitely many functions $f: (0, \infty) \rightarrow (0, \infty)$ satisfying (1).

Problem 6.12. (Bulgaria 2003) Find all $a > 0$ for which there exists a function $f: \mathbb{R} \rightarrow \mathbb{R}$ having the following two properties:

(i) $f(x) = ax + 1 - a$ for all $x \in [2, 3]$;

(ii) $f(f(x)) = 3 - 2x$ for all $x \in \mathbb{R}$.

Chapter 7

Constructive Problems

7.1 Examples

These problems involve explicit constructions of functions or other "objects", or inductive arguments. We did not include here only problems that ask the reader specifically to construct a function given some conditions. We also added problems in which the reader has to recover the function step by step, thus gradually *constructing* the solution for the problem. Another type of argument might be one where the reader constructs a sequence of numbers that is important to the solution. Many of the problems of this type may fit into other sections of the book, this is why the section has much less problems than it could have. The reader will meet many exercises of this category throughout the entire book.

Example 7.1. Let k be an even positive integer. Find the number of all functions $f: \mathbb{N}_0 \rightarrow \mathbb{N}_0$ such that $f(f(n)) = n + k$ for all $n \in \mathbb{N}_0$.

Solution. We have

$$f(n + k) = f(f(f(n))) = f(n) + k$$

and it follows by induction on m that

$$f(n + km) = f(n) + km$$

for all $n, m \in \mathbb{N}_0$. Thus f is determined by its values on the set $\{0, 1, \dots, k-1\}$.

Now take an arbitrary integer $p, 0 \leq p \leq k-1$, and let $f(p) = kq + r$, where $q \in \mathbb{N}_0$ and $0 \leq r \leq k-1$. Then

$$p + k = f(f(p)) = f(kq + r) = f(r) + kq.$$

Hence either $q = 0$ or $q = 1$ and therefore either $f(p) = r, f(r) = p + k$ or $f(p) = r + k, f(r) = p$. In both cases we have $p \neq r$ which shows that f defines a pairing of the set $A = \{0, 1, \dots, k-1\}$. More precisely we pick the pair (p, r) if $f(p) = r, f(r) = p + k$ and (r, p) if $f(r) = p, f(p) = r + k$. This partitions A into *ordered* pairs. Note that different functions define different pairings of A .

Conversely, any pairing of A defines a function $f: \mathbb{N}_0 \rightarrow \mathbb{N}_0$ with the given property in the following way. We define f on A by setting $f(p) = r, f(r) = p + k$ for any pair (p, r) of the given pairing and $f(n) = f(q) + ks$ for $n \geq k+1$, where q and s are respectively the quotient and the remainder of n in the division by k .

Thus the number of the functions with the given property is equal to that of all pairings of the set A . It is easy to see that this number is equal to $\frac{k!}{(k/2)!}$.

Remark. The above solution shows that if k is an odd positive integer then there are no functions $f: \mathbb{N}_0 \rightarrow \mathbb{N}_0$ such that

$$f(f(n)) = n + k$$

for all $n \in \mathbb{N}_0$. For $k = 1987$ this problem was given at IMO 1987.

Example 7.2. Find all functions $f: \mathbb{N} \rightarrow \mathbb{R} \setminus \{0\}$ which satisfy

$$f(1) + f(2) + \dots + f(n) = f(n)f(n+1)$$

for all $n \in \mathbb{N}$.

Solution. If we try to set $f(x) = cx$ we compute that $c = \frac{1}{2}$. However the condition of the problem provides a clear recurrence relation for f , therefore there are as many solutions as possible values for $f(1)$. So set $f(1) = a$. Then setting $n = 1$ in the condition we get $a = af(2)$ and as $a \neq 0$ we get $f(2) = 1$. Then setting $n = 2$ we get $f(3) = a + 1$. Setting $n = 3$ we get

$f(4)(a+1) = a+1 + (a+1)$ so $f(4) = 2$ as $a+1 = f(3) \neq 0$. Now we see a pattern: for even numbers k we have $f(k) = \frac{k}{2}$, whereas for odd numbers k we have an additional a , and we can suppose that

$$f(k) = \left\lfloor \frac{k}{2} \right\rfloor + (k \bmod 2)a.$$

Let's now prove this hypothesis by induction on k . Clearly we have to consider two cases according to the parity of k .

a) $k = 2n$. Then we have

$$f(1) + f(2) + \dots + f(k) = f(k)f(k+1)$$

or

$$\frac{1}{2} + \frac{2}{2} + \dots + \frac{2n}{2} + n \left(a - \frac{1}{2} \right) = nf(2n+1).$$

So

$$\frac{2n(2n+1)}{4} + na - \frac{n}{2} = nf(2n+1)$$

which gives us $f(2n+1) = n + a$, as desired.

b) $k = 2n+1$. This case is absolutely analogous.

Hence all desired functions are of form $f(k) = \left\lfloor \frac{k}{2} \right\rfloor + (k \bmod 2)a$ for some a . They clearly satisfy the conditions of the problem provided that a is not a negative integer (in this case $f(-2a+1) = 0$).

Example 7.3. Find all functions $f: \mathbb{N}_0 \rightarrow \mathbb{N}_0$ that satisfy $f(0) = 1$ and

$$f(n) = f\left(\left\lfloor \frac{n}{a} \right\rfloor\right) + f\left(\left\lfloor \frac{n}{a^2} \right\rfloor\right)$$

for all $n \in \mathbb{N}$, where $a \geq 2$ is a fixed integer.

Solution. Partition $\mathbb{N}$ into the sets

$$S_k = \{a^k, a^k + 1, \dots, a^{k+1} - 1\}, \quad k = 0, 1, 2, \dots$$

We see that if $n \in S_k$ then $\lfloor \frac{n}{a} \rfloor \in S_{k-1}$, $\lfloor \frac{n}{a^2} \rfloor \in S_{k-2}$ (for $k \geq 2$). Next we see that if $n \in S_0$, then $f(n) = 2$ and if $n \in S_1$ then $f(n) = 3$. So we can easily prove by induction that f is constant on each S_k . If we let $g(k)$ be the value of f on S_k , then $g(k) = g(k-1) + g(k-2)$ for $k \geq 2$. It's clear now that $g(k) = F_{k+2}$ where $(F_n)_{n \in \mathbb{N}_0}$ is the Fibonacci sequence. So $f(n) = F_{\lfloor \log_a n \rfloor + 2}$ for $n \geq 1$.

Example 7.4. Find all functions $f: \mathbb{Z} \rightarrow \mathbb{Z}$ that satisfy the relation

$$f(m+n) + f(m)f(n) = f(mn+1)$$

whenever m, n are integers.

Solution. By setting $m = n = 1$ we get $f(2) + f(1)^2 = f(2)$ so $f(1) = 0$. Now let $m = 0$ to get $f(n) + f(0)f(n) = 0$. Therefore either $f(0) = -1$ or f is identically zero. Excluding this trivial case we get $f(0) = -1$. Now take $m = -1$ to get $f(n-1) + f(-1)f(n) = f(1-n)$ so

$$f(n-1) - f(1-n) = -f(-1)f(n)$$

and we have two cases:

a) $f(-1) = 0$. In this case we have $f(-x) = f(x)$. First, let us try to settle the uniqueness of f . Set $f(2) = f(-2) = a$. Then by setting $m = 2, n = -2$ we compute $f(3) = f(-3) = a^2 - 1$. Set $m = 2, n = -3$ to get $f(5) = f(2)f(3) = a(a^2 - 1)$. Set $m = n = 2$ to compute $f(4) = a^3 - a^2 - a$. Set $m = 3, n = -3$ to get $f(8) = a^4 - 2a^2$. Next by setting $m = 4, n = -4$ and $m = 6, n = -2$ we get $f(15) = f^2(4) - 1 = f(6) + f(8)f(2)$ which allows us to deduce that $f(6) = a^3 + a^2 - 1$. Set $m = 2, n = 3$ and get $f(7) = f(5) + f(2)f(3) = 2a(a^2 - 1)$. Also by setting $m = 4, n = -2$ we get $f(7) = f(2) + f(2)f(4) = a^4 - a^3 - a^2 + a$. Thus we get $2a^2(a^2 - 1) = a^4 - a^3 - a^2 - a = 0$ or $a(a-1)(a+1)(a-3) = 0$ so we have four values for a . We can guess that all values of f can be determined from the value of a only. Indeed, let us prove by induction on $|n|$ that $f(n)$ is uniquely determined by $f(2) = a$ for each n . We see this holds true for any n with $|n| \leq 8$. Now assume this holds whenever $|n| \leq k-1$ and let us prove it for $n = k \geq 9$ ($n = -k$ is the same). We seek numbers $0 < x < u < v < y$

with $x + y = k, xy = uv$. Then applying the condition for $m = x, n = y$ and $m = u, n = v$ we get

$$f(x + y) + f(x)f(y) = f(xy + 1) = f(uv + 1) = f(u + v) + f(u)f(v).$$

So

$$f(k) = f(x + y) = f(u + v) + f(u)f(v) - f(x)f(y)$$

and we can easily prove that $u, v, x, y, u + v < k$. Thus $f(k)$ is determined and the induction is finished ($u + v < k = x + y$ because $(u - x) + (v - y) = u - x + \left(\frac{uv}{u} - \frac{uv}{x}\right) = \frac{(u - x)(x - v)}{vx} < 0$). Now if $k = (2a + 1)2^b$ for $a > 0$ then we can set $x = 2^b, y = 2^{b+1}a, u = 2^{b+1}, v = 2^b a$. If k is a power of 2 then either $k = 3p + 1$ and $x = 1, y = 3p, u = 3, v = p$ or $k = 3p + 2$ with $x = 2, y = 3p, u = 3, v = 2p$.

So at most one f stems from each value of a - in particular finding a function that satisfies the condition implies this is the only one (for the corresponding value of a). Surprisingly, each of the four values of a gives a solution f , so the general set of solutions is quite unexpected. Let us analyze them one by one:

i) $a = 0$. By substituting we get

$$f(0) = f(3) = f(6) = -1, f(1) = f(2) = f(4) = f(5) = f(7) = f(8) = 0.$$

We guess the solution $f(x) = -1$ when x is divisible by 3 and $f(x) = 0$ otherwise. Indeed it satisfies the conditions: If both m, n are divisible by 3 we get the identity $-1 + 1 = 0$ as $3|m + n$ but $mn + 1$ is not divisible by 3; if one of m, n is divisible by 3 and the other is not we get $0 + 0 = 0$ as neither $mn + 1$ nor $m + n$ are divisible; finally if neither of m, n is divisible by 3 the either they give the same residue mod 3, in which $m + n$ and $mn + 1$ are not divisible by 3 and we get $0 + 0 = 0$ or they give different residues mod 3 which means $m + n, mn + 1$ are divisible by 3 so $1 + 0 = 1$.

ii) $a = 1$. By substituting we get

$$f(2) = f(6) = 1, f(4) = f(8) = -1, f(1) = f(3) = f(5) = f(7) = -1$$

so we propose $f(4k + 2) = 1, f(4k) = -1, f(2k + 1) = 0$. Indeed, if m, n are both odd then $m + n$ and $mn + 1$ give the same residue modulo 4 as

$mn + 1 - m - n = (m - 1)(n - 1)$ is a product of two even integers, so the equality holds. If one of m, n is odd and the other is even then $mn + 1, m + n$ are both odd and again the equality holds. If both m, n are even then either they give the same residue mod 4 in which we have the identity $-1 + 1 = 0$ or they give different residues which implies $1 + (-1) = 0$, again true.

iii) $a = -1$. By substituting we get

$$f(2) = f(4) = f(6) = f(8) = -1, f(1) = f(3) = f(5) = f(7) = 0$$

so we guess $f(2k) = -1, f(2k + 1) = 0$. Indeed, if at least one of m, n is odd then $(mn + 1) - (m + n) = (m - 1)(n - 1)$ is even so the identity holds true, while when they are both even we get the equality $-1 + 1 = 0$.

iv) $a = 3$. We compute $f(3) = 8, f(4) = 15$ and so on, guessing

$$f(x) = x^2 - 1.$$

Indeed,

$$\begin{aligned} (m + n)^2 - 1 + (m^2 - 1)(n^2 - 1) &= m^2 + n^2 + 2mn - 1 + m^2n^2 - m^2 - n^2 + 1 \\ &= m^2n^2 + 2mn = (mn + 1)^2 - 1. \end{aligned}$$

This case is exhausted.

b) $f(-1) \neq 0$. Then we have

$$f(n) = -\frac{f(n-1) - f(-(n-1))}{f(-1)}.$$

Therefore we get

$$f(-n) = \frac{f(n+1) - f(-n-1)}{f(-1)}.$$

If we set $a_n = f(n) - f(-n)$, we observe by subtracting the previous two results the nice recursive identity $a_n = a(a_{n-1} + a_{n+1})$ where $a = -\frac{1}{f(-1)}$. We can write it as $a_{n+1} = ba_n - a_{n-1}$ where $b = \frac{1}{a} = -f(-1)$. Also $a_0 = 0, a_1 = b$. Then $a_2 = b^2, a_3 = b^3 - b, a_4 = b^4 - 2b^2, a_5 = b^5 - 3b^3 + b$ and so on. Next $f(n) = \frac{a_{n-1}}{b}$. Hence

$$f(2) = 1, f(3) = b, f(4) = b^2 - 1, f(5) = b^3 - 2b, f(6) = b^4 - 3b^2 + 1.$$

Setting $m = n = 2$ we get $f(4) + f^2(2) = f(5)$ or $b^2 - 1 + 1 = b^3 - 2b$ so $b^2 = b^3 - 2b$ and as $b \neq 0$ we get $b^2 = b + 2$ so $b = 2$ or $b = -1$.

i) $b = 2$. In this case we can guess the identity $f(x) = x - 1$. It holds by induction as we prove that $a_n = 2n$ by induction (another proof can be obtained by using the uniqueness of the solution that we deduced in case a)). Indeed $f(x) = x - 1$ satisfies the condition as

$$(m + n - 1) + (m - 1)(n - 1) = mn = (mn + 1) - 1.$$

ii) $b = -1$. We compute that

$$f(-1) = 1, f(0) = -1, f(1) = 0, f(2) = 1,$$

$$f(3) = -1, f(4) = 0, f(5) = 1, f(6) = -1.$$

We conjecture that $f(3k) = -1, f(3k + 1) = 0, f(3k + 2) = 1$. Indeed, this can be either proven by showing inductively that $b_{3k} = 1, b_{3k+1} = 0, b_{3k+2} = 1$, or by using again the uniqueness argument after we verify that f satisfies our equation:

If $3|m, 3|n$ then we get $-1 + 1 = 0$. When $3|m, 3|n - 1$ or $3|n, 3|m - 1$ then we get $0 + 0 = 0$. $3|m, 3|n - 2$ or $3|n, 3|m - 2$ then we get $1 + (-1) = 0$. $3|m - 1, 3|n - 1$ then $1 + 0 = 1$. $3|m - 1, 3|n - 2$ or $3|m - 2, 3|n - 1$ then we get $0 - 1 = -1$. $3|m - 2, 3|n - 2$ then $0 + 1 = 1$. All the identities are true so this function is indeed a solution.

Concluding, we have plenty of different solutions:

$$f(x) = x^2 - 1, f(x) = x - 1, f(x) = (x + 1) \pmod{2}, f(x) = x \pmod{3} - 1,$$

$$f(x) = (x + 1) \pmod{2}(x \pmod{4} + 1), f(x) = (x \pmod{3})^2 - 1.$$

Example 7.5. Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ that satisfy

$$f(-x) = -f(x), f(x + 1) = f(x) + 1$$

for any $x \in \mathbb{R}$ and

$$f\left(\frac{1}{x}\right) = \frac{f(x)}{x^2}$$

for any $x \in \mathbb{R} \setminus \{0\}$.

Solution. All the conditions are in one variable. In this case, some graph theory helps us understand the path to the solution. Consider the reals as vertices of a graph, and connect x with $x+1$, $-x$, $\frac{1}{x}$. The conditions link two values of the function in two vertices joined by an edge. So if we pick up a x_0 , we can deduce from $f(x_0)$ the values of f on C where C is the set of numbers connected to x_0 by some chain of edges. Now we can get a condition if and only if there is a cycle somewhere. So finding a cycle would impose a condition on $f(x_0)$ and maybe would exactly find the value of $f(x_0)$. So let us try to construct such a cycle for any x . After some tries we see

$$x \rightarrow x+1 \rightarrow \frac{1}{x+1} \rightarrow -\frac{1}{x+1} \rightarrow 1 - \frac{1}{x+1} = \frac{x}{x+1} \rightarrow \frac{x+1}{x} = 1 + \frac{1}{x} \rightarrow \frac{1}{x} \rightarrow x.$$

Set $f(x) = y$. Then

$$\begin{aligned} f(x+1) &= y+1, \quad f\left(\frac{1}{x+1}\right) = \frac{y+1}{(x+1)^2}, \quad f\left(-\frac{1}{x+1}\right) = -\frac{y+1}{(x+1)^2}, \\ f\left(\frac{x}{x+1}\right) &= \frac{x^2+2x-y}{(x+1)^2}, \quad f\left(\frac{x+1}{x}\right) = \frac{x^2+2x-y}{x^2}, \\ f\left(\frac{1}{x}\right) &= \frac{2x-y}{x^2}, \quad f(x) = 2x-y. \end{aligned}$$

So $y = 2x - y$ thus $y = x$. Note that we need to have $x \neq 0, -1$ in order not to divide by zero. This is no problem for us, as $f(0) + 1 = f(1)$ and we know that $f(1) = 1$ so $f(0) = 0$, also $f(-1) = -f(1) = 1$ hence $f(x) = x$ for all x , and it satisfies the condition.

Example 7.6. Find all functions $f: \mathbb{Q}^+ \rightarrow \mathbb{Q}^+$ that satisfy

$$f(x) + f\left(\frac{1}{x}\right) = 1 \text{ and } f(1+2x) = \frac{f(x)}{2}$$

for any $x \in \mathbb{Q}^+$.

Solution. First, let us try to guess the function. It is natural to suppose

$$f(x) = \frac{ax+b}{cx+d}.$$

The condition $f(x) + f\left(\frac{1}{x}\right) = 1$ now translates as $\frac{(ax+b)}{(cx+d)} + \frac{bx+a}{dx+c} = 1$ and we conclude that $c = d$. We can assume $c = d = 1$ otherwise divide a, b by c to see

$$\frac{ax+b}{cx+c} = \frac{\frac{a}{c}x + \frac{b}{c}}{x+1}.$$

Next we need to have $\frac{ax+b}{x+1} + \frac{bx+a}{x+1} = 1$ which is possible for $a+b=1$.

Now the condition $f(1+2x) = \frac{f(x)}{2}$ means $\frac{a(1+2x)+b}{2x+2} = \frac{1}{2} \frac{ax+b}{x+1}$ or $a(1+2x)+b = ax+b$. Hence $a=0$ and $f(x) = \frac{1}{x+1}$ which satisfies the conditions. The idea here is that all the values of f are positive. So we try to prove that if $f(x) \neq \frac{1}{x+1}$ then some value of f should be negative. Indeed, set $g(x) = f(x) - \frac{1}{x+1}$. Then $g\left(\frac{1}{x}\right) = -g(x)$, $g(1+2x) = \frac{g(x)}{2}$. As

$$g(x) + 1 > g(x) + \frac{1}{x+1} = f(x) > 0$$

we see that $g(x) > -1$ and as $g(x) = -g\left(\frac{1}{x}\right)$ we deduce $g(x) < 1$ so $|g(x)| < 1$.

1. Note that the second condition can be rewritten as $g\left(\frac{x-1}{2}\right) = 2g(x)$ for $x > 1$. If $g(x) = a \neq 0$ we find by induction numbers x_n such that $|g(x_n)| = 2^n a$. We set $x_0 = x$. Now assume we have found x_k . Then $x_k \neq 1$ as $g(1) = 0$. If $x_k > 1$ then $g\left(\frac{x_k-1}{2}\right) = 2g(x_k)$ and we set $x_{k+1} = \frac{x_k-1}{2}$.

If $0 < x_k < 1$ then $\frac{1}{x_k} > 1$, $g\left(\frac{1}{x_k}\right) = -g(x_k)$ therefore $g\left(\frac{\frac{1}{x_k}-1}{2}\right) = -2g(x_k)$

and we set $x_{k+1} = \frac{\frac{1}{x_k}-1}{2}$. Then if we find k such that $2^k|a| > 1$, we obtain a contradiction. Thus $g(x) = 0$ for all x and $f(x) = \frac{1}{x+1}$.

Example 7.7. Find all continuous functions $f: \mathbb{R} \rightarrow \mathbb{R}$ that satisfy

$$f(1 + x^2) = f(x)$$

for all $x \in \mathbb{R}$.

Solution. Set $g(x) = 1 + x^2$. As g is even we see that

$$f(x) = f(g(x)) = f(g(-x)) = f(-x)$$

so f is even, thus we need to find f only on $[0; \infty)$. Now we know that $f(g^{(k)}(x)) = f(x)$. Also $g(x)$ is increasing and $g(x) > x$ as $1 + x^2 > x$. Set $x = 0$ to get $f(0) = f(1)$. Next set $x_k = g^{(k)}(0)$ so that $x_0 = 0, x_1 = 1$. Then g maps $[x_{k-1}; x_k]$ into $[x_k; x_{k+1}]$ so $g^{(k)}$ maps $[0; 1]$ into $[x_k; x_{k+1}]$. As $g(x) > 0$ is increasing and $g(x) > 1$ we cannot establish any condition between $f(x)$ and $f(y)$ for $0 < x < y < 1$ because we cannot link x and y by operating with g : if $g^{(k)}(x) = g^{(l)}(y)$ then as $g^{(k)}(x) \in (x_k; x_{k+1}); g^{(l)}(y) \in (x_l; x_{l+1})$ we conclude $k = l$ and by injectivity $x = y$. Thus we may construct f as follows: define f a continuous function on $[0; 1]$ with $f(0) = f(1)$ and extend f to $(0, \infty)$ by setting $f(g^{(k)}(x)) = f(x)$ and then to $\mathbb{R}$ by setting $f(-x) = -f(x)$. Indeed, f satisfies $f(1 + x^2) = f(x)$. Moreover it's continuous: the graphs of f on $[x_k; x_{k+1}]$ are continuous as they are the composition of the continuous functions f on $[0; 1]$ and $(g^{(k)})^{-1}$ on $[x_k; x_{k+1}]$. As $f(x_k) = f(x_{k+1})$ the continuous graphs of f on intervals $[x_k; x_{k+1}]$ unite to form a continuous curve, and reflecting it with respect to the y axis we get the continuous graph of f .

The next example is a generalization of Problem A1 from Putnam 1999.

Example 7.8. Find all polynomials $f(x)$, $g(x)$ and $h(x)$ such that

$$|f(x)| - |g(x)| + h(x) = \begin{cases} -1 & \text{for } x < -1 \\ 3x + 2 & \text{for } -1 \leq x \leq 0 \\ -2x + 2 & \text{for } x > 2 \end{cases}$$

Solution. The function

$$p(x) = \begin{cases} -1 & \text{for } x < -1 \\ 3x + 2 & \text{for } -1 \leq x \leq 0 \\ -2x + 2 & \text{for } x > 2 \end{cases}$$

can be written in the form

$$p(x) = a|x + 1| + b|x| + cx + d$$

and one can find easily that $a = \frac{3}{2}$, $b = -\frac{5}{2}$, $c = -1$, $d = \frac{1}{2}$. Hence we have to find all polynomials $f(x)$, $g(x)$, $h(x)$ such that

$$|f(x)| - |g(x)| + h(x) = \frac{3}{2}|x + 1| - \frac{5}{2}|x| - x + \frac{1}{2} \quad (1)$$

for all $x \in \mathbb{R}$.

Without loss of generality we may assume that

$$\lim_{x \rightarrow -\infty} f(x) = \lim_{x \rightarrow -\infty} g(x) = -\infty.$$

Then there is $x_0 < -1$ such that $f(x) < 0$ and $g(x) < 0$ for $x < x_0$. Hence

$$-f(x) + g(x) + h(x) = -1 \quad (2)$$

for $x < x_0$ and since $f(x)$, $g(x)$, $h(x)$ are polynomials we conclude that (2) is fulfilled for all $x \in \mathbb{R}$. Now we shall show that $\lim_{x \rightarrow +\infty} f(x) = \lim_{x \rightarrow +\infty} g(x) = +\infty$.

If $\lim_{x \rightarrow +\infty} f(x) = \lim_{x \rightarrow +\infty} g(x) = -\infty$ we get as above that $-f(x) + g(x) + h(x) = -2x + 2$ for all $x \in \mathbb{R}$ which contradicts (2). Suppose that $\lim_{x \rightarrow +\infty} f(x) = -\infty$ and $\lim_{x \rightarrow +\infty} g(x) = +\infty$. The same reasoning as above shows that

$$-f(x) - g(x) + h(x) = -2x + 2 \quad (3)$$

and using (2) we get that $g(x) = x - \frac{3}{2}$, $h(x) = f(x) - x + \frac{1}{2}$. Hence (2) takes the form

$$|f(x)| + f(x) = \frac{3}{2}|x + 1| - \frac{5}{2}|x| + \left|x - \frac{3}{2}\right|. \quad (4)$$

For $x \in (-1, 0)$ we obtain from (4) that $|f(x)| + f(x) = 3x + 3 > 0$. Hence $f(x) > 0$ and $f(x) = \frac{3x + 3}{2}$ which shows that $f(x) = \frac{3x + 3}{2}$ for any $x \in \mathbb{R}$.

The same reasoning for $x \in \left(0, \frac{3}{2}\right)$ gives $f(x) = \frac{3-2x}{2}$ for any $x \in \mathbb{R}$, a contradiction.

Thus $\lim_{x \rightarrow +\infty} f(x) = \lim_{x \rightarrow +\infty} g(x) = +\infty$ and we get that

$$f(x) - g(x) + h(x) = -2x + 2 \quad (5)$$

for all $x \in \mathbb{R}$. Now (2) and (5) give

$$h(x) = -x + \frac{1}{2}, \quad g(x) = f(x) + x - \frac{3}{2}$$

and we have to find all polynomials $f(x)$ such that

$$|f(x)| - \left|f(x) + x - \frac{3}{2}\right| = \frac{3}{2}|x+1| - \frac{5}{2}|x| \quad (6)$$

for all $x \in \mathbb{R}$. We shall show that for $x \leq -1$ the identity (6) is equivalent to the inequality $f(x) \leq 0$. Indeed, suppose first that $x \leq -1$ and $f(x) + x - \frac{3}{2} > 0$. Then $f(x) > \frac{3}{2} - x > 0$ and it follows from (6) that $\frac{3}{2} - x = x - \frac{3}{2}$, a contradiction. Hence $f(x) + x - \frac{3}{2} \leq 0$ and (6) takes the form $|f(x)| + f(x) = 0$, i.e. $f(x) \leq 0$. Conversely, if $f(x) \leq 0$ then $f(x) + x - \frac{3}{2} \leq 0$ since $x \leq -1$ and (6) is satisfied.

The same reasoning shows that a polynomial $f(x)$ is a solution of (6) if and only if it satisfies the following inequalities:

$$\begin{aligned} f(x) &\leq 0 && \text{for } x \leq -1, \\ 0 \leq f(x) &\leq \frac{3}{2} - x && \text{for } -1 \leq x \leq 0, \\ f(x) &\geq \frac{3}{2} - x && \text{for } 0 \leq x \leq \frac{3}{2}, \\ f(x) &\geq 0 && \text{for } x \geq \frac{3}{2}. \end{aligned} \quad (7)$$

Hence all solutions of the problem with $\lim_{x \rightarrow -\infty} f(x) = \lim_{x \rightarrow -\infty} g(x) = -\infty$ are given by the polynomials $f(x)$, $g(x) = f(x) + x - \frac{3}{2}$, $h(x) = -x + \frac{1}{2}$, where

$f(x)$ is a polynomial of odd degree satisfying the inequalities (7). Finally, let us note that the inequalities (7) imply that $f(-1) = 0$ and $f(0) = \frac{3}{2}$. Thus if $\deg f = 1$ we have that $f(x) = \frac{3}{2}(x+1)$. Moreover, for any odd number $2n+1$, $n \geq 1$, one can construct infinitely many polynomials $f(x)$ of degree $2n+1$ satisfying (7). For example, take

$$f(x) = \frac{3}{2}(x+1) + a(x+1)x^{2n},$$

where $0 < a \leq 1$ is a constant.

7.2 Constructing Functions by Iterations

There is a class of functional equations, most of them on $\mathbb{N}$, that ask to find a function from its iterate. For example one could ask for all functions g such that $g^{(n)}(x) = f(x)$, where f is a given function, or at least for an example of such a function. These equations can be solved by constructing the orbits $O(x) = (x, f(x), f(f(x)), \dots)$ of f and investigating the relations determined by g on these orbits. If the orbit $O(x)$ is a finite set we define the *order* of f at x , denoted $\text{ord}_f(x)$ to be the least positive integer m such that $f^{(m)}(x) = x$. Note that if the order exists, then $f^{(k)}(x) = x$ if and only if $\text{ord}_f(x) \mid k$, by a standard argument using the minimality of the order.

For positive integers m, k denote $E_m(k)$ to be k/s where s is the largest divisor of k which is relatively prime to m ; that is, $E_m(k)$ is the largest divisor of k whose prime divisors all divide m . The reason for introducing this quantity is the following observation:

Lemma. If $\text{ord}_{f^{(k)}}(x) = m$, then $\text{ord}_f(x)$ exists and is divisible by $E_m(k) \cdot m$.

Proof. Note first that $f^{(km)}(x) = x$, so the order of f at x exists and divides km . Let now l be the least common multiple of $\text{ord}_f(x)$ and k . Then $(f^{(k)})^{(\frac{l}{k})}(x) = x$ so $m \mid \frac{l}{k}$ thus $mk \mid l$. Let p be a prime dividing m , say with exponent $\alpha > 0$ and let $\beta \geq 0$ be the exponent of p in k . Then $p^{\alpha+\beta}$ is the exponent of p in mk , which is strictly larger than the exponent of p in k , therefore since mk divides the least common multiple of $\text{ord}_f(x)$ and k , we get $p^{\alpha+\beta} \mid \text{ord}_f(x)$. But the exponent of p in $E_m(k) \cdot m$ is precisely $p^{\alpha+\beta}$.

It remains to take the product over all p dividing m and the proof of the lemma is finished. $\square$

Conventionally, we also let the order of x be infinite if the orbit of x is infinite, and write $\text{ord}(x) = \infty$.

By convention, we write $E_\infty(k) = k$.

For a function $f: S \rightarrow S$, define the set $S_1(f)$ to consist of all elements x that have finite order. The above lemma tells us that if $S_1(f^{(k)}) = S_1(f)$.

Orbits are particularly useful in investigating compositions of injective functions, because if $f: S \rightarrow S$ then S breaks into orbits of f . Such orbits can be of three types. Type 1 is finite orbits, they partition the set $F(f)$. Type 2 is orbits that are finite in one direction - such an orbit is a set $\{x_0, x_1, x_2, \dots\}$ where $x_{i+1} = f(x_i)$ and $x_0 \notin \text{Im}(f)$. Type 3 are orbits that are infinite in both directions - such an orbit is a sequence $(x_i)_{i \in \mathbb{Z}}$ where $x_{i+1} = f(x_i)$ for all $i \in \mathbb{Z}$. The union of orbits of type 1 is $S_1(f)$. Call the union of orbits of type 2 $S_2(f)$, and the union of orbits of type 3 be $S_3(f)$. It is easy to see that if $f^{(k)} = g$ then $S_1(f) = S_1(g)$, $S_2(f) = S_2(g)$ and $S_3(f) = S_3(g)$. $S_1(f)$ is the disjoint union of the finite orbits; $S_2(f)$ is the disjoint union of the *maximal* orbits that are finite in one direction, and $S_3(f)$ is the disjoint union of orbits that are infinite in both directions.

Theorem. *Let S be a set and $f: S \rightarrow S$ a function. Then the equation $g^{(k)} = f$ has a solution $g: S \rightarrow S$ if and only if the following conditions are satisfied:*

- *For every $m > 1$, the number of elements of S where f has order m , is either infinite or divisible by $E_m(k) \cdot m$.*
- *The number of orbits of type 2 is infinite or divisible by k .*
- *The number of orbits of type 3 is infinite or divisible by k .*

Proof. It is enough to investigate the equation for S_1, S_2, S_3 separately since they would coincide for f and g if $g^{(k)} = f$.

Let's deal with S_1 first, so we assume that every element has finite order.

Note that that number is a priori divisible by m because the set of elements of order m can be partitioned into disjoint orbits of size m : $\{a, f(a), \dots, f^{(m)}(a)\}$. Assume $g^{(k)} = f$. Then clearly, g commutes with f

hence if a has order m so does $g(a)$. Thus the set of elements of S where f has order m can be partitioned into orbits of g . The size of the orbit $\{a, g(a), \dots\}$ equals $\text{ord}_g(a)$. Because $\text{ord}_f(a) = m$ by the lemma $\text{ord}_g(a)$ exists and is divisible by $E_m(k) \cdot m$.

Conversely, assume this divisibility holds. We may include $m = 1$ as well as that divisibility is always true.

Let S_m be the elements of order m . Then they can be partitioned into orbits of f of size m and the number of such orbits is either infinite or divisible by $k' = E_m(k)$. Clumping sets of $E_m(k)$ orbits together we partition S_m into sets of the form

$$T = \{a_1, a_2, \dots, a_{k'}, f(a_1), \dots, f(a_{k'}), \dots, \dots, f^{(m-1)}(a_1), \dots, f^{(m-1)}(a_{k'})\}.$$

We construct g on T . Let $s = \frac{k}{k'}$, then s is co-prime to m so there is some c such that $cs \equiv 1 \pmod{m}$. It suffices to construct g such that $g^{(k')} = f^c$ on T because then $g^{(k)} = f^{(cs)} = f$ as on T , $f^{(m)}$ is the identity. Such a g may be defined by

$$\begin{aligned} g(a_1) &= a_2, g(a_2) = a_3, \dots, g(a_{k'-1}) = a_{k'}, \\ g(a_{k'}) &= f^c(a_1), g(f^c(a_1)) = f^c(a_2), \dots \end{aligned}$$

In general we define for $i < k'$,

$$g(f^{(rc)}(a_i)) = f^{(rc)}(a_{i+1}) \text{ and } g(f^{(rc)})(a_{k'}) = f^{((r+1)c)}(a_1).$$

It is verifiable that g is defined on all of T and $g^{(k)} = f$. Defining g on all such T , and for all m , yields a solution.

Next, let's work with S_2 , i.e. assume that S is a disjoint union of type 2 orbits. If $g^{(k)} = f$ the each type 2 orbit of g breaks into k type 2 orbits of f ; if $(x_0, x_1, x_2, \dots)$ is a type 2 orbit of g , then $(x_0, x_k, x_{2k}, \dots), (x_1, x_{k+1}, \dots), \dots, (x_{k-1}, x_{2k-1}, \dots)$ are type 2 orbits of f . Therefore, the type 2 orbits of f group into k -tuples which means their number is either infinite or divisible by k . Conversely, if the number of type 2 orbits of f is divisible by k or is infinite, then they can be grouped in k -tuples

and each k -tuple can be assembled into an orbit of g : if $a_1, a_2, \dots, a_k$ are the starting elements of k such orbits, define

$$g(a_1) = a_2, \dots, g(a_{k-1}) = a_k, g(a_k) = f(a_1),$$

$$g(f(a_1)) = f(a_2), \dots, g(f(a_{k-1})) = f(a_k),$$

$$g(f(a_k)) = f(f(a_1)), g(f(f(a_1))) = f(f(a_2)), \dots$$

The set S_3 is dealt with similarly. $\square$

If the function f is not injective, we cannot apply the above method anymore. However we can still deduce that if $g^{(k)} = f$ then the number of elements of order m is divisible by $E_m(k)$, as above. Ideas from the above theorem can be adapted to non-injective functions in some special cases.

The next example is a nice application of these ideas and the Möbius inversion formula.

Let $k, n \in \mathbb{N}$. Define W to be the set of roots of unity of order not coprime to n , and $S = \mathbb{C} - W$. The equation $g^{(k)} = x^n$ has a solution $g: S \rightarrow S$ if and only if for every prime q and positive integer m if q^m divides k then q^{m+1} divides $n^q - n$.

Solution. Let $f(x) = x^n$. The set $S_1 = S_1(f)$ consists of zero and all roots of unity in S , these are the roots of order coprime to n . These are all elements w such that $w^{n^m-1} = 1$ for some $m \in \mathbb{N}$. If a_d is the number of elements of exact order d (with respect to the function $f(x) = x^n$) then the number of elements x such that $f^{(m)}(x) = x$ is $\sum_{d|m} a_d$. But $f^{(m)}(x) = x^{n^m}$ which has exactly n^m fixed points. By the Möbius inversion formula, the number of elements of exact order m is

$$p_m(n) = \sum_{d|m} \mu(d) n^{\frac{m}{d}}.$$

So, it remains to prove that $E_m(k) \cdot m \mid p_m(n)$ is equivalent to $q^{m+1} \mid n^q - n$ if $q^m \mid k$.

One implication is direct because $p_q(n) = n^q - n$ and if $q^m \mid k$ then $q^m \mid E_q(k)$.

Conversely, assume $q^{m+1} \mid n^q - n$ whenever $q^m \mid k$. We shall then prove that $E_m(k) \cdot m$ divides $p_m(n)$ for any m .

Let q be a divisor of m , we want to prove that the exponent of q in $p_m(n)$ equals at least the exponent of q in $E_m(k) \cdot m$. If $q \nmid k$ this is true because $m \mid p_m(n)$ as $p_m(n)$ is the number of elements of exact order m which partition into orbits of m elements each, and $q \mid E_m(k)$.

Otherwise, assume $q \mid k$, let α be the exponent of q in k .

Then $q^{\alpha+1} \mid n^q - n = p_q(n)$. Let β be the exponent of q in m . Note that

$$p_q(x^q)/p_q(x) = x^{q-1} \cdot ((x^{q-1})^{q-1} + \dots + x^{q-1} + 1)$$

which is divisible by q as either x is divisible by q or else $x^{q-1} \cong 1 \pmod{q}$ and then $((x^{q-1})^{q-1} + \dots + x^{q-1} + 1)$ is divisible by q . It follows that the exponent of q in $p_q(x^q)$ is strictly larger than the exponent of q in $p_q(x)$. From here we immediately deduce $q^{\alpha+\beta} \mid p_q(n^{q^{\beta-1}})$. Since $p_q(x) \mid p_q(x^r)$, we deduce that $q^{\alpha+\beta}$ divides $p_q(n^r)$ whenever $q^{\beta-1}$ divides r .

Finally, the following property of p follows immediately from the multiplicativity of μ :

$$p_{ab}(n) = \sum_{d \mid a} \mu(a) p_b(n^{\frac{a}{d}})$$

We apply this to $a = \frac{m}{q^\beta}, b = q^\beta$ to obtain

$$p_m(n) = \sum_{d \mid \frac{m}{q^\beta}} p_{q^\beta}(n^{\frac{m}{q^\beta \cdot d}})$$

It remains to see that $p_{q^\beta}(n^{\frac{m}{q^\beta \cdot d}})$ equals $p_q(n^{\frac{m}{qd}})$ which is divisible by $q^{\alpha+\beta}$ by the above. Hence $q^{\alpha+\beta} \mid E_m(n)$ which implies the conclusion as $\alpha + \beta$ is the exponent of q in $E_m(k) \cdot m$.

Thus if our condition above is satisfied, the equation $g^{(k)} = x^n$ has a solution when restricted to S_1 and viceversa. It remains to deal with $S \setminus S_1$. We claim that the equation $g^{(k)} = f$ has a solution on $S \setminus S_1$. The function is not one-to-one but we can still deal with its "orbits". Let x be equivalent to y of $x^{n^{m_1}} = y^{n^{m_2}}$ for some $m_1, m_2 \in \mathbb{N}$. Then $S \setminus S_1$ decomposes into infinitely many equivalence classes. Each equivalence class can be described as follows: let T be the set of all w such that $w^{n^m} = 1$ for

some $m > 0$. Choose an sequence $(a_i)_{i \in \mathbb{Z}}$ with $a_i^n = a_{i+1}$; then an equivalence class consists of numbers of form wa_i for $w \in T$. Moreover each number can be uniquely represented as wa_i because a_i is not in S_1 . We can now combine k equivalence classes for f to construct g : if the equivalence classes are given by sequences $(a_{1,i})_{i \in \mathbb{Z}}, (a_{2,i})_{i \in \mathbb{Z}}, \dots, (a_{k,i})_{i \in \mathbb{Z}}$ then define $g(wa_{1,i}) = wa_{2,i}, g(wa_{2,i}) = wa_{3,i}, \dots, g(wa_{k-1,i}) = wa_{k,i}, g(wa_{k,i}) = wa_{1,i+1}$. It is easy to see that $g^{(k)} = f$. This finishes the proof.

Example 7.9. Show that there are infinitely many functions $f: \mathbb{Z} \rightarrow \mathbb{Z}$ for which $f(f(k)) = -k$ whenever $k \in \mathbb{Z}$.

Solution. We may set $f(0) = 0$. Then $\mathbb{Z} \setminus \{0\}$ can be divided into an infinite number of pairs $(a_1, -a_1), (a_2, -a_2), \dots$, where $a_1, a_2, \dots$, is some enumeration of $\mathbb{N}$. We can then set

$$f(a_{2k}) = a_{2k+1}, f(a_{2k+1}) = -a_{2k}, f(-a_{2k}) = -a_{2k+1}, f(-a_{2k+1}) = a_{2k}$$

and check that the condition is verified.

Example 7.10. Find all functions $f: \mathbb{N} \rightarrow \mathbb{N}$ that verify

$$f(f(n)) = an$$

for all $n \in \mathbb{N}$, where $a \in \mathbb{N}$ is fixed.

Solution. If $a = 1$, then $f(f(x)) = x$. So f is an involution and is obtained by paring all the natural numbers into pairs and mapping one element of a pair into another. Next, suppose $a > 1$. If $f(x) = y$ then $f(y) = ax$, $f(ax) = f(f(y)) = ay$ and we prove by induction on k the following statement: (*) $f(a^k x) = a^k y$, $f(a^k y) = a^{k+1} x$. Let S be the set of all positive integers not divisible by a . Every positive integer can be represented uniquely as $a^k b$ where $b \in S$. Now let $s \in S$ and $f(s) = a^k t$ where $t \in S$. If we set $u = f(t)$, then using (*) we get $f(a^k t) = a^k u$. But $f(a^k t) = f(f(s)) = as$ therefore $a^k u = as$ so as s is not divisible by a we get either $k = 1, u = s$ or $k = 0, u = as$. In the first case $f(t) = s, f(s) = at$ and in the second case $f(s) = t, f(t) = as$. In any case, f maps one of s, t into another. Therefore S splits into pairs (x, y) that satisfy $f(x) = y, f(y) = ax$, hence by (*) $f(a^k x) = a^k y, f(a^k y) = a^{k+1} x$. It's clear that all such functions satisfy our requirements.

Example 7.11. (Romanian TST 1991) Let $n \geq 2$ be a positive integer, $a, b \in \mathbb{Z} \setminus \{0, 1\}$. Show that there exist infinitely many functions $f: \mathbb{Z} \rightarrow \mathbb{Z}$ such that $f^{(n)}(x) = ax + b$ for all $x \in \mathbb{Z}$. Show that for $a = 1$ there exists b such that $f^{(n)}(x) = ax + b$ has no solutions.

Solution. The second part of the problem is already known for us when $n = 2$, b is odd, and the procedure is the same. If $b = n - 1$ and we let $a_i = f_i(0)$ ($a_{i+n} = a_i + b$), then for some $0 \leq i < j \leq n - 1$ we have $a_i \equiv a_j \pmod{b}$, thus $a_j = a_i + hb$. Hence $a_{i+hn} = a_j$, and thus $a_{r+hn+i-j} = a_r$ for all sufficiently large r , which in turn as $hn+i-j \neq 0$ implies $a_{r+n(hn+i-j)} = a_r$, a contradiction to the conclusion that $a_{r+n(hn+i-j)} = a_r + b(hn+i-j)$. Actually with one more effort one can prove that f exists if and only if $n|b$.

Now let us turn to the first part, which seems more challenging but bears also some similarity to the simpler case $a = 1, n = 2$. Let $g(x) = ax + b$. First, consider the case $a \neq -1$ (it is special because $g(g(x)) = x$ in this case, whereas in the general case $|g^{(n)}(x)|$ tends to infinity for almost all x). It follows from Example 6.1,(b) that

$$g^{(n)}(x) = a^n \left(x - \frac{b}{a-1} \right) + \frac{b}{a-1}.$$

Particularly this guarantees our claim that $|g^{(n)}(x)|$ tends to infinity for almost all x , particularly for all x except maybe $x = \frac{b}{a-1}$ (if it is an integer). Now call the orbit $O(x) = \{x, g(x), g^{(2)}(x), \dots, g^{(n)}(x), \dots\}$ of x maximal if it is not a proper subset of another orbit (to put it otherwise, if $x \neq g(y)$ for all $y \in \mathbb{Z}$). We claim that maximal orbits constitute a partition of $\mathbb{N} \setminus \left\{ -\frac{b}{a-1} \right\}$.

Indeed, first pick up a number $n \neq -\frac{b}{a-1}$. Then $n = g^{(k)}(m)$ is equivalent to

$$n = a^k \left(m + \frac{b}{a-1} \right) - \frac{b}{a-1},$$

or $(a-1)n + b = a^k((a-1)m + b)$. So take k be the greatest power of a dividing $(a-1)n + b$ and let

$$s = \frac{(a-1)m + b}{a^k}.$$

Then s is not divisible by a and moreover $s - b$ is divisible by $a - 1$. Hence if we set $m = \frac{s - b}{a - 1} + b$ then m is an integer and the equation $g(t) = m$ has no

solutions in $\mathbb{N}$ (because otherwise $at = m - b = \frac{s - b}{a - 1}$ so $s - b$ is divisible by a).

Thus $O(m)$ is the desired maximal orbit. Next, let us prove that two distinct maximal orbits do not intersect. If $O(x)$ and $O(y)$ intersect for $x \neq y$ then $g^{(m)}(x) = g^{(n)}(y)$ for some $m \neq n$. Without loss of generality $m \geq n$. Then as g is invertible on $\mathbb{R}$ we deduce $g^{(m-n)}(x) = y$; hence $O(y) \subset O(x)$ contradicting the fact that the orbit $O(y)$ is maximal. Now consider all the maximal orbits (there are infinitely many of them since every element x such that the equation $g(y) = x$ has no solutions in $\mathbb{N}$ generates such an orbit). We can group them into n -tuples and define f on each such an n -tuple $(O(x_1), O(x_2), \dots, O(x_n))$. Then we define $f(g^{(k)}(x_i)) = g^{(k)}(x_{i+1})$ for $i = 1, 2, \dots, n-1$ and $f(g^{(k)}(x_n)) = g^{(k+1)}(x_1)$. Define also $f\left(-\frac{b}{a-1}\right) = -\frac{b}{a-1}$. Then it is easy to see that f satisfies our requirements.

Let us investigate now the case $a = -1$. In this case, $\mathbb{N} \setminus \{\frac{b}{2}\}$ splits into infinitely many disjoint pairs (x, y) with $x + y = b$. Again we can group the pairs into n -tuples and define f on each n -tuple $((x_1, y_1), \dots, (x_n, y_n))$ as $f(x_i) = x_{i+1}, f(y_i) = y_{i+1}$ for $i = 1, 2, \dots, n-1$ and $f(x_n) = y_1, f(y_n) = x_1$. Define $f\left(\frac{b}{2}\right) = \frac{b}{2}$ if necessary. Again we see that f satisfies the conditions.

Finally, as we can group the orbits or the pairs into n -tuples in infinitely many ways, we have in both cases infinitely many such functions. It can be also proven that all functions with the desired property are of form we found.

The following is a generalization of Example 7.7.

Example 7.12. Given a real number a determine all continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x) = f(x^2 + a)$ for all $x \in \mathbb{R}$.

Solution. Since $f(x) = f(-x)$ it suffices to determine $f(x)$ for $x \geq 0$.

Set $\alpha(x) = x^2 + a$ and for any $x \in \mathbb{R}$ define the sequence $\{a_n(x)\}_{n=1}^\infty$ by $a_1(x) = \alpha(x)$, $a_{n+1}(x) = \alpha(a_n(x))$, $n \geq 1$. Since $f(x) = f(\alpha(x))$ we see that $f(x) = f(a_n(x))$ for all $n \geq 1$. The equation $\alpha(x) = x$ has a double root if and only if $a = \frac{1}{4}$, two distinct real roots if and only if $a < \frac{1}{4}$ and no real roots if

and only if $a > \frac{1}{4}$.

Case 1. $a = \frac{1}{4}$.

In this case $x \leq \alpha(x) \leq \frac{1}{2}$ for $x \in [0, \frac{1}{2}]$ and $\alpha(x) = x$ if and only if $x = \frac{1}{2}$. Hence for every $x \in [0, \frac{1}{2}]$ the sequence $\{a_n(x)\}_{n=1}^{\infty}$ is increasing and bounded above by $\frac{1}{2}$. Hence it converges and if $a(x) = \lim_{n \rightarrow \infty} a_n(x)$ then

$$a(x) = \lim_{n \rightarrow \infty} a_{n+1}(x) = \lim_{n \rightarrow \infty} \alpha(a_n(x)) = \alpha\left(\lim_{n \rightarrow \infty} a_n(x)\right) = \alpha(a(x)),$$

i.e. $a(x) = \frac{1}{2}$. Thus

$$f(x) = \lim_{n \rightarrow +\infty} f(a_n(x)) = f\left(\lim_{n \rightarrow +\infty} a_n(x)\right) = f\left(\frac{1}{2}\right)$$

for any $x \in [0, \frac{1}{2}]$.

For any $x > \frac{1}{2}$ set $\beta(x) = \alpha^{-1}(x) = \sqrt{x - \frac{1}{4}}$ and define the sequence $\{b_n(x)\}_{n=1}^{\infty}$ by $b_1(x) = \beta(x)$, $b_{n+1}(x) = \beta(b_n(x))$, $n \geq 1$.

Since $f(\beta(x)) = f(\alpha(\beta(x))) = f(x)$ we get that $f(x) = f(b_n(x))$ for any $n \geq 1$. Since $x \geq \beta(x) \geq \frac{1}{2}$ the sequence $\{b_n(x)\}_{n=1}^{\infty}$ is decreasing and bounded below by $\frac{1}{2}$ and the same reasoning as above shows that $f(x) = f\left(\frac{1}{2}\right)$ for $x \in \left[\frac{1}{2}, +\infty\right)$. Hence in this case f is a constant function.

Case 2. $a < \frac{1}{4}$.

In this case the equation $\alpha(x) = x$ has two distinct real roots

$$x_1 = \frac{1 - \sqrt{1 - 4a}}{2} \quad \text{and} \quad x_2 = \frac{1 + \sqrt{1 - 4a}}{2}.$$

If $x \leq x_1$ the sequence $\{a_n(x)\}_{n=1}^\infty$ is increasing and converges to x_1 and if $x_1 < x < x_2$ it is decreasing and converges to x_1 . Hence the function $f(x)$ is constant on the interval $[0, x_2]$.

Similarly, if $x \geq x_2$ the sequence $\{b_n(x)\}_{n=1}^\infty$ defined by

$$\beta(x) = \alpha^{-1}(x) = \sqrt{x - a}$$

is decreasing and converges to x_2 . Hence $f(x)$ is constant on the interval $[x_2, +\infty)$ and since $f(x) = f(-x)$ we conclude that $f(x)$ is constant on $\mathbb{R}$.

Case 3. $a > \frac{1}{4}$.

In this case for every $x \in \mathbb{R}$ the sequence $\{a_n(x)\}_{n=1}^\infty$ is strictly increasing and unbounded. Then as in Example 7.7 we can describe all functions $f(x)$ satisfying the given condition as follows. Let g be any continuous function on the interval $[0, a_1(0)] = [0, a]$ such that $g(0) = g(a)$. It is easily checked that for every $x > 0$ there exists a unique $y \in [0, a]$ and a nonnegative integer n such that $x = a_n(y)$. We define $f(x) = g(y)$ and extend f to negative x by $f(x) = f(-x)$. It can be checked that any f defined in this way satisfies the given functional equation and that all its continuous solutions can be represented in this form.

Remark. Cases 1 and 2 in the proof above follow by the following general result [12]: Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function such that

$$\lim_{x \rightarrow \infty} (g(x) - x) = \infty$$

and the set of fixed points of g is finite and nonempty. Then any continuous function $f : \mathbb{R} \rightarrow \mathbb{R}$ with $f(g(x)) = f(x)$ for all $x \in \mathbb{R}$ is constant.

Example 7.13. Prove that there is no function $f : \mathbb{N} \rightarrow \mathbb{N}$ such that

$$f^{(f(n))}(n) = n + 1$$

for all $n \in \mathbb{N}$.

Solution. Consider the sequence $\{a_n\}_{n=1}^\infty$ defined by $a_1 = 1$ and $a_n = f(a_{n-1})$ for $n \geq 2$. Note first that this sequence contains all positive integers. This follows by induction since if $a_n = x$, then

$$a_{n+f(x)} = f^{(f(x))}(a_n) = f^{(f(x))}(x) = x + 1.$$

Also all terms of the sequence are distinct. To see this suppose that $a_n = a_m$, where $n \neq m$. Then the sequence is periodic and hence finite, which contradicts to the fact proved above. Moreover $\{a_n\}_{n=1}^{\infty}$ is an increasing sequence since $a_n + 1 = a_{n+f(a_n)}$ and $a_n + 1$ appears in the sequence later than a_n . All proved above shows that $a_n = n$ for all n . Hence $f(n) = n + 1$ and this function obviously does not satisfy the given condition.

7.3 Practice Problems

Problem 7.1. Let $2 \leq k \in \mathbb{N}$. Find all functions $f: \mathbb{N}_0 \rightarrow \mathbb{N}_0$ that satisfy $f(0) = 0$ and

$$f(n) = 1 + f\left(\left\lfloor \frac{n}{k} \right\rfloor\right)$$

for all $n \in \mathbb{N}$.

Problem 7.2. Find all functions $f: \mathbb{N} \rightarrow \mathbb{N}$ that satisfy $f(1) = 2$ and

$$f(n+1) = \lfloor 1 + f(n) + \sqrt{1 + f(n)} \rfloor - \lfloor \sqrt{f(n)} \rfloor$$

for all $n \in \mathbb{N}$.

Problem 7.3. Find all functions $f: \mathbb{Z} \rightarrow \mathbb{Z}$ that satisfy

$$f(m+n) + f(mn) = f(m)f(n) + 1$$

for all $m, n \in \mathbb{Z}$.

Problem 7.4. Find all functions $f: \mathbb{Z} \rightarrow \mathbb{Z}$ that verify

$$f(f(k+1) + 3) = k$$

for all $k \in \mathbb{Z}$.

Problem 7.5. Find all functions $f: \mathbb{N} \rightarrow \mathbb{N}$ satisfying

$$f(m + f(n)) = n + f(m + k)$$

for all $m, n \in \mathbb{N}$, where $k \in \mathbb{N}$ is fixed.

Problem 7.6. (IMO Shortlist 1995) Does there exist a sequence $F(1), F(2), F(3), \dots$ of nonnegative integers that simultaneously satisfies the following three conditions?

- (a) Each of the integers $0, 1, 2, \dots$ occurs in the sequence.
- (b) Each positive integer occurs in the sequence infinitely often.
- (c) For all $n \geq 2$, $F(F(n^{163})) = F(F(n)) + F(F(361))$.

Problem 7.7. Find all functions $f: \mathbb{N}_0 \rightarrow \mathbb{R}$ that satisfy $f(4) = f(2) + 2f(1)$ and

$$f\left(\binom{n}{2} - \binom{m}{2}\right) = f\left(\binom{n}{2}\right) - f\left(\binom{m}{2}\right)$$

for $n > m$.

Problem 7.8. Find all functions $f: \mathbb{Q}^+ \rightarrow \mathbb{Q}^+$ that satisfy

$$f(x) + f\left(\frac{1}{x}\right) = 1 \text{ and } f(f(x)) = \frac{f(x+1)}{f(x)}$$

for all $x \in \mathbb{Q}^+$.

Problem 7.9. Let g and h be continuous real-valued bijections on the reals. It is also known that g has no a fixed point (i.e. $g(x) \neq x$ for all $x \in \mathbb{R}$). Construct a continuous function $f: \mathbb{R} \rightarrow \mathbb{R}$ such that $f(g(x)) = h(f(x))$.

Problem 7.10. Let $g: \mathbb{C} \rightarrow \mathbb{N}$ be a given function, $a \in \mathbb{C}$ and w the primitive cubic root of unity. Find all functions $f: \mathbb{C} \rightarrow \mathbb{C}$ such that

$$f(z) + f(wz + a) = g(z)$$

for all $z \in \mathbb{C}$.

Problem 7.11. Given a positive integer n find all monotonic functions $f: \mathbb{R} \rightarrow \mathbb{R}$ that satisfy $f^{(n)}(x) = -x$ for all $x \in \mathbb{R}$.

Problem 7.12. (Singapore 1996) Show that there exist functions $f: \mathbb{N} \rightarrow \mathbb{N}$ such that $f(f(n)) = n^2$ for all $n \in \mathbb{N}$.

Problem 7.13. Prove that for each integer $n \geq 2$ there are infinitely many functions $f: \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}$ such that $f^{(n)}(x) = \frac{1}{x}$.

Problem 7.14. Show that there are infinitely many functions $f : \mathbb{R} \setminus \{-1\} \rightarrow \mathbb{R} \setminus \{-1\}$ such that

$$f\left(\frac{x}{x+1}\right) = \frac{f(x)}{f(x)+1}$$

for all $x \in \mathbb{R} \setminus \{-1\}$.

Problem 7.15. Determine all continuous functions $f : (0, \infty) \rightarrow (0, \infty)$ such that $f(f(x)) = x$ and $f(x+1) = \frac{f(x)}{f(x)+1}$ for all $x \in (0, \infty)$.

Problem 7.16. (Bulgaria 2009) Determine all real numbers a for which there exists:

- (i) a function $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(0) = a$ and $f(f(x)) = x^{2009}$ for all $x \in \mathbb{R}$;
- (ii) a continuous function satisfying the conditions in (i).

Problem 7.17. Let $f : \mathbb{N} \rightarrow \mathbb{N}$ be a function satisfying

$$f(f(n)) = 4n - 3 \text{ and } f(2^n) = 2^{n+1} - 1$$

for all $n \in \mathbb{N}$. Find $f(993)$. What values can $f(2007)$ take?

Problem 7.18. (Romania TST 1989) Let $\mathbf{F}$ be the set of all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ such that $f(0) = 10$ and $f(f(x)) - 2f(x) + x = 0$ for all $x \in \mathbb{N}$. Find the set $A = \{f(1989) | f \in \mathbf{F}\}$.

Problem 7.19. Find all continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(1-x) = 1 - f(f(x))$$

for all $x \in \mathbb{R}$.

Chapter 8

D'Alembert Equation

8.1 D'Alembert Equation

It is well-known that the trigonometric functions

$$f(x) = \sin x \text{ and } g(x) = \cos x$$

satisfy several identities which are a natural source of nice and sometimes difficult functional equations. A typical example is the system

$$\begin{cases} f(x+y) = f(x)g(y) + g(x)f(y) \\ g(x+y) = g(x)g(y) - f(x)f(y) \end{cases}$$

whose continuous solutions (c.f. [2]) have the form $f(x) \equiv g(x) \equiv 0$, or $f(x) = e^{ax} \sin bx$, $g(x) = e^{ax} \cos bx$, where $a, b \in \mathbb{R}$ are arbitrary constants.

The purpose of this chapter is to show that the trigonometric functions $\sin ax$ and $\cos ax$ can be characterized by single functional equations suggested by some well-known trigonometric identities.

It is well-known that the trigonometric function $\cos x$ satisfies the identity

$$\cos(x+y) + \cos(x-y) = 2 \cos x \cos y$$

for all $x, y \in \mathbb{R}$. This suggests considering the functional equation

$$f(x+y) + f(x-y) = 2f(x)f(y) \tag{8.1}$$

which has been first studied by d'Alembert [4] (c.f. also [2]) in connection with the axiomatic foundations of Mechanics. Here we shall determine the continuous solutions of (8.1) for functions defined on $[0, \infty)$ or on the whole $\mathbb{R}$. To do this, let us first note that the hyperbolic cosine function

$$\cosh x = \frac{e^x + e^{-x}}{2}$$

is also a solution of d'Alembert equation. In fact, we shall prove that all continuous solutions of (8.1) are determined by the functions $\cos x$ and $\cosh x$.

Theorem 8.1. Let $f : [0, \infty) \rightarrow \mathbb{R}$ be a continuous function satisfying d'Alembert equation for all $x \geq y \geq 0$. Then either $f(x) = 0$, $f(x) = \cos ax$ or $f(x) = \cosh ax$, where $a \in \mathbb{R}$ is an arbitrary constant.

Proof. It is clear that the zero-function $f(x) \equiv 0$ is a solution of (8.1) and next we shall assume that $f(x) \not\equiv 0$. Then setting $y = 0$ in (8.1) we get that $f(0) = 1$ and now setting $x = y = \frac{t}{2}$ gives

$$f(t) + 1 = 2f\left(\frac{t}{2}\right)^2 \quad (8.2)$$

for all $t \in [0, \infty)$.

Since the function $f(x)$ is continuous and $f(0) = 1$ it follows that there exists $c > 0$ such that $f(x) > 0$ for $x \in [0, c]$.

Next we shall distinguish the two cases $f(c) \leq 1$ and $f(c) > 1$.

Case 1. Suppose that $f(c) \leq 1$. Then there exists $\alpha \in \left[0, \frac{\pi}{2}\right]$ such that $f(c) = \cos \alpha$. Now setting $t = c$ in (8.2) we get that

$$f\left(\frac{c}{2}\right) = \sqrt{\frac{1+f(c)}{2}} = \sqrt{\frac{1+\cos \alpha}{2}} = \cos \frac{\alpha}{2}.$$

Proceeding in the same way, we prove easily by induction on n that

$$f\left(\frac{c}{2^n}\right) = \cos \frac{\alpha}{2^n}$$

for all non-negative integers n .

On the other hand, setting $x = \frac{mc}{2^n}$, $y = \frac{c}{2^n}$ in (8.1) we see that

$$f\left(\frac{(m+1)c}{2^n}\right) + f\left(\frac{(m-1)c}{2^n}\right) = 2f\left(\frac{mc}{2^n}\right)f\left(\frac{c}{2^n}\right)$$

and it follows by induction on m that

$$f\left(\frac{mc}{2^n}\right) = \cos \frac{m\alpha}{2^n}$$

for all non-negative integers m and n . Hence $f(\delta c) = \cos \delta \alpha$ for all binary fractions δ (i.e. fractions of the form $\delta = \frac{m}{2^n}$).

Now take an arbitrary nonnegative real number t . Then it is well-known that there exists a sequence $\{\delta_n\}_{n=1}^{\infty}$ of binary fractions such that $\lim_{n \rightarrow \infty} \delta_n = t$. Hence the continuity of $f(x)$ implies that

$$f(tc) = \lim_{n \rightarrow \infty} f(\delta_n c) = \lim_{n \rightarrow \infty} \cos \delta_n \alpha = \cos t\alpha.$$

Now setting $x = tc$ and $a = \frac{\alpha}{c}$ we get that $f(x) = \cos ax$ for all $x \in [0, \infty)$.

Case 2. Let $f(c) > 1$. Then there exists $\alpha > 1$ such that $f(c) = \cosh \alpha$ and proceeding in the same way as in Case 1 we get that $f(x) = \cosh ax$ for all $x \in [0, \infty)$.

Conversely, it is easy to check that the functions $\cos ax$ and $\cosh ax$ satisfy the equation (8.1) and the theorem is proved. $\square$

Corollary 8.1. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function satisfying d'Alembert equation. Then either $f(x) = 0$, $f(x) = \cos ax$ or $f(x) = \cosh ax$, where $a \in \mathbb{R}$ is an arbitrary constant.

Proof. It follows from Theorem 8.1 that either $f(x) = 0$, $f(x) = \cos ax$ or $f(x) = \cosh ax$ for all $x \in [0, \infty)$. On the other hand setting $x = 0$ in (8.1) we get that $f(y) = f(-y)$ for all $y \in \mathbb{R}$. Hence the corollary follows from the fact that the above three functions are even. $\square$

Note that the function $\cosh ax$ is unbounded for $a \neq 0$. Hence Theorem 8.1 implies that all bounded continuous solutions of d'Alembert equation (8.1) are the functions $f(x) = 0$ and $f(x) = \cos ax$, $a \in \mathbb{R}$.

8.2 Polynomial Recurrences and Continuity

We have seen Cauchy's equation in Chapter 1, where a simple idea was needed to solve the equation on $\mathbb{Q}$, and then continuity to extend it on $\mathbb{R}$. Remark that what we actually did was to set $a_n = f(n)$ and apply the simple recurrence relation $a_n + a_1 = a_{n+1}$ to compute the sequence (a_n) effectively. In this paragraph, we shall consider functional equations that give rise to more complicated polynomial recurrences, which can be handled using the continuity of functions.

Namely, the main theorem of this section is the following:

Theorem A. *Let a) $f: \mathbb{R} \rightarrow \mathbb{C}$; b) $f: (0, \infty) \rightarrow \mathbb{C}$ be a continuous function with the following property:*

For each a) $x \in \mathbb{R}$; b) $x \in (0, \infty)$, there exists a number $w \in \mathbb{C}$ (depending on x) and polynomials $f_1, f_2, f_3 \in \mathbb{C}[X]$ such that

$$f(nx) = f_1(n)w^n + f_2(n)w^{-n} + f_3(n)$$

for all a) $n \in \mathbb{N}$; b) $n \in \mathbb{Z}$. Then

$$f(x) = p(x)e^{tx} + q(x)e^{-tx} + r(x)$$

for some $t \in \mathbb{C}$ and $p, q, r \in \mathbb{C}[X]$.

Remark. As one sees, the theorem has two versions, for f defined on all of $\mathbb{R}$ and for f only defined on $(0, \infty)$. Correspondingly, one needs to have $n \in \mathbb{Z}$ in the second case, because with $n \in \mathbb{N}$ one can not compare the negative and positive values of x .

Proof. We use the following helpful result:

Lemma. If

$$\sum_{i=1}^k p_i(n)w_i^n = \sum_{i=1}^m q_i(n)r_i^n$$

for any $n \in \mathbb{N}$, where r_i and w_i are two sequences of distinct numbers, and p_i, q_i are polynomials, then (q_i, r_i) are some permutation of (p_i, w_i) .

Proof of the lemma. Assume not. Then we can write

$$0 = \sum_{i=1}^k p_i(n)w_i^n - \sum_{i=1}^m q_i(n)r_i^n = \sum_{i=1}^l s_i(n)u_i^n,$$

where s_i are not zero and u_i are different numbers. Then the generating function of this recurrent sequence is also zero. However we know that the generating function of it can be written as

$$\sum_{i=1}^l \frac{f_i(x)}{(x - u_i)^{\deg(s_i+1)}},$$

where $f_i \neq 0, \deg(f_i) \leq \deg(s_i)$ and is not zero (if we multiply it by $(x - u_i)^{\deg(s_i+1)}$, then we get all terms divisible by $(x - u_i)^{\deg(s_i)+1}$ except the term $f_i(x)$, so the sum cannot be zero). $\square$

Let us return to the theorem. The general strategy is as follows: note that by letting $p_i(t) = f_i(\frac{t}{x})$ and by taking a suitable $t \in \mathbb{C}$ such that $e^{tx} = w$ one transforms the equation into $f(nx) = p_1(nx)e^{nxt} + p_2(nx)e^{-nxt} + p_3(nx)$.

That is, the following relation R_x holds for all z in the set $\{nx\}$:

$$f(z) = p_1(z)e^{tz} + p_2(z)e^{-tz} + p_3(z)$$

We will then choose different values x, x' that are commensurable (that is $\frac{x}{x'} \in \mathbb{Q}$) so that the sets $\{nx\}$ and $\{nx'\}$ intersect in a set of the form $\{nx''\}$ where x'' is the *least common multiple* of x and x' . We then use the lemma to compare the two relations R_x and $R_{x'}$, to conclude that p_1, p_2, p_3 coincide and that the two values of t do, too. This will allow to extend the relation R_x to the entire $\mathbb{Q}$ or $\mathbb{Q}^+$ (depending on whether we are in the case a) or b)) and then by continuity for the entire domain of definition. Essentially, this idea is exactly the idea in the solution of the Cauchy's problem, with added technical sophistication brought in by the more complicated relation R_x .

We shall consider only $x = \frac{1}{k}$ for $k \in \mathbb{N}$. If for all rational x we have $f_1 = f_3 = 0$ then $f(nx) = f_3(n) = p_3(nx)$ for all n , if we define

$$p_3(t) = f_3\left(\frac{t}{x}\right).$$

Moreover p_3 does not depend on x , because if x, x' are rational, p_3 is defined for x and p'_3 for x' then for $\frac{x}{x'} = \frac{p}{q}$ we get $qnx = pnx'$ so

$$p_3(qnx) = p'_3(qnx') = f(qnx)$$

for all n so p_3 coincides with p'_3 in infinitely many points whence $p_3 = p'_3$. Hence $f(x) = p_3(x)$ for all rational x . As f is continuous we get $f(x) = p_3(x)$ for all x and we are done in this easier case.

Next note (as mentioned above) that for any x , the condition

$$f(nx) = f_1(n)w^n + f_2(n)w^{-n} + f_3(n)$$

is equivalent to

$$f(nx) = p_1(nx)w^n + p_2(nx)w^{-n} + p_3(nx),$$

where we set $p_1(t) = f_1\left(\frac{t}{x}\right), p_2(t) = f_2\left(\frac{t}{x}\right), p_3(t) = f_3\left(\frac{t}{x}\right)$. Let's work with p_1, p_2, p_3 instead (as they are in the same variable as f), and let's call this relation R'_x .

Assume for some special value $x = x_0$ we have p_1 or p_2 non-zero. Then we can assume $w \neq 1$ for this x_0 , because if $w = 1$ we can set $p_3 = p_1 + p_2 + p_3$, $p_1 = p_2 = p_0$. Without loss of generality $x_0 = 2$ (we'll see why we take 2 instead of say 1 later). This can be performed by scaling: we work with the function $g(x) = f\left(\frac{2}{x_0} \cdot x\right)$ which will satisfy the exact same condition but now will have 2 as the sought special value.

Let w_k be the value of w defined for $x = \frac{1}{k}$. One should note that w_k is not unique: changing it by a multiple $2\pi i k$ will give the same exact value for w^{nx} therefore we feel free to change w_k by such a multiple and the relation R'_x will still hold.

Claim. We can pick up $w_1, \dots, w_i$ in such a way that the relation $w_{kl}^l = w_k$ holds for any $kl \leq i$.

Proof of claim. We induct on i . Assume we have proven this for $i < k$ and let's prove it for $i = k$. So we need to pick up w_k in such a way that

$$f\left(\frac{n}{k}\right) = p_1\left(\frac{n}{k}\right)w_k^n + p_2\left(\frac{n}{k}\right)\left(\frac{1}{w_k}\right)^n + p_3\left(\frac{n}{k}\right)$$

and $w_k^{\frac{k}{d}} = w_d$ for any $d \mid k$. By the condition of the problem, there exists a w_k satisfying the first condition (but it may not be unique). We begin by taking any value of w_k satisfying the first condition.

We have $f\left(\frac{n}{k}\right) = p_1\left(\frac{n}{k}\right)w_k^n + p_2\left(\frac{n}{k}\right)w_k^{-n} + p_3\left(\frac{n}{k}\right)$ and if $d \mid n$ then

$$f\left(\frac{n}{d}\right) = q_1\left(\frac{n}{d}\right)w_d^n + q_2\left(\frac{n}{d}\right)w_d^{-n} + q_3\left(\frac{n}{d}\right).$$

But

$$f\left(\frac{n}{d}\right) = p_1\left(\frac{n}{d}\right)w_k^{\frac{k}{d}n} + p_2\left(\frac{n}{d}\right)w_k^{-\frac{k}{d}n} + p_3\left(\frac{n}{d}\right).$$

Moreover, note that p_1 or p_2 are not zero. If they were, then in particular for $n = 2km$ we would have $f(2m) = p_3(2m)$ but this is impossible as $f(2m)$ is of the form $f_1(m)w^m + f_2(m)w^{-m} + f_3(m)$ for $w \neq 1$, and f_1 and f_2 not both zero; this would contradict the lemma.

The condition $p_1 \neq 0$ or $p_2 \neq 0$ allows us to apply the lemma we deduce $w_k^{\frac{k}{d}} = w_d$ or $w_k^{\frac{k}{d}} = \frac{1}{w_d}$.

If $w_k^{\frac{k}{d}} = w_d$ for all $d \mid k$ then w_k satisfies the desired condition. If $w_k^{\frac{k}{d}} = \frac{1}{w_d}$ then we have $p_1 = q_2, p_2 = q_1$ for any d so we can replace w_k by $\frac{1}{w_k}$ and exchange p_1 and p_2 to ensure the condition. The troublesome case is when for some d the first condition occurs, and for some d the second occurs. However, this case cannot happen. Indeed, assume that, without loss of generality, the first condition occurs for $d = k$ and the second for some other d . Then $w_k^k = w_1, w_k^{\frac{k}{d}} = \frac{1}{w_d}$. Now raise the latter equality to power d . As $w_d^d = 1$ by induction step, we get $w_k^k = \frac{1}{w_1}$ hence $w_1 = \frac{1}{w_1}$ so $w_1^2 = 1$. But this implies exactly that w defined for 2 is equal to 1, contradicting our original assumption (this is why we took $x = 2$, because we need to square). The induction is finished. $\square$

Also it is rather clear that p_1, p_2, p_3 do not really depend on k , as the identity

$$f(n) = p_1(n)w_1^n + p_2(n)w_1^{-n} + p_3(n)$$

compared with

$$f(n) = f\left(\frac{nk}{k}\right) = p'_1(n)w_k^{nk} + p'_2(n)w_k^{-nk} + p'_3(n) = p'_1(n)w_1^n + p'_2(n)w_1^{-n} + p_3$$

gives $p'_1 = p_1, p'_2 = p_2, p_3 = p'_3$ by our lemma. So we have chosen p_1, p_2, p_3 and w_i such that $w_{kl}^l = w_k$. We now need to show that $w_i = e^{\frac{t}{i}}$ for some $t \in \mathbb{C}$. Then we will have

$$f(x) = p_1(x)e^{tx} + p_2(x)e^{-tx} + p_3(x)$$

for all $x \in \mathbb{Q}$, thus for all $x \in \mathbb{R}$ by continuity, which is what we desired. This last step is more technical, and requires continuity.

Let $a_k = \ln(w_k)$ (there are many logarithms in the field of complex numbers, but they all differ by a multiple of $2\pi i$, choose one of them). Then $ka_k - a_1 = r_k 2\pi i$ where r_k is an integer, because $w_k^k = w_1$. We may assume $-\frac{k}{2} \leq r_k \leq \frac{k}{2}$ otherwise subtract from a_k a suitable multiple of $2\pi i$. Then $a_k = \frac{a_1}{k} + \frac{r_k}{k} \cdot 2\pi i$. Now as p_1, p_2 are not both identically zero, there is an integer m such that $p_1(m) \neq 0$ or $p_2(m) \neq 0$. As $\frac{1}{k} \rightarrow 0$,

$$f\left(m + \frac{1}{k}\right) \rightarrow f(m) = p_1(m)w_1^m + p_2(m)w_1^{-m} + p_3(m)$$

which implies that

$$\begin{aligned} p_1\left(m + \frac{1}{k}\right) w_1^{m+\frac{1}{k}} e^{\frac{r_k i}{k}} + p_2\left(m + \frac{1}{k}\right) w_1^{-(m+\frac{1}{k})} e^{-\frac{r_k i}{k}} + p_3\left(m + \frac{1}{k}\right) \\ \rightarrow p_1(m)w_1^m + p_2(m)w_1^{-m} + p_3(m). \end{aligned}$$

This readily implies $\lim_{k \rightarrow \infty} \frac{r_k}{k} = 0$ (we keep in mind that $|\frac{r_k}{k}| \leq \frac{1}{2}$). We also know that $la_{lk} - a_k$ is a multiple of $2\pi i$ which implies that $r_{kl} - r_k$ is a multiple of k . Particularly $r_{2k} - r_k$ is a multiple of k . But for all sufficiently big k_0 we have $\frac{|r_n|}{n} < \frac{1}{3}$ for $n \geq k_0$ therefore

$$\frac{|r_{2k} - r_k|}{k} \leq 2 \frac{|r_{2k}|}{2k} + \frac{|r_k|}{k} < 2 \frac{1}{3} + \frac{1}{3} = 1$$

hence $r_{2k} - r_k = 0$. We then conclude $r_{2^m k} = r_k$ for all m when $k \geq k_0$. Now pick up $k \geq k_0, m > k$. Then $r_{2^m k} = r_k$ but $2^m |r_{2^m k} - r_{2^m}|$. As

$$|r_{2^m k}| = |r_k| < \left| \frac{k}{3} \right|, \quad |r_{2^m}| < \frac{2^m}{3}$$

we conclude that $|r_{2^m k} - r_{2^m}| < \frac{k}{2} + \frac{2^m}{3} < 2^m$ so $r_k = r_{2^m}$. Analogously we conclude $r_{k+1} = r_{2^m}$ so $r_k = r_{k+1}$ and thus r_k is eventually constant. Now we claim that r_k can be made in fact constant by changing a_k by an irrelevant $2\pi i$ multiple. Indeed, let a be the eventual value of (the initial) r_k and let m be the smallest integer with $a_m \neq a$. Then $m|r_m - r_{2^m}|$ so we can change a_m by $2\pi i \cdot \frac{r_{2^m} - r_m}{m}$ to have $r_m = a$. We keep doing this operation until $r_k = a$ for all $k \geq 1$. Then by setting $t = a_1 + 2a\pi i$ we ensure $w_k = \frac{t}{k}$.

The proof of the theorem is finished. We should also note that if

$$f(x) = p_1(x)e^{tx} + p_2(x)e^{-tx} + p_3(x),$$

then the representation

$$f(nx_0) = q_1(n)w^n + q_2(n)\left(\frac{1}{w}\right)^n + r(n)$$

must match the representation of f . E.g. if $w \neq \frac{1}{w}$ we must have $w = e^{tx_0}$ (or $w = e^{-tx_0}$), $p_1(nx_0) = q_1(n)$, $p_2(nx_0) = q_2(n)$, $p_3(nx_0) = q_3(n)$. If

$$w = \frac{1}{w} = -1$$

then

$$e^{tx_0} = -1, \quad p_1(nx_0) + p_2(nx_0) = q_1(n) + q_2(n), \quad p_3(nx_0) = q_3(n)$$

and finally if $w = \frac{1}{w} = -1$ then

$$e^{tx_0} = 1, \quad p_1(nx_0) + p_2(nx_0) + p_3(nx_0) = q_1(n) + q_2(n) + q_3(n).$$

This fact follows from comparing the values of the sequence

$$f(nx_0) = p_1(nx_0)e^{ntx_0} + p_2(nx_0)e^{-ntx_0} + p_3(nx_0) = q_1(n)w^n + q_2(n)\left(\frac{1}{w}\right)^n + r(n)$$

and using the helpful result from the proof of the theorem. $\square$

To see how powerful Theorem A is, let us consider some really hard functional equations which are solved in a bite by it.

Example 8.1. Find all continuous functions $f: \mathbb{R} \rightarrow \mathbb{R}$ that satisfy the relation

$$f(x+y) + af(x) + f(x-y) = 0$$

for all $x, y \in \mathbb{R}$ and some non-zero $a \in \mathbb{R}$.

Solution. If we set $u_n = f(nx_0)$, then the above condition readily implies $u_{n+1} + au_n + u_{n-1} = 0$. This is a quadratic recurrence. If its characteristic equation has two distinct roots r_1, r_2 then $u_n = c_1 r_1^n + c_2 r_2^n$ for some c_1, c_2 . Now Theorem A tells us that $f(x) = \alpha e^{tx} + \beta e^{-tx}$. Then by the remark at the end of the proof of Theorem A we can deduce $\alpha = c_1, \beta = c_2, e^{tx_0} = r_1$. But this is impossible as $e^{tx_0} = r_1$ cannot hold for any x_0 unless $r_1 = 1$. Thus $r_1 = r_2 = \pm 1$. The case $r_1 = r_2 = \pm 1$ implies $u_n = (-1)^n(cn + d)$ hence $f(x) = e^{tx}(\frac{c}{x_0}x + d)$ and we must have $e^{tx_0} = -1$ for any x_0 , again impossible. Thus $r_1 = r_2 = 1, a = -2$ and we get $u_n = cn + d$. So $f(x)$ is linear and for $a = -2$ any linear function satisfies the condition of the problem.

Example 8.2. (D'Alembert functional equation) Find all continuous functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x+y) + f(x-y) = 2f(x)f(y)$$

for all $x, y \in \mathbb{R}$.

Solution. We have already seen a more traditional solution of this equation in Section 1. This does not stop us, of course, from presenting a completely different solution here.

The function $f \equiv 0$ clearly satisfies the equation. Now if $f \not\equiv 0$, plugging $y = 0$ yields $2f(x)(1 - f(0)) = 0$, i.e. $f(0) = 1$. Let us set $a_n = f(nx)$. Then by setting $y = nx$ we get $a_{n-1} + a_{n+1} = 2aa_n$, where $a = a_1 = f(x)$. This is a polynomial recurrence with characteristic equation $x^2 - 2ax + 1 = 0$, which has roots $w, \frac{1}{w}$, where $w = a + \sqrt{a^2 - 1}$ is either a real number or a complex number of absolute value 1. Then $a_n = cw^n + dw^{-n}$ if $w \neq \frac{1}{w}$, and $a_n = (an + b)w^n$ otherwise. In any case, we are in the condition of Theorem A and hence we establish that

$$f(x) = p(x)e^{tx} + q(x)e^{-tx} + r(x)$$

for some $t \neq 0$.

Now neither p, q or r may have degree more than 1, because then we could find an x such that

$$f(nx) = p_1(n)w^n + q_1(n)\frac{1}{w^n} + r_1(n)$$

(specifically when tx is not a multiple of $2\pi i$), and p_1, q_1, r_1 have the same degrees as p, q, r , but this contradicts the representation of $f(x)$ obtained above. The same reasoning shows that if $w \neq \frac{1}{w}$ for at least one x , then p, q, r are actually constants. Thus $f(x) = pe^{tx} + qe^{-tx} + r$. Then $r = 0$ because the free term in the representation of $a_n = cw^n + dw^{-n}$. So $f(x) = pe^{tx} + qe^{-tx}$ for $t \neq 0$. Also note that as w was either real or of absolute value 1, t must be either real or completely imaginary. In the first case we get p, q real and in the second case they are conjugates of each other.

Now let us pin down p and q . For this, let's return to the representation

$$a_n = cw^n + dw^{-n} \text{ for } w \neq \frac{1}{w}.$$

As $a_0 = 1, a_1 = a = \frac{w + \frac{1}{w}}{2}$ we get $c + d = 1, cw + \frac{d}{w} = \frac{w + \frac{1}{w}}{2}$. This is a linear system which has solutions $c = d = \frac{1}{2}$. Hence $p = q = \frac{1}{2}$ thus $f(x) = \frac{e^{tx} + e^{-tx}}{2}$. If t is real, this is just the hyperbolic cosine: $\cosh(tx)$. If $t = is$ is purely imaginary this is the cosine $\cos(sx)$. Both these functions satisfy the equation. Now if $w = \frac{1}{w}$ for all x , then $w^2 = 1$ hence $f(2nx) = a_{2n} = 2an + b$. Now we set $n = 0$ to get $b = 1$. Hence we have a representation of form $f(nx) = f\left(2n\frac{x}{2}\right) = anx + 1$ for all x , hence $f(x) = ax + 1$. We check: $f(x + y) + f(x - y) = a(x + y) + 1 + a(x - y) + 1 = 2(ax + 1)$ but $2f(x)f(y) = 2(ax + 1)(ay + 1)$ and from here $ay + 1 = 1$ hence $a = 0$ and f is identically 1.

D'Alembert equation is considered only for real-valued functions. Our method, however, allows us to solve it for functions $f: \mathbb{R} \rightarrow \mathbb{C}$. In fact, with some more care, we can solve the problem for functions $f: \mathbb{C} \rightarrow \mathbb{C}$. The main

idea is to take $f_{\alpha,\beta}: \mathbb{R} \rightarrow \mathbb{C}$ be $f_{\alpha,\beta}(x) = f(\alpha x + \beta)$ and thus applying our method write $f(\alpha x + \beta)$ as $c_{\alpha,\beta}e^{t_{\alpha,\beta}\alpha x} + d_{\alpha,\beta}e^{t_{\alpha,\beta}\alpha x}$. It remains to mutually compare the $t_{\alpha,\beta}$ with t_{α_1,β_1} if $\alpha x + \beta = \alpha_1 x + \beta_1$ and draw the conclusions. The reader is invited to think about this problem for him/herself.

Theorem A in fact follows with little effort from the following more powerful result whose proof is, in essence, the same as the proof of Theorem A; there are a little more technicalities to consider in the general case. For the functional equations encountered here, however, we will not have to go beyond the statement of Theorem A.

Theorem B. Assume $f: (0, \infty) \rightarrow \mathbb{C}$ (respectively $f: \mathbb{R} \rightarrow \mathbb{C}$) is a continuous function such that for every $x \in (0, \infty)$ (respectively $x \in \mathbb{R}$) there are polynomials $f_1, \dots, f_k$ and complex numbers $w_1, \dots, w_k$ depending on x , with $k \leq m$ where m is a universal constant not depending on x , such that

$$f(nx) = f_1(n)w_1^n + \dots + f_k(n)w_k^n$$

for all $n \in \mathbb{N}$ (respectively $n \in \mathbb{Z}$). Then there is an integer $k \leq m$, numbers $t_1, \dots, t_k \in \mathbb{C}$ and polynomials $p_1, \dots, p_k \in \mathbb{C}[X]$ such that

$$f(x) = p_1(x)e^{t_1x} + \dots + p_k(x)e^{t_kx}.$$

for all x .

Proof. The general strategy is as follows: note that by letting

$$p_i(t) = f_i\left(\frac{t}{x}\right)$$

and by taking suitable $t_i \in \mathbb{C}$ such that $e^{t_ix} = w_i$ one transforms the equation into

$$f(nx) = p_1(nx)e^{nxt_1} + p_2(nx)e^{nxt_2} + \dots + p_k(nx)e^{nxt_k}.$$

That is, the following relation R_x , which resembles the target identity more, holds for all z in the set $\{nx \mid n \in \mathbb{N}\}$:

$$f(z) = p_1(z)e^{t_1z} + p_2(z)e^{t_2z} + \dots + p_k(z)e^{t_kz}$$

We will then choose different values x, x' that are commensurable (that is $\frac{x}{x'} \in \mathbb{Q}$) so that the sets $\{nx\}$ and $\{nx'\}$ intersect in a set of the form $\{nx''\}$

where x'' is the *least common multiple* of x and x' . We then use the lemma to compare the two relations R_x and $R_{x'}$, to conclude that $p_1, p_2, \dots, p_k$ coincide and that the corresponding values of t_i do, too. This will allow to extend the relation R_x to the entire $\mathbb{Q}$ or $\mathbb{Q}^+$ (depending on whether we are in the case a) or b)) and then by continuity for the entire domain of definition. Essentially, this idea is exactly the idea in the solution of the Cauchy's problem, with added technical sophistication brought in by the more complicated relation R_x . Remark that each t_i is not unique - it is only so up to a multiple of $2\pi i$, thus we will have to choose it carefully (in fact this is the most difficult technicality of the proof). There are also increased technical difficulties compared to the proof of Theorem A because there are more variables to consider - these technical difficulties reduce to knowledge of some linear algebra.

We can of course assume that the numbers $w_1, \dots, w_k$ are distinct (otherwise merge equal values together) and the polynomials p_i are non-zero (otherwise just forget them). We then choose x_0 for which k is largest possible. It is enough to assume $x_0 = 1$: we can work with $f_1(x) = f(t_0 x)$ instead to reduce to this case. In that case we can just assume $k = m$.

We will now set $x = \frac{1}{k}$, $k \in \mathbb{N}$ and thus only concentrate on the values of f on $\mathbb{Q}^+$; the rest of the values will follow by continuity. (Notice the resemblance with the proof of Cauchy's theorem!)

More exactly, setting $x = \frac{1}{k}$ we get

$$f\left(\frac{n}{k}\right) = q_1\left(\frac{n}{k}\right)v_1^n + \dots + q_j\left(\frac{n}{k}\right)v_j^n,$$

where $j \leq n$. In particular if we replace n by nk we obtain

$$f(n) = q_1(n)v_1^{kn} + \dots + q_j(n)v_j^{kn}.$$

Comparing this with the identity

$$f(n) = p_1(n)w_1^n + \dots + p_k(n)w_k^n$$

we deduce (by permuting the v_j if necessary) that $j = k$, $p_i = q_i$ and $v_i^k = w_i$ using the lemma (note that if $j > k$ we would not be able to deduce that as

some of the v_i^k could become equal but this is impossible in our case as we need at least k different terms).

It remains thus to prove that $v_i = e^{\frac{t_i}{k}}$ for some t_i independent of k . This last step is the most technical, and requires continuity (beyond the passage from $\mathbb{Q}$ to $\mathbb{R}$).

We thus know that for $x = \frac{1}{k}$, v_i can be written as $e^{a_k^{(i)}}$; let also $w_i = e^{a_1^{(i)}}$. The condition $v_i^k = w_i$ means $e^{ka_k^{(i)}} = e^{a_1^{(i)}}$ so that $ka_k^{(i)} - a_1^{(i)} = r_k^{(i)} \cdot 2\pi i$ is a multiple of $2\pi i$. Since $a_k^{(i)}$ is unique up to a multiple of $2\pi i$ we may choose $a_k^{(i)}$ such that $|r_k^{(i)}| \leq \frac{k}{2}$.

Then $a_k^{(i)} = \frac{a_1^{(i)}}{k} + \frac{r_k^{(i)}}{k} \cdot 2\pi i$ with $\frac{r_k^{(i)}}{k} \in \left[-\frac{1}{2}, \frac{1}{2}\right]$.

In fact we can prove that $\frac{r_k^{(i)}}{k} \rightarrow 0$ as $k \rightarrow \infty$. For this, we observe that $f(n + \frac{1}{k}) \rightarrow f(n)$ for any fixed value of n .

In particular this means that

$$p_1\left(n + \frac{1}{k}\right) w_1^n \cdot e^{\frac{a_1^{(1)}}{k}} \cdot e^{\frac{r_k^{(1)}}{k} \cdot 2\pi i} + \dots + p_m\left(n + \frac{1}{k}\right) w_m^n \cdot e^{\frac{a_1^{(m)}}{k}} \cdot e^{\frac{r_k^{(m)}}{k} \cdot 2\pi i}$$

converges to

$$p_1(n)w_1^n + p_2(n)w_2^n + \dots + p_m(n)w_m^n.$$

Now $p_i\left(n + \frac{1}{k}\right)$ converges to $p_i(n)$ and $e^{\frac{a_1^{(i)}}{k}}$ converges to 1.

We therefore deduce

$$p_1(n)w_1^n e^{\frac{r_k^{(1)}}{k} \cdot 2\pi i} + \dots + p_m(n)w_m^n e^{\frac{r_k^{(m)}}{k} \cdot 2\pi i} \rightarrow p_1(n)w_1^n + \dots + p_m(n)w_m^n.$$

Observe now that the vectors

$$(p_1(n)w_1^n, p_2(n)w_2^n, \dots, p_m(n)w_m^n)$$

for all values of $n \in \mathbb{N}$ must span $\mathbb{C}^n$. Otherwise there would be come c_i such that

$$\sum c_i p_i(n)w_i^n = 0$$

which cannot happen by the lemma.

It follows that we can find values $n_1, n_2, \dots, n_m$ for which matrix $(p_i(n_j)w_i^{n_j})$ is non-singular. In particular, the linear map $T: \mathbb{C}^n \rightarrow \mathbb{C}^n$ given by

$$T(x_1, \dots, x_m) = (p_1(n_1)w_1^{n_1}x_1 + p_2(n_1)w_2^{n_1}x_2 + \dots + p_m(n_1)w_m^{n_1}x_m, \dots, \\ p_1(n_m)w_1^{n_m}x_1 + p_2(n_m)w_2^{n_m}x_2 + \dots + p_m(n_m)w_m^{n_m}x_m)$$

has an inverse map which is also linear so continuous.

Because we know that

$$T(e^{\frac{r_k^{(1)}}{k} \cdot 2\pi i}, \dots, e^{\frac{r_k^{(1)}}{k} \cdot 2\pi i}) \rightarrow T(1, 1, \dots, 1)$$

we apply T^{-1} and deduce

$$(e^{\frac{r_k^{(1)}}{k} \cdot 2\pi i}, \dots, e^{\frac{r_k^{(1)}}{k} \cdot 2\pi i}) \rightarrow (1, 1, \dots, 1)$$

hence $e^{\frac{r_k^{(i)}}{k} \cdot 2\pi i} \rightarrow 1$. Since $\frac{r_k^{(i)}}{k} \in [-\frac{1}{2}, \frac{1}{2}]$ this implies $\frac{r_k^{(i)}}{k} \rightarrow 0$, as desired.

Next, we can compare $f\left(\frac{n}{k}\right)$ for $x = \frac{1}{k}$ and $x = \frac{1}{kl}$ to deduce that $a_k^{(i)} - la_{kl}^{(i)}$ is a multiple of $2\pi i$ - that is $r_k^{(i)} - r_{lk}^{(i)}$ is an integer divisible by k .

In particular $r_{2k}^{(i)} - r_k^{(i)}$ is divisible by k but since $\frac{r_k^{(i)}}{k}, \frac{r_{2k}^{(i)}}{2k} \rightarrow 0$, this is only possible if $r_{2k}^{(i)} = r_k^{(i)}$ (once k is sufficiently large). Thus if $k \geq k_0$ then $r_{2^m k}^{(i)} = r_k^{(i)}$.

Also if $k \geq k_0$ we choose some $2^p \geq k_0$ to deduce $2^p \mid r_{2^p k}^{(i)} - r_{2^p}^{(i)} = r_k^{(i)} - r_{2^p}^{(i)}$ which similarly implies $r_k^{(i)} = r_{2^p}^{(i)}$ once k is large enough. Since p does not depend on k we deduce $r_k^{(i)}$ is eventually constant.

We now claim we can change $a_k^{(i)}$ by a multiple of $2\pi i$ to make all $r_k^{(i)}$ constant (for $k > 0$).

Indeed say $r_k^{(i)}$ is the last value distinct from the eventual value of $r_{k+1}^{(i)}$ so $r_{k+1}^{(i)} = r_{k+2}^{(i)} = \dots = a_i$. Then $r_k^{(i)} - r_{2k}^{(i)} = r_k^{(i)} - a_i$ is divisible by k , say it

equals sk . We then replace $a_k^{(i)}$ by $a_k^{(i)} + 2 \cdot 2\pi i$ to make $r^{(i)}$ equal to a_i . Repeat until we reach $k = 0$. We then just let $t_i = a_1 + a_0 \cdot 2\pi i$ to obtain $v_k^{(i)} = e^{\frac{t_i}{k}}$.

Then, relations $R_{\frac{1}{k}}$ all agree and give the identity

$$f(x) = p_1(x)e^{t_1x} + p_2(x)e^{t_2x} + \dots + p_m(x)e^{t_mx}$$

for all $x \in \mathbb{Q}^+$. By continuity, this holds for all $x \in (0, \infty)$.

The case $x \in \mathbb{R}$ is similar. $\square$

Remark. The above theorem can also be extended to functions defined on the entire domain $\mathbb{R}$ without the stronger requirement to allow all $n \in \mathbb{Z}$. In that case, working with f restricted on $(0, \infty)$ and f restricted to $(-\infty, 0)$ we conclude that f can be written in the above form on $(0, \infty)$, and in another way on $(-\infty, 0)$ provided that the two expressions evaluate to the same number at 0. However if we require f to be smooth for example, then all derivatives would also have to agree - which using the lemma implies that the two expressions are the same so in this case the conclusion follows.

Here is how one can deduce Theorem A and other results as easy corollaries of the above theorem.

Proposition. Let $f: (0, \infty) \rightarrow \mathbb{C}$ be a continuous function.

a) (Theorem A) If

$$f(nx) = p_1(n)w^n + p_2(n)w^{-n} + p_3(n), \forall n \in \mathbb{N}$$

for every fixed $x \in (0, \infty)$ and some $w \in \mathbb{C}, p_1, p_2, p_3 \in \mathbb{C}[X]$ (depending on x) then

$$f(x) = f_1(x)e^{tx} + f_2(x)e^{-tx} + f_3(x)$$

for some $t \in \mathbb{C}, f_1, f_2, f_3 \in \mathbb{C}[X]$.

b) (easier version of Theorem A) If

$$f(nx) = p_1(n)w^n + p_2(n), \forall n \in \mathbb{N}$$

for every fixed $x \in (0, \infty)$ and some $w \in \mathbb{C}, p_1, p_2 \in \mathbb{C}[X]$, depending on x , then

$$f(x) = f_1(x)e^{tx} + f_2(x)$$

for some $t \in \mathbb{C}, f_1, f_2 \in \mathbb{C}[X]$.

Proof. a) Applying Theorem B for $m = 3$ we write

$$f(x) = f_1(x)e^{t_1x} + f_2(x)e^{t_2x} + f_3(x)e^{t_3x}$$

so that

$$f(nx) = f_1(nx)(e^{t_1x})^n + f_2(nx)(e^{t_2x})^n + f_3(nx)(e^{t_3x})^n.$$

Comparing this with the expression given by the condition and using the lemma, we deduce that one of the $e^{t_1x}, e^{t_2x}, e^{t_3x}$ must be 1 (unless p_3 is always 0 in which case we can take $m = 2$ and get rid of one variable) for any x . This of course implies that one of t_1, t_2, t_3 must be 0. Further, we deduce that $e^{t_1x} \cdot e^{t_2x}$ must be 1 for all x which implies $t_2 = -t_1$ thus

$$f(x) = f_1(x)e^{t_1x} + f_2(x)e^{-t_1(x)} + f_3(x)$$

which is what we want.

b) Apply Theorem B for $m = 2$ and write $f(x) = f_1(x)e^{t_1x} + f_2(x)e^{t_2x}$, so

$$f(nx) = f_1(nx)(e^{t_1x})^n + f_2(nx)(e^{t_2x})^n.$$

Comparing this with the expression we want, one of e^{t_1x}, e^{t_2x} must be 0 whatever x we choose, which implies that for example $t_2 = 0$ which translates to the conclusion. $\square$

Finally, note that one can make other small refinements. For example, if we know that all polynomials p_i have degrees less than d then so must the polynomials f_i . Conversely, if for some x one of the p_i has degree larger than d so must one of the f_i . Further, if there are at least k distinct w_i for some x then there must be at least k distinct t_i . All these follow easily from comparing the statement of the theorem with the conclusion and using the lemma.

8.3 Practice Problems

Problem 8.1. Find all solutions to the functional equation

$$f(x+y) + f(x-y) = 2f(x)\cos(y)$$

for any $x, y \in \mathbb{R}$.

Problem 8.2. Find all continuous functions $f, g: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x - y) = f(x)f(y) + g(x)g(y)$$

for any $x, y \in \mathbb{R}$.

Problem 8.3. Find all continuous functions $f, g: \mathbb{R} \rightarrow \mathbb{R}$ that satisfy

$$f(x + y) + f(x - y) = 2f(x)g(y)$$

for any $x, y \in \mathbb{R}$.

Problem 8.4. Find all continuous functions $f, g, h, k: \mathbb{R} \rightarrow \mathbb{R}$ that satisfy $f(x + y) + g(x - y) = 2h(x)k(y)$ for any $x, y \in \mathbb{R}$.

Problem 8.5. Find all smooth functions $f: \mathbb{R} \rightarrow \mathbb{R}$ that satisfy

$$f(x + y) + 2f(x)f(y) = f'(x)f(y) + f(x)f'(y)$$

for any $x, y \in \mathbb{R}$.

Problem 8.6. Find all continuous functions $f: \mathbb{R} \rightarrow \mathbb{R}$ that satisfy the following condition: whenever $x + y + z$ is an integer multiple of 2π ,

$$f^2(x) + f^2(y) + f^2(z) - 2f(x)f(y)f(z) = 1$$

Chapter 9

The Aczél–Gołąb–Schinzel Equation

9.1 Theory and Examples

A useful approach in studying various functional equations is to use the algebraic structure of the set of periods of a function. In general it is difficult to describe this set but in some cases the functional equation under consideration provides additional information about the set of periods which is crucial for finding its solution. Here we shall apply this approach to find all continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x + yf(x)) = f(x)f(y) \quad (9.1)$$

for all $x, y \in \mathbb{R}$. This functional equation has been first considered by Aczél [1] and Gołąb-Schinzel [10](c.f. also [3]) and it is still an object of intensive study (c.f. e.g. [6],[9]). We refer the reader to [2] for the history and various generalizations of Aczél–Gołąb–Schinzel (AGS) equation.

Recall that the set of periods of a function $f : \mathbb{R} \rightarrow \mathbb{R}$ is defined by

$$\mathcal{P} = \{\alpha \in \mathbb{R} \mid f(x + \alpha) = f(x) \text{ for all } x \in \mathbb{R}\}.$$

It is easy to check that $\mathcal{P}$ is an additive group, i.e. $0 \in \mathcal{P}$ and if $\alpha, \beta \in \mathcal{P}$, then $-\alpha \in \mathcal{P}$ and $\alpha + \beta \in \mathcal{P}$. Indeed, this is equivalent to prove that $\alpha - \beta \in \mathcal{P}$ for

all $\alpha, \beta \in \mathcal{P}$ which follows from the identity

$$f(x + \alpha - \beta) = f((x + \alpha - \beta) + \beta) = f(x + \alpha) = f(x)$$

for all $x \in \mathbb{R}$.

Example 9.1. It is well-known that the trigonometric functions $\sin x$ and $\cos x$ are periodic with period 2π . Moreover one can easily show that their sets of periods coincide with the set $\mathcal{P} = \{2k\pi | k \in \mathbb{Z}\}$. As for the functions $\tan x$ and $\cot x$, defined on the set $\mathbb{R} \setminus \{k\pi | k \in \mathbb{Z}\}$ and $\mathbb{R} \setminus \{(2k+1)\frac{\pi}{2} | k \in \mathbb{Z}\}$, respectively we have that $\mathcal{P} = \{k\pi | k \in \mathbb{Z}\}$.

Example 9.2. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be the Dirichlet function defined by

$$f(x) = \begin{cases} 0 & \text{if } x \text{ is rational number} \\ 1 & \text{if } x \text{ is irrational number.} \end{cases}$$

We shall show that the set $\mathcal{P}$ of periods of f is the set $\mathbb{Q}$ of rational numbers. Indeed, let $\alpha \in \mathcal{P}$, i.e. $f(x + \alpha) = f(x)$ for all $x \in \mathbb{R}$. Setting $x = 0$ we get that $f(\alpha) = f(0) = 0$. Hence $\alpha \in \mathbb{Q}$. Conversely, let $\alpha \in \mathbb{Q}$. Then $f(x + \alpha) = f(x)$ for all $x \in \mathbb{R}$ since the number $x + \alpha$ is rational if and only if x is rational. Hence $\mathcal{P} = \mathbb{Q}$.

In general, the set of periods of a function might be very complicated but as we shall see for continuous functions it has a relatively simple structure.

Theorem 9.1. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function. Then either $\mathcal{P} = \mathbb{R}$ (i.e. f is a constant function), or there exists $\alpha \in \mathbb{R}$ such that $\mathcal{P} = \{\alpha k | k \in \mathbb{Z}\}$.

Proof. Since f is a continuous function it follows that $\mathcal{P}$ is a closed subset of $\mathbb{R}$ which means that if $\alpha_1, \alpha_2, \dots \in \mathcal{P}$ and $\lim \alpha_n = \alpha$, then $\alpha \in \mathcal{P}$. Hence the theorem follows from the well-known description of the closed additive subgroups of $\mathbb{R}$. For the sake of completeness we shall prove it. Suppose that $\mathcal{P} \neq \mathbb{R}$. Since $\mathcal{P} = -\mathcal{P}$ and $\mathcal{P}$ is closed it follows that there is an open interval $(a, b) \subset \mathbb{R} \setminus \mathcal{P}$ for some $b > a \geq 0$. Set $c = \inf\{a' \in [0, a] | (a', b) \subset \mathbb{R} \setminus \mathcal{P}\}$. Since the set $\mathcal{P}$ is closed, we see that $c \in \mathcal{P}$ and then $(0, b - c) = (c, b) - \{c\} \subset \mathbb{R} \setminus \mathcal{P}$. Set now $\alpha = \sup\{\beta \geq b - c | (0, \beta) \subset \mathbb{R} \setminus \mathcal{P}\}$. If $\alpha = \infty$ then $\mathcal{P} = \{0\}$.

Otherwise, $(0, \alpha) \subset \mathbb{R} \setminus \mathcal{P}$ and $\alpha \in \mathcal{P}$. Now it follows by induction on n that $n\alpha \in \mathcal{P}$, $(n\alpha, (n+1)\alpha) \subset \mathbb{R} \setminus \mathcal{P}$ for any nonnegative integer n . Since $\mathcal{P} = -\mathcal{P}$ we conclude that $(k\alpha, (k+1)\alpha) \in \mathbb{R} \setminus \mathcal{P}$ for any $k \in \mathbb{Z}$ and therefore $\mathcal{P} = \{k\alpha/k \in \mathbb{Z}\}$. $\square$

We are now ready to determine all continuous solutions of AGS functional equation (9.1).

Theorem 9.2. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function satisfying the AGS functional equation (9.1). Then either $f(x) \equiv 0$, $f(x) = ax + 1$, or $f(x) = \max(ax + 1, 0)$, where $a \in \mathbb{R}$ is an arbitrary constant.

First Proof. Let $\mathcal{P}$ be the set of periods of f . If $\mathcal{P} = \mathbb{R}$ then f is a constant function and it follows from (9.1) that either $f(x) = 0$ or $f(x) = 1$. Hence it follows from Theorem 9.1 that we can assume that $\mathcal{P} = \{\alpha k | k \in \mathbb{Z}\}$ for some $\alpha \in \mathbb{R}$ and $f(x) \not\equiv 0$.

Now we shall show that (9.1) implies the following property of $\mathcal{P}$: if $f(x_1) = f(x_2) \neq 0$, then $x_2 - x_1 \in \mathcal{P}$. Indeed, for any $x \in \mathbb{R}$ there exists $y \in \mathbb{R}$ such that $x = x_1 + f(x_1)y$. Then it follows from (9.1) that

$$\begin{aligned} f(x) &= f(x_1 + f(x_1)y) = f(x_1)f(y) = f(x_2)f(y) = f(x_2 + f(x_2)y) = \\ &= f(x_2 + f(x_1)y) = f(x + x_2 - x_1), \end{aligned}$$

i.e. $x_2 - x_1 \in \mathcal{P}$.

Since $f(x) \not\equiv 0$ setting $y = 0$ in (9.1) gives $f(0) = 1$ and the continuity of $f(x)$ implies that there exists $y_0 \neq 0$ such that $f(y_0) \neq 0$. Suppose now that $f(x) \neq 0$ for some $x \in \mathbb{R}$. Then it follows from (9.1) that

$$0 \neq f(x)f(y_0) = f(x + y_0f(x)) = f(y_0 + f(y_0)x)$$

and the property of $\mathcal{P}$ proved above implies that $t = x - y_0 + y_0f(x) - xf(y_0) \in \mathcal{P}$. We shall show that $t = 0$. If $t \neq 0$ then $\alpha \neq 0$ and the function $f(x)$ is periodic with period α . Hence it takes all its values in the interval $[0, \alpha]$ and the Extremal value theorem shows that $f(x)$ has an absolute maximum that is attained at a point $x_0 \in [0, \alpha]$. Since $f(x_0) \neq 0$ it follows that there exist $x_1, x_2 \in \left(x_0 - \frac{\alpha}{3}, x_0 + \frac{\alpha}{3}\right)$ such that $x_1 < x_0 < x_2$ and $f(x_1) = f(x_2) \neq 0$. Hence $x_2 - x_1 \in \mathcal{P}$, i.e. $x_2 - x_1 = k\alpha$ for some $k \in \mathbb{Z}$. But this is a contradiction

since $0 < x_2 - x_1 < \frac{2\alpha}{3}$. Thus $t = 0$ and we conclude that if $f(x) \neq 0$ then $x - y_0 + y_0 f(x) - x f(y_0) = 0$, i.e. $f(x) = ax + 1$, where $a = \frac{f(y_0) - 1}{y_0}$. Since the constant function $f(x) = 1$ is a solution of (9.1) we may further assume that $a \neq 0$.

Consider the zero-set $\text{Ker } f = \{x \in \mathbb{R} | f(x) = 0\}$ of f . Since $f(x) \neq 0$ it follows that $\text{Ker } f \neq \mathbb{R}$. Note also that $\text{Ker } f \neq \emptyset$ since otherwise $f(x) = ax + 1$ for all $x \in \mathbb{R}$ and $f\left(-\frac{1}{a}\right) = 0$, a contradiction.

Suppose now that $\text{Ker } f$ contains a closed interval $[a, b]$, $b > a$. Set

$$m = \inf\{c \in \mathbb{R} | [c, b] \subset \text{Ker } f\} \text{ and } M = \sup\{c \in \mathbb{R} | [a, c] \subset \text{Ker } f\}.$$

Then at least one of the numbers m and M is finite since otherwise $\text{Ker } f = \mathbb{R}$. Assume first that both of them are finite numbers. Then the continuity of f implies that $[m, M] \subset \text{Ker } f$. Moreover, there exists a sequence $\{x_n\}_{n=1}^{\infty}$ such that $x_n < m$, $\lim_{n \rightarrow \infty} x_n = m$ and $f(x_n) \neq 0$. Hence

$$0 = f(m) = \lim_{n \rightarrow \infty} f(x_n) = \lim_{n \rightarrow \infty} (ax_n + 1) = am + 1,$$

i.e. $m = -\frac{1}{a}$. Analogously $M = -\frac{1}{a}$, a contradiction.

Let now $m = -\infty$ and M be a finite number. Then we see as above that $M = -\frac{1}{a}$ and $f(x) \neq 0$ for $x > -\frac{1}{a}$. Hence $f(x) = 0$ for $x \leq -\frac{1}{a}$ and $f(x) = ax + 1$ for $x \geq -\frac{1}{a}$. Since $f(0) = 1$ it follows that $a > 0$ and therefore $f(x) = \max(ax + 1, 0)$ for all $x \in \mathbb{R}$.

The same arguments show that if $M = +\infty$ then $m = -\frac{1}{a} > 0$ and we have again $f(x) = \max(ax + 1, 0)$ for all $x \in \mathbb{R}$.

It remains to consider the case when $\text{Ker } f$ does not contain an interval. Then for any $x \in \text{Ker } f$ there exists a sequence $\{x_n\}_{n=1}^{\infty}$ such that $x_n \notin \text{Ker } f$ for any n and $\lim_{n \rightarrow \infty} x_n = x$. Hence

$$0 = f(x) = \lim_{n \rightarrow \infty} f(x_n) = \lim_{n \rightarrow \infty} (ax_n + 1) = ax + 1$$

which shows that $\text{Ker } f = \left\{ -\frac{1}{a} \right\}$. Therefore in this case $f(x) = ax + 1$ for all $x \in \mathbb{R}$ and the theorem is proved.

Second Proof. Obviously $f = 0$ satisfies the condition. Next set $y = 0$ to get $f(x) = f(x)f(0)$. Thus if $f \not\equiv 0$, then $f(0) = 1$. If $f(x) = 1 + ax$, we get

$$\begin{aligned} f(x + yf(x)) &= f(x + y(1 + ax)) = f(x + y + ax) = 1 + ax + ay + a^2x \\ &= (1 + ax)(1 + ay) = f(x)f(y). \end{aligned}$$

If $f(x) \neq 1$ and $y = \frac{x}{1 - f(x)}$, then $x + yf(x) = y$. Hence the condition says

that $f(x)f(y) = f(y)$. As $f(x) \neq 1$ we get $f\left(\frac{x}{1 - f(x)}\right) = 0$. So if $f \not\equiv 1$, then the set A of t for which $f(t) = 0$ is not empty. Now if $t \in A, x \notin A$, then set $y = \frac{t - x}{f(x)}$ to get $x + yf(y) = t$. From here we get $0 = f(t) = f(x)f(y)$.

So $f(y) = 0$ and $\frac{t - x}{f(x)} \in A$. If $\frac{t - x}{f(x)}$ is a constant, then f is linear but linear

functions were already investigated by us. Otherwise $\frac{t - x}{f(x)}$ is a continuous non-constant function so A contains infinitely many numbers. Without loss of generality we may assume that A contains infinitely many positive numbers (the second case is analogous). Let $b = \inf A \cap (0, \infty)$. As A is closed, $b \in A$ but $[0; b)$ does not intersect A . As $f(0) = 1$ we deduce that f is positive on $[0; b)$. Thus if $x \in [0; b)$, then $g(x) = \frac{c - x}{f(x)} \in A$ where $c \in A, c > b$. But

$g(0) = b, \lim_{x \rightarrow b} g(x) = \infty$ and hence we conclude that $[b; \infty) \subset A$. Then if $x \notin A$, we get $h_x(y) = x + yf(x) \in A$ if and only if $y \in A$. Therefore $h_x(y) \notin A$ as $y < b$ but $h_x(b) \in A$ which is possible only when $h_x(b)$ is a boundary point of A as A is closed. But $h_x(b)$ is continuous in x . As b cannot be written as a limit of boundary points of A except b itself, we conclude that $h_x(b) = b$ for all $x \in [0; b]$. So $x + bf(x) = b$ and hence $f(x) = \frac{b - x}{b}$ for $x \in [0; b]$. If A contains negative points, let $-d \in A, d > 0$. Then set $y = -d$ to get $f\left(x - \frac{(b - x)d}{b}\right) = 0$. But if x is sufficiently close to b and less than

b , then $0 < x - \frac{(b-x)d}{b} < b$, a contradiction. Hence A contains no point in $(-\infty, 0)$ and by continuity of h on $(-\infty; b)$ we conclude h is constant so $f(x) = \frac{b-x}{b}$ for $x \leq b$, $f(x) = 0$ for $x \geq b$. If A contains negative numbers, we get analogously that $f(x) = \frac{b-x}{b}$ for $x \geq b$, $f(x) = 0$ for $x \leq b$, where b is negative. These two functions can be checked to verify the problem. Together with $f = 0$ and $f = 1 + ax$ they form the answer set. $\square$

9.2 Practice Problems

Problem 9.1. (IMO 1968) Let $a > 0$ be a real number and $f(x)$ a real function defined on all of $\mathbb{R}$, satisfying for all $x \in \mathbb{R}$

$$f(x+a) = \frac{1}{2} + \sqrt{f(x) - f(x)^2}.$$

- (a) Prove that the function f is periodic;
- (b) Give an example of such a nonconstant function for $a = 1$.

Problem 9.2. Is it possible to express the function $f(x) = 2^x + 3^x + 9^x$ as the sum of finitely many periodic functions?

Problem 9.3. Let k be a positive real number and $f : (0, \infty) \rightarrow (0, \infty)$ be a function such that

$$f(x)f(y) = kf(x + yf(x))$$

for all $x, y \in (0, \infty)$.

- (a) Prove that f is an increasing function.
- (b) Determine all functions satisfying the above functional equation.

Problem 9.4. Let $a > 0$, $a \neq 1$ and $b \neq 0$ be real numbers. Find all functions $f : [0, \infty) \rightarrow (0, \infty)$ such that

$$f(x)f(y) = af(x + yf(x)^b)$$

for all $x, y \in [0, \infty)$.

Problem 9.5. Given a positive integer n find as many nonconstant continuous solutions $f : \mathbb{R} \rightarrow \mathbb{R}$ as you can of the functional equation

$$f(x)f(y) = f(x + yf(x)^n)$$

for all $x, y \in \mathbb{R}$.

Chapter 10

Arithmetic Functional Equations

10.1 Theory and Examples

It can be debated what exactly should be the category of "arithmetic" functional equations. We decided, for this book, to classify a functional equation as an arithmetic one if the solution relies on the properties of divisibility and factorization of integers. Naturally, we will only have functions on $\mathbb{N}$, $\mathbb{Z}$ or at most $\mathbb{Q}$ here, so not $\mathbb{R}$ or $\mathbb{C}$.

A good knowledge of arithmetic functions may help the reader to solve arithmetic functional equations, because for many of them one has to guess the solution first. For this reason, we list some of the more important theoretical notions and the most famous arithmetic functions. In the following, $p_1^{\alpha_1} \cdots p_k^{\alpha_k}$ will always denote the Euclidean factorization of a positive integer as a product of powers of distinct primes.

Recall that a function is called multiplicative if $f(xy) = f(x)f(y)$ for all x, y in the domain of f . In the case of functions over $\mathbb{N}$, $\mathbb{Z}$ or $\mathbb{Q}$ (we shall call such functions arithmetic) however, this is called *strong* or *total* multiplicativity. The reason for that is the following alternate important definition of multiplicativity:

Definition. An arithmetic function f is called multiplicative if it satisfies $f(xy) = f(x)f(y)$ for *relatively prime* numbers x, y . (In the case of $\mathbb{Q}$, *relatively prime* means there are no primes appearing in the numerator or denominator of both numbers written in lowest terms).

It is immediate that if $x = \prod p_i^{\alpha_i}$ is the decomposition of x as a product of primes, then $f(x) = \prod f(p_i^{\alpha_i})$ for f multiplicative. For f strongly multiplicative one has $f(x) = \prod f(p_i)^{\alpha_i}$.

Here are some important non-trivial examples of multiplicative functions (none of them are strongly multiplicative):

- The Euler totient function $\phi: \mathbb{N} \rightarrow \mathbb{N}$ whose value at n is the number of integers between 1 and n that are relatively prime to n . The Chinese Remainder Theorem implies that ϕ is multiplicative - but it is not strongly multiplicative. In fact, it is known that

$$f\left(\prod p_i^{\alpha_i}\right) = \prod (p_i - 1)p_i^{\alpha_i - 1}.$$

An important arithmetic property of the Euler totient function is that for a co-prime to n , $a^{\phi(n)} \equiv 1 \pmod{n}$

- The Möbius function $\mu: \mathbb{N} \rightarrow \mathbb{N}$ defined as follows:

$$\begin{cases} \mu(n) = 0 & p^2 \mid n \\ \mu(n) = (-1)^k & n = p_1 p_2 \dots p_k \end{cases}$$

In other words, μ takes the value 0 at all integers n that are not square-free (that is divisible by a non-trivial square) and ± 1 on integers n that are square-free, depending on the parity of the number of their prime divisors.

- The function $\tau: \mathbb{N} \rightarrow \mathbb{N}$ which calculates the number of divisors of $n \in \mathbb{N}$ (including 1 and itself). If the Euclidean decomposition of n is $n = p_1^{\alpha_1} \dots p_k^{\alpha_k}$ then

$$\tau(n) = (\alpha_1 + 1)(\alpha_2 + 1) \dots (\alpha_k + 1)$$

which implies multiplicativity.

- The function $\sigma: \mathbb{N} \rightarrow \mathbb{N}$ which calculates the sum of all divisors of $n \in \mathbb{N}$. For $n = p_1^{\alpha_1} \dots p_k^{\alpha_k}$ one has

$$\sigma(n) = (1 + p_1 + \dots + p_1^{\alpha_1}) \dots (1 + p_k + \dots + p_k^{\alpha_k}) = \frac{p_1^{\alpha_1+1} - 1}{p_1 - 1} \dots \frac{p_k^{\alpha_k+1} - 1}{p_k - 1}.$$

This formula follows immediately from the following theorem applying to the function $f(x) = 1$.

Theorem. *If $f: \mathbb{N} \rightarrow \mathbb{N}$ is a multiplicative function then for any $n \in \mathbb{N}$, the function $F(n) = \sum_{d|n} f(d)$ satisfies*

$$F(p_1^{\alpha_1} \dots p_k^{\alpha_k}) = (1 + f(p_1) + \dots + f(p_1^{\alpha_1})) \dots (1 + f(p_k) + \dots + f(p_k^{\alpha_k}))$$

If, in addition f is strongly multiplicative, the latter rewrites as

$$\frac{f(p_1)^{\alpha_1+1} - 1}{f(p_1) - 1} \dots \frac{f(p_k)^{\alpha_k+1} - 1}{f(p_k) - 1}.$$

Note that the above product makes sense only for $f(p_i) \neq 1$; if $f(p_i) = 1$, the factor

$$\frac{f(p_i)^{\alpha_i+1} - 1}{f(p_i) - 1} = 1 + f(p_i) + \dots + f(p_i)^{\alpha_i}$$

is replaced by the number $\alpha_i + 1$.

Proof. Clearing the brackets in the right-hand side and using multiplicativity to pull all numbers inside a single application of f rewrites the right-hand side as $\sum_{0 \leq m_i \leq \alpha_i} f(p_1^{m_1} \dots p_k^{m_k})$ which is the same as the left-hand side. If f is multiplicative, then $f(p_i^{m_i}) = f(p_i)^{m_i}$ from where the conclusion follows. (It should be noted that $f(1)=1$ follows immediately). $\square$

The application $f \rightarrow F$, where $F(n) = \sum_{d|n} f(d)$ has an inverse which involves the Möbius function:

Theorem. *If $F: \mathbb{N} \rightarrow \mathbb{R}$ is a function (where $\mathbb{R}$ can be $\mathbb{Z}, \mathbb{C}, \mathbb{Q}, \mathbb{R}$ or more generally any ring) then there is a unique function f for which $F(n) = \sum_{d|n} f(d)$, and that function is given by*

$$f(n) = \sum_{d|n} \mu\left(\frac{n}{d}\right) F(d).$$

Proof. For any n , the relation $F(n) = \sum_{d|n} f(d)$ expresses $f(n)$ in terms of the values $f(1), \dots, f(n-1)$ and $F(n)$. Also $f(1) = F(1)$ if we set $n = 1$, and now the uniqueness of f follows by induction. It remains to verify that the above example for f satisfies the condition.

Indeed, one has

$$\sum_{d|n} f(d) = \sum_{d|n, d'|d} \mu\left(\frac{d}{d'}\right) F(d') = \sum_{d'|d} \sum_{m|\frac{n}{d'}} \mu(m) F(d'),$$

where the last identification is obtained by letting m be $\frac{d}{d'}$. We thus rewrite the required sum as

$$\sum_{d'|n} E\left(\frac{n}{d'}\right) F(d'),$$

where

$$E(x) = \sum_{d|x} \mu(d).$$

It remains to check that $E(1) = 1$ and $E(x) = 0$ for $x > 1$. That $E(1) = 1$ is clear. That $E(x) = 0$ for $x > 1$ follows from the previous theorem since

$$E(p_1^{\alpha_1} \dots p_k^{\alpha_k}) = (1 + \mu(p_1) + \dots + \mu(p_1^{\alpha_1})) \dots (1 + \mu(p_k) + \dots + \mu(p_k^{\alpha_k}))$$

and the term $1 + \mu(p_1) + \dots + \mu(p_1^{\alpha_1})$ is $1 - 1 + 0 + \dots + 0 = 0$ for $\alpha_1 > 0$. $\square$

Note that this theorem does not require any arithmetic property of f ; however it is often used for functions f with nice arithmetic attributes.

The theory of multiplicative functions whose domain is $\mathbb{Z}$ or $\mathbb{Q}$ does not essentially differ from the theory of multiplicative functions on $\mathbb{N}$. This is because such a function is determined reasonably well by its behavior on $\mathbb{N}$: $f\left(\frac{p}{q}\right)$ will be $\frac{f(p)}{f(q)}$ for $\frac{p}{q}$ an irreducible fraction, and $f(-n)$ will be $f(-1)f(n)$.

The following are some examples of arithmetic functional equations.

Example 10.1. Prove that there is no bijection $f : \mathbb{N} \rightarrow \mathbb{N}_0$ such that

$$f(mn) = f(m) + f(n) + 3f(m)f(n)$$

for all $m, n \in \mathbb{N}$.

Solution. Suppose that there is a bijection $f : \mathbb{N} \rightarrow \mathbb{N}_0$ with the given property. Set $g(n) = 3f(n) + 1$ and $S = \{3n + 1 | n \in \mathbb{N}_0\}$. Then $g : \mathbb{N} \rightarrow S$ is a bijection such that $g(mn) = g(m)g(n)$ for any $m, n \in \mathbb{N}$. (i.e. a strongly

multiplicative bijection) In particular, $g(1) = g^2(1)$, i.e. $g(1) = 1$. Let $p, q, r \in \mathbb{N}$ be such that $g(p) = 4$, $g(q) = 10$ and $g(r) = 25$. Since any of the numbers 4, 10 and 25 is not a product of two numbers from the set $S \setminus \{1\}$, it follows from the multiplicativity of g that p , q and r are distinct prime numbers. On the other hand,

$$g(pr) = g(p)g(r) = 10^2 = g^2(q) = g(q^2)$$

and so $pr = q^2$, a contradiction.

Example 10.2. Show that there exists a unique function $f: \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ which satisfies the following three conditions:

- i) $f(x, y) = f(y, x)$ for all $x, y \in \mathbb{N}$;
- ii) $f(x, x) = x$ for all $x \in \mathbb{N}$;
- iii) $(y - x)f(x, y) = yf(x, y - x)$ for all $x, y \in \mathbb{N}, y > x$.

Solution. The structure of the conditions reminds us of the Euclidean Algorithm. But there are these superfluous factors of $y - x$ and x in part iii). We can get rid of them by some algebra. Let us set $g(x, y) = \frac{xy}{f(x, y)}$. We then get $g(x, y) = g(y, x)$, $g(x, x) = x$ and $g(x, y) = g(x, y - x)$, $y > x$. We know that these relations are the ones which establish the function $\gcd(x, y)$, by the Euclidean Algorithm, and therefore $g(x, y) = \gcd(x, y)$. So

$$f(x, y) = \frac{xy}{\gcd(x, y)} = \text{lcm}(x, y).$$

Example 10.3. Find all functions $f: \mathbb{N} \rightarrow \mathbb{R}$ for which we have $f(1) = 1$ and

$$\sum_{d|n} f(d) = n$$

whenever $n \in \mathbb{N}$.

Solution. Again a little mathematical culture helps us: an example of such a function is Euler's totient function ϕ . So let us try to prove that $f = \phi$. As ϕ is multiplicative, let's firstly show that f is multiplicative, i.e. $f(mn) = f(m)f(n)$ whenever $(m, n) = 1$. We do this by induction on $m + n$. Note

that when one of m, n is 1 this is clearly true. Now assume that $m, n > 1$, $(m, n) = 1$. Then the condition written for mn gives us $\sum_{d|mn} f(d) = mn$. But any $d|mn$ can be written uniquely as $d = d_1 d_2$, where $d_1|m$ and $d_2|n$. If $d < mn$ then $d_1 + d_2 < m + n$ and by the induction hypothesis we get $f(d) = f(d_1 d_2) = f(d_1) f(d_2)$ for $d < mn$. Therefore

$$\begin{aligned} mn &= \sum_{d|mn} f(d) = \sum_{d|mn, d < mn} f(d) + f(mn) \\ &= \sum_{d_1|m, d_2|n} f(d_1) f(d_2) - f(m) f(n) + f(mn) \\ &= \left(\sum_{d|m} f(d) \right) \left(\sum_{d|n} f(d) \right) - f(m) f(n) + f(mn) = mn - f(m) f(n) + f(mn), \end{aligned}$$

so $f(mn) = f(m) f(n)$, as desired. So it suffices to compute f for powers of primes. Let p be a prime. Then writing the condition for $n = p^k$ we get $f(1) + f(p) + \dots + f(p^k) = p^k$. Subtracting this from the analogous condition for $n = p^{k+1}$ we get $f(p^{k+1}) = p^{k+1} - p^k = \phi(p^{k+1})$, and now the relation $f = \phi$ follows from the multiplicativity. It remains to verify that $\sum_{d|n} \phi(d) = n$. There are many proofs of this. One of the shortest is evaluating the numbers of subunitary (and unitary) non-zero fractions with denominator n . On one hand, this number is clearly n . On the other hand writing every fraction in lowest terms will produce exactly $\phi(d)$ fractions with denominator d for every $d|n$. Also, according to the theorem proved in the previous section the function f that satisfies the required condition is computed to be

$$\sum_{d|n} \mu\left(\frac{n}{d}\right) \cdot d.$$

This expression turns out to simplify to $\phi(n)$ - because $\mu(x) \cdot \frac{1}{x}$ is multiplicative, the latter sum evaluates to

$$\begin{aligned} n \cdot \left(1 + \mu(p_1) \frac{1}{p_1} + \mu(p_1^2) \frac{1}{p_1^2} + \dots \right) \cdot \dots \cdot \left(1 + \mu(p_k) \frac{1}{p_k} + \mu(p_k^2) \frac{1}{p_k^2} + \dots \right) \\ = n \left(1 - \frac{1}{p_1} \right) \cdot \dots \cdot \left(1 - \frac{1}{p_k} \right) \end{aligned}$$

which is precisely $\phi(n)$.

Example 10.4. (Turkey 1995) Find all surjective functions $f: \mathbb{N} \rightarrow \mathbb{N}$ such that $m \mid n$ if and only if $f(m) \mid f(n)$ for any $m, n \in \mathbb{N}$.

Solution. Note first that f is a bijection, as if $f(k) = f(l)$ for $k < l$ then $f(l) \mid f(k)$ so $l \mid k$ impossible. Since 1 divides every number, $f(1) = 1$. Now we also see that $f(n)$ has as many divisors as n because f provides a bijection between the set of divisors of n and the set of divisors of $f(n)$. Next, let us prove that f is multiplicative. If $(m, n) = 1$, then $(f(m), f(n)) = 1$ because if $f(e) = d \mid (f(m), f(n))$ then $e \mid m, e \mid n$ so $e = 1$. Hence if $(m, n) = 1$, then $f(m) \mid f(mn), f(n) \mid f(mn)$ so $f(m)f(n) \mid f(mn)$. As $f(m)$ has as many divisors as m , $f(n)$ as many divisors as n and $f(m), f(n)$ are co-prime, $f(m)f(n)$ has as many divisors as mn . Hence $f(mn) = f(m)f(n)$ and thus f is multiplicative. Now if p is prime then $f(p)$ must also be a prime, and the converse is also true, so f is a bijection on the set of all prime numbers. We prove that if n is a prime power, then $f(p)$ is a prime power of the same exponent. Indeed, we have just proven the basis. Now if we have proven that $f(p^k) = q^k$, then $q^k \mid f(p^{k+1})$ so $f(p^{k+1}) = q^{k+r}M$ where M is a number co-prime to q . As $f(p^{k+1})$ has $k+2$ divisors, we must have $(k+r+1)t = k+2$, where t is the number of divisors of M . If $t \geq 2$, then this is impossible. So $t = 1, M = 1$ and $k+r+1 = k+2$ hence $f(p^{k+1}) = q^{k+1}$ as desired. Now using the multiplicativity of f we deduce

$$f\left(\prod p_i^{k_i}\right) = \prod q_i^{k_i},$$

where $q_i = f(p_i)$. Any f defined by this relation and by a bijection on the set of primes clearly satisfies the relation, and so this is the form of all solutions.

Example 10.5. (IMO Shortlist 2007) Find all surjective functions $f: \mathbb{N} \rightarrow \mathbb{N}$ such that for every $m, n \in \mathbb{N}$ and every prime p , the number $f(m+n)$ is divisible by p if and only if $f(m) + f(n)$ is divisible by p .

Solution. Clearly $f(1) = 1$, as a prime factor of $f(1)$ would divide all values of f .

For a prime p let $g(p)$ be the smallest positive integer n such that $p \mid f(n)$. As f is surjective, g is well defined. Clearly, $g(p) > 1$ and p divides $f(g(p))$

and so p divides $f(n)$ whenever $g(p)$ divides n . Using Euclidean division and the minimality of $g(p)$, we deduce that the converse is true: if $p|f(n)$ then $g(p)$ divides n .

Next, we prove that there cannot be a prime dividing $f(n+1) - f(n)$ for some positive integer n . Suppose p divides $f(n+1) - f(n)$. By surjectivity, there is m such that $p|f(n) + f(m)$, so $p|f(m+n)$ and also $p|f(m+n+1)$. Therefore $g(p)$ divides $m+n$ and $m+n+1$, a contradiction. Thus

$$|f(n+1) - f(n)| = 1 \text{ for all } n.$$

From the previous steps we have $f(2) = 2$. Let us prove that $f(n) = n$ by induction. Assume that $f(k) = k$ for $k \leq n$ and that $f(n+1) = n-1$. But $n+1 = f(n) + f(1)$ and $n-1 = f(n+1)$ have the same prime factors, which can only be all equal to 2 and so $n+1, n-1$ are both powers of 2, that differ by 2. Thus $n-1 = 2$ and $n = 3$. In this case we obtain $f(4) = 2$. But then $|f(5) - 2| = 1$ and on the other hand $f(5) = f(2+3)$ is a multiple of 5. This is a contradiction, so $f(n+1) \neq n-1$ and so $f(n+1) = n+1$.

Example 10.6. (G. Dospinescu) Find all functions $f : \mathbb{N} \rightarrow \mathbb{Z}$ with the properties:

- i) $f(a) \geq f(b)$ whenever a divides b ;
- ii) $f(ab) + f(a^2 + b^2) = f(a) + f(b)$ for all $a, b \in \mathbb{N}$.

Solution. By considering the function $f(x) - f(1)$, we may assume that $f(1) = 0$, so $f(n) \leq 0$ for all n by the first condition. By the second condition $f(2) = 0$.

A key point is that $f(p) = 0$ for all primes $p \equiv 1 \pmod{4}$. Indeed, take such a prime p and consider a positive integer a such that $p|a^2 + 1$. Such a number always exists by classical results in number theory. By the second relation with $b = 1$ we deduce that $f(a^2 + 1) = 0$ and by the first condition $f(p) \geq f(a^2 + 1) = 0$. But we saw that $f(p) \leq 0$ and so $f(p) = 0$. We claim now that if n is a product of primes (not necessarily distinct) congruent to 1 mod 4, then $f(n) = 0$. Indeed, it is enough to observe that if $f(a) = f(b) = 0$ then $f(ab) + f(a^2 + b^2) = 0$ and $f(ab), f(a^2 + b^2) \leq 0$ and so $f(ab) = 0$. This allows us to prove the claim by induction on the number of primes dividing n .

Another key point is that $f(ab) = f(a) + f(b)$ if $\gcd(a, b) = 1$. Indeed, $a^2 + b^2$ is then a product of primes congruent to 1 mod 4, except for a possible power of 2. But since we saw that $f(2) = 0$, we may apply the argument in the last part of the previous paragraph to conclude that $f(a^2 + b^2) = 0$ and so $f(ab) = f(a) + f(b)$.

Finally, let us try to compute $f(p^k)$ for a prime p . We saw that if $f(a) = f(b) = 0$ then $f(ab) = 0$, so $f(p^k) = 0$ if $p = 2$ or if $p = 1 \pmod{4}$, so we may assume that $p = 3 \pmod{4}$. By taking $b = a^k$ in the second relation and taking into account that $f(a^k) \geq f(a^{k+1})$ and $f(a) \geq f(a^2 + a^{2k})$, we deduce that both of these inequalities are equalities and so $f(a^k) = f(a^{k+1})$ for all a and k and so $f(p^k) = f(p)$.

Putting everything together we deduce that if $n = p_1^{k_1} \dots p_r^{k_r}$ for some distinct primes $p_1, \dots, p_r$ and $k_1, \dots, k_r$ positive integers, then $f(n) = f(p_1) + \dots + f(p_r)$ and each $f(p_i)$ is 0 if $p_i = 2$ or $p_i = 1 \pmod{4}$. This determines f uniquely if we fix the values of $f(p)$ for all primes $p = 3 \pmod{4}$. This gives us a whole family of possible solutions and the last step will be to check that we can allow arbitrary values at these primes.

So, choose any function g defined on the set of primes $p = 3 \pmod{4}$ and define $f(1) = f(2) = 0$ and $f(p) = g(p)$ if $p = 3 \pmod{4}$, $f(p) = 0$ for the other primes p and extend f to all positive integers by $f(p_1^{k_1} \dots p_r^{k_r}) = f(p_1) + \dots + f(p_r)$. We have to check that f is a solution. But the first relation is clear and the second one follows by considering the prime factorization of $a, b, \gcd(a, b)$ and using the fact that for $\gcd(a, b) = 1$ the prime factors of $a^2 + b^2$ are all congruent 2 or 1 mod 4, on which f vanishes.

Example 10.7. (P. Erdős) Find all increasing multiplicative functions $f : \mathbb{N} \rightarrow \mathbb{R}$.

Solution. We first show that $f(p^{n+1}) = f(p)f(p^n)$ for all positive integers n and primes p . To see this, note that we have

$$\frac{f(xp - 1)}{f(x)} \leq \frac{f(p^{n+1})}{f(p^n)} \leq \frac{f(xp + 1)}{f(x)}$$

for all $(x, p) = 1$, and so

$$f(p) \cdot \frac{f(x - p)}{f(x)} \leq \frac{f(p^{n+1})}{f(p^n)} \leq f(p) \cdot \frac{f(x + p)}{f(x)},$$

for all $(x, p) = 1$. Now, if

$$\frac{f(p^{n+1})}{f(p)f(p^n)} = r,$$

then $f(x + kp) \geq r^k f(x)$ for all positive integers k . Also, for some sufficiently large x , $f(2)f(x) = f(2x) > f(x + kp)$ as the function is increasing, so $f(2) > r^k$ for all integers k , and so $r \leq 1$. Similarly, $r \geq 1$, therefore

$$\frac{f(p^{n+1})}{f(p)f(p^n)} = r = 1$$

for all p, n , which proves that indeed $f(p^n) = f(p)^n$ for all p, n .

Now, suppose that $f(n)$ is not of the form n^a for some constant a . Then there exist integers l, m so that

$$\frac{\ln l}{\ln m} \neq \frac{\ln f(l)}{\ln f(m)}.$$

But note that if the $LHS > RHS$, then there is a rational number

$$\frac{\ln l}{\ln m} > \frac{s}{t} > \frac{\ln f(l)}{\ln f(m)},$$

so that $l^t > m^s$ but $f(m^s) = f(m)^s > f(l)^t = f(l^t)$, thus leading to a contradiction; similarly, we get a contradiction if the RHS is greater. Hence, we conclude that the only functions satisfying the requirements of the hypothesis are the functions of form $f(n) = n^a$ with constant a .

Example 10.8. (IMO 1990) Construct a function $f: \mathbb{Q}^+ \rightarrow \mathbb{Q}^+$ such that

$$yf(xf(y)) = f(x)$$

for all $x, y \in \mathbb{Q}^+$.

Solution. Let f be a function with the given properties. Then setting $x = 1$ gives $yf(f(y)) = f(1)$. Hence the function f is injective and from $f(f(1)) = f(1)$ we get $f(1) = 1$. Thus

$$yf(f(y)) = 1. \tag{1}$$

Now replacing x by xy and y by $f(y)$ in the given identity gives

$$f(xy) = f(x)f(y). \quad (2)$$

Conversely, it is easy to check that any function $f: \mathbb{Q}^+ \rightarrow \mathbb{Q}^+$ satisfying (1) and (2) satisfies the given condition as well.

It follows by (2) that $1 = f(1) = f(y)f\left(\frac{1}{y}\right)$ and using (2) again we get that $f\left(\frac{x}{y}\right) = \frac{f(x)}{f(y)}$. This shows that if the function f is defined on the set P of the prime numbers, then it has a unique continuation on $\mathbb{Q}^+$ given by: $f(1) = 1$; if $n \geq 2$ is an integer and $n = p_1^{\alpha_1} \dots p_k^{\alpha_k}$ is its canonical representation as a product of primes, then $f(n) = f(p_1)^{\alpha_1} f(p_2)^{\alpha_2} \dots f(p_k)^{\alpha_k}$; if $r = \frac{m}{n} \in \mathbb{Q}^+$ then $f(r) = \frac{f(m)}{f(n)}$. If moreover f satisfies (1) for all $p \in P$, then one checks easily that (1) and (2) are satisfied for all $x, y \in \mathbb{Q}^+$.

Hence we have to construct a function $f: P \rightarrow \mathbb{Q}^+$ which satisfies (1). To this end let $p_1 < p_2 < \dots$ be the sequence of prime numbers. Set $f(p_{2n-1}) = p_{2n}$ and $f(p_{2n}) = \frac{1}{p_{2n-1}}$ for all $n \geq 1$. Then it is obvious that $f(f(p)) = \frac{1}{p}$ for all $p \in P$, i.e. the identity (1) is satisfied.

Example 10.9. (IMO 1998) Consider all functions $f: \mathbb{N} \rightarrow \mathbb{N}$ such that

$$f(n^2 f(m)) = m f(n)^2$$

for all $m, n \in \mathbb{N}$. Find the least possible value of $f(1998)$.

Solution. Let f be an arbitrary function satisfying the given condition. Set $f(1) = a$. Then setting $n = 1$ and $m = 1$ gives $f(f(m)) = a^2 m$ and $f(an^2) = f(n)^2$. Thus

$$f(m)^2 f(n)^2 = f(m)^2 f(an^2) = f(m^2 f(f(an^2))) = f(m^2 a^3 n^2) = f(amn)^2,$$

i.e. $f(m)f(n) = f(amn)$. In particular, $f(am) = af(m)$ and therefore

$$af(mn) = f(m)f(n). \quad (1)$$

We shall prove that a divides $f(n)$ for all $n \in \mathbb{N}$. Let p be a prime number and let $\alpha \geq 0, \beta \geq 0$ be the degrees of p in the canonical representations of a and $f(n)$, respectively. It follows by induction from (1) that $f(n)^k = a^{k-1}f(n^k)$ for all $k \in \mathbb{N}$. Hence $k\beta \geq (k-1)\alpha$, which implies that $\beta \geq \alpha$. This shows that a divides $f(n)$.

Set $g(n) = \frac{f(n)}{a}$. Then $g: \mathbb{N} \rightarrow \mathbb{N}$ satisfies the conditions

$$g(mn) = g(m)g(n) \quad (2)$$

$$g(g(m)) = m \quad (3)$$

for all $m, n \in \mathbb{N}$. Conversely, given a function g with the above properties then for any $a \in \mathbb{N}$ the function $f(n) = ag(n)$ satisfies the given condition.

Setting $m = n = 1$ in (2) gives

$$g(1) = 1. \quad (4)$$

Now we shall prove that $g(p) \in P$ for all $p \in P$. Indeed, let $p \in P$ and $g(p) = uv$. Then it follows from (3) and (2) that $p = g(g(p)) = g(uv) = g(u)g(v)$. We may assume that $g(u) = 1$. Hence $u = g(g(u)) = g(1) = 1$ and therefore $g(p) \in P$. Now let $n \geq 2$ and $n = p_1^{\alpha_1} \dots p_k^{\alpha_k}$ be the canonical representation of n as a product of prime numbers. Then it follows from (2) that

$$g(n) = g(p_1)^{\alpha_1} \dots g(p_k)^{\alpha_k}. \quad (5)$$

Hence we have proved that any function $f: \mathbb{N} \rightarrow \mathbb{N}$ with the given property is uniquely determined by its value $f(1)$ and a function $g: P \rightarrow P$ such that $g(g(p)) = p$, i.e. by a pairing of the set of prime numbers.

We shall show now that the least possible value of $f(1998)$ is 120. We have that

$$f(1998) = f(2 \cdot 3^3 \cdot 37) = f(1)g(2)g(3)^3g(37).$$

Since $g(2), g(3)$ and $g(37)$ are different prime numbers it follows that

$$g(2)g(3)^3g(37) \geq 3 \cdot 2^3 \cdot 5 = 120,$$

i.e. $f(1998) \geq 120$. To construct a function f satisfying the given condition and $f(1998) = 120$ set $a = f(1) = 1, g(2) = 3, g(3) = 2, g(5) = 37, g(37) = 5$

and $g(p) = p$ for all prime numbers $p \neq 2, 3, 5, 37$. Then $g(g(p)) = p$ for all $p \in P$ and as we said above these data determine uniquely a function $f: \mathbb{N} \rightarrow \mathbb{N}$ with desired properties.

Example 10.10. (G. Dospinescu) Find the least nonnegative integer n for which there exists a nonconstant function $f: \mathbb{Z} \rightarrow [0, \infty)$ with the following properties:

- a) $f(xy) = f(x)f(y)$;
- b) $2f(x^2 + y^2) - f(x) - f(y) \in \{0, 1, 2, \dots, n\}$

for all $x, y \in \mathbb{Z}$. For this n find all functions with the above properties.

Solution. Note first that for $n = 1$ there are functions satisfying a) and b). Indeed, for any prime p with $p \equiv 3 \pmod{4}$ define $f_p: \mathbb{Z} \rightarrow [0, \infty)$ by $f_p(x) = 0$ if $p \nmid x$ and $f_p(x) = 1$, otherwise. Then a) follows from the fact that if $p \mid xy$ then $p \mid x$ or $p \mid y$. On the other hand it is well known that $p \mid x^2 + y^2$ iff $p \mid x$ and $p \mid y$, and this implies b). We shall show now that if f is nonconstant and satisfies a) and b) then $n > 0$. Suppose $n = 0$. Then $2f(x^2 + y^2) = f(x) + f(y)$ and hence $2f(x)^2 = 2f(x^2 + 0) = f(x) + f(0)$. In particular, $f(0)^2 = f(0)$. If $f(0) = 1$ then a) implies that $f \equiv 1$, so $f(0) = 0$. Consequently $2f(x)^2 = f(x)$ for every $x \in \mathbb{Z}$. This together with a) imply that $f(x)^2 = f(x^2) = 2f(x^2)^2 = 2f(x)^4$. In particular, $2f(x) \neq 1$ for all x and therefore $f \equiv 0$, a contradiction. So, $n = 1$ is the least integer with required properties.

We will prove now that if $n = 1$, then each nonconstant function f satisfying a) and b) is of the form f_p , or the function equal to 1 at nonzero integers and 0 at 0. We already know that $f(0) = 0$. Since $f(1)^2 = f(1)$ and f is nonconstant we have $f(1) = 1$. Also,

$$2f(x)^2 - f(x) = 2f(x^2 + 0) - f(x) - f(0) \in (0, 1)$$

for every $x \in \mathbb{Z}$. Thus $f(x) \in (0, 1)$. We have $f(-1)^2 = f(1) = 1$, so $f(-1) = 1$. Then $f(-x) = f(-1)f(x) = f(x)$ and it follows from a) that it suffices to find $f(p)$ for any prime p . Suppose there is $x > 0$ with $f(x) = 0$. Since $x \neq 1$ it follows that for some prime divisor p of x we have $f(p) = 0$. Suppose that there is another prime q for which $f(q) = 0$. Then $2f(p^2 + q^2) \in (0, 1)$ shows that $f(p^2 + q^2) = 0$. Hence for all integers a and b we have

$$0 = 2f(a^2 + b^2)f(x^2 + y^2) = 2f((ap + bq)^2 + (aq - bp)^2).$$

On the other hand $0 \leq f(x) + f(y) \leq 2f(x^2 + y^2)$ and the above identities show that $f(ap + bq) = f(aq - bp) = 0$. But p and q are relatively prime and by Bezout's lemma there are integers a and b such that $aq - bp = 1$. Then $1 = f(1) = f(aq - bp) = 0$, a contradiction. So, there is only one prime p for which $f(p) = 0$. Suppose that $p = 2$. Then $f(x) = 0$ for x even and $2f(x^2 + y^2) = 0$ for x, y odd. Hence $f(x) = f(y) = 0$ for all odd x and y , a contradiction since f is not constant. Suppose that $p \equiv 1 \pmod{4}$. Then it is well-known that there are positive integers a and b such that $p = a^2 + b^2$ and we have $f(a) = f(b) = 0$. But $\max(a, b) > 1$ and there is a prime q that divides it. Therefore $f(q) = 0$, a contradiction since $q < p$. Hence $p \equiv 3 \pmod{4}$ and we have that $f(x) = 0$ if x is divisible by p and $f(x) = 1$ if not. Hence $f = f_p$.

10.2 Practice Problems

Problem 10.1. Find all non-decreasing functions $f: \mathbb{Z} \rightarrow \mathbb{Z}$ that satisfy

$$f(k) + f(k+1) + \dots + f(k+n-1) = k$$

for all $k \in \mathbb{Z}$, and a fixed $n \in \mathbb{N}$.

Problem 10.2. (IMO 2012) Find all functions $f: \mathbb{Z} \rightarrow \mathbb{Z}$ such that

$$f(x)^2 + f(y)^2 + f(z)^2 = 2f(x)f(y) + 2f(y)f(z) + 2f(z)f(x)$$

for all integers x, y, z that satisfy $x + y + z = 0$.

Problem 10.3. Find all functions $f: \mathbb{N} \rightarrow [1; \infty)$ for which we have $f(2) = 4$, $f(mn) = f(m)f(n)$ and $\frac{f(n)}{n} \leq \frac{f(n+1)}{n+1}$ for all $m, n \in \mathbb{N}$.

Problem 10.4. Is there a strictly increasing function $f: \mathbb{N} \rightarrow \mathbb{N}$ such that $f(2) = 3$ and $f(mn) = f(m)f(n)$ for all $m, n \in \mathbb{N}$?

Problem 10.5. (AMM 2001) Find all completely multiplicative functions $f: \mathbb{N} \rightarrow \mathbb{C}$ such that the function

$$F(n) = \sum_{k=1}^n f(k)$$

is also completely multiplicative.

Problem 10.6. (AMM 1998) Find all functions $f: \mathbb{N}^2 \rightarrow \mathbb{N}$ that satisfy

$$f(n, n) = n, \quad f(m, n) = f(n, m) \quad \text{and} \quad \frac{f(m, n+m)}{f(m, n)} = \frac{n+m}{n}$$

for all $m, n \in \mathbb{N}$.

Problem 10.7. Let $f: \mathbb{N}_0 \rightarrow \mathbb{N}_0$ be a function such that $f(0) = 1$ and

$$f(n) = f\left(\left\lfloor \frac{n}{2} \right\rfloor\right) + f\left(\left\lfloor \frac{n}{3} \right\rfloor\right)$$

whenever $n \in \mathbb{N}$. Show that $f(n-1) < f(n)$ if and only if $n = 2^k 3^h$ for some $k, h \in \mathbb{N}_0$.

Problem 10.8. Find all functions $f: \mathbb{N} \rightarrow \mathbb{Z}$ verifying

$$f(mn) = f(m) + f(n) - f(\gcd(m, n))$$

for all $m, n \in \mathbb{N}$.

Problem 10.9. Find all functions $f: \mathbb{N} \rightarrow \mathbb{N}$ such that $m^2 + f(n)$ divides $f(m)^2 + n$ for all $m, n \in \mathbb{N}$.

Problem 10.10. Let p be a prime number. Find all functions $f: \mathbb{Z} \rightarrow \mathbb{Z}$ that satisfy:

- a) if p divides $m - n$ then $f(m) = f(n)$;
- b) $f(mn) = f(m)f(n)$

for all $m, n \in \mathbb{Z}$.

Problem 10.11. (Iran TST 2005) Find all functions $f: \mathbb{N} \rightarrow \mathbb{N}$ such that:

- 1) there exist $k \in \mathbb{N}$ and a prime p such that $f(n+p) = f(n)$ for all $n \geq k$;
- 2) if m divides n then $f(m+1)$ divides $f(n)+1$.

Problem 10.12. (IMO 2011) Let $f: \mathbb{Z} \rightarrow \mathbb{N}$ be a function such that $f(m-n)$ divides $f(m) - f(n)$ for all integers m, n . Prove that, for all integers m and n with $f(m) \leq f(n)$, the number $f(n)$ is divisible by $f(m)$.

Problem 10.13. (Iran TST 2008) Let k be a positive integer. Find all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ such that $f(m) + f(n)$ divides $(m + n)^k$ for all $m, n \in \mathbb{N}$.

Problem 10.14. (IMO Shortlist 2011) Let $n \geq 1$ be an odd integer. Determine all functions $f : \mathbb{Z} \rightarrow \mathbb{Z}$ such that $f(x) - f(y)$ divides $x^n - y^n$ for all $x, y \in \mathbb{Z}$.

Problem 10.15. (MR 2009, N. Tung) Let k be a positive integer. Find all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ such that $f(a) + f(b)$ divides $a^k + b^k$ for all $a, b \in \mathbb{N}$.

Problem 10.16. (IMO Shortlist 2004) A function $f : \mathbb{N} \rightarrow \mathbb{N}$ is such that for all $m, n \in \mathbb{N}$ the number $(m^2 + n)^2$ is divisible by $f(m)^2 + f(n)$. Prove that $f(n) = n$ for all $n \in \mathbb{N}$.

Problem 10.17. (IMO Shortlist 1996) Find a bijection $f : \mathbb{N}_0 \rightarrow \mathbb{N}_0$ that satisfies

$$f(3mn + m + n) = 4f(m)f(n) + f(m) + f(n)$$

for all $m, n \in \mathbb{N}_0$.

Problem 10.18. (IMO 2010) Determine all functions $g : \mathbb{N} \rightarrow \mathbb{N}$ such that $(g(m) + n)(m + g(n))$ is a perfect square for all $m, n \in \mathbb{N}$.

Problem 10.19. (IMO Shortlist 2008) For every $n \in \mathbb{N}$ let $\tau(n)$ denote the number of (positive) divisors of n . Find all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ with the following properties:

(i) $\tau(f(x)) = x$;

(ii) $f(xy)$ divides $(x - 1)y^{xy-1}f(x)$

for all $x, y \in \mathbb{N}$.

Problem 10.20. (IMO Shortlist 2007) Find all surjective functions $f : \mathbb{N} \mapsto \mathbb{N}$ such that for every $m, n \in \mathbb{N}$ and every prime p , the number $f(m + n)$ is divisible by p if and only if $f(m) + f(n)$ is divisible by p .

Chapter 11

Binary and Other Bases

11.1 Theory and Examples

Every man is accustomed with the decimal base. It is the natural way of representing numbers in terms of powers of ten: we can encode every positive integer as $\overline{a_n a_{n-1} \dots a_1 a_0}$, where a_i are *digits* in the set $\{0, 1, 2, \dots, 9\}$, and this number is precisely $a_n \cdot 10^n + a_{n-1} \cdot 10^{n-1} + \dots + a_1 \cdot 10 + a_0$. If we assume $a_n \neq 0$, then this representation is unique, and we can extend it by convention to negative integers and non-integer real numbers. For a mathematician, of course, there is nothing special about the number 10, and one can similarly write every positive integer uniquely as $\overline{b_n b_{n-1} \dots b_1 b_0}_{(b)}$ meaning $b_n \cdot b^n + b_{n-1} \cdot b^{n-1} + \dots + b_1 \cdot b + b_0$, where $b_i \in \{0, 1, \dots, b-1\}$, $b_n \neq 0$ and b is a positive integer greater than 1. This is called the representation of a number in base b . A common base is the binary base, for $b = 2$. It is so important because there are only two digits: 0 and 1 (*yes* and *no*), and is used a lot in computers.

The whole theory of bases is relevant to the topic of functional equations for the following reason: one can take a function which somehow depends on the expansions of a number in some base (for example the sum of digits of a number in base 10), then give you some of its properties, and ask you to find the function! The examples we will consider here are precisely of this type. Usually, it is clear what base to take: if the base $1, b, b^2, \dots$, then b should most probably appear somewhere in the condition of the problem. But if the base

is of a more complicated type, it might be a bit more difficult to understand it.

We write any positive integer in base b as a unique combination of the numbers $b^n, n \geq 0$, and we can have at most $b - 1$ repetitions of the same number ($b_i \leq b - 1$). What if we want to write every number as a unique combination of the numbers from an increasing sequence $0 < a_0 < a_1 < \dots < a_n < \dots$, and we require that each a_i can repeat at most c_i times ($c_i \in \mathbb{N}$)? What are the sequences $(a_i)_{i \in \mathbb{N}}$ and $(c_i)_{i \in \mathbb{N}}$ that allow us to do this (in other words, $(a_i)_{i \in \mathbb{N}}$ is *some kind* of a base)? There are two conditions to check: representability and uniqueness.

a) Representability. Clearly $a_1 > 1$. Therefore if we want to represent 1 in the desired form, we need $a_0 = 1$. Now, $c_0 + 1 = c_0 a_0 + 1$ cannot be represented only using terms equal to $a_0 = 1$ (because we can only have c_0 of them, and no more). Therefore we need at least one term which is a_1 or larger, which is only possible when $a_1 \leq c_0 + 1$. Similarly, $c_0 a_0 + c_1 a_1 + 1$ cannot be represented using only terms equal to a_0 or a_1 , so we need at least one term of a_2 or larger, possible only when $a_2 \leq c_0 a_0 + c_1 a_1 + 1$. Continuing analogously, we need $a_k \leq c_0 a_0 + c_1 a_1 + \dots + c_{k-1} a_{k-1} + 1$ for all $k \geq 1$. We actually claim this conditions are sufficient:

(*) The sequences of positive integers $a_0 < a_1 < \dots < a_n < \dots$ and $c_0, c_1, \dots, c_n, \dots$, allow us to represent any positive integer as a finite sum $\sum_{i=0}^m a_i b_i$ where $0 \leq b_i \leq c_i$, if and only if $a_0 = 1$ and $a_k \leq c_0 a_0 + c_1 a_1 + \dots + c_{k-1} a_{k-1} + 1$ for every $k > 1$.

We have already shown the necessity. Now assume the condition is satisfied. We shall prove by induction that any number less than or equal to $c_0 a_0 + c_1 a_1 + \dots + c_k a_k$ can be represented (not necessarily in a unique way) as $a_0 b_0 + a_1 b_1 + \dots + a_k b_k$, where $a_i \leq c_i$. Since $a_0 = 1$, the basis $k = 0$ is clear. Now assume we have proven it for $k = m$, and let $r \leq c_0 a_0 + \dots + c_{m+1} a_{m+1}$. We then try to subtract from r the largest possible number of terms equal to a_{m+1} , thus having $r = a_{m+1} b_{m+1} + r'$ where either $r' < a_{m+1}$, $b_{m+1} \leq c_{m+1}$ or $b_{m+1} = c_{m+1}$. If $r' < a_{m+1}$ then $r' \leq a_0 c_0 + \dots + a_m c_m$ from the condition, and if $b_{m+1} = c_{m+1}$ then $r' = r - c_{m+1} a_{m+1} \leq a_0 c_0 + \dots + a_m c_m$. So in any case $r' \leq a_0 c_0 + \dots + a_m c_m$ thus it can be represented as $a_0 b_0 + a_1 b_1 + \dots + a_m b_m$ by the induction hypothesis, meaning $r = a_0 b_0 + \dots + a_m b_m + a_{m+1} b_{m+1}$ as

desired.

b) *Uniqueness.* We have established a bunch of inequalities needed for representability. Now we claim that for a representation of a number in the desired form to be unique, we need them to actually become equalities.

(**) The sequences of positive integers $a_0 < a_1 < \dots < a_n < \dots$ and $c_0, c_1, \dots, c_n, \dots$, allow us to represent any positive integer uniquely as a finite sum $\sum_{i=0}^m a_i b_i$ where $0 \leq b_i \leq c_i$ if and only if $a_0 = 1$ and $a_k = c_0 a_0 + c_1 a_1 + \dots + c_{k-1} a_{k-1} + 1$ for every $k > 1$.

If $a_k < c_0 a_0 + c_1 a_1 + \dots + c_{k-1} a_{k-1} + 1$, then $a_k \leq c_0 a_0 + c_1 a_1 + \dots + c_{k-1} a_{k-1}$. So from the proof of (*), we know it can be represented as $a_0 b_0 + a_1 b_1 + \dots + a_{k-1} b_{k-1}$. But it can also be represented as a_k , so we get two distinct representations for the same number. Thus the equalities are needed. Now assume that they hold and suppose $N = \sum_{i=0}^m a_i b_i = \sum_{i=0}^m a_i b'_i$ are two distinct representations. Subtracting them one from another, we get a sum $u_0 a_0 + u_1 a_1 + \dots + u_m a_m = 0$ where not all the u_i are zero and $|u_i| \leq c_i$. We may assume $u_m \neq 0$ else ignore the zero terms, and $u_m > 0$ or else change sign. Then we get $u_m a_m = -u_{m-1} a_{m-1} - \dots - u_0 a_0 \leq c_{m-1} a_{m-1} + \dots + c_0 a_0 = u_m - 1$, and this is of course impossible for $a_m \geq 1$. Thus we get a contradiction.

The reader may check that for any integer $b \geq 2$ the sequences $a_i = b^i$ and $c_i = b - 1$ satisfy the conditions (**), and this validates their status as *bases*. However, there may be some other bases. Particularly, if we take into consideration only the condition (*), we will get a *base* in which every number can be represented, but not uniquely. The representation may be made unique by trying to expand the big terms into smaller terms as much as possible, or viceversa. A common example is the so-called Fibonacci base. Consider the Fibonacci sequence defined by $F_1 = 1, F_2 = 2, F_{n+2} = F_n + F_{n+1}$. We immediately see that $F_{n+2} \leq F_{n+1} + F_n + \dots + F_1$, and thus setting $a_i = F_{i+1}, c_i = 1$ and applying (*) we deduce that every number can be represented as a sum of distinct Fibonacci numbers, but not necessarily in a unique way (as the identities $F_{n+2} = F_{n+1} + F_n = F_{n+1} + F_{n-1} + F_{n-2}$ show).

Let us start with a functional equation whose solution uses the binary representation of integers.

Example 11.1. Find all functions $f: \mathbb{N}_0 \rightarrow \mathbb{N}_0$ such that $f(0) = 0$ and

$$f(2n+1) = f(2n) + 1 = f(n) + 1$$

for any $n \in \mathbb{N}_0$.

Solution. The statement suggests to look at the binary expansion of n . As $f(2n+1) = f(n) + 1$ and $f(2n) = f(n)$ it is straightforward to observe and check that $f(n)$ is the number of ones (or the sum of digits) of the binary representation of n .

Example 11.2. (Iberoamerican Olympiad 1989) Find all functions $f: \mathbb{N} \rightarrow \mathbb{N}$ such that $f(1) = 1$, $f(2n) = 3f(n)$ and $f(2n+1) = 3f(n) + 1$ for all $n \in \mathbb{N}$.

Solution. We guess that f somehow depends on the binary expansion of n , and the right-hand sides $3f(n)$, $3f(n) + 1$ suggest that f might have something to do with base 3 as well! Now you do not need much to guess that f is obtained by just writing n in base 2 and reading the result in base 3, i.e. if $k_0 < k_1 < \dots < k_m$, then $f(2^{k_0} + 2^{k_1} + \dots + 2^{k_m}) = 3^{k_0} + 3^{k_1} + \dots + 3^{k_m}$. Once we have guessed the solution, it is quite easy to prove that $f(n)$ is indeed n written in base 2 and read in base 3.

Example 11.3. Find all functions $f: \mathbb{N} \rightarrow \mathbb{N}$ for which $f(1) = 1$, $f(2n) < 6f(n)$ and $3f(n)f(2n+1) = f(2n)(3f(n) + 1)$ for all $n \in \mathbb{N}$.

Solution. Rewrite the main condition as

$$3f(n)(f(2n+1) - f(2n)) = f(2n).$$

Hence $f(2n+1) - f(2n) > 0$ and as $f(2n) < 6f(n)$ we deduce

$$f(2n+1) - f(2n) < 2.$$

Thus $f(2n+1) - f(2n) = 1$ which implies $f(2n) = 3f(n)$. Now it follows from Example 11.2 that the solution of the problem is the function $f(n)$ obtained by writing n in base 2 and reading the result in base 3.

Example 11.4. (IMO Shortlist 2000) The function $f: \mathbb{N}_0 \rightarrow \mathbb{N}_0$ satisfies $f(4n) = f(2n) + f(n)$, $f(4n+2) = f(4n) + 1$, $f(2n+1) = f(2n) + 1$ for all $n \in \mathbb{N}_0$. Show that the number of non-negative integers n such that $f(4n) = f(3n)$ and $n < 2^m$ is $f(2^{m+1})$.

Solution. The condition suggests us look at the binary decomposition of n . Firstly as $f(4n) = f(2n) + f(n)$ we can easily deduce that $f(2^k) = F_{k+1}$, where $(F_n)_{n \in \mathbb{N}}$ is the Fibonacci sequence. Indeed, setting $n = 0$ we get $f(0) = 0$ thus $f(1) = 1, f(2) = 1$. Now the conditions $f(4n+2) = f(4n) + 1$, $f(2n+1) = f(2n) + 1$ may suggest some sort of additivity for f , at least $f(a+b) = f(a) + f(b)$ when a do not share digits in base 2. And this is indeed the case if we look at some small particular values of f . So we conjecture this assertion, which would mean that $f(n)$ is actually n transferred from base 2 into *Fibonacci base*, i.e. $f(b_k 2^k + \dots + b_0) = b_k F_{k+1} + \dots + b_0$. This is easily accomplished by induction on n : if $n = 4k$ then $f(n) = f(2k) + f(k)$, if $n = 2k + 1$ then $f(n) = f(2k) + 1$ and if $n = 4k + 2$ then $f(n) = f(4k) + 1$ and the verification is direct.

As we found f , let us turn to the final question. It asks when $f(4n) = f(3n)$. Actually f should be some sort of increasing function, so we could suppose $f(3n) \leq f(4n)$. Indeed this holds true if we check some particular cases, with equality sometimes. Now what connects $4n$ and $3n$? The condition says us that $f(4n) = f(2n) + f(n)$ but we have $3n = 2n + n$. So we can suppose that $f(a+b) \leq f(a) + f(b)$ and look for equality cases. We work of course in binary. The addition of two binary numbers can be thought of as adding their corresponding digits pairwise, and then repeating a number of times the following operation: if we have reached a 2 in some position, replace it by a zero and add a 1 to the next position. (Note that we will never have digits greater than two if we eliminate the 2 at the highest level at each step). For example $3 + 9 = 11_2 + 1001_2 = 1012_2$ and then we remove the 2 to get 1020_2 and again to get $1100_2 = 12$ so $3 + 9 = 12$. We can extend f to sequences of 0's, 1's and 2's by setting $f(b_k, \dots, b_0) = b_k F_{k+1} + b_{k-1} F_k + \dots + b_0$. Then we can see that if S is the sequence obtained by adding a and b component-wise (as vectors), then $f(s) = f(a) + f(b)$. And we need to prove that the operation of removing a 2 does not increase f . Indeed, if we remove a 2 from position k and add a 1 to position $k+1$ the f changes by $F_{k+2} - 2F_{k+1}$. This value is never positive and is actually zero only for $k = 0$. So f indeed is not increased by this operation (which guarantees the claim that $f(a+b) \leq f(a) + f(b)$), and moreover it is not decreased by it only if the operation consists of removing the 2 at the units position. So $f(a+b) = f(a) + f(b)$ if and only if by adding

them component-wise we either reach no transfer of unity, or have only one transfer at the lowest level. Hence $f(4n) = f(3n)$ if and only if adding $2n + n$ we can reach at most a transfer at the lowest level. But a transfer at the lowest level cannot occur as the last digit of $2n$ is 0. So $f(4n) = f(3n)$ if and only if by adding $2n$ and n we have no transfer, i.e. $2n$ and n don't share an one digit in the same position. But as the digits of $2n$ are just the digits of n shifted one position, this is possible if and only if n has no two consecutive one's in its binary representation. So we need to prove that there are exactly $f(2^{m+1}) = F_{m+2}$ such numbers less than 2^m . Let we have $g(m)$ such numbers. Then $g(0) = 1, g(1) = 2$. Now note that if n is such a number and $n \geq 2^{m-1}$ then $n = 2^{m-1} + n'$ where $n' < 2^{m-2}$ (as it cannot have a one in position $m-1$ that would conflict with the leading unity), so we have $g(m-2)$ possibilities for this case. For $n < 2^{m-1}$ we have $g(m-1)$ possibilities. So $g(m) = g(m-1) + g(m-2)$ and an induction finishes the problem.

11.2 Practice Problems

Problem 11.1. Find all functions $f: \mathbb{N}_0 \rightarrow \{0, 1\}$ for which

$$f(0) = 0, f(2n) = f(n), f(2n+1) = 1 - f(2n)$$

for all $n \in \mathbb{N}_0$.

Problem 11.2. Find all functions $f: \mathbb{N} \rightarrow \mathbb{N}$ for which

$$f(1) = 1, f(2n) = 2f(n) - 1, f(2n+1) = 2f(n) + 1$$

for all $n \in \mathbb{N}$.

Problem 11.3. Find all functions $f: \mathbb{N} \rightarrow \mathbb{R}$ for which

$$f(2n+1) = f(2n) + 1 = 3f(n) + 1$$

for all $n \in \mathbb{N}$.

Problem 11.4. (IMO 1988) Find all functions $f: \mathbb{N} \rightarrow \mathbb{N}$ that satisfy

$$f(1) = 1, f(3) = 3 \text{ and } f(2n) = f(n),$$

$$f(4n+1) = 2f(2n+1) - f(n), \quad f(4n+3) = 3f(2n+1) - 2f(n)$$

for all $n \in \mathbb{N}$.

Problem 11.5. Find all strictly increasing functions $f : \mathbb{N} \rightarrow \mathbb{N}$ such that $f(f(n)) = 3n$ for all $n \in \mathbb{N}$.

Problem 11.6. Let a and b be nonnegative integers with $a > b$. Prove that there is a function $f : \mathbb{N} \rightarrow \mathbb{N}$, such that $f(f(n)) = an + b$ for all $n \in \mathbb{N}$.

Problem 11.7. (IMO 1978) Let $(f(n))$ be a strictly increasing sequence of positive integers: $0 < f(1) < f(2) < f(3) < \dots$. Of the positive integers not belonging to the sequence, the n th in order of magnitude is $f(f(n)) + 1$. Determine $f(240)$.

Problem 11.8. (IMO 1993) Determine whether there exists a strictly increasing function $f : \mathbb{N} \rightarrow \mathbb{N}$ such that $f(1) = 2$ and $f(f(n)) = f(n) + n$ for all $n \in \mathbb{N}$.

Problem 11.9. The function f is defined on the set of rational numbers in the interval $[0, 1]$ as follows:

$$f(0) = 0, f(1) = 1, f(x) = \begin{cases} \frac{f(2x)}{4} & \text{if } x \in (0, \frac{1}{2}) \\ \frac{3}{4} + \frac{f(2x-1)}{4} & \text{if } x \in [\frac{1}{2}, 1) \end{cases}$$

Find $f(x)$ in terms of the binary representation of x .

Problem 11.10. (IMO Shortlist 1983) Let $f : [0, 1] \rightarrow \mathbb{R}$ be a continuous function that satisfies

$$f(x) = \begin{cases} bf(2x) & \text{if } x \in [0, \frac{1}{2}] \\ b + (1-b)f(2x-1) & \text{if } x \in [\frac{1}{2}, 1] \end{cases},$$

where $b = \frac{1+c}{2+c}$, $c > 0$. Show that $0 < f(x) - x < c$ for all $x \in (0, 1)$.

Chapter 12

Geometric Functional Equations

12.1 Fundamental Theorem of Affine Geometry

By a *geometric* functional equation we mean a problem asking for a description of functions (mappings) defined on the plane (space) and satisfying some geometric conditions. A typical example is the Fundamental theorem of affine geometry stating that any bijection of the plane that maps a straight line to a straight line is an affine transformation. In this section we present an elementary proof of this theorem (proposed first by Mikusiński [2], [14]) which uses the additive Cauchy equation. Then in Section 12.2 we consider various examples of geometric functional equations.

Denote by $\mathbb{R}^2$ the plane together with a coordinate system Oxy . Then a transformation $A : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is called affine if it has the form

$$A : (x, y) \mapsto (a_1x + b_1y + c_1, a_2x + b_2y + c_2),$$

where $a_i, b_i, c_i, i = 1, 2$ are real constants. Note that A is the composition of the translation

$$T_A : (x, y) \mapsto (x + c_1, y + c_2)$$

and the linear transformation

$$L_A : (x, y) \mapsto (a_1x + b_1y, a_2x + b_2y).$$

Hence any affine transformation preserves the collinearity of points, i.e. it maps any straight line to a straight line. It is easy to check that an affine transformation $A : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is a bijection iff the linear transformation L_A is so, i.e. when $a_1b_2 - a_2b_1 \neq 0$. In this case the converse transformation A^{-1} is also an affine transformation. Next we shall use the following fact whose proof we leave as an exercise to the reader.

Fact. For any two (non-degenerate) triangles $M_1M_2M_3$ and $N_1N_2N_3$ in the plane there exists a unique bijective affine transformation $A : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ such that $A(M_i) = N_i$, $1 \leq i \leq 3$.

Hint. Reduce the problem to the case when $M_1 = N_1 = O$ by using appropriate translations.

Now we are ready to prove the Fundamental theorem of affine geometry.

Theorem 12.1. Any bijection of the plane that maps a straight line to a straight line is an affine transformation.

Proof. Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a bijection that maps any straight line to a straight line. Since the points $O = (0, 0)$, $O_1 = (1, 0)$ and $O_2 = (0, 1)$ are not collinear then the points $T(O)$, $T(O_1)$ and $T(O_2)$ are not collinear too. Hence it follows from the Exercise that there is a bijective affine transformation A such that $A(T(O)) = O$, $A(T(O_1)) = O_1$ and $A(T(O_2)) = O_2$. Set $S = A \circ T$ and $S(M) = S(x, y) = (F(x, y), G(x, y))$ for $M = (x, y) \in \mathbb{R}^2$. Then $S : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is a bijection that maps any straight line to a straight line and

$$F(0, 0) = G(0, 0) = 0, F(1, 0) = G(0, 1) = 1, F(0, 1) = G(1, 0) = 0.$$

For every $x \in \mathbb{R}$ the points $(0, 0)$, $(1, 0)$ and $(x, 0)$ are collinear, so the points $S(0, 0) = (0, 0)$, $S(1, 0) = (1, 0)$ and $S(x, 0) = (F(x, 0), G(x, 0))$ are collinear too. Hence $G(x, 0) = 0$. Analogously, considering the points $(0, 0)$, $(0, 1)$ and $(0, y)$ we get that $F(0, y) = 0$.

Since S is a bijection it maps parallel lines into parallel lines. In particular, S maps the vertices of any parallelogram into the vertices of a parallelogram.

Consider an arbitrary parallelogram with vertices

$$O = (0, 0), M_1 = (x_1, y_1), M_2 = (x_2, y_2) \text{ and } M_3 = (x_1 + x_2, y_1 + y_2).$$

Set

$$N_1 = S(M_1) = (F(x_1, y_1), G(x_1, y_1)), N_2 = S(M_2) = (F(x_2, y_2), G(x_2, y_2))$$

and

$$N_3 = S(M_3) = (F(x_1 + x_2, y_1 + y_2), G(x_1 + x_2, y_1 + y_2)).$$

Since $S(O) = O = (0, 0)$ it follows that $ON_1N_3N_2$ is a parallelogram, i.e. $\overrightarrow{ON_1} + \overrightarrow{ON_2} = \overrightarrow{ON_3}$. Hence

$$F(x_1, y_1) + F(x_2, y_2) = F(x_1 + x_2, y_1 + y_2),$$

$$G(x_1, y_1) + G(x_2, y_2) = G(x_1 + x_2, y_1 + y_2)$$

for all $x_1, x_2, y_1, y_2 \in \mathbb{R}$. Setting $x_1 = x, y_1 = 0, x_2 = 0, y_2 = y$ in the above identities we get that

$$F(x, y) = F(x, 0) + F(0, y) = F(x, 0),$$

$$G(x, y) = G(x, 0) + G(0, y) = G(0, y)$$

for all $x, y \in \mathbb{R}$. Now setting $f(x) = F(x, 0), g(y) = G(0, y)$ and using again the above identities we see that $S(x, y) = (f(x), g(y))$, where $f, g : \mathbb{R} \rightarrow \mathbb{R}$ are additive functions with $f(1) = g(1) = 1$.

Now let x, y be arbitrary nonzero real numbers. Consider the points

$$M = (1, 0), N = (x, 0), P = (0, y) \text{ and } Q = (0, xy).$$

Then

$$M_1 = S(M) = (1, 0), N_1 = S(N) = (f(x), 0),$$

$$P_1 = S(P) = (0, g(y)) \text{ and } Q_1 = S(Q) = (0, g(xy)).$$

Since the lines MP and NQ are parallel it follows that the lines M_1P_1 and N_1Q_1 are parallel too. Hence $g(xy) = f(x)g(y)$. Setting $y = 1$ we get that $g(x) = f(x)$ for $x \neq 0$. Since f and g are additive functions we have that

$f(0) = g(0) = 0$, i.e. $g(x) = f(x)$ for any $x \in \mathbb{R}$. Hence $f(xy) = f(x)f(y)$ for all $x, y \in \mathbb{R}$ and Corollary 1.1 shows that $f(x) = x$ for all $x \in \mathbb{R}$ since $f(1) = 1$. Therefore $A \circ T = S$ is the identity transformation and we conclude that $T = A^{-1}$ is an affine transformation. $\square$

Note that Theorem 12.1 has various generalizations. For example, its conclusion is true even for bijections of the plane preserving collinearity of the points lying on four lines, or on three lines and a point.

As noted in [2] the approach of Mikusiński to prove Theorem 12.1 led to considering the so-called *conditional functional equations*. Here is an example, known as Mikusiński's equation.

Example 12.1. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function such that

$$f(x + y) = f(x) + f(y)$$

for all $x, y \in \mathbb{R}$ with $f(x + y) \neq 0$. Prove that f is an additive function.

Solution. We shall divide the solution into several steps. Set

$$\text{Ker } f = \{x \in \mathbb{R} \mid f(x) = 0\}.$$

Step 1. $\text{Ker } f$ is an additive subgroup of $\mathbb{R}$.

We have to prove that if $x, y \in \text{Ker } f$ then $x - y \in \text{Ker } f$. To do this we first note that if $x, y \in \text{Ker } f$ then $x + y \in \text{Ker } f$ since otherwise $0 = f(x) + f(y) = f(x + y) \neq 0$, a contradiction. Hence it is enough to prove that if $x \in \text{Ker } f$ then $-x \in \text{Ker } f$. Suppose the contrary, i.e. $-x \notin \text{Ker } f$. Then

$$f(-x) = f(-2x + x) = f(-2x) + f(x) = f(-2x) = f(-x - x) = 2f(-x).$$

Hence $f(-x) = 0$, a contradiction.

Step 2. $f(2x) = 2f(x)$ for any $x \in \mathbb{R}$.

If $x \in \text{Ker } f$ then it follows from Step 1 that $2x \in \text{Ker } f$ and $f(2x) = 2f(x) = 0$. If $x, 2x \notin \text{Ker } f$ then

$$f(2x) = f(x + x) = f(x) + f(x) = 2f(x).$$

Hence it remains to prove that the case $f(2x) = 0$, $f(x) \neq 0$ is impossible. Suppose that for some $x_0 \in \mathbb{R}$ we have $f(2x_0) = 0$ but $f(x_0) \neq 0$. Let $x \notin \text{Ker } f$ and $f(x) \neq f(x_0)$. Then

$$f(x) = f(x - x_0 + x_0) = f(x - x_0) + f(x_0). \quad (1)$$

Hence $f(x - x_0) \neq 0$ and therefore

$$f(x - x_0) = f(x - 2x_0 + x_0) = f(x - 2x_0) + f(x_0). \quad (2)$$

On the other hand

$$f(x) = f(x - 2x_0 + 2x_0) = f(x - 2x_0) + f(2x_0) = f(x - 2x_0)$$

and it follows from (1) and (2) that $f(x_0) = 0$, a contradiction.

Hence $f(x) = f(x_0)$ for any $x \notin \text{Ker } f$.

But $f(x_0) = f(x) = f\left(\frac{x}{2}\right) + f\left(\frac{x}{2}\right) = 2f\left(\frac{x}{2}\right) = 2f(x_0)$, a contradiction since $f(x_0) \neq 0$.

Step 3. $f(-x) = -f(x)$ for any $x \in \mathbb{R}$.

If $x \in \text{Ker } f$ then $-x \in \text{Ker } f$ (Step 1) and $f(-x) = -f(x) = 0$. If $x \notin \text{Ker } f$ we have

$$f(x) = f(2x - x) = f(2x) + f(-x) = 2f(x) + f(-x),$$

i.e. $f(-x) = -f(x)$.

Now we are ready to prove that the function f is additive. If $x, y \in \text{Ker } f$ then $x + y \in \text{Ker } f$ (Step 1) and $f(x) + f(y) = f(x + y) = 0$.

Suppose now that $y \notin \text{Ker } f$. Then

$$f(y) = f(-x + y + x) = f(-x) + f(y + x) = -f(x) + f(y + x),$$

i.e. $f(x) + f(y) = f(x + y)$. The case $x \notin \text{Ker } f$ is similar and the additivity of f is proved.

12.2 Examples

In this section we consider some typical examples of geometric functional equations.

Example 12.2. Let f be a real function on the plane such that

$$f(A) + f(B) + f(C) + f(D) = 0$$

for any square $ABCD$. Prove that f is identically zero.

Solution. Given a point P in the plane consider a square $ABCD$ with center P . Denote by E, F, G, H the mid points of the segments AB, BC, CD, DA . Then f satisfies the identities

$$\begin{aligned} f(A) + f(E) + f(P) + f(H) &= 0 \\ f(B) + f(F) + f(P) + f(E) &= 0 \\ f(C) + f(G) + f(P) + f(F) &= 0 \\ f(D) + f(H) + f(P) + f(G) &= 0 \end{aligned}$$

If we add up these identities we get

$$\begin{aligned} 0 &= 4f(P) + (f(A) + f(B) + f(C) + f(D)) \\ &\quad + 2(f(E) + f(F) + f(G) + f(H)) = 4f(P), \end{aligned}$$

i.e. $f(P) = 0$.

Example 12.3. A bijection of the plane maps circles to circles. Does it map lines to lines?

Solution. Yes! Let f be a bijection of the plane that maps circles to circles. For any point X in the plane we let $f(X) = X'$. We first note that if the images A', B', C' of three points A, B, C are collinear then A, B, C are also collinear. Indeed, suppose that A, B, C are not collinear. Then they lie on a circle. Their images also lie on a circle and are not collinear, a contradiction. Now let A, B, C be three points on a line g . Consider the circles c_1 and c_2 with diameters AB and AC . Their images c'_1, c'_2 are also circles which touch

at A' . Note that $A'B'$ is not tangent to c'_1 because it contains B' , hence it is not tangent to c'_2 either because the tangents to c'_1, c'_2 at A' coincide. Thus $A'B'$ has a common point with c'_2 which is different from A' . Its inverse image must lie on c_2 and it follows from the above remark that it coincides with C . Thus C' lies on the line $A'B'$.

Example 12.4. (USAMO 2001) Every point in the plane is assigned a real number, so that for any three points which are not collinear, the number assigned to the incenter is the mean of the numbers assigned to the three points. Show that the same number is assigned to every point.

Solution. Let $f(P)$ be the number assigned to a point P in the plane. Let P, Q be arbitrary points and R , a point on the segment PQ . Its position will be determined later. Take AA' perpendicular to PQ with R its midpoint. Take a rectangle $ACFA'$ on the opposite side of AA' to P and $AA' : AC = 3 : 2$. Let B be the midpoint of AC and B' the midpoint of $A'F$. Take points D, E on CF , so that $AB = BC = CD = DE = EF = FB' = B'A'$. Finally take X on the ray RP . We will choose the lengths XP and RA so that P is the incenter of $\triangle XAA'$ and Q is the incenter of $\triangle XBB'$. Suppose we can do that. Then the incircles of $\triangle ACD$ and $\triangle BCE$ coincide, so $f(A) + f(D) = f(B) + f(E)$. Hence $f(D) - f(E) = f(B) - f(A)$. Similarly, the incircles of $\triangle A'EF$ and $\triangle B'DF$ coincide, so $f(D) - f(E) = f(A') - f(B')$ and we get

$$f(P) = \frac{f(A) + f(A') + f(X)}{3} = \frac{f(B) + f(B') + f(X)}{3} = f(Q).$$

That is all we need, since it shows that the same number is assigned to two arbitrary points. So it remains to show that XP and RA can be chosen as claimed. The incenter of an isosceles triangle with base $2a$ and height h is $\frac{ah}{a + \sqrt{a^2 + h^2}}$ above the base (if the distance is x , then by similar triangles $\frac{x}{h-x} = \frac{a}{\sqrt{a^2 + h^2}}$). So if we take $XR : RA = 3 : 2$ and $RA : PR = \frac{1 + \sqrt{5}}{2}$, then P is the incenter of $\triangle XAA'$.

Example 12.5. (AOPS). Let P and D be the sets of the points and of the lines of the (Euclidean) plane, respectively. Find, with proof, whether there

exists a bijective function $f : P \rightarrow D$ such that for arbitrary three collinear points A, B, C the lines $f(A), f(B), f(C)$ are either parallel or concurrent.

Solution. We shall prove that such a function does not exist. Let us suppose the contrary, i.e. that such a bijective function $f : P \rightarrow D$ exists. First, we shall prove the following preliminary result:

Lemma. If the lines d_1, d_2, d_3 have a common point M and $d_i = f(B_i)$, $B_i \in P$, $i = 1, 2, 3$, then the points B_1, B_2, B_3 are collinear.

Proof of lemma. Indeed, if $B_1B_2B_3$ is a triangle, then the image of any point C , $C \in B_iB_j$, $i \neq j$, is a line $d = f(C)$ through M . It is clear that any point C of the plane is collinear with two different points C_1, C_2 belonging to the sides B_iB_j of the triangle. Then, for any point C of the plane its image must be a line $d = f(C)$ through M . This contradicts the assumption that the map f is onto. This proves the Lemma. $\square$

In the same way it can be proved that if the lines d_1, d_2, d_3 are parallel and $d_i = f(B_i)$, then the points B_1, B_2, B_3 are collinear. From the above two results it follows that f defines a bijection between the points of a line d and the lines of a pencil of lines through a point or a pencil of parallel lines (hence, a pencil of lines).

Let us consider two parallel lines d, d' such that the images of the points $B \in d$ are the lines of a pencil $\mathcal{P}$ of parallel lines. The images of the points $B' \in d'$ are the lines of a pencil $\mathcal{P}'$.

If $\mathcal{P}'$ is a pencil of lines through a point M then it contains a line p belonging to the pencil $\mathcal{P}$, $p \in \mathcal{P}$. It follows that p is the image of a point belonging to d and of another point belonging to d' ; this is a contradiction. Therefore $\mathcal{P}'$ is also a pencil of parallel lines.

Now, let d'' be a line which intersects d and such that the images of the points belonging to d'' are a pencil $\mathcal{P}''$ of lines through a point. For any line δ , $\delta \parallel d''$, the images of the points of δ are a pencil Π of lines through a point. The pencils $\mathcal{P}''$ and Π have a common line l and hence $l = f(B)$ where $B \in d'' \cap \delta$; this is a contradiction. Thus, it follows that a function with the property given in the hypothesis does not exist.

Example 12.6. (Romania TST 2000) Let S be the set of interior points of a sphere and C be the set of interior points of a circle in the space. Does there

exist a function $f : S \rightarrow C$ such that $|AB| \leq |f(A)f(B)|$, for all $A, B \in S$. (Here $|XY|$ is the Euclidean distance between the points X and Y .)

Solution. We shall show that such a function does not exist. To do this assume that a function $f : S \rightarrow C$ has the desired property and for any point A in the space we let $f(A) = A'$. Inscribe a cube in the sphere and suppose without loss of generality that its edge has length 1. For any positive integer n partition the cube into n^3 small cubes by planes parallel to its faces. The set of vertices of these cubes has $(n+1)^3$ points denoted by $A_1, A_2, \dots, A_{(n+1)^3}$. For all $i \neq j$ we have $|A_i A_j| \geq \frac{1}{n}$ and therefore $|A'_i A'_j| \geq \frac{1}{n}$. Hence the discs D_i centered at A'_i and with radius $\frac{1}{2n}$ are disjoint and contained in the disc

C' , the image of C under dilation with ratio $\frac{r + \frac{1}{n}}{r}$, where r is the radius of C . Then it follows that

$$\sum_{i=1}^{(n+1)^3} [D_i] = (n+1)^3 \frac{\pi}{4n^2} \leq [C'] = \pi \left(r + \frac{1}{2n}\right)^2$$

for all $n \in \mathbb{N}$, a contradiction.

12.3 Practice Problems

Problem 12.1. Find all continuous functions $f : \mathbb{R}^n \rightarrow \mathbb{R}$ that satisfy the identity

$$f(x_1, x_2, \dots, x_n) + f(y_1, y_2, \dots, y_n) = f(x_1 + y_1, \dots, x_n + y_n)$$

for all $(x_1, x_2, \dots, x_n), (y_1, y_2, \dots, y_n) \in \mathbb{R}^n$.

Problem 12.2. Let $n \geq 3$ be a positive integer. Find all continuous functions $f : [0; 1] \rightarrow \mathbb{R}$ for which $f(x_1) + f(x_2) + \dots + f(x_n) = 1$ whenever $x_1, x_2, \dots, x_n \in [0; 1]$ and $x_1 + x_2 + \dots + x_n = 1$.

Problem 12.3. Prove that $f : \mathbb{R} \rightarrow \mathbb{R}$ is an additive function if and only if it satisfies one of the following conditions:

- (a) $f(x+y) - f(x-y) = 2f(y)$;
- (b) $f(xy+x+y) = f(xy) + f(x) + f(y)$;
- (c) $f(xy+x+y) + f(xy-x-y) = 2(f(x) + f(y))$;
- (d) $f(x+y)^2 = (f(x) + f(y))^2$,

for all $x, y \in \mathbb{R}$.

Problem 12.4. (German TST 2009) Find all functions $f : \mathbb{R} \mapsto \mathbb{R}$ such that if $x^3 + f(y) \cdot x + f(z) = 0$, then $f(x)^3 + y \cdot f(x) + z = 0$.

Problem 12.5. (CTSJ Competition, Romania 2008) Let O be a fixed point in the plane $\mathbb{R}^2$ and $f : \mathbb{R}^2 \setminus \{O\} \rightarrow \mathbb{R}$ be a function such that

$$f(A) - f(B) + f(C) - f(D) = 0$$

for any four distinct points $A, B, C, D \in \mathbb{R}^2 \setminus \{O\}$ with $\triangle AOB \sim \triangle COD$. Prove that f is a constant.

Problem 12.6. Let f be a real function on the plane such that

$$f(A_1) + f(A_2) + \cdots + f(A_n) = 0$$

for each regular n -gon $A_1 A_2 \dots A_n$. Prove that f is identically zero.

Problem 12.7. (Romania TST 1996) Let $n \geq 3$ be an integer and $p \geq 2n - 3$ be a prime. Let M be a set of n points in the plane no three of which are collinear and $f : M \rightarrow \{0, 1, \dots, p-1\}$ be a function such that:

- (i) f is zero for only one point of M ;
- (ii) if $A, B, C \in M$ are distinct points and (ABC) is the circumcircle of triangle ABC , then

$$\sum_{P \in M \cap (ABC)} f(P) \equiv 0 \pmod{p}.$$

Show that all points of M lie on the same circle.

Problem 12.8. (Romania TST 1997) Let S be a set of $n \geq 4$ points in the plane, not all lying on a circle, and no three of them collinear. Let $f : S \rightarrow \mathbb{R}$ be a function such that for each circle C containing three or more points from S we have

$$\sum_{P \in C \cap S} f(P) = 0.$$

Prove that f is identically 0.

Problem 12.9. Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a map such that if $|XY| = 1$, then $|f(X)f(Y)| = 1$. Prove that f is an isometry, i.e. $|f(X)f(Y)| = |XY|$ for all $X, Y \in \mathbb{R}^2$.

Chapter 13

Approximating by Linear Functions

13.1 Theory and Examples

There are some functional equations on $\mathbb{N}$ that seem untouchable. Sometimes however one can prove that their solutions are unique and in this case guessing the function would be very helpful. Very often the solutions are linear, thus it is natural to try $f(x) = cx$, where c is an integer. But sometimes c can be rational or even irrational, and we can have solutions of the form $f(x) = \lfloor cx \rfloor$. To overpass this difficulty, in this section we use relations $f(x) \sim cx$ meaning that $f(x) - cx$ is "small" in absolute value compared to x , where "small" may have different interpretations.

The most natural relation $f(x) \sim cx$ of the above type is when the function $|f(x) - cx|$ is bounded. Another interpretation of $f(x) \sim cx$ when $c \neq 0$ could be if $\left| \frac{f(x)}{cx} \right|$ tends to 1 as x goes to infinity. Then $\sqrt{x^2 + 1} \sim x$ according to both definitions, while $2x + \log(x) \sim 2x$ according to the second definition because

$$\lim_{x \rightarrow \infty} \frac{2x + \log(x)}{2x} = 1 + \lim_{x \rightarrow \infty} \frac{\log(x)}{2x} = 1$$

but not according to the first.

No matter which definition one chooses, one can perform certain operations with this relation:

- if $f_1(x) \sim c_1x$ and $f_2(x) \sim c_2x$ then $f_1(x) + f_2(x) \sim (c_1 + c_2)x$ (unless $c_1 + c_2 = 0$ for the second relation);
- if $f(x) \sim cx$ then $af(x) \sim acx$.

More generally we can write $f(x) \sim g(x)$ if either $|f(x) - g(x)|$ is bounded or $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = 1$. If $|f_1 - g_1|$ is bounded and $|f_2 - g_2|$ is bounded then

so is $|(f_1 + f_2) - (g_1 + g_2)|$, and if $\frac{f_1}{g_1} \rightarrow 1$, $\frac{f_2}{g_2} \rightarrow 1$ then $\frac{f_1 f_2}{g_1 g_2} \rightarrow 1$. This

shows that approximations can be added and multiplied. However one has to be somewhat careful about that. For example, $|f_1 - g_1|$ and $|f_2 - g_2|$ being bounded does not imply $|f_1 f_2 - g_1 g_2|$ being bounded. Also $\frac{f_1}{g_1} \rightarrow 1$ and $\frac{f_2}{g_2} \rightarrow 1$

does not necessarily imply that $\frac{f_1 + f_2}{g_1 + g_2} \rightarrow 1$ - but this is true in the case when

$g_1, g_2 > 0$ because then $\frac{f_1 + f_2}{g_1 + g_2}$ is between $\frac{f_1}{g_1}$ and $\frac{f_2}{g_2}$.

Because one can perform certain operations with estimates, one can guess c from the given equation and then look at some initial case to guess the exact formula. More precisely, if we expect f to be close to a linear function cx for some c , then plugging in cx instead of f in the functional equation should give something close to a true identity. These considerations are best understood through examples.

Example 13.1. Find all increasing functions $f: \mathbb{N} \rightarrow \mathbb{N}$ such that the only positive integers which are not in the image of f are those of form $f(n) + f(n+1)$, $n \in \mathbb{N}$.

Solution. First, let us assume $f(x) \sim cx$ - meaning for example that

$$|f(x) - cx| < A \text{ for some } A > 0.$$

Let us compute the effective value of c . If $f(n) = m$ then there are exactly $m - n$ natural numbers up to m that are not values of f . Therefore we conclude that they are exactly $f(1) + f(2), \dots, f(m-n) + f(m-n+1)$. Hence $f(m-n) + f(m-n+1) < m < f(m-n+1) + f(m-n+2)$. Now as $f(x) \sim cx$

we conclude $m \sim cn$. So we get $2c(m - n) \sim m$ or $2c(c - 1)n \sim cn$ for all n which means $2c - 2 = 1$, so $c = \frac{3}{2}$. Hence we can make the assumption that $f(x) = \left\lfloor \frac{3}{2}x + a \right\rfloor$ for some a . Let us search for a . Clearly $f(1) = 1, f(2) = 2$ as $1, 2$ must necessarily belong to $\text{Im}f$. Then 3 does not belong to $\text{Im}f$ hence $f(3) \geq 4$ so $f(2) + f(3) \geq 6$. Thus 4 belongs to $\text{Im}f$ and $f(3) = 4$. We continue to $f(4) = 5, f(5) = 7$ and so on. So $\left\lfloor \frac{3}{2} + a \right\rfloor = 1, \lfloor 3 + a \rfloor = 2$ which implies $a \in \left[-\frac{1}{2}; 0\right)$. Note that for all a, b in this interval, $\left\lfloor \frac{3}{2}x + a \right\rfloor = \left\lfloor \frac{3}{2}x + b \right\rfloor$. So we can assume $a = -\frac{1}{2}$ and infer $f(n) = \left\lfloor \frac{3n - 1}{2} \right\rfloor$. To prove it, we wish to show first that $\left\lfloor \frac{3n - 1}{2} \right\rfloor$ satisfies the conditions. Indeed,

$$\left\lfloor \frac{3n - 1}{2} \right\rfloor + \left\lfloor \frac{3(n + 1) - 1}{2} \right\rfloor = \left\lfloor \frac{3n - 1}{2} \right\rfloor + 1 + \left\lfloor \frac{3n}{2} \right\rfloor = 3n + 1$$

by Hermite's Identity, and we need to prove that the only numbers that are not of form $\left\lfloor \frac{3n - 1}{2} \right\rfloor$ are those that give remainder 1 after division by 3. Indeed, if $n = 2k$ then $\left\lfloor \frac{3n - 1}{2} \right\rfloor = 3k - 1$ and if $n = 2k + 1$ then $\left\lfloor \frac{3n - 1}{2} \right\rfloor = 3k$, and the conclusion is straightforward.

The fact that $f(n) = \left\lfloor \frac{3n - 1}{2} \right\rfloor$ stems now from the inductive assertion that f is unique. Indeed, if we have determined $f(1), f(2), \dots, f(n - 1)$ then we have determined all $f(1) + f(2), f(2) + f(3), \dots, f(n - 2) + f(n - 1)$. Then $f(n)$ must be the least number which is greater than $f(n - 1)$ and not among $f(1) + f(2), f(2) + f(3), \dots, f(n - 2) + f(n - 1)$. This is because if m is this number and $f(n) \neq m$ then $f(n) > m$ and then m does not belong neither to $\text{Im}(f)$ nor to the set $\{f(n) + f(n + 1) | n \in \mathbb{N}\}$, a contradiction. Hence $f(n)$ is computed uniquely from the previous values of f and thus f is unique.

Example 13.2. Find all functions $f: \mathbb{N} \rightarrow \mathbb{N}$ such that

$$f(f(n)) + f(n + 1) = n + 2$$

for all $n \in \mathbb{N}$.

Solution. First, note that $f(n+1) \leq n+1$ and $f(f(n)) \leq n+1$. Hence $f(k) \leq k$ for $k > 1$. If we try to set $f(x) \sim cx$, then we get $f(f(n)) \sim c^2n$.

Thus $c^2n + cn \sim n$, so $c^2 + c = 1$ and hence $c = \frac{\sqrt{5}-1}{2}$. Let us compute now $f(1)$. Assume that $f(1) = a$. Then $f(a) + f(2) = 3$. We have two cases:

a) $f(2) = 1, f(a) = 2$. Then $a \geq 3$. Set $n = a-1$ to get $f(f(a-1)) + 2 = a$ and so $f(f(a-1)) = a-2$. This is possible only when $f(a-1) = a-2, f(a-2) = a-2$. Then set $n = a-2$ to get $a-2 + a-2 = a$ so $a = 4$. But this contradicts the fact that $f(2) = 1$ as we got $f(a-2) = 2$. So this case is impossible.

b) $f(2) = 2, f(a) = 1$. We claim that $a = 1$. Indeed, assume for sake of contradiction that $a \geq 3$. Set $n = a-1$. Then $f(f(a-1)) + a = a+1$ and so $f(f(a-1)) = 1$. Let $f(a-1) = b$. Then $f(b) = 1$ and so $b \geq 3$. Set $n = b-1$ to get $f(f(b-1)) = b$. As $f(k) \leq k$ for $k \geq 2$ this is possible only when $f(b-1) = 1$ so $b = a$. But then $f(a-1) = a$ which is impossible. So $f(1) = 1$.

Thus we can conclude that $f(k) \leq k$ for any k and we can determine $f(n)$ by strong induction on n : if we have found $f(1), f(2), \dots, f(k)$, then by setting $n = k+1$ in the given relation we compute $f(k+1)$. So f is uniquely determined and it remains to find an example. As $f(1) = 1, f(2) = 2$ we get $f(3) = 2$. Then $f(4) = 3, f(5) = 4, f(6) = 4$. We may conjecture that

$$f(x) = \lfloor cx \rfloor + 1, \quad c = \frac{\sqrt{5}-1}{2},$$

as this holds for $x = 1, 2, 3, 4, 5, 6$. Indeed, we prove that $f(x) = \lfloor cx \rfloor + 1$ satisfies the condition. Since

$$f(f(n)) + f(n+1) = \lfloor c(\lfloor cn \rfloor + 1) \rfloor + 1 + \lfloor cn + c \rfloor + 1,$$

we need to show that

$$\lfloor c(\lfloor cn \rfloor + 1) \rfloor + \lfloor cn + c \rfloor = n.$$

We prove this by induction on n by showing that either

$$\lfloor c(n+1) + c \rfloor = \lfloor cn + c \rfloor + 1$$

and

$$\lfloor c\lfloor c(n+1) \rfloor + c \rfloor = \lfloor c\lfloor cn \rfloor + c \rfloor$$

or

$$\lfloor c(n+1) + c \rfloor = \lfloor cn + c \rfloor$$

and

$$\lfloor c\lfloor c(n+1) \rfloor + c \rfloor = \lfloor c\lfloor cn \rfloor + c \rfloor + 1.$$

Indeed, set $x = cn$. Then

$$\lfloor c(n+1) + c \rfloor - \lfloor cn + c \rfloor = \lfloor x + 2c \rfloor - \lfloor x + c \rfloor.$$

It equals 0 when $\{x + c\} < 1 - c$ and 1 otherwise.

Thus it is 0 when $1 - c < \{x\} < 2 - 2c$ and 1 otherwise. On the other hand,

$$\lfloor c\lfloor c(n+1) \rfloor + c \rfloor - \lfloor c\lfloor cn \rfloor + c \rfloor = \lfloor c\lfloor x + c \rfloor + c \rfloor - \lfloor c\lfloor x \rfloor + c \rfloor.$$

Let $\{x\} = t$. If $t < 1 - c$, then $t < c$

$$\lfloor c\lfloor x + c \rfloor + c \rfloor - \lfloor c\lfloor x \rfloor + c \rfloor = \lfloor c\lfloor x \rfloor + c \rfloor - \lfloor c\lfloor x \rfloor + c \rfloor = 0.$$

If $2 - 2c > t > 1 - c$, then

$$\begin{aligned} \lfloor c\lfloor x + c \rfloor + c \rfloor - \lfloor c\lfloor x \rfloor + c \rfloor &= \lfloor c(x - t + 1) + c \rfloor - \lfloor c(x - t) + c \rfloor \\ &= \lfloor cx + 2c - ct \rfloor - \lfloor cx + c - ct \rfloor. \end{aligned}$$

Now $cx = c^2n = n - cn = n - x$ and we substitute to get

$$\lfloor n - x + 2c - ct \rfloor - \lfloor n - x + c - ct \rfloor = \lfloor 2c - ct - x \rfloor - \lfloor c - ct - x \rfloor.$$

As $\{x\} = t$, it equals

$$\lfloor 2c - (c+1)t \rfloor - \lfloor c - (c+1)t \rfloor.$$

Now $t \in (1 - c; 2 - 2c)$ and so $2c - (c+1)t \in (2c - 2(1 - c^2); 2c - (1 - c^2)) = (0; c)$ while $c - (c+1)t \in (-c; 0)$ and the difference is 1. Finally, if $t > 2 - 2c$, then we get $c - 1 < 2c - (c+1)t < 0$ and $-1 < c - (c+1)t < -c$, and the difference is 0. The condition we seek therefore holds and $f(x) = \lfloor cx \rfloor + 1$.

We end with an equation whose solution is not an approximation of cx but the method is the same. This time the answer is an approximation of $c\sqrt{x}$. This is to show that the main idea of this chapter is not necessarily limited to approximating by linear functions.

Example 13.3. Find all non-decreasing functions $f: \mathbb{N} \rightarrow \mathbb{N}$ such that:

- i) $f(1) = 1$;
 - ii) $f(n + f(n)) = f(n) + 1$;
 - iii) $f(m)f(n) \leq f(2mn + m + n)$
- for all $m, n \in \mathbb{N}$.

Solution. One begins by guessing the solution. Since

$$f(n + f(n) + 1) = f(n) + 1$$

this cannot be satisfied by a linear equation because it grows slower, so we can try $f(x) = cx^\alpha$.

Note that $c(x + cx^\alpha)^\alpha$ has Taylor expansion $cx^\alpha + \alpha c^2 x^{2\alpha-1} + \dots$ from where one immediately tries $\alpha = \frac{1}{2}, c = \sqrt{2}$ thus we try $f(n) \sim \sqrt{2n}$.

The last condition gives an even better approximation - note that the function $f(x) = 2x + 1$ satisfies the condition $f(m)f(n) = f(2mn + m + n)$, in particular so does the function $f(x) = \sqrt{2x + 1}$. Thus we suspect $f(n)$ is close to $\sqrt{2n + 1}$.

We can prove that $f(n) \leq \sqrt{2n + 1}$. Indeed, set $g(n) = \frac{f(n)}{\sqrt{2n + 1}}$ then we have $g(2mn + m + n) \geq g(m)g(n)$. In particular if $g(n) > 1$ then

$$g(2n^2 + 2n) \geq g^2(n), \quad g(2(2n^2 + 2n)^2 + 2(2n^2 + 2n)) \geq g^4(n)$$

and so on, meaning that $g(n)$ is unbounded. However we can prove it is bounded because of the condition $f(n + f(n) + 1) = f(n) + 1$. Note that as $n + 1 < n + f(n) + 1$ this implies that $f(n + 1) \leq f(n) + 1$ so f is surjective. Let a_i be the largest number x for which $f(x) = i$. Then as

$$f(a_i + f(a_i) + 1) = f(a_i) + 1 = i + 1$$

we deduce

$$a_{i+2} > a_i + f(a_i) + 1 = a_i + i + 1$$

so $a_{i+2} \geq a_i + i + 2$. This readily implies $a_i \geq \frac{i^2}{4}$ by a quick strong induction, particularly if $f(x) = i$ then $x \geq a_{i-1} \geq \frac{(i-1)^2}{4}$ which implies that $f(x) \leq \sqrt{4x + 1}$. This shows that $g(n)$ is bounded.

The contradiction means that $f(n) \leq \lfloor \sqrt{2n+1} \rfloor$. We will now show that it is actually equal to that, by induction on n . The base is true.

Assume that this was true for all $n \leq k$. If $k+1 = m + f(m)$ then $f(k+1) = f(m) + 1$. However $k+1$ will be $m + \lfloor \sqrt{2m+1} \rfloor + 1$ and it turns out that $\lfloor \sqrt{2(k+1)+1} \rfloor = \lfloor \sqrt{2m+1} \rfloor + 1$. This is because $2(k+1)+1$ is $2m+1+2\lfloor \sqrt{2m+1} \rfloor+2$ which is greater than $(\sqrt{2m+1}+1)^2$ so that $f(2k+1) \geq \lfloor \sqrt{2m+1} \rfloor + 1$. On the other hand, if $\lfloor \sqrt{2m+1} \rfloor = x$ then $2m+1 < (x+1)^2$ so $2(k+1)+1 \leq x^2+2x+2x+2 < (x+2)^2$ so $f(2k+1) < x+2$. So in this case we are done. Assume now $k+1$ is not of the form $m + f(m)$. Because f increases by at most 1, $m + f(m)$ increases by at most two so this can only happen when $m + f(m) = k$. Then $f(k+1) \geq f(k)$ so $f(k+1) \geq \lfloor \sqrt{2k+1} \rfloor$. Since it is also at most $\lfloor \sqrt{2(k+1)+1} \rfloor$ it may only contradict our claim if $\lfloor \sqrt{2k+1} \rfloor < \lfloor \sqrt{2(k+1)+1} \rfloor$ which means $2k+2$ or $2k+3$ is a perfect square. Recall that we have $m + f(m) < k+1 < m+1 + f(m+1)$ so $f(m+1) > f(m)$ which means $2m+2$ or $2m+3$ is a perfect square. Call this perfect square x^2 . Then $2k+2 = 2m+2+2x-2$ which is then x^2+2x-2 or x^2+2x-1 - that is not a perfect square. Similarly, $2k+3$ is either x^2+2x-1 or x^2+2x , this is again not a perfect square.

Note that x^2+2x-2 can be a perfect square for $x=1$ only. But this case does not happen because $x > 1$.

The solution of the problem is now complete.

13.2 Practice Problems

Problem 13.1. (IMO 1993) Determine whether there exists a strictly increasing function $f: \mathbb{N} \rightarrow \mathbb{N}$ such that $f(1) = 2$ and $f(f(n)) = f(n) + n$ for all $n \in \mathbb{N}$.

Problem 13.2. (IMO 1979) Find all increasing functions $f: \mathbb{N} \rightarrow \mathbb{N}$ with the property that all natural numbers which are not in the image of f are those of form $f(f(n)) + 1, n \in \mathbb{N}$.

Problem 13.3. Find all increasing functions $f: \mathbb{N} \rightarrow \mathbb{N}$ such that the only natural numbers which are not in the image of f are those of form $2n + f(n), n \in \mathbb{N}$.

Problem 13.4. Find all functions $f: \mathbb{N} \rightarrow \mathbb{N}$ that satisfy

$$f(1) = 1, \quad f(n+1) = f(n) + 2$$

if $f(f(n) - n + 1) = n$ and $f(n+1) = f(n) + 1$ otherwise.

Problem 13.5. (China TST 2006) Suppose that a function f defined on nonnegative integers satisfies $f(0) = 0$, $f(n) = n - f(f(n-1))$ for $n \geq 1$. Find all real polynomials g such that $f(n) = \lfloor g(n) \rfloor$, $n = 0, 1, 2, \dots$.

Chapter 14

Extremal Element Method

14.1 Theory and Examples

The extremal element method is used in many areas of mathematics and exploits the property of certain sets to contain a largest/smallest element. Knowing the largest/smallest element α of a set can tell us certain things about the set; conversely the property of being larger (smaller) than anything else in the set tells us something about α . This is especially useful in functional equations because they relate the values of the function at different points to one another.

The reasoning above is made clear by some examples, but before we proceed with them let's provide the rigorous foundation by recalling some fundamental theorems about the structure of our favorite sets.

Definition. Let A be a subset of $\mathbb{R}$. The *largest element* of A is defined to be $a \in A$ such that whenever $x \in A$, $x \leq a$. Similarly, the *smallest element* of A is an element b such that whenever $x \in A$, $x \geq b$.

The largest and smallest element of a set are also called the maximum and minimum, written max and min.

It is clear that the largest element is unique, and so is the smallest element. But it does not always need to exist - for example the set of real numbers itself does not have a largest element, nor a smallest one.

Definition. A set A is called *bounded from above* if there is an element $a \in \mathbb{R}$ such that $x < a$ for any $x \in A$. Similarly, A is *bounded from below* if there is an element $a \in \mathbb{R}$ such that $x > a$ for any $x \in A$. A set that is bounded from above and below is called bounded.

Note that A is bounded if and only if the set $\{|x| : x \in A\}$ is bounded from above. A is bounded from above/below if and only if the set $c \cdot A$ is bounded from above/below if $c > 0$ or below/above if $c < 0$, and a set has the same boundedness properties as any of its translates. There are a lot of other similar properties, but they are easy to deduce whenever need comes.

Clearly a set that is unbounded from above does not have a largest element, and a set that is unbounded from below does not have a smallest element. Thus we better deal with bounded sets. But even this does not solve our problem - the set $(0, 1)$, for example, does not have a smallest or a largest element (nor does any open set, for that matter). They should be 0 and 1 but they do not belong to the set. They do belong to the closure of the set, though.

Note that if A is bounded from above, then the set $\{y : y \geq x, \forall x \in A\}$ is non-empty. It is also closed, therefore it can only be a semi-interval of the form $[y_0, \infty)$, and y_0 is its smallest element.

Definition/Proposition. Let A be a non-empty subset of $\mathbb{R}$. If A is bounded from above, then there is a smallest element y such that $x \leq y$ for all $x \in A$. This element is called the *supremum* of A , written $\sup(A)$. This element is in the closure of A , i.e. it can be written as an (increasing) limit of elements from A . Similarly, if A is bounded from below, then the set of elements y such that $x \geq y$ for any $x \in A$ has a largest element, and it is called the *infimum* of A , or $\inf(A)$. It can be written as a decreasing limit of elements of A .

By convention, we write $\sup(A) = +\infty$ if A is unbounded from above, and $\inf(A) = -\infty$ if A is unbounded from below. Also, if $\emptyset$ is the empty set we convene to write $\sup(\emptyset) = -\infty$, $\inf(\emptyset) = +\infty$. This is justified by the following observation: if $A \subset B$, then $\sup(A) \leq \sup(B)$ and $\inf(A) \geq \inf(B)$. Since $\emptyset$ is a subset of any other set, $\sup(\emptyset)$ must be smaller than any real number and $\inf(\emptyset)$ must be larger than any real number. This seems to contradict the general principle that $\inf(A) \leq \sup(A)$ because any element of A is between those two, but there is not element in $\emptyset$.

We will not prove the above proposition here, it can be found in introduc-

tory calculus textbooks.

Proposition. a) If A has a largest element then it equals the supremum, $\max(A) = \sup(A)$. Similarly, if the smallest element exists then it equals the infimum. In fact, A has a largest element if and only if $\sup(A) \in A$, and it has a smallest element if and only if $\inf(A) \in A$.

b) Any finite set has a maximum and minimum.

c) Any subset of $\mathbb{Z}$ (or generally a discrete subset of $\mathbb{R}$) that is bounded above has a largest element, and if it is bounded below it has a smallest element.

We leave the proof to the reader. Note that part c) in particular, implies that every set of positive integers has a smallest element; this is one of the founding principles of mathematical induction.

The extremal element method precisely uses these notions we introduced. Usually, we take the smallest/largest value of f (i.e. the maximum/minimum element of $\text{Im}(f)$), or $\sup/\inf$ of the image of f . Or we may use auxiliary expressions, like $f(x) - x$ or the like, and see to minimize/maximize them.

Here come the examples.

Example 14.1. Find all functions $f : \mathbb{Z} \rightarrow \mathbb{N}$ such that

$$6f(k+3) - 3f(k+2) - 2f(k+1) - f(k) = 0$$

for any $k \in \mathbb{Z}$.

Solution. The first inconvenience in solving this problem is that we have f defined on $\mathbb{Z}$, not on $\mathbb{N}$, so computing f inductively would lead to induction in both directions. The second is the 6 before $f(k+3)$ which implies that if we would try to compute f inductively, we could obtain non-integer numbers. That is why we must use another idea. As surprisingly the range of f is in $\mathbb{N}$ (not in $\mathbb{Z}$ like usually when functions are defined on $\mathbb{Z}$), we might try to use some of the properties of $\mathbb{N}$ that distinguish it from $\mathbb{Z}$. A first that comes to our mind is that any subset of $\mathbb{N}$ contains a minimal element, and indeed as $6 = 1 + 2 + 3$ this is the best idea. So let $a = \min\{\text{Im}f\}$ and $f(x) = a$. Then writing the condition for $k = x - 3$ we get

$$6a = 3f(x-1) + 2f(x-2) + f(x-3) \geq 3a + 2a + a = 6a.$$

So equalities hold everywhere thus

$$f(x-1) = f(x-2) = f(x-3) = a.$$

A straightforward induction now shows that $f(y) = a$ whenever $y \leq x$. Another simple induction using the recursive relation shows that $f(y) = a$ for $y > x$ (the condition written for $k = x-2$ gives us $f(x+1) = a$ and so on). It is clear that the constant functions satisfy the given condition.

Example 14.2. (IMO Shortlist 2007) Consider the functions $f : \mathbb{N} \rightarrow \mathbb{N}$ which satisfy the condition

$$f(m+n) \geq f(m) + f(f(n)) - 1,$$

for all $m, n \in \mathbb{N}$. Find all possible values of $f(2007)$.

Solution. We shall prove that the desired numbers are $1, 2, \dots, 2008$.

Note first that if $f(n) > n$ for some n , then setting $m = f(n) - n$, implies $f(f(n) - n) \leq 1$. We shall prove that $f(n) \leq n+1$ for all n . It is clear that f is an increasing function. Assume that $f \not\equiv 1$ and denote by a the least positive integer for which $f(a) > 1$. Suppose that there exist positive numbers of the form $f(n) - n$. All such numbers are less than a and therefore there exists a largest number b of the form $b = f(k) - k$. Then

$$2k + b \geq f(2k) \geq f(k) + f(f(k)) - 1 \geq 2f(k) - 1 = 2(k + b) - 1$$

and thus $b \leq 1$, which completes the proof.

On the other hand, if $1 \leq j \leq 2007$, then the functions

$$f_j(n) = \begin{cases} 1, & n < 2007 \\ j, & n = 2007 \\ n, & n > 2007 \end{cases} \quad \text{and} \quad f(n) = \begin{cases} n, & 2007 \nmid n \\ n+1, & 2007 | n \end{cases}$$

satisfy the given condition.

Example 14.3. (Romania TST 1986) Let $f, g : \mathbb{N} \rightarrow \mathbb{N}$ be functions such that f is surjective, g is injective and $f(n) \geq g(n)$ for all $n \in \mathbb{N}$. Prove that $f(n) = g(n)$ for each $n \in \mathbb{N}$.

Solution. Suppose, by way of contradiction, that $f(n) \neq g(n)$ for some $n \in \mathbb{N}$. Then it follows from the hypothesis that $f(n) > g(n)$. So the subset $A = \{g(n) \mid f(n) > g(n)\}$ of $\mathbb{N}$ is nonempty, and hence it has a least element. Denote this element by m_0 and let $m_0 = g(n_0)$ for some $n_0 \in \mathbb{N}$. Note that n_0 is unique, because g is injective. Since f is surjective, there exists $n_1 \in \mathbb{N}$ for which $m_0 = f(n_1)$. Clearly $n_0 \neq n_1$, because

$$f(n_0) > g(n_0) = m_0 = f(n_1).$$

The hypothesis implies that $m_0 = f(n_1) \geq g(n_1)$, and we have two possibilities. If $f(n_1) = g(n_1)$, then $g(n_1) = m_0 = g(n_0)$, which contradicts the injectivity of g . If $f(n_1) > g(n_1)$, then $g(n_1) \in A$. Since $f(n_1) = m_0$, it turns out that $g(n_1)$ is less than the least element m_0 of A , again a contradiction.

Example 14.4. (IMO Shortlist 2010) Suppose that f and g are two functions defined on the set $\mathbb{N}$ of positive integers and taking positive integer values. Suppose also that $f(g(n)) = f(n) + 1$ and $g(f(n)) = g(n) + 1$ for all $n \in \mathbb{N}$. Prove that $f(n) = g(n)$ for all $n \in \mathbb{N}$.

Solution. Denote by $f^{(k)}$ and $g^{(k)}$ the k -th iterates of f and g , respectively. Observe that

$$f(g^{(k)}(x)) = f(g^{(k-1)}(x)) + 1 = \cdots = f(x) + k$$

for all positive integers k . Similarly

$$g(f^{(k)}(x)) = g(x) + k.$$

Now let a and b be the minimal values attained by f and g , respectively, and let $f(n_f) = a, g(n_g) = b$. Then we have

$$f(g^{(k)}(n_g)) = a + k, g(f^{(k)}(n_f)) = b + k,$$

so $\text{Im}(f) = \{a, a+1, \dots\}$ and $\text{Im}g = \{b, b+1, \dots\}$. Next, note that $f(x) = f(y)$ implies $g(x) = g(f(x)) - 1 = g(f(y)) - 1 = g(y)$ and the converse is also true. Now we say that x and y are *similar* and write $x \sim y$ if $f(x) = f(y)$ (equivalently, $g(x) \sim g(y)$). For every $x \in \mathbb{N}$ define $[x] = \{y \in \mathbb{N} \mid x \sim y\}$. Note that $[x] = [y]$ iff $y \in [x]$. Now we investigate the structure of the sets $[x]$.

Claim 1. If $f(x) \sim f(y)$ then $x \sim y$, i.e. $f(x) = f(y)$. Consequently, each class $[x]$ contains at most one element from $\text{Im}(f)$, as well as from $\text{Im}(g)$.

Proof of claim. If $f(x) \sim f(y)$ then $g(x) = g(f(x)) - 1 = g(f(y)) - 1 = g(y)$ and $x \sim y$. The second statement follows now from the structure of $\text{Im}f$ and $\text{Im}g$. $\square$

Claim 2. For each $x \in \mathbb{N}$, we have $[x] \subseteq \{1, 2, \dots, b-1\}$ if and only if $f(x) = a$ and $[x] \subseteq \{1, 2, \dots, a-1\}$ if and only if $g(x) = b$.

Proof of claim. We will prove that $[x] \not\subseteq \{1, 2, \dots, b-1\}$ iff $f(x) > a$. Note that $f(x) > a$ implies that there exists some y satisfying $f(y) = f(x) - 1$, so $f(g(y)) = f(y) + 1 = f(x)$, and hence $x \sim g(y) \geq b$. Conversely, if $b \leq c \sim x$ then $c = g(y)$ for some $y \in \mathbb{N}$, which in turn implies

$$f(x) = f(g(y)) = f(y) + 1 \geq a + 1,$$

and hence $f(x) > a$. $\square$

Claim 2 implies that there exists exactly one class contained in $\{1, 2, \dots, a-1\}$ (that is, the class $[n_g]$), as well as exactly one class contained in $\{1, 2, \dots, b-1\}$ (the class $[n_f]$). Assume for a moment that $a \leq b$. Then $[n_g]$ is contained in $\{1, 2, \dots, b-1\}$ as well, hence it coincides with $[n_f]$. So, we get that

$$f(x) = a \Leftrightarrow g(x) = b \Leftrightarrow x \sim n_f \sim n_g.$$

Claim 3. $a = b$.

Proof of claim. By *Claim 2*, we have $[a] \neq [n_f]$, so $[a]$ should contain some element $a_1 \geq b$ by *Claim 2* again. If $a \neq a_1$, then $[a]$ contains two elements $\geq a$ which is impossible by *Claim 1*. Therefore $a = a_1 \geq b$. Similarly, $b \geq a$. $\square$

Now we are ready to prove the problem statement. First we establish the following

Claim 4. For every nonnegative integer d we have

$$f^{(d+1)}(n_f) = g^{(d+1)}(n_f) = a + d.$$

Proof of claim. Induction on d . For $d = 0$ the statement follows from *Claim 3* and the equivalence mentioned just before it. For $d > 1$ from the induction hypothesis we have

$$f^{(d+1)}(n_f) = f(g^{(d)}(n_f)) = f(n_f) + d = a + d.$$

The equality $g^{(d+1)}(n_f) = a + d$ is analogous. $\square$

Finally, for each $x \in \mathbb{N}$ we have $f(x) = a + d$ for some $d \geq 0$, so

$$f(x) = f(g^{(d)}(n_f))$$

and hence $x \sim g^{(d)}(n_f)$. It follows that

$$g(x) = g(g^{(d)}(n_f)) = g^{(d+1)}(n_f) = a + d = f(x)$$

by *Claim 4*. $\square$

Remark. Using similar arguments as in the proof above one can describe all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ such that $f(f(n)) = f(n) + 1$ for all $n \in \mathbb{N}$. For each such function there exists n_0 such that $f(n) = n + 1$ for all $n \geq n_0$, while for each $n < n_0$, $f(n)$ is an arbitrary positive integer greater than or equal to n_0 (these numbers may be different for different $n < n_0$).

The extremal element method is most naturally used in $\mathbb{N}$, in which every subset has a minimal element. But it can also be used in $\mathbb{R}$, for example. Of course, not every subset of $\mathbb{R}$ has a minimal element, however we know that every subset which is bounded below has an infimum and every subset which is bounded above has a supremum. These two notions work as a kind of extremal element for us.

Example 14.5. Find all functions $f : \mathbb{R} \rightarrow [0; 2]$ such that

$$f(x+1) = \sqrt{\frac{f(x)+1}{2}}$$

for any $x \in \mathbb{R}$.

Solution. We shall prove that $f(x) = 1$ for all $x \in \mathbb{R}$. Indeed, let us prove first that $f(x) \geq 1$ for all $x \in \mathbb{R}$. Since $\text{Im}(f)$ is bounded below, we can set $c = \inf(\text{Im}(f)) \geq 0$. To prove $f(x) \geq 1$ we just prove that $c \geq 1$. Indeed, assume that $c < 1$. By the definition of infimum, for a given $\epsilon > 0$ there exists x_0 such that $c \leq f(x_0) < c + \epsilon$. In particular, we may take $\epsilon = \frac{1-c}{2}$ and get

$$c \leq f(x_0) < \frac{c+1}{2}.$$

Therefore, by setting $x = x_0 - 1$ in the identity of the problem we get

$$f(x_0) = \sqrt{\frac{f(x_0 - 1) + 1}{2}}.$$

Thus

$$f(x_0 - 1) = 2f(x_0)^2 - 1 < 2\left(\frac{c+1}{2}\right)^2 - 1 = \frac{c^2 + 2c - 1}{2}.$$

Now $\frac{c^2 + 2c - 1}{2} < c$ for $c < 1$, and this contradicts the definition of the infimum of f . Similarly, we prove that $f(x) \leq 1$ by setting $s = \sup(\text{Im}(f))$. We need $s \leq 1$ and indeed assume $s > 1$. Like above, we can choose x_0 such that $s - \frac{s-1}{2} < f(x_0) \leq s$ thus $\frac{s+1}{2} < f(x_0) \leq s$ and then we set $x = x_0 - 1$ to get

$$f(x_0 - 1) = 2f(x_0)^2 - 1 \geq 2\left(\frac{s+1}{2}\right)^2 - 1 = \frac{s^2 + 2s - 1}{2} > s \text{ as } s > 1.$$

This contradicts the assumption that $s = \sup(\text{Im}(f))$. Thus we have established that $f(x) \leq 1$ and $f(x) \geq 1$ simultaneously, hence the only possibility is $f(x) = 1$.

Example 14.6. (Bulgaria 2007) Find all $a \in \mathbb{R}$ for which there is a nonconstant function $f : (0, 1] \rightarrow \mathbb{R}$ such that

$$a + f(x + y - xy) + f(x)f(y) \leq f(x) + f(y)$$

for all $x, y \in (0, 1]$.

Solution. Replacing x and y by $1-x$ and $1-y$ and setting $g(x) = 1 - f(1-x)$, we get $g(xy) \geq g(x)g(y) + a$, $x, y \in \Delta = [0, 1)$. We shall prove that the desired a are $a < \frac{1}{4}$.

Let $g(xy) \geq g(x)g(y) + \frac{1}{4}$. Then $g(0) \geq g^2(0) + \frac{1}{4}$, i.e. $\left(g(0) - \frac{1}{2}\right)^2 \leq 0$ and hence $g(0) = \frac{1}{2}$. Now $g(0) \geq g(0)g(x) + \frac{1}{4}$ implies that $g(x) \leq \frac{1}{2}$. On the

other hand, since $g(x^2) \geq \frac{1}{4}$, then $m = \inf_{\Delta} g \geq \frac{1}{4}$. For any $\varepsilon > 0$, there is $x \in \Delta$ such that $m + \varepsilon > g(x)$. Then $m + \varepsilon > g(x) \geq g^2(\sqrt{x}) + \frac{1}{4} \geq m^2 + \frac{1}{4}$ implies that $m \geq m^2 + \frac{1}{4}$, i.e. $m = \frac{1}{2}$. Hence $g = \frac{1}{2}$. Thus for all $a \geq \frac{1}{4}$ there is no function satisfying the given conditions.

Now we shall show that for every $a < \frac{1}{4}$ there is a function with the required properties. It is enough to consider the case $0 < a < \frac{1}{4}$. The equation $t^2 - t + a = 0$ has roots $0 < t_1 < t_2$. Choosing $b \in (t_1, t_2)$, then $b > c = b^2 + a$. So for any function $g : \mathbb{R} \rightarrow (c, b)$ one has that $g(xy) > c > g(x)g(y) + a$.

Example 14.7. (Russia 2005) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a bounded function such that

$$f(x+y)^2 \geq f(x)^2 + 2f(xy) + f(y)^2$$

for all $x, y \in \mathbb{R}$. Prove that $-2 \leq f(x) \leq 0$ for all $x \in \mathbb{R}$.

Solution. First, we shall prove that $|f(x)| \leq 2$ for all x . Let $M = \sup_{x \neq 0} |f(x)|$. Then there is a sequence $x_1, x_2, \dots$ of non-zero real numbers such that $|f(x_n)| \rightarrow M$ and the inequalities

$$M^2 \geq f(2x_n)^2 \geq 2f(x_n)^2 + 2f(x_n^2) \geq 2f(x_n)^2 - 2M \rightarrow 2M^2 - 2M$$

show that $M^2 \geq 2M^2 - 2M$. Hence $M(M-2) \leq 0$, i.e. $M \leq 2$.

It remains to show that $f(x) \leq 0$ for all x . Assume the contrary, i.e. $f(a) > 0$ for some a . We have that

$$f(x + a/x) \geq f(x)^2 + 2f(a) + f(a/x)^2, \quad x \neq 0.$$

Choose $n_0 \in \mathbb{N}$ such that $2n_0 f(a) > M^2$. Since $\lim_{x \rightarrow +\infty} (x + a/x) = +\infty$, we can find $a_1 > 0$ such that if $a_{n+1} = a_n + a/a_n$, $n \in \mathbb{N}$, then $a_n > 0$ for $n \leq n_0$ (in particular, the sequence is well-defined up to a_{n_0+1}). Now the inequality

$$f(a_{n+1})^2 \geq f(a_n)^2 + 2f(a)$$

implies that

$$M^2 \geq f(a_{n_0+1})^2 \geq f(a_1)^2 + 2n_0f(a) > M^2,$$

a contradiction.

Remark. Note that the given inequality becomes an equality for the constant functions 0 and -2 , as well as for the unbounded function x .

Example 14.8. Find all functions $f : [0, \infty) \rightarrow [0, \infty)$ such that

$$f(y)f(xf(y)) = f(x+y)$$

for all $x, y \in [0, \infty)$.

Solution. Setting $x = 0$ gives $f(0)f(y) = f(y)$. Hence $f(0)^2 = f(0)$. If $f(0) = 0$, then $f(y) = 0$ for all $y \in [0, \infty)$. Hence we may assume that $f(0) = 1$. We shall consider two cases.

Case 1. $f(y) = 0$ for some $y > 0$. Set $a = \inf\{y > 0 : f(y) = 0\}$. For any $x > a$ there is y such that $a < y < x$ and $f(y) = 0$. Then

$$f(x) = f(y)f((x-y)f(y)) = 0.$$

So, if $a = 0$ then $f(x) = 0$ for all $x > 0$.

Suppose now that $a > 0$ and let $0 < y < a$. For any $\varepsilon > 0$ we have

$$f(y)f((a+\varepsilon-y)f(y)) = f(a+\varepsilon) = 0.$$

Hence $(a+\varepsilon-y)f(y) \geq a$ since $f(y) > 0$. On the other hand

$$f\left(\frac{a+\varepsilon}{f(y)} + y\right) = f(a+\varepsilon)f(y) = 0$$

and therefore $\frac{a+\varepsilon}{f(y)} + y \geq a$. Now letting $\varepsilon \rightarrow 0$ in the above two inequalities gives $f(y) = \frac{a}{a-y}$ for any $0 < y < a$. In particular $f\left(\frac{a}{2}\right) = 2$ and setting $x = y = \frac{a}{2}$ in the given equation gives $2f(a) = f(a)$, i.e. $f(a) = 0$.

Case 2. $f(y) > 0$ for all $y > 0$. First we shall show that $f(y) \leq 1$ for any $y \geq 0$. Indeed, assume that $f(y) > 1$ for some $y > 0$. Then we set

$$x = \frac{y}{f(y) - 1}$$

in the given equation and get $f(y) = 1$, a contradiction.

The inequality $f(y) \leq 1$ together with the given equation shows that the function f is decreasing.

Suppose that $f(y) = 1$ for some $y > 0$. Then $f(x+y) = f(x)$ for any $x \geq 0$ and using the fact that the function f is decreasing we get that $f(x) = 1$ for all $x \geq 0$.

It remains to consider the case when $f(y) < 1$ for any $y > 0$. Then the function f is strictly decreasing and hence injective. Now the identity

$$\begin{aligned} f(y)f(xf(y)) &= f(x+y) = f(xf(y) + y + x(1-f(y))) \\ &= f(xf(y))f((y+x(1-f(y))))f(xf(y)) \end{aligned}$$

implies that

$$y = (y+x(1-f(y)))f(xf(y)).$$

Setting $y = 1$, $xf(1) = z$ and $\frac{f(1)}{1-f(1)} = a$ we get that $f(z) = \frac{a}{a+z}$ for any $z > 0$.

Thus all the functions we have found are the following:

$$f(x) = 0, f(x) = 1, f(x) = \frac{a}{x+a},$$

$$f(x) = \begin{cases} 1 & \text{for } x = 0 \\ 0 & \text{for } x > 0, \end{cases}$$

$$f(x) = \begin{cases} \frac{a}{a-x} & \text{for } 0 \leq x < a \\ 0 & \text{for } x \geq a, \end{cases}$$

where $a > 0$ is an arbitrary constant. It is easy to check that the first four functions are solutions of the problem. Now we shall show that the fifth function also gives a solution.

If $x, y \geq 0$ and $x + y < a$ then

$$f(y)f(xf(y)) = \frac{a}{a-y}f\left(\frac{ax}{a-y}\right) = \frac{a}{a-y} \cdot \frac{a-y}{a-(x+y)} = f(x+y).$$

Let $x, y \geq 0$ and $x + y \geq a$. Then $f(x+y) = 0$. If $y \geq 0$ then $f(y) = 0$. If $y < a$ then $\frac{ax}{a-y} \geq a$ and $xf(y) \geq a$. Hence $f(xf(y)) = 0$. Therefore in both cases we have

$$f(y)f(xf(y)) = 0 = f(x+y).$$

Remark. The above problem with the additional conditions $f(2) = 0$ and $f(x) \neq 0$ for $0 \leq x < 2$ is a problem from IMO 1986 (see Example 4.4). In this case the only solution is the function

$$f(x) = \begin{cases} \frac{2}{2-x} & \text{for } 0 \leq x < 2 \\ 0 & \text{for } x \geq 2. \end{cases}$$

14.2 Practice Problems

Problem 14.1. Find all functions $f: \mathbb{N} \rightarrow \mathbb{N}$ such that

$$f(f(f(n))) + f(f(n)) + f(n) = 3n$$

for all $n \in \mathbb{N}$.

Problem 14.2. Find all bijections

$$f, g, h: \mathbb{N} \rightarrow \mathbb{N}$$

for which

$$f(n)^3 + g(n)^3 + h(n)^3 = 3ng(n)h(n)$$

whenever $n \in \mathbb{N}$.

Problem 14.3. (IMO 1977) Let $f: \mathbb{N} \rightarrow \mathbb{N}$ be a function such that

$$f(n+1) > f(f(n)) \text{ for all } n \in \mathbb{N}.$$

Show that $f(n) = n$ for all $n \in \mathbb{N}$.

Problem 14.4. Find all injective functions $f: \mathbb{N} \rightarrow \mathbb{N}$ such that

$$f(f(m) + f(n)) = f(f(m)) + f(n)$$

for all $m, n \in \mathbb{N}$.

Problem 14.5. (BMO 2002) Find all functions $f: \mathbb{N} \rightarrow \mathbb{N}$ such that

$$2n + 2001 \leq f(f(n)) + f(n) \leq 2n + 2002$$

for all $n \in \mathbb{N}$.

Problem 14.6. (IMO 1972) Let f and g be real-valued functions defined for all real values of x and y , and satisfying the equation

$$f(x + y) + f(x - y) = 2f(x)g(y)$$

for all x, y . Prove that if f is not identically zero, and if $|f(x)| \leq 1$ for all x then $|g(y)| \leq 1$ for all y .

Problem 14.7. (Bulgaria 2008) Find all real numbers a for which there exists a function $f: (0, \infty) \rightarrow (0, \infty)$ such that

$$3f(x)^2 = 2f(f(x)) + ax^4$$

for all $x > 0$.

Problem 14.8. Is there a bounded function $f: \mathbb{R} \rightarrow \mathbb{R}$ such that $f(1) = 1$ and

$$f\left(x + \frac{1}{x^2}\right) = f(x) + f\left(\frac{1}{x}\right)^2$$

for all $x \neq 0$.

Problem 14.9. (G. Dospinescu) Find all integers k for which there is a function $f: \mathbb{N} \rightarrow \mathbb{N}$ such that

$$f(f(f(n))) = f(n + 1) + k$$

for all positive integers n .

Chapter 15

Fixed Points

15.1 Theory and Examples

A fixed point of a map f from a set X to itself is an element $x \in X$ such that $f(x) = x$. One may ask why these points are important. Of course a map may not have fixed points at all, but there are some situations where they come handy.

- The most common solution to a functional equation is the function $f(x) = x$, and in this case one has to prove that every point is a fixed point of f .

- The iterates of a map f at a fixed point x are all the same $f^{(m)}(x) = x$, no matter how many times we take f . Fixed points and iterates of a function often combine with its continuity or monotonicity. If one takes, for example, the orbit $O(x) = \{x, f(x), f^{(2)}(x), \dots\}$ of a function f at x , and it happens that this sequence converges to some point a , then a must be a fixed point, as long as f is continuous.

- Fixed points can often help show that a given functional equation has no solutions, usually when it involves iterates and compositions of functions because fixed points behave well with composition of functions. For example, if we want to solve the functional equation $f(f(x)) = g(x)$ then every fixed point of f is a fixed point of g . We can often reach a contradiction by just investigating the fixed points of f .

In short, fixed points can often carry important information about the map under consideration.

We start with a problem whose solution relies on the fact that if f is a map of a set X to itself, then X is the disjoint union of the orbits of f . We leave the proof of this fact to the reader.

Example 15.1. Let X be a finite set with n elements and $f : X \rightarrow X$ be a map with $f^{(p)}(x) = x$ for all $x \in X$, where p is a prime. Prove that if $n \equiv 1 \pmod{p}$, then f has a fixed point.

Solution. All points of X are fixed points for the p th iterate of f , so the length of each orbit of f is a divisor of p , i.e. it is either 1 or p . Since X is the disjoint union of the orbits of f and $n \equiv 1 \pmod{p}$ we see that at least one of the orbits of f is of length 1, i.e. f has a fixed point.

Example 15.2. (IMO 1996) Find all functions $f : \mathbb{N}_0 \rightarrow \mathbb{N}_0$ such that

$$f(m + f(n)) = f(f(m)) + f(n)$$

for all $m, n \in \mathbb{N}_0$.

Solution. Setting $m = n = 0$ gives $f(0) = 0$ and therefore $f(f(n)) = f(n)$, i.e. $f(n)$ is a fixed point of f for any $n \in \mathbb{N}_0$. Hence the given identity is equivalent to

$$f(0) = 0 \text{ and } f(m + f(n)) = f(m) + f(n).$$

It is obvious that the zero function is a solution of the problem.

Now suppose that $f(a) \neq 0$ for some $a \in \mathbb{N}$ and denote by b the least non-zero fixed point of f . Then

$$2b = 2f(b) = f(b + f(b)) = f(2b)$$

and it follows by induction that $f(nb) = nb$ for any $n \in \mathbb{N}_0$. If $b = 1$ then $f(n) = n$ for any $n \in \mathbb{N}_0$ and this function is also a solution of the problem. Hence we may assume that $b \geq 2$. Let c be an arbitrary fixed point of f . Then $c = kb + r$, where $k \in \mathbb{N}_0, 0 \leq r < b$, and we get

$$kb + r = c = f(c) = f(kb + r) = f(f(kb)) + f(r) = kb + f(r).$$

Thus $f(r) = r$ and therefore $r = 0$ since $r < b$. Hence any fixed point of f has the form kb . Now the identity $f(f(i)) = f(i)$ implies that $f(i) = bn_i$ for all $i, 0 \leq i < b$, where $n_i \in \mathbb{N}_0$ and $n_0 = 0$. Thus if $n = kb + i$ then $f(n) = (k + n_i)b$. Conversely, it is easily checked that for any fixed integers $b \geq 2, n_0 = 0$ and $n_1, n_2, \dots, n_{b-1} \in \mathbb{N}_0$ the function $f(n) = \left(\left[\frac{n}{b}\right] + n_i\right)b$ has the given property.

Example 15.3. (IMO 1983) Find all functions $f: (0, \infty) \rightarrow (0, \infty)$ such that:

- (i) $f(xf(y)) = yf(x)$ for all $x, y \in (0, \infty)$;
- (ii) $\lim_{x \rightarrow +\infty} f(x) = 0$.

Solution. It follows from (i) that $f(xf(x)) = xf(x)$ for all $x > 0$. Then it follows by induction on n that if $f(a) = a$ for some $a > 0$ then $f(a^n) = a^n$ for all $n \in \mathbb{N}$. Note also that $a \leq 1$ since otherwise

$$\lim_{n \rightarrow \infty} f(a^n) = \lim_{n \rightarrow \infty} a^n = +\infty,$$

a contradiction to (ii).

On the other hand $a = f(1 \cdot a) = f(1 \cdot f(a)) = af(1)$. Hence

$$1 = f(1) = f(a^{-1}a) = f(a^{-1}f(a)) = af(a^{-1}),$$

i.e. $f(a^{-1}) = a^{-1}$. Thus we have as above that $f(a^{-n}) = a^{-n}$ for all $n \in \mathbb{N}$ and therefore $a^{-1} \leq 1$.

In conclusion, the only $a > 0$ such that $f(a) = a$ is $a = 1$. Hence the identity $f(xf(x)) = xf(x)$ implies that $f(x) = \frac{1}{x}$ for all $x > 0$. It is easy to check that this function satisfies the conditions (i) and (ii) of the problem.

Example 15.4. (IMO 1994) Let S be the set of all real numbers greater than -1 . Find all functions $f: S \rightarrow S$ such that:

- (i) $f(x + f(y) + xf(y)) = y + f(x) + yf(x)$ for all $x, y \in S$;
- (ii) $\frac{f(x)}{x}$ is strictly increasing in the intervals $(-1, 0)$ and $(0, +\infty)$.

Solution. If $x = y > -1$ we have from (i) that

$$f(x + (1 + x)f(x)) = x + (1 + x)f(x). \quad (1)$$

On the other hand, (ii) implies that the equation $f(x) = x$ has at most one solution in each of the intervals $(-1, 0)$ and $(0, +\infty)$.

Suppose that $f(a) = a$ for some $a \in (-1, 0)$. Then (1) implies that

$$f(a^2 + 2a) = a^2 + 2a$$

and therefore $a^2 + 2a = a$ since $a^2 + 2a = (a+1)^2 - 1 \in (-1, 0)$. Hence $a = -1$ or $a = 0$, contradiction. The same arguments show that the equation $f(x) = x$ has no solutions in the interval $(0, +\infty)$.

Then we conclude from (1) that $x + (1+x)f(x) = 0$, i.e. $f(x) = -\frac{x}{1+x}$ for all $x > -1$. It is easy to check that this function satisfies the conditions (i) and (ii) of the problem.

Example 15.5. (Tournament of the towns 1996) Prove that there is no function $f: \mathbb{R} \rightarrow \mathbb{R}$ such that $f(f(x)) = x^2 - 1996$ for all $x \in \mathbb{R}$.

Solution. We shall prove the following more general result.

Proposition. Let $g(x)$ be a quadratic function such that the equation $g(g(x)) = x$ has at least three different real roots. Then there is no function $f: \mathbb{R} \rightarrow \mathbb{R}$ such that $f(f(x)) = g(x)$ all $x \in \mathbb{R}$.

Proof. The fixed points of $g(x)$ are also fixed points of the fourth degree polynomial $h(x) = g(g(x))$. Hence it follows by the given conditions that $g(x)$ has one or two real fixed points. Indeed, if, for example, $g(x) > x$ for all $x \in \mathbb{R}$, then $g(g(x)) > g(x) > x$ and h has no fixed points. Denote the fixed points of $g(x)$ by x_1 and x_2 (possibly $x_1 = x_2$). Then $h(x)$ has one or two real fixed points, different from x_1 and x_2 . Denote them by x_3 and x_4 (possibly $x_3 = x_4$). The identity

$$f(g(x)) = f(f(f(x))) = g(f(x))$$

implies that $\{f(x_1), f(x_2)\} = \{x_1, x_2\}$. On the other hand, we have

$$f(f(g(x))) = f(g(f(x))) \text{ and } f(f(f(g(x)))) = f(f(g(f(x)))) ,$$

i.e. $f(h(x)) = h(f(x))$. Hence $\{f(x_3), f(x_4)\} \in \{x_1, x_2, x_3, x_4\}$. Suppose that $f(x_l) = x_k$ for some $k \in \{1, 2\}$ and $l \in \{3, 4\}$. Then

$$x_l = h(x_l) = f(f(f(f(x_l)))) = f(g(x_k)) = f(x_k) \in \{x_1, x_2\},$$

a contradiction. Hence $f(x_3) = x_3$ if $x_3 = x_4$ and $\{f(x_3), f(x_4)\} = \{x_3, x_4\}$ if $x_3 \neq x_4$. In both cases we have $g(x_3) = f(f(x_3)) = x_3$, a contradiction. Thus the proposition is proved. $\square$

Turning back to the problem we note that the equation

$$g(g(x)) = (x^2 - 1996)^2 - 1996 = x$$

has four different real roots since

$$(x^2 - 1996)^2 - 1996 - x = (x^2 - 1996 - x)(x^2 + x - 1995).$$

Remark. Consider a quadratic function $g(x) = ax^2 + bx + c$. Then

$$g(g(x)) - x = (ax^2 + (b-1)x + c)(a^2x^2 + a(b+1)x + ac + b + 1).$$

Therefore the four roots of the equation $g(g(x)) = x$ are equal to

$$\frac{1-b+\sqrt{D}}{2a}, \frac{1-b-\sqrt{D}}{2a}, \frac{-1-b+\sqrt{D-4}}{2a} \text{ and } \frac{-1-b-\sqrt{D-4}}{2a},$$

where $D = (b-1)^2 - 4ac$. All these roots are real if and only if $D \geq 4$. If $D > 4$ then all the roots are different whereas for $D = 4$ one of them is equal to $\frac{3-b}{2a}$ and the other three are equal to $-\frac{1+b}{2a}$.

The proposition proved above says that if $D > 4$, then there are no functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that $f(f(x)) = g(x)$ for all $x \in \mathbb{R}$. On the other hand, if $D \leq 4$ then there are infinitely many continuous functions $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfying the above equation. Moreover, it is noticed in [21] that the same also holds true for the general equation $f^{(n)}(x) = g(x)$, where $n \geq 2$.

15.2 Practice Problems

Problem 15.1. Prove that every increasing function $f: [0, 1] \rightarrow [0, 1]$ has a fixed point.

Problem 15.2. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function such that

$$f(f(x)) = x^3 + \frac{3}{4}x$$

for all $x \in \mathbb{R}$. Prove that there exist three distinct real numbers a, b, c such that $f(a) + f(b) + f(c) = 0$.

Problem 15.3. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function such that for every $x \in \mathbb{R}$ there is a positive integer $n = n(x)$ for which $f^{(n)}(x) = 1$. Show that f has a fixed point.

Problem 15.4. (M. Tetiva) Find all continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(f(x)) = 3f(x) - 2x$$

for all $x \in \mathbb{R}$.

Chapter 16

Functional Equations for Polynomials

16.1 Theory and Examples

Polynomials represent a very special class of functions which are not only very useful in basically all areas of mathematics, but also have certain kinds of properties that distinguish them from other functions. It is our aim here to illustrate how these distinctive properties help us solve functional equations which have polynomials as the unknowns.

Let $K = \mathbb{Z}, \mathbb{Q}, \mathbb{R}$ or $\mathbb{C}$. A polynomial $p \in K[X]$ of degree $n \geq 0$ is a (formal) sum $p(X) = a_n X^n + a_{n-1} X^{n-1} + \dots + a_0$, where $a_i \in K$ and $a_n \neq 0$. The degree of a polynomial is denoted by $\deg(p)$ and we may conventionally set $\deg(0) = -\infty$.

To each polynomial is naturally associated a polynomial function p that takes a number x to $a_n x^n + a_{n-1} x^{n-1} + \dots + a_0$, called p evaluated at x and written $p(x)$. While formally a polynomial and a polynomial function are two different gadgets, we will not distinguish between them, and call both of them simply *polynomials*.

We can define addition of polynomials by adding the coefficients of the corresponding powers of X . We can define multiplication of polynomials, according to the laws $x^m \cdot x^n = x^{m+n}$, and to the law of distributivity of

addition and multiplication. We say that $f|g$ (f divides g) in $K[X]$ if there is another polynomial $h \in K[X]$ such that $f = g \cdot h$. It is also easy to see that $\deg(fg) = \deg(f) + \deg(g)$ while $\deg(f + g) \leq \max(\deg(f), \deg(g))$.

If $p(r) = 0$, then r is called a zero of p . Bezout's Theorem states that if $p(r) = 0$ then $x - r|p(x)$ thus $p(x) = (x - r)q(x)$, where the degree of q is then $\deg(p) - 1$. A zero r is called a root of multiplicity k if $(x - r)^k|p(x)$ but $(x - r)^{k+1} \nmid p(x)$.

It is easy to deduce from Bezout's Theorem that if $\deg(p) = n$, then p has at most n roots counted with multiplicity (unless $p = 0$). As a corollary, we have the identity principle: two polynomials are identical (as formal sums) if their polynomial functions coincide for infinitely many x . Indeed, if $p(x) \neq q(x)$ then $p - q$ is not zero hence it can only have at most $\deg(p - q)$ zeros, thus a finite number of zeros. Therefore two distinct polynomials can coincide only on a finite set. Also, as $\deg(p - q) \leq \max(\deg(p), \deg(q))$, it suffices for the polynomials to coincide on at least $\max(\deg(p), \deg(q)) + 1$ values.

Next, we have the Fundamental Theorem of Algebra stating that every complex polynomial p has a complex zero r . Therefore $p = (x - r)q$ and then applying the Fundamental Theorem of Algebra to q and continuing the process, we get a unique factorization

$$p(x) = c(x - r_1)(x - r_2) \dots (x - r_n).$$

In general, if K is not $\mathbb{C}$ but $\mathbb{R}, \mathbb{Z}$ or $\mathbb{Q}$ we will not always factor the polynomial into linear factors. We can of course regard it as a complex polynomial, and factor it into complex linear factors, but then these factors may not necessarily lie in $K[X]$.

We call a polynomial f irreducible if f does not decompose as the product of two non-constant polynomials. In $\mathbb{C}[X]$, only linear polynomials are irreducible. In $\mathbb{R}[X]$, linear polynomials and quadratics with negative discriminant are irreducible, while in $\mathbb{Q}[X]$ and $\mathbb{Z}[X]$ we can have irreducible polynomials of any degree.

It is known that there exists some sort of Euclidean Algorithm for polynomials, and therefore a theory of divisibility can be constructed for polynomials, where irreducible polynomials take the place of primes. We will not go deeper into this theory, just use the fact that if f, g are relatively prime polynomials

(i.e. have no common irreducible factor) and $f|gh$, then $f|h$. Note that two polynomials are relatively prime if and only if they have no a common complex zero.

It is not our aim here to describe in much detail all the properties of polynomials. Our focus is to illustrate how some of them can be used to solve functional equations whose unknowns are polynomials. We start with an example which is hard, but is particularly rewarding because it illustrates so many of the common approaches to solving functional equations involving polynomials.

Example 16.1. Find all nonconstant polynomials $P \in \mathbb{R}[X]$ such that

$$P(x^3 + 1) = P(x + 1)^3$$

for all $x \in \mathbb{R}$.

First Solution. Set $P(x + 1) = Q(x) = a_k x^k + a_{k-1} x^{k-1} + \cdots + a_0$, $a_k \neq 0$. Then $Q(x^3) = Q(x)^3$ for all $x \in \mathbb{R}$, i.e.

$$a_k x^{3k} + a_{k-1} x^{3k-3} + \cdots + a_0 = (a_k x^k + a_{k-1} x^{k-1} + \cdots + a_0)^3.$$

Suppose that $a_m \neq 0$ for some $m < k$ and take the largest m with this property. Then $3m < 2k + m$ and comparing the coefficient of x^{2k+m} in both sides, we get $0 = 3a_k^2 a_m$, a contradiction. Hence $a_k x^{3k} = Q(x^3) = Q(x)^3 = a_k^3 x^{3k}$ and so $a_k = \pm 1$. It follows that $P(x) = a_k(x - 1)^k$ for all $x \in \mathbb{R}$, where $a_k = \pm 1$.

Second Solution. The given condition implies that

$$P(3^{3^n} + 1) = P(3^{3^{n-1}} + 1)^3 = \cdots = P(4)^{3^n}$$

for all $n \in \mathbb{N}$. Set $P(x) = a_k x^k + a_{k-1} x^{k-1} + \cdots + a_0$, $a_k \neq 0$. Then

$$\lim_{x \rightarrow \infty} \frac{P(x)}{a_k x^k} = 1$$

and we get

$$1 = \lim_{n \rightarrow \infty} \frac{P(3^{3^n} + 1)}{a_k (3^{3^n} + 1)} = \lim_{n \rightarrow \infty} \frac{P(3^{3^n} + 1)}{a_k 3^{3^n}} = \frac{1}{a_k} \lim_{n \rightarrow \infty} \left(\frac{P(4)}{3^k} \right)^{3^n},$$

i.e.

$$a_k = \lim_{n \rightarrow \infty} \left(\frac{P(4)}{3^k} \right)^{3^n}. \quad (1)$$

On the other hand, comparing the leading coefficients in both sides of the given identity gives $a_k = a_k^3$, i.e. $a_k = \pm 1$. Hence (1) implies that $P(4) = \pm 3^k$ and we conclude that

$$P(3^{3^n} + 1) = P(4)^{3^n} = a_k (3^{3^n})^k$$

Thus $P(x) = a_k(x-1)^k$ for all $x = 3^{3^n} + 1$, $n \in \mathbb{N}$, and the identity principle tells us that $P(x) = a_k(x-1)^k$ for all $x \in \mathbb{R}$, where $a_k = \pm 1$.

Example 16.2. (Romania 2001) Find all polynomials $P \in \mathbb{R}[X]$ such that

$$P(x)P(2x^2 - 1) = P(x^2)P(2x - 1)$$

for all $x \in \mathbb{R}$.

First Solution. It is obvious that constant polynomials are solutions to the problem. Suppose now that $\deg P = n \geq 1$. Then $P(2x-1) = 2^n P(x) + R(x)$, where either $R \equiv 0$ or $\deg R = m < n$. Assume that $R \not\equiv 0$. It follows from the given identity that $P(x)(2^n P(x^2) + R(x^2)) = P(x^2)(2^n P(x) + R(x))$, i.e. $P(x)R(x^2) = P(x^2)R(x)$ for all $x \in \mathbb{R}$. Hence $n + 2m = 2n + m$, i.e. $n = m$, a contradiction. Thus $R \equiv 0$ and $P(2x-1) = 2^n P(x)$. Set $Q(x) = P(x+1)$. Then

$$Q(2x) = 2^n Q(x) \quad (1)$$

for any $x \in \mathbb{R}$. Set

$$Q(x) = \sum_{k=0}^n a_k x^{n-k}.$$

Then comparing the coefficients of x^{n-k} on both sides of (1) gives

$$a_k 2^{n-k} = 2^n a_k,$$

i.e. $a_k = 0$ for $k \geq 1$. Hence $Q(x) = a_0 x^n$ and therefore $P(x) = a_0(x-1)^n$.

Second Solution. Suppose that $P \not\equiv 0$ and set

$$P(x) = \sum_{k=0}^n a_k x^{n-k}$$

where $n = \deg P$ and $a_0 \neq 0$. Then

$$\sum_{k=0}^n a_k x^{n-k} \sum_{k=0}^n a_k (2x^2 - 1)^{n-k} = \sum_{k=0}^n a_k x^{2(n-k)} \sum_{k=0}^n a_k (2x - 1)^{n-k}.$$

Comparing the coefficients of x^{3n-k} , $k \geq 1$, on both sides gives

$$a_k a_0 + R_1(a_0, \dots, a_{k-1}) = a_0 a_k 2^{n-k} + R_2(a_0, \dots, a_{k-1})$$

where R_1 and R_2 are polynomials of $k-1$ variables. Hence a_k is determined uniquely by $a_0, \dots, a_{k-1}$. This shows that for given a_0 and n there is at most one polynomial satisfying the given condition. On the other hand it is easy to check that the polynomials $P(x) = a_0(x-1)^n$ are solutions and therefore they give all the solutions to the problem.

Third solution. Suppose that the polynomial $P(x)$ has a complex zero $\alpha \neq 1$. Of all these zeroes take that for which the number $|\alpha - 1| \neq 0$ is the least possible. Let β be a complex number such that $\alpha = 2\beta^2 - 1$. Setting $x = \pm\beta$ in the given equation we see that either $P\left(\frac{\alpha+1}{2}\right) = 0$ or $P(2\beta-1) = P(-2\beta-1) = 0$. The inequality $\left|\frac{\alpha+1}{2} - 1\right| < |\alpha - 1|$ shows that $P\left(\frac{\alpha+1}{2}\right) \neq 0$, i.e. $P(2\beta-1) = P(-2\beta-1) = 0$. Then

$$2|(\beta-1)(\beta+1)| = |\alpha-1| \leq \min(|(2\beta-1)-1|, |(-2\beta-1)-1|)$$

and $\beta \neq \pm 1$ imply that $\max(|\beta-1|, |\beta+1|) \leq 1$, i.e. $\beta = 0$. Hence $\alpha = -1$ and therefore $P(x) = (x+1)^k Q(x)$, where $k \geq 1$ and $Q(-1) \neq 0$. Substituting in the given identity gives

$$(x+1)^k x^k Q(x) Q(2x^2-1) = (x^2+1) Q(x^2) Q(2x-1).$$

Setting $x = 0$ in this identity gives $Q(0) = 0$ since $Q(-1) \neq 0$. Thus $P(0) = 0$ which contradicts the choice of $\alpha = -1$ since $|-1-1| > |0-1|$. Hence all the

zeros of the polynomial $P(x)$ are equal to 1 and therefore $P(x) = a_0(x-1)^n$ for some real constant a_0 .

Example 16.3. (Bulgaria 2001) Find all polynomials $P \in \mathbb{R}[X]$ such that

$$P(x)P(2x^2+1) = P(x^2)(P(2x+1)-4)$$

for all $x \in \mathbb{R}$.

First Solution. This solution is similar to the first solution to the previous problem. It is obvious that $P \equiv 0$ is a solution. Suppose now that $P \not\equiv 0$. Then $P(2x+1) = 2^n P(x) + R(x)$ where $n = \deg P$ and either $R \equiv 0$ or $\deg R = m < n$. It follows from the given identity that

$$P(x)R(x^2) = P(x^2)(R(x) - 4x).$$

Hence $R \not\equiv 0$ since otherwise $P \equiv 0$. Suppose that $m \geq 2$. Then comparing the degrees of both sides gives $n + 2m = 2n + m$, i.e. $n = m$, a contradiction. Thus $m \leq 1$ and $1 \geq k = \deg(R(x) - 4x)$. Now the equality $n + 2m = 2n + k$ shows that $n = 2, m = 1, k = 0$ and therefore $P(x)$ is a quadratic function such that

$$P(2x+1) = 4P(x) + 4x + c. \quad (1)$$

On the other hand, setting $x = 1$ in the given identity gives $P(1) = 0$, i.e. $P(x) = a(x-1)(x-b)$. Substituting this in (1) implies that $P(x) = x^2 - 1$ and one checks easily that this polynomial is a solution to the problem. Thus $P \equiv 0$ or $P(x) = x^2 - 1$.

Second Solution. First, we shall show that if P is a nonconstant solution then all the zeros of the polynomial $P(x)$ are real. Assume the contrary and let α be a complex zero of $P(x)$ with argument $\varphi \in (0, 2\pi)$. Since the coefficients of $P(x)$ are real it follows that $\bar{\alpha}$ is also a zero of $P(x)$. Hence we may assume that $\varphi \in (0, \pi)$ and that φ is the least possible argument of a complex zero of $P(x)$. It follows from the given identity that at least one of the complex numbers $\sqrt{|\alpha|} \left(\cos \frac{\varphi}{2} + i \sin \frac{\varphi}{2} \right)$ and $2\alpha + 1$ is a zero of $P(x)$. This is a contradiction since the arguments of both numbers are less than φ .

Suppose now that $\deg P = n > 2$ and set

$$P(x) = \sum_{k=0}^n a_k x^{n-k}.$$

Then comparing the coefficients of x^{3n-1} on both sides of the given identity we get $2^n a_0 a_1 = a_0(n \cdot 2^{n-1} a_0 + 2^{n-1} a_1)$, i.e. $\frac{a_1}{a_0} = n$. Hence the sum of all zeros of $P(x)$ is equal to $-n$ (Vieta's formulas). Since 1 is a zero of $P(x)$ (set $x = 1$ in the given identity) we conclude that the least zero α of $P(x)$ is less than -1 . On the other hand, it is easily seen that at least one of the numbers $i\sqrt{-\alpha}$ and $2\alpha + 1$ is a zero of $P(x)$. This is a contradiction since $i\sqrt{-\alpha}$ is not a real number and $2\alpha + 1 < \alpha$.

Finally, we see as in the first solution that if $\deg P \leq 2$ then either $P \equiv 0$ or $P(x) = x^2 - 1$.

Example 16.4. Let $\{P_n\}_{n=1}^\infty$ be the sequence of polynomials defined by:

$$P_1(x) = x, P_{n+1}(x) = P_n(x)^2 + 1, \quad n \geq 1.$$

Prove that a polynomial P satisfies the identity

$$P(x^2 + 1) = P(x)^2 + 1$$

if and only if P belongs to the above sequence.

Solution. Let P satisfy the given identity. Then $P(x)^2 = P(-x)^2$, hence either $P(x) = P(-x)$ or $P(x) = -P(-x)$. It follows that $P(x) \equiv P(-x)$ or $P(x) \equiv -P(-x)$. In the second case we get $P(0) = 0$ and an easy induction shows that $P(n^2 + 1) = n^2 + 1$ for all $n \in \mathbb{N}$ in the sequence $n_0 < n_1 < \dots$ given by $n_0 = 0, n_{k+1} = n_k^2 + 1$. Hence the identity principle implies that $P(x) = x$ for all $x \in \mathbb{R}$ and this polynomial belongs to the given sequence. In the first case it follows easily that $P(x) = Q(x^2)$, where Q is a polynomial. Then

$$Q((x^2 + 1)^2) = P(x^2 + 1) = P(x)^2 + 1 = Q(x^2)^2 + 1$$

and setting $R(x) = Q(x - 1)$ we see that $R(y^2 + 1) = R(y)^2 + 1$ for $y = x^2 + 1$. Hence $R(y^2 + 1) = R(y)^2 + 1$ for all $y \in \mathbb{R}$. Thus

$$P(x) = R(x^2 + 1) = R(x)^2 + 1$$

where $\deg R = \frac{\deg P}{2}$ and the polynomial R satisfies the given identity. Conversely, if R is a polynomial satisfying the given identity then the same is true for the polynomial $P(x) = R(x^2 + 1)$. Now the statement of the problem follows by induction on $\deg P$.

Example 16.5. Find all polynomials $P \in \mathbb{Q}[X]$ that satisfy

$$P(x) = P\left(\frac{-x + \sqrt{3(1-x^2)}}{2}\right)$$

whenever $|x| \leq 1$.

Solution. We have

$$P(x) = P\left(\frac{-x + \sqrt{3(1-x^2)}}{2}\right) = Q(x) + \sqrt{3(1-x^2)}R(x), \quad (1)$$

where $Q, R \in \mathbb{Q}[x]$. This can be proven by Newton's Binomial Formula. It follows that $R = 0$ and $Q = P$ since $\sqrt{1-x^2}$ is not a rational function. (Why?) Then

$$P(x) = P\left(\frac{-x + \sqrt{3(1-x^2)}}{2}\right) = P\left(\frac{-x - \sqrt{3(1-x^2)}}{2}\right).$$

The condition holds for all x with $|x| \leq 1$ and we may extend it to all $x \in \mathbb{C}$. Let

$$r(x) = \frac{-x + \sqrt{3(1-x^2)}}{2}.$$

Then $r^{(3)}(x) = 1$ which can be checked manually or if one notes that

$$r(\cos t) = \cos\left(t + \frac{2\pi}{3}\right).$$

If w is a zero of P , then so is $r(w)$ from (1) and then $P(r(r(w))) = 0$. Hence $(x-w)(x-r(w))(x-r(r(w)))|P$. But

$$Q_w(x) = (x-w)(x-r(w))(x-r(r(w))) = \left(x^3 - \frac{3}{4}x - w^3\right)$$

satisfies our conditions and hence so does $\frac{P}{Q_w}$. Continuing this operation we shall reach a constant at some point hence

$$P(x) = \prod Q_w(x) = R\left(x^3 - \frac{3}{4}x\right),$$

where $R(x) = \prod (x - w^3)$. We must have $R \in \mathbb{Q}[x]$ otherwise if a_k is the irrational coefficient at the smallest power k then the coefficient of x^k in $R\left(x^3 - \frac{3}{4}x\right)$ would be irrational. It is clear from the proof that for any $R \in \mathbb{Q}[x]$ the polynomial $P(x) = R\left(x^3 - \frac{3}{4}x\right)$ satisfies our hypothesis.

Example 16.6. Find all polynomials $P \in \mathbb{C}[X]$ with only real zeros that satisfy the identity

$$P(x)P(-x) = P(x^2 - 1)$$

for all $x \in \mathbb{R}$.

Solution. If r is a zero of P , then by setting $x = r$ we conclude that $g(r) = r^2 - 1$ is also a zero of P . Then $g(g(r))$ is also a zero of P and so on. As we may have a finite number of zeros, we may encounter a zero for a second time, so $g^{(n)}(s) = s$ for some $n \in \mathbb{N}$. Now let us find r . We have $g(r) - r = (r - u)(r - v)$, where $u = \frac{1 - \sqrt{5}}{2}$, $v = \frac{1 + \sqrt{5}}{2}$. Then $g(g(r)) - r = r(r + 1)(r - u)(r - v)$. If $r < -1$, then set $x = \sqrt{1 + r}$ to obtain that $\pm\sqrt{1 + r}$ is a zero of P . But it is not real so this case is not possible. If $r = -1$, then $g(r) = 0$, $g(g(r)) = -1$ and so $x(x + 1) | P$. If $r \in (-1; u)$, then $g(r) \in (u; 0)$ and $g(g(r)) \in (-1; u)$. But $g(g(r)) - r = r(r + 1)(r - u)(r - v) < 0$ so $g(g(r)) < r$. We repeat the reasoning with $g(g(r))$ and so on to obtain an infinite decreasing sequence of zeros of P in $(-1; u)$, a contradiction. If $r = u$, then $u - x | P$. If $r \in (u; 0)$, then $g(r) \in (-1; u)$ and we have shown no zero can occur in $(-1; u)$. If $r = 0$, then $g(r) = -1$ and $x(x + 1) | P$. If $0 < r < v$, then $\pm\sqrt{1 + r}$ is a zero of P . As P has no zeros less than -1 , it follows that $\sqrt{1 + r}$ is a zero of P . Also $r < \sqrt{1 + r}$, $\sqrt{1 + r} < v$ and we can build an increasing sequence of zeros of P in $(0; v)$. If $r = v$, then $v - x | P$. If $r > v$ then $g(r) > v$ is a zero of P and

continuing this operation we get an infinite increasing set of zeros of P greater than v . So all zeros of P can be $-1, 0, u, v$. As $x(x+1), u-x, v-x$ all satisfy the condition, we can divide P by any of them and repeat the reasoning to get that

$$P(x) = x^m(x+1)^m(u-x)^q(v-x)^r.$$

If P is a constant, then $P = 0$ or $P = 1$.

Example 16.7. Find all polynomials $f \in \mathbb{R}[X]$ such that

$$f(x^2 + x) = f(x)f(x+1)$$

for all $x \in \mathbb{R}$.

Solution. First note that the identity principle implies that

$$f(z^2 + z) = f(z)f(z+1)$$

for all $z \in \mathbb{C}$. Hence if z is a complex zero of f , then so is $z^2 + z$. Note also that

$$f(z-1)f(z) = f((z-1)^2 + z - 1) = f(z^2 - z)$$

and if z is a zero of f , then so is $z^2 - z$. On the other hand, the triangle inequality implies that

$$|z^2 + z| + |z^2 - z| \geq 2|z| \quad (1)$$

with equality only if $z = 0$. Suppose now that $z \neq 0$. Then (1) implies that

$$\max\{|z^2 + z|, |z^2 - z|\} > |z|$$

and we may generate an infinite sequence of distinct zeros which is a contradiction. Hence either f is a constant polynomial, or 0 is the only zero of f . Thus $f \equiv 0$, $f \equiv 1$ or $f(x) = x^n$ for some $n \in \mathbb{N}$.

Example 16.8. Let f be a rational function in x that satisfies

$$f(x) = f\left(\frac{1}{x}\right)$$

for all $x \neq 0$. Prove that f is a rational function in $x + \frac{1}{x}$.

Solution. For a polynomial P with $P(x) \neq 0$ set

$$P^*(x) = x^{\deg(P)} P\left(\frac{1}{x}\right).$$

We directly prove that $(PQ)^* = P^*Q^*$ and if $P(x) = a_n x^n + \dots + a_0$ then $P^*(x) = a_0 x^n + \dots + a_n$, where $a_0 a_n \neq 0$. Next, if $P = a_{2n} x^{2n} + \dots + a_0$ is a polynomial of degree $2n$ that satisfies $P = P^*$, we conclude $a_{n+k} = a_{n-k}$ and thus

$$\frac{P(x)}{x^n} = a_n + \sum_{i=1}^n a_{n-i} \left(x^i + \frac{1}{x^i}\right).$$

Now for all $k \in \mathbb{N}$ we see that $x^k + \frac{1}{x^k}$ is a polynomial in $x + \frac{1}{x}$. This is proven by induction on n : if we set $q_n = x^n + \frac{1}{x^n}$, then $q_{n+1} = q_n q_1 - q_{n-1}$ and from here it is clear how to show that q_n is a polynomial in q_1 . Therefore $\frac{P(x)}{x^n}$ is a polynomial in $x + \frac{1}{x}$. Finally let $f = \frac{g}{h}$ where g, h are co-prime polynomials. Set $\deg(g) = k, \deg(h) = l$. We can assume g, h are monic. We distinguish two cases:

a) $h(0) \neq 0$. Then let $g = x^m g_1(x)$, where $g_1(0) \neq 0$. We have

$$\frac{x^m g_1(x)}{h(x)} = \frac{g_1\left(\frac{1}{x}\right)}{x^m h\left(\frac{1}{x}\right)}$$

thus

$$x^{2m-k} g_1(x) h^*(x) = \frac{h(x) g_1^*(x)}{x^{l-m}}$$

so $x^{l+m-k} g_1(x) h^*(x) = g_1^*(x) h(x)$.

We get $l + m = k$ and $g_1(x) h^*(x) = g_1^*(x) h(x)$. Now as $(g_1, h) = 1$ we conclude $g_1 | g_1^*$, so $g_1^* = c g_1$ and $h^* = c h$. Then the zeros of g_1 group into pairs $w, \frac{1}{w}$ which consist of different numbers unless $w = \pm 1$. Hence the free coefficient of g_1 is ± 1 depending on whether 1 is a root of g_1 . So $c = \pm 1$. If $c = -1$, then 1 is a root of g_1 and analogously a root of h . This is impossible

since g and h are co-prime. So $c = 1$. Also note that $\deg(g_1) = \deg(h_1) - 2m$. We can suppose $\deg(g_1)$ and $\deg(h)$ are even because otherwise we can multiply g_1, h by $x + 1$ and still have $g_1^* = g_1, h^* = h$ because of the multiplicativity of $*$ and since $(x + 1)^* = x + 1$. Therefore $\frac{g_1(x)}{x^{\frac{1}{2}\deg(g_1)}}$ and $\frac{h(x)}{x^{\frac{1}{2}\deg(h)}}$ are polynomials in $x + \frac{1}{x}$ as we have proven above. Thus

$$\frac{g_1(x)}{h(x)} x^{\frac{1}{2}(\deg(h) - \deg(g_1))} = x^m \frac{g_1(x)}{h(x)} = f(x)$$

is a rational function in $x + \frac{1}{x}$, as desired.

b) $h(0) = 0$. Then $g(0) \neq 0$ as $(g, h) = 1$ and we repeat the arguments of a) for $\frac{1}{f}$.

Example 16.9. (Bulgaria 2006) Find all polynomials $P, Q \in \mathbb{R}[X]$ such that

$$\frac{P(x)}{Q(x)} - \frac{P(x+1)}{Q(x+1)} = \frac{1}{x(x+2)}$$

for all $x \in \mathbb{R}$ where the identity is defined.

Solution. Set $R(x) = \frac{P(x)}{Q(x)}$. Then

$$\begin{aligned} R(x) - R(x+n) &= \sum_{i=0}^{n-1} (R(x+i) - R(x+i+1)) = \sum_{i=0}^{n-1} \frac{1}{(x+i)(x+i+2)} \\ &= \frac{1}{2} \sum_{i=0}^{n-1} \left(\frac{1}{x+i} - \frac{1}{x+i+2} \right) = \frac{1}{2} \left(\frac{1}{x} + \frac{1}{x+1} - \frac{1}{x+n} - \frac{1}{x+n+1} \right). \end{aligned}$$

Therefore,

$$\lim_{n \rightarrow \infty} R(x+n) = R(x) - \frac{1}{2x} - \frac{1}{2(x+1)}.$$

Since this limit does not depend of x (why?), we conclude that

$$R(x) = c + \frac{1}{2x} + \frac{1}{2(x+1)}.$$

Thus

$$\frac{P(x)}{Q(x)} = \frac{P_0(x)}{Q_0(x)},$$

where

$$P_0(x) = x + \frac{1}{2} + cx(x+1), \quad Q_0(x) = x(x+1).$$

Since P_0 and Q_0 are relatively prime we have that $P(x) = R(x)P_0(x)$ and $Q(x) = R(x)Q_0(x)$, where R is an arbitrary nonzero polynomial and $c \in \mathbb{R}$. Conversely, the polynomials of these forms satisfy the given condition.

Remark. One can show the following:

Let $a \in \mathbb{R}$ and R be a rational function with real coefficients such that

$$R(x) - R(x+1) = \frac{1}{x(x+a)}$$

for all $x \neq 0, -a$. Then a is a nonzero integer. Moreover, if $a > 0$, then

$$R(x) = c + \frac{1}{a} \sum_{i=0}^{a-1} \frac{1}{x+i},$$

and if $a < 0$, then

$$R(x) = c - \frac{1}{a} \sum_{i=a}^{-1} \frac{1}{x+i}.$$

Example 16.10. (IMO 2004) Find all polynomials $f \in \mathbb{R}[X]$ such that

$$f(x-y) + f(y-z) + f(z-x) = 2f(x+y+z)$$

for all $x, y, z \in \mathbb{R}$ with $xy + yz + zx = 0$.

Solution. Set $x = y = 0$. Then $xy + yz + zx = 0$ and we get

$$f(0) + f(-z) + f(z) = 2f(z).$$

In particular, $3f(0) = 2f(0)$ which implies that $f(0) = 0$ and then $f(z) = f(-z)$ for all $z \in \mathbb{R}$. Hence the polynomial f is even, i.e. $f(x) = g(x^2)$ for some $g \in \mathbb{R}[x]$. Thus

$$g((x-y)^2) + g((y-z)^2) + g((z-x)^2) = 2g((x+y+z)^2) \quad (1)$$

for all $x, y, z \in \mathbb{R}$ with $xy + yz + zx = 0$. For each $r \in \mathbb{R}$ set $x = r(1 + \sqrt{3})$, $y = r$ and $z = r(1 - \sqrt{3})$. Then $xy + yz + zx = 0$ and it follows from (1) that

$$2g(3r^2) + g(12r^2) = 2g(9r^2). \quad (2)$$

It is clear that $f \equiv 0$ satisfies the given condition. Otherwise, set $n = \deg g$. Then comparing the leading coefficients in (2) gives $2 \cdot 3^n + 12^n = 2 \cdot 9^n$, i.e. $2 + 4^n = 2 \cdot 3^n$. Hence $n = 1$ or $n = 2$, since it follows by induction on n that $2 + 4^n > 2 \cdot 3^n$ for $n \geq 3$. Thus $g(x) = a_0 + a_1x + a_2x^2$ and $f(x) = g(x^2) = a_0 + a_1x^2 + a_2x^4$. Since $f(0) = 0$, we get that $f(x) = a_1x^2 + a_2x^4$ and it is easy to check that these polynomials satisfy the given condition.

16.2 Fermat's Theorem for Polynomials

As we noticed in the previous chapter the polynomials have many properties in common with positive integers such as Division theorem, unique decomposition into prime factors etc. This suggests considering functional equations for polynomials that are analogs of Diophantine equations. In general, these functional equations can be solved much easier than the respective Diophantine equations and this is best manifested by the Fermat equation $x^n + y^n = z^n$. The famous conjecture by Fermat that for $n \geq 3$ this equation has no solution in positive integers remained open for more than 300 years. It was proved by Andrew Wiles [26] in 1995 who used deep results from various mathematical fields. Our purpose here is to show that in contrast the non-solvability of the Fermat equation $f^n + g^n = h^n$ for polynomials can be proved by using elementary methods.

Theorem 16.1. The equation $f^n + g^n = h^n$, $n \geq 3$, has no solution in pairwise coprime polynomials at least one of which is not a constant.

We shall present two different proofs of the theorem following respectively [20] and [15].

First proof. This proof is an easy consequence of the following general result (cf. e.g. [16], [17]):

Theorem (Mason-Stothers). *Let a, b, c be pairwise co-prime polynomials with $a + b + c = 0$. Then their degrees are not greater than $N(abc) - 1$, where $N(abc)$ is the number of the distinct zeros of the polynomial abc .*

Proof of the Theorem. Set $p = \frac{a}{c}$ and $q = \frac{b}{c}$. Then $p + q + 1 = 0$ and differentiating this equation we get $p' = -q'$. Hence

$$\frac{b}{a} = \frac{q}{p} = -\frac{p'/p}{q'/q}$$

Let

$$a(x) = A \prod_{i=1}^k (x - \alpha_i)^{a_i}, \quad b(x) = B \prod_{i=1}^l (x - \beta_i)^{b_i}, \quad c(x) = C \prod_{i=1}^m (x - \gamma_i)^{c_i}.$$

Then it is easy to check that

$$\begin{aligned} \frac{p'}{p} &= \sum_{i=1}^k \frac{a_i}{x - \alpha_i} - \sum_{i=1}^m \frac{c_i}{x - \gamma_i}, \\ \frac{q'}{q} &= \sum_{i=1}^l \frac{b_i}{x - \beta_i} - \sum_{i=1}^m \frac{c_i}{x - \gamma_i}. \end{aligned}$$

Denote by d the monic polynomial whose zeros are the different zeros of the polynomial abc . Then $\frac{dp'}{p}$ and $\frac{dq'}{q}$ are polynomials whose degrees are not greater than $\deg d - 1 = N(abc) - 1$. On the other hand

$$\frac{b}{a} = -\frac{dp'/p}{dq'/q}$$

and since the polynomials a and b are co-prime it follows that a and b divide respectively $\frac{dp'}{p}$ and $\frac{dq'}{q}$. Hence their degrees are not greater than $N(abc) - 1$. The same reasoning shows that the statement is true for the polynomial c and the Theorem is proved. $\square$

Now we are ready to prove Theorem 16.1. To do this set $\alpha = \deg f$, $\beta = \deg g$, $\gamma = \deg c$ and apply the Mason-Stothers theorem to the polynomials $a = f^n$, $b = g^n$ and $c = -h^n$. Since a , b , c are pairwise co-prime, $\deg a = n\alpha$, $\deg b = n\beta$, $\deg c = n\gamma$, and $N(abc) = N((fgh)^n) = N(fgh) \leq \alpha + \beta + \gamma$ we get that $n\alpha \leq \alpha + \beta + \gamma - 1$, $n\beta \leq \alpha + \beta + \gamma - 1$, $n\gamma \leq \alpha + \beta + \gamma - 1$. Adding this inequalities yields $n(\alpha + \beta + \gamma) \leq 3(\alpha + \beta + \gamma - 1) < 3(\alpha + \beta + \gamma)$ and therefore $n < 3$. $\square$

Second proof. This proof uses the Fermat method of infinite descent. More precisely we shall show that if f , g , h are polynomials satisfying the conditions of the theorem, then there are polynomials f_0 , g_0 , h_0 with the same properties but such that

$$\max(\deg f_0, \deg g_0, \deg h_0) < \max(\deg f, \deg g, \deg h).$$

To do this, note first that the Fermat equation can be written in the form

$$f^n + g^n + (\varepsilon h)^n = 0,$$

where ε is a complex number with $\varepsilon^n = -1$. This shows that without loss of generality we may assume that $\max(\deg f, \deg g, \deg h) = \deg h = m$. Denote by $\varepsilon_1, \varepsilon_2, \dots, \varepsilon_n$ the n -th roots of -1 . Then the identity

$$z^n + 1 = (z - \varepsilon_1)(z - \varepsilon_2) \cdots (z - \varepsilon_n)$$

implies that

$$h^n = f^n + g^n = (f - \varepsilon_1 g)(f - \varepsilon_2 g) \cdots (f - \varepsilon_n g).$$

Since the polynomials $f - \varepsilon_i g$ are pairwise co-prime and their product is n -th power of a polynomial it follows from the unique factorization of polynomials into irreducible factors that $f - \varepsilon_i g = u_i^n$, $1 \leq i \leq n$, where u_i is a polynomial. Without loss of generality we may assume that $\max_{1 \leq i \leq n} (\deg u_i) = \deg u_1 = m_1$.

We shall show that $0 < m_1 < m$. It is obvious that $m_1 \leq m$. If $m_1 = m$, then the polynomials $f - \varepsilon_i g$, $2 \leq i \leq n$, are constants. Set $f - \varepsilon_2 g = A$, $f - \varepsilon_3 g = B$. Then subtracting these identities gives $g = \frac{B - A}{\varepsilon_2 - \varepsilon_3}$, which

contradicts to the condition that the polynomial g is not constant. Hence $m_1 < m$. The above reasoning shows also that $m_1 > 0$.

Set $c_1 = \frac{\varepsilon_2 - \varepsilon_3}{\varepsilon_2 - \varepsilon_1}$, $c_2 = \frac{\varepsilon_1 - \varepsilon_3}{\varepsilon_1 - \varepsilon_2}$. Then $c_1 + c_2 = 1$ and $c_1\varepsilon_1 + c_2\varepsilon_2 = \varepsilon_3$. Hence $c_1u_1^n + c_2u_2^n = c_1(f - \varepsilon_1g) + c_2(f - \varepsilon_2g) = f - \varepsilon_3g = u_3^n$ and setting $f_0 = \sqrt[n]{c_1}u_1$, $g_0 = \sqrt[n]{c_2}u_2$, $h_0 = u_3$ we get $f_0^n + g_0^n = h_0^n$. Moreover, it is easy to check that the polynomials f_0, g_0, h_0 are pairwise co-prime and at least one of them is not constant. Repeating the same reasoning for f_0, g_0, h_0 and so on, we obtain a strictly decreasing sequence of positive integers, a contradiction. $\square$

As noted in [20] the Mason-Stothers theorem has many interesting consequence. Here we shall consider only two of them.

Theorem 16.2. (Davenport) Let f and g be co-prime polynomials of positive degree and $h = f^3 - g^2 \neq 0$. Then $\deg f \leq 2 \deg h - 2$.

Proof. If $\deg f^3 \neq \deg g^2$, then

$$\deg(f^3 - g^2) \geq \deg f^3 = 3 \deg f \geq \frac{\deg f}{2} + 1.$$

Hence we may assume that $\deg f^3 = \deg g^2 = 6k$. Consider the polynomials $F = f^3, G = g^2$ and $H = F - G$. According to the Mason-Stothers theorem

$$\max(\deg F, \deg G, \deg H) \leq N(FGH) - 1 \leq \deg F + \deg G + \deg H - 1.$$

Hence $6k \leq 2k + 3k + \deg H - 1$, i.e. $\deg H \geq k + 1 = \frac{\deg f}{2} + 1$. $\square$

Theorem 16.3. The equation $f^a + g^b = h^c$ has solutions in pairwise co-prime polynomials if and only if (a, b, c) is a permutation of one of the triples $(1, m, n)$, $m, n \geq 1$, $(2, 2, n)$, $n \geq 2$, $(2, 3, 3)$, $(2, 3, 4)$ and $(2, 3, 5)$.

Proof. Without loss of generality we may assume that $2 \leq a \leq b \leq c$. Set $\alpha = \deg f$, $\beta = \deg g$ and $\gamma = \deg h$. Then it follows from the Mason-Stothers theorem that

$$\alpha a \leq \alpha + \beta + \gamma - 1, \tag{1}$$

$$\beta b \leq \alpha + \beta + \gamma - 1, \quad (2)$$

$$\gamma c \leq \alpha + \beta + \gamma - 1. \quad (3)$$

Hence

$$a(\alpha + \beta + \gamma) \leq \alpha a + \beta b + \gamma c \leq 3(\alpha + \beta + \gamma - 1) < 3(\alpha + \beta + \gamma)$$

and $a < 3$. Since $a \geq 2$ we conclude that $a = 2$. Now the inequality (1) is equivalent to

$$\alpha \leq \beta + \gamma - 1 \quad (4)$$

and adding up (2), (3) and (4) gives

$$\beta b + \gamma c \leq 3(\beta + \gamma) + \alpha - 3.$$

This together with (4) implies that

$$b(\beta + \gamma) \leq \beta b + \gamma c \leq 3(\beta + \gamma) + \alpha - 3 \leq 4(\beta + \gamma) - 3,$$

i.e. $b < 4$. Thus $b = 2$ or $b = 3$ and we have to prove that if $b = 3$, then $c \leq 5$. For $b = 3$ we get from (2) that $2\beta \leq \alpha + \gamma - 1$ which together with (4) gives $\beta \leq 2\gamma - 2$ and $\alpha \leq 3\gamma - 3$. Now it follows from (3) that

$$\gamma c \leq \alpha + \beta + \gamma - 1 \leq 6\gamma - 6,$$

i.e. $c \leq 5$.

To complete the proof of the theorem it remains to show that the given equation has a solution for each of the triples $(2, 2, 4)$, $(2, 3, 3)$, $(2, 3, 4)$ and $(2, 3, 5)$. The following examples are taken from [13]:

$$\left(\frac{x^4 + 1}{2}\right)^2 + \left(\frac{i(x^4 - 1)}{2}\right)^2 = x^4,$$

$$\left(\sqrt{12i\sqrt{3}}(x^5 - x)\right)^2 + \left(x^4 - 2i\sqrt{3}x^2 + 1\right)^3 = \left(x^4 + 2i\sqrt{3}x^2 + 1\right)^3,$$

$$\begin{aligned}
& (x^2 - 33x^8 - 33x^4 + 1)^2 + (-x^8 - 14x^4 - 1)^3 = \left(\sqrt{i\sqrt{108}}(x^5 - x) \right)^4, \\
& (x^{30} + 522(x^{25} - x^5) - 10005(x^{20} + x^{10}) + 1)^2 + \\
& (-x^{20} + 228(x^{15} - x^5) - 494x^{10} - 1)^3 = \left(\sqrt[5]{1728}x(x^{10} + 11x^5 - 1) \right)^5. \quad \square
\end{aligned}$$

We refer the reader to [13] where the relation of polynomials satisfying the equation $f^a + g^b = h^c$ and regular polyhedra is discussed.

16.3 Practice Problems

Problem 16.1. Let n be a positive integer. Find all polynomials P with real coefficients such that

$$P\left(x + \frac{1}{n}\right) + P\left(x - \frac{1}{n}\right) = 2P(x)$$

for all $x \in \mathbb{R}$.

Problem 16.2. (MR 2008, O. Furdui) Find all monic polynomials P and Q with real coefficients such that

$$P(1) + P(2) + \cdots + P(n) = Q(1 + 2 + \cdots + n)$$

for all $n \geq 1$.

Problem 16.3. Find all polynomials $F, G \in \mathbb{R}[X]$ such that

$$F(G(x)) = F(x)G(x)$$

for all $x \in \mathbb{R}$.

Problem 16.4. (Putnam 2010) Find all pairs of polynomials $P(x)$ and $Q(x)$ with real coefficients for which

$$P(x)Q(x+1) - P(x+1)Q(x) = 1$$

for all $x \in \mathbb{R}$.

Problem 16.5. (Bulgaria 2001) Find all polynomials $P \in \mathbb{R}[X]$ such that

$$P(x)P(x+1) = P(x^2)$$

for all $x \in \mathbb{R}$.

Problem 16.6. (IMO Shortlist 1979) Find all polynomials $P \in \mathbb{R}[X]$ such that

$$P(x)P(2x^2) = P(2x^3 + x)$$

for all $x \in \mathbb{R}$.

Problem 16.7. (Romania 1990) Find all polynomials $P \in \mathbb{R}[X]$ such that

$$2P(2x^2 - 1) = P(x)^2 - 2$$

for all $x \in \mathbb{R}$.

Problem 16.8. Let $k, l \in \mathbb{N}$. Find all nonconstant polynomials $P \in \mathbb{R}[X]$ such that

$$xP(x-k) = (x-l)P(x)$$

for all $x \in \mathbb{R}$.

Problem 16.9. (H. Shapiro) Find all polynomials $P \in \mathbb{R}[X]$ that satisfy the identity

$$P(x)P(x+1) = P(x^2 + x + 1)$$

for all $x \in \mathbb{R}$.

Problem 16.10. Find all polynomials $P \in \mathbb{C}[X]$ that satisfy the identity

$$P(x)P(-x) = P(x^2)$$

for all $x \in \mathbb{C}$.

Problem 16.11. (Putnam 2003) Do there exist polynomials $A(x)$, $B(x)$, $C(y)$, $D(y)$ such that

$$1 + xy + x^2y^2 = A(x)C(y) + B(x)D(y)$$

holds identically?

Problem 16.12. Find all polynomials $P \in \mathbb{C}[x]$ such that

$$P(x^2 - y^2) = P(x - y)P(x + y)$$

for all $x, y \in \mathbb{C}$.

Problem 16.13. Let $a, b \in \mathbb{R}$, $(a, b) \neq (0, 0)$. Find all polynomials $P(x, y) \in \mathbb{R}[x, y]$ that satisfy

$$P(x + a, y + b) = P(x, y)$$

for all $x, y \in \mathbb{R}$.

Problem 16.14. (IMO 1975) The function $f(x, y)$ is a homogeneous polynomial of degree n in x and y . If $f(1, 0) = 1$ and for all a, b, c ,

$$f(a + b, c) + f(b + c, a) + f(c + a, b) = 0,$$

prove that $f(x, y) = (x - 2y)(x + y)^{n-1}$.

Problem 16.15. (MR 6, 2008, M. Beckeanu, T. Dimitrescu) Let $P \in \mathbb{Z}[X]$ be a polynomial such that $P(1) = P(-1)$. Prove that there is a polynomial $Q(x, y)$ with integer coefficients such that

$$P(t) = Q(t^2 - 1, t^3 - t)$$

for all $t \in \mathbb{R}$.

Problem 16.16. Let $P \in \mathbb{R}[X]$ be a polynomial such that $P(\sin t) = P(\cos t)$ for all real t . Prove that there exists a polynomial $Q \in \mathbb{R}[X]$ such that $P(x) = Q(x^4 - x^2)$ for all $x \in \mathbb{R}$.

Problem 16.17. (G. Dospinescu) Find all polynomials $f(x, y, z)$ with real coefficients such that

$$f\left(a + \frac{1}{a}, b + \frac{1}{b}, c + \frac{1}{c}\right) = 0$$

whenever $abc = 1$.

Problem 16.18. Prove that the equation $f^4 + g^4 = h^2$ has only trivial solutions in $\mathbb{C}[X]$.

Problem 16.19. Prove that there are no nonconstant polynomials $f, g \in \mathbb{C}[X]$ such that $f^3 - g^2$ is a nonzero constant.

Problem 16.20. Prove that the equation $f^2 = g^3 + g$ has only trivial solutions in rational functions with complex coefficients.

Chapter 17

Functional Inequalities

17.1 Examples

This section deals with problems whose conditions involve inequalities, rather than equalities, involving the values of a certain function. An inequality is much weaker than an equality, it does not pin down a number exactly. So it is more difficult to determine a function from an inequality condition. But in certain cases, we can.

One obvious possibility is when the equation has no solutions at all - then we just seek to obtain a contradiction. Another is to use a substitution to turn the inequality into an equality - for example if we get a condition of the form $E^2 \leq 0$ then this means $E = 0$.

The last, and most important, way is to guess a solution f_0 and then to show $f = f_0$. If $f(x) \neq f_0(x)$ when we use this to produce a counter-example to the inequality. For example, assume that $f: \mathbb{R} \rightarrow \mathbb{R}$ is a function such that $f(2x) = 2f(x)$ and $|f(x) - x| \leq 1$ for all $x \in \mathbb{R}$. We can guess the solution $f(x) = x$ and this is the only solution: if $f(x_0) \neq x_0$ then $f(x_0) = x_0 + \epsilon$ for $\epsilon \neq 0$. Then $f(2^n x_0) = 2^n x_0 + 2^n \epsilon$ and choose n such that $2^n |\epsilon| > 1$. This will contradict $|f(2^n x_0) - 2^n x_0| \leq 1$. Thus $f(x_0) = x_0$.

Notice that we also have an equality to accompany the inequality in the condition, this is often the case. But sometimes we can do with just an inequality.

Example 17.1. (Vietnam 1991) Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$\frac{1}{2}f(xy) + \frac{1}{2}f(xz) - f(x)f(yz) \geq \frac{1}{4}$$

for all $x, y, z \in \mathbb{R}$.

Solution. If we set $x = y = z = 1$ the condition is equivalent to

$$-\left(f(1) - \frac{1}{2}\right)^2 \geq 0$$

possible only for $f(1) = \frac{1}{2}$. Next set $y = z = 1$ to get $f(x) - \frac{1}{2}f(x) \geq \frac{1}{4}$ so $f(x) \geq \frac{1}{2}$. Set $y = z = \frac{1}{x}, x \neq 0$ so get $f(1) - f(x)f\left(\frac{1}{x^2}\right) \geq \frac{1}{4}$. Hence $f(x)f\left(\frac{1}{x^2}\right) \leq \frac{1}{4}$. But $f(x) \geq \frac{1}{2}, f\left(\frac{1}{x^2}\right) \geq \frac{1}{2}$ and so $f(x)f\left(\frac{1}{x^2}\right) \geq \frac{1}{4}$. Thus the equality holds and $f(x) = \frac{1}{2}$ for $x \neq 0$. Next set $x = y = z = 0, x \neq 0$ to get $f(0) - f(0)^2 \geq \frac{1}{4}$, i.e. $f(0) = \frac{1}{2}$. As we have proven $f(x) \geq \frac{1}{2}$ we get $f(0) = \frac{1}{2}$. So $f(x) = \frac{1}{2}$ which satisfies the condition.

Example 17.2. Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x + 19) \leq f(x) + 19,$$

$$f(x + 94) \geq f(x) + 94$$

for all $x \in \mathbb{R}$.

Solution. It follows by induction that for all $m, n \in \mathbb{N}$ we have

$$f(x + 19m) \leq f(x) + 19m,$$

$$f(x + 94n) \geq f(x) + 94n.$$

In particular, $f(x + 95) = f(x + 19.5) \leq f(x) + 95$ and

$$f(x + 95) = f(x + 1 + 94) \geq f(x + 1) + 94$$

which imply that $f(x+1) \leq f(x)+1$. We have also $f(x+94.18) \geq f(x)+94.18$ and $f(x+1+19.89) \leq f(x+1)+19.89$. But $18.94 = 1692 = 19.89+1$ and the above inequalities imply that $f(x+1) \geq f(x)+1$. Thus $f(x+1) = f(x)+1$ and we get by induction on n that $f(x+n) = f(x)+n$ for all $x \in \mathbb{R}$ and $n \in \mathbb{N}$. For every real number x denote by $[x]$ and $\{x\}$ its integral and fractional part, respectively. Then $f(x) = f(\{x\} + [x]) = f(\{x\}) + [x]$ which shows that every function f satisfying the given inequalities has the form $f(x) = g(\{x\}) + [x]$, where $g : [0, 1] \rightarrow \mathbb{R}$ is an arbitrary function.

Example 17.3. (Belarus 1997) Let $f : (0, \infty) \rightarrow (0, \infty)$ be a function such that $f(2x) \geq x + f(f(x))$ for all $x \in (0, \infty)$. Prove that $f(x) \geq x$ for all $x \in (0, \infty)$.

Solution. First note that $f(x) > \frac{x}{2}$. Suppose that $f(x) > ax$ for all $x \in (0, \infty)$, where $a > 0$ is a constant. Then

$$f(x) \geq \frac{x}{2} + f\left(f\left(\frac{x}{2}\right)\right) > \frac{x}{2} + af\left(\frac{x}{2}\right) > \frac{1+a^2}{2}x.$$

This shows that $f(x) > a_n x$ for all $n \in \mathbb{N}$, where the sequence $\{a_n\}_{n=1}^{\infty}$ is defined by

$$a_1 = \frac{1}{2}, a_{n+1} = \frac{1+a_n^2}{2}, n \geq 1.$$

Then

$$a_{n+1} - a_n = \frac{(1-a_n)^2}{2} \geq 0,$$

i.e. the sequence is monotone increasing. Moreover, it follows by induction on n that $a_n < 1$ for all $n \in \mathbb{N}$. Hence the sequence is convergent and denote by a its limit. Then $a = \frac{1+a^2}{2}$, i.e. $a = 1$. Now letting $n \rightarrow \infty$ in the inequality $f(x) > a_n x$ gives $f(x) \geq x$.

Example 17.4. Find all functions $f : [1, \infty) \rightarrow [1, \infty)$ given that

$$f(x) \leq 2(1+x) \text{ and } xf(x+1) = f(x)^2 - 1$$

for all $x \geq 1$.

Solution. We can guess the solution $f(x) = x + 1$ and now we try to prove it is the only one. As in many other situations we assume that $f(x_0) \neq x_0 + 1$ and try to obtain a x such that $f(x) < 1$ or $f(x) > 2(1 + x)$. Indeed, we observe that $xf(x + 1) = f(x)^2 - 1$ can be interpreted as a recursive relation on $a_n = f(n + x_0)$ by $a_{n+1} = \frac{a_n^2 - 1}{n + x_0}$. Consider now $b_n = \frac{a_n}{n + 1 + x_0}$. Then

$$b_{n+1} = \frac{(n + 1 + x_0)^2 b_n^2 - 1}{(n + x_0)(n + 2 + x_0)} = b_n^2 + \frac{b_n^2 - 1}{(n + 2)(n + 2 + x_0)}.$$

If $b_0 > 1$ then we prove by induction that $b_n > 1$ and then $b_{n+1} > b_n^2$ which implies that $b_n > 2$ for some n . Hence $f(n + x_0) > 2(1 + n + x_0)$, contradiction. If $b_0 < 1$ then we prove by induction that $b_n < 1$ and therefore $b_{n+1} < b_n^2$. So $b_n < b_0^{2^n}$ and $\frac{1}{b_n} > \left(\frac{1}{b_0}\right)^{2^n}$. However $\frac{1}{b_n} = \frac{n + 1 + x_0}{f(n + x_0)} < n + 1 + x_0$ and we get $\left(\frac{1}{b_0}\right)^{2^n} < n + 1 + x_0$. On the other hand, $\frac{1}{b_0} > 1$ and setting $\frac{1}{b_0} = 1 + x$, $x > 0$ gives $1 + 2^n x < (1 + x)^{2^n} < n + 1 + x_0$, i.e. $0 < x < \frac{n + x_0}{2^n}$ for all $n \in \mathbb{N}$. But this is a contradiction since $\lim_{n \rightarrow \infty} \frac{n + x_0}{2^n} = 0$.

So $b_0 = 1$ and hence $f(x_0) = x_0 + 1$. As x_0 was picked at random, $f(x) = x + 1$.

Example 17.5. Let $f: [0, 1] \rightarrow \mathbb{R}$ be a function such that $f(0) = f(1) = 0$ and

$$f\left(\frac{x + y}{2}\right) \leq f(x) + f(y)$$

for all $x, y \in [0, 1]$.

(i) Prove that $f(x) \geq 0$ for all $x \in [0, 1]$ and that f has infinitely many zeros.

(ii) Construct a nonconstant function f with the given properties.

Solution. (i) Suppose that $f(\alpha) = -a < 0$ for some $\alpha \in (0, 1)$. Then

$$f\left(\frac{\alpha}{2}\right) \leq f(\alpha) + f(0) = -a,$$

$$f\left(\frac{\alpha}{4}\right) \leq f\left(\frac{\alpha}{2}\right) + f(0) \leq -a,$$

$$f\left(\frac{3\alpha}{4}\right) \leq f\left(\frac{\alpha}{2}\right) + f(\alpha) \leq -2a < -a.$$

Since

$$f\left(\frac{\alpha}{2}\right) \leq f\left(\frac{\alpha}{2} - \frac{\alpha}{2^n}\right) + f\left(\frac{\alpha}{2} + \frac{\alpha}{2^n}\right),$$

$$f\left(\frac{\alpha}{2} - \frac{\alpha}{2^{n+1}}\right) \leq f\left(\frac{\alpha}{2}\right) + f\left(\frac{\alpha}{2} - \frac{\alpha}{2^n}\right)$$

$$f\left(\frac{\alpha}{2} + \frac{\alpha}{2^{n+1}}\right) \leq f\left(\frac{\alpha}{2}\right) + f\left(\frac{\alpha}{2} + \frac{\alpha}{2^n}\right),$$

it follows by induction on $n \geq 1$ that

$$f\left(\frac{\alpha}{2}\right) \leq -2^{n-1}a, \quad f\left(\frac{\alpha}{2} - \frac{\alpha}{2^n}\right) \leq -2^{n-1}a, \quad f\left(\frac{\alpha}{2} + \frac{\alpha}{2^n}\right) < -2^{n-1}a.$$

Since the first inequality cannot hold for all $n \in \mathbb{N}$, we get a contradiction.

It follows also by induction on n that $f\left(\frac{k}{2^n}\right) = 0$ for all $n \in \mathbb{N}$ and $k = 0, 1, \dots, 2^n$.

(ii) Define $f(x) = 0$ if $x = \frac{k}{2^n}$, $n \in \mathbb{N}$, $k = 0, 1, \dots, 2^n$, and $f(x) = 1$ otherwise. Then it is easy to check that f has the required properties.

Example 17.6. Prove that there is no a function $f: \mathbb{R} \rightarrow \mathbb{R}$ such that $f(0) > 0$ and

$$f(x+y) \geq f(x) + yf(f(x))$$

for all $x, y \in \mathbb{R}$.

Solution. Suppose that there is a function f with the given properties. If $f(f(x)) \leq 0$ for all $x \in \mathbb{R}$ then

$$f(x+y) \geq f(x) + yf(f(x)) \geq f(x)$$

for all $y \leq 0$ and the function f is decreasing. Now the inequalities $f(0) > 0 \geq f(f(x))$ imply $f(x) > 0$ for all x , a contradiction to $f(f(x)) \leq 0$. Hence there exists z such that $f(f(z)) > 0$. Then the inequality

$$f(z+x) \geq f(z) + xf(f(z))$$

shows that $\lim_{x \rightarrow +\infty} f(x) = +\infty$ and therefore $\lim_{x \rightarrow \infty} f(f(x)) = +\infty$. In particular, there exist $x, y > 0$ such that

$$f(x) \geq f(f(x)) > 1, \quad y \geq \frac{x+1}{f(f(x))-1}, \quad f(f(x+y+1)) \geq 0.$$

Then

$$f(x+y) \geq f(x) + yf(f(x)) \geq x+y+1$$

and therefore

$$\begin{aligned} f(f(x+y)) &\geq f(x+y+1) + (f(x+y) - (x+y+1))f(f(x+y+1)) \\ &\geq f(x+y+1) \geq f(x+y) + f(f(x+y)) \\ &\geq f(x) + yf(f(x)) + f(f(x+y)) > f(f(x+y)), \end{aligned}$$

a contradiction.

Remark. Note that the only function $f: \mathbb{R} \rightarrow \mathbb{R}$ with $f(0) = 0$ and satisfying the given inequality is the constant 0. Indeed, as in the second part of the above solution we conclude that $f(f(x)) \leq 0$ for all $x \in \mathbb{R}$. On the other hand, setting $x = 0$ in the initial inequality gives $f(y) \geq 0$ for all y . Hence $f(x+y) \geq f(x)$ for all $x, y \in \mathbb{R}$ which easily implies that $f(x) = 0$ for all x .

It is not known to the authors if there is a function $f: \mathbb{R} \rightarrow \mathbb{R}$ with $f(0) < 0$ and satisfying the required inequality.

Example 17.7. (Bulgaria 2008) Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x+y^2) \geq (y+1)f(x) \text{ for all } x, y \in \mathbb{R}.$$

First Solution. Since $f(x) \geq 0 \cdot f(x-1) = 0$, then

$$f(x+y^2) \geq (y+1)f(x) \geq f(x)$$

for $y \geq 0$ and hence f is an increasing function. Let $a_0 = b_0 = 0$,

$$a_n = \sum_{i=1}^n \frac{1}{i^2} \text{ and } b_n = \sum_{i=1}^n \frac{1}{i}.$$

Since $f(x + y^2) - f(x) \geq yf(x)$, then

$$f(x + a_{k+1}) - f(x + a_k) \geq \frac{f(x + a_k)}{k + 1} \geq \frac{f(x)}{k + 1}.$$

Adding up these inequalities for $k = 0, 1, \dots, n - 1$ gives

$$f(x + a_n) - f(x) \geq b_n f(x).$$

It is well-known that $a_n < 2$ for all n and $b_n \rightarrow \infty$ (prove!). Then

$$f(x + 2) \geq (1 + b_n)f(x)$$

and letting $n \rightarrow \infty$ shows that $f(x) \leq 0$. On the other hand, $f(x) \geq 0$ and hence $f(x) = 0$ for all x . Obviously this function satisfies the given condition.

Remark. After proving non-negativity and monotonicity of f , non-positivity of f can be also shown in the following way. We have that

$$\frac{f(x + y^2) - f(x)}{y^2} \geq \frac{f(x)}{y}, \quad y \neq 0.$$

It is well-known that any monotone function is differentiable almost everywhere – in particular, for all $x \in A$, where A is an unbounded from above subset of $\mathbb{R}$. For $x \in A$ and $y \rightarrow 0+$, it follows that $f(x) \leq 0$. It remains to use the fact that for each $x \in \mathbb{R}$ there is $y > 0$ such that $x + y^2 \in A$ and then $0 \geq f(x + y^2) \geq (y + 1)f(x) \geq f(x)$.

Second Solution. We prove as above that $f(x) \geq 0$. For $x > 0$ and $k = 0, 1, \dots, n - 1$ we have that

$$f\left(x\left(2 - \frac{k}{n}\right)\right) \geq \left(1 + \sqrt{\frac{x}{n}}\right) f\left(x\left(2 - \frac{k+1}{n}\right)\right).$$

Multiplying these inequalities gives

$$f(2x) \geq \left(1 + \sqrt{\frac{x}{n}}\right)^n f(x).$$

Since $\left(1 + \frac{t}{n}\right)^n \rightarrow e^t$, then $\left(1 + \sqrt{\frac{x}{n}}\right)^n \rightarrow +\infty$ for $x > 0$. Hence $f(x) \leq 0$ for $x > 0$. For each x we may choose $y > 0$ such that $x + y^2 > 0$ and using the given condition we get that $f(x) \leq 0$. It follows that $f(x) = 0$ for all x .

17.2 Practice Problems

Problem 17.1. (Romania 2011) Find all functions $f : [0, 1] \rightarrow \mathbb{R}$ such that

$$|x - y|^2 \leq |f(x) - f(y)| \leq |x - y|$$

for all $x, y \in [0, 1]$.

Problem 17.2. (Romania TST 2007). Prove that any function $f : \mathbb{Q} \rightarrow \mathbb{R}$ such that

$$|f(x) - f(y)| \leq (x - y)^2$$

for all $x, y \in \mathbb{Q}$ is constant.

Problem 17.3. (Bulgaria 2008) Let $f : (0, \infty) \rightarrow (0, \infty)$ be a function such that

$$2f(x^2) \geq xf(x) + x$$

for all $x > 0$. Prove that $f(x^3) \geq x^2$ for all $x > 0$.

Problem 17.4. (China 1990) Let $f : [0, \infty) \rightarrow \mathbb{R}$ be a function such that

$$f(x)f(y) \leq y^2 f\left(\frac{x}{2}\right) + x^2 f\left(\frac{y}{2}\right)$$

for all $x, y \geq 0$ and $|f(x)| \leq M$ for $0 \leq x \leq 1$, where M is a fixed constant. Prove that $f(x) \leq x^2$ for all $x \geq 0$.

Problem 17.5. (G. Dospinescu) Find all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ such that

$$f(n+1) > \frac{f(n) + f(f(n))}{2}$$

for all $n \in \mathbb{N}$.

Problem 17.6. (Romania 2001) Prove that there is no function $f : (0, \infty) \rightarrow (0, \infty)$ such that

$$f(x+y) \geq f(x) + yf(f(x))$$

for all $x, y \in (0, \infty)$.

Problem 17.7. Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x) \geq x + 1 \text{ and } f(x + y) \geq f(x)f(y)$$

for all $x, y \in \mathbb{R}$.

Problem 17.8. Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x + y) \geq (y + 1)f(x)$$

for all $x, y \in \mathbb{R}$.

Problem 17.9. (Hungary 2008). Determine all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfying for all real numbers x and y the relation $f(xy) \leq xf(y)$.

Problem 17.10. (Bulgaria 1998) Prove that there is no function $f: (0, \infty) \rightarrow (0, \infty)$ such that

$$f(x)^2 \geq f(x + y)(f(x) + y)$$

for all $x, y \in (0, \infty)$.

Problem 17.11. Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ that are continuous at 0 and

$$f(2x) - 3x^2 - x \leq f(x) \leq f(3x) - 8x^2 - 2x$$

for all $x \in \mathbb{R}$.

Problem 17.12. (ZIMO 2009) Find all real numbers a such that there exists a function $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfying the inequality

$$x + af(y) \leq y + f(f(x))$$

for all $x, y \in \mathbb{R}$.

Problem 17.13. (IMO 2009) Find all functions $f: \mathbb{N} \rightarrow \mathbb{N}$ such that $x, f(y)$ and $f(y + f(x) - 1)$ are the sides of a triangle for all $x, y \in \mathbb{N}$.

Problem 17.14. (IMO 2011) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be such a function that

$$f(x + y) \leq yf(x) + f(f(x))$$

for all $x, y \in \mathbb{R}$. Prove that $f(x) = 0$ for all $x \leq 0$.

Problem 17.15. (IMAR 2009) Show that for each function $f : (0, \infty) \rightarrow (0, \infty)$ there exist $x, y \in (0, \infty)$ such that $f(x + y) < yf(f(x))$.

Problem 17.16. (IMO Shortlist 2009) Prove that for each function $f : \mathbb{R} \rightarrow \mathbb{R}$ there exist $x, y \in \mathbb{R}$ such that $f(x - f(y)) > yf(x) + x$.

Chapter 18

Miscellaneous Problems

18.1 Inductive Arguments

Problem 18.1. Find all functions $f: \mathbb{N} \rightarrow \mathbb{R}$ such that $f(1) \neq 0$ and

$$f(1)^2 + f(2)^2 + \dots + f(n)^2 = f(n)f(n+1)$$

for all $n \in \mathbb{N}$.

Problem 18.2. Find all functions $f: \mathbb{N} \rightarrow \mathbb{N}$ such that

$$f(1)^3 + f(2)^3 + \dots + f(n)^3 = (f(1) + f(2) + \dots + f(n))^2$$

for all $n \in \mathbb{N}$.

Problem 18.3. (IMO 1981) Assume that $f(x, y)$ is defined for all nonnegative integers x and y , and that the following equations are satisfied:

- (i) $f(0, y) = y + 1$;
 - (ii) $f(x + 1, 0) = f(x, 1)$;
 - (iii) $f(x + 1, y + 1) = f(x, f(x + 1, y))$.
- Determine $f(4, 1981)$.

Problem 18.4. (Putnam 1992) Prove that $f(n) = 1 - n$ is the only integer-valued function defined on the integers that satisfies the following conditions:

- (i) $f(f(n)) = n$ for all integers n ;
- (i) $f(f(n+2)+2) = n$ for all integers n ;
- (iii) $f(0) = 1$.

Problem 18.5. Find all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ such that

$$f^{(19)}(n) + 97f(n) = 98n + 232$$

for all $n \in \mathbb{N}$.

Problem 18.6. (IMO Shortlist 2011) Determine all pairs (f, g) of functions from the set of positive integers to itself that satisfy

$$f^{(g(n)+1)}(n) + g^{(f(n))}(n) = f(n+1) - g(n+1) + 1$$

for every positive integer n .

Problem 18.7. (Canada 2002) Find all functions $f : \mathbb{N}_0 \rightarrow \mathbb{N}_0$ such that

$$mf(n) + nf(m) = (m+n)f(m^2+n^2)$$

for all $m, n \in \mathbb{N}_0$.

Problem 18.8. Find all functions $f : \mathbb{Z} \rightarrow \mathbb{Z}$ that obey

$$f(m+n) + f(mn-1) = f(m)f(n) + 2$$

for all $m, n \in \mathbb{Z}$.

Problem 18.9. (Nordic Contest 1999) A function $f : \mathbb{N} \rightarrow \mathbb{R}$ satisfies for some positive integer m the conditions

$$f(m) = f(1995), f(m+1) = 1996, f(m+2) = 1997 \text{ and } f(n+m) = \frac{f(n)-1}{f(n)+1}$$

for all $n \in \mathbb{N}$. Prove that $f(n+4m) = f(n)$ and find the least m for which such a function f exists.

Problem 18.10. Suppose $f: \mathbb{Q} \rightarrow \{0, 1\}$ is such that $f(1) = 1, f(0) = 0$ and if $f(x) = f(y)$, then $f\left(\frac{x+y}{2}\right) = f(x) = f(y)$. Prove that $f(x) = 1$ whenever $x \geq 1$.

Problem 18.11. (Nordic Contest 1998) Find all functions $f: \mathbb{Q} \rightarrow \mathbb{Q}$ such that

$$f(x+y) + f(x-y) = 2(f(x) + f(y))$$

for all $x, y \in \mathbb{Q}$.

Problem 18.12. (Iran 1995) Find all functions $f: \mathbb{Z} \setminus \{0\} \rightarrow \mathbb{Q}$ such that

$$f\left(\frac{x+y}{3}\right) = \frac{f(x) + f(y)}{2}$$

for all $x, y \in \mathbb{Z} \setminus \{0\}$ such that $\frac{x+y}{3} \in \mathbb{Z} \setminus \{0\}$.

Problem 18.13. (IMO 1982) The function $f(n)$ is defined for all positive integers n and takes on nonnegative integer values. Also, for all m, n ,

$$f(m+n) - f(m) - f(n) = 0 \text{ or } 1;$$

$$f(2) = 0, f(3) > 0 \text{ and } f(9999) = f(3333).$$

Determine $f(1982)$.

Problem 18.14. (Romania TST 2004) Find all injective functions $f: \mathbb{N} \rightarrow \mathbb{N}$ such that

$$f(f(n)) \leq \frac{n + f(n)}{2}$$

for all $n \in \mathbb{N}$.

Problem 18.15. (Romania 1986) Find all surjective functions $f: \mathbb{N} \rightarrow \mathbb{N}$ such that

$$f(n) \geq n + (-1)^n$$

for all $n \in \mathbb{N}$.

Problem 18.16. Find all functions $f: \mathbb{N} \rightarrow \mathbb{N}$ such that $f(1) = 1$ and

$$f(n+1) = \left[f(n) + \sqrt{f(n)} + \frac{1}{2} \right]$$

for all $n \in \mathbb{N}$.

Problem 18.17. (IMO Shortlist 2001) Find all functions $f: \mathbb{N}_0^3 \rightarrow \mathbb{R}$ that satisfy $f(p, q, r) = 0$ if $pqr = 0$ and

$$\begin{aligned} f(p, q, r) = 1 + \frac{1}{6} & (f(p+1, q-1, r) + f(p-1, q+1, r) + f(p-1, q, r+1) \\ & + f(p+1, q, r-1) + f(p, q+1, r-1) + f(p, q-1, r+1)) \end{aligned}$$

otherwise.

Problem 18.18. (IMO Shortlist 1988) Find all functions $f: \mathbb{N} \rightarrow \mathbb{N}$ such that

$$f(f(m) + f(n)) = m + n$$

for all $m, n \in \mathbb{N}$.

Problem 18.19. Find all functions $f: \mathbb{N} \rightarrow \mathbb{N}$ that satisfy

$$f(m^2 + f(n)) = f(m)^2 + n$$

for all $m, n \in \mathbb{N}$.

Problem 18.20. Find all functions $f: \mathbb{N}_0 \rightarrow \mathbb{N}_0$ that satisfy

$$f(f(m)^2 + f(n)^2) = m^2 + n^2$$

for all $m, n \in \mathbb{N}_0$.

Problem 18.21. (BMO 2009) Find all functions $f: \mathbb{N} \rightarrow \mathbb{N}$ such that

$$f(f(m)^2 + 2f(n)^2) = m^2 + 2n^2$$

for all $m, n \in \mathbb{N}$.

Problem 18.22. (Korea 1998) Find all functions $f: \mathbb{N}_0 \rightarrow \mathbb{N}_0$ that satisfy

$$2f(m^2 + n^2) = f(m)^2 + f(n)^2$$

for all $m, n \in \mathbb{N}_0$.

Problem 18.23. Find all functions $f: \mathbb{Z} \rightarrow \mathbb{R}$ satisfying

$$f(a^3 + b^3 + c^3) = f(a)^3 + f(b)^3 + f(c)^3$$

whenever $a, b, c \in \mathbb{Z}$.

Problem 18.24. Find all functions $f: \mathbb{Q}^+ \rightarrow \mathbb{Q}^+$ such that

$$f(m^{2010} + f(n)) = m^{10} f(m^{50})^{40} + n$$

for all $m, n \in \mathbb{Q}^+$.

Problem 18.25. (IMO Shortlist 2010) Determine all functions $f: \mathbb{Q}^+ \rightarrow \mathbb{Q}^+$ which satisfy the equation

$$f(f(x)^2 y) = x^3 f(xy)$$

for all $x, y \in \mathbb{Q}^+$.

18.2 Basic Properties of Functions

Problem 18.26. (Iran 1998) Let $f: (0, \infty) \rightarrow (0, \infty)$ be a decreasing function that satisfies

$$f(x + y) + f(f(x) + f(y)) = f(f(x + f(y)) + f(y + f(x)))$$

for all $x, y \in (0, \infty)$. Show that $f(f(x)) = x$.

Problem 18.27. (Bulgaria 2011) Determine all functions $f: \mathbb{Q} \rightarrow \mathbb{R}$ such that

$$(f(x) + f(y) - 2f(xy)) \cdot (f(x) + f(z) - 2f(xz)) \geq 0$$

for all $x, y, z \in \mathbb{Q}$.

Problem 18.28. Find all functions $f : \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}$ such that

$$f(x^2)(f(x)^2 + f\left(\frac{1}{y^2}\right)) = 1 + f\left(\frac{1}{xy}\right)$$

for all $x, y \in \mathbb{R} \setminus \{0\}$.

Problem 18.29. (Bulgaria 2011) Prove that for every nonconstant function $f : \mathbb{Q}^+ \rightarrow \mathbb{R}$ there exist $x, y, z \in \mathbb{Q}^+$, such that

$$(f(x) + f(y) - 2f(xy)) \cdot (f(x) + f(z) - 2f(xz)) < 0.$$

Problem 18.30. Is there a function $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(xy) = \max\{f(x), y\} + \min\{f(y), x\}$$

for all $x, y \in \mathbb{R}$?

Problem 18.31. Prove that there is no function $f : (0, \infty) \rightarrow (0, \infty)$ satisfying the following conditions:

- (i) $f(x + f(y)) = f(x)f(y)$ for all $x, y > 0$;
- (ii) there are at most finitely many x with $f(x) = 1$.

Problem 18.32. (IMO Shortlist 1997) Prove that if the function $f : \mathbb{R} \rightarrow \mathbb{R}$ is such that $|f(x)| \leq 1$ and

$$f(x) + f\left(x + \frac{13}{42}\right) = f\left(x + \frac{1}{6}\right) + f\left(x + \frac{1}{7}\right)$$

for all $x \in \mathbb{R}$, then it is periodic.

Problem 18.33. (Romania TST 2007) Let $f \in \mathbb{R}[X]$ be a polynomial of degree n . Prove that:

- (i) f can not be written as a sum of at most n periodic functions;
- (ii) if $n = 1$, then f can be written as a sum of two real periodic functions;
- (iii) if $n = 1$ and f is the sum of two periodic functions, then they are unbounded on every interval;
- (iv) f can be written as a sum of $n + 1$ periodic functions;
- (v) there are real functions which can't be written as a finite sum of periodic functions.

Problem 18.34. (Kiev 2007) Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x^2 - f(y)^2) = xf(x) - y^2$$

for all $x, y \in \mathbb{R}$.

Problem 18.35. (IMO Shortlist 2005) Determine all functions $f : (0, \infty) \rightarrow (0, \infty)$ such that

$$f(x)f(y) = 2f(x + yf(x))$$

for all positive real numbers x and y .

Problem 18.36. (India 2008) Find all functions $f : (0, \infty) \rightarrow (0, \infty)$ such that $f(x + f(y)) = yf(xy + 1)$ for all $x, y \in (0, \infty)$.

Problem 18.37. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x + y)f(x - y) = f(x^2) - f(y^2)$$

for all $x, y \in \mathbb{R}$.

Problem 18.38. Find all functions $f : (0, \infty) \rightarrow (0, \infty)$ such that

$$f(x)f(yf(x)) = f(x + y)$$

for all $x, y \in (0, \infty)$.

Problem 18.39. (G. Dospinescu) Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(xf(y)) + f(yf(x)) = 2xy$$

for all $x, y \in \mathbb{R}$.

Problem 18.40. (Japan 2009) Find all functions $f : (0, \infty) \rightarrow (0, \infty)$ such that

$$f(x^2) + f(y) = f(x^2 + y + xf(4y))$$

for all $x, y \in (0, \infty)$.

Problem 18.41. (G. Dospinescu) Find all functions $f : (0, \infty) \rightarrow (0, \infty)$ such that

$$f(1 + xf(y)) = yf(x + y)$$

for all $x, y \in (0, \infty)$.

Problem 18.42. (Ukraine TST 2007) Find all functions $f : \mathbb{Q} \rightarrow \mathbb{Q}$ such that

$$f(x^2 + y + f(xy)) = 3 + (x + f(y) - 2)f(x)$$

for all $x, y \in \mathbb{Q}$.

Problem 18.43. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfy the equations $f(x^2) = f(x)^2$ and $f(x + 1) = f(x) + 1$. Prove that $f(x) = x$ for all $x \in \mathbb{R}$.

Problem 18.44. Determine all monotone functions $f : [0, \infty) \rightarrow \mathbb{R}$ such that

$$f(x + y) - f(x) - f(y) = f(xy + 1) - f(xy) - f(1),$$

for all $x, y \geq 0$ and $f(3) + 3f(1) = 3f(2) + f(0)$.

Problem 18.45. (IMO Shortlist 2009) Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(xf(x + y)) = f(yf(x)) + x^2$$

for all $x, y \in \mathbb{R}$.

Problem 18.46. (IMO Shortlist 2008) Let $f : \mathbb{R} \rightarrow \mathbb{N}$ be a function which satisfies

$$f\left(x + \frac{1}{f(y)}\right) = f\left(y + \frac{1}{f(x)}\right)$$

for all $x, y \in \mathbb{R}$. Prove that the function f is not surjective.

18.3 Continuous Functions

Problem 18.47. Find all continuous functions $f : (0, \infty) \rightarrow (0, \infty)$ such that

$$f(x) + \frac{1}{f(x)} = x + \frac{1}{x}$$

for all $x \in (0, \infty)$.

Problem 18.48. (Bulgaria 2006) Let $f : (0, \infty) \rightarrow (0, \infty)$ be a function such that

$$f(x+y) - f(x-y) = 4\sqrt{f(x)f(y)}$$

for all $x > y > 0$.

- a) Prove that $f(2x) = 4f(x)$ for all $x \in (0, \infty)$.
- b) Find all such functions f .

Problem 18.49. Find all continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfying

$$f(f(x)y - xf(y)) = yf(f(x)) - f(x)f(y)$$

for all $x, y \in \mathbb{R}$.

Problem 18.50. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function satisfying the equation

$$f(x+y) = f(x) + f(y)$$

for all real x and y . Prove that:

- (a) if f is continuous then it is a constant;
- (b) there are noncontinuous solutions of the given functional equation.

Problem 18.51. Prove that there is no continuous function $f : [0, 1] \rightarrow \mathbb{R}$ such that

$$f(x) + f(x^2) = x$$

for all $x \in [0, 1]$.

Problem 18.52. Determine all continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x + f(x)) = f(x)$$

for all $x \in \mathbb{R}$.

Problem 18.53. (Tuymaada 2003) Find all continuous functions $f : (0, \infty) \rightarrow \mathbb{R}$ that satisfy

$$f\left(x + \frac{1}{x}\right) + f\left(y + \frac{1}{y}\right) = f\left(x + \frac{1}{y}\right) + f\left(y + \frac{1}{x}\right)$$

for all $x, y \in (0, \infty)$.

Problem 18.54. Find all continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(xf(y)) + f(yf(x)) = \frac{1}{2}f(2x)f(2y)$$

for all real values x, y .

Problem 18.55. (IMC 2008). Find all continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x) - f(y)$ is rational for all reals x and y such that $x - y$ is rational.

18.4 The Odd and Even Parts of Functions

Problem 18.56. Find all continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x + y) + f(x)f(y) = f(xy + 1)$$

for all $x, y \in \mathbb{R}$.

Problem 18.57. Find all continuous functions $f, g, h : \mathbb{R} \rightarrow \mathbb{R}$ that obey

$$f(x + y) + g(xy) = h(x)h(y) + 1$$

for all $x, y \in \mathbb{R}$.

18.5 Constructive Problems

Problem 18.58. (Romania Shortlist 2001) (a) Let $g, h : \mathbb{Z} \rightarrow \mathbb{Z}$ be one-to-one functions. Show that the function $f : \mathbb{Z} \rightarrow \mathbb{Z}$ defined by $f(x) = g(x)h(x)$ for all $x \in \mathbb{Z}$ is not surjective.

(b) Let $f : \mathbb{Z} \rightarrow \mathbb{Z}$ be a surjective function. Show that there exist surjective functions $g, h : \mathbb{Z} \rightarrow \mathbb{Z}$ such that $f(x) = g(x)h(x)$ for all $x \in \mathbb{Z}$.

Problem 18.59. (G. Dospinescu) Find all functions f defined on the non-negative reals with values in the same set such that

$$f([f(x)]) + \{f(x)\} = x$$

for all $x \geq 0$. Here $[a]$ and $\{a\}$ are the integer and fractional parts of a real number a .

Problem 18.60. (IMO Shortlist 1997) Do there exist functions $f, g : \mathbb{R} \rightarrow \mathbb{R}$ such that:

- a) $f(g(x)) = x^2$ and $g(f(x)) = x^4$;
- b) $f(g(x)) = x^2$ and $g(f(x)) = x^3$

for all $x \in \mathbb{R}$?

Problem 18.61. Find a solution to the equations:

- a) $f(x^2) = 1 + f(x)$ for all $x \in (0, \infty)$;
- b) $f(x+1) = f(x)^2$ for all $x \in \mathbb{R}$.

Problem 18.62. (IMC 2003). (a) Show that for each function $f : \mathbb{Q} \times \mathbb{Q} \rightarrow \mathbb{R}$ there exists a function $g : \mathbb{Q} \rightarrow \mathbb{R}$ such that

$$f(x, y) \leq g(x) + g(y)$$

for all $x, y \in \mathbb{Q}$.

(b) Find a function $f : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ for which there is no function $g : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x, y) \leq g(x) + g(y)$$

for all $x, y \in \mathbb{R}$.

Problem 18.63. (IMO Shortlist 2008) Let $S \subseteq \mathbb{R}$ be a set of real numbers. We say that a pair (f, g) of functions from S into S is a Spanish Couple on S , if they satisfy the following conditions:

- (i) Both functions are strictly increasing, i.e. $f(x) < f(y)$ and $g(x) < g(y)$ for all $x, y \in S$ with $x < y$;
- (ii) The inequality $f(g(g(x))) < g(f(x))$ holds for all $x \in S$.

Decide whether there exists a Spanish Couple

- (a) on the set $S = \mathbb{N}$ of positive integers;
- (b) on the set $S = (a - \frac{1}{b} : a, b \in \mathbb{N})$.

18.6 Functional Equations Using Special Groups

Problem 18.64. (IMO Shortlist 2001) Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfying

$$f(xy)(f(x) - f(y)) = (x - y)f(x)f(y)$$

for all $x, y \in \mathbb{R}$.

Problem 18.65. Find all functions $f : \mathbb{Q} \rightarrow \mathbb{Q}$ such that

$$f(x + y + f(x)) = x + f(x) + f(y)$$

for all $x, y \in \mathbb{Q}$.

18.7 Density

Problem 18.66. (IMO Shortlist 2003) Find all functions $f : (0, \infty) \rightarrow (0, \infty)$, which are increasing on the segment $[1, \infty)$ and such that

$$f(xyz) + f(x) + f(y) + f(z) = f(\sqrt{xy})f(\sqrt{yz})f(\sqrt{zx})$$

for all $x, y, z \in (0, \infty)$.

Problem 18.67. Show that there exists a continuous function $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x) + f(2x) + f(3x) = 0$$

for all $x \in \mathbb{R}$.

Problem 18.68. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(xy(x + y)) = f(x)f(y)(f(x) + f(y))$$

for all $x, y \in \mathbb{R}$.

Problem 18.69. (Tuymaada 2006) Find all functions $f : (0, \infty) \rightarrow (0, \infty)$ with the following two properties:

(i) $f(x + 1) = f(x) + 1$;

(ii) $f\left(\frac{1}{f(x)}\right) = \frac{1}{x}$

for all $x \in (0, \infty)$.

18.8 Iterations

Problem 18.70. Find all functions $f: \mathbb{R} \setminus \{0; 1\} \rightarrow \mathbb{R}$ satisfying

$$f(x) + f\left(\frac{1}{1-x}\right) = \frac{2(1-2x)}{x(1-x)}$$

for all x in the domain of f .

Problem 18.71. (China 1988) For any integer $n \geq 3$, let $f(n)$ denote the least positive integer which is not a divisor of n . For each integer $n \geq 3$ find the positive integer k such that $f^{(k)}(n) = 2$.

Problem 18.72. For a given positive integer m , find all positive integers a such that there exist functions $f, g: \mathbb{N} \rightarrow \mathbb{N}$ with g bijective and

$$f^{(m)}(x) = g(x) + a$$

for all $x \in \mathbb{N}$.

Problem 18.73. (Bulgaria 1996) Find all strictly monotone functions $f: (0, \infty) \rightarrow (0, \infty)$ such that

$$f\left(\frac{x^2}{f(x)}\right) = x$$

for all $x \in (0, \infty)$.

Problem 18.74. Let $a, b \in (0, 1/2)$ and $f: \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function such that

$$f(f(x)) = af(x) + bx$$

for all $x \in \mathbb{R}$. Prove that there exists a real constant c such that $f(x) = cx$ for all $x \in \mathbb{R}$.

Problem 18.75. Find all continuous functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that the function $f(f(x)) - 2f(x) + x$ is constant.

Problem 18.76. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function such that

$$f^{(n)}(x) = x$$

for some integer $n \geq 2$ and all $x \in \mathbb{R}$. Prove that:

- (i) if n is odd, then $f(x) = x$ for all $x \in \mathbb{R}$;
- (ii) if n is even, then $f^{(2)}(x) = x$ for all $x \in \mathbb{R}$.

Problem 18.77. (Romania TST 2011) Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous, decreasing function, such that $g(\mathbb{R}) = (-\infty, 0)$. Prove that there are no continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f^{(k)} = g$ for an integer $k \geq 2$.

Problem 18.78. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function such that

$$f^{(2)}(x) = x$$

for all $x \in \mathbb{R}$. Prove that either $f(x) = x$ for all $x \in \mathbb{R}$, or f has the form $f(x) = f_0(x - p) + p$, where $p \in \mathbb{R}$ and

$$f_0(x) = \begin{cases} g(x) & \text{if } x \geq 0 \\ g^{-1}(x) & \text{if } x < 0 \end{cases}$$

for some continuous function $g : \mathbb{R} \rightarrow \mathbb{R}$ such that $g(0) = 0$, $g(x) < 0$ for $x > 0$, $g(x)$ is decreasing for $x > 0$ and $g^{-1}(x)$ is the inverse function to $g(x)$.

Problem 18.79. (IMO Shortlist 2009) Let $P(x)$ be a non-constant polynomial with integer coefficients. Prove that there is no function $T : \mathbb{Z} \rightarrow \mathbb{Z}$ such that the number of integers x with $T^{(n)}(x) = x$ is equal to $P(n)$ for all $n \geq 1$, where $T^{(n)}$ is the n -th iterate of T .

18.9 Discrete Subharmonic Functions

Problem 18.80. Let $f : \mathbb{Z} \rightarrow \mathbb{R}$ be a discrete subharmonic function, i.e.

$$f(k) \leq \frac{f(k-1) + f(k+1)}{2}$$

for all $k \in \mathbb{Z}$. Prove that if f is bounded above, then it is constant.

Problem 18.81. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$4f(x) = 2f^2(x+1) + 2f^2(x-1) + 1$$

for all $x \in \mathbb{R}$.

Problem 18.82. Let $f : \mathbb{Z}^2 \rightarrow \mathbb{R}$ be a discrete subharmonic function, i.e.

$$f(x, y) \leq \frac{f(x+1, y) + f(x, y+1) + f(x-1, y) + f(x, y-1)}{4}$$

for all $x, y \in \mathbb{Z}$. Prove that if f is bounded above, then it is constant.

Problem 18.83. Find all positive constants c such that

$$g(x, y) = \ln(x^2 + y^2 + c)$$

is a discrete subharmonic function on $\mathbb{Z}^2$.

Problem 18.84. Every lattice point in the space is assigned a real number from the interval $(0, 1)$ so that each of these numbers is equal to the arithmetic mean of the numbers assigned to the six neighboring lattice points. Prove that all numbers are equal.

Chapter 19

Hints and Solutions

19.1 Cauchy's Equations

Problem 1.1. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be an additive function such that $f(1) = 0$. Then for each non-empty interval $I \subset \mathbb{R}$ we have $f(I) = f(\mathbb{R})$.

Solution. It follows from Proposition 1.1 that for each rational number q we have $f(q) = qf(1) = 0$. Let $y_0 = f(x_0) \in f(\mathbb{R})$ and $I = [a, b]$, $a < b$. Take a rational number $q \in [a - x_0, b - x_0]$. Then $x_0 + q \in I$ and

$$f(x_0 + q) = f(x_0) + f(q) = f(x_0) = y_0.$$

Thus $y_0 = f(x_0 + q) \in f(I)$.

Problem 1.2. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be an additive function. Prove that:

- (a) either $\text{Ker } f = \{0\}$ or $\text{Ker } f$ is a dense subset of $\mathbb{R}$;
- (b) either $\text{Im } f = \{0\}$ or $\text{Im } f$ is a dense subset of $\mathbb{R}$.

Solution. (a) If $\text{Ker } f \neq \{0\}$ then there exists $x_0 \neq 0$ such that $f(x_0) = 0$. Consider the set $A = \{qx_0 \mid q \in \mathbb{Q}\}$. Then $A \in \text{Ker } f$ since

$$f(qx_0) = qf(x_0) = 0.$$

On the other hand A is a dense subset of $\mathbb{R}$ since for any open interval (a, b) the interval $\left(\frac{a}{x_0}, \frac{b}{x_0}\right)$ contains a rational number q , i.e. $qx_0 \in (a, b)$. Hence $\text{Ker } f$ is also a dense subset of $\mathbb{R}$.

(b) If $\text{Im } f \neq \{0\}$ then there exists x_0 such that $y_0 = f(x_0) \neq 0$. Consider the set $B = \{qy_0/q \in \mathbb{Q}\}$. Then $B \subset \text{Im } f$ since $qy_0 = qf(x_0) = f(qx_0)$. It follows as in (a) that B is a dense subset of $\mathbb{R}$ and therefore $\text{Im } f$ is also a dense subset of $\mathbb{R}$.

Problem 1.3. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a non-injective additive function. Prove that for each $a \in \mathbb{R}$ the set $F_a = \{x \in \mathbb{R} \mid f(x) = f(a)\}$ is dense in $\mathbb{R}$.

Solution. Since f is non-injective there exist $x_1 \neq x_2$ such that $f(x_1) = f(x_2)$, i.e. $f(x_1 - x_2) = 0$. Set $x_0 = x_1 - x_2$ and consider the set $A = \{a + qx_0 \mid q \in \mathbb{Q}\}$. Then as in the solution of Problem 1.2, (a) it follows that A is a dense subset of $\mathbb{R}$. On the other hand $f(a + qx_0) = f(a) + f(qx_0) = f(a) + qf(x_0) = f(a)$. Thus $A \subset F_a$ and F_a is a dense subset of $\mathbb{R}$.

Problem 1.4. (Romania TST 2006) Given $r, s \in \mathbb{Q}$, determine all functions $f : \mathbb{Q} \rightarrow \mathbb{Q}$ such that

$$f(x + f(y)) = f(x + r) + y + s$$

for all $x, y \in \mathbb{Q}$.

Solution. Set $g(x) = f(x) - r - s$. Then

$$\begin{aligned} g(x + g(y)) &= g(x + f(y) - r - s) = f((x + r - s) + f(y)) - r - s \\ &= f(x - s) + (y - s) - r - s = g(x - s) + y + s \end{aligned}$$

and

$$g^2(x + g(y)) = g(y + s + g(x - s)) = g(y) + x - s + s = x + g(y).$$

Hence $g^2 = id$, so g is one-to-one. Then replacing y by $g(y)$ we get

$$g(x + y) = g(x - s) + g(y) + s.$$

We have also $g(y + x) = g(y - s) + g(x) + s$, hence

$$g(x) - g(x - s) = g(y) - g(y - s) = k,$$

a constant. This gives $g(x + y) = g(x) + g(y) + (s - k)$ and therefore

$$h(x) = g(x) + s - k$$

is an additive function on $\mathbb{Q}$. By Proposition 1.1 we get that $h(x) = xh(1)$ for all $x \in \mathbb{Q}$ and therefore g has the form $g(x) = px + q$. Plugging this into the given relation and comparing the coefficients we obtain that $p = 1, q = 0$, or $p = -1, q = -2s$. Thus we have two solutions $f(x) = x + r + s$ and $f(x) = -x + r - s$.

Problem 1.5. Find all functions $f : \mathbb{Q} \rightarrow \mathbb{Q}$ such that

$$f(x + f(x) + 2y) = 2x + 2f(f(y))$$

for all $x, y \in \mathbb{Q}$.

Solution. Setting $x = y = 0$ in the given equation we get $f(f(0)) = 0$. Now if $f(u) = 0$, then setting $x = u, y = 0$ gives $u = 0$, hence $f(0) = 0$. Thus for $x = 0$ we get $f(2y) = 2f(f(y))$ for all $y \in \mathbb{R}$ and the given equation can be written as

$$f(x + y + f(y)) = 2x + f(y) \quad (1)$$

for all $x, y \in \mathbb{R}$. In particular, $f(x + f(x))$ and we get from (1) that

$$f(x + f(x) + y + f(y)) = 2x + 2y.$$

Now setting $y = -x$ gives $f(f(x) + f(-x)) = 0$, i.e. $f(-x) = -f(x)$ and the same setting in (1) implies that $f(f(x)) + f(x) = 2x$. Hence the function $f(x) + x$ is surjective. But we can write (1) as

$$f((x + f(x)) + y) = f(x + f(x)) + f(y)$$

and we conclude that f is an additive function on $\mathbb{Q}$. Thus $f(x) = xf(1)$ and plugging in the initial equation shows that $f(x) = x$ is the only solution.

Problem 1.6. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be an additive function such that

$$f\left(\frac{1}{x}\right) = \frac{f(x)}{x^2}$$

for all $x \neq 0$. Prove that $f(x) = cx$, $c \in \mathbb{R}$.

Solution. Since f is an additive function we have $f(x - y) = f(x) - f(y)$ for all $x, y \in \mathbb{R}$. Let $x \neq 0, 1$. Then

$$\frac{1}{x-1} - \frac{1}{x} = \frac{1}{x(x-1)}$$

and we get

$$f\left(\frac{1}{x-1}\right) - f\left(\frac{1}{x}\right) = f\left(\frac{1}{x(x-1)}\right).$$

Hence

$$\frac{f(x-1)}{(x-1)^2} - \frac{f(x)}{x^2} = \frac{f(x(x-1))}{x^2(x-1)^2}$$

which is equivalent to the identity

$$x^2 f(x-1) - (x-1)^2 f(x) = f(x^2 - x).$$

Thus

$$x^2(f(x) - f(1)) - (x-1)^2 f(x) = f(x^2) - f(x)$$

and therefore

$$f(x^2) = 2xf(x) - x^2 f(1). \quad (1)$$

Now replace x by $x + x^{-1}$ in this identity to obtain

$$f(x^2 + x^{-2} + 2) = 2(x + x^{-1})f(x + x^{-1}) - (x^2 + x^{-2} + 2)f(1).$$

Thus

$$f(x^2) + x^{-4}f(x^2) + f(2) = 2xf(x) + 4x^{-1}f(x) + 2x^{-3}f(x)$$

and using (1) again we get

$$f(x) = \left(\frac{f(2) + 2f(1)}{4}\right)x$$

for $x \neq 0, 1$. But $f(2) = 2f(1)$ and therefore $f(x) = f(1)x$ which holds also for $x = 0, 1$.

Problem 1.7. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be an additive function such that $f(1) = 1$ and

$$f(x)f\left(\frac{1}{x}\right) = 1$$

for all $x \neq 0$. Prove that $f(x) = x$.

Solution. We shall show that f is a bounded function on the interval $\left(0, \frac{1}{2}\right)$.

To do this note first that if $y \geq 2$ then there is $x > 0$ such that $y = x + \frac{1}{x}$. Then

$$(f(y))^2 = (f(x) + f\left(\frac{1}{x}\right))^2 \geq 4f(x)f\left(\frac{1}{x}\right) = 4$$

and therefore $|f(y)| \geq 2$. Now let $y \in \left(0, \frac{1}{2}\right)$. Then $\frac{1}{y} \geq 2$ and we get

$$|f(y)| = \left(\left|f\left(\frac{1}{y}\right)\right|\right)^{-1} \leq \frac{1}{2}.$$

Hence it follows from Theorem 1.1, (ii) that $f(x) = x$.

Problem 1.8. Determine all continuous functions $f : (0, \infty) \rightarrow (0, \infty)$ such that

$$f(xy) = xf(y) + yf(x)$$

for all $x, y \in (0, \infty)$.

Solution. Let $g(x) = f(x)/x$. Then we get the Logarithmic Cauchy equation $g(xy) = g(x) + g(y)$ and Corollary 1.2 implies that $g(x) = a \lg x$. Thus

$$f(x) = ax \lg x,$$

where a is a constant.

Problem 1.9. Let $\mathbb{Z}(\sqrt{2}) = \{a + b\sqrt{2} \mid a, b \in \mathbb{Z}\}$. Find all increasing additive functions $f : \mathbb{Z}(\sqrt{2}) \rightarrow \mathbb{R}$.

Solution. Since $f(0) = f(0) + f(0)$, $f(0) = 0$ and $f(x) \geq 0$ for $x \geq 0$. Set $f(1) = u$, $f(\sqrt{2}) = v$. Then it follows from Proposition 1.1 in Chapter 1 that $f(a + b\sqrt{2}) = au + bv$ for all $a, b \in \mathbb{Z}$. Since f is increasing, for every positive integer n we have that

$$f(\lfloor n\sqrt{2} \rfloor) \leq f(n\sqrt{2}) \leq f(\lfloor n\sqrt{2} \rfloor + 1),$$

so that

$$\lfloor n\sqrt{2} \rfloor u \leq nv \leq (\lfloor n\sqrt{2} \rfloor + 1)u.$$

Therefore

$$\begin{aligned}
 \left(\sqrt{2} - \frac{1}{n}\right)u &\leq \left(\frac{\lfloor n\sqrt{2} \rfloor}{n}\right)u \\
 &\leq v \\
 &\leq \frac{1}{n}(\lfloor n\sqrt{2} \rfloor + 1)u \\
 &\leq \left(\sqrt{2} + \frac{1}{n}\right)u.
 \end{aligned}$$

Letting $n \rightarrow \infty$ we get $\sqrt{2}u \leq v \leq \sqrt{2}u$, i.e. $v = \sqrt{2} \cdot u$. Thus

$$f(a + b\sqrt{2}) = (a + b\sqrt{2})u,$$

i.e. $f(x) = f(1)x$ for all $x \in \mathbb{Z}(\sqrt{2})$. Hence all the functions with the required properties have the form $f(x) = cx$, where $c \geq 0$.

Problem 1.10. Let $D \subset \mathbb{R}$ be a non-empty set such that $x + y \in D$ for all $x, y \in D$ and let $f : D \rightarrow \mathbb{R}$ be an increasing additive function. Prove that $f(x) = ax$, $x \in D$, where $a \geq 0$ is a constant.

Solution. The problem is trivial if $D = \{0\}$ since $f(0) = 0$. Hence we may assume that there is $x_0 \in D$ with $x_0 > 0$ (the case $x_0 < 0$ is analogous). Set $a = \frac{f(x_0)}{x_0}$. Then it follows by induction that $f(nx_0) = anx_0$, $n \in \mathbb{N}$. On the other hand for any $x \in D$ there is $n_0 \in \mathbb{N}$ such that $x + n_0x_0 > 0$. Moreover for n sufficiently large there is $a_n \in \mathbb{N}$ such that

$$a_n x_0 \leq n(x + n_0 x_0) \leq (a_n + 1)x_0. \quad (1)$$

Hence $f(a_n x_0) \leq f(n(x + n_0 x_0)) \leq f((a_n + 1)x_0)$, i.e.

$$a \frac{a_n}{n} x_0 \leq f(x + n_0 x_0) \leq a \frac{a_n + 1}{n} x_0 \quad (2)$$

for any sufficiently large $n \in \mathbb{N}$. But it follows from (1) that

$$\lim_{n \rightarrow \infty} x_0 a_n / n = x + n_0 x_0$$

and we get from (2) that $f(x + n_0x_0) = a(x + n_0x_0)$. Hence

$$f(x) = f(x + n_0x_0) - f(n_0x_0) = a(x + n_0x_0) - an_0x_0 = ax$$

for all $x \in D$.

Problem 1.11. Find all continuous functions $f, g : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$\begin{cases} f(x+y) = f(x)f(y) + g(x)g(y) \\ g(x+y) = f(x)g(y) + g(x)f(y) \end{cases}$$

for all $x, y \in \mathbb{R}$.

Solution. The system is equivalent to

$$\begin{cases} f(x+y) + g(x+y) = (f(x) + g(x))(f(y) + g(y)) \\ f(x+y) - g(x+y) = (f(x) - g(x))(f(y) - g(y)) \end{cases}$$

Set $h(x) = f(x) + g(x)$, $t(x) = f(x) - g(x)$. Then $h(x+y) = h(x)h(y)$ and $t(x+y) = t(x)t(y)$ for all $x, y \in \mathbb{R}$. It follows from Corollary 1.2 that $h = 0$ or $h = a^x$ and $t(x) = 0$ or $t(x) = b^x$, where $a, b > 0$. Hence all the solutions of the system are

$$f = g = 0; \quad f(x) = g(x) = \frac{1}{2}a^x, \quad f(x) = -g(x) = \frac{1}{2}b^x;$$

$$f(x) = \frac{a^x + b^x}{2}, \quad g(x) = \frac{a^x - b^x}{2}.$$

Problem 1.12. (Romania TST 1996) Let $a \in \mathbb{R}$ and $f_1, f_2, \dots, f_n : \mathbb{R} \rightarrow \mathbb{R}$ be additive functions such that

$$f_1(x)f_2(x) \dots f_n(x) = ax^n$$

for all $x \in \mathbb{R}$. Prove that:

- (a) if $a \neq 0$, then $f_1, f_2, \dots, f_n$ are linear functions;
- (b) if $a = 0$ then there exists i such that $f_i \equiv 0$.

Solution. For any additive function $f : \mathbb{R} \rightarrow \mathbb{R}$ we have $f(0) = 0$ and $f(m) = mf(1)$ for all $m \in \mathbb{Z}$. Set $f_i(1) = c_i$, $1 \leq i \leq n$, and let $x \in \mathbb{R}$ be arbitrary. Then for any integer m we have

$$\prod_{i=1}^n f_i(1 + mx) = \prod_{i=1}^n [c_i + mf_i(x)] = a(1 + mx)^n.$$

Consider the polynomials

$$P_x(t) = \prod_{i=1}^n (c_i + f_i(x)t),$$

$$Q_x(t) = a(1 + tx)^n.$$

(a) If $a \neq 0$, then

$$a = \prod_{i=1}^n f_i(1) = \prod_{i=1}^n c_i \neq 0$$

and therefore $c_i \neq 0$ for all $1 \leq i \leq n$. In this case the polynomials $P_x(t)$ and $Q_x(t)$ are not identically zero and $P_x(t) \equiv Q_x(t)$. Hence the unique decomposition in linear factors of polynomials shows that there exist real numbers b_i , $1 \leq i \leq n$, such that $c_i + f_i(x)t = b_i(1 + xt)$ for all $t \in \mathbb{R}$. Thus $f_i(x) = b_i x$ for all $x \in \mathbb{R}$, i.e. f_i is a linear function.

(b) Let $a = 0$. Then

$$\prod_{i=1}^n f_i = 0$$

and we shall prove by induction on n that there exists i such that $f_i(x) = 0$ for all $x \in \mathbb{R}$. For $n = 1$ the statement is obviously true. Suppose it is true for n and let $f_1, f_2, \dots, f_n, f_{n+1}$ be additive functions such that

$$\prod_{i=1}^n f_i(x) = 0$$

for all $x \in \mathbb{R}$. Assume that there exists $x_0 \in \mathbb{R}$ such that $f_{n+1}(x_0) \neq 0$. For a fixed $y \in \mathbb{R}$ and every m we have

$$0 = \prod_{i=1}^n f_i(x_0 + my) = \prod_{i=1}^n [f_i(x_0) + mf_i(y)],$$

hence the polynomial

$$\prod_{i=1}^n [f_i(x_0) + tf_i(y)]$$

is identically zero. Since $f_{n+1}(x_0) + tf_{n+1}(y)$ is not identically zero we get that

$$\prod_{i=1}^n [f_i(x_0) + tf_i(y)] \equiv 0$$

which obviously implies that

$$\prod_{i=1}^n f_i(y) \equiv 0.$$

Hence $f_i(y) \equiv 0$ for some $1 \leq i \leq n$ by the inductive hypothesis.

Problem 1.13. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function such that

$$|f(x+y) - f(x) - f(y)| \leq 1$$

for all $x, y \in \mathbb{R}$. Prove that there exists an additive function $g : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$|f(x) - g(x)| \leq 1$$

for all $x \in \mathbb{R}$.

Solution. For a fixed $x \in \mathbb{R}$ consider the sequence $a_n = f(nx)$, $n \geq 1$. Then $|a_{m+n} - a_m - a_n| \leq 1$ for all $m, n \in \mathbb{N}$. It follows easy by induction that $|a_{mn} - ma_n| \leq m - 1$ and $|a_{mn} - na_m| \leq n - 1$. Hence $|ma_n - na_m| < m + n$, i.e. $|\frac{a_m}{m} - \frac{a_n}{n}| \leq \frac{1}{m} + \frac{1}{n}$. This shows that $\left\{\frac{a_n}{n}\right\}$ is a Cauchy sequence and thus it converges. For every $x \in \mathbb{R}$ define $g(x)$ as the limit of $f(nx)/n$. Then g is an additive function since $|f(n(x+y))/n - f(nx)/n - f(ny)/n| \leq 1/n$ for every $n \in \mathbb{N}$. The inequality $|f(x) - g(x)| \leq 1$ follows from $|f(nx) - nf(x)| \leq n - 1$ which can be proved by induction on n , using the given inequality.

19.2 Generalized Cauchy Equations

Problem 2.1. Find all functions $f : (1, 4) \rightarrow \mathbb{R}$ such that

$$f(x+y) = f(x) + f(y)$$

for all $x, y \in (1, 2)$.

Solution. If $x, y \in (1, 2)$ then $\frac{x+y}{2} \in (1, 2)$ and

$$f(x+y) = f\left(2\frac{x+y}{2}\right) = f\left(\frac{x+y}{2}\right) + f\left(\frac{x+y}{2}\right) = 2f\left(\frac{x+y}{2}\right).$$

Hence f satisfies Jensen's equation and it follows from Theorem 2.1 that $f(x) = g(x) + a$, where $g : \mathbb{R} \rightarrow \mathbb{R}$ is an additive function and a is a constant.

Let $z \in (2, 4)$. Then $z = x + y$, where $x, y \in (1, 2)$ and we get

$$f(z) = f(x+y) = f(x) + f(y) = g(x) + a + g(y) + a = g(x+y) + 2a = g(z) + 2a.$$

Hence the desired functions have the form

$$f(x) = \begin{cases} g(x) + a & \text{if } x \in (1, 2) \\ b & \text{if } x = 2 \\ g(x) + 2a & \text{if } x \in (2, 4) \end{cases}$$

where $g : \mathbb{R} \rightarrow \mathbb{R}$ is an additive function and a, b are arbitrary constants.

Problem 2.2. Let $f : (1, 6) \rightarrow \mathbb{R}$ be a function such that $f(x+y) = f(x) + f(y)$ for all $x, y \in (1, 3)$. Prove that there is an additive function $g : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x) = g(x)$ for all $x \in (1, 6)$.

Solution. We get as in the solution of Problem 2.1 that

$$f(x) = \begin{cases} g(x) + a, & x \in (1, 3) \\ g(x) + 2a, & x \in (2, 6) \end{cases}$$

where $g : \mathbb{R} \rightarrow \mathbb{R}$ is an additive function and a is a constant. Then

$$g(5/2) + a = f(5/2) = g(5/2) + 2a,$$

i.e. $a = 0$. Hence $f(x) = g(x)$ for $x \in (1, 6)$.

Problem 2.3. Find all increasing functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f\left(\frac{x+y}{3}\right) = \frac{f(x) + f(y)}{2}$$

for all $x, y \in \mathbb{R}$.

Solution. We have

$$\frac{f\left(\frac{x+y}{2}\right) + f\left(\frac{x+y}{2}\right)}{2} = f\left(\frac{x+y}{3}\right) = \frac{f(x) + f(y)}{2}$$

and from here we deduce that f satisfies the Jensen equation

$$\frac{f(x) + f(y)}{2} = f\left(\frac{x+y}{2}\right).$$

Since f is increasing, it follows that $f(x) = ax + b$. Hence

$$a\frac{x+y}{3} + b = a\frac{x+y}{2} + b$$

and therefore $a = 0$. Thus only the constant functions satisfy the conditions of the problem.

Problem 2.4. Find all pairs of continuous functions $f, g: \mathbb{R} \rightarrow \mathbb{R}$ that satisfy

$$f(x) + f(y) = g(x+y)$$

for all $x, y \in \mathbb{R}$.

Solution. We remark that $f(x+y) + f(0) = f(x) + f(y) = g(x+y)$. Hence $f(x) - f(0) + f(y) - f(0) = f(x+y) - f(0)$, so $f(x) - f(0)$ is an additive function, therefore $f(x) = cx + d$ for some c and hence $g(x) = cx + 2d$. Or we can just use Pexider's equation (see Chapter 1) for g, f, f to get

$$g(x) = ax + b + c, \quad f(x) = ax + b, \quad f(x) = ax + c$$

thus $b = c$ and $f(x) = ax + c, g(x) = ax + 2c$.

Problem 2.5. Find all continuous functions $f, g, k : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x + y) = g(x) \cdot k(y) + x + 2y$$

for all $x, y \in \mathbb{R}$.

Solution. The given equation can be written as $F(x + y) = g(x) \cdot k(y) + y$, where $F(x) = f(x) - x$. This is equation (2.13) with $h(y) = y$. Suppose first that $F(x)$ is not constant. Then it follows from Theorem 2.3 that $y = ay + c$, i.e. $a = 1, c = 0$. Thus $f(x) = F(x) + x = 2x + b$, $g(x) = \frac{x + b}{d}$ and $k(y) = d$, where b and $d \neq 0$ are constants. If $F(x)$ is a constant, i.e. $f(x) = x + a$, then it is easy to see that $g(x) = b$ and $k(y) = \frac{a - y}{b}$, where a and $b \neq 0$ are constants.

Problem 2.6. Find all continuous functions $f, g, k : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(xy) = g(x)k(y) + y^2.$$

Solution. This is the equation (2.13) with $k(y) = y^2$. Suppose first that f is a nonconstant function. Then it follows from Theorem 2.4 that $y^2 = (c - b)y^t + b$ and we get $c = 1, b = 0, t = 2$. Hence

$$f(x) = ax^2, \quad d(x) = \frac{ax^2 - 1}{d}, \quad k(y) = dy^2,$$

where $a \neq 0, d \neq 0$ are constants. If $f(x) \equiv a$, then it is easy to see that $g(x) \equiv b$ and $k(y) = \frac{a - y^2}{b}$, where a and $b \neq 0$ are constants.

Problem 2.7. Determine all continuous solutions of the following functional equations:

$$(i) \quad f(\sqrt{xy}) = \frac{f(x) + f(y)}{2}, \quad x, y \in (0, \infty);$$

$$(ii) \quad f(\sqrt{xy}) = \sqrt{\frac{f^2(x) + f^2(y)}{2}}, \quad x, y \in (0, \infty);$$

$$(iii) \quad f(\sqrt{xy}) = \frac{2}{\frac{1}{f(x)} + \frac{1}{f(y)}}, \quad x, y \in (0, \infty);$$

$$(iv) \quad f\left(\sqrt{\frac{x^2 + y^2}{2}}\right) = \frac{f(x) + f(y)}{2}, \quad x, y \in \mathbb{R};$$

$$(v) \quad f\left(\sqrt{\frac{x^2 + y^2}{2}}\right) = \sqrt{\frac{f^2(x) + f^2(y)}{2}}, \quad x, y \in \mathbb{R};$$

$$(vi) \quad f\left(\sqrt{\frac{x^2 + y^2}{2}}\right) = \frac{2}{\frac{1}{f(x)} + \frac{1}{f(y)}}, \quad x, y \in \mathbb{R};$$

$$(vii) \quad f\left(\frac{2}{\frac{1}{x} + \frac{1}{y}}\right) = \frac{f(x) + f(y)}{2}, \quad x, y \in (0, \infty);$$

$$(viii) \quad f\left(\frac{2}{\frac{1}{x} + \frac{1}{y}}\right) = \sqrt{f(x)f(y)}, \quad x, y \in (0, \infty);$$

$$(ix) \quad f\left(\frac{2}{\frac{1}{x} + \frac{1}{y}}\right) = \sqrt{\frac{f^2(x) + f^2(y)}{2}}, \quad x, y \in (0, \infty).$$

Solution. (i) Set $g(x) = f(10^x)$. We obtain $f(x) = a \log x + b$.

(ii) Set $g(x) = f^2(10^x)$. Then

$$g\left(\frac{x+y}{2}\right) = f^2\left(\sqrt{10^x \cdot 10^y}\right) = \frac{f^2(10^x) + f^2(10^y)}{2} = \frac{g(x) + g(y)}{2}.$$

It follows from Theorem 2.1 that $g(x) = ax + b$. But $g(x) \geq 0$ for all $x \in \mathbb{R}$ and we conclude that $a = 0$, $b \geq 0$. Hence $f(x) = b$, where $b \geq 0$ is a constant.

(iii) Set $g(x) = \frac{1}{f(10^x)}$. Then $g(x) \neq 0$ for all $x \in \mathbb{R}$ and it is easy to check that

$$g\left(\frac{x+y}{2}\right) = \frac{g(x) + g(y)}{2}.$$

Theorem 2.1 implies that $g(x) = ax + b$ and since $g(x) \neq 0$ we get $a = 0$, $b \neq 0$. Thus $f(x) = b$, where b is a nonzero constant.

(iv) Consider the function $g : [0, \infty) \rightarrow \mathbb{R}$ defined by $g(x) = f(\sqrt{x})$. Then for all $x, y \in [0, \infty)$ we have

$$\begin{aligned} g\left(\frac{x+y}{2}\right) &= g\left(\frac{(\sqrt{x})^2 + (\sqrt{y})^2}{2}\right) = f\left(\sqrt{\frac{(\sqrt{x})^2 + (\sqrt{y})^2}{2}}\right) \\ &= \frac{f(\sqrt{x}) + f(\sqrt{y})}{2} = \frac{g(x) + g(y)}{2}. \end{aligned}$$

Hence it follows from Theorem 2.1 that $g(x) = ax + b$ and

$$f(x) = g(x^2) = ax^2 + b$$

for $x \in [0, \infty)$. On the other hand setting $x = y$ in the given equation we get $f(x) = f(|x|)$ for all $x \in \mathbb{R}$. Thus $f(x) = ax^2 + b$ for all $x \in \mathbb{R}$.

(v) Set $g(x) = f^2(\sqrt{x})$, $x \in [0, \infty)$. Then

$$g\left(\frac{x+y}{2}\right) = \frac{g(x) + g(y)}{2}$$

and Theorem 2.1 implies that $g(x) = ax + b$, where $a, b \geq 0$ since $g(x) \geq 0$ for $x \in [0, \infty)$. Thus $f(x) = \sqrt{ax^2 + b}$ for $x \in [0, \infty)$. Moreover, for all $x \in \mathbb{R}$ we have $f(|x|) = |f(x)|$ and therefore $f^2(x) = ax^2 + b$ for all $x \in \mathbb{R}$. Suppose that $b = 0$. Then the continuity of f implies that either $f(x) = \sqrt{ax^2}$ or $f(x) = -\sqrt{ax^2}$ for all $x > 0$. Hence the only solutions in this case are $f(x) = \sqrt{d}x$ and $f(x) = \sqrt{d}|x|$, where $d \geq 0$ is an arbitrary constant.

If $b > 0$ consider the sets

$$A = \{x \in \mathbb{R} \mid f(x) = \sqrt{ax^2 + b}\} \text{ and } B = \{x \in \mathbb{R} \mid f(x) = -\sqrt{ax^2 + b}\}.$$

Then A and B are closed subsets of $\mathbb{R}$ such that $A \cup B = \mathbb{R}$, $A \cap B = \emptyset$.

Since $[0, \infty) \subset A$ we get that $A = \mathbb{R}$ and therefore $f(x) = \sqrt{ax^2 + b}$ for all $x \in \mathbb{R}$.

Hence all solutions of the given equation are the functions $f(x) = ax$, $f(x) = a|x|$ and $f(x) = \sqrt{ax^2 + b}$, where $a \geq 0$ and $b > 0$ are arbitrary constants.

(vi) Set $g(x) = \frac{1}{f(\sqrt{x})}$ for $x \in [0, \infty)$. Then for all $x, y \in [0, \infty)$ we have

$$g\left(\frac{x+y}{2}\right) = \frac{1}{f\left(\sqrt{\frac{x+y}{2}}\right)} = \frac{\frac{1}{f(\sqrt{x})} + \frac{1}{f(\sqrt{y})}}{2} = \frac{g(x) + g(y)}{2}.$$

It follows from Theorem 2.1 that $g(x) = ax + b$ and therefore $f(x) = ax^2 + b$ for $x \in [0, \infty)$. Moreover $ab > 0$ since $g(x) \neq 0$. On the other hand the given equation implies that $f(x) = f(|x|) = ax^2 + b$ for all $x \in \mathbb{R}$.

(vii) Set $g(x) = f\left(\frac{1}{x}\right)$ and use Theorem 2.1. It follows that

$$f(x) = \frac{a}{x} + b, \quad a, b \in \mathbb{R}.$$

(viii) Set $g(x) = f\left(\frac{1}{x}\right)$ and use Example 2.7. The answer is

$$f(x) = A \cdot \frac{1}{ax}, \quad A \geq 0, \quad a > 0.$$

(ix) Set $g(x) = f^2\left(\frac{1}{x}\right)$. The solutions are $f(x) = \sqrt{\frac{a}{x} + b}$, $a, b \geq 0$.

Problem 2.8. For a real number r , define the function

$$f_r: (0, \infty) \rightarrow (0, \infty) \text{ by } f_r(x) = \begin{cases} x^r & r \neq 0 \\ \ln(x) & r = 0 \end{cases}$$

(in the case $r = 0$, the range is $\mathbb{R}$) and let g_r be its inverse, i.e.

$$g_r(x) = \begin{cases} x^{\frac{1}{r}} & r \neq 0 \\ e^x & r = 0 \end{cases}$$

Show that the continuous solutions $f: (0, \infty)$ to the equation

$$f(M_k(x, y)) = M_n(f(x), f(y))$$

have the form

$$g_n(af_k(x) + b).$$

Solution. We observe that

$$M_k(x, y) = g_k\left(\frac{f_k(x) + f_k(y)}{2}\right).$$

From here, we immediately see that if $f(M_k(x, y)) = M_n(f(x), f(y))$ then if we set $h = f_n \circ f \circ g_k$, i.e. we let $f(x) = g_n(h(f_k(x)))$ we deduce that

$$g_n h\left(\frac{f_k(x) + f_k(y)}{2}\right) = g_n\left(\frac{h(f_k(x)) + h(f_k(y))}{2}\right)$$

which means that

$$h\left(\frac{f_k(x) + f_k(y)}{2}\right) = \frac{h(f_k(x)) + h(f_k(y))}{2}.$$

Setting $u = f_k(x), v = f_k(y)$ we get

$$h\left(\frac{u + v}{2}\right) = \frac{h(u) + h(v)}{2}.$$

This is Jensen's equation, so $h(t) = at + b$ is linear, and therefore

$$f(x) = g_n(af_k(x) + b),$$

as desired. (Note that not all a, b will yield solutions: we need to verify that $af_k(x) + b \in (0, \infty)$ for all x . However, all solutions are of this type.)

19.3 Reducing to Cauchy

Problem 3.1. (Bulgaria 1994) Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$xf(x) - yf(y) = (x - y)f(x + y)$$

for all $x, y \in \mathbb{R}$.

Solution. It follows from the given equation that

$$(x + y)f(x + y) - yf(y) = xf(x + 2y).$$

Subtracting these two identities we get

$$f(x) + f(x + 2y) = 2f(x + y)$$

which is equivalent to

$$f(x) + f(y) = 2f\left(\frac{x + y}{2}\right) \quad (1)$$

for all $x, y \in \mathbb{R}$. Set $b = f(0)$. Then

$$f(x) + b = 2f\left(\frac{x}{2}\right)$$

which together with (1) gives

$$f(x) + f(y) = f(x + y) + b.$$

Now using the given equation we get

$$x(f(y) - b) = y(f(x) - b).$$

Hence $f(x) - b = x(f(1) - b)$, i.e. $f(x) = ax + b$, where a and b are constants.

Conversely, it is easily checked that all linear functions satisfy the given equation.

Problem 3.2. Find all functions $f: \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}$ such that for all nonzero numbers x, y with $x \neq y$ we have

$$f(y) - f(x) = f(y)f\left(\frac{x}{x - y}\right).$$

Solution. Consider the function $g(x) = f\left(\frac{1}{x}\right)$. The equation becomes

$$g(y) - g(x) = g(y)g\left(1 - \frac{x}{y}\right).$$

Take $y = 1$. Then for x different from 0 and 1 we have

$$g(1) - g(x) = g(1)g(1 - x).$$

Thus if $g(1) = 0$ then we obtain the zero function. Suppose now that $g(1) \neq 0$. Then we get

$$g\left(1 - \frac{x}{y}\right) = 1 - \frac{g\left(\frac{x}{y}\right)}{g(1)}$$

which together with the first identity for g gives

$$g(y)g\left(\frac{x}{y}\right) = g(x)g(1)$$

for all x, y non-zero and different. Thus

$$g(y)g\left(1 - \frac{x}{y}\right) = g(1)g(y - x)$$

and we have for different and non-zero x, y the relations

$$g(y) - g(x) = g(1)g(y - x) \text{ and } g(xy)g(1) = g(x)g(y).$$

Thus $g(u + v) = g(u) + g(1)g(v)$ for u, v non-zero with non-zero sum. Swap them and we will find that if $g(1) \neq 1$ then g is the zero function, a contradiction. Hence $g(1) = 1$ and therefore g is an additive and multiplicative function. Thus $g(x) = x$ by Corollary 1.2 and therefore $f(x) = \frac{1}{x}$.

Problem 3.3. (AMM 2001, Bulgaria 2003) Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x^2 + y + f(y)) = 2y + f(x)^2$$

for all $x, y \in \mathbb{R}$.

Solution. It follows by the given condition that f is a surjective function. Moreover, $f(x)^2 = f(-x)^2$. Let a be such that $f(a) = 0$. Then $f(-a) = 0$. Setting $x = 0$, $y = \pm a$ in the given identity we get $0 = f(\pm a) = (f(0))^2 \pm 2a$, i.e. $a = 0$. Now substitute $y = -\frac{f(x)^2}{2}$ to obtain $f(x^2 + y + f(y)) = 0$ and hence $y + f(y) = -x^2$. Thus, $y + f(y)$ spans all the non-positive real numbers. Since $f(0) = 0$, setting $y = 0$ and $x = 0$ respectively in the given condition we obtain $f(x^2) = f(x)^2 \geq 0$ and $f(y + f(y)) = 2y$. Setting $z = x^2$, $t = y + f(y)$ we conclude that $f(z + t) = f(z) + f(t)$ for all $z \geq 0 \geq t$. Then for $z = -t$ we obtain that $f(-t) = -f(t)$ and it is easy to see that $f(z + t) = f(z) + f(t)$ for arbitrary z and t . Since $f(x) \geq 0$ for $x \geq 0$, it follows from Theorem 1.1 that $f(x) = x$ for all x and this function obviously satisfies the given identity.

Problem 3.4. (Vietnam TST 2004) Find all real values of a for which there exists one and only one function $f : \mathbb{R} \mapsto \mathbb{R}$ satisfying the equation

$$f(x^2 + y + f(y)) = f(x)^2 + ay$$

for all $x, y \in \mathbb{R}$.

Hint. If $a = 0$ we have two different constant solutions, so suppose that $a \neq 0$ and f is a solution. Then the same arguments as in the solution of Problem 3.3 show that $f(x) = x$ and $a = 2$. So, for all $a \neq 0, 2$ there are no such functions and the answer is $a = 2$.

Problem 3.5. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ that satisfy

$$f(x + y) + f(xy) = f(x)f(y) + 1$$

for all $x, y \in \mathbb{R}$.

Solution. We can guess $f(x) = x + 1$ as a solution. If $x = y = 0$ we get $2f(0) = f(0)^2 + 1$ so $f(0) = 1$. If we set $y = 1$, we get $f(x + 1) + f(x) = f(x)f(1) + 1$. Set $f(1) = a$. Then $f(x + 1) = (a - 1)f(x) + 1$. If $a = 1$ we obtain $f(x + 1) = 1$, i.e. $f \equiv 1$. Otherwise we get $1 = f(-1 + 1) = (a - 1)f(-1) + 1$ and so $f(-1) = 0$. Also $f(2) = a(a - 1) + 1 = a^2 - a + 1$, $f(3) = (a - 1)(a^2 - a + 1) + 1 = a^3 - 2a^2 + 2a$, $f(4) = (a - 1)(a^3 - 2a^2 + 2a) + 1 = a^4 - 3a^3 + 4a^2 - 2a + 1$. Now set $x = y = 2$ into the condition to get $2f(4) = f(2)^2 + 1$. So $2(a^4 - 3a^3 + 4a^2 - 2a + 1) =$

$a^4 - 2a^3 + 3a^2 - 2a + 2$, i.e. $a^4 - 4a^3 + 5a^2 - 2a = 0$ and thus $a(a-2)(a-1)^2 = 0$. As $a \neq 1$ we either have $a = 2$ or $a = 0$. If $a = 0$ we get $f(x+1) = 1 - f(x)$ hence $f(x+2) = f(x)$. Then set $y = 2$ into the given condition to get $f(x+2) + f(2x) = f(2)f(x) + 1 = f(x) + 1$. As $f(x+2) = f(x)$ we have $f(2x) = 1$ and then for $x = \frac{1}{2}$ we get $f(1) = 1$, a contradiction. So $a = 2$ and then $f(x+1) = f(x) + 1$. Hence $f(x) = x + 1$ for $x \in \mathbb{Z}$. Now as $x+1$ is a solution, we may suppose $f(x) = g(x) + 1$ and try to prove $g(x) = x$. The condition transforms to $g(x+y) + g(xy) = g(x)g(y) + g(x) + g(y)$. Then $g(x) = x$ for $x \in \mathbb{N}$ and $g(x+1) = g(x) + 1$. If we set $y = k \in \mathbb{N}$ then $g(x+k) + g(kx) = kg(x) + g(x) + k$ so $g(kx) = kg(x)$. Then $y = x$ gives $g(x^2) + g(2x) = g(x)^2 + 2g(x)$ and as $g(2x) = 2g(x)$ we get $g(x^2) = g(x)^2$. So $g \geq 0$ on $(0, \infty)$. Now replace y by $y+1$ to get $g(x+y+1) + g(xy+x) = g(x)g(y+1) + g(x) + g(y+1)$. As $g(x+y+1) = g(x+y) + 1$, $g(y+1) = g(y) + 1$ and $g(x+y) + g(xy) = g(x)g(y) + g(x)$, subtracting these two relations we deduce $g(xy+x) = g(xy) + g(x)$. Now if $u, v \neq 0$, we set $y = \frac{v}{u}$, $x = u$ to get $g(u+v) = g(u) + g(v)$ for $u, v \neq 0$. As $g(0) = 0$ we conclude that g is additive. Since $g \geq 0$ on $\mathbb{R}$, then $g(x) = cx$. But $g(x) = x$ for $x \in \mathbb{N}$ and thus $c = 1$. So we have two solutions $f(x) = x + 1$ and $f(x) = 1$.

Problem 3.6. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x+y) + f(x)f(y) = f(xy) + f(x) + f(y)$$

for all $x, y \in \mathbb{R}$.

Solution. Let $f(1) = a$. Then $f(2) = 3a - a^2$ and $f^2(2) = 2f(2)$ implies that $(3a - a^2)^2 = 2(3a - a^2)$, i.e. $a(a-1)(a-2)(a-3) = 0$.

Case 1. $a = 0$. Then setting $y = 1$ gives $f(x+1) = 2f(x)$. So $f(x+y+1) + f(x)f(y+1) = f(xy+x) + f(x) + f(y+1)$ and therefore $2f(x+y) + 2f(x)f(y) = f(xy+x) + f(x) + 2f(y)$. This together with the given identity implies that $2f(xy) + f(x) = f(xy+x)$. Set $x \neq 0$ and $y = \frac{1}{x}$ in this equality to get $f(x+1) = f(x)$ and since $f(x+1) = 2f(x)$ we see that $f(x) = 0$ for all $x \neq 0$. Now setting $x = 1, y = -1$ in the given conditions shows that $f(0) = 0$ and so $f(x) = 0$ for all x which is a solution of our equation.

Case 2. $a = 1$. Then setting $y = 1$ gives $f(x+1) = f(x) + 1$ and so $f(n) = n$ for all $n \in \mathbb{Z}$. Then $f(x+y+1) + f(x)f(y+1) = f(xy+x) + f(x) + f(y+1)$ and so $f(x+y) + f(x)f(y) = f(xy+x) + f(y)$. Subtracting the initial identity

from this equality, we get $f(xy + x) = f(xy) + f(x)$. Now setting $x = u \neq 0$ and $y = \frac{v}{u}$ we get $f(u + v) = f(u) + f(v)$ for all $u \neq 0, v$ and, since $f(0) = 0$ we see that $f(x + y) = f(x) + f(y)$ for all x, y . Now the given identity takes the form $f(xy) = f(x)f(y)$ and it follows from Corollary 1.1 that $f(x) = x$ which is a solution of our equation.

Case 3. $a = 2$. Then setting $y = 1$ we get $f(x + 1) = 2$. So $f(x) = 2$ for all x which is a solution of our equation.

Case 4. $a = 3$. Then setting $x = y = 1$ gives $f(2) = 0$. Now the given identity for $y = 1$ takes the form $f(x + 1) = 3 - f(x)$ and so $f(x + 2) = 3 - f(x + 1) = f(x)$. But now setting $y = 2$ we get $f(x + 2) = f(2x) + f(x)$. So $f(2x) = 0$, which is impossible since, for example, $f(2 \cdot \frac{1}{2}) = 3$.

Thus the given equation has three solutions: $f(x) = 0, f(x) = 2$ and $f(x) = x$.

Problem 3.7. (IMO Shortlist 2007) Find all functions $f : (0, \infty) \rightarrow (0, \infty)$ such that

$$f(x + f(y)) = f(x + y) + f(y)$$

for all $x, y \in (0, \infty)$.

Solution. First, we shall show that $f(y) > y$ for all $y \in (0, \infty)$. It is clear that $f(y) \neq y$, since otherwise it follows from the condition that $f(y) = 0$, a contradiction. If $f(y) < y$ for some y , then setting $x = y - f(y)$, we get the contradiction $f(2y - f(y)) = 0$.

Let now $g(x) = f(x) - x$. Then $g > 0$ and setting $t = x + y$ in the condition, we get that $g(t + g(y)) = g(t) + y$ for any $t > y > 0$. This equality implies that g is an injective function.

On the other hand, if $t > x + y$, then

$$g(t + g(x) + g(y)) = g(t + g(x)) + y = g(t) + x + y = g(t + g(x + y)).$$

Since g is injective, it follows that $g(x + y) = g(x) + g(y)$, i.e. g is an additive function. As we already know (Theorem 1.1), this and $g > 0$ imply that $g(x) = cx$. Then $f(x) = (c + 1)x$ and a direct verification leads to the only solution of the problem, namely $f(x) = 2x$.

Problem 3.8. (Romania 2009) Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x^3 + y^3) = xf(x^2) + yf(y^2)$$

for all $x, y \in \mathbb{R}$.

Solution. Setting $y = 0$ gives $f(x^3) = xf(x^2)$ and it follows that

$$f(x^3 + y^3) = f(x^3) + f(y^3).$$

Hence f is an additive function on $\mathbb{R}$ and therefore $f(nx) = nf(x)$ for every positive integer n and real number x . For every solution f of the given functional equation and every constant c the function cf is also a solution, so we may assume that $f(1) = 1$, or $f(1) = 0$. Let $f(1) = 1$. Since f is additive we get from $f((x+1)^3) = (x+1)f((x+1)^2)$ that $2f(x^2) + f(x) = 2xf(x) + x$. Now replacing x by $x+1$ we get

$$2f((x+1)^2) + f(x+1) = (2x+2)f(x+1) + x+1$$

and using again that f is additive we obtain the identity

$$2f(x^2) + 3f(x) = 2xf(x) + 3x.$$

This together with the identity $2f(x^2) + f(x) = 2xf(x) + x$ gives $f(x) = x$. Now consider the case $f(1) = 0$. Using the same arguments as above we get first that $2f(x^2) + f(x) = 2xf(x)$ and then replacing x by $x+1$ gives $f = 0$. Thus all solutions of the given functional equation are $f(x) = cx$, where c is a real constant.

Problem 3.9. Let n be a positive integer. Find all monotonic functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x + f(y)) = f(x) + y^n$$

for all $x, y \in \mathbb{R}$.

Solution. Set $y = 0$ to get $f(x + f(0)) = f(x)$. If $f(0) \neq 0$, then $f(x)$ is a periodic monotonic function, so is constant, which is impossible. Hence $f(0) = 0$ and plugging $x = 0$ in the given equation we get $f(f(y)) = y^n$. Now $f(x)$ monotonic implies $f(f(x))$ increasing and so n must be odd.

Setting $x = 0, y = f(x)$ in the given equation we get $f(x^n) = f(x)^n$. Hence replacing y by $f(y)$ gives $f(x + y^n) = f(x) + f(y)^n = f(x) + f(y^n)$ and since n is odd we get $f(x + y) = f(x) + f(y)$. But f is monotonic and by Theorem

1.1 we conclude that $f(x) = ax$. Plugging back in the given equation, we get $n = 1$ and $a = \pm 1$.

Hence the equation has solutions only for $n = 1$ and in this case they are $f(x) = x$ and $f(x) = -x$.

Problem 3.10. Denote by T the set of real numbers greater than 1. Given a positive integer n find all functions $f: T \rightarrow \mathbb{R}$ such that

$$f(x^{n+1} + y^{n+1}) = x^n f(x) + y^n f(y)$$

for all $x, y \in T$.

Solution. Setting $x = y$ gives $f(2x^{n+1}) = 2x^n f(x)$. Hence

$$2f(x^{n+1} + y^{n+1}) = f(2x^{n+1}) + f(2y^{n+1}),$$

i.e.

$$2f(x + y) = f(2x) + f(2y)$$

for all $x, y > 1$. Then

$$f(x + y) + f(z) = f\left(2\frac{x + y}{2}\right) + f\left(2\frac{z}{2}\right) = 2f\left(\frac{x + y + z}{2}\right) = f(x) + f(z + y)$$

for all $x, z > 2$ and $y > 0$. This shows that the function $f(x + y) - f(x)$, where $x > 2, y > 0$, depends only on y . Set $g(y) = f(x + y) - f(x)$. Then

$$\frac{f(2x)}{2} + \frac{f(2y)}{2} = f(x + y) = f(x) + g(y),$$

for $x > 2, y > 1$ and therefore $f(2x) = 2f(x) + a$, where a is a constant. Now it follows by induction on k that

$$f(2^k) = 2^{k-2}f(4) + a(2^{k-2} - 1)$$

for all $k \geq 2$. Hence

$$f(2 \cdot 4^{n+1}) = f(2^{2n+3}) = 2^{2n+1}f(4) + a(2^{2n+1} - 1) = 2 \cdot 4^n f(4) + a(2^{2n+1} - 1).$$

On the other hand, we have

$$f(2 \cdot 4^{n+1}) = 2 \cdot 4^n f(4)$$

and therefore $a = 0$. Then

$$f(x + y) = f(x) + f(y), f(x^{n+1}) = x^n f(x),$$

for all $x, y > 2$. It follows by induction on k that $f(kx) = kf(x)$ for all $k \in \mathbb{N}$.

In particular, for all $s \in \mathbb{N}$ we have $f(3^s) = 3^{s-1}f(3) = 3^s \cdot c$ where $c = \frac{f(3)}{3}$.

Thus for all $x > 2$ we get

$$f(x + 3^s) = f(x) + f(3^s) = f(x) + 3^s \cdot c. \quad (1)$$

Set $k = 3^s$, where $s \in \mathbb{N}$. Then

$$\begin{aligned} \sum_{j=0}^{n+1} \binom{n+1}{j} f(x^j) k^{n+1-j} &= f((x+k)^{n+1}) \\ &= (x+k)^n f(x+k) = \sum_{j=0}^n \binom{n}{j} x^j k^{n-j} (f(x) + kc), \end{aligned}$$

where in the last identity we have used (1). Comparing the coefficients of k^n on both sides we get $(n+1)f(x) = f(x) + nxc$, i.e. $f(x) = xc$ for all $x > 2$. Now let $x > 1$ and take $y > 2$. Then

$$c(x^{n+1} + y^{n+1}) = x^n f(x) + y^n f(y) = x^n f(x) + cy^{n+1}$$

which shows that $f(x) = cx$. Thus the solutions to the problem are the functions $f(x) = cx$, where $c \in \mathbb{R}$.

Problem 3.11. (Russia 1993) Find all functions $f : (0, \infty) \rightarrow (0, \infty)$ such that

$$f(x^y) = f(x)^{f(y)}$$

for all $x, y \in (0, \infty)$.

Solution. The function $f \equiv 1$ clearly satisfies the condition. We shall show now that the function $f(x) = x$ is the only nonconstant solution of the problem. Let $f(a) \neq 1$ for some $a > 0$. Then

$$f(a)^{f(xy)} = f(a^{xy}) = f(a^x)^{f(y)} = f(a)^{f(x)f(y)},$$

i.e. $f(xy) = f(x)f(y)$. Hence

$$f(a)^{f(x+y)} = f(a^{x+y}) = f(a^x)f(a^y) = f(a)^{f(x)+f(y)},$$

i.e. $f(x+y) = f(x) + f(y)$. Now Corollary 1.1 implies that $f(x) = cx$, where $c > 0$. Hence $cx^y = (cx)^{cy}$. In particular $c = c^{cy}$ for all $y > 0$ which shows that $c = 1$.

Problem 3.12. (generalization of Example 15.3) Find all functions $f : (0, \infty) \rightarrow (0, \infty)$ which are bounded from above on an interval and such that

$$f(xf(y)) = yf(x)$$

for all $x, y \in (0, \infty)$.

Solution. For any $z > 0$ set $x = \frac{z}{f(1)}$ and $y = 1$. Then

$$f(f(z)) = f(f(xf(1))) = f(1.f(x)) = xf(1) = z,$$

i.e. $f(f(z)) = z$. Hence $f(xy) = f(xf(f(y))) = f(y)f(x)$. Set

$$g(x) = \lg f(10^x), \quad x \in \mathbb{R}.$$

Then

$$\begin{aligned} g(x) + g(y) &= \lg f(10^x) + \lg f(10^y) = \lg f(10^x)f(10^y) \\ &= \lg f(10^x \cdot 10^y) = \lg f(10^{x+y}) = g(x+y), \end{aligned}$$

i.e. $g(x+y) = g(x) + g(y)$ for all $x, y \in \mathbb{R}$. Moreover the function $g(x)$ is bounded from above on an interval since $f(x)$ is so. Then Theorem 1.1 implies that $g(x) = cx$ for all $x \in \mathbb{R}$, where $c \in \mathbb{R}$. Hence $\lg f(10^x) = cx$, i.e. $f(10^x) = (10^x)^c$ showing that $f(x) = x^c$ for all $x \in (0, \infty)$. Now the given equation reads as $(xf(y))^c = yx^c$, i.e. $x^c \cdot y^{c^2} = yx^c$. Hence $c^2 = 1$, i.e. $c = \pm 1$. Thus $f(x) = x$ for all $x \in (0, \infty)$ or $f(x) = \frac{1}{x}$ for all $x \in (0, \infty)$.

Problem 3.13. (generalization of Example 15.4) Let S be the set of all real numbers greater than -1 . Find all functions $f : S \rightarrow S$ which are bounded from above on an interval and such that

$$f(x + f(y) + xf(y)) = y + f(x) + yf(x)$$

for all $x, y \in S$.

Solution. Replacing x and y by $x - 1$ and $y - 1$, respectively we get

$$f(x(f(y-1)+1)-1) = y(f(x-1)+1)-1$$

for all $x, y > 0$. Hence the function $g(x) = f(x-1)+1$ satisfies the functional equation $g(xg(y)) = yg(x)$ for $x, y \in (0, \infty)$. It follows from Problem 3.12 that either $g(x) = x$ or $g(x) = \frac{1}{x}, x \in (0, \infty)$. Hence the solutions of the problem are the functions $f(x) = x$ and $f(x) = -\frac{x}{1+x}$.

Problem 3.14. (IMO 2002) Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$(f(x) + f(z))(f(y) + f(t)) = f(xy - zt) + f(xt + yz)$$

for all $x, y, z, t \in \mathbb{R}$.

Solution. Setting $y = z = t = 0$ gives $2(f(x) + f(0))f(0) = 2f(0)$. Hence $2(f(0))^2 = f(0)$ and it follows that either $f(0) = \frac{1}{2}$ or $f(0) = 0$. If $f(0) = \frac{1}{2}$ then $f(x) = \frac{1}{2}$ for all $x \in \mathbb{R}$ and this function is a solution of the problem.

Suppose now that $f(0) = 0$. Setting $z = t = 0$ gives $f(x)f(y) = f(xy)$. In particular, $f^2(1) = f(1)$, i.e. either $f(1) = 0$ or $f(1) = 1$. If $f(1) = 0$ then $f(x) = f(x)f(1) = 0$ for all $x \in \mathbb{R}$ and this function is also a solution of the problem.

So we may assume that $f(1) = 1$. Setting $x = 0, y = t = 1$ gives $f(z) = f(-z)$, i.e. f is an even function. Then the identity $f(x^2) = f^2(x)$ shows that $f(x) \geq 0$ for all x . Now setting $x = t, y = z$ we get

$$(f(x) + f(y))^2 = f(x^2 + y^2).$$

Consider the function $g(x) = \sqrt{f(x)}$. Then $g(x^2) = \sqrt{f(x^2)} = f(x) = g^2(x)$ and the above identity implies

$$g(x^2) + g(y^2) = g(x^2 + y^2).$$

Thus g is a non-negative even function which is additive on $(0, \infty)$ and $g(1) = 1$ (this function is also multiplicative on $(0, \infty)$). Now it follows from Theorem

1.5 that $g(x) = x$ for $x \in (0, \infty)$ and the fact that g is even shows that $g(x) = |x|$ for all $x \in \mathbb{R}$. Thus $f(x) = x^2$ and it is easy to check that this function satisfies the given condition (in fact, we have to check the Lagrange identity).

Problem 3.15. (Romania TST 2007) Find all functions $f: \mathbb{R} \rightarrow [0, \infty)$ that satisfy

$$f(x^2 + y^2) = f(x^2 - y^2) + f(2xy)$$

for all $x, y \in \mathbb{R}$.

Solution. For $x = y = 0$ we obtain $f(0) = 0$. For $x = 0$ we obtain

$$f(y^2) = f(-y^2);$$

hence f is an even function and it is sufficient to find it for positive numbers. Let a, b be positive real numbers. Then the algebraic system

$$x^2 - y^2 = a, \quad 2xy = b$$

always has real solutions. It is sufficient, for proving this, to observe that the solutions can be obtained as the intersection points of two hyperbolas.

Taking a solution (x, y) of the above system, one obtains

$$x^2 + y^2 = \sqrt{a^2 + b^2}.$$

Then, for all positive numbers a and b , the function f satisfies

$$f(a) + f(b) = f(\sqrt{a^2 + b^2}).$$

Let $g: [0, \infty) \rightarrow [0, \infty)$ be defined $g(a) = f(\sqrt{a})$. Then

$$g(a^2) + g(b^2) = g(a^2 + b^2),$$

for all $a, b \in \mathbb{R}$. For $x \geq 0, y \geq 0$ we set $a = \sqrt{x}, b = \sqrt{y}$ in the above identity and get $g(x + y) = g(x) + g(y)$. Thus, by Theorem 1.5, it follows that $g(x) = kx$ and $f(x) = kx^2$, for all $x \geq 0$. $\square$

Problem 3.16. Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ that satisfy

$$f(y + zf(x)) = f(y) + xf(z)$$

for all $x, y, z \in \mathbb{R}$.

Solution. If $f \not\equiv 0$, then $f(x_1) = f(x_2)$ implies that $x_1 = x_2$. So f is injective. Also f is surjective as if we fix y, z with $f(z) \neq 0$ then the RHS of the given condition is a linear function. Thus f is a bijection. Now if $x = 0$, then $f(y + zf(0)) = f(y)$ and the injectivity of f implies that $y + zf(0) = y$ for all z which is possible only for $f(0) = 0$. Next if we set $y = 0, x = 1$, we get $f(zf(1)) = f(z)$ hence $zf(1) = z$ for all z so $f(1) = 1$. Next set $y = 0, z = x$ to get $f(f(x)) = x$. Hence if we set $y = 0$ and replace x by $f(x)$, we get $f(zx) = f(z)f(x)$ and so f is multiplicative. If we set $z = 1, x \rightarrow f(x)$, we get $f(y + x) = f(y) + f(x)$ so f is additive. As we know from Corollary 1.1 the only additive and multiplicative functions are $f(x) = 0$ and $f(x) = x$ and they are the only solutions of the given equation.

Problem 3.17. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x)f(yf(x) - 1) = x^2f(y) - f(x)$$

for all real numbers x and y .

Solution. Setting $y = 0$ we get $f(x)(1 + f(-1)) = x^2f(0)$. If $f(-1) \neq -1$, this implies $f(x) = ax^2$ and plugging in original equation we see that $a = 0$, i.e. $f(x) = 0$ for all $x \in \mathbb{R}$.

So let us assume from now on that $f(x)$ is not identically zero. Then $f(-1) = -1$ and $f(0) = 0$.

We shall show now that f is surjective. If $f(x) = 0$, then $x^2f(y) = 0$ and so $x = 0$. Setting $y = x$ and $x = -1, y = -xf(x)$ in the given equation we get respectively

$$f(x)f(xf(x) - 1) = x^2f(x) - f(x) \text{ and } f(xf(x) - 1) = -f(-xf(x)) - 1.$$

Thus

$$f(x)f(xf(x) - 1) = -f(x)f(-xf(x)) - f(x) \text{ and } f(x)f(-xf(x)) = -x^2f(x).$$

So $f(-xf(x)) = -x^2$ for all $x \in \mathbb{R}$ and we get that every non-positive real number belongs to $f(\mathbb{R})$. Let now $x > 0$ and u be such that $f(u) = -(x + 1)$. Then setting $x = -1, y = -u - 1$ in the given equation we get

$$-f(u) - 1 = f(-u - 1)$$

and so $x = f(u+1)$. Thus the positive reals also belong to $f(\mathbb{R})$ and therefore $f(x)$ is surjective.

Our next step is to prove that f is an additive function. To do this we set $x = -1, y = -yf(x)$ in the given equation to obtain that

$$f(yf(x) - 1) = -f(-yf(x)) - 1$$

and so

$$f(x)f(yf(x) - 1) = -f(x)f(-yf(x)) - f(x).$$

Comparing with initial identity we get $x^2f(y) = -f(x)f(-yf(x))$. Now we set $y = \frac{1}{f(x)} - y$ in the given equation and get

$$f(x)f(-yf(x)) = x^2f\left(\frac{1}{f(x)} - y\right) - f(x).$$

Thus

$$-x^2f(y) = x^2f\left(\frac{1}{f(x)} - y\right) - f(x)$$

and setting $y = 0$ we get $0 = x^2f\left(\frac{1}{f(x)}\right) - f(x)$. Hence

$$-x^2f(y) = x^2f\left(\frac{1}{f(x)} - y\right) - x^2f\left(\frac{1}{f(x)}\right)$$

and so

$$-f(y) = f\left(\frac{1}{f(x)} - y\right) - f\left(\frac{1}{f(x)}\right)$$

for all $x \neq 0$. Since $f(x)$ is surjective, $\frac{1}{f(x)}$ can take any non zero value and we get $f(z - y) = f(z) - f(y)$ for all $y, z \neq 0$. Swapping y and z , we see that $f(-x) = -f(x)$ and so $f(z - y) = f(z) - f(y)$ for all z, y .

Now we shall show that $f(x)f(\frac{1}{x}) = 1$ for all $x \neq 0$. Let $x \neq 0$. Then setting $x = -1, y = f(x)$ in the initial identity we get $f(-f(x) - 1) = -f(f(x)) - 1$ and so $f(x)f(-f(x) - 1) = -f(x)f(f(x)) - f(x)$. Comparing with the identity

obtained by setting $y = 1$ in the given condition we see that $f(f(x)) = \frac{x^2}{f(x)}$. Let then $x \neq 0$ be such that $f(x) \neq 0$. Then setting $y = \frac{1}{f(x)}$ in the given equation we obtain that $x^2 f\left(\frac{1}{f(x)}\right) = f(x)$. Thus $f\left(\frac{1}{f(x)}\right) = \frac{f(x)}{x^2}$ and therefore $f(f(x))f\left(\frac{1}{f(x)}\right) = 1 \forall x \neq 0$. Now using the fact that the function $f(x)$ is surjective we get that $f(x)f\left(\frac{1}{x}\right) = 1$ for all $x \neq 0$. Hence it follows from Problem 1.7 that $f(x) = x$.

Hence we have two solutions: $f(x) = 0$ and $f(x) = x$.

Problem 3.18. (IMO Shortlist 2003) Find all nondecreasing functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that $f(0) = 0$, $f(1) = 1$, and

$$f(x) + f(y) = f(x)f(y) + f(x + y - xy)$$

for all $x < 1$ and $y > 1$.

Solutions. Set $g(x) = f(x + 1) - 1$. The function g is nondecreasing and $g(0) = 0, g(-1) = -1$. Let $r < 0$ and $s > 0$. Then $r = x - 1$ and $s = y - 1$ for $x < 1$ and $y > 1$. Hence

$$\begin{aligned} g(-rs) &= g(-(x-1)(y-1)) = f(x+y-xy) - 1 = f(x) + f(y) - f(x)f(y) - 1 \\ &= -(f(x) - 1)(f(y) - 1) = -g(r)g(s) \end{aligned}$$

which shows that $g(rs) = -g(-r)g(s)$ for all $r, s > 0$.

We first consider the case where $g(1) = 0$. Then for all $s > 0$ we have $g(s) = -g(-s)g(1) = 0$. So every nondecreasing function g such that $g(-1) = -1$ and $g(s) = 0$ for $s \geq 0$ will satisfy the functional equation for g . Next consider the case where $g(1) \neq 0$. Note that for all $x > 0$, $g(x) = -g(-x)g(1)$. Set $h(x) = g(x)/g(1)$. Then

$$h(xy) = g(xy)/g(1) = -g(-x)g(y)/g(1) = h(x)h(y).$$

Moreover, $h(0) = 0$ and $h(1) = 1$. Because h is monotonic we conclude from Theorem 1.11 that there exists some $k > 0$ such that $h(x) = x^k$ for all

$x > 0$. Thus $g(x) = cx^k$, where $c = g(1)$. Furthermore for $x > 0$ we have $g(-x) = -g(x)/g(1) = -x^k$. Thus in the first case the solutions for f are given by

$$f(x) = \begin{cases} 1, & \text{for } x \geq 1, \\ 0, & \text{for } x = 0, \\ \text{any}, & \text{for } x < 0. \end{cases}$$

In the second case the full family of solutions is

$$f(x) = \begin{cases} 1 + c(x-1)^k, & \text{for } x > 1, \\ 1, & \text{for } x = 1, \\ 1 - c(1-x)^k, & \text{for } x < 1. \end{cases}$$

Problem 3.19. Let n be an integer greater than 1. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x^n + 2f(y)) = f(x)^n + y + f(y)$$

for all $x, y \in \mathbb{R}$.

Solution. We shall consider two cases according to the parity of n .

Case 1. n is even.

If $f(a) = f(b)$, then the given identity implies $a = b$ and so $f(x)$ is injective. It follows also that $f(x)^n = f(-x)^n$ and so $f(-x) = -f(x)$ since $f(x)$ is injective. As a consequence, $f(0) = 0$ and $f(x^n) = f(x)^n$. Now the given equation may be written as $f(x^n + 2f(y)) = f(x)^n + y + f(y)$, so

$$f(x + 2f(y)) = f(x) + y + f(y)$$

for all $x \geq 0$. Now replacing y by $-y$ we get $f(x - 2f(y)) = f(x) - y - f(y)$ and so

$$f(-x + 2f(y)) = f(-x) + y + f(y).$$

In particular, $f(x - 2f(x)) = -x$ which shows that $f(x)$ is surjective. We have also that $f(2f(y)) = y + f(y)$ and therefore $f(x + 2f(y)) = f(x) + f(2f(y))$. Hence the surjectivity of f implies $f(x + y) = f(x) + f(y)$. Since n is even the

identity $f(x^n) = f(x)^n$ implies that $f(x) > 0, \forall x > 0$ and by Theorem 1.1 it follows that $f(x) = ax$. Plugging this in original equation, we get $f(x) = x$.

Case 2. n is odd.

First we shall prove that $f(x+y) = f(x) + f(y) - f(0)$. Indeed, it follows from the given equation that

$$f(x^n + 2f(y)) = f(x^n + 2f(0)) + y + f(y) - f(0)$$

and, since n is odd we get

$$f(x + 2f(y) - 2f(0)) = f(x) + y + f(y) - f(0).$$

Let then $g(x) = f(x) - f(0)$. This last equation becomes

$$g(x + 2g(y)) = g(x) + y + g(y)$$

for all x, y . On the other hand the identity $f(x + 2f(y)) = f(x) + y + f(y)$ obtained in *Case 1* implies $g(-x - 2g(-x)) = x$ and so $g(x)$ is surjective. We also get from it that $g(x + 2g(y)) = g(x) + y + g(y)$ and $g(2g(y)) = y + g(y)$. Subtracting these two identities, we get $g(x + 2g(y)) = g(x) + g(2g(y))$ and so, since g is surjective $g(x + y) = g(x) + g(y)$. Thus $f(x + y) = f(x) + f(y) - f(0)$ as desired.

Our next step is to prove that $f(0) = 0$. We have

$$f(x^n + 2f(0)) = f(x)^n + f(0)$$

and $f(x^n + 2f(0)) = f(x^n) + f(2f(0)) - f(0)$, so $f(x^n) = f(x)^n + c$ where $c = 2f(0) - f(2f(0))$. Then $f(x + y) = f(x) + f(y) - f(0)$ implies

$$f(px) = pf(x) - (p-1)f(0)$$

for all x , and all $p \in \mathbb{Z}$. So

$$f(p^n x^n) = p^n f(x^n) - (p^n - 1)f(0) = p^n (f(x)^n + c) - (p^n - 1)f(0)$$

But

$$f(p^n x^n) = f(px)^n + c = (pf(x) - (p-1)f(0))^n + c$$

and we get

$$p^n(f(x)^n + c) - (p^n - 1)f(0) = (pf(x) - (p - 1)f(0))^n + c.$$

Hence the polynomial

$$S(z) = z^n(f(x)^n + c) - (z^n - 1)f(0) - (zf(x) - (z - 1)f(0))^n - c$$

has infinitely many zeros (since $S(p) = 0$, $\forall p \in \mathbb{Z}$), and so it is the zero polynomial. Then the coefficient of z in this polynomial is 0, i.e. $f(0)^{n-1}(f(x) - f(0)) = 0$. But the constant polynomials are not solutions and we get $f(0) = 0$ and $c = 0$.

Now we are ready to prove that $f(x) = x$. As we saw $f(x)$ satisfies the following identities:

$$f(x + y) = f(x) + f(y), \quad f(x^n) = f(x)^n \quad \text{and} \quad 2f(f(x)) = x + f(x).$$

From the second identity above, we get $f(1) \in \{-1, 0, 1\}$, and using the first and the third identities, it follows $f(1) = 1$. Hence $f(p) = p$ for all $p \in \mathbb{Z}$. Then

$$f((x + p)^n) = f\left(\sum_{k=0}^n \binom{n}{k} x^k p^{n-k}\right) = \sum_{k=0}^n \binom{n}{k} f(x^k) p^{n-k}.$$

But

$$f((x + p)^n) = (f(x) + p)^n = \sum_{k=0}^n \binom{n}{k} f(x)^k p^{n-k}$$

and therefore

$$\sum_{k=0}^n \binom{n}{k} (f(x)^k - f(x^k)) p^{n-k} = 0.$$

Thus the polynomial

$$T(z) = \sum_{k=0}^n \binom{n}{k} (f(x)^k - f(x^k)) z^{n-k}$$

has infinitely many roots and so is identically zero. Since n is odd and $n > 1$, we get $n > 2$ and we can look at the coefficient of z^{n-2} to get $f(x^2) = f(x)^2$.

Hence $f(x) > 0$ for all $x > 0$ and Theorem 1.1 implies that $f(x) = xf(1) = x$.

Problem 3.20. (China TST 2009) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function such that for any pairwise distinct real numbers a, b, c, d satisfying the identity

$$\frac{a-b}{b-c} + \frac{a-d}{d-c} = 0,$$

the values $f(a), f(b), f(c), f(d)$ are pairwise distinct and

$$\frac{f(a)-f(b)}{f(b)-f(c)} + \frac{f(a)-f(d)}{f(d)-f(c)} = 0.$$

Prove that f is a linear function.

Solution. We may assume that $f(0) = 0$ and $f(1) = 1$ (consider a constant multiple of the function $f(x) - f(0)$). Moreover, one can prove easily that if $f(a) = 0$, then $a = 0$. Now we set $a = 0$ in the condition to get that if distinct nonzero real numbers b, c, d satisfy $\frac{1}{b} + \frac{1}{d} = \frac{2}{c}$, then

$$\frac{1}{f(b)} + \frac{1}{f(d)} = \frac{2}{f(c)}.$$

This suggests to consider a new function $g : \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R} \setminus \{0\}$, defined by

$$g(x) = \frac{1}{f\left(\frac{1}{x}\right)}.$$

Then it is easy to check that for distinct nonzero real numbers x and y with $x + y \neq 0$ we have $g(x) + g(y) = 2g\left(\frac{x+y}{2}\right)$ and $g(1)=1$. Now we shall prove that g is monotonic. This follows from the following

Lemma. If $\frac{1}{x} + \frac{1}{y} \neq \frac{1}{z}$ for nonzero real numbers $x > y > z$, then $g(x) - g(z)$ and $g(y) - g(z)$ are both positive or negative.

Proof of lemma. Set

$$a = x, \quad c = y, \quad b = z + \sqrt{(x-z)(y-z)}, \quad d = z - \sqrt{(x-z)(y-z)}.$$

According to $x > y > z$, we have $a > b > c > d$. If b or d is 0, then $z^2 = (x-z)(y-z)$, which leads to $\frac{1}{x} + \frac{1}{y} = \frac{1}{z}$, a contradiction. So, a, b, c, d are distinct nonzero real numbers satisfying the condition in the original problem. Thus we get

$$\frac{f(a) - f(b)}{f(b) - f(c)} + \frac{f(a) - f(d)}{f(d) - f(c)} = 0,$$

which can be written as

$$\lfloor 2g(a) - g(b) - g(d) \rfloor \lfloor 2g(c) - g(b) - g(d) \rfloor = g(b) - g(d)^2.$$

Since $g(b) + g(d) = 2g\left(\frac{b+d}{2}\right)$ we get

$$\left\lfloor g(a) - g\left(\frac{b+d}{2}\right) \right\rfloor \left\lfloor g(c) - g\left(\frac{b+d}{2}\right) \right\rfloor = \frac{\lfloor g(b) - g(d) \rfloor^2}{4}.$$

So

$$\lfloor g(x) - g(z) \rfloor \lfloor g(y) - g(z) \rfloor = \frac{\lfloor g(b) - g(d) \rfloor^2}{4} \geq 0$$

and the Lemma is proved.

Now we show two ways to finish the solution of the problem. Since g is monotonic and satisfies Jensen's equation on $(0, \infty)$ and $(-\infty, 0)$ it follows from Theorem 2.1 and Theorem 1.1 that the restrictions of g on $(0, \infty)$ and $(-\infty, 0)$ are linear functions. This shows that the restrictions of f on the above intervals have the form $\frac{x}{px+q}$ and using the given condition we see easily that $f(x) = x$.

Another way to prove that $f(x) = x$ is the following. We get from Jensen's equation that $g(z) = z$ for all nonzero rational numbers. Assume that $g(x) \neq x$ for a real number x . If $g(x) > x$, then we can find a rational number y satisfying $x < y < g(x)$ and a rational number z satisfying $g(z) > g(x)$ and $\frac{1}{z} + \frac{1}{y} \neq \frac{1}{x}$. So $g(z) = z$ and $g(y) = y$. Since $z > y > x$ but $g(z) - g(x)$ is positive while $g(y) - g(x)$ is negative we obtain a contradiction to the Lemma. In the case $g(x) < x$ we similarly obtain a contradiction.

19.4 Substitutions

Problem 4.1. Find all continuous functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x+y) - f(x) - f(y) = xy(x+y)$$

for all $x, y \in \mathbb{R}$.

Solution. We can guess the solution $f(x) = \frac{x^3}{3}$. Thus the function

$$g(x) = f(x) - \frac{x^3}{3}$$

is additive and continuous. It follows from Theorem 1.1 that $g(x) = cx$ and

$$f(x) = \frac{x^3}{3} + cx.$$

Problem 4.2. Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(xy + 2x + 2y) = f(x)f(y) + 2x + 2y$$

for all $x, y \in \mathbb{R}$.

Solution. Substitute $y = -2$ to get $f(-4) = f(x)f(-2) + 2x - 4$. We therefore get $f(-2) \neq 0$ (or else $2x - 4 = f(-4)$ for all $x \in \mathbb{R}$), and then

$$f(x) = \frac{4 + f(-4) - 2x}{f(-2)}.$$

Thus f is linear and we set $f(x) = ax + b$. We have

$$a(xy + 2x + 2y) + b = (ax + b)(ay + b) + 2x + 2y,$$

i.e. $(a - a^2)xy + (2a - ab - 2)(x + y) + (b - b^2) = 0$. So $a^2 = a$, $a(2 - b) = 2$ and $b = b^2$ which gives $a = 1$, $b = 0$ and $f(x) = x$.

Problem 4.3. (USAMO 2002) Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x^2 - y^2) = xf(x) - yf(y)$$

for all $x, y \in \mathbb{R}$.

Solution. Setting $y = 0$ gives $f(x^2) = xf(x)$ which implies $f(0) = 0$. If $x \neq 0$ then $xf(x) = f(x^2) = -xf(-x)$, i.e. $f(-x) = -f(x)$. Hence $f(-x) = -f(x)$ and $f(x^2 - y^2) = f(x^2) - f(y^2)$ for all $x, y \in \mathbb{R}$. These two equations imply that

$$f(x) + f(y) = f(x + y)$$

for all $x, y \in \mathbb{R}$. Note that the above equation together with $f(x^2) = xf(x)$ is equivalent to the given equation. For any x and $y = 1 - x$ we get

$$\begin{aligned} f(x) - f(y) &= f(x - y) = f(x^2 - y^2) = xf(x) - yf(y) \\ &= xf(x) - (1 - x)f(y) = x(f(x) + f(y)) - f(y) = xf(1) - f(y), \end{aligned}$$

i.e. $f(x) = xf(1)$. Hence $f(x) = cx$, where $c \in \mathbb{R}$.

Problem 4.4. (IMO Shortlist 2005) Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x + y) + f(x)f(y) = f(xy) + 2xy + 1$$

for all $x, y \in \mathbb{R}$.

Solution. It is easy to check that the functions $f(x) = 2x - 1$, $f(x) = -x - 1$ and $f(x) = x^2 - 1$ satisfy the given conditions. We shall prove that these are the only solutions of the problem.

Setting $y = 1$ gives

$$f(x + 1) = af(x) + 2x + 1, \tag{1}$$

where $a = 1 - f(1)$. Then replace y by $y + 1$ in the initial condition and use (1) to expand $f(x + y + 1)$ and $f(y + 1)$. The result is

$$a(f(x + y) + f(x)f(y)) + (2y + 1)(1 + f(x)) = f(x(y + 1)) + 2xy + 1,$$

and therefore

$$a(f(xy) + 2xy + 1) + (2y + 1)(1 + f(x)) = f(x(y + 1)) + 2xy + 1.$$

Now set $x = 2t$ and $y = -\frac{1}{2}$ to obtain

$$a(f(-t) - 2t + 1) = f(t) - 2t + 1.$$

Replacing t by $-t$ gives also

$$a(f(t) + 2t + 1) = f(-t) + 2t + 1.$$

We now eliminate $f(-t)$ from the last two equations. Then

$$(1 - a^2)f(t) = 2(1 - a)^2t + a^2 - 1.$$

Note that $a \neq -1$ (or else $8t = 0$ for any t , which is false). If additionally, $a \neq 1$, then $1 - a^2 \neq 0$; therefore

$$f(t) = 2\frac{1-a}{1+a}t - 1.$$

Setting $t = 1$ and recalling that $f(1) = 1 - a$, we get $a = 0$ or $a = 3$, which gives the first two solutions.

Let $a = 1$. Then f is an even function. Set now $y = x$ and $y = -x$ in the original equation. It follows that

$$f(2x) + f^2(x) = f(x^2) + 2x^2 + 1, \quad f(0) + f^2(x) = f(x^2) - 2x^2 + 1,$$

respectively. Subtracting gives $f(2x) = 4x^2 + f(0)$. Set $x = 0$ in (1). Since $f(1) = 1 - a = 0$, this yields $f(0) = -1$. Hence $f(2x) = (2x)^2 - 1$, i.e., $f(x) = x^2 - 1$. This completes the solution.

Problem 4.5. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(f(x) - y^2) = f(x)^2 - 2f(x)y^2 + f(f(y))$$

for all $x, y \in \mathbb{R}$.

Solution. The function $f(x) = 0$ is a solution and from now on we assume that f is not identically zero.

Denote $A = f(\mathbb{R})$. Then setting $x = y = 0$ we get $f(f(0)) = f(0)^2 + f(f(0))$ and so $f(0) = 0$. Now set $y = 0$ to get $f(f(x)) = f(x)^2$ and so $f(x) = x^2$ for all $x \in A$.

The given equation may be written as

$$f(f(x) - y^2) = f(x)^2 - 2f(x)y^2 + f(y)^2.$$

Hence if $y \in A$ we get that $f(f(x) - f(y)) = f(x)^2 - 2f(x)f(y) + f(y)^2$ and so $f(f(x) - f(y)) = (f(x) - f(y))^2$ for all $x \in \mathbb{R}, y \in A$.

Let u be such that $f(u) \neq 0$. Then $f(f(u)) = f(u)^2$ which shows that $f(v) > 0$, where $v = f(u)$. Now take $t \leq 0$ and s such that $s^2 = \frac{f(v)^2 - t}{2f(v)}$.

Then $f(f(v) - s^2) = f(f(s)) = t$ and so any non positive real number t can be written as $t = f(x) - f(y)$ with $x \in \mathbb{R}$ and $y \in A$. But we know that $f(f(x) - f(y)) = (f(x) - f(y))^2$ for all $x \in \mathbb{R}, y \in A$, so $f(x) = x^2$ for all $x \leq 0$. Hence $x \in A$ for all $x \geq 0$ since $x = f(-\sqrt{x})$. We also know that $f(x) = x^2$ for all $x \in A$ and we get $f(x) = x^2$ for all $x \geq 0$. Thus $f(x) = x^2$ for all x which indeed is a solution.

Problem 4.6. Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x - f(y)) = f(x) + x \cdot f(y) + f(f(y))$$

for all $x, y \in \mathbb{R}$.

Solution. Let $f(x)$ be a solution that is not identically zero. Then there is u such that $f(u) \neq 0$ and the given identity implies that

$$f\left(\frac{x - f(f(u))}{f(u)} - f(u)\right) - f\left(\frac{x - f(f(u))}{f(u)}\right) = x$$

Hence any real number u can be written as $f(a) - f(b)$ for some $a, b \in \mathbb{R}$.

The given equation implies

$$f(f(x)) = \frac{f(0) - f(x)^2}{2}$$

and

$$f(f(x) - f(y)) = f(f(x)) + f(x)f(y) + f(f(y))$$

Hence

$$\begin{aligned} f(f(x) - f(y)) &= \frac{f(0) - f(x)^2}{2} + f(x)f(y) + \frac{f(0) - f(y)^2}{2} \\ &= f(0) - \frac{(f(x) - f(y))^2}{2}. \end{aligned}$$

So $f(u) = a - \frac{u^2}{2}$ for any real u which may be written as $f(x) - f(y)$ for some x, y and hence for every real u , according to the remark above.

Now, you just have to plug $f(x) = a - \frac{x^2}{2}$ in the original equation to get $a = 0$ and so the two solutions to this problem: $f(x) = 0$ and $f(x) = -\frac{x^2}{2}$.

Problem 4.7. Find all continuous functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x+y) - f(x-y) = 2f(xy+1) - f(x)f(y) - 4$$

for all $x, y \in \mathbb{R}$.

Solution. If we set $x = 0$, we get $f(y) - f(-y) = 2f(1) - f(0)f(y) - 4$. If we set $y = 0$, we get $2f(1) - f(x)f(0) - 4 = 0$. So if f is not a constant, then $f(0) = 0$. Hence $f(1) = 2$ and from the first relation we get $f(y) - f(-y) = 0$. So f is even. (Note that if $f = c$, then $2c - c^2 - 4 = 0$ which has no real roots and so f cannot be a constant). Next set $y = 1$ to get

$$f(x+1) - f(x-1) = 2f(x+1) - 2f(x) - 4$$

and so $f(x+1) - 2f(x) + f(x-1) = 4$. From here we deduce by induction that $f(n) = 2n^2$ for all $n \in \mathbb{N}$ and that $f(x+n) = 2n^2 + bn + c$ for some b, c depending only on x and not on $n \in \mathbb{N}$. For $x = \frac{1}{2}$ we have $b = 0$ because

$$f\left(\frac{1}{2}\right) = f\left(-\frac{1}{2}\right).$$

Next set $y = \frac{1}{2}$ to get

$$f\left(x + \frac{1}{2}\right) - f\left(x - \frac{1}{2}\right) = 2f\left(\frac{x}{2} + 1\right) - f(x)f\left(\frac{1}{2}\right) - 4.$$

If now $x = 2k$, where $k \in \mathbb{N}$, we get

$$\left(2\left(2k + \frac{1}{2}\right)^2 + c\right) - \left(2\left(2k - \frac{1}{2}\right)^2 + c\right) = 2(k+1)^2 - 8k^2\left(\frac{1}{2} + c\right) - 4$$

and we deduce that $c = 0$. So

$$f\left(k + \frac{1}{2}\right) = 2\left(k + \frac{1}{2}\right)^2.$$

Now we prove by induction on k that $f(t) = 2t^2$ for all $t = \frac{a}{2^k}$ where $a, k \in \mathbb{N}$. The base is proven. For the induction step, write the previously obtained relation

$$f\left(x + \frac{1}{2}\right) - f\left(x - \frac{1}{2}\right) = 2f\left(\frac{x}{2} + 1\right) - f(x)f\left(\frac{1}{2}\right) - 4.$$

Now if $t = \frac{a}{2^k}$ where $k > 1$ and a is odd then taking $x = 2(t - 1)$ and plotting it into the relation we get $f(t) = 2t^2$ as desired. As the numbers $\frac{a}{2^k}$ are dense in $(0, \infty)$ and f is even, then $f(x) = 2x^2$. Note that this function satisfies the given condition.

Problem 4.8. (IMO Shortlist 2000) Find all pairs of functions $f, g: \mathbb{R} \rightarrow \mathbb{R}$ that satisfy

$$f(x + g(y)) = xf(y) - yf(x) + g(x)$$

for all $x, y \in \mathbb{R}$.

Solution. If $f \equiv 0$, then so is g . Now assume that $f \not\equiv 0$. Let us prove that g takes the value 0. Set $x = 0$ to get $f(g(y)) = -yf(0) + g(0)$. In particular, f takes the value 0 because if $f(0) \neq 0$, then $-yf(0) + g(0)$ is surjective. Next replace x by $g(x)$ to obtain

$$f(g(x) + g(y)) = g(x)f(y) - yf(0) + xyf(0) + g(g(x)).$$

Swapping x and y we get

$$f(g(x) + g(y)) = f(x)g(y) - xf(0) + xyf(0) + g(g(y)).$$

Therefore

$$g(x)f(y) - yf(0) + f(f(x)) = g(y)f(x) - xf(0) + g(g(x)). \quad (1)$$

Now set $y = t$ with $f(t) = 0$ to get

$$-tf(0) + g(g(x)) = g(t)f(x) - xf(0) + g(g(t))$$

so $g(g(x)) = c - ax + uf(x)$, where $c = tf(0) + g(g(t))$, $a = f(0)$, $u = g(t)$.

If we substitute into (1), we get

$$g(x)f(y) + uf(x) = g(y)f(x) + uf(y)$$

and hence

$$(g(x) - u) = \frac{g(y) - u}{f(y)}f(x)$$

if $f(y) \neq 0$. Taking a fixed value of y at which f does not vanish we get $g(x) = kf(x) + u$ or g depends linearly on f . If $k = 0$, then g is a constant and $f(x + u) = xf(y) - yf(x) + u$. So setting $y = x$ gives us $f(x + u) = u$ and then $f = g = u$. Now assume that $k \neq 0$. We have $g(g(x)) = c - ax + uf(x)$ and also $f(g(x)) = -xf(0) + g(0)$. But

$$g(g(x)) = kf(g(x)) + u = k(-xf(0) + g(0)) + u = -kf(0)x + kg(0) + u.$$

So $c - ax + uf(x) = -kf(0)x + kg(0) + u$. If $u \neq 0$, then we express f as a linear function hence so is g . If $u = 0$, then $g(t) = 0$ and $-kf(0)x + g(0) = c - ax$. Hence $f(0) = a = -kf(0)$, $c = g(0)$. Then set $y = t$ to get

$$f(x) = -tf(x) + kf(x) = (k - t)f(x)$$

and so $k - t = 1$. Now the original condition can be rewritten as

$$f(x + kf(y)) = xf(y) - yf(x) + kf(x).$$

In particular, we deduce that f is injective as if $f(y_1) = f(y_2)$, then by setting $y = y_1, y_2$ we conclude that $y_1f(x) = y_2f(x)$ for all x , i.e. $y_1 = y_2$. If $f(0) = 0$, then $f(g(y)) = g(0)$. So the injectivity of f implies that g is a constant and hence so is f . Otherwise we get

$$f(kf(y)) = -yf(0) + g(0) = (k - y)f(0).$$

Now replace x by $kf(x)$ to get

$$f(kf(x) + kf(y)) = kf(x)f(y) + (k - y)f(kf(x)) = kf(x)f(y) + (k - y)^2 f(0).$$

By symmetry we deduce also

$$f(kf(x) + kf(y)) = kf(x)f(y) + (k - x)^2 f(0)$$

and so $(k - x)^2 f(0) = (k - y)^2 f(0)$ for all x, y which is impossible.

We have thus proven that f, g are linear functions. Set

$$f(x) = ax + b, \quad g(x) = cx + d.$$

We substitute to get

$$f(x + cy + d) = x(ay + b) - y(ax + b) + cx + d$$

or

$$ax + acy + ad + b = (b - d)x - by + d + cx$$

so $(a + d - b - c)x + (ac + b)y + ad + b - d = 0$. So $b = -ac, d = b + c - a = c - ac - a$. Then $ad + b - d = 0$ so $a(c - ac - a) - ac - c + ac + a = 0$ or $ac - a^2c - a^2 - c + a = 0$ so $c(a^2 - a + 1) = a(1 - a)$ and hence $c = \frac{a(1 - a)}{a^2 - a + 1}$.

Problem 4.9. (Bulgaria 2008) Find all continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$ that satisfy

$$(1 - f(x)f(y))f(x + y) = f(x) + f(y) + 2f(x)f(y)$$

for all $x, y \in \mathbb{R}$.

Solution. First, assume that $f(x_0) = -1$ for some x_0 . Setting $x = y = x_0/2$ in the given condition we get $(f(x_0/2) + 1)^2 = 0$, i.e. $f(x_0/2) = -1$. Hence $f(x_0/2^n) = -1$ for all $n \in \mathbb{N}$. Since f is continuous (at 0), it follows that $f(0) = -1$. Set now $y = 0$ in the initial condition to get $(f(x) + 1)^2 = 0$, i.e. $f(x) = -1$. Obviously $f \equiv -1$ is a solution.

Let now $f(x) \neq -1$ for all x . If $g(x) = \frac{f(x)}{f(x) + 1}$, then the given condition takes the form $g(x + y) = g(x) + g(y)$. Since g is continuous, it follows that

$g(x) = ax$ for some $a \in \mathbb{R}$. But $g(x) \neq 1$ for all x and hence $a = 0$, i.e. $f \equiv 0$. This function is also a solution.

Problem 4.10. Find all continuous functions $f : (-1, 1) \rightarrow \mathbb{R}$ such that

$$f\left(\frac{x+y}{1-xy}\right) = f(x) + f(y)$$

for all $x, y \in (-1, 1)$.

Solution. Set $g(\alpha) = f(\tan \alpha)$, $\alpha \in \left(-\frac{\pi}{4}, \frac{\pi}{4}\right)$. Then

$$\begin{aligned} g(\alpha + \beta) &= f(\tan(\alpha + \beta)) = f\left(\frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}\right) \\ &= f(\tan \alpha) + f(\tan \beta) = g(\alpha) + g(\beta). \end{aligned}$$

Hence Theorem 2.1 implies (see the solution of Problem 2.2) that $g(\alpha) = c\alpha$, where c is a constant. Thus $f(\tan \alpha) = c\alpha$ and setting $\alpha = \arctan x$, $x \in (-1, 1)$ we get that $f(x) = c \arctan x$.

Problem 4.11. Find all continuous functions $f : (a; b) \rightarrow \mathbb{R}$ that satisfy $f(xyz) = f(x) + f(y) + f(z)$ whenever $xyz, x, y, z \in (a; b)$, where $1 < a^3 < b$.

Solution. Let $a = e^k, b = e^l$ with $0 < 3k < l$. Consider the function $g : (0; l - k) \rightarrow \mathbb{R}$ such that $g(t) = f(e^{k+t})$. The condition rewrites as $g(u + v + w + 2k) = g(u) + g(v) + g(w)$ whenever $u + v + w < l - 3k$. Particularly $g(u + v + 2k) = g(u) + g(v) + g(0) = g(u + v) + 2g(0)$ for $u + v < l - 2k$. (Note that $g(0)$ is actually not defined but we can define $g(0) = \frac{1}{3}g(2k)$ and observe that when $c \rightarrow 0$ we have $g(3c + 2k) \rightarrow g(2k) = 3g(0)$; so $g(x) \rightarrow g(0)$ as $x \rightarrow 0$.) The condition $g(u) + g(v) + g(0) = g(u + v) + 2g(0)$ is then obtained by taking $c \rightarrow 0$ in $g(u) + g(v) + g(2c) = g(u + v) + 2g(c)$. Then the function $g - g(0)$ is additive and continuous on $[0; l - 3k]$ which implies $g(x) = cx + d$ on $[0; l - 3k]$. Then if $t < l - 3k$, we have

$$g(2k + t) = g\left(\frac{t}{3}\right) + g\left(\frac{t}{3}\right) + g\left(\frac{t}{3}\right) = ct + 3d.$$

So $f(x) = c \ln x + d$ for $x \in (a; \frac{b}{a^2})$ and $f(x) = x \ln x + 3d$ for $x \in (a^3; b)$. Note that if $b < a^5$, then the condition says nothing about f on $\left(\frac{b}{a^2}; a^3\right)$ except

that it is continuous: if $xyz \in (a; b)$ then $xyz > a^3$ and $x, y, z < \frac{b}{a^2}$ because $b > xyz > xa^2, ya^2, za^2$. So in this case every continuous function that satisfies $f(x) = c \ln x + d$ for $x \in \left(a; \frac{b}{a^2}\right)$ and $f(x) = c \ln x + 3d$ for $x \in (a^3; b)$ is a solution. If $b \geq a^5$, then $\frac{b}{a^2} > a^3$ and hence $f\left(\frac{b}{a^2}\right) = c \ln\left(\frac{a}{b^2}\right) + d$ from one side, and $c \ln\left(\frac{a}{b^2}\right) + 3d$ from the other side. So $d = 0$ and since the intervals $\left(a; \frac{b}{a^2}\right]$ and $[a^3; b)$ cover (a, b) we see that $f(x) = c \ln x$ is the only solution.

Problem 4.12. Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ that obey

$$f(f(x) + y) = f(x^2 - y) + 4f(x)y$$

for all $x, y \in \mathbb{R}$.

Solution. We can guess the solution $f(x) = x^2$ together with the trivial one $f \equiv 0$. By setting $y = 0$ we get $f(f(x)) = f(x^2)$. If f would be injective, we would immediately conclude that $f(x) = x^2$. Assume that $f \not\equiv 0$ and $f(x_1) = f(x_2)$. Then substituting $x = x_1, x = x_2$ into the condition we get $f(x_1^2 - y) = f(x_2^2 - y)$. So $f(t) = f(t + b)$, where $b = x_1^2 - x_2^2$. If $b \neq 0$, then we can find t_0 such that $(t_0 + b)^2 - t_0^2 = a$, where $0 \leq a \leq \frac{b^2}{2}$ is a fixed number.

Thus setting $x_1 = t_0, x_2 = t_0 + b$ we get $f(t + a) = f(t)$. Hence if $|x - y| \leq \frac{b^2}{2}$ we can assert that $f(x) = f(y)$. Hence f is a constant function and substituting into the condition we get $f \equiv 0$, a contradiction. Thus $f(x) = f(y)$ can hold only if $y = \pm x$. Returning to $f(f(x)) = f(x^2)$ we deduce $f(x) = \pm x^2$ for any x . Now assume for some $x_0 \neq 0$ we have $f(x_0) = -x_0^2$. Then setting $x = x_0$ we get $f(-x_0^2 + y) = f(x_0^2 - y) - 4x_0^2y$. Therefore

$$f(-x_0^2 + y) - f(x_0^2 - y) = -4x_0^2y.$$

As $f(-x_0^2 + y) = \pm(x_0^2 - y)^2, f(x_0^2 - y) = \pm(x_0^2 - y)^2$, we have

$$f(-x_0^2 + y) - f(x_0^2 - y) = 0, \quad 2(x_0^2 - y)^2$$

or $-2(x_0 - y)^2$. So for any y it follows that $-4x_0^2y \in \{0, 2(x_0^2 - y)^2, -2(x_0 - y)^2\}$. However this can hold for at most 5 values of y , a contradiction. So $f(x) = x^2$.

Problem 4.13. (China 1991) Find all functions $f: [0, 1] \times [0, 1] \rightarrow \mathbb{R}$ such that:

- i) $f(x, 1) = x$;
- ii) $f(1, y) = y$;
- iii) $f(zx, zy) = z^k f(x, y)$;
- iv) $f(f(x, y), z) = f(x, f(y, z))$

for all $x, y, z \in [0, 1]$ and some positive constant k .

Solution. It follows from iv) that $f(0, 0) = 0$. Let now $x \leq y, y > 0$. Then

$$f(x, y) = f\left(y \frac{x}{y}, y\right) = y^k f\left(\frac{x}{y}, 1\right) = y^k \frac{x}{y} = xy^{k-1}.$$

If $y \leq x, x > 0$ then we get in the same way that $f(x, y) = yx^{k-1}$. Hence $f(x, y) = \min\{x, y\} \max\{x, y\}^{k-1}$. This function clearly satisfies conditions i)-iii) and we have to check iv). Assume that $x < y = z$. Then

$$f(f(x, y), z) = f(xy^{k-1}, z) = f(xy^{k-1}, y).$$

If $x < (\frac{1}{y})^{2-k}$, then $xy^{k-1} < y$ and hence

$$f(f(x, y), z) = xy^{k-1}y^{k-1} = xy^{2k-2}.$$

On the other hand, $f(x, f(y, z)) = f(x, y^k)$ and if $x < y^k$, then

$$f(x, f(y, z)) = xy^{k(k-1)}.$$

So $xy^{2k-2} = xy^{k(k-1)}$ for all $x, y \in [0, 1]^2$ that satisfy $x < y^k, x < y^{2-k}$. This is possible only when $k(k-1) = 2k-2$ so $k = 1$ or $k = 2$. If $k = 1$, then $f(x, y) = \min\{x, y\}$ and if $k = 2$, then $f(x, y) = xy$. It is obvious that both functions satisfy iv).

Problem 4.14. Find all continuous functions $f: \mathbb{R} \rightarrow \mathbb{R}$ that satisfy

$$3f(2x+1) = f(x) + 5x$$

for all $x \in \mathbb{R}$.

Solution. If we seek $f(x) = ax + b$, then we must have

$$3(2ax + a + b) = (a + 5)x + b$$

hence $3b + 3a = b, 6a = a + 5$, i.e. $a = 1, b = -\frac{3}{2}$. Then if we set

$$g(x) = f(x) + \frac{3}{2} - x,$$

we get $3g(x) = g(2x+1)$. If we let $h(x) = g(x-1)$, we get $3h(x+1) = h(2x+2)$ or $h(x) = \frac{1}{3}h(2x)$. Thus $h(x) = \frac{1}{3^n}h\left(\frac{x}{2^n}\right)$. Since $\frac{x}{2^n} \rightarrow 0$ as $n \rightarrow \infty$, then $f\left(\frac{x}{2^n}\right) \rightarrow f(0)$ and so $\frac{1}{3^n}f\left(\frac{x}{2^n}\right) \rightarrow 0$. Hence $h = 0$ and thus $f(x) = x - \frac{3}{2}$.

Problem 4.15. (IMO Shortlist 2002) Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(f(x) + y) = 2x + f(f(y) - x)$$

for all $x, y \in \mathbb{R}$.

Solution. Setting $y = -f(x)$ in the given equation we get

$$f(f(-f(x)) - x) = -2x + f(0) \text{ for all } x \in \mathbb{R}.$$

Hence the function $f(x)$ is surjective since the function $-2x + f(0)$ takes all possible real values when x ranges over $\mathbb{R}$. Hence there is $a \in \mathbb{R}$ such that $f(a) = 0$. Setting $x = a$ in the given equation we get

$$f(y) = 2a + f(f(y) - a)$$

for all $y \in \mathbb{R}$. Set $z = f(y) - a$. Then $f(z) = z - a$ for all $z \in \mathbb{R}$ since the function $f(y) - a$ takes all possible real values. Conversely, it is easy to check that for all $a \in \mathbb{R}$ the function $f(x) = x - a$ satisfies the given condition.

Problem 4.16. (Bulgaria 1996) Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(f(x) + xf(y)) = xf(y + 1)$$

for all $x, y \in \mathbb{R}$.

Solution. Set $f(0) = a$. Then it follows from the given equation for $x = 0$ that $f(a) = 0$ and setting $y = a$ gives

$$f(f(x)) = xf(a+1). \quad (1)$$

We shall consider two cases.

Case 1. Let $f(a+1) = 0$. Then $f(f(x)) = 0$ for all $x \in \mathbb{R}$. Suppose that $f(y+1) \neq 0$ for some $y \in \mathbb{R}$. Then the given equation shows that the function $f(x)$ is surjective. Hence the function $f(f(x))$ is also surjective which contradicts $f(f(x)) \equiv 0$. Thus $f(x) \equiv 0$.

Case 2. Let $f(a+1) \neq 0$. Then it follows from (1) that the function $f(x)$ is injective (if $f(x_1) = f(x_2)$ then $x_1f(a+1) = f(f(x_1)) = f(f(x_2)) = x_2f(a+1)$, i.e. $x_1 = x_2$). Setting $x = 1$ in the initial condition gives

$$f(f(1) + f(y)) = f(y+1),$$

i.e. $f(1) + f(y) = y+1$. In particular, $f(1) + f(1) = 1+1$, i.e. $f(1) = 1$. Hence $f(y) = y$ for all $y \in \mathbb{R}$.

Thus there are two functions satisfying the given functional equation: $f(x) = 0$ and $f(x) = x$.

Problem 4.17. (BMO 1997, BMO 2000) Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(xf(x) + f(y)) = f(x)^2 + y$$

for all $x, y \in \mathbb{R}$.

Solution. Setting $x = 0$ gives $f(f(y)) = f(0)^2 + y$. In particular, $f(f(b)) = 0$ for $b = -f(0)^2$. Then

$$f(f(y)) = f(f(b)f(f(b)) + f(y)) = f(f(b))^2 + y = y$$

which shows that $f(0) = 0$ and $f(f(y)) = y$ for all $y \in \mathbb{R}$. Now setting $y = 0$ in the given equation we get $f(xf(x)) = f(x)^2$. Hence

$$f(x)^2 = f(xf(x)) = f(f(x)f(f(x))) = f(f(x))^2 = x^2,$$

i.e. $f(x) = \pm x$ for all $x \in \mathbb{R}$. Suppose that $f(x) = x$ and $f(y) = -y$ for some x and y . Then we get from the given equation that $\pm(x^2 - y) = x^2 + y$ which implies $x = 0$ or $y = 0$.

Thus the only solutions of the problem are the functions $f(x) = x$ and $f(x) = -x$.

Problem 4.18. (Ukraine 2003) Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(xf(x) + f(y)) = x^2 + y$$

for all $x, y \in \mathbb{R}$.

Hint. See the solution of the previous problem!

Problem 4.19. Find all functions $f : (0, \infty) \rightarrow \mathbb{R}$ such that

$$f(x + y) = f(x^2 + y^2)$$

for all $x, y \in (0, \infty)$.

Solution. Let u, v be such that $v\sqrt{2} > u \geq v > 0$ and let $w \in \left[\frac{u^2}{2}, v^2\right]$.

Then $u^2 > w \geq \frac{u^2}{2}$ and so the system $x + y = u, xy = \frac{u^2 - w}{2}$ has at least two positive roots x_1, y_1 and plugging them in the given identity we get $f(u) = f(w)$.

If $v^2 > w \geq \frac{v^2}{2}$ then the system $x + y = v, xy = \frac{v^2 - w}{2}$ has at least two positive roots x_2, y_2 and plugging them in the given identity we get

$$f(v) = f(w).$$

So $f(u) = f(v)$ for all u, v such that $v\sqrt{2} > u \geq v > 0$ and it is easy to conclude from this that $f(x)$ is constant.

Hence the only solutions are the constant functions.

Problem 4.20. (IMO Shortlist 2004) Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfying the equation

$$f(x^2 + y^2 + 2f(xy)) = f(x + y)^2$$

for all $x, y \in \mathbb{R}$.

Solution. Let us make the substitution $z = x + y, t = xy$. Given $z, t \in \mathbb{R}$, x, y are real if and only if $4t \leq z^2$. Define $g(x) = 2(f(x) - x)$. Then the given functional equation transforms into

$$f(z^2 + g(t)) = f(z)^2 \tag{1}$$

for all $t, z \in \mathbb{R}$ with $z^2 \geq 4t$. Let us set $c = g(0) = 2f(0)$. Substituting $t = 0$ in (1) gives

$$f(z^2 + c) = f(z)^2 \quad (2)$$

for all $z \in \mathbb{R}$. If $c < 0$, then taking z such that $z^2 + c = 0$, we obtain from (2) that $f(z)^2 = \frac{c}{2}$, a contradiction; hence $c \geq 0$. We also observe from (2) that

$$x > c \Rightarrow f(x) \geq 0. \quad (3)$$

If g is a constant function it follows easily that $c = 0$ and therefore $f(x) = x$, which is a solution. Suppose g is nonconstant, and let $a, b \in \mathbb{R}$ be such that $g(a) - g(b) = d > 0$. For some sufficiently large K and each $u, v \geq K$ with $v^2 - u^2 = d$ the equality $u^2 + g(a) = v^2 + g(b)$ implies by (1) and (3) that $f(u) = f(v)$. This further leads to

$$g(u) - g(v) = 2(u - v) = \frac{d}{u + (u^2 + d)^{1/2}}.$$

Therefore every value from some suitable chosen segment $[\delta, 2\delta]$ can be expressed as $g(u) - g(v)$, with u and v bounded above by some M . Consider arbitrary x, y with $y > x \geq 2M^{1/2}$ and $\delta < y^2 - x^2 < 2\delta$. By the above considerations, there exist $u, v \leq M$ such that $g(u) - g(v) = y^2 - x^2$, i.e. $x^2 + g(u) = y^2 + g(v)$. Since $x^2 \geq 4u$ and $y^2 \geq 4v$, (1) leads to $(f(x))^2 = (f(y))^2$. Moreover, if we assume that $4M \geq c^2$, we conclude from (3) that $f(x) = f(y)$. Since this holds for all $x, y \geq 2M^{1/2}$ with $y^2 - x^2 \in [\delta, 2\delta]$, it follows that $f(x)$ is eventually constant, say $f(x) = k$ for $x \geq N = 2M^{1/2}$. Setting $x > N$ in (2) we obtain $k^2 = k$, so $k = 0$ or $k = 1$. By (2) we have $f(-z) = |f(z)|$, and thus $|f(z)| \leq 1$ for all $z \leq -N$. Hence $g(u) = 2f(u) - 2u \geq -2 - 2u$ for $u \leq -N$, which implies that g is unbounded. Hence for each z there exists t such that $z^2 + g(t) > N$, and consequently $f(z)^2 = f(z^2 + g(t)) = k = k^2$. Therefore $f(z) = \pm k$ for each z . If $k = 0$, then $f(x) \equiv 0$, which is clearly a solution. Assume $k = 1$. Then $c = 2f(0) = 2$ (because $c \geq 0$), which together with (3) implies $f(x) = 1$ for all $x \geq 2$. Suppose that $f(t) = -1$ for some $t < 2$. Then $t - g(t) = 3t + 2 > 4t$. If also $t - g(t) \geq 0$, then for some $z \in \mathbb{R}$ we have $z^2 = t - g(t)$, which by (1) leads to $(f(z))^2 = f(z^2 + g(t)) = f(t) = -1$, which is impossible. Hence $t - g(t) < 0$,

giving us $t < -2/3$. On the other hand, if X is any subset of $(-\infty, -2/3)$, the function f defined by $f(x) = -1$ for $x \in X$ and $f(x) = 1$ otherwise satisfies the equation. To sum up, the solutions are $f(x) = x$, $f(x) = 0$ and all functions of the form described above.

19.5 Symmetrization and Additional Variables

Problem 5.1. Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ that satisfy

$$f(xy) = f(x)y^2 + f(y) - 1$$

for all $x, y \in \mathbb{R}$.

Solution. The left-hand side is symmetric, so we swap x and y to get

$$f(x)y^2 + f(y) - 1 = f(y)x^2 + f(x) - 1$$

or $f(x)(y^2 - 1) = f(y)(x^2 - 1)$ which yields $\frac{f(x)}{x^2 - 1} = \frac{f(y)}{y^2 - 1}$ for $x, y \neq \pm 1$.

This implies $f(x) = c(x^2 - 1)$ for all $x \neq \pm 1$. This holds for $x = \pm 1$ as well because $f(\pm 1)(y^2 - 1) = f(y)((\pm 1)^2 - 1) = 0$ so $f(\pm 1) = 0$. We substitute $f(x) = c(x^2 - 1)$ back into the equation to see $cx^2y^2 - c = c^2y^2 - cy^2 + cy^2 - 1$ which means $c = 1$ so $f(x) = x^2 - 1$ is the solution.

Problem 5.2. (MR 2012) Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfying

$$f(y + f(x)) = f(x)f(y) + f(f(x)) + f(y) - xy$$

for all $x, y \in \mathbb{R}$.

Solution. For $y = 0$ we have $f(x)f(0) + f(0) = 0$, so either $f(0) = 0$ or $f(x) = -1$ for all x . But the latter is not possible thus $f(0) = 0$. Now replace y by $f(y)$ in the given equation to make its left-hand side symmetric in x and y . Then the right-hand side should be also symmetric and we get

$$\frac{f(f(x)) + x}{x} = \frac{f(f(y)) + y}{y} = k.$$

Thus $f(f(x)) = kf(x) - x$ and this equation is also satisfied for $x = 0$. Now the given equation becomes

$$f(y + f(x)) = f(x)f(y) + kf(x) - x + f(y) - xy.$$

Let $b = f(-1)$. Substituting $y = -1$ in the equation above we have

$$f(f(x) - 1) = (b + k)f(x) + b$$

and then replacing y by $f(y) - 1$ in it we get

$$\begin{aligned} f(f(x) + f(y) - 1) &= ((k+b)f(y) + b)f(x) + kf(x) - x + (k+b)f(y) + b - xf(y) + x \\ &= (k+b)[f(x)f(y) + f(x) + f(y)] + b - xf(y). \end{aligned}$$

This equation clearly implies $xf(y) = yf(x)$, so

$$\frac{f(x)}{x} = \frac{f(y)}{y}$$

for all $x, y \neq 0$. Hence $f(x) = cx$ which is also true for $x = 0$. Now substituting $f(x) = cx$ to the given equation we easily get $c = \pm 1$. Thus $f(x) = -x$ and $f(x) = x$ are the only solutions.

Problem 5.3. Find all functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$xf(x) - yf(y) = (x - y)f(x + y)$$

for all $x, y \in \mathbb{R}$.

Solution. We add a new variable to get the relation

$$xf(x) - zf(z) = (x - z)f(x + z).$$

We also have

$$xf(x) - zf(z) = xf(x) - yf(y) + yf(y) - zf(z) = (x - y)f(x + y) + (y - z)f(y + z).$$

Therefore

$$(x - z)f(x + z) = (x - y)f(x + y) + (y - z)f(y + z).$$

If we want $x + z = u, x + y = 1, y + z = 0$ by solving the system of equations we get $x = \frac{u+1}{2}, y = \frac{1-u}{2}, z = \frac{u-1}{2}$ and thus our condition becomes $f(u) = uf(1) + (1-u)f(0)$. Hence $f(x) = ax + b$ is a linear function and the linear functions satisfy the condition.

Problem 5.4. Find all pairs of functions $f, g : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x) - f(y) = (x - y)(g(x) + g(y))$$

for all $x, y \in \mathbb{R}$.

Solution. The identity

$$(f(x) - f(y)) + (f(y) - f(z)) + (f(z) - f(x)) = 0$$

gives us an invaluable insight. We get

$$(x - y)(g(x) + g(y)) + (y - z)(g(y) + g(z)) + (z - x)(g(z) + g(x)) = 0.$$

Setting $y = 1, z = 0$ we get $g(x) = -(x - 1)g(0) + xg(1)$, i.e. $g(x) = ax + b$. We then get $f(x) - f(y) = (x - y)(a(x + y) + 2b)$ and setting $y = 0$ we obtain $f(x) = ax^2 + 2bx + c$. It is straightforward to check that for every $a, b, c \in \mathbb{R}$ these functions satisfy the problem conditions.

Problem 5.5. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x + y) = f(x)f(y)f(xy)$$

for all $x, y \in \mathbb{R}$.

Solution. If $f(u) = 0$ for some u , then $f(x + u) = 0$ and we deduce $f(x) = 0$ for all x . If $f \not\equiv 0$, then f is non-zero on $\mathbb{R}$. We add a new variable z by expanding:

$$f(x+y+z) = f(x)f(y+z)f(xy+yz) = f(x)f(y)f(z)f(yz)f(xy)f(xz)f(x^2yz).$$

Now as the LHS side is symmetric by swapping x and y we get the relation

$$f(x+y+z) = f(x)f(y+z)f(xy+yz) = f(x)f(y)f(z)f(yz)f(xy)f(xz)f(xy^2z).$$

Hence $f(x^2yz) = f(xy^2z)$. Then picking up $u, v \neq 0$ and setting $x = u, y = v, z = \frac{1}{uv}$ we get $x^2yz = u, y^2xz = v$ and thus $f(u) = f(v)$. So f is constant on $\mathbb{R} \setminus \{0\}$. If $f = c$, then $c = c^3$ so $c = 0$ or $c = \pm 1$. In any case by setting $y = -x \neq 0$ we get $f(0) = c^3 = c$. Thus $f \equiv 0, f \equiv -1$ or $f \equiv 1$. All of these functions satisfy the given condition.

Problem 5.6. Find all functions $f : \mathbb{Q} \rightarrow \mathbb{Q}$ such that

$$f(xy) = f(x)f(y) - f(x+y) + 1$$

for all $x, y \in \mathbb{Q}$.

Solution. Setting $x = y = 0$ we get $f^2(0) - 2f(0) + 1 = 0$, i.e. $f(0) = 1$. Now setting $x = 1, y = -1$ gives $f(-1) = f(-1)f(1)$ and therefore $f(-1) = 0$, or $f(1) = 1$.

Case 1. Let $f(-1) = 0$. Then replacing y by yz in the given identity we get

$$\begin{aligned} f(xyz) &= f(x)f(yz) - f(x+yz) + 1 \\ &= f(x)(f(y)f(z) - f(y+z) + 1) - f(x+yz) + 1. \end{aligned}$$

On the other hand replacing x by xy and y by z in the initial identity gives

$$\begin{aligned} f(xyz) &= f(xy)f(z) - f(xy+z) + 1 \\ &= (f(x)f(y) - f(x+y) + 1)f(z) - f(xy+z) + 1 \end{aligned}$$

and the two identities above imply that

$$f(x)f(y+z) - f(x) + f(x+yz) = f(z)f(x+y) - f(z) + f(xy+z). \quad (1)$$

Now setting $x = 1, z = -1$ and replacing y by $y+1$ we get

$$f(y)(1 - f(1)) = f(-y) - f(1) \quad (2)$$

for all $y \in \mathbb{Q}$. In particular, for $y = 1$ we obtain that $f(1)(2 - f(1)) = 0$, i.e. either $f(1) = 0$ or $f(1) = 2$. Suppose that $f(1) = 0$. Then (2) shows that $f(y) = f(-y)$ and replacing y by $-y$ in the initial identity we get

$$f(xy) = f(x)f(y) - f(x-y) + 1.$$

Hence $f(x+y) = f(x-y)$ for all $x, y \in \mathbb{Q}$ which shows that f is a constant. But this is a contradiction because $f(0) = 1$ and $f(1) = 0$. Thus $f(1) = 2$ and we get from (2) that $f(y) + f(-y) = 2$. Set $g(x) = f(x) - 1$. Then the initial identity can be written as

$$g(xy) = g(x) + g(y) + g(x)g(y) - g(x+y).$$

On the other hand $g(y) = -g(-y)$ and replacing y by $-y$ in the above identity we get

$$-g(xy) = g(x) - g(y) - g(x)g(y) - g(x-y).$$

Therefore

$$2g(x) = g(x+y) + g(x-y)$$

for all $x, y \in \mathbb{Q}$. Now it follows by induction on n that $g(nx) = ng(x)$ for all $n \in \mathbb{N}$ and $x \in \mathbb{Q}$. Since every $r \in \mathbb{Q}$ can be written as $r = \frac{p}{q}$, where $p \in \mathbb{Z}$ and $q \in \mathbb{N}$ we get that $qg(r) = g(qr) = g(p) = pg(1)$. Hence $g(x) = xg(1)$ for all $x \in \mathbb{Q}$ and therefore $f(x) = x + 1$.

Case 2. Let now $f(1) = 1$. Then setting $y = -1$ and $z = 1$ in (1) and using the fact that $f(0) = 1$ we get $f(1-x) = 1$ and therefore $f(x) \equiv 1$.

Thus all solutions of the problem are the functions $f(x) = x + 1$ and $f(x) = 1$.

Problem 5.7. Find all continuous functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$\text{a) } f(x+y) + f(xy-1) = f(x) + f(y) + f(xy);$$

$$\text{b) } f(x+y) + f(xy) = f(x) + f(y) + f(xy+1)$$

for all $x, y \in \mathbb{R}$.

Solution. a) If $g(x) = f(x) - f(x-1)$ then we have

$$f(x+y) - f(x) - f(y) = g(xy).$$

From here

$$f(x+y+z) = f(x) + f(y+z) + g(xy+xz) = f(x) + f(y) + f(z) + g(yz) + g(xy+xz).$$

We have encountered this sort of equation before, establishing that g is linear. Hence $f(x+1) - f(x) = cx + d$. Setting $y = 1$ we get

$$f(x+1) - f(x) - f(1) = g(x) = cx + d$$

so $f(0) = 0$. Then setting $y = 0$ we get $g(0) = 0$ so $g(x) = cx$. We then have $f(x+y) - f(x) - f(y) = cxy$. This identity implies that $f(x) - \frac{c}{2}x^2$ is a continuous additive function. Hence $f(x) = \frac{c}{2}x^2 + dx$ and we have

$$f(x) - f(x-1) = \frac{c}{2}(2x-1) + d.$$

Thus $\frac{c}{2} = d$ and $f(x) = dx^2 + dx$.

b) Set $g(x) = f(x+1) - f(x)$. Then $f(x+y) - f(x) - f(y) = g(xy)$. Then

$$f(x+y+z) - f(x+y) - f(z) = g(xz+yz)$$

so $f(x+y+z) - f(x) - f(y) - f(z) = g(xz+yz) + g(xy)$. Due to the symmetry among x, y, z we conclude that

$$\begin{aligned} f(x+y+z) - f(x) - f(y) - f(z) &= g(xz+yz) + g(xy) \\ &= g(xz+xy) + g(yz) = g(xy+yz) + g(xz). \end{aligned}$$

Now if we set $a = xy, b = yz, c = xz$, we get

$$g(a+b) + g(c) = g(a+c) + g(b) = g(b+c) + g(a).$$

The condition that $a = xy, b = xz, c = yz$ can be satisfied if $abc > 0$ by setting

$$x = \frac{\sqrt{abc}}{c}, \quad y = \frac{\sqrt{abc}}{b}, \quad z = \frac{\sqrt{abc}}{a}.$$

Thus we get $g(a+b) + g(c) = g(a+c) + g(b) = g(b+c) + g(a)$ for $abc > 0$. Now we claim that $g(x+y) + g(0) = g(x) + g(y)$ for $xy \neq 0$. Indeed, either for $z < 0$ or for $z > 0$ we have $xyz > 0$ thus $g(x+y) + g(z) = g(x+z) + g(y)$. Now taking $z \rightarrow 0$ we obtain $g(x+y) + g(0) = g(x) + g(y)$. This also holds by continuity even when $xy = 0$. Hence the function $g(x) - g(0)$ is additive thus $g(x) = ax + b$ is a linear function. Hence $f(x+y) - f(x) - f(y) = axy + b$. Now if we set $h = f(x) - \frac{a}{2}x^2 + b$, then we see that h is additive, so $h(x) = cx$. We conclude that f is a polynomial of degree at most 2 and let $f(x) = ax^2 + bx + c$. Then

plugging in the given equation we get $a+b+c=0$. Hence $f(x) = ax^2+bx-a-b$ and all these functions clearly satisfy the condition.

Problem 5.8. (Hosszu's functional equation) Show that a function $f : \mathbb{R} \rightarrow \mathbb{R}$ which satisfies the identity

$$f(x+y-xy) + f(xy) = f(x) + f(y)$$

is an additive function plus some constant.

Solution. This problem is of the form $f(a) + f(b) = f(c) + f(d)$, where $a+b=c+d$. The inconvenience is that (a, b) and (c, d) are linked to each other, so we cannot state that $f(a) + f(b) = f(c) + f(d)$ whenever $a+b=c+d$. We try to eliminate the inconvenience by adding a new variable and symmetrizing:

$$\begin{aligned} f(x) + f(y) + f(z) &= f(x+y-xy) + f(xy) + f(z) \\ &= f(x+y-xy) + f(xy+z-xyz) + f(xyz). \end{aligned}$$

By symmetry it also equals

$$f(x+z-xz) + f(xz+y-xyz) + f(xyz)$$

and

$$f(y+z-yz) + f(yz+x-xyz) + f(xyz).$$

We thus deduce that

$$\begin{aligned} f(x+y-xy) + f(xy+z-xyz) &= f(x+z-xz) + f(xz+y-xyz) \\ &= f(y+z-yz) + f(yz+x-xyz). \end{aligned}$$

This is again an equation of the form $f(a) + f(b) = f(c) + f(d)$, where $a+b=c+d$, but this time the restraints are milder. Indeed, we can find x, y, z with $(1-x)(1-y)=a$, $(1-z)(1-xy)=b$, $(1-x)(1-z)=c$. For comfort we set $u=1-x$, $v=1-y$, $w=1-z$ to get $uv=a$, $w(u+v-uv)=b$, $uw=c$. Hence $w = \frac{c}{u}$, $v = \frac{a}{u}$ and $\frac{c}{u} \left(u + \frac{a}{u} - a\right) = b$ or $(c-b)u^2 - acu + ac = 0$. For this equation to have a non-zero solution we need to have a positive discriminant so $a^2c^2 - 4ac(c-b) \geq 0$ or $(ac-2c)^2 + 4(abc-c^2) \geq 0$. When $abc > c^2$ this

is certainly true. Now consider (a, b) and (c, d) with $a + b = c + d$. If ab, cd have the same sign then we can find such an e sufficiently small in absolute value such that $abe > e^2, cde > e^2$. Setting $e_1 = a + b - e = c + d - e$ we get $f(a) + f(b) = f(e) + f(e_1) = f(c) + f(d)$ and so $f(a) + f(b) = f(c) + f(d)$. So $f(a) + f(b) = f(c) + f(d)$ when $a + b = c + d$ and $abcd > 0$. Next if we have $a + b = c + d = s \notin [0; 4]$ and $abcd < 0$ then there are u, v such that $u + v = s$, $u + v - uv < 0$ (just take $u = v = \frac{s}{2}$). Then $u + v = uv + (u + v - uv)$ by $uvuv(u + v - uv) = u^2v^2(u + v - uv) < 0$. We may assume that $ab > 0, cd < 0$. Then if $s > 0$, we have

$$f(a) + f(b) = f(u) + f(v) = f(uv) + f(u + v - uv) = f(c) + f(d)$$

and if $s < 0$ we have

$$f(c) + f(d) = f(u) + f(v) = f(uv) + f(u + v - uv) = f(a) + f(b).$$

Finally, even if $s \in [0; 4]$ (we actually cannot have $s = 0$ and $abcd < 0$ as $abcd = a^2c^2$), we can find a sufficiently large e such that

$$a + c + e, b + d + e, a + b + 2e, c + d + 2e > 4$$

and state $f(a + e) + f(b + e) = f(c + e) + f(d + e)$ while

$$f(a) + f(c + e) = f(c) + f(a + e)$$

and

$$f(b) + f(d + e) = f(d) + f(b + e).$$

Hence $f(c + e) - f(c) = f(a + e) - f(a)$, $f(b + e) - f(b) = f(d + e) - f(d)$ and $f(a + e) + f(b + e) = f(c + e) + f(d + e)$ implies $f(a) + f(b) = f(c) + f(d)$ in this last case.

So $f(a) + f(b) = f(c) + f(d)$ whenever $a + b = c + d$ and $abcd \neq 0$. Now what if $abcd = 0$? If one of the numbers is zero, say d , we have to prove that $f(a) + f(b) = f(a + b) + f(0)$ when $ab \neq 0$. If $a + b \notin [0; 4]$, we deduce the existence of $x, y \neq 0$ such that $xy = x + y = a + b$. Thus $f(a) + f(b) = f(x) + f(y) = f(x + y - xy) + f(xy) = f(a + b) + f(0)$. If

$a+b \in [0; 4]$, we find again a sufficiently large e such that $a+b+e, 2a+b+e > 4$ and we conclude $f(a+b+e)+f(0) = f(a+e)+f(b)$ while $f(a+e)+f(a+b) = f(a)+f(a+b+e)$. Hence adding these expressions and canceling common terms we get $f(a+b)+f(0) = f(a)+f(b)$ as desired. If two of the numbers are zero, then either we get $(a, b) = (c, d)$ or one of (a, b) and (c, d) is $(0, 0)$. The first case is clear. In the second case we need to prove $f(a)+f(-a) = 2f(0)$ for $a \neq 0$. In this case $f(2a)+f(0) = 2f(a)$ while $f(2a)+f(-a) = f(a)+f(0)$. Subtracting this two relations we get $f(0)-f(-a) = f(a)-f(0)$ so $f(a)+f(-a) = 2f(0)$. If three of a, b, c, d are zero, the fourth is also zero and the conclusion is clear.

We have established that $f(a)+f(b) = f(c)+f(d)$ whenever $a+b = c+d$. Then $f(a+b)+f(0) = f(a)+f(b)$ thus $f(x)-f(0)$ is an additive function and the conclusion follows.

19.6 Iterations and Recursive Relations

Problem 6.1. Find the n^{-th} iterate of the function:

(a) $f(x) = 4x(1-x), x \in [0, 1]$;

(b) $f(x) = ax^2 + bx + c, a \neq 0, 4ac = b^2 - 2b$.

Solution. (a) Set $\varphi(x) = \arcsin \sqrt{x} \in [0, \frac{\pi}{2}]$ and $g(x) = 2x$. Then

$$\varphi^{-1}(x) = \sin^2 x$$

and we get

$$\begin{aligned} \varphi^{-1}(g(\varphi(x))) &= \sin^2(2 \arcsin \sqrt{x}) = (2 \sin(\arcsin \sqrt{x}) \cos(\arcsin \sqrt{x}))^2 \\ &= (2\sqrt{x} \cdot \sqrt{1-x})^2 = f(x) \end{aligned}$$

Hence

$$f^{(n)}(x) = \varphi^{-1}(g^{(n)}(\varphi(x))) = \sin^2(2^n \arcsin \sqrt{x}).$$

(b) Set $\varphi(x) = x + \frac{b}{2a}$. Then $\varphi^{-1}(x) = x - \frac{b}{2a}$ and we get that

$$\begin{aligned} g(x) &= \varphi(f(\varphi^{-1}(x))) \\ &= a(x - \frac{b}{2a})^2 + b(x - \frac{b}{2a}) + \frac{b^2 - 2b}{4a} + \frac{b}{2a} \\ &= ax^2. \end{aligned}$$

Thus

$$f^{(n)}(x) = \varphi^{-1}(g(\varphi(x))) = a^{2^n-1} \left(x + \frac{b}{2a}\right)^{2^n} - \frac{b}{2a}.$$

Problem 6.2. Find all functions $f: \mathbb{Q}_+ \rightarrow \mathbb{R}$ that satisfy

$$f(x+y) + f(x-y) = (2^y + 2^{-y})f(x)$$

for all $x, y \in \mathbb{Q}_+$.

Solution. Let $t \in \mathbb{Q}_+$ and define $f(nt) = a_n$. If we set $x = nt, y = t$ we get the second-degree recursive relation $a_{n+1} + a_{n-1} = (2^t + 2^{-t})a_n$. The characteristic polynomial $x^2 - (2^t + 2^{-t})x + 1$ has 2^t and 2^{-t} as zeros, and therefore $a_n = c_1(2^t)^n + c_2(2^{-t})^n$, i.e. $f(nt) = c_1 2^{nt} + c_2 2^{-nt}$. We then claim that $f(x) = c_1 2^x + c_2 2^{-x}$ for all $x \in \mathbb{Q}_+$. Indeed, assume that

$$f(x_0) \neq c_1 2^{x_0} + c_2 2^{-x_0}.$$

Then there exist c'_1, c'_2 such that $f(mx_0) = c'_1 2^{mx_0} + c'_2 2^{-mx_0}$ for every $m \in \mathbb{N}$. If now $\frac{t}{x_0} = \frac{m}{n}$ then $a = mx_0 = nt$ and thus $f(a) = c_1 2^a + c_2 2^{-a} = c'_1 2^a + c'_2 2^{-a}$ and similarly $f(2a) = c_1 2^{2a} + c_2 2^{-2a} = c'_1 2^{2a} + c'_2 2^{-2a}$. There is only one solution to the system $x2^a + y2^{-a} = f(a), x2^{2a} + y2^{-2a} = f(2a)$ and therefore $c_1 = c'_1, c_2 = c'_2$ as desired. It is easy to check that the function

$$f(x) = c_1 2^x + c_2 2^{-x}$$

satisfies the requirements of the problem.

Problem 6.3. (Bulgaria 1996) Find all functions $f: \mathbb{Z} \rightarrow \mathbb{Z}$ such that

$$3f(n) - 2f(f(n)) = n$$

for all $n \in \mathbb{Z}$.

Solution. For a given n set

$$a_0 = n, a_{k+1} = f(a_k), \quad k \geq 0.$$

Then the given identity implies

$$3a_{k+1} - 2a_{k+2} = a_k, \quad k \geq 0.$$

The characteristic equation of this recurrence is $3x - 2x^2 = 1$ with roots 1 and $\frac{1}{2}$. Hence

$$a_k = c_0 \cdot 1^k + c_1 \cdot \left(\frac{1}{2}\right)^k,$$

where $c_0 = 2a_1 - a_0$ and $c_1 = 2(a_0 - a_1)$. Since a_k and c_0 are integers, it follows that 2^k divides c_1 for any $k \geq 0$. Hence $c_1 = 0$ and $a_k = c_0 = 2a_1 - a_0$ for any $k \geq 0$. In particular, $a_1 = a_0$, i.e. $f(n) = n$.

Problem 6.4. (MOSP 2001) Find all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ such that

$$f(f(f(n))) + 6f(n) = 3f(f(n)) + 4n + 2001$$

for all $n \in \mathbb{N}$.

Solution. Let $n \in \mathbb{N}$. Set $x_0 = n$ and $x_{k+1} = f(x_k)$, $k \geq 0$, then the given equation becomes the linear recursive equation

$$x_{k+3} - 3x_{k+2} + 6x_{k+1} - 4x_k = 2001.$$

Under the change $y_k = x_k - 667k$, it reduces to the homogeneous equation

$$y_{k+3} - 3y_{k+2} + 6y_{k+1} - 4y_k = 0.$$

The associated characteristic equation being

$$Y^3 - 3Y^2 + 6Y - 4 = (Y - 1)(Y^2 - 2Y + 4) = 0$$

with roots $1, \exp(\pi/3), \exp(-\pi/3)$.

The solution in $\mathbb{R}$ of the homogeneous equation is given by

$$y_k = a(n) + 2^k \left(b(n) \cos \frac{k\pi}{3} + c(n) \sin \frac{k\pi}{3} \right),$$

where a, b, c do not depend on k . But

$$b(n) \cos \frac{k\pi}{3} + c(n) \sin \frac{k\pi}{3}$$

cannot be a rational number for all k (why?), so the solution in $\mathbb{N}$ of the homogeneous equation corresponds to $b = c = 0$ and $y_k = a(n)$ with $a : \mathbb{N} \rightarrow \mathbb{N}$ independent of k . It follows that $x_k = a(n) + 667k$. Since $n = x_0 = a(n)$ we get $f(n) = x_1 = n + 667$.

Problem 6.5. (Putnam 1988) Prove that there exists a unique function $f : (0, \infty) \rightarrow (0, \infty)$ such that

$$f(f(x)) + f(x) = 6x$$

for all $x \in (0, \infty)$.

Solution. For a given $x \in (0, \infty)$ set

$$a_0 = x, a_{k+1} = f(a_k), \quad k \geq 0.$$

Then we obtain the recurrence relation

$$a_{k+2} + a_{k+1} = 6a_k, \quad k \geq 0.$$

Its characteristic equation $x^2 + x = 6$ has roots 2 and -3 . Hence

$$a_k = c_0 2^k + c_1 (-3)^k,$$

where $c_0 = \frac{3a_0 + a_1}{5}$ and $c_1 = \frac{2a_0 - a_1}{5}$. Note that $\lim_{k \rightarrow \infty} \frac{3^k}{2^k} = \infty$. Hence $\lim_{k \rightarrow \infty} a_{2k} = -\infty$ if $c_1 < 0$ and $\lim_{k \rightarrow \infty} a_{2k+1} = -\infty$ if $c_1 > 0$. Hence $c_1 = 0$, i.e. $a_{k+1} = 2a_k$ and therefore $f(x) = 2x$ for all $x \in (0, \infty)$.

Problem 6.6. Find all functions $f : (0, \infty) \rightarrow (0, \infty)$ such that

$$f(f(f(x))) + f(f(x)) = 2x + 5$$

for all $x \in (0, \infty)$.

Solution. Using the same notation as in the solution of the previous problem we have

$$a_{k+3} + a_{k+2} = 2a_k + 5, \quad k \geq 0. \quad (1)$$

Subtracting this equality from

$$a_{k+4} + a_{k+3} = 2a_{k+1} + 5$$

we get

$$a_{k+4} = a_{k+2} + 2a_{k+1} + 2a_k, \quad k \geq 0.$$

The characteristic equation

$x^4 = x^2 + 2x + 2$ can be written as $(x-1)^2(x^2 + 2x + 2) = 0$, i.e. it has a double root 1 and two complex roots $\sqrt{2} \left(\cos \frac{\pi}{4} \pm i \sin \frac{\pi}{4} \right)$. Hence

$$a_k = c_0 + c_1 k + 2^{\frac{k}{2}} \left(c_2 \cos \frac{k\pi}{4} + c_3 \sin \frac{k\pi}{4} \right),$$

where the constants $c_n, 0 \leq n \leq 3$ are real and depend only on the first four terms of the sequence. Considering the subsequences with indexes congruent respectively to 0, 2, 4, 6 modulo 8 we conclude, as in the solution of the previous problem, that respectively $c_2 \geq 0, c_3 \geq 0, c_2 \leq 0, c_3 \leq 0$, i.e. $c_2 = c_3 = 0$. Thus $a_{k+1} = a_k + c_1$ and using (1) we get $c_1 = 1$. Hence $f(x) = x + 1$ which is a solution of the problem.

Problem 6.7. Find all continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfying

$$f(2x + 1) = f(x)$$

for all $x \in \mathbb{R}$.

Solution. For any $x \in \mathbb{R}$ define $x_0 = x, x_{n+1} = \frac{x_n - 1}{2}, n \geq 0$. Then

$$x_{n+1} + 1 = \frac{x_n + 1}{2}$$

showing that $(x_n + 1)_{n \geq 0}$ is a geometric progression with ratio $\frac{1}{2}$. Hence $\lim_{n \rightarrow \infty} (x_n + 1) = 0$, i.e. $\lim_{n \rightarrow \infty} x_n = -1$. On the other hand

$$f(x_n) = f(2x_{n+1} + 1) = f(x_{n+1}), n \geq 0$$

and we get $f(x_n) = f(x_0) = f(x)$ for any $n \geq 0$. Thus

$$f(x) = \lim_{n \rightarrow \infty} f(x_n) = f(\lim_{n \rightarrow \infty} x_n) = f(-1)$$

showing that f is a constant.

Problem 6.8. Find all continuous functions $f: \mathbb{R} \rightarrow \mathbb{R}$ that satisfy

$$f(f(x)) = f(x) + 2x$$

for all $x \in \mathbb{R}$.

First Solution. It is clear that f is injective. Now we prove that it is surjective. Assume for contradiction that $f(x) \neq a$ for all $x \in \mathbb{R}$. Then as f is continuous we either have $f(x) > a$ or $f(x) < a$ for all $x \in \mathbb{R}$. If $f(x) > a$, then we deduce from $f(f(x)) - f(x) = 2x$ that $f(x) > a - 2x$. Since f is injective and continuous it is either increasing or decreasing. If f is increasing, then $f(-n) \geq 2n + a$ for $n \in \mathbb{N}$ and hence $f(0) > f(-n) \geq 2n + a$ for any $n \in \mathbb{N}$ which is impossible. If f is decreasing, then $f(f(x)) \leq f(a)$ hence $f(x) \leq f(a) - 2x < a$ when x is sufficiently large, a contradiction. Assume now that $f(x) < a$. We then deduce $f(x) < a - 2x$, $f(f(x)) < 2x - a$. If f is increasing, then as $f(x) < x$ for $x > a$ we get $2x = f(f(x)) - f(x) < 0$ for $x > a$ which is impossible if $x > 0$. If f is decreasing, then $f(n) < a - 2n$ for $n \in \mathbb{N}$ and hence $f(0) < a - 2n$ for all $n \in \mathbb{N}$, a contradiction. So we have proven that f is injective and surjective and thus it has an inverse g . If we try the usual pattern: setting $a_n = f_n(x)$ and using the recurrence $a_{n+2} = a_{n+1} + 2a_n$, this will lead us nowhere because we cannot link the recurrence with the continuity of f , as the roots of the equation are -1 and 2 ; so the members of the sequence will get larger and sparser with increasing n . However we may use the backwards formula: if $y = g(x)$, then $2y + x = f(x)$ and so $y = \frac{f(x) - x}{2}$. If we set $a_n = g^{(n)}(x)$, then a_n satisfies the recurrence $a_{n+2} = \frac{a_n}{2} - \frac{a_{n+1}}{2}$. The characteristic equation $x^2 + \frac{x}{2} - \frac{1}{2} = 0$ has roots $-1, \frac{1}{2}$ and thus $a_n = u(-1)^n + v\left(\frac{1}{2}\right)^n$. Hence the odd terms of the sequence converge to $-u$ and the even to u . As $f(a_{n+1}) = a_n$ using the continuity we deduce $f(u) = -u$, $f(-u) = u$. Now u can be computed from the initial values: if $a_0 = f(x)$, $a_1 = x$ we have $u + v = f(x)$, $-u + \frac{v}{2} = x$, so $u = \frac{f(x) - 2x}{3}$. Let $h(x) = \frac{f(x) - 2x}{3}$ and $A = \text{Im}(h)$. As h is continuous, A must be connected. Also A is symmetric with respect to the origin, because if $t \in A$,

then $f(t) = -t$, hence $\frac{f(t) - 2t}{3} = -t$ and so $-t \in A$. Note also that $A \neq \emptyset$. Hence either A consists of a single point 0 and $f(x) = 2x$ or A contains an interval $(-a; a)$. We prove that if A contains an interval then $A = \mathbb{R}$. Without loss of generality we may assume that $A = [-a; a]$ or $A = (-a; a)$. Suppose $h(x) = t \neq 0$, $|t| < a$. As $a_{2n} \rightarrow t$, $a_{2n-1} \rightarrow -t$, for some n we have $a_{2n} \in A$ and so $a_{2n-1} \in A$. Thus we have $a_{2n} = -a_{2n-1}$ and we conclude by induction on k that $a_{2n-2k} = a_{2n}$, $a_{2n-2k-1} = a_{2n-1}$. So $f(x) = -x$. We have only one problem: if A is $[-a; a]$ and $t = \pm a$. Then we cannot say $a_{2n} \in A$ for some n as t is on the boundary of A . We can handle it like this. Let B be the set of all x for which $f(x) \neq -x$. Then $B = (-\infty; -a) \cup (a; +\infty)$. Note that B can be split in two sets: $C = \{x | h(x) = a\}$ and $D = \{x | h(x) = -a\}$. Both sets are closed intervals therefore either $C = B$, $D = B$, $C = (-\infty; -a)$, $D = (a; +\infty)$ or $D = (-\infty; -a)$, $C = (a; +\infty)$. However $f(x) = 2x + 3a$ if $x \in C$ and $f(x) = 2x - 3a$ if $x \in D$. Thus if $(a; +\infty) \subset C$, we get $f(x_0) = 2x_0 + 3a$ for $x_0 > a$ and taking the limit in a we get $f(a) = 5a$ which is impossible. Hence $(a; +\infty) \subset D$. Analogously $(-\infty; -a) \subset C$. Thus we conclude $f(x) = -x$ if $|x| \leq a$, $f(x) = 2x - 3a$ if $x > a$ and $f(x) = 2x + 3a$ if $x < -a$. Now if $x > 2a$, then $f(x) = 2x - 3a$, $f(f(x)) = 2f(x) - 3a = 4f(x) - 9a$ but $2x + f(x) = 4x - 3a \neq f(f(x))$ for $a > 0$. So $A = \mathbb{R}$ and the solutions are $f(x) = 2x$ and $f(x) = -x$.

Second Solution. As in the first solution we conclude that the function $f : \mathbb{R} \rightarrow \mathbb{R}$ is one-to-one and $f(0) = 0$. Since f is continuous it is either strictly increasing or strictly decreasing.

Case 1. Let f be strictly increasing. Set $h(x) = f(x) - 2x$. Then $h(0) = 0$ and

$$h(f(x)) = f(f(x)) - 2f(x) = f(x) + 2x - 2f(x) = -h(x).$$

Thus

$$h(f^{(2)}(x)) = -h(f(x)) = h(x)$$

and we get by induction on n that

$$h(f^{(2n)}(x)) = h(x), x \in \mathbb{R}, n \in \mathbb{N}.$$

Let $f^{-1}(x)$ be the inverse function of $f(x)$. Then setting $x = f^{(-2n)}(y)$ we get

$$h(y) = h(f^{(-2n)}(y)), y \in \mathbb{R}.$$

On the other hand

$$2f^{(-n-2)}(x) = f^{(-n)}(x) - f^{(-n-1)}(x)$$

and we get by induction that

$$|f^{(-n-2)}(x)| \leq \frac{|f^{(-n)}(x)|}{2}$$

and therefore

$$|f^{(-2n)}(x)| \leq \frac{|x|}{2^n}, n \in \mathbb{N}.$$

Thus

$$\begin{aligned} h(x) &= h(f^{(-2n)}(x)) = \lim_{x \rightarrow \infty} h(f^{(-2n)}(x)) \\ &= h(\lim_{x \rightarrow \infty} f^{(-2n)}(x)) = h(0) = 0. \end{aligned}$$

Thus $h(x) \equiv 0$ and $f(x) = 2x$.

Case 2. Let now $f(x)$ be a strictly decreasing function. This time we set $h(x) = f(x) + x$. Then

$$h(f(x)) = f(f(x)) + f(x) = 2f(x) + 2x = 2h(x)$$

and we get by induction on n that

$$h(f^{(n)}(x)) = 2^n h(x), n \in \mathbb{N}.$$

On the other hand it follows from the initial condition that

$$|f^{(2)}(x)| \leq 2|x|, x \in \mathbb{R}$$

and we conclude by induction that

$$|f^{(2n)}(x)| \leq 2^n |x|, x \in \mathbb{R}.$$

Hence $h(x) = f(f(x)) - x \in (-|x|, |x|)$ and therefore

$$|h(x)| = \frac{|h(f^{(2n)}(x))|}{2^{2n}} \leq \frac{|f^{(2n)}(x)|}{2^{2n}} \leq \frac{2^n |x|}{2^{2n}} = \frac{|x|}{2^n}.$$

Hence $h(x) \equiv 0$ and $f(x) = -x$.

Problem 6.9. (IMO Longlist 1979) Find all monotonic bijections $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x) + f^{-1}(x) = 2x$$

for all $x \in \mathbb{R}$.

Solution. Set $g(x) = f(x) - x$. First, we shall prove by induction that

$$f(x + kg(x)) = x + (k + 1)g(x), \quad k \in \mathbb{Z}.$$

By definition, this is true for $k = 0$. Suppose that

$$f(x \pm (k - 1)g(x)) = x \pm kg(x), \quad k \in \mathbb{Z}.$$

Then

$$\begin{aligned} f(x \pm kg(x)) &= 2(x \pm kg(x)) - f^{-1}(x \pm kg(x)) \\ &= 2(x \pm kg(x)) - f^{-1}(x \pm (k - 1)g(x)) = x \pm (k + 1)g(x) \end{aligned}$$

and the induction is completed.

Assume now that $g(x) < g(y)$ for some $x, y \in \mathbb{R}$. Then we may choose $k \in \mathbb{N}$ such that $x - kg(x) > y - kg(y)$. Note that f is an increasing function; otherwise f and f^{-1} are decreasing functions and hence $2x = f(x) + f^{-1}(x)$ is such a function, a contradiction. Thus the n -th iterate $f^{(n)}$ of f is an increasing function, too. It follows that

$$f^{(n)}(x - kg(x)) \geq f^{(n)}(y - kg(y)).$$

Since

$$f^{(n)}(x - kg(x)) = f^{(n-1)}(x + (1 - k)kg(x)) = \cdots = x + (n - k)g(x)$$

and similarly

$$f^{(n)}(y - kg(y)) = y + (n - k)g(y),$$

we get that

$$x + (n - k)g(x) \geq y + (n - k)g(y).$$

But this is not true for all n large enough, since otherwise

$$\frac{x}{n-k} + g(x) \geq \frac{y}{n-k} + g(y)$$

and letting $n \rightarrow \infty$ we get $g(x) \geq g(y)$, a contradiction.

So $g(x)$ is a constant. Thus $f(x) = x + c$, $c \in \mathbb{R}$, and all such functions satisfy the given condition.

Problem 6.10. Find all functions $f: \mathbb{Z} \rightarrow \mathbb{Z}$ that satisfy

$$f(m+n) + f(mn-1) = f(m)f(n) + k$$

for all $m, n \in \mathbb{Z}$ and a given $k \in \mathbb{Z}$.

Solution. The cases $k = 0$ and $k = 2$ are investigated in Example 6.6 and Problem 18.8, respectively. So we shall assume that $k \neq 0, 2$. If $f = c$ is constant, then $2c = c^2 + k$ so $(c-1)^2 = 1-k$ hence $f(x) = 1 \pm \sqrt{1-k}$ if $\sqrt{1-k}$ is an integer. So assume f is not constant. Set $m = 0$ to get $f(n)(1-f(0)) = k - f(-1)$, possible only for $f(0) = 1, f(-1) = k$. Next set $m = -1$ to get $f(n-1) + f(-n-1) = kf(n) + k$. If we replace n by $-n$, the LHS does not change hence neither does the RHS. So f is even (recall that $k \neq 0$). Therefore we can write $f(n-1) + f(n+1) = kf(n) + k$. If $k = -1$, we get $f(n-1) + f(n) + f(n+1) = -1$ and we deduce $f(3k) = 1, f(3k \pm 1) = -1$ which satisfies the equation. If $k = 1$, we get $f(n+1) + f(n-1) = f(n) + 1$. In this case we prove by induction on $|n|$ that $f(n) = 1$ so f is constant. If $k = -2$, then we have $f(n-1) + 2f(n) + f(n+1) = -2$ so $a_n = f(n) + \frac{1}{2}$ satisfies the equation $a_{n-1} + 2a_n + a_{n+1} = 0$. Its characteristic equation $x^2 + 2x + 1 = 0$ has a double root -1 , thus $a_n = (an + b)(-1)^n$. As f is even $a_n = a_{-n}$ which is possible only for $a = b = 0$. So $a_n = 0$ and hence f is identically $-\frac{1}{2}$, which is not possible. Finally if $|k| > 2$, then set $a_n = f(n) + \frac{k}{k-2}$. It satisfies the condition $a_{n-1} - ka_n + a_{n+1} = 0$ with characteristic equation $x^2 - kx + 1 = 0$. This equation has two roots r and $\frac{1}{r}$, where $|r| > 1$. Then we find a_n to be $cr^n + d(\frac{1}{r})^n$. As f is even, $a_n = a_{-n}$ so $cr^n + \frac{d}{r^n} = \frac{c}{r^n} + dr^n$ or $(c-d)\left(r^n - \frac{1}{r^n}\right)$. Hence $c = d \neq 0$ as the function is not constant. Then

$f(n) \sim cr^n$ for $n \rightarrow \infty$. We let $m = n \rightarrow \infty$ to get $f(2n) + f(n^2 - 1) = f^2(n) + k$. But $f(2n) + f(n^2 - 1) \sim cr^{n^2-1}$, $f^2(n) + k \sim c^2r^{2n}$, and for $n \rightarrow \infty$ we get a contradiction. So there are no solutions in this case.

Problem 6.11. (Belarus 1998) Prove that:

a) if $a \leq 1$ then there is no function $f : (0, \infty) \rightarrow (0, \infty)$ such that

$$f\left(f(x) + \frac{1}{f(x)}\right) = x + a \quad (1)$$

for all $x \in (0, \infty)$;

b) if $a > 1$, then there are infinitely many functions $f : (0, \infty) \rightarrow (0, \infty)$ satisfying (1).

Solution. a) Suppose that $f : (0, \infty) \rightarrow (0, \infty)$ satisfies (1). Set

$$g(x) = f(x) + \frac{1}{f(x)}.$$

If $f(x) > 1$ then $y = f(x) - a > 0$ and $f(g(y)) = f(x)$. But (1) implies that the function f is injective and therefore $x = g(y) \geq 2$. Hence $f(x) < 1$ for any $x \in (0, 2)$ with at most one exception. For any $x \in (0, 2)$ we have

$$y = \frac{1}{f(x)} - a > 0, z = f(g(y)) \geq 2, f(z) = \frac{1}{f(x)},$$

$$z + a = f(g(z)) = f(g(x)) = x + a.$$

Thus $2 \leq z = x < 2$, a contradiction.

b) Let $a > 1$ and f be an arbitrary strictly increasing continuous function on the interval $[0, 2]$ such that $f(0) = 1$ and $f(2) = a$. Then (1) defines a unique strictly increasing continuous function f on $[0, \infty)$. Indeed, let

$$g(x) = f(x) + \frac{1}{f(x)}$$

and define the sequence $\{a_n\}_{n=0}^{\infty}$ by: $a_0 = 0, a_n = g(a_{n-1}), n \in \mathbb{N}$. Suppose that f has been defined as a strongly increasing and continuous function on

the interval $(a_{n-1}, a_n]$ greater than 1. Then it is easy to check that $g(x)$ is a strictly increasing and continuous function on this interval with

$$g(a_{n-1}) = a_n, g(a_n) = a_{n+1}.$$

Hence for any $x \in (a_n, a_{n+1}]$ there is a unique $y \in (a_{n-1}, a_n]$ such that $x = g(y)$ and set $f(x) = y + a$. It follows by induction that the function $f(x)$ is defined on $[0, \infty)$ and it satisfies (1).

Problem 6.12. (Bulgaria 2003) Find all $a > 0$ for which there exists a function $f: \mathbb{R} \rightarrow \mathbb{R}$ having the following two properties:

- (i) $f(x) = ax + 1 - a$ for all $x \in [2, 3]$;
- (ii) $f(f(x)) = 3 - 2x$ for all $x \in \mathbb{R}$.

Solution. Set $h(x) = f(x + 1) - 1$. Then the conditions (i) and (ii) can be written as $h(x) = ax$ for $x \in [1, 2)$ and $h(h(x)) = -2x$ for all $x \in \mathbb{R}$. Then $h(-2x) = h(h(h(x))) = -2h(x)$; in particular $h(0) = 0$. It follows by induction that $h(4^n x) = 4^n h(x)$ for all $n \in \mathbb{Z}$ and hence $h(x) > 0$ for $x \in [4^n, 2 \cdot 4^n)$. On the other hand, $0 > -2x = h(h(x)) = h(ax)$ for $x \in [1, 2)$ and therefore $[a, 2a) \subset [2 \cdot 4^k, 4^{k+1})$ for a $k \in \mathbb{Z}$. Thus $a = 2 \cdot 4^k$. Conversely, let $a = 2 \cdot 4^k$ for some integer k . Then one checks easily that the function

$$h(x) = \begin{cases} ax & \text{for } x \in [4^n, 2 \cdot 4^n), \\ -\frac{2x}{a} & \text{for } x \in [2 \cdot 4^n, 4^{n+1}), \\ 0 & \text{for } x = 0, \\ ax & \text{for } x \in (-4^{n+1}, -2 \cdot 4^n], \\ -\frac{2x}{a} & \text{for } x \in (-2 \cdot 4^n, -4^n], \end{cases}$$

where n ranges over the integers has the desired properties. One can easily prove that this is the only function with the given properties.

19.7 Constructive Problems

Problem 7.1. Let $2 \leq k \in \mathbb{N}$. Find all functions $f: \mathbb{N}_0 \rightarrow \mathbb{N}_0$ that satisfy $f(0) = 0$ and

$$f(n) = 1 + f\left(\left\lfloor \frac{n}{k} \right\rfloor\right)$$

for all $n \in \mathbb{N}$.

Solution. This is seen as a recurrence. To compute $f(n)$, we need to compute $f\left(\left\lfloor \frac{n}{k} \right\rfloor\right)$ first. To compute $f\left(\left\lfloor \frac{n}{k} \right\rfloor\right)$ we need to compute

$$f\left(\left\lfloor \frac{\left\lfloor \frac{n}{k} \right\rfloor}{k} \right\rfloor\right) = f\left(\left\lfloor \frac{n}{k^2} \right\rfloor\right)$$

and so on. Thus if $n \geq k^{r-1}$, by repeating this argument, we get

$$f(n) = r + f\left(\left\lfloor \frac{n}{k^r} \right\rfloor\right).$$

Thus if $k^r \leq n < k^{r+1}$ then $f(n) = r + 1 + f(0) = r + 1$, so $f(n) = 1 + \lfloor \log_k n \rfloor$ for $n > 0$. It clearly satisfies the condition.

Remark. We have used in the proof the identity

$$\left\lfloor \frac{\left\lfloor \frac{x}{m} \right\rfloor}{n} \right\rfloor = \left\lfloor \frac{x}{mn} \right\rfloor,$$

where $m, n \in \mathbb{N}$. Indeed, if $a = \left\lfloor \frac{x}{mn} \right\rfloor, b = \left\lfloor \frac{x}{m} \right\rfloor$ then $mna \leq x < mna + mn$, thus $na \leq \frac{x}{m} < na + n$. Hence $na \leq b < n(a + 1)$, so $\left\lfloor \frac{b}{n} \right\rfloor = a$, and we are done. The problem could be solved without this identity, if we would look at the representation of x in base k .

Problem 7.2. Find all functions $f: \mathbb{N} \rightarrow \mathbb{N}$ that satisfy $f(1) = 2$ and

$$f(n+1) = \lfloor 1 + f(n) + \sqrt{1 + f(n)} \rfloor - \lfloor \sqrt{f(n)} \rfloor$$

for all $n \in \mathbb{N}$.

Solution. As $\lfloor \sqrt{1 + f(n)} \rfloor = \lfloor \sqrt{f(n)} \rfloor$ unless $1 + f(n)$ is a perfect square, we deduce that $f(n+1) = f(n) + 1$ unless $f(n) + 1$ is a perfect square. Thus f jumps over the perfect squares, and $f(n)$ is the n -th number in the list of numbers not perfect squares. To find an explicit expression for f , assume that

$f(n) = k$. Then there are $\lfloor \sqrt{k} \rfloor$ perfect squares less than k so $k - \lfloor \sqrt{k} \rfloor$ numbers which are not perfect squares. As k is the n -th number we get $k - \lfloor \sqrt{k} \rfloor = n$ so $k - \sqrt{k} < n < k - \sqrt{k} + 1$. We claim that $k = n + \left\lfloor \sqrt{n} + \frac{1}{2} \right\rfloor$. Indeed $n + \left\lfloor \sqrt{n} + \frac{1}{2} \right\rfloor$ is not a perfect square, for if $n + \left\lfloor \sqrt{n} + \frac{1}{2} \right\rfloor = m^2$ then we deduce $n < m^2$, so $\left\lfloor \sqrt{n + \frac{1}{2}} \right\rfloor \leq m - 1$ and $n \geq m^2 - m + 1$. Then $\left\lfloor \sqrt{n} + \frac{1}{2} \right\rfloor \geq m$ which implies $n + \left\lfloor \sqrt{n} + \frac{1}{2} \right\rfloor \geq m^2 + 1$, a contradiction. Next we have to prove that

$$\left\lfloor \sqrt{n + \left\lfloor \sqrt{n} + \frac{1}{2} \right\rfloor} \right\rfloor = \left\lfloor \sqrt{n} + \frac{1}{2} \right\rfloor.$$

Indeed, if $m(m-1) \leq n \leq m(m+1)$ then $\left\lfloor \sqrt{n} + \frac{1}{2} \right\rfloor = m$. So

$$n + \left\lfloor \sqrt{n} + \frac{1}{2} \right\rfloor = n + m,$$

hence $m^2 \leq n + \left\lfloor \sqrt{n} + \frac{1}{2} \right\rfloor \leq m^2 + 2m$ and

$$\left\lfloor \sqrt{n + \left\lfloor \sqrt{n} + \frac{1}{2} \right\rfloor} \right\rfloor = m.$$

Problem 7.3. Find all functions $f: \mathbb{Z} \rightarrow \mathbb{Z}$ that satisfy

$$f(m+n) + f(mn) = f(m)f(n) + 1$$

for all $m, n \in \mathbb{Z}$.

Solution. Set $m = 0, n = 0$ to get $2f(0) = f^2(0) + 1$ so $f(0) = 1$. Next set $m = -1, n = 1$ to get $f(0) + f(-1) = f(1)f(-1) + 1$ so $f(-1)(f(1) - 1) = 0$. If $f(1) = 1$ then set $m = 1$ to get $f(n+1) + f(n) = f(n) + 1$. Hence $f \equiv 1$ which

satisfies our condition. If $f(-1) = 0$, set $m = -1$ to get $f(n-1) + f(-n) = 1$. Also set $m = 1$ to get $f(n+1) = f(n)(f(1) - 1) + 1$. If $a = f(1) - 1$ we get $f(n+1) = af(n) + 1$. From here $(a-1)f(n+1) - 1 = a((a-1)f(n) - 1)$. We conclude that

$$(a-1)f(n+k) - 1 = a^k((a-1)f(n) - 1).$$

So $a^k | (a-1)f(n) - 1$ for any k which is possible only when $f(n) = \frac{1}{a-1}$ or $a = \pm 1$. If $f(n) = \frac{1}{a-1}$, then $f(1) = a + 1 = \frac{1}{a-1}$. So $a^2 = 2$ which is impossible. Hence $a = \pm 1$. If $a = 1$, we get $f(n+1) = f(n) + 1$ so $f(n) = n + 1$ by induction on $|n|$, which obeys the equation as

$$(m+n+1) + (mn+1) = (m+1)(n+1) + 1.$$

If $a = -1$ we get $f(n+1) = 1 - f(n)$. We conclude by induction on $|n|$ that $f(n) = 1$ if n is even and $f(n) = 0$ if n is odd. Indeed, if m, n have the same parity then $m+n$ is even so $f(m+n) = 1$ and also $f(m)f(n) = f(mn)$. So the condition holds, while if one of m, n is even and the other odd we get $f(m+n) = 0, f(mn) = 1, f(m)f(n) = 0$ and so the condition holds again. Hence all solutions are given by

$$f(n) = 1, \quad f(n) = n + 1 \quad \text{and} \quad f(n) = 1 - n \pmod{2}.$$

Problem 7.4. Find all functions $f: \mathbb{Z} \rightarrow \mathbb{Z}$ that verify

$$f(f(k+1) + 3) = k$$

for all $k \in \mathbb{Z}$.

Solution. Let us start by noting that f is injective, as if $f(m) = f(n)$ then plugging $k = m - 1, n - 1$ we get $m = n$. Therefore if we set $k = f(n)$ we get $f(f(f(n) + 1) + 3) = f(n)$ and the injectivity implies $f(f(n) + 1) + 3 = n$ or $f(f(n) + 1) = n - 3$. By plugging $k = n - 3$ in the condition we get $f(f(n - 2) + 3) = n - 3$ and the injectivity gives us $f(n - 2) + 3 = f(n) + 1$ so $f(n) = f(n - 2) + 2$. From here we deduce that if $f(0) = a, f(1) = b$

then $f(2n) = 2n + a$, $f(2n + 1) = 2n + b$. Also from the given condition f is surjective so a and b have distinct parities and we encounter two cases:

a) a is even and b is odd. Plugging $k = 2n$ we get $f(f(2n + 1) + 3) = 2n$ or $f(2n + b + 3) = 2n$ so $2n + b + 3 + a = 2n$ hence $a + b + 3 = 0$. Plugging $k = 2n - 1$ we get $f(f(2n) + 3) = 2n - 1$ or $f(2n + a + 3) = 2n - 1$ so $2n + a + 2 + b = 2n - 1$ and again $a + b + 3 = 0$. Conversely, if $a + b + 3 = 0$, then f defined by $f(2n) = 2n + a$, $f(2n + 1) = 2n + b$ satisfies the condition.

b) a is odd and b is even. Then plugging $k = 2n$ we deduce

$$f(2n + b + 3) = 2n$$

so $2n + 2b + 2 = 2n$ hence $b = -1$ which contradicts the evenness of b .

To conclude, all solutions are given by $f(2n) = 2n + a$, $f(2n + 1) = 2n + b$ where a is even, b is odd and $a + b + 3 = 0$.

Problem 7.5. Find all functions $f: \mathbb{N} \rightarrow \mathbb{N}$ satisfying

$$f(m + f(n)) = n + f(m + k)$$

for all $m, n \in \mathbb{N}$, where $k \in \mathbb{N}$ is fixed.

Solution. It is clear that f is injective (if $f(n_1) = f(n_2)$ taking $n = n_1, n_2$, the LHS side of the condition stays the same, while the RHS is $n_1 + f(m + k)$, respectively $n_2 + f(m + k)$, so $n_1 = n_2$). Now let us symmetrize the condition by replacing n by $f(n + k)$. We get $f(m + f(f(n + k))) = f(n + k) + f(m + k)$. Swapping m and n we get $f(n + f(f(m + k))) = f(n + k) + f(m + k)$ and the injectivity of f gives us $n + f(f(m + k)) = m + f(f(n + k))$, or if we take

$$x = m + k, \quad y = n + k$$

we deduce $f(f(x)) - x = f(f(y)) - y = a$ so $f(f(x)) = x + a$. Then $a \geq 0$. Now set $n \rightarrow f(n)$ to get

$$f(m + n + a) = f(n) + f(m + k).$$

Interchanging m, n we get

$$f(m + n + a) = f(m) + f(n + k)$$

so $f(n+k) - f(n) = f(m+k) = b$. So $f(n+k^2) = f(n) + bk$ thus

$$f(f(n+k^2)) = f(f(n) + bk) = f(f(n)) + b^2.$$

But $f(f(n+k^2)) = n+k^2+a$, $f(f(n)) = n+a$ which implies $b^2 = k^2$; so $b = k$ as $b \geq 0$. Therefore $f(m+n+a) = f(m) + f(n) + k$. This Cauchy-type equation can be solved in a usual way:

$$f(m+n+a) = f(m+n-1) + f(1) + k = f(m) + f(n) + k$$

and so $f(m+n-1) + f(1) = f(m) + f(n)$. Now set $m = 2$ to get

$$f(n+1) + f(1) = f(2) + f(n),$$

and therefore f is a linear function. If $f(x) = cx + d$ then substituting into the condition we get $c(m+cn+d) + d = n + c(m+k) + d$ so $c(cn+d-k) = n$ which is possible only for $c = 1$ (as $c \geq 0$) and $d = k$. So $f(x) = x + k$ and it satisfies the given condition.

Problem 7.6. (IMO Shortlist 1995) Does there exist a sequence $F(1), F(2), F(3), \dots$ of nonnegative integers that simultaneously satisfies the following three conditions:

- (a) each of the integers $0, 1, 2, \dots$ occurs in the sequence;
- (b) each positive integer occurs in the sequence infinitely often;
- (c) $F(F(n^{163})) = F(F(n)) + F(F(361))$ for all $n \geq 2$?

First Solution. Letting $F(1) = 0$ and $F(361) = 1$, condition (c) becomes

$$F(F(n^{163})) = F(F(n)), n \geq 2.$$

Now let $F(n) = n$ for $n = 2, 3, \dots, 360$ and inductively define $F(n)$ for $n \geq 362$ as follows: $F(n) = F(m)$ if $n = m^{163}$, $m \in \mathbb{N}$ and $F(n)$ = the least number not in the set $\{F(k) | k < n\}$, otherwise. It is now easy to check that the sequence $F(1), F(2), F(3), \dots$ satisfies the given conditions.

Second Solution. Another example of such a sequence is as follows. If

$$n = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_k^{\alpha_k}$$

is the factorization of n into primes, we put

$$F(n) = \alpha_1 + \alpha_2 + \cdots + \alpha_k$$

and $F(1) = 0$. Conditions (a) and (b) are obviously satisfied, while (c) follows from

$$F(F(n^{163})) = F(163F(n)) = F(F(n)) + 1$$

because 163 is a prime, and $F(F(361)) = F(F(19^2)) = F(2) = 1$.

Problem 7.7. Find all functions $f: \mathbb{N}_0 \rightarrow \mathbb{R}$ that satisfy $f(4) = f(2) + 2f(1)$ and

$$f\left(\binom{n}{2} - \binom{m}{2}\right) = f\left(\binom{n}{2}\right) - f\left(\binom{m}{2}\right)$$

for $n > m$.

Solution. The functions $f(x) = cx$, $c \in \mathbb{R}$, clearly satisfy the condition. Set $f(1) = a$ and $f(2) = b$. It is natural to try to prove that $b = 2a$. To do this, we have to compute some values of f . If

$$S = \left\{ \binom{n}{2} \mid n \in \mathbb{N} \right\} = \{0, 1, 3, 6, 10, 15, 21, 28, \dots\},$$

then $f(x - y) = f(x) - f(y)$ for all $x, y \in S$. Now we can find

$$f(3) = f(2) + f(1) = a + b, \quad f(4) = 2a + b, \quad f(6) = 2f(3) = 3(a + b),$$

$$f(5) = f(6) - f(1) = a + 2b, \quad f(10) = f(6) + f(4) = 4a + 3,$$

$$f(7) = f(10) - f(3) = 3a + 2b.$$

If we continue like this, we will not find a contradiction, a fact that leads us to the conclusion that there are more functions that satisfy the condition. If we look attentively at the list of computed values, we see that

$$f(3k) = k(a + b), \quad f(3k + 1) = (k + 1)a + kb, \quad f(3k + 2) = ka + (k + 1)b$$

so we are led to looking at the residues of n modulo 3, and find another example $g(x) = x \bmod 3$ where we set $2 \bmod 3 = -1$. It satisfies the conclusion because S does not contain numbers congruent to 2 modulo 3. So $f(x) = cx + dg(x)$

also satisfies the conclusion. Now we can try to prove that these are the only solutions. Let $c = \frac{a+b}{3}$, $d = \frac{2(a-b)}{3}$, and let $h(x) = f(x) - cx - dg(x)$. It satisfies the conclusion and also $h(1) = h(2) = h(3) = \dots = h(7) = 0$. We need to prove that $h(x) = 0$ for all x . We do this by induction on x . Suppose that we have $h(x) = 0$ for all $x = 1, 2, \dots, n$ and let us prove that $h(n+1) = 0$. First, as $\binom{m+1}{2} - \binom{m}{2} = m$ we deduce that $h\left(\binom{k}{2}\right) = 0$ for $k \leq n+1$, and we need to prove that $h\left(\binom{n+2}{2}\right) = 0$. It could be done if we would find $k, x, y \leq n+1$ such that

$$\binom{n+2}{2} - \binom{n+2-k}{2} = \binom{x}{2} - \binom{y}{2}.$$

This relation is equivalent to $k(2n+3-k) = (x-y)(x+y-1)$. Now let us find a $k \leq 3$ such that $3|2n+3-k$. Then

$$k(2n+3-k) = 3k \left(\frac{2n+3-k}{3} \right)$$

and $3k, \frac{2n+3-k}{3}$ have opposite parities. Hence we can set

$$2x = \frac{2n+3-k}{3} + 3k, \quad 2y = \frac{2n+3-k}{3} - 3k.$$

We only need to verify that $x, y \leq n+1$ (we could also have $y < 0$ but then $\binom{y}{2} = \binom{-y+1}{2}$, so there is no problem here) which is equivalent to $2n+4 > \frac{2n+3-k}{3} + 3k$ or $4n+9 > 8k$. As $k \leq 3$ this is satisfied when $n \geq 4$. Since we have already $h(1), h(2), h(3), h(4)$ zero, we are done.

Problem 7.8. Find all functions $f: \mathbb{Q}^+ \rightarrow \mathbb{Q}^+$ that satisfy

$$f(x) + f\left(\frac{1}{x}\right) = 1 \text{ and } f(f(x)) = \frac{f(x+1)}{f(x)}$$

for all $x \in \mathbb{Q}^+$.

Solution. We have already encountered a function that satisfies

$$f(x) + f\left(\frac{1}{x}\right) = 1,$$

namely $f(x) = \frac{1}{x+1}$. There are of course many of them, but this seems more comfortable, as it indeed satisfies the condition. So let us prove $f(x) = \frac{1}{x+1}$. If we set $x = 1$, we get $f(1) + f(1) = 1$ and so $f(1) = \frac{1}{2}$. Let A be the set of all $x \in \mathbb{Q}^+$ that satisfy $f(x) = \frac{1}{x+1}$. If $x \in A$ then

$$f\left(\frac{1}{x}\right) = 1 - f(x) = 1 - \frac{1}{x+1} = \frac{x}{x+1} = \frac{1}{1+\frac{1}{x}}.$$

Also writing the second condition for x yields $f\left(\frac{1}{x+1}\right) = \frac{f(x+1)}{\frac{1}{x+1}}$ so

$$f\left(\frac{1}{x+1}\right) = (x+1)f(x+1).$$

As $f\left(\frac{1}{x+1}\right) = 1 - f(x+1)$ we get $(x+2)f(x+1) = 1$, so $f(x+1) = \frac{1}{x+2}$.

Thus if $x \in A$ then $\frac{1}{x}$ and $x+1$ also belong to A . It remains to show that every positive rational number $\frac{p}{q}$ can be obtained from 1 by means of operations $x \rightarrow \frac{1}{x}, x \rightarrow x+1$. This can be proven by induction on $p+2q \geq 3$.

For $p+2q = 3, p = q = 1$ and the basis holds true. Next if $p > q$ then $\frac{p-q}{q}$ obeys the induction step and the operation $x \rightarrow x+1$ turns $\frac{p-q}{q}$ to $\frac{p}{q}$. If $p < q$ then $\frac{q}{p}$ obeys the induction step and the operation $x \rightarrow \frac{1}{x}$ turns $\frac{q}{p}$ to $\frac{p}{q}$.

Problem 7.9. Let g and h be continuous real-valued bijections on the reals. It is also known that g has no a fixed point (i.e. $g(x) \neq x$ for all $x \in \mathbb{R}$). Construct a continuous function $f: \mathbb{R} \rightarrow \mathbb{R}$ such that $f(g(x)) = h(f(x))$.

Solution. Since g is a continuous bijection, it must be either increasing or decreasing. Assume, for example, that g is increasing. As the function $g(x) - x$ is continuous and non-vanishing we cannot simultaneously have $g(s) < s$ and $g(t) > t$. Assume $g(x) > x$ for all $x \in \mathbb{R}$ (the second case is similar). Define $a_0 = 0, a_1 = g(a_0), a_i = g(a_{i-1})$ for $i > 0$ and $a_i = g^{-1}(a_{i+1})$ for $i < 0$. Then we have $a_i < a_{i+1}$ for all i . Since g is increasing, it provides a bijection between the sets $[a_{i-1}; a_i]$ and $[a_i; a_{i+1}]$. We need $f(a_1) = h(f(a_0))$. Now consider any continuous function f on the interval $[a_0; a_1]$ such that $f(a_1) = h(f(a_0))$ (graphically it is a curve connecting (a_0, t) with $(a_1, h(t))$ where $t = f(a_0)$). We claim that f can be extended to the whole real line in a unique way. Indeed, we can extend f to $[a_1; a_2]$: any x in $[a_1; a_2]$ can be written uniquely as $g(y)$ where $y \in [a_0; a_1]$, then we set $f(x) = h(f(y))$. In this way, we ensure that the condition $f(g(x)) = h(f(x))$ holds for all $x \in [a_0; a_1]$. We similarly extend f to $[a_2; a_3]$, then to $[a_3, a_4]$ and so on. Since a_i is increasing, if $(a_n)_{n>0}$ converges to a then $g(a) = a$ which is impossible. Thus a_n tends to infinity meaning that we have defined f on the whole $[a_0; +\infty)$ and the condition $f(g(x)) = h(f(x))$ holds for all $x \in [a_0; +\infty)$. We similarly extend f to $[a_{-1}; a_0]$ by setting $f(x) = h^{-1}(y)$, where $y = g(x) \in [a_0; a_1]$. Then we proceed by induction to define f on the intervals $[a_{i-1}; a_i]$ for $i < 0$. We know that a_i tends to $-\infty$ as i decreases, so we define f to the whole $\mathbb{R}$. The function f satisfies the conditions of the problem and it is uniquely defined by its restriction to the interval $[a_0; a_1]$.

Remark. If g has a fixed point x_0 , then plugging $x = x_0$ yields

$$f(x_0) = h(f(x_0)),$$

so h has also a fixed point $y = h(x_0)$. Conversely, if y is a fixed point then $f(x) = y$ satisfies the condition. It is much more interesting to find all f , and the method to do this is similar to the one above. In this case $\mathbb{R}$ will split in intervals where $g(x) - x$ is constant and in intervals where $g(x) - x$ is positive or negative. The reader is encouraged to investigate this more general problem and reach the complete solution.

Problem 7.10. Let $g: \mathbb{C} \rightarrow \mathbb{N}$ be a given function, $a \in \mathbb{C}$ a given constant and w a primitive cubic root of unity. Find all functions $f: \mathbb{C} \rightarrow \mathbb{C}$ such that

$$f(z) + f(wz + a) = g(z)$$

for all $z \in \mathbb{C}$.

Solution. In this case, we iterate $wz + a$ rather than f . Let $h(z) = wz + a$. Then $f(z) + f(h(z)) = g(z)$. Now

$$h(h(z)) = w(wz + a) + a = w^2z + (w + 1)a,$$

$$h(h(h(z))) = w(w^2z + (w + 1)a) + a = w^3z + (w^2 + w + 1)a = z.$$

Therefore $h(h(h(z))) = z$. So

$$f(z) + f(h(z)) = g(z), \quad f(h(z)) + f(h(h(z))) = g(h(z)),$$

$$f(h(h(z))) + f(z) = f(h(h(z))) + f(h(h(h(z)))) = g(h(h(z))).$$

If we solve the obtained non-singular linear system in $f(z), f(f(z)), f(f(f(z)))$, we get

$$f(z) = \frac{g(z) + g(w^2z + (w + 1)a) - g(wz + a)}{2},$$

and similar expressions for $f(h(z))$ and $f(h(h(z)))$. It is easy to see that this f satisfies the condition.

Problem 7.11. Given a positive integer n find all monotonic functions $f: \mathbb{R} \rightarrow \mathbb{R}$ that satisfy $f^{(n)}(x) = -x$ for all $x \in \mathbb{R}$, where $f^{(n)}$ is the n -th iterate of f .

Solution. The function f cannot be increasing since otherwise $f^{(n)}(x) = -x$ would be increasing too, a contradiction. Thus f is decreasing. Set

$$a_m = f^{(m)}(x).$$

Since $f^{(2n)}(x) = -f^{(n)}(x) = x$ the sequence $A = \{a_k | k \in \mathbb{N}\}$ is periodic of period $2n$ so is finite. Let a_k be the smallest term of A . Then as f is decreasing we conclude that $a_{k+1} = f(a_k)$ is the largest term of A . We have $a_{i+n} \geq a_k$ and hence $a_i = -a_{i+n} \leq -a_k$, so $a_{k+n} = -a_k$ is the largest term of

A , i.e. $f(a_k) = -a_k$. Also $f(a_{k+1})$ should be the smallest term of A . Hence $f(-a_k) = a_k$ and we conclude by induction on p that $a_{k+p} = (-1)^p a_k$. Since A is periodic, then $a_{k+p} = a_0 = x$ for some p and therefore $f(x) = -x$. This function satisfies the conditions if and only if n is odd.

Problem 7.12. (Romanian TST 1979, Singapore 1996) Show that there exist functions $f: \mathbb{N} \rightarrow \mathbb{N}$ such that $f(f(n)) = n^2$ for all $n \in \mathbb{N}$.

Solution. The problem presents no difficulty to us once we have enough experience in solving this kind of problems. If m is not a perfect square, define a *chain* $C(m)$ as the sequence $(x_n)_{n \in \mathbb{N}}$ with $x_i = m^{2^i}$. The chains partition all $\mathbb{N} \setminus \{1\}$ (1 presents no problem as we can merely set $f(1) = 1$ and ignore this number). Then by pairing the infinitely many chains: $(C(x_1), C(y_1))$, $(C(x_2), C(y_2))$, ... and setting $f(x_i^{2^k}) = y_i^{2^k}$, $f(y_i^{2^k}) = x_i^{2^{k+1}}$ we construct the desired function. It presents no difficulty proving that all solutions can be written in such a form.

Problem 7.13. Prove that for each integer $n \geq 2$ there are infinitely many functions $f: \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}$ such that $f^{(n)}(x) = \frac{1}{x}$.

Solution. Let $A_1, A_2, \dots, A_n$ be a partition of the set $\mathbb{R} \setminus \{0\}$ such that if $x \in A_i$ then $\frac{1}{x} \in A_i$ and there exists a bijection $f_i: A_i \rightarrow A_{i+1}$ for all $i = 1, 2, \dots, n-1$. Then define $f(x) = f_i(x)$ for $x \in A_i$ and

$$f(f_{n-1}(f_{n-2}(\dots(f_1(x))))) = \frac{1}{x}$$

for $x \in A_1$. Since every $y \in A_n$ can be represented uniquely as

$$y = f_{n-1}(f_{n-2}(\dots(f_1(x))))$$

for some $x \in A_1$, it follows that f is well defined on $\mathbb{R} \setminus \{0\}$ and $f^{(n)}(x) = \frac{1}{x}$.

Problem 7.14. Show that there are infinitely many functions $f: \mathbb{R} \setminus \{-1\} \rightarrow \mathbb{R} \setminus \{-1\}$ such that

$$f\left(\frac{x}{x+1}\right) = \frac{f(x)}{f(x)+1}$$

for all $x \in \mathbb{R} \setminus \{-1\}$.

Solution. Set $\alpha(x) = \frac{x}{x+1}$. Then the given equation takes the form $f(\alpha(x)) = \alpha(f(x))$ and its obvious solutions are the iterations $\alpha^{(n)}(x)$ of $\alpha(x)$. Since $\alpha^{(n)}(x) = \frac{x}{nx+1}$ we see that the functions $f(x) = \frac{x}{cx+1}$, where c is a constant satisfy the given equation. With a little more effort one can prove that all rational functions satisfying the given equation have the above form.

Problem 7.15. Determine all continuous functions $f : (0, \infty) \rightarrow (0, \infty)$ such that $f(f(x)) = x$ and $f(x+1) = \frac{f(x)}{f(x)+1}$ for all $x \in (0, \infty)$.

Solution. We first show that $f(1) = 1$. Indeed, using the identity

$$f(x+1) = \frac{f(x)}{f(x)+1}$$

it follows by induction on n that $f(n+1) = \frac{a}{1+an}$, where $a = f(1)$. Hence

$$f\left(\frac{a}{1+an}\right) = f(f(n+1)) = n+1$$

and we get that

$$f\left(\frac{1+(n+1)a}{1+na}\right) = f\left(1 + \frac{a}{1+na}\right) = \frac{f\left(\frac{a}{1+na}\right)}{1+f\left(\frac{a}{1+na}\right)} = \frac{n+1}{n+2}.$$

Then letting $n \rightarrow \infty$ and using the continuity of f we get $f(1) = 1$.

Now consider the set $M = \left\{x \in \mathbb{Q}^+ \mid f(x) = \frac{1}{x}\right\}$. Then $1 \in M$ and the given identities for $f(x)$ imply that if $x \in M$ then $x+1 \in M$ and $\frac{1}{x} \in M$. We shall show that M contains the set $\mathbb{Q}^+$ of positive rationals. Note first that M contains the positive integers since $1 \in M$ and $x+1 \in M$ for $x \in M$. Hence M contains the positive rationals of the form $\frac{n}{2}$ since $\frac{1}{2} \in M$ and $\frac{2m+1}{2} = \frac{1}{2} + m \in M$ for all positive integers m . Suppose that M contains

the positive rationals of the form $\frac{n}{j}$ for all $1 \leq j \leq k$. To prove that $\frac{n}{k+1} \in M$ denote by p the remainder of n when divided by $k+1$. Then $n = m(k+1) + p$ and if $p = 0$ we have $\frac{n}{k+1} = m \in M$. Otherwise $0 < p \leq k$ and $\frac{k+1}{p} \in M$. Hence $\frac{p}{k+1} \in M$ and $\frac{n}{k+1} = m + \frac{p}{k+1} \in M$. It follows by induction that M contains the positive rationals $\mathbb{Q}^+$. Thus $f(x) = \frac{1}{x}$ for $x \in \mathbb{Q}^+$ and using the continuity of $f(x)$ we extend this identity to all $x \in (0, \infty)$.

Problem 7.16. (Bulgaria 2009) Determine all real numbers a for which there exists:

- (i) a function $f: \mathbb{R} \rightarrow \mathbb{R}$ such that $f(0) = a$ and $f(f(x)) = x^{2009}$ for all $x \in \mathbb{R}$;
- (ii) a continuous function satisfying the conditions in (i).

Solution. (i) Since $f(x)^{2009} = f(f(f(x))) = f(x^{2009})$ we get $a^{2009} = a$, i.e. $a = 0, \pm 1$. Let $\{b, c, d\} = \{-1, 0, 1\}$ and define a function $f: \mathbb{R} \rightarrow \mathbb{R}$ as follows:

$$f(x) = \frac{1}{x} \text{ for } |x| > 1,$$

$$f(x) = \frac{1}{x^{2009}} \text{ for } 0 < |x| < 1,$$

$$f(b) = b, \quad f(c) = d, \quad f(d) = c.$$

Then $f(f(x)) = x^{2009}$ for every $x \in \mathbb{R}$ and therefore the desired a are $a = 0, \pm 1$.

(ii) Note first that f is an injective function (if $f(x) = f(y)$, then $x^{2009} = y^{2009}$, i.e. $x = y$.) Suppose that $a = f(0) = -1$. Then $f(-1) = f(f(0)) = 0$. Hence for $e = f(1)$ we have $e \neq 0, -1$ and $e^{2009} = e$, i.e. $e = 1$. Since $f(0)f(1) = -1 < 0$ the Intermediate Value theorem implies that $f(x) = 0$ for some $x \in (0, 1)$ which contradicts the injectivity of f . Analogously $a \neq 1$. For $a = 0$ we can define a function f with the desired properties as follows: $f(x) = x^{\sqrt{2009}}$ for $x > 0$, $f(0) = 0$ and $f(x) = -(-x)^{\sqrt{2009}}$ for $x < 0$.

Problem 7.17. Let $f: \mathbb{N} \rightarrow \mathbb{N}$ be a function satisfying

$$f(f(n)) = 4n - 3 \text{ and } f(2^n) = 2^{n+1} - 1$$

for all $n \in \mathbb{N}$. Find $f(993)$. What values can $f(2007)$ take?

Solution. It is clear that $f(x) = 2x - 1$ satisfies the condition. So we try to prove that $f(993) = 1995$. Now if we can prove $f(t) = 2t - 1$ by using the first condition we prove that $f(2t - 1) = 4t - 3$, $f(4t - 3) = 8t - 7$ and so on, so $f(t_n) = 2t_n - 1$ for t_n being recurrently defined as

$$t_0 = t, \quad t_{k+1} = 2t_k - 1.$$

We know that $f(2^n) = 2^{n+1} - 1$. So it seems natural to try to write $993 = t_n$, where $t_0 = 2^k$ for some k . Let us try to go backwards:

$$993 = 2 \cdot 497 - 1, \quad 497 = 2 \cdot 249 - 1, \quad 249 = 2 \cdot 125 - 1, \quad 125 = 2 \cdot 63 - 1, \quad 63 = 2 \cdot 32 - 1$$

and 32 is a power of two. This procedure works for any number of the form $2^m(2^n - 1) + 1$. But if we try to obtain $f(2007)$ by this procedure we fail because $2007 = 2 \cdot 1004 - 1$ and at 1004 we stop. Now we try to construct a different function than $2x - 1$ to answer the second question negatively. Consider a number $x > 1$ which is not $1 \pmod{4}$. Then we can define a sequence $S(x) = (x_n)_{n \in \mathbb{N}}$ for $x_1 = x$, $x_{k+1} = 4x_k - 3$, and call x the ancestor of the sequence. It is easy to prove that f is injective and that these sequences partition $\mathbb{N} \setminus \{1\}$. Let $g(x) = 4x - 3$ and $g^{(n)}(x)$ be g iterated n times. Now we can prove that if $f(n) = m$ then $f(m) = g(n)$ then $f(g(n)) = g(m)$ and so on, to get $f(g^{(k)}(n)) = g^{(k)}(m)$. Hence f provides a pretty comfortable intuitive mapping between these sequences, which we shall define now. Let x, y be ancestors of two sequences and assume $f(x) = y_m$. Let $f(y) = u$. Then $f(g^{(m)}(y)) = g^{(m)}(f(y)) = g^{(m)}(u)$ as proven just before. However $f(g^{(m)}(y)) = f(y_m) = g(x)$. So $g(x) = g^{(m)}(u)$ and hence $x = g^{(m-1)}(u)$ (if $m > 0$). As $x \not\equiv 1 \pmod{4}$, this is only possible for $m = 1$ or $m = 0$. In the first case we have $f(y) = x$ and in the second $f(x) = y$. In any case we see that f maps one ancestor into another ancestor, and then f can be defined recursively without contradiction on the sequences of these ancestors. In the condition we have defined f for the powers of two and for powers of two minus one. However we have infinitely many ancestors left which we can pair up as we want, constructing many different functions. In particular, pairing up 2007 with different ancestors we can obtain different values for $f(2007)$, so $f(2007)$ can not be uniquely determined. Now if 2007 is paired with another ancestor u then either $f(2007) = u$ or $f(u) = 2007$ and $f(2007) = 4u - 3$. Hence all

possible values for $f(2007)$ are either u or $4u - 3$ where u is a number not $1 \pmod{4}$ and not a power of two or one less a power of two.

Problem 7.18. (Romania TST 1989) Let $\mathbf{F}$ be the set of all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ such that $f(0) = 10$ and $f(f(x)) - 2f(x) + x = 0$ for all $x \in \mathbb{N}$. Find the set $A = \{f(1989) | f \in \mathbf{F}\}$.

Solution. It follows by induction that for every $n, x \in \mathbb{N}$ we have

$$f^{(n)}(x) = nf(x) - (n-1)x.$$

Since $f^{(n)}(x) \geq 0$ we get that $f(x) \geq \left(1 - \frac{1}{n}\right)x$ for all $n \in \mathbb{N}$. Hence $f(x) \geq x$ for every $x \in \mathbb{N}$. Set $f(x) = x + r(x)$. Then $r(x) \geq 0$ for all $x \in \mathbb{N}$ and it follows by induction from the identity above that $f^{(n)}(x) = x + nr(x)$. In particular, if $f(0) = 10$ then $f^{(n)}(0) = 10n$. Let $f(1989) = 1989 + r$. If r is co-prime with 10 then there exist a and b , $0 \leq a \leq 9$, such that $1989 + ar = 10b$. Hence

$$f^{(a)}(1989) = 1989 + ar = 10b = f^{(b)}(0).$$

Since obviously any function $f \in \mathbf{F}$ is injective it is easy to check that the above identity is not possible. Thus r is not co-prime with 10 and $\gcd(r, 10) \in \{2, 5, 10\}$. Hence it follows that either $f(1989) = 1989 + 2m$ or $f(1989) = 1989 + 5m$, or $f(1989) = 1989 + 10m$. This shows that the set A consists of all odd integers greater than 1989 and all even integers greater than 1989 which are congruent to 4 $\pmod{5}$. Let us note that the functions

$$f(x) = \begin{cases} x + 10 & \text{if } x \text{ is even} \\ x + 2m & \text{if } x \text{ is odd} \end{cases}$$

and

$$f(x) = \begin{cases} x + 10 & \text{if } x \text{ is even} \\ x + 5m & \text{if } x \text{ is odd} \end{cases}$$

satisfy the conditions of the problem. Thus

$$A = \{2k + 1 | k \geq 994\} \cup \{5l + 4 | l \geq 339\}.$$

Problem 7.19. Find all continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(1-x) = 1 - f(f(x))$$

for all $x \in \mathbb{R}$.

Solution. Let $g(x) = \frac{1}{2} - f\left(\frac{1}{2} - x\right)$. Then the equation becomes

$$-g\left(x - \frac{1}{2}\right) = g\left(g\left(\frac{1}{2} - x\right)\right)$$

and we have to find all continuous functions $g : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$g(g(x)) = -g(-x).$$

First we shall prove that $g(x) = x$ for all $x \in g(\mathbb{R}) = A$. The identity above shows that $a \in A \implies -a \in A$ and so, since $g(x)$ is continuous either $A = \mathbb{R}$, or $A = [-u, u]$ for some $u \geq 0$.

We have

$$g(g(g(g(x)))) = -g(-g(g(x))) = -g(g(-x)) = g(x)$$

and so $g(g(g(x))) = x$ for all $x \in A$. Let then $g_r : A \rightarrow A$ be the restriction of g to A . Then $g_r(g_r(g_r(x))) = x$ for all $x \in A$ and so g_r is a bijection, so is monotonous. On the other hand the identity $g(g(x)) = -g(-x)$ implies $g_r(g_r(x)) = -g_r(-x)$, so $-g_r(-x)$ is increasing and therefore $g_r(x)$ is increasing. Then $g_r(a) > a$ implies $g_r(g_r(a)) > g_r(a) > a$ and so $a = g_r(g_r(g_r(a))) > g_r(a) > a$, which is impossible. Thus $g_r(a) \leq a$. But $g_r(a) < a$ implies $g_r(g_r(a)) < g_r(a) < a$ and so $a = g_r(g_r(g_r(a))) < g_r(a) < a$, a contradiction. Hence $g_r(x) = x$ for all $x \in A$ and the original equation becomes $g(x) = -g(-x)$ (since $g(g(x)) = g_r(g(x)) = g(x)$).

Thus the continuous solutions of the equation $g(g(x)) = -g(-x)$ are the odd continuous functions $g(x)$ such that $g(x) = x$ for all $x \in g(\mathbb{R})$. A simple way to build them is the following. If $g(\mathbb{R}) = \mathbb{R}$, then $g(x) = x$ for all x . If $g(\mathbb{R}) = [-u, u]$ for some $u \geq 0$, then let any continuous function $h : [u, +\infty) \rightarrow [-u, +u]$ such that $h(u) = u$ and define $g(x)$ as follows: $g(x) = -h(-x)$ for

all $x \in (-\infty, -u)$, $g(x) = x$ for all $x \in [-u, +u]$ and $g(x) = h(x)$ for all $x \in (u, +\infty)$. Notice that $u = 0$ gives the solution $g(x) = 0$.

Finally, all solutions of the original equation are given by

$$f(x) = \frac{1}{2} - g\left(\frac{1}{2} - x\right),$$

where $g(x)$ is a function described above. Here are some examples:

1). $g(x) = 0$ gives the solution $f(x) = \frac{1}{2}$;

2). $g(x) = x$ gives the solution $f(x) = x$;

3). $u = 1$ and $h(x) = \sin\left(\frac{\pi}{2}x\right)$ gives the solution: If $\left|x - \frac{1}{2}\right| < 1$, then

$$f(x) = x \text{ and if } \left|x - \frac{1}{2}\right| \geq 1, \text{ then } f(x) = \frac{1}{2} + \sin\left(\frac{\pi}{2}x - \frac{\pi}{4}\right).$$

4). $u = 2$ and $h(x) = 4\left|2\left\{\frac{x-2}{8}\right\} - 1\right| - 2$ gives the solution

$$f(x) = \frac{5}{2} - 4\left|2\left\{\frac{-2x-3}{16}\right\} - 1\right|.$$

Here $\{a\}$ is the fractional part of a real number a .

19.8 D'Alembert's Equation

Problem 8.1. Find all solutions to the functional equation

$$f(x+y) + f(x-y) = 2f(x) \cos y$$

for all $x, y \in \mathbb{R}$.

Solution. Substituting $(x, y) = (0, t)$, $\left(\frac{\pi}{2} + t, \frac{\pi}{2}\right)$, $\left(\frac{\pi}{2}, \frac{\pi}{2} + t\right)$ we get

$$f(t) + f(-t) = 2a \cos t, f(\pi + t) + f(t) = 0, f(\pi + t) + f(-t) = -2b \sin t,$$

where $a = f(0)$, $b = f\left(\frac{\pi}{2}\right)$. Hence $f(t) = a \cos t + b \sin t$ and these functions satisfy the conditions.

Problem 8.2. Find all continuous functions $f, g : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x - y) = f(x)f(y) + g(x)g(y)$$

for all $x, y \in \mathbb{R}$.

Solution. Set $y = x$ to conclude that $f^2(x) + g^2(x) = c = f(0)$. In particular, for $x = 0$ we get $c^2 + g^2(0) = c$ so $0 \leq c \leq 1$. If $c < 1$ we set $y = 0$ to get $f(x) = cf(x) + g(0)g(x)$ so $g(x) = \frac{1-c}{g(0)}f(x)$. Then

$$c = f^2(x) + g^2(x) = f^2(x) \left(1 + \frac{(1-c)^2}{g^2(0)} \right) = f^2(x) \left(1 + \frac{(1-c)^2}{c - c^2} \right) = \frac{f^2(x)}{c}$$

and so $f^2(x) = c^2$. Thus $f(x) = \pm c$ and by continuity $f(x) = c, g(x) = \sqrt{c - c^2}$ or $g(x) = -\sqrt{c - c^2}$. We have left to investigate the case $c = 1$ in which case we get $f(0) = 1, g(0) = 0, f^2(x) + g^2(x) = 1$. This already suggests the sine-cosine formula. If we replace x by y in the given equation we get that f is even. If we set $y \rightarrow -y$ we get

$$f(x + y) = f(x)f(y) + g(x)g(-y)$$

and analogously $f(x + y) = f(x)f(y) + g(-x)g(y)$. Thus

$$g(x)g(-y) = g(y)g(-x)$$

so $g(-y) = g(y)\frac{g(-x)}{g(x)}$ if $g(x) \neq 0$. If $g(x) = 0$ for all x we immediately conclude that $f(x) = 1$. If $g(x_0) \neq 0$, then as

$$f^2(x_0) + g^2(x_0) = 1 = f^2(-x_0) + g^2(-x_0)$$

and f is even, we get $\frac{g(-x_0)}{g(x_0)} = \pm 1$. Hence

$$g(-y) = g(y)\frac{g(-x_0)}{g(x_0)} = \pm g(y)$$

for all $y \in \mathbb{R}$ and thus g is either even or odd. If g is even by setting $y \rightarrow -y$ we get $f(x+y) = f(x)f(y) + g(x)g(y) = f(x-y)$ and so f is constant. If f is not constant, then g is odd. We can now set $y \rightarrow -y$ to get

$$f(x+y) + f(x-y) = 2f(x)f(y).$$

This is D'Alembert's equation, and since the function $f(x)$ is continuous, it follows that $f(x) = \cos \alpha x$, $\alpha \in \mathbb{R}$. Now the given equation implies that $g(x)g(y) = \sin \alpha x \sin \alpha y$. Then either

$$g(x) = \sin \alpha x \text{ or } g(x) = -\sin \alpha x = \sin(-\alpha x).$$

Hence all continuous solutions of the given equation have the form

$$f(x) = \cos \alpha x, \quad g(x) = \sin \alpha x, \quad \alpha \in \mathbb{R}.$$

Problem 8.3. Find all continuous functions $f, g: \mathbb{R} \rightarrow \mathbb{R}$ that satisfy

$$f(x+y) + f(x-y) = 2f(x)g(y)$$

for all $x, y \in \mathbb{R}$.

Solution. This resembles D'Alembert's equation, but it's a generalized version of it. We try to reduce it to D'Alembert's equation. If f is identically zero, then g could be any function. Otherwise, if $f(x_0) \neq 0$ we get $f(x_0+y) + f(x_0-y) = 2f(x_0)g(y) = f(x_0-y) + f(x_0+y) = 2f(x_0)g(-y)$ so g is an even function. Now set $y = 0$ to get $g(0) = 1$. We now distinguish two cases:

a) $f(0) = 0$. In this case set $x = 0$ to get $f(y) + f(-y) = 0$ and so f is an odd function. Therefore

$$f(x+y) + f(x-y) = 2f(x)g(y)$$

and

$$f(x+y) - f(x-y) = f(y+x) + f(y-x) = 2g(x)f(y).$$

Hence

$$f(x+y) = f(x)g(y) + g(x)f(y), \quad f(x-y) = f(x)g(y) - g(x)f(y).$$

This resembles the formula for the sine of sum, with f for sine and g for cosine (this is confirmed by the oddness of f and evenness of g). We only lack the cosine of the sum formula, so let us replace x by $x + y$ into our second formula to get

$$f(x) = f(x+y)g(y) - g(x+y)f(y) = (f(x)g(y) + g(x)f(y))g(y) - g(x+y)f(y).$$

Hence

$$g(x+y) = g(x)g(y) - \frac{f(x)(1-g^2(y))}{f(y)}.$$

This is not really the formula we expected, but if we replace y by $-y$ we get

$$g(x-y) = g(x)g(y) + \frac{f(x)(1-g^2(y))}{f(y)}.$$

So $g(x+y) + g(x-y) = 2g(x)g(y)$ for $f(y) \neq 0$. Thus g can be found from D'Alembert's equation but we have a problem: if $f(y) = 0$, then the condition $g(x+y) + g(x-y) = 2g(x)g(y)$ may not hold. If there is a sequence y_n that tends to y and $f(y_n) \neq 0$, then it holds by continuity. Otherwise there is an interval I containing y such that $f \equiv 0$ on I . Let l be the length of I . If we denote by $X - Y$ the set $\{x - y | x \in X, y \in Y\}$, then the formula

$$f(x-y) = f(x)g(y) - f(y)g(x)$$

says that if $f \equiv 0$ on X and Y , then $f \equiv 0$ on $X - Y$. Hence $f \equiv 0$ on $I - I = (-l, l)$. Then $f \equiv 0$ on $(l, -l) - (l, -l) = (2l, -2l)$ and so on, $f \equiv 0$ on $(2^k l, -2^k l)$ for any k . So $f \equiv 0$ on $\mathbb{R}$, a contradiction. Therefore D'Alembert's equation is satisfied by g for all x and y . Hence $g(x) = \cos(ax)$ or $g(x) = \cosh(ax)$ for some a . Without loss of generality we may assume that $g(x) = \cos(ax)$; the second case being handled analogously. For $a = 0$ we have that $g = 1$ and we get that $f(x+y) = f(x) + f(y)$. Hence $f(x) = cx$, $c \in \mathbb{R}$. Now suppose that $a \neq 0$. If now $f(x) = b_x \sin(ax)$ for $f(x) \neq 0$, then we prove by induction on n that $f(nx) = b_x \sin(anx)$, because $c \sin(ax)$ and $\cos(ax)$ satisfy our initial condition. Now we claim $b_x = b_y$ for all x, y . If $\frac{x}{y} \in \mathbb{Q}$, then there is a z such that $x = mz, y = nz$ for $m, n \in \mathbb{N}$ and thus $b_x = b_z = b_y$.

If $\frac{x}{y} \notin \mathbb{Q}$, then there is a sequence x_n such that $\frac{x_n}{x} \in \mathbb{Q}, x_n \rightarrow y$. Then $b_{x_n} = x$ and using the continuity of f we get $f(y) = b_y \sin(ay)$; so $b_x = b_y$. We thus conclude that $f(x) = b_{x_0} \sin(ax)$ when $f(x) \neq 0$. If $f(x) = 0$, then there is a sequence $x_n \rightarrow x$ such that $f(x_n) \neq 0$ and by continuity we get $f(x) = b_{x_0} \sin(ax)$. Therefore $f(x) = b \sin(ax), g(x) = \cos(ax)$. It is clear they satisfy our condition. In the case $g(x) = \cosh(ax)$ we get analogously $f(x) = b \sinh(ax)$.

b) $f(0) \neq 0$. We employ the case a). Set

$$f^+(x) = f(x) + f(-x), \quad f^-(x) = f(x) - f(-x).$$

If f satisfies the condition then $-f$ also satisfies the condition. Hence so do f^+, f^- (the condition is linear in f). But f^- falls into case a). Therefore $g(x) = \cos(ax), f^-(x) = b \sin(ax)$ or $g(x) = \cosh(ax), f^-(x) = b \sinh(ax)$. We may also suppose $f^+(0) = 1$ because we can multiply or divide f by any constant we want. Now we have $f^+(x+y) + f^+(x-y) = 2f^+(x)g(y)$ and $f^+(x+y) + f^+(y-x) = 2g(x)f^+(y)$ if x, y change places. As f^+ is even, then $f^+(y-x) = f^+(x-y)$ and thus $f^+(x)g(y) = f^+(y)g(x)$. In particular, $f^+(x)g(0) = f^+(0)g(x)$. As $f^+(0) = g(0) = 1$ we have $f^+ = g$.

If we combine all the cases, we get

$$f(x) = bx + c, g(x) = 1, \quad f(x) = c \sin(ax) + d \cos(ax), \quad g(x) = \cos(ax);$$

$$f(x) = c \sinh(ax) + d \cosh(ax), g(x) = \cosh(ax).$$

All of these pairs of functions satisfy the condition.

Problem 8.4. Find all continuous functions $f, g, h, k: \mathbb{R} \rightarrow \mathbb{R}$ that satisfy $f(x+y) + g(x-y) = 2h(x)k(y)$ for all $x, y \in \mathbb{R}$.

Solution. This is a generalization of Problem 8.3. First, we try to eliminate g . Replace y by $-y$ to get $f(x-y) + g(x+y) = 2h(x)k(-y)$. Now subtracting these conditions and setting $u = f - g, l(x) = k(x) - k(-x)$ we get

$$u(x+y) - u(x-y) = 2h(x)l(y) \quad (*)$$

From here we may find u explicitly by using Theorem A from Section 8.2. Indeed, if we set $a_n = f(nx)$ then setting nx instead of x, x instead of y we get

$a_{n+1} - a_{n-1} = 2h(nx)l(x)$. If $l(x) = 0$ then $a_{2n} = a_0$ thus $f(2nx)$ is constant. If $l(x) \neq 0$ then taking $y = 2x$ we deduce $a_{n+2} - a_{n-2} = 2h(nx)l(2x)$ thus $a_{n+2} - a_{n-2} = b(a_{n+1} - a_{n-1})$, where $b = \frac{l(2x)}{l(x)}$. The characteristic polynomial of this recurrence is

$$x^4 - b(x^3 - x) - 1 = (x+1)(x-1)(x^2 - bx + 1).$$

If $b \neq \pm 2$, this polynomial has four different zeros $1, -1, w, \frac{1}{w}$. Thus

$$a_n = \alpha + \beta(-1)^n + \gamma w^n + \theta \frac{1}{w^n}.$$

So

$$a_{2n} = \alpha + \beta + \gamma w^n + \theta \frac{1}{w^n}.$$

If $b = 2$, the polynomial has a triple zero 1 and we have $a_n = p(n) + c(-1)^n$, where p is a polynomial of degree at most 2. So $u(2nx) = p(2nx)$ where p is some polynomial of degree at most 2. Finally, if $b = -2$, we get a triple zero -1 and a single zero -1 and similarly $u(nx) = p(2nx)$ where p is some polynomial of degree at most 2. Thus we can apply Theorem A to establish that either $u(x)$ is a polynomial of degree at most 2 or

$$u(x) = \alpha e^{ax} + \beta e^{-ax} + \gamma. \quad (1)$$

Now if we add the equalities in the beginning instead of subtracting them we get

$$v(x+y) + v(x-y) = 2h(x)j(y), \quad (**)$$

where $j(y) = k(y) + k(-y)$. We proceed exactly like in the previous problem to establish that v is either linear or $\alpha' e^{a'y} + \beta' e^{-a'y}$. If v is linear then so is h . Therefore u cannot be of form $\alpha e^{ax} + \beta e^{-ax}$ for $a \neq 0$ because then

$$l(y) = \frac{u(x+y) - u(x-y)}{2h(x)} = \frac{\alpha e^{ax} - \beta e^{ax}}{2h(x)}(e^{ay} - e^{-ay})$$

and does depend on x . So u is a polynomial of degree 2 and hence so are

$$f = \frac{u+v}{2}, \quad g = \frac{v-u}{x}.$$

If $f(x) = ax^2 + bx + c, g(x) = -ax^2 + dx + e$ (the coefficients of x^2 are complementary because they come only from u), then we have

$$\begin{aligned} f(x+y) - g(x-y) &= a((x+y)^2 - (x-y)^2) + b(x+y) + d(x-y) + c + e \\ &= 4axy + (b+d)x + (b-d)y + c + e. \end{aligned}$$

As h is linear, $h(x) = kx + l$, then $k(y)$ is also linear. Thus

$$k(x)h(y) = (kx + l)(my + n) = mnxy + knx + lmy + ln.$$

Therefore

$$a = \frac{mn}{4}, \quad b = \frac{kn + lm}{2}, \quad c = \frac{kn - lm}{2}, \quad c + e = ln.$$

Next suppose that v is of form $\alpha'e^{a'x} + \beta'e^{-a'x}$ then we conclude

$$h(x) = c(\alpha'e^{a'x} + \beta'e^{-a'x}).$$

In this case $a = a'$ or $a = -a'$ because then looking at (*) we get

$$k(y) = \frac{u(x+y) - u(x-y)}{h(x)}.$$

As k does not depend on x but h depends exponentially on $a'x$ we readily conclude that u must be of form $\alpha e^{ax} + \beta e^{-ax} + \gamma$ and then $a = \pm a'x$ in order for $h(x)$ to cancel in the expression for $k(y)$. We substitute u, v into f, g to get $f(x) = ke^{ax} + le^{-ax} + m, g(x) = re^{ax} + se^{-ax} - m$. Then

$$f(x+y) + g(x-y) = (ke^{ay} + re^{-ay})e^{ax} + (le^{-ay} + se^{ay})e^{-ax} = h(x)k(y).$$

If we avoid the trivial case $k = r = l = s = 0$, then h, k cannot be identically zero. Thus selecting $x = x_0$ with $h(x_0) \neq 0$ we can express $k(y)$ as

$$k_1 e^{ay} + k_2 e^{-ay},$$

and choosing $y = y_0$ with $k(y_0) \neq 0$ we can select

$$h(x) = h_1 e^{ax} + h_2 e^{-ax}.$$

Thus substituting we get

$$k = h_1 k_1, \quad l = k_2 h_2, \quad r = h_1 k_2, \quad s = h_2 k_1$$

to obtain the parametric representation for f, g, h, k .

Problem 8.5. Find all smooth functions $f : \mathbb{R} \rightarrow \mathbb{R}$ that satisfy

$$f(x+y) + 2f(x)f(y) = f'(x)f(y) + f(x)f'(y)$$

for any $x, y \in \mathbb{R}$.

Solution. We take a leaf out of the *Symmetrization and additional variables* section. In the case when $f(x)$ and $f(y)$ are non-zero one can divide by $f(x)f(y)$ and transform the condition into

$$\frac{f(x+y)}{f(x)f(y)} = \frac{f'(x)}{f(x)} + \frac{f'(y)}{f(y)} - 2$$

This immediately implies that

$$\frac{f(x+y)}{f(x)f(y)} + \frac{f(z+t)}{f(z)f(t)} = \frac{f'(x)}{f(x)} + \frac{f'(y)}{f(y)} + \frac{f'(z)}{f(z)} + \frac{f'(t)}{f(t)} - 4$$

whence by symmetry

$$\frac{f(x+y)}{f(x)f(y)} + \frac{f(z+t)}{f(z)f(t)} = \frac{f(x+t)}{f(x)f(t)} + \frac{f(y+z)}{f(y)f(z)}$$

and multiplying by $f(x)f(y)f(z)f(t)$ yields

$$f(x+y)f(z)f(t) + f(z+t)f(x)f(y) = f(x+t)f(y)f(z) + f(y+z)f(x)f(t)$$

One does not need $f(x), f(y), f(z), f(t)$ to be non-zero to obtain this identity; it is just more intuitive to obtain this way.

To obtain it in general, one just needs to avoid division. Instead, amplify the condition for x, y by $f(z)f(t)$ and add it with the condition for z, t multiplied by $f(x)f(y)$, and note that the RHS is symmetric.

One now replaces (x, y, z, t) by $(nx, x, x, -x)$ and the condition becomes

$$a_{n+1}a_1a_{-1} + a_na_{-1}a_0 = a_{n+1}a_1^2 + a_na_2a_{-1},$$

where $a_k = f(kx)$.

This is a quadratic recurrence; applying Theorem B means that

$$f(x) = p(x)e^{ax} + q(x)e^{bx}$$

for some a, b . Now note that because the recurrence is quadratic, either p, q are constant or $q = 0$ and p is at most linear; otherwise choosing a suitable x will make a_k too complicated to satisfy a quadratic recurrence.

We are therefore left to verify the two options.

First, we check $f(x) = c_1e^{ax} + c_2e^{bx}$ with $a \neq b, c_1, c_2 \neq 0$. Let's plug it into the condition. In the LHS, the coefficient of e^{ax+by} is $2c_1c_2$ whereas in the RHS, the coefficient of e^{ax+by} is $(a+b)c_1c_2$. Hence we obtain $a+b=2$. On the other hand, the coefficient of e^{ax+ay} in the LHS is $c_1+2c_1^2$ and in the RHS is $2ac_1^2$. Thus we get $c_1 = \frac{1}{2(a-1)}$ and similarly $c_2 = \frac{1}{2(b-1)}$. If we let $a = 1+s, b = 1-s$ with $s \neq 0$. We obtain

$$f(x) = \frac{e^{(1+s)x} - e^{(1-s)x}}{2s}$$

Conversely, every such function satisfies the condition because we worked by equivalence.

Now we check $f(x) = (ux+v)e^{ax}$ by plugging it into the condition. The coefficient of xye^{ax+ay} in the LHS is $2u^2$, while in the RHS it is $2u^2a$ hence either $a=1$ or $u=0$. If $u=0$ we get $f(x) = ve^{ax}$. We get

$$ve^{ax+ay} + 2v^2e^{ax+ay} = 2vae^{ax+ay}$$

so $2v^2 + v = 2va$. One the either gets $v=0$ in which case $f=0$ or $v = a - \frac{1}{2}$,

i.e. $f(x) = \left(a - \frac{1}{2}\right)e^{ax}$. Both satisfy the condition. If $a=1$ then

$$f(x) = (ux+v)e^x.$$

We get

$$(ux+uy+v)e^{x+y} + 2(ux+v)(uy+v)e^{x+y}$$

$$= (ux + u + v)(uy + v)e^{x+y} + (uy + u + v)(ux + v)e^{x+y}$$

hence $ux + uy + v = u(ux + v + uy + v)$ i.e. $v = 2uv, u = u^2$ so either $u = 0$ in which case we get the solution $f = 0$ obtained before or $u = 1$ in which case $v = 2v$ so $v = 0$ and we get $f(x) = xe^x$ which satisfies the condition.

In conclusion, the solutions are the following functions:

$$f(x) = 0, f(x) = \frac{e^{(1+s)x} - e^{(1-s)x}}{2s}, f(x) = \left(a - \frac{1}{2}\right)e^{ax}, f(x) = xe^x.$$

Problem 8.6. Find all continuous functions $f: \mathbb{R} \rightarrow \mathbb{R}$ that satisfy the following condition: whenever $x + y + z$ is an integer multiple of 2π ,

$$f^2(x) + f^2(y) + f^2(z) - 2f(x)f(y)f(z) = 1.$$

Solution. It is well-known that

$$\cos^2(u) + \cos^2(v) + \cos^2(w) - 2\cos u \cos v \cos w = 1$$

is $u + v + w$ is a multiple of 2π , so the cosine function satisfies the condition. We will next show every solution is related to the cosine function.

If we regard the above identity as a quadratic equation in $f(z)$, its discriminant is $4(f^2(x) - 1)(f^2(y) - 1)$. Since the equation has a solution, namely $f(z)$, it follows that the discriminant is non-negative and thus $|f(x)| \geq 1$ for all x , or $|f(x)| \leq 1$ for all x .

We now prove that $f(x) = f(x + 2\pi)$ for all x . Suppose this were not true so $f(x_0) \neq f(x_0 + 2\pi)$. If we pick y, z such that $x_0 + y + z$ is a multiple of 2π then it follows that $f(x_0), f(x_0 + 2\pi)$ satisfy the same quadratic equation, namely the equation $t^2 - 2f(y)f(z)t + f^2(y) + f^2(z) - 1 = 0$. This means that either $f(x_0) = f(x_0 + 2\pi)$ or $f(x_0) \neq f(x_0 + 2\pi)$ and in the second case the discriminant is strictly positive and $f(x_0) + f(x_0 + 2\pi) = 2f(y)f(z), f(x_0)f(x_0 + 2\pi) = f^2(y) + f^2(z) - 1$. In that case, the discriminant $4(f^2(y) - 1)(f^2(z) - 1)$ equals $(f(x_0) - f(x_0 + 2\pi))^2$. This is independent of the choice of y, z and hence we conclude that for all y, z such that $x_0 + y + z$ is a multiple of 2π we have $f(x_0) + f(x_0 + 2\pi) = 2f(y)f(z), f(x_0)f(x_0 + 2\pi) = f^2(y) + f^2(z) - 1$. This clearly shows $f(y)f(z), f^2(y) + f^2(z)$ is independent of y, z and hence

$\{f^2(y), f^2(z)\} = \{f^2(y_1), f^2(z_1)\}$ whenever $x_0 + y + z, x_0 + y_1 + z_1$ are integer multiples of 2π , by Vietta's relations. In particular, the set $\{f^2(y) \mid y \in \mathbb{R}\}$ has at most two elements, which means that f^2 is constant since the range of a continuous function is an interval. If $f^2 \equiv a$ then either $a = 0$ or $a > 0$ and then $f(x) = \pm\sqrt{a}$. Since $f(y)f(z)$ must be constant for all y, z with $x_0 + y + z = 2\pi$, the second case implies by continuity that $f \equiv \sqrt{a}$ or $f \equiv -\sqrt{a}$. In either case, f is constant and that contradicts the assumption that $f(x_0) \neq f(x_0 + 2\pi)$.

We have thus proved that f is periodic with period 2π .

Now let's plug in $x = y = z = 0$ and obtain that $f(0)$ satisfies the cubic equation $3t^2 - 2t^3 = 1$, whose roots are 1 and $-\frac{1}{2}$. We investigate these cases separately.

- If $f(0) = 1$ then we plug in $y = -z$ and obtain $(f(y) - f(-y))^2 = 0$ so f is an even function. In particular, letting since $f(x+y) = f(-x-y)$ and $x+y+(-x-y) = 0$ we deduce that $f(x+y)$ is a root of the equation $t^2 - 2f(x)f(y)t + f^2(x) + f^2(y) - 1 = 0$. The same equation is satisfied by $f(x-y)$ for the same reason (we use $f(y) = f(-y)$) and hence like before, either $f(x+y) = f(x-y)$ or $f(x+y) + f(x-y) = 2f(x)f(y)$. In particular, if we set $x = y$ we deduce that $f(2x) = 1$ or $f(2x) = 2f^2(x) - 1$. If $f(2x) = f(-2x) = 1$ then letting $y+z+2x$ be a multiple of 2π , we deduce as above that $f(y) = f(z)$ and this implies that $f(2x+y) = f(-2x-y) = f(y)$ so f is periodic of period $2x$. The set of periods of f is either everything or of form $\{na \mid n \in \mathbb{Z}\}$ for a the smallest positive non-zero period. In the first case, f is constant so $f \equiv 1$. In the second case, for $|x| < \frac{a}{2}$ we have $f(2x) \neq 1$ so $f(2x) = 2f^2(x) - 1$. Note also that $a = \frac{2\pi}{k}$ since 2π is a period of f as well. As in the proof of D'Alembert Equation, the identity $f(2x) = 2f^2(x) - 1$ on $(-\frac{a}{2}, \frac{a}{2})$ implies $f(x) = \cos(cx)$ or $\cosh(cx)$ for some fixed $c \in \mathbb{R}$. Since f is even, we deduce $f(x) = \cos(cx)$ and moreover c is a multiple of k since f is $\frac{2\pi}{k}$ -periodic. Using periodicity, this holds for all $x \in \mathbb{R}$. Such a function satisfies the conditions.

- If $f(0) = -\frac{1}{2}$ then we note that the function $\arccos$ is continuous on $[-1, 1)$ so the function $g(x) = \arccos(f(x))$ is continuous around 0; it makes sense because $|f(x)| \leq 1$ for all x since $|f(0)| < 1$ (see the beginning of the proof). It is also well-known that

$$\cos^2(u) + \cos^2(v) + \cos^2(w) - 2\cos(u)\cos(v)\cos(w) = 1$$

holds if and only if $w = \pm u \pm v$ (modulo 2π) because the solutions to the quadratic trinomial $\cos^2(u) + \cos^2(v) + t^2 - 2\cos(u)\cos(v)t = 1$ are $\cos(u \pm v)$. In particular, if $u + v + w$ is a multiple of 2π then so is $g(u) \pm g(v) \pm g(w)$. Since $g(0) = \frac{2\pi}{3}$, by continuity we conclude that around 0, $u + v + w = 0$ implies $g(u) + g(v) + g(w) = 2\pi$. In particular, on a small interval $(-2\epsilon, 2\epsilon)$ we conclude that

$$\begin{aligned} g(u+v) &= 2\pi - g(-u) - g(-v) = 2\pi - (2\pi - g(u) - g(0)) - (2\pi - g(v) - g(0)) \\ &= g(u) + g(v) - \frac{2\pi}{3} \end{aligned}$$

and continuity implies that $g(x) = kx + \frac{2\pi}{3}$.

We now prove by induction on n that

$$f(x) = \cos\left(kx + \frac{2\pi}{3}\right)$$

when $x \in (-n\epsilon, n\epsilon)$; the basis $n = 2$ is true. Say now this was true for $n = m$, and let's prove it for $n = m + 1$. Choose $m\epsilon \leq x < (m+1)\epsilon$; the case of negative x is similar. If we choose $u, v \in (0, m\epsilon)$ such that $u + v = x$ then setting $y = -u, z = -v$ we deduce that

$$f(x) \in \left\{ \cos\left(kx + \frac{2\pi}{3}\right), \cos(k(u-v)) \right\}.$$

Since the cosine function is not constant around any point, and the range of possible values of $u - v$ is a more than a point, $\cos(u - v) = f(x)$ cannot hold for all u, v for which $u + v = x$ which means $f(x) = \cos(kx + \frac{2\pi}{3})$. The induction step is finished. Therefore $f(x) = \cos(kx + \frac{2\pi}{3})$ for all $x \in \mathbb{R}$. Since f is 2π -periodic, k must be an integer. Conversely, such functions satisfy the conditions since $x + y + z$ is an integer multiple of 2π , so is

$$\left(kx + \frac{2\pi}{3}\right) + \left(ky + \frac{2\pi}{3}\right) + \left(kz + \frac{2\pi}{3}\right).$$

The set of solutions is

$$\{f(x) = \cos(kx) \mid k \in \mathbb{Z}\} \cup \left\{f(x) = \cos\left(kx + \frac{2\pi}{3}\right) \mid k \in \mathbb{Z}\right\}.$$

19.9 The Aczel-Golab-Schinzel Equation

Problem 9.1. (IMO 1968) Let $a > 0$ be a real number and $f(x)$ a real function defined on all of $\mathbb{R}$, satisfying for all $x \in \mathbb{R}$

$$f(x+a) = \frac{1}{2} + \sqrt{f(x) - f(x)^2}.$$

- (a) Prove that the function f is periodic;
- (b) Give an example of such a nonconstant function for $a = 1$.

Solution. (a) We shall show that $2a$ is a period of f . Indeed

$$\begin{aligned} f(x+2a) &= \frac{1}{2} + \sqrt{f(x+a)(1-f(x+a))} = \\ &= \frac{1}{2} + \sqrt{\left(\frac{1}{2} + \sqrt{f(x) - f(x)^2}\right)\left(\frac{1}{2} - \sqrt{f(x) - f(x)^2}\right)} = \\ &= \frac{1}{2} + \sqrt{\frac{1}{4} - f(x) + f(x)^2} = \frac{1}{2} + \sqrt{\left(f(x) - \frac{1}{2}\right)^2} = \frac{1}{2} + f(x) - \frac{1}{2} = f(x), \end{aligned}$$

since $f(x) \geq \frac{1}{2}$ for all $x \in \mathbb{R}$.

- (b) The following function satisfies the conditions of the problem:

$$f(x) = \begin{cases} 1/2 & \text{if } 2n \leq x < 2n+1 \\ 1 & \text{if } 2n+1 \leq x < 2n+2 \end{cases}$$

for $n = 0, 1, 2, \dots$

Problem 9.2. Can we express the function $f(x) = 2^x + 3^x + 9^x$ as the sum of finitely many periodic functions?

Solution. Suppose that $f = f_1 + f_2 + \dots + f_n$, where f_i has a period $t_i > 0$. Consider the sum

$$S = \sum_{\epsilon_j \in \{0,1\}} (-1)^{n-\sum_j \epsilon_j} f\left(\sum_j \epsilon_j t_j\right).$$

Then a direct computation shows that

$$S = \prod_j (2^{t_j} - 1) + \prod_j (3^{t_j} - 1) + \prod_j (9^{t_j} - 1) > 0.$$

on the other hand using the periodicity of f_i we get

$$\begin{aligned} S &= \sum_{\epsilon_j \in \{0,1\}} (-1)^{n - \sum_j \epsilon_j} f_i \left(\sum_j \epsilon_j t_j \right) \\ &= \sum_{\epsilon_j \in \{0,1\}, j \neq i} (-1)^{n - \sum_{j \neq i} \epsilon_j} f_i \left(\sum_{j \neq i} \epsilon_j t_j \right) - f_i \left(\sum_{j \neq i} \epsilon_j t_j + t_i \right) = 0, \end{aligned}$$

a contradiction.

Problem 9.3. Let k be a positive real number and $f : (0, \infty) \rightarrow (0, \infty)$ be a function such that

$$f(x)f(y) = kf(x + yf(x))$$

for all $x, y \in (0, \infty)$.

(a) Prove that f is an increasing function.

(b) Determine all functions satisfying the above functional equation.

Solution. (a) Suppose that $f(x) < f(z)$ for some $x > z > 0$. Then

$$y = \frac{x - z}{f(z) - f(x)} > 0$$

and $x + yf(x) = z + yf(z)$. Hence

$$f(x)f(y) = kf(x + yf(x)) = kf(z + yf(z)) = f(z)f(x),$$

i.e. $f(y) = f(z)$, a contradiction.

(b) Since $x + yf(x) > x$ it follows from (a) that

$$f(x)f(y) = kf(x + yf(x)) \geq kf(x),$$

i.e. $f(y) \geq k$ for all $y > 0$. Suppose that $f(y_0) = k$ for some $y_0 > 0$. Then

$$k^2 = f^2(y_0) = kf((k+1)y_0)$$

and we get that $f((k+1)y_0) = k$. Now it follows by induction on n that

$$f((k+1)^n y_0) = k$$

for all non-negative integers n . On the other hand $\lim_{n \rightarrow \infty} (k+1)^n y_0 = \infty$ and since f is an increasing function with $f(x) \geq k$ for all $x \in \mathbb{R}$ we conclude that $f \equiv k$.

Suppose now that $f(y) > k$ for all $y > 0$. If $z > x$ setting $y = \frac{z-x}{f(x)}$ we get that $kf(x) < f(x)f(y) = kf(x+yf(x)) = kf(z)$, which shows that f is a strictly increasing function. Then the identity

$$kf(x+f(x)) = f(x)f(1) = f(1)f(x) = kf(1+xf(1))$$

implies that $x+f(x) = 1+xf(1)$, i.e. $f(x) = cx+1$, where $c > 0$. Now it is easy to check that such a function is a solution of the given equation only for $k = 1$. Thus the desired functions are the following: $f(x) = k$ if $k \neq 1$ and $f(x) = cx+1, c \geq 0$ if $k = 1$.

Problem 9.4. Let $a > 0, a \neq 1$ and $b \neq 0$ be real numbers. Find all functions $f : [0, \infty) \rightarrow (0, \infty)$ such that

$$f(x)f(y) = af(x+yf(x)^b)$$

for all $x, y \in [0, \infty)$.

Solution. Setting $x = y = 0$ we get $f(0) = a$. Now setting $x = 0$ gives $f(y) = f(a^b \cdot y)$. On the other hand one can show as in Problem 9.3 that f is an increasing function. Since $a^b \neq 1$ we conclude that $f \equiv a$.

Problem 9.5. Given a positive integer n find as many nonconstant continuous solutions $f : \mathbb{R} \rightarrow \mathbb{R}$ as you can to the functional equation

$$f(x)f(y) = f(x+yf(x)^n)$$

as you can.

Solution. It is easy to check that if n is odd then the functions

$$f(x) = (cx+1)^{\frac{1}{n}} \text{ and } f(x) = (\max(cx+1, 0))^{\frac{1}{n}},$$

where $c \neq 0$ are nonconstant continuous solutions and if n is even then the functions of the second class are solutions. It can be proved [7] that these are the only nonconstant continuous solutions of the given functional equation.

19.10 Arithmetic Functional Equations

Problem 10.1. Find all non-decreasing functions $f: \mathbb{Z} \rightarrow \mathbb{Z}$ which satisfy

$$f(k) + f(k+1) + \dots + f(k+n-1) = k$$

for all $k \in \mathbb{Z}$, and a fixed $n \in \mathbb{N}$.

Solution. Subtracting the condition for k from the condition for $k+1$ we get $f(k+n) = f(k)+1$. Therefore f is determined by its values on $\{0, 1, \dots, n-1\}$ and the relation $f(k) = \lfloor \frac{k}{n} \rfloor + f(k \bmod n)$. As f is non-decreasing and $f(n) = f(0)+1$, we see that there is $0 \leq m \leq n-1$ such that $f(0) = f(1) = \dots = f(m)$, $f(0)+1 = f(m+1) = \dots = f(n)$. Now by writing the condition for $k=0$ we get $nf(0) + (n-m-1) = 0$. Hence n divides $m+1$ which implies $m = n-1$ thus $f(0) = f(1) = \dots = f(n-1) = 0$. It is now clear that $f(k) = \lfloor \frac{k}{n} \rfloor$. This value clearly satisfies the condition, as it is a consequence of Hermite's Identity

$$\lfloor x \rfloor + \left\lfloor x + \frac{1}{n} \right\rfloor + \dots + \left\lfloor x + \frac{n-1}{n} \right\rfloor = \lfloor nx \rfloor.$$

Note that we have also proven the identity by induction during the proof.

Problem 10.2. (IMO 2012) Find all functions $f: \mathbb{Z} \rightarrow \mathbb{Z}$ such that

$$f(x)^2 + f(y)^2 + f(z)^2 = 2f(x)f(y) + 2f(y)f(z) + 2f(z)f(x)$$

for all integers x, y, z that satisfy $x + y + z = 0$.

Solution. Setting $x = y = z = 0$ gives $f(0) = 0$. Then for $z = 0$ and $y = -x$ we get $f(x) = f(-x)$. Now the given condition can be written as

$$f(x)^2 + f(y)^2 + f(x+y)^2 = 2f(x)f(y) + 2f(x+y)(f(x) + f(y)), \quad x, y \in \mathbb{Z}.$$

Let $f(1) = k$. If $k = 0$, then, putting $y = 1$, it follows that $f(x+1) = f(x)$, i.e. $f \equiv 0$.

Let $k \neq 0$. For $x = y$ we get $f(2x) = 0$ or $f(2x) = 4f(x)$. In particular, $f(2) = 0$ or $f(2) = 4k$.

If $f(2) = 0$, then for $y = 2$ we obtain $f(x) = f(x + 2)$. Hence

$$f(x) = \begin{cases} 0, & x - \text{even} \\ k, & x - \text{odd}. \end{cases}$$

Let $f(2) = 4k$. If $f(4) = 0$, then, plugging $y = 4$, we have $f(x + 4) = f(x)$. In particular, $f(3) = f(-1) = f(1) = k$. Therefore

$$f(x) = \begin{cases} 0, & x = 4x_1 \\ k, & x - \text{odd} \\ 4k & x = 4x_1 + 2 \end{cases}$$

Finally, let $f(4) = 4f(2) = 16k$. Then the substitutions $x = 1, y = 2$ and $x = 1, y = 3$ lead to $f(3) \in \{k, 9k\} \cap \{9k, 25k\} = 9k$ (since $k \neq 0$). Hence $f(n) = kn^2$, $n = 1, 2, 3$. Assume that $f(n) = kn^2$ for some integer $n \geq 3$. The substitutions $x = 1, y = n, x = 2, y = n - 1$ show that

$$f(n + 1) \in \{k(n + 1)^2, k(n - 1)^2\} \cap \{k(n + 1)^2, k(n - 3)^2\} = k(n + 1)^2$$

(since $k \neq 0$ and $n \neq 2$). It follows by induction that $f(x) = kx^2$ for any $x \in \mathbb{N}$. Since f is an even function and $f(0) = 0$, we conclude that $f(x) = kx^2$ for any $x \in \mathbb{Z}$.

Direct verifications show that all the functions we have found satisfy the given condition.

Problem 10.3. Find all functions $f: \mathbb{N} \rightarrow [1; \infty)$ for which we have $f(2) = 4$, $f(mn) = f(m)f(n)$ and $\frac{f(n)}{n} \leq \frac{f(n+1)}{n+1}$ for all $m, n \in \mathbb{N}$.

Solution. It is clear that the function g defined by $g(n) = \frac{f(n)}{n}$ is increasing and multiplicative. Also $g(2) = \frac{f(2)}{2} = 2$ and it follows from Example 10.7 that g is the identity function. Thus $f(n) = n^2$ for all $n \in \mathbb{N}$.

Problem 10.4. Is there a strictly increasing function $f: \mathbb{N} \rightarrow \mathbb{N}$ such that $f(2) = 3$ and $f(mn) = f(m)f(n)$ for all $m, n \in \mathbb{N}$?

Solution. No! Suppose that there is a function with the given properties. Set $f(3) = k$. Then $3^3 = f^3(2) = f(2^3) < f(3^2) = k^2$ and we get that $k > 5$. On the other hand $2^5 > 3^3$ and therefore $3^5 = f^5(2) = f(2^5) > f(3^3) = k^3$.

Hence $k < 7$ and we conclude that $k = 6$. Now $f(6561) = f(3^8) = 6^8$ and $f(8192) = f(2^{13}) = 3^{13}$. Thus $6^8 < 3^{13}$ and we get $256 = 2^8 < 3^5 = 243$, a contradiction.

Problem 10.5. (AMM 2001) Find all completely multiplicative functions $f : \mathbb{N} \rightarrow \mathbb{C}$ such that the function

$$F(n) = \sum_{k=1}^n f(k)$$

is also completely multiplicative.

Solution. There are two such functions: the function that is identically 1, and the function f such that $f(1) = 1$ and $f(n) = 0$ for $n \geq 2$. For $k > 1$, we have $f(2k) = f(2)f(k)$ and

$$\begin{aligned} f(2k-1) &= F(2k) - F(2k-2) - f(2k) = F(2)(F(k) - F(k-1)) - f(2k) \\ &= (1 + f(2))f(k) - f(2)f(k) = f(k). \end{aligned}$$

Therefore, each value $f(n)$ is a power of $f(2)$. Furthermore,

$$f(2) = f(3) = f(5) = f(9) = f(3)^2 = f(2)^2.$$

Thus $f(2) \in \{0, 1\}$, and the result follows.

Problem 10.6. (AMM 1998) Find all functions $f : \mathbb{N}^2 \rightarrow \mathbb{N}$ that satisfy

$$f(n, n) = n, \quad f(m, n) = f(n, m) \quad \text{and} \quad \frac{f(m, n+m)}{f(m, n)} = \frac{n+m}{n}$$

for all $m, n \in \mathbb{N}$.

Solution. We can immediately see the algorithm of finding f , because the three conditions follow exactly the three possible steps of the Euclidean Algorithm. Indeed, following the Euclidean Algorithm by means of the second and third identity we shall reach from (m, n) the pair (d, d) , where $d = \gcd(m, n)$.

Also we note that all three steps preserve the quantity $\frac{f(m, n)}{mn}$. Therefore

$$\frac{f(m, n)}{mn} = \frac{f(d, d)}{d^2} = \frac{d}{d^2} = \frac{1}{d}.$$

So $f(m, n) = \frac{mn}{d} = \text{lcm}(m, n)$. Indeed, lcm satisfies trivially the first two conditions. For the third one we use the well-known identities $\text{gcd}(m, n+m) = \text{gcd}(m, n)$ and $\text{gcd}(m, n)\text{lcm}(m, n) = mn$. Thus

$$\frac{\text{lcm}(m, n+m)}{\text{lcm}(m, n)} = \frac{n+m}{n}.$$

Problem 10.7. Let $f: \mathbb{N}_0 \rightarrow \mathbb{N}_0$ be a function such that $f(0) = 1$ and

$$f(n) = f\left(\left\lfloor \frac{n}{2} \right\rfloor\right) + f\left(\left\lfloor \frac{n}{3} \right\rfloor\right)$$

whenever $n \in \mathbb{N}$. Show that $f(n-1) < f(n)$ if and only if $n = 2^k 3^h$ for some $k, h \in \mathbb{N}_0$.

Solution. The solution is by induction (recall f is non-decreasing by the same induction). The basis for $n \leq 6$ is easy to check. Now let us perform the induction step. For $\lfloor \frac{n}{2} \rfloor$ and $\lfloor \frac{n}{3} \rfloor$, the residue of n modulo 6 matters. So we distinguish 6 cases:

a) $n = 6k$. Then $f(n) = f(2k) + f(3k)$ while $f(n-1) = f(2k-1) + f(3k-1)$. So $f(n-1) < f(n)$ if and only if $f(2k-1) < f(2k)$ or $f(3k-1) < f(3k)$ thus $2k$ or $3k$ is of form $2^i 3^j$, which is equivalent to $n = 6k$ being of the same form.

b) $n = 6k + 1$. In this case n is not of form $2^i 3^j$, and

$$f(n-1) = f(n) = f(2k) + f(3k).$$

c) $n = 6k + 2$. Then $f(n) = f(3k+1) + f(2k)$ while $f(n-1) = f(3k) + f(2k)$ and $f(n-1) < f(n)$ if and only if $3k+1$ is of form $2^i 3^j$, which is equivalent to $n = 6k + 2$ being of the same form.

d) $n = 6k + 3$. Then $f(n) = f(3k+1) + f(2k+1)$ and

$$f(n-1) = f(3k+1) + f(2k),$$

so $f(n-1) < f(n)$ if and only if $f(2k) < f(2k+1)$ or $2k+1 = 2^i 3^j$, which is equivalent to $6k+3 = 2^i 3^{j+1}$.

e) $n = 6k + 4$. We have

$$f(n) - f(n-1) = (f(3k+2) + f(2k+1)) - (f(3k+1) + f(2k+1))$$

$$= f(3k+2) - f(3k+1)$$

which is possible if and only if $3k+2$ is of form $2^i 3^j$, or the same condition for $n = 2(3k+2)$.

f) $n = 6k+5$. Like in case b) we have $f(n) = f(n-1)$ and n is not of the desired form, since it's neither even nor divisible by 3.

The induction is finished.

Problem 10.8. Find all functions $f: \mathbb{N} \rightarrow \mathbb{Z}$ verifying

$$f(mn) = f(m) + f(n) - f(\gcd(m, n))$$

for all $m, n \in \mathbb{N}$.

Solution. Again we smell some dependency between f and the prime decomposition of n . Let us set $m = p^k$, $n = p$, $k > 0$, where p is a prime. Then $f(p^{k+1}) = f(p^k) + f(p) - f(p) = f(p^k)$, so $f(p^k) = f(p)$ for any $k > 0$. Next consider two co-prime numbers m, n . Then $f(mn) = f(m) + f(n) - f(1)$ and we easily deduce the more general version

$$f(m_1 m_2 \dots m_k) = f(m_1) + f(m_2) + \dots + f(m_k) - (k-1)f(1)$$

when $m_1, m_2, \dots, m_k$ are pairwise co-prime. Particularly if

$$n = \prod_{i=1}^m p_i^{k_i}$$

(p_i are distinct primes) then

$$f(n) = \sum_{i=1}^m f(p_i^{k_i}) - (m-1)f(1) = \sum_{i=1}^m f(p_i) - (m-1)f(1).$$

A more comfortable setting is obtained if we work with $f(x) - f(1)$. Indeed, if we denote $g(n) = f(n) - f(1)$, then we get $g(n) = \sum_{p|n} g(p)$, where the sum is taken over all prime divisors of n . Note that any function g with this property is uniquely determined by its values at prime numbers. Conversely,

let $f(n) = g(n) + f(1)$, where g has the above property. It is straightforward to check that this function satisfies the condition: If

$$m = \prod_{i=1}^k p_i^{k_i} \prod_{i=1}^l q_i^{m_i}, \quad n = \prod_{i=1}^j r_i^{j_i} \prod_{i=1}^l q_i^{n_i},$$

where p_i, q_i, r_i are distinct primes and $k_i, m_i, j_i, n_i > 0$ then

$$\gcd(m, n) = \prod_{i=1}^l q_i^{\min\{m_i, n_i\}}$$

and we have

$$f(mn) = \sum_{i=1}^k g(p_i) + \sum_{i=1}^l g(q_i) + \sum_{i=1}^j g(r_i) + f(1),$$

$$f(m) = \sum_{i=1}^k g(p_i) + \sum_{i=1}^l g(q_i) + \sum_{i=1}^j g(r_i) + f(1),$$

$$f(n) = \sum_{i=1}^j g(r_i) + \sum_{i=1}^l g(q_i) + g(r_i) + f(1),$$

$$f(\gcd(m, n)) = \sum_{i=1}^l g(q_i) + f(1).$$

Hence

$$f(mn) = f(m) + f(n) - f(\gcd(m, n)).$$

Remark. To obtain an example of a function f satisfying the condition of Problem 10.8 take the function g defined on the primes by $g(2) = 1$ and $g(p) = 0$ if $p \geq 3$. Then it follows that $g(n) = 1$ if n is even and $g(n) = 0$ if n is odd. Setting $f(1) = 0$ we get the following function: $f(n) = 1$ if n is even and $f(n) = 0$ if n is odd.

Problem 10.9. Find all functions $f: \mathbb{N} \rightarrow \mathbb{N}$ such that $m^2 + f(n)$ divides $f(m)^2 + n$ for all $m, n \in \mathbb{N}$.

Solution. Since $n^2 + f(n) | f(n)^2 + n$ it follows that $f(n)^2 + n \geq n^2 + f(n)$ which implies that $f(n) \geq n$. Now take arbitrary $p, q \in \mathbb{N}$ and let

$$n > \max(f(p)^2 - 2p^2, f(q)^2 - 2q^2).$$

Then

$$f(p)^2 + n < 2(p^2 + n) \leq 2(p^2 + f(n))$$

and since $p^2 + f(n) | f(p)^2 + n$ we get $f(p)^2 + n = p^2 + f(n)$. Analogously $f(q)^2 + n = q^2 + f(n)$ and therefore $f(p)^2 - p^2 = f(q)^2 - q^2$ for all $p, q \in \mathbb{N}$. So $f(n) = \sqrt{n^2 + a}$ for all $n \in \mathbb{N}$, where a is a nonnegative integer. But $f(n) \in \mathbb{N}$ for all $n \in \mathbb{N}$ and we conclude easily that $a = 0$, i.e. $f(n) = n$ which indeed is a solution.

Problem 10.10. Let p be a prime number. Find all functions $f: \mathbb{Z} \rightarrow \mathbb{Z}$ that satisfy:

- a) if p divides $m - n$ then $f(m) = f(n)$;
- b) $f(mn) = f(m)f(n)$

for all $m, n \in \mathbb{Z}$.

Solution. We see that $f(n)$ takes one of the p values $f(0), f(1), \dots, f(p-1)$. Therefore we can pick up an a for which $|f(a)|$ has the maximal value. Then $|f(a^2)| = |f(a)|^2 \leq |f(a)|$. Thus $|f(a)| \leq 1$ and hence $f(n) \in \{-1, 0, 1\}$. Now if we set $m = 0$ into b), we get $f(0)(f(n) - 1) = 0$. So either $f \equiv 1$ (which is a solution) or $f(0) = 0$ and hence $f(n) = 0$ whenever n is divisible by p . If $f(k) = 0$ for k not divisible by p , then for any n we can find l such that $p | kl - n$. So $f(n) = f(kl) = f(k)f(l) = 0$, i.e. $f \equiv 0$ which is a solution. Assume that $f \not\equiv 0$. Then $f(n) \neq 0$ if n is not divisible by p . Now if n is a quadratic residue modulo p , then there is a b such that $p | n - b^2$ so $f(n) = f(b^2) = f(b)^2 = 1$. Next pick up a quadratic non-residue r . Then for every n which is a non-quadratic residue modulo p we can find b such that $p | n - rb^2$. Hence $f(n) = f(rb^2) = f(r)f(b)^2 = f(r)$. If $f(r) = 1$ then $f(n) = 1$ for all n not divisible by p . If $f(r) = -1$ then f is 1 for quadratic residues and -1 for quadratic non-residues.

In conclusion, there are four solutions to our problem: $f \equiv 1$, $f \equiv 0$, $f(n) = 1$ for n not divisible by p , and $f(n) = 0$ for n divisible by p , and the Legendre symbol [22].

Problem 10.11. (Iran TST 2005) Find all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ such that:

- 1) there exist $k \in \mathbb{N}$ and a prime p such that $f(n+p) = f(n)$ for all $n \geq k$;
- 2) if m divides n then $f(m+1)$ divides $f(n)+1$.

Solution. (by Omid Hatami) Suppose $n \geq k$ and p doesn't divide $n-1$. Then there is k that $n-1 \mid n+kp$. So $f(n) \mid f(n+kp)+1$ but we know that $f(n) = f(n+kp)$, Therefore $f(n) \mid 1$ and $f(n) = 1$. Consider an arbitrary $n \neq 1$, We know that $n-1 \mid (n-1)kp$, So $f(n) \mid f((n-1)kp)+1 = 2$. So for every $n \neq 1$, $f(n) \in \{1, 2\}$. Now we have two cases:

a) $f(n) = 2, \forall n \geq k$ and $p \mid n-1$. Consider n such that p does not divide $n-1$. There is m such that $n-1 \mid m$ and $p \mid m-1$. So $f(n) \mid f(m)+1 = 3$ and $f(n) = 1$. Hence $f(n) = 1$ for all $n \geq k$ or m doesn't divide n and is arbitrarily defined for every $n < k$ and $p \mid n-1$.

b) $f(n) = 1, \forall n \geq k$ and $p \mid n-1$. In this case $f(n) = 1, \forall n \geq k$, and if we set $S = \{a \mid f(a) = 2\}$ then it doesn't exist $m, n \in S$ such that $m-1 \mid n$. So all functions with the problem properties are defined like this: Suppose S is a finite set of positive integers such that there doesn't exist $m, n \in S$ with $m-1 \mid n$. Then for $n > 1$, $f(n) = 2$ iff $n \in S$ and $f(1)$ is arbitrarily defined with the condition that $f(2) \mid f(1)+1$.

Problem 10.12. (IMO 2011) Let $f : \mathbb{Z} \rightarrow \mathbb{N}$ be a function such that $f(m-n)$ divides $f(m) - f(n)$ for all integers m, n . Prove that, for all integers m and n with $f(m) \leq f(n)$, the number $f(n)$ is divisible by $f(m)$.

Solution. Set $n = 0$ and $m = 0$ to get, respectively that $f(m) \mid f(0)$ and $f(-n) \mid f(n)$ for all $m, n \in \mathbb{Z}$. Thus $f(n) \mid f(-n)$ and $f(-n) \mid f(n)$ which shows that $f(n) = f(-n)$ for all $n \in \mathbb{Z}$. Now we get from the initial condition that $f(m-n) \mid f(m) - f(n)$, $f(m) \mid f(n) - f(n-m)$ and $f(n) \mid f(m) - f(m-n)$ for all $m, n \in \mathbb{Z}$. Hence denoting $\{a, b, c\} = \{f(n), f(m), f(m-n)\}$ such that $a \geq b \geq c$ we see that the three divisibilities are equivalent to $a \mid b-c$, $b \mid c-a$ and $c \mid a-b$. But $a > b-c \geq 0$ (since $a \geq b, c > 0$) and therefore $b = c$. Thus $b \mid a$ which shows that if $f(m) \geq f(n)$ then either $f(m) = f(n)$ or $f(n) \mid f(m)$.

Problem 10.13. (Iran TST 2008) Let k be a positive integer. Find all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ such that $f(m) + f(n)$ divides $(m+n)^k$ for all $m, n \in \mathbb{N}$.

Solution. The first step is to prove that f is injective. Note that if $f(a) = f(b)$, then $\gcd(a+n, b+n) \neq 1$ for all n , as $(n+a)^k, (n+b)^k$ are both multiples of $f(a) + f(n)$. But then for all primes $p > b$ we have $\gcd(a-b+p, p) \neq 1$, so $p|a-b$ and so $a=b$. Next, we show that $f(a+1) - f(a) = \pm 1$ for all a . Suppose we can find a and a prime p that divides $f(a+1) - f(a)$. Choose l such that $p^l > k$ and observe that $f(p^l - b) + f(b) | p^{lk}$, so $p | f(b) + f(p^l - b)$. But then $p | f(b+1) + f(p^l - b) | (p^l + 1)^k$, a contradiction. Combining the first two steps with the fact that f always takes positive values (and so cannot decrease forever), we deduce that $f(a+1) - f(a) = 1$ for all a and so $f(n) = n + f(1) - 1$. Then we get $2(f(1) - 1) + m + n | (m+n)^k$ for all m, n and by choosing $m+n$ a large prime we get $f(1) = 1$ and so f is the identity.

Problem 10.14. (IMO Shortlist 2011) Let $n \geq 1$ be an odd integer. Determine all functions $f : \mathbb{Z} \rightarrow \mathbb{Z}$ such that $f(x) - f(y)$ divides $x^n - y^n$ for all $x, y \in \mathbb{Z}$.

Solution. (by P. Mebane) We shall show that there exists a positive integer d dividing n , an integer a , and an $e = \pm 1$ such that $f(x) = e \cdot x^d + a$ for all x . It can be easily checked that all of these solutions work.

First, notice that if $f(x)$ is a solution, then $f(x) + a$ is a solution for any a . So, we may assume that $f(0) = 0$. Now take $x = 1, y = 0$ in the given equation. We get $f(1) | 1$, so $f(1) = \pm 1$. If $f(x)$ is a solution, then $-f(x)$ is a solution, so we may assume that $f(1) = 1$. We also get $f(-1) | 1$ by taking $x = -1$ and $y = 0$. Now take $x = -1$ and $y = 1$. We get $f(-1) - 1 | 2$, so $f(-1) = -1$.

Consider any odd prime p . We have $f(p) - f(0) | p^n$, so $f(p)$ is either p^d or $-p^d$ for some d . First consider the case that $f(p) = -p^d$. We have $f(p) - f(-1) | p^n + 1$, so $p^d - 1 | p^n + 1$. If we write $n = qd + r$ using the division algorithm, we get $p^d - 1 | p^r + 1$. We know $0 \leq r < d$, so $0 < p^r + 1 < p^d - 1$ since $p > 2$, a contradiction. So this case is impossible.

Then we must have $f(p) = p^d$. Again write $n = qd + r$ using the division algorithm. We get $p^d - 1 | (-1)^q p^r - 1$. Since $0 \leq r < d$, this is only possible if $r = 0$. Therefore, d divides n .

Let b be an arbitrary integer, and let q be a prime number greater than $b^n + 2|f(b)|^n$. We know $f(q) = q^d$ for some d dividing n . Consider $x = b$ and

$y = q$. We get $f(b) - q^d \mid b^n - q^n$. Writing q^n as $(q^d)^{n/d}$, we get

$$f(b) - q^d \mid b^n - (f(b))^{n/d}.$$

Suppose that $b^n - (f(b))^{n/d}$ is nonzero. Then we know that

$$|b^n - (f(b))^{n/d}| \leq b^n + |f(b)|^n < q - |f(b)|^n \leq q^d - f(b).$$

This is a contradiction, so therefore $b^n - (f(b))^{n/d} = 0$, and $f(b) = b^d$. If we apply this process to two arbitrary integers b, c , taking the same large q for both of them, we obtain that $f(b) = b^d$ and $f(c) = c^d$ for the same d . Therefore there exists a single d dividing n such that $f(x) = x^d$ for all integers x . Reversing our transformations in the second paragraph, we get the set of solutions described in the first paragraph, so we are done.

Problem 10.15. (MR 2009, N. Tung) Let k be a positive integer. Find all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ such that $f(a) + f(b)$ divides $a^k + b^k$ for all $a, b \in \mathbb{N}$.

Solution. (by G. Dospinescu) Clearly $f(a) \mid a^k$ and this implies that $f(1) = 1$ and if p is a prime then $f(p) = p^{\alpha_p}$ for some $\alpha_p \geq 0$. But $f(p) + f(1) \mid p^k + 1 \Rightarrow p^{\alpha_p} + 1 \mid p^k + 1$. Suppose that $\alpha_p > 0$ and let $k = q\alpha_p + r, 0 \leq r < \alpha_p$. Then $p^{\alpha_p} + 1 \mid (p^{\alpha_p})^q \cdot p^r + 1$ and therefore $p^{\alpha_p} + 1 \mid (-1)^q p^r + 1$. But $|(-1)^q p^r + 1| < p^{\alpha_p} + 1$ and this implies $(-1)^q p^r + 1 = 0$. Hence $r = 0$ and q is odd. So $\frac{k}{\alpha_p}$ is

an odd number. Keep p fixed and write $f(a) + p^{\alpha_p} \mid a^k + p^k, \forall a \geq 1$. Since $\frac{k}{\alpha_p}$ is odd we get

$$a^k + p^k \equiv a^k + (p^{\alpha_p})^{\frac{k}{\alpha_p}} \equiv a^k + (-f(a))^{\frac{k}{\alpha_p}} \equiv a^k - f(a)^{\frac{k}{\alpha_p}} \pmod{(f(a) + p^{\alpha_p})}$$

so that $p^{\alpha_p} + f(a) \mid a^k - f(a)^{\frac{k}{\alpha_p}}$ and this is for all primes p and all $a \geq 1$.

Lemma. For all p we have $f(p) \geq p$, i.e. $\alpha_p > 0$.

Proof of lemma. Suppose that $f(p) = 1$ for some p . Then $f(a) + 1 \mid a^k + 1$ and $f(a) + 1 \mid a^k + p^k, \forall a$. So $f(a) + 1 \mid p^k - 1$ for all $a \in \mathbb{N}$ and therefore f is bounded. But $f(2^{2^n}) + 1 \mid 2^{k \cdot 2^n} + 1$ and any prime factor of $2^{k \cdot 2^n} + 1$ is of the form $l2^{n+1} + 1$, a contradiction. $\square$

Now fix $a \in \mathbb{N}$ and take $p > a^k + f(a)^k$. As $p^{\alpha_p} + f(a) \mid a^k - f(a)^{\frac{k}{\alpha_p}}$ then $a^k = f(a)^{\frac{k}{\alpha_p}}$, i.e. $f(a) = a^{\alpha_p}$ if p is large enough. So the sequence $(\alpha_p)_p$ is eventually constant, say $\alpha_p = d$ for $p \geq p_0$ and some d such that $\frac{k}{d}$ is odd. We saw above that $\forall a \geq 1$, $f(a) = a^{\alpha_p} = a^d$ and these are the solutions.

Problem 10.16. (IMO Shortlist 2004) A function $f: \mathbb{N} \rightarrow \mathbb{N}$ is such that for all $m, n \in \mathbb{N}$ the number $(m^2 + n)^2$ is divisible by $f(m)^2 + f(n)$. Prove that $f(n) = n$ for all $n \in \mathbb{N}$.

Solution. For $m = n = 1$ we obtain that $f(1)^2 + f(1)$ divides $(1^2 + 1)^2 = 4$ showing that $f(1) = 1$. Next, we prove that $f(p - 1) = p - 1$ for each prime p . By the hypothesis for $m = 1$ and $n = p - 1$ we get that $f(p - 1) + 1$ divides p^2 , so $f(p - 1)$ equals either $p - 1$ or $p^2 - 1$. If $f(p - 1) = p^2 - 1$, then $f(1) + f(p - 1)^2 = p^4 - 2p^2 + 2$ divides $(1 + (p - 1)^2)^2 < p^4 - 2p^2 + 2$, a contradiction. Hence $f(p - 1) = p - 1$. Now for an arbitrary $n \in \mathbb{N}$ it follows that $A = f(n) + (p - 1)^2$ divides $(n + (p - 1)^2)^2 \equiv (n - f(n))^2 \pmod{A}$, hence A divides $(n - f(n))^2$ for any prime p . Taking p large enough, we can obtain A to be greater than $(n - f(n))^2$, which implies that $f(n) = n$ for every n .

Problem 10.17. (IMO Shortlist 1996) Find a bijection $f: \mathbb{N}_0 \rightarrow \mathbb{N}_0$ that satisfies

$$f(3mn + m + n) = 4f(m)f(n) + f(m) + f(n)$$

for all $m, n \in \mathbb{N}_0$.

Solution. If we denote $g(3k + 1) = f(k)$, then the condition becomes

$$g((3m + 1)(3n + 1)) = 4g(3m + 1)g(3n + 1) + g(3m + 1) + g(3n + 1).$$

Next if we set $4g(x) + 1 = h(x)$ the condition is rewritten as

$$h((3m + 1)(3n + 1)) = h(3m + 1)h(3n + 1).$$

Hence we have to construct a multiplicative bijection h of A into B , where

$$A = \{3k + 1 \mid k \in \mathbb{N}\}, B = \{4k + 1 \mid k \in \mathbb{N}\}.$$

We set $h(1) = 1$ and consider the set U of all primes of form $3k - 1$, the set V of all primes of form $3k + 1$, the set X of all primes of form $4k - 1$ and

the set Y of all primes of form $4k + 1$. All these four sets are infinite but countable. So we can provide a bijection h between U and X and between V and Y . Then we extend it by multiplicativity to the whole A . We shall prove this is the required bijection. Indeed, assume that $3k + 1 = \prod p_i^{a_i} \prod q_i^{b_i}$ where $p_i \in U, q_i \in V$. Then p_i are $-1 \pmod{3}$, q_i are $1 \pmod{3}$, so $\sum a_i$ must be even. Then $h(3k + 1) = \prod h(p_i)^{a_i} \prod h(q_i)^{b_i}$ where $h(p_i) \in X, h(q_i) \in Y$. As $h(p_i)$ is $-1 \pmod{4}$, $h(q_i)$ is $1 \pmod{4}$ and $\sum a_i$ is even we conclude that $h(3k + 1)$ is $1 \pmod{4}$, so $h(3k + 1) \in B$. We can analogously prove the reverse implication. Assume that $4k + 1 = \prod p_i^{a_i} \prod q_i^{b_i}$, where $p_i \in X, q_i \in Y$. As p_i are $-1 \pmod{4}$ but q_i are $1 \pmod{4}$, the sum $\sum a_i$ must be even. Then $x = \prod h^{-1}(p_i)^{a_i} \prod h^{-1}(q_i)^{b_i}$ satisfies $h(x) = 4k + 1$. Moreover as $h^{-1}(p_i)$ are $-1 \pmod{3}$, $h^{-1}(q_i)$ are $1 \pmod{3}$ and $\sum a_i$ is even, we conclude that x is $1 \pmod{3}$, so $x \in A$. Finally h is injective because of the uniqueness of the decomposition of a positive integer into product of primes.

Problem 10.18. (IMO 2010) Determine all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ such that $(f(m) + n)(m + f(n))$ is a perfect square for all $m, n \in \mathbb{N}$.

Solution. We shall prove that the solutions are all functions of the form $f(n) = n + c$, where c is a nonnegative integer. First, it is clear that any such a function is a solution and we have to prove that there are no other solutions. To do this we start with the following

Lemma. Suppose that $p \mid f(k) - f(l)$ for some prime p and positive integers k, l . Then $p \mid k - l$.

Proof of lemma. Suppose first that $p^2 \mid f(k) - f(l)$, so $f(l) = f(k) + p^2 a$ for some integer a . Take some positive integer $D > \max\{f(k), f(l)\}$ which is not divisible by p and set $n = pD - f(k)$. Then the positive integers $n + f(k) = pD$ and $n + f(l) = pD + (f(l) - f(k)) = p(D + pa)$ are both divisible by p but not by p^2 . Now, applying the problem conditions, we get that both numbers $(f(k) + n)(f(n) + k)$ and $(f(l) + n)(f(n) + l)$ are perfect squares divisible by p and thus by p^2 . This means that the multipliers $f(n) + k$ and $f(n) + l$ are also divisible by p , therefore $p \mid (f(n) + k) - (f(n) + l) = k - l$. On the other hand, if $f(k) - f(l)$ is divisible by p but not by p^2 , then choose the same number D and set $n = p^3 D - f(k)$. Then the positive integers $f(k) + n = p^3 D$ and $f(l) + n = p^3 D + (f(l) - f(k))$ are respectively divisible by p^3 (but not by p^4) and by p (but not by p^2). Hence in analogous way we obtain that the numbers

$f(n) + k$ and $f(n) + l$ are divisible by p , therefore $p \mid k - l$. Thus the lemma is proved. $\square$

We turn to the problem. First, suppose that $f(k) = f(l)$ for some $k, l \in \mathbb{N}$. Then by the lemma $k - l$ is divisible by every prime, so $k = l$. Therefore the function f is injective. Next, consider the numbers $f(k)$ and $f(k + 1)$. Since the number $(k + 1) - k = 1$ has no prime divisors, by Lemma the same holds for $f(k + 1) - f(k)$, i.e. $|f(k + 1) - f(k)| = 1$. Now, let $f(2) - f(1) = q, |q| = 1$. Then we prove by induction that $f(n) = f(1) + q(n - 1)$. The base for $n = 1, 2$ holds by the definition of q . For the step, if $n > 1$ we have $f(n + 1) = f(n) \pm q = f(1) + q(n - 1) \pm q$. Since $f(n) \neq f(n - 2) = f(1) + q(n - 2)$, we get $f(n) = f(1) + qn$, as desired. Finally, we have $f(n) = f(1) + q(n - 1)$. Then q cannot be -1 since otherwise for $n \geq f(1) + 1$ we have $f(n) \leq 0$ which is impossible. Hence $q = 1$ and $f(n) = (f(1) - 1) + n$ for each $n \in \mathbb{N}$ and $f(1) - 1 \geq 0$, as desired.

Problem 10.19. (IMO Shortlist 2008) For every $n \in \mathbb{N}$ let $\tau(n)$ denote the number of (positive) divisors of n . Find all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ with the following properties:

- (i) $\tau(f(x)) = x$;
- (ii) $f(xy)$ divides $(x - 1)y^{xy-1}f(x)$

for all $x, y \in \mathbb{N}$.

Solution. There is a unique solution: the function $f : \mathbb{N} \rightarrow \mathbb{N}$ defined by $f(1) = 1$ and

$$f(n) = p_1^{p_1^{\alpha_1}-1} p_2^{p_2^{\alpha_2}-1} \cdots p_k^{p_k^{\alpha_k}-1},$$

where $n = p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k}$ is the prime factorization of $n > 1$. Direct verification shows that this function meets the requirements.

Conversely, let $f : \mathbb{N} \rightarrow \mathbb{N}$ satisfy the conditions of the problem. Then (i) for $x = 1$ gives $\tau(f(1)) = 1$, so $f(1) = 1$. Notice that (i) implies also that the function f is injective. Next we shall use also the well-known formula

$$\tau(p_1^{\alpha_1} p_2^{\alpha_2} \cdots p_k^{\alpha_k}) = (\alpha_1 + 1)(\alpha_2 + 1) \cdots (\alpha_k + 1).$$

Let p be a prime. Since $\tau(f(p)) = p$, the formula just mentioned yields $f(p) = q^{p-1}$ for some prime q ; in particular $f(2) = q^{2-1} = q$ is a prime. We prove that $f(p) = p^{p-1}$ for all primes p . Suppose that p is odd and

$f(p) = q^{p-1}$ for some prime q . Applying (ii) first with $x = 2, y = p$ and then with $x = p, y = 2$ shows that $f(2p)$ divides both $(2-1)p^{2p-1}f(2) = p^{2p-1}f(2)$ and $(p-1)2^{2p-1}f(p) = (p-1)2^{2p-1}q^{p-1}$. If $q \neq p$ then the odd prime p does not divide $(p-1)2^{2p-1}q^{p-1}$, hence the greatest common divisor of $p^{2p-1}f(2)$ and $(p-1)2^{2p-1}q^{p-1}$ is a divisor of $f(2)$. Thus $f(2p)$ divides $f(2)$ which is a prime. As $f(2p) > 1$, we obtain $f(2p) = f(2)$, a contradiction to the injectivity of f . So, $q = p$, i.e. $f(p) = p^{p-1}$. For $p = 2$ the same argument with $x = 2, y = 3$ and $x = 3, y = 2$ shows that $f(6)$ divides both $3^5f(2)$ and $2^6f(3)$. If the prime $f(2)$ is odd then $f(6)$ divides $3^2 = 9$, so $f(6) = 1, 3$ or 9 . But then $6 = \tau(f(6)) = \tau(1), \tau(3)$ or $\tau(9)$ which is a contradiction since $\tau(1) = 1, \tau(3) = 2$ and $\tau(9) = 3$. In conclusion $f(2) = 2$. Now we show that for each $n > 1$ the prime divisors of $f(n)$ are among the ones of n . Indeed, let p be the least prime divisor of n . Apply (ii) with $x = p$ and $y = n/p$ to obtain that $f(n)$ divides $(p-1)y^{n-1}f(p) = (p-1)y^{n-1}p^{p-1}$. Write $f(n) = lP$, where l is co-prime to n and P is a product of primes dividing n . Since l divides $(p-1)y^{n-1}p^{p-1}$ and is co-prime to $y^{n-1}p^{p-1}$, it divides $p-1$. Hence $\tau(l) \leq l < p$. But (i) gives $n = \tau(f(n)) = \tau(lP) = \tau(l)\tau(P)$ as l and P are co-prime. Therefore $\tau(l)$ is a divisor of n less than p , meaning that $l = 1$ and proving the claim. If p is a prime and $a \geq 1$, by the above, the only prime factor of $f(p^a)$ is p , so $f(p^a) = p^b$ for some $b \geq 1$. Thus (i) yields $p^a = \tau(f(p^a)) = \tau(p^b) = b+1$, i.e. $f(p^a) = p^{p^a-1}$. Now consider the general case when $n = p_1^{\alpha_1} p_2^{\alpha_2} \dots p_k^{\alpha_k}$. We saw that the prime factorization of $f(n)$ has the form $f(n) = p_1^{\beta_1} p_2^{\beta_2} \dots p_k^{\beta_k}$. For $i = 1, 2, \dots, k$ set $x = p_i^{\alpha_i}$ and $y = n/x$ in (ii) to infer that $f(n)$ divides $(p_i^{\alpha_i} - 1)y^{n-1}f(p_i^{\alpha_i})$. Hence $p_i^{\beta_i}$ divides $f(p_i^{\alpha_i}) = p_i^{p_i^{\alpha_i}-1}$ since it is co-prime to $(p_i^{\alpha_i} - 1)y^{n-1}$. So $b_i \leq p_i^{\alpha_i} - 1$ for all $i = 1, 2, \dots, k$. Combined with (i), these conclusions imply

$$\begin{aligned} p_1^{\alpha_1} p_2^{\alpha_2} \dots p_k^{\alpha_k} &= n = \tau(f(n)) = \tau(p_1^{\beta_1} p_2^{\beta_2} \dots p_k^{\beta_k}) \\ &= (\beta_1 + 1)(\beta_2 + 1) \dots (\beta_k + 1) \leq p_1^{\alpha_1} p_2^{\alpha_2} \dots p_k^{\alpha_k}. \end{aligned}$$

Hence all inequalities $b_i \leq p_i^{\alpha_i} - 1$ must be equalities, implying that the only solution is that given above.

Problem 10.20 (IMO Shortlist 2007) Find all surjective functions $f : \mathbb{N} \mapsto \mathbb{N}$ such that for every $m, n \in \mathbb{N}$ and every prime p , the number $f(m+n)$ is

divisible by p if and only if $f(m) + f(n)$ is divisible by p .

Solution. We shall show that $f(n) = n$. Suppose that f satisfies the problem conditions.

Lemma. For each prime p and all $x, y \in \mathbb{N}$, we have $x \equiv y \pmod{p}$ iff $f(x) \equiv f(y) \pmod{p}$. Moreover, $p \mid f(x)$ iff $x \mid p$.

Proof. Consider an arbitrary prime p . Since f is surjective, there exists some $x \in \mathbb{N}$ such that $p \mid f(x)$. Let

$$d = \min(x \in \mathbb{N} : p \mid f(x)).$$

By induction on k , we obtain that $p \mid f(kd)$ for all $k \in \mathbb{N}$. The base is true since $p \mid f(d)$. If $p \mid f(kd)$ and $p \mid f(d)$, then by the problem condition, $p \mid f(kd + d) = f((k+1)d)$ as required. Suppose that there exists an $x \in \mathbb{N}$ such that $d \nmid x$ but $p \mid f(x)$. Let

$$y = \min(x \in \mathbb{N} : d \nmid x, p \mid f(x)).$$

By the choice of d , we have $y > d$, and $y - d$ is a positive integer not divisible by d . Then $p \nmid f(y - d)$, while $p \mid f(d)$ and $p \mid f(d + (y - d)) = f(y)$. This contradicts the problem condition. Hence there is no such x , and $p \mid f(x)$ iff $d \mid x$. Take arbitrary $x, y \in \mathbb{N}$ such that $x \equiv y \pmod{d}$. We have $p \mid f(x + (2xd - x)) = f(2xd)$; moreover, since $d \mid 2xd + (y - x) = y + (2xd - x)$, we get $p \mid f(y + (2xd - x))$. Then by the problem condition $p \mid f(x) + f(2xd - x)$, $p \mid f(y) + f(2xd - x)$, and hence $f(x) \equiv -f(2xd - x) \equiv f(y) \pmod{p}$. Assume now that $f(x) \equiv f(y) \pmod{p}$. Again we have $p \mid f(x) + f(2xd - x)$ which by our assumptions implies that $p \mid f(x) + f(2xd - x) + (f(y) - f(x)) = f(y) + f(2xd - x)$. Hence by the problem condition $p \mid f(y) + f(2xd - x)$ and $0 \equiv y + (2xd - x) \equiv y - x \pmod{d}$. Thus we have proved that $x \equiv y \pmod{d}$ iff $f(x) \equiv f(y) \pmod{p}$ and we are left to show that $p = d$. To do this note first that the numbers $f(1), f(2), \dots, f(d)$ have distinct residues modulo p , hence $p \geq d$. On the other hand, f is surjective, so there exist $x_1, x_2, \dots, x_p \in \mathbb{N}$ such that $f(x_i) = i$ for $i = 1, 2, \dots, p$. Hence all these x_i 's have distinct residues modulo d which implies that $d \geq p$. Thus $d = p$ which proves the lemma.

Now we prove that $f(n) = n$ by induction on n . If $n = 1$ then by the lemma $p \nmid f(1)$ for all primes p , so $f(1) = 1$. Suppose now that $n > 1$ and denote

$k = f(n)$. Note that there is a prime $q \mid n$, so by the lemma $q \mid k$ and $k > 1$. If $k > n$ then $k - n + 1 > 1$, and there exists a prime $p \mid k - n + 1$. We have $k \equiv n - 1 \pmod{p}$. By the induction hypothesis we have $f(n - 1) = n - 1 \equiv k = f(n) \pmod{p}$. Now, by the lemma we obtain $n - 1 \equiv n \pmod{p}$, a contradiction. Analogously, if $k < n$, then $f(k - 1) = k - 1$ by the induction hypothesis. Moreover, $n - k + 1 > 1$, so there is a prime $p \mid n - k + 1$ and $n \equiv k - 1 \pmod{p}$. By the lemma we get again that $k = f(n) \equiv f(k - 1) = k - 1 \pmod{p}$, which is false. Thus $k = n$, so $f(n) = n$. The function $f(n) = n$ obviously satisfies the conditions of the problem.

19.11 Binary and Other Bases

Problem 11.1. Find all functions $f: \mathbb{N}_0 \rightarrow \{0, 1\}$ for which

$$f(0) = 0, f(2n) = f(n), f(2n + 1) = 1 - f(2n)$$

for all $n \in \mathbb{N}_0$.

Hint. Show that $f(n)$ is the sum of the binary digits of n modulo 2.

Problem 11.2. Find all functions $f: \mathbb{N} \rightarrow \mathbb{N}$ for which

$$f(1) = 1, f(2n) = 2f(n) - 1, f(2n + 1) = 2f(n) + 1$$

for all $n \in \mathbb{N}$.

Hint. Let $n = \overline{1a_{k-1} \dots a_1 a_0}_{(2)}$ be the binary representation of n . Show that $f(n) = \overline{a_{k-1} \dots a_1 a_0}_{(2)}$.

Problem 11.3. Find all functions $f: \mathbb{N} \rightarrow \mathbb{R}$ for which

$$f(2n + 1) = f(2n) + 1 = 3f(n) + 1$$

for all $n \in \mathbb{N}$.

Solution. The problem is pretty much solved if we look at the binary representation of n . Let $f(1) = r$ and $n = \overline{1a_{k-1} \dots a_1 a_0}_{(2)}$. Then it is easy to prove by induction on k that $f(n) = r3^k + a_{k-1}3^{k-1} + \dots + 3a_1 + a_0$.

Problem 11.4. (IMO 1978) Find all functions $f: \mathbb{N} \rightarrow \mathbb{N}$ that satisfy

$$f(1) = 1, f(3) = 3 \text{ and } f(2n) = f(n),$$

$$f(4n+1) = 2f(2n+1) - f(n) \text{ and } f(4n+3) = 3f(2n+1) - 2f(n)$$

for all $n \in \mathbb{N}$.

Solution. Again the function is uniquely determined, and the key to finding it should be the binary representation of n . By direct computation we get that $f(1) = 1$, $f(2) = 1$, $f(3) = 3$, $f(4) = 1$, $f(5) = 5$, $f(6) = 3$, $f(7) = 7$, $f(8) = 1$, $f(9) = 9$, $f(10) = 5$, $f(11) = 13$. Up to $f(11)$ we could conjecture that $f(n)$ is obtained from n by deleting the zeroes at the end. However if we write 11 in base 2 we get $11 = 1011_2$ and $f(11) = 13 = 1101_2$. So it's natural to suppose that $f(n)$ is obtained by reversing the digits of n . Indeed, this is confirmed by all the previously computed values of f , since all numbers less than 11, if we delete their last zeroes, become palindromes. This claim is easy to verify by strong induction. Indeed assume that it is true for all numbers less than $k-1$ and let us prove it for k . Of course we need to consider three cases according to the three cases of the condition:

a) k is even. In this case $f(k) = f\left(\frac{k}{2}\right)$ and the claim follows from the induction step.

b) k is of form $4n+1$. Assume that n has r digits, and let $m = f(n)$ be obtained from n by reversing its binary digits. Then the number obtained from k by reversing its digits is $2^{r+1} + m$. Also $f(2n+1) = 2^r + m$ so $f(k) = 2f(2n+1) - f(n) = 2^{r+1} + 2m - m = 2^{r+1} + m$ and the claim holds in this case.

c) k is of form $4k+3$. Again assuming that n has k digits, and $m = f(n)$ then the inverse of k is $2^{r+1} + 2^r + m$, while

$$f(k) = 3f(2n+1) - 2f(n) = 3(2^r + m) - 2m = 2^{r+1} + 2^r + m,$$

and this case is true, too.

Problem 11.5. Find all strictly increasing functions $f : \mathbb{N} \rightarrow \mathbb{N}$ such that $f(f(n)) = 3n$ for all $n \in \mathbb{N}$.

Solution. We have $f(f(1)) = 3$ and so $f(1) \neq 1$. Hence $f(1) > 1$ and then $f(f(1)) > f(1)$. So $f(1) < 3$ and thus $f(1) = 2$. It follows that $f(2) = 3$. Then $f(3) = f(f(2)) = 6$ and we get by induction on k that $f(3^k) = 2 \cdot 3^k$, $f(2 \cdot 3^k) = 3^{k+1}$. Then $f(3) = 6$, $f(6) = 9$ and as f is increasing we get $f(4) = 7$, $f(5) = 8$.

Thus $f(7) = 12, f(8) = 15$. We then come to the following conjecture: If $3^k \leq n < 3^{k+1}$, then $f(n) = n + 3^k$ for $n \leq 2 \cdot 3^k$, $f(n) = 3(n - 3^k)$ for $n \geq 2 \cdot 3^k$. Indeed, this function is increasing and the assumption holds for $k = 1$. We now reason by induction. Assume that it holds for $k = m - 1$ and let us prove it for $k = m$. Assume $3^m \leq n < 2 \cdot 3^m$. If $n = 3s$, then $f(n) = f(3s) = f(f(f(s))) = 3f(s) = 3(s + 3^{m-1}) = n + 3^m$ by the induction hypothesis. If $3s < n < 3s + 3$ then $f(3s) < f(3s + 1) < f(3s + 2) < f(3s + 3)$. But we have proven $f(3s) = 3^m + 3s, f(3s + 3) = 3^m + 3s + 3$. As f is increasing, we can only have $f(3s + 1) = 3^m + 3s + 1, f(3s + 2) = 3^m + 3s + 2$ and so $f(n) = n + 3^m$. If now $2 \cdot 3^m \leq n < 3^{m+1}$, then $3^m \leq n - 3^m < 2 \cdot 3^{m+1}$. Thus $f(n - 3^m) = n$ so $f(n) = f(f(n - 3^m)) = 3(n - 3^m)$. The induction step is proven and so we have found f .

Note that if we use base 3 then $f(\overline{1a_0a_1 \dots a_{k(3)}}) = \overline{2a_0a_1 \dots a_{k(3)}}$ and

$$f(\overline{2a_0a_1 \dots a_{k(3)}}) = \overline{1a_0a_1 \dots a_{k(3)}0}_{(3)}.$$

Problem 11.6. Let a and b be nonnegative integers with $a > b$. Prove that there is a function $f : \mathbb{N} \rightarrow \mathbb{N}$ such that $f(f(n)) = an + b$ for all $n \in \mathbb{N}$.

Solution. If $a = 1$, then $b = 0$ and the function $f(n) = n$ satisfies the condition. If $a > 1$ then any positive integer n can be written in base a as

$$n = \dots c_n d_n \underbrace{b \dots bb}_{k_n},$$

where $k_n \geq 0, c_n, d_n \in [0, a), d_n \neq b$. The set of ordered pairs $\{(c_n, d_n) | n \in \mathbb{N}\}$ has $a(a - 1)$ elements which is an even number. Hence we can divide it into ordered pairs (c_n, d_n) and (c'_n, d'_n) which are called respectively *left neighbor* and *right neighbor*. Define a function $f : \mathbb{N} \rightarrow \mathbb{N}$ in the following way:

$$f(n) = \begin{cases} \overline{\dots c'_n d'_n \underbrace{b \dots bb}_{k_n}}, & \text{if } (c_n, d_n) \text{ is a left neighbor of } (c'_n, d'_n); \\ \overline{\dots c'_n d'_n \underbrace{b \dots bb}_{k_n+1}}, & \text{if } (c_n, d_n) \text{ is a right neighbor of } (c'_n, d'_n). \end{cases}$$

We shall prove that this function satisfies the condition of the problem. Write n in base a as $n = \dots c_n d_n \underbrace{b \dots bb}_{k_n}$.

- If (c_n, d_n) is a left neighbor of (c'_n, d'_n) , then (c'_n, d'_n) is a right neighbor of (c_n, d_n) and

$$\begin{aligned} f(f(n)) &= f(f(\overline{\dots c_n d_n \underbrace{b \dots bb}_{k_n}})) = f(\overline{\dots c'_n d'_n \underbrace{b \dots bb}_{k_n}}) \\ &= \overline{\dots c_n d_n \underbrace{b \dots bb}_{k_n+1}} = an + b. \end{aligned}$$

- If (c_n, d_n) is a right neighbor of (c'_n, d'_n) , then (c'_n, d'_n) is a left neighbor of (c_n, d_n) and

$$\begin{aligned} f(f(n)) &= f(f(\overline{\dots c_n d_n \underbrace{b \dots bb}_{k_n}})) = f(\overline{\dots c'_n d'_n \underbrace{b \dots bb}_{k_n+1}}) \\ &= \overline{\dots c_n d_n \underbrace{b \dots bb}_{k_n+1}} = an + b. \end{aligned}$$

Hence f has the desired property.

Remark. If a and b are arbitrary positive integers then a function $f : \mathbb{N} \rightarrow \mathbb{N}$ with $f(f(n)) = an + b$ may not exist, see Example 7.1.

Problem 11.7. (IMO 1978) Let $(f(n))$ be a strictly increasing sequence of positive integers: $0 < f(1) < f(2) < f(3) < \dots$. Of the positive integers not belonging to the sequence, the n -th in order of magnitude is $f(f(n)) + 1$. Determine $f(240)$.

First Solution. Since the n^{th} missing number (gap) is $f(f(n)) + 1$ and $f(f(n))$ is a member of the sequence, there are exactly $n - 1$ gaps less than $f(f(n))$. This leads to

$$f(f(n)) = f(n) + n - 1. \quad (1)$$

Since 1 is not a gap, we have $f(1) = 1$. The first gap is $f(f(1)) + 1 = 2$. Two consecutive integers cannot both be gaps (the predecessor of a gap is of the form $f(f(m))$). Hence $f(2) = 3$ and a repeated application of (1) gives

$$f(3) = 3 + 1 = 4, \quad f(4) = 4 + 2 = 6, \quad f(6) = 9, \quad f(9) = 14,$$

$$f(14) = 22, f(22) = 35, f(35) = 56, f(56) = 90,$$

$$f(90) = 145, f(145) = 234, f(234) = 378.$$

Also, $f(f(35)) + 1 = 91$ is a gap, so $f(57) = 92$. Then by (1),

$$f(92) = 148, f(148) = 239, f(239) = 386.$$

Finally, $f(f(148)) + 1 = 387$ is a gap, so $f(240) = 388$.

Second Solution. As above we arrive at formula (1). Then by simple induction it follows that $f(F_n + 1) = F_{n+1} + 1$, where F_k is the Fibonacci sequence ($F_1 = F_2 = 1$).

We now prove by induction on n that $f(F_n + x) = F_{n+1} + f(x)$ for all x with $1 \leq x \leq F_{n-1}$. This is trivially true for $n = 0, 1$. Supposing it holds for $n - 1$, we shall prove it for n . If $x = f(y)$ for some y , then by the inductive assumption and (1) we get

$$\begin{aligned} f(F_n + x) &= f(F_n + f(y)) = f(f(F_{n-1} + y)) \\ &= F_n + f(y) + F_{n-1} + y - 1 = F_{n+1} + f(x). \end{aligned}$$

If $x = f(f(y)) + 1$ is a gap, then

$$f(F_n + x - 1) + 1 = F_{n+1} + f(x - 1) + 1$$

is a gap also and

$$\begin{aligned} F_{n+1} + f(x) + 1 &= F_{n+1} + f(f(f(y))) + 1 \\ &= f(F_n + f(f(y))) + 1 = f(f(F_{n-1} + f(y))) + 1. \end{aligned}$$

Hence

$$f(F_n + x) = F_{n+1} + f(x - 1) + 2 = F_{n+1} + f(x).$$

We know that each positive integer x can be written as

$$x = F_{k_1} + F_{k_2} + \cdots + F_{k_r},$$

where $0 < k_r \neq 2, k_i \geq k_{i+1} + 2$. Thus

$$f(x) = F_{k_1+1} + F_{k_2+1} + \cdots + F_{k_r+1}.$$

In particular, $240 = 233 + 5 + 2 = F_{13} + F_5 + F_3$ and therefore

$$f(240) = F_{14} + F_6 + F_4 = 377 + 8 + 3 = 388.$$

Remark. It can be shown as in Problem 13.1 that $f(x) = \lfloor cx \rfloor$, where

$$c = \frac{\sqrt{5} + 1}{2}.$$

Problem 11.8. (IMO 1993) Determine whether there exists a strictly increasing function $f : \mathbb{N} \rightarrow \mathbb{N}$ such that $f(1) = 2$ and $f(f(n)) = f(n) + n$ for all $n \in \mathbb{N}$.

Hint. Consider the base determined by the Fibonacci type sequence $\{F_n\}_{n=0}^\infty$ defined by

$$F_0 = 1, F_1 = 2, F_{n+1} = F_n + F_{n-1}, n \geq 1.$$

Then define a function $f : \mathbb{N} \rightarrow \mathbb{N}$ in the following way. If

$$n = F_{i_1} + F_{i_2} + \dots + F_{i_k}$$

then

$$f(n) = F_{i_1+1} + F_{i_2+1} + \dots + F_{i_k+1}.$$

This function is well defined since the above representation of n is unique. Prove that it satisfies the problem conditions.

Remark. Problem 11.8 is considered also in Chapter 13 (Problem 13.1), where a different solution is given.

Problem 11.9. The function f is defined on the set of rational numbers in the interval $[0, 1]$ as follows:

$$f(0) = 0, f(1) = 1, f(x) = \begin{cases} f(2x)/4 & \text{if } x \in (0, 1/2) \\ 3/4 + f(2x - 1)/4 & \text{if } x \in [1/2, 1) \end{cases}$$

Find $f(x)$ in terms of the binary representation of x .

Solution. Let $x = 0.x_1x_2\dots$ be the binary representation of x . If $x_1 = 0$ then $x < \frac{1}{2}$ and

$$f(x) = 0.x_1x_1 + \frac{1}{4}f(0.x_2x_3\dots).$$

If $x_1 = 1$, then $a \geq \frac{1}{2}$ and

$$f(x) = 0.x_1x_1 + \frac{1}{4}f(0.x_2x_3\dots).$$

Continuing in the same way we get $f(x) = 0.x_1x_1x_2x_2\dots$.

Problem 11.10. (IMO Shortlist 1983) Let $f : [0, 1] \rightarrow \mathbb{R}$ be a continuous function that satisfies

$$f(x) = \begin{cases} bf(2x) & \text{if } x \in [0, 1/2] \\ b + (1-b)f(2x-1) & \text{if } x \in [1/2, 1] \end{cases},$$

where $b = \frac{1+c}{2+c}$, $c > 0$. Show that $0 < f(x) - x < c$ for all $x \in (0, 1)$.

Solution. First suppose that the binary representation of x is finite:

$$x = 0.a_1a_2\dots a_n = \sum_{j=1}^n a_j 2^{-j},$$

where $a_j = 0$ or 1 . We shall prove by induction on n that

$$f(x) = \sum_{j=1}^n b_0\dots b_{j-1}a_j,$$

where

$$b_k = \begin{cases} b & \text{if } a_k = 0 \\ 1-b & \text{if } a_k = 1 \end{cases}.$$

(Here $a_0 = 0$.) Indeed, by the recursive relation, if $a_1 = 0$ then

$$f(x) = bf\left(\sum_{j=1}^{n-1} a_{j+1}2^{-j}\right) = b \sum_{j=1}^{n-1} b_1\dots b_j a_{j+1}$$

hence

$$f(x) = \sum_{j=0}^{n-1} b_0\dots b_j a_{j+1}$$

as $b_0 = b_1 = b$. If $a_1 = 1$ then

$$f(x) = b + (1-b)f\left(\sum_{j=1}^{n-1} a_{j+1}2^{-j}\right) = \sum_{j=0}^{n-1} b_0 \dots b_j a_{j+1},$$

as $b_0 = b, b_1 = 1-b$.

Clearly, $f(0) = 0, f(1) = 1, f(1/2) = b > 1/2$. For $k \geq 2$ set

$$v = x + 2^{-n-k+1} \text{ and } u = x + 2^{-n-k} = (v+x)/2.$$

Then

$$f(v) = f(x) + b_0 \dots b_n b^{k-2} \text{ and } f(u) = f(x) + b_0 \dots b_n \cdot b^{k-1} > (f(v) + f(x))/2.$$

This means that the point $(u, f(u))$ lies above the line joining $(x, f(x))$ and $(v, f(v))$. By induction every point $(x, f(x))$, where x has a finite binary expansion, lies above the line joining $(0, 0)$ and $(1/2, b)$ if $0 < x < 1/2$, or above the line joining $(1/2, b)$ and $(1, 1)$ if $1/2 < x < 1$. It follows immediately that $f(x) > x$. For the second inequality, observe that

$$\begin{aligned} f(x) - x &= \sum_{j=1}^{\infty} (b_0 \dots b_{j-1} - 2^{-j}) a_j \\ &< \sum_{j=1}^{\infty} (b^j - 2^{-j}) a_j < \sum_{j=1}^{\infty} (b^j - 2^{-j}) = \frac{b}{1-b} - 1 = c. \end{aligned}$$

By continuity, these inequalities also hold for x with infinite binary representation.

19.12 Geometric Functional Equations

Problem 12.1. Find all continuous functions $f: \mathbb{R}^n \rightarrow \mathbb{R}$ that satisfy the identity

$$f(x_1, x_2, \dots, x_n) + f(y_1, y_2, \dots, y_n) = f(x_1 + y_1, \dots, x_n + y_n)$$

for all $(x_1, x_2, \dots, x_n), (y_1, y_2, \dots, y_n) \in \mathbb{R}^n$.

Solution. If we set $f_i(x) = f(0, 0, \dots, 0, x, 0, \dots, 0)$, where x is in the i -th position, then $f(x_1, x_2, \dots, x_n) = f_1(x_1) + f_2(x_2) + \dots + f_n(x_n)$. Now as f is additive so are f_i (just set $x_k = 0, y_k = 0$ for $k \neq i$ to get $f_i(x_i) + f_i(y_i) = f_i(x_i + y_i)$). Moreover, since f is continuous, so are f_i . Therefore $f_i(x) = c_i x$. Hence $f(x_1, x_2, \dots, x_n) = c_1 x_1 + \dots + c_n x_n$ and this function clearly satisfies the condition.

Problem 12.2. Let $n \geq 3$ be a positive integer. Find all continuous functions $f: [0; 1] \rightarrow \mathbb{R}$ for which $f(x_1) + f(x_2) + \dots + f(x_n) = 1$ whenever $x_1, x_2, \dots, x_n \in [0; 1]$ and $x_1 + x_2 + \dots + x_n = 1$.

Solution. If $x, y \in [0; 1]$ and $x + y \leq 1$, we conclude that

$$f(x) + f(y) + f(1 - x - y) + f(0) + \dots + f(0) = 1$$

and

$$f(x + y) + f(0) + f(1 - x - y) + f(0) + \dots + f(0) = 1.$$

Thus $f(x) + f(y) = f(x + y) + f(0)$ and so $f(x) = ax + b$. It follows that

$$f(x_1) + f(x_2) + \dots + f(x_n) = a(x_1 + x_2 + \dots + x_n) + nb = a + nb$$

for $x_1 + x_2 + \dots + x_n = 1$. Hence $a = 1 - nb$ and $f(x) = (1 - nb)x + b, b \in \mathbb{R}$.

Problem 12.3. Prove that $f: \mathbb{R} \rightarrow \mathbb{R}$ is an additive function if and only if it satisfies one of the following conditions:

- (a) $f(x + y) - f(x - y) = 2f(y)$;
- (b) $f(xy + x + y) = f(xy) + f(x) + f(y)$;
- (c) $f(xy + x + y) + f(xy - x - y) = 2(f(x) + f(y))$;
- (d) $f(x + y)^2 = (f(x) + f(y))^2$,

for all $x, y \in \mathbb{R}$.

Solution. (a) Setting $y = 0$ we get $f(0) = 0$ and then setting $x = y$ gives $f(2x) = 2f(x)$. Also setting $x = 0$ we get $f(y) - f(-y) = 2f(y)$, i.e. $f(-y) = -f(y)$. Now for arbitrary $t, s \in \mathbb{R}$ set $x + y = t, y - x = s$. Then

$$f(t) + f(s) = f(x + y) - f(x - y) = 2f(y) = 2f\left(\frac{t + s}{2}\right) = f(t + s).$$

Hence the function f is additive.

(b) Setting respectively $x = y = 0$ and $y = -1$ we get $f(0) = 0$ and $f(-x) = -f(x)$, $x \in \mathbb{R}$. Now replace x and y by $-x$ and $-y$ to see that $f(xy - x - y) = f(xy) - f(x) - f(y)$. Thus

$$f(xy + x + y) + f(xy - x - y) = 2f(xy). \quad (\star)$$

Note that if $a \leq 0$ and $xy = a$, then $x + y$ runs over the whole $\mathbb{R}$ since the equation $b = x + y = x + \frac{a}{x} \Leftrightarrow x^2 - bx + a = 0$ has real roots ($b^2 - 4a \geq 0$).

Hence setting $xy = u$, $x + y = v$ in $(\star)$ gives

$$f(u + v) + f(u - v) = 2f(u) \quad (\star\star)$$

for any $u \leq 0$ and $v \in \mathbb{R}$. Now using the identity $f(-x) = -f(x)$ we see that $(\star\star)$ holds for arbitrary $u, v \in \mathbb{R}$. Hence (a) shows that f is an additive function.

(c) Plugging $x = y = 0$ yields $f(0) = 0$, whereas $y = 0$ yields $f(-x) = -f(x)$, for all reals x . Now for $y = 1$ and $y = -1$ we get respectively

$$f(2x + 1) = 2f(x) + f(1) \text{ and } f(2x - 1) = 2f(x) - f(1).$$

Furthermore, denote $x * y = xy + x + y$ and notice that this operation is associative, that is, $x * (y * z) = (x * y) * z$. So, by plugging $x = 1$, we have that $y * 1 = 2y + 1$, and $f(y * 1) = 2f(y) + f(1)$ for all y . Hence using the above identities we get

$$\begin{aligned} f(x * (y * 1)) &= f(x * (2y + 1)) = f(x(2y + 1) - x - 2y - 1) + 2f(x) + 2f(2y + 1) \\ &= f(2(xy - y) - 1) + 2f(2y + 1) + 2f(x) = 2(f(xy - y) + f(x) + 2f(y) + f(1)) \end{aligned}$$

and

$$\begin{aligned} f((x * y) * 1) &= 2f(x * y) + f(1) = 2(2(f(x) + 2f(y)) + 2f(xy - x - y) + f(1)) \\ &= 2f(xy - x - y) + 2f(x) + 2f(y) + f(1). \end{aligned}$$

Now, note that the last two relations and the associativity of " $*$ " yield

$$f(xy - x - y) + f(x) = f(xy - y),$$

for all $x, y \in \mathbb{R}$. This means that $f(u+v) = f(u) + f(v)$ for all real numbers u, v , which can be written in the form $u = x, v = xy - x - y$, that is there exists $x, y \in \mathbb{R}$ so that $x = u$ and $y = \frac{v+u}{u-1}$; this happens if $u \neq 1$ or $u = 1 = -v$.

It remains to check that $f(1+v) = f(1) + f(v)$, for all $v \in \mathbb{R}$. This follows from the identity $f(2x+1) = 2f(x+1) - f(1) = 2f(x) + f(1)$.

(d) We have $f(x+y) = \pm(f(x) + f(y))$ and therefore $f(2x) = \pm 2f(x)$ for any $x \in \mathbb{R}$. In particular $f(0) = 0$. Suppose that there is $x_0 \in \mathbb{R}$ such that $f(2x_0) \neq 2f(x_0)$, i.e. $f(2x_0) = -2f(x_0)$. Then

$$f(3x_0) = f(2x_0 + x_0) = \varepsilon_1 (f(2x_0) + f(x_0)) = \varepsilon_1 (-2f(x_0) + f(x_0)) = -\varepsilon_1 f(x_0),$$

where $\varepsilon_1 \in \{-1, 1\}$. Hence

$$f(4x_0) = f(3x_0 + x_0) = \varepsilon_2 (f(3x_0) + f(x_0)) = \varepsilon_2 (1 - \varepsilon_1) f(x_0),$$

where $\varepsilon_2 \in \{-1, 1\}$. On the other hand

$$f(4x_0) = f(2x_0 + 2x_0) = 2\varepsilon_3 f(2x_0) = -4\varepsilon_3 f(x_0),$$

where $\varepsilon_3 \in \{-1, 1\}$, and we get $f(x_0) = 0$ since $\varepsilon_2(1 - \varepsilon_1) \neq -4\varepsilon_3$. Thus $f(2x_0) = -2f(x_0) = 0 = 2f(x_0)$, which shows that $f(2x) = 2f(x)$ for all $x \in \mathbb{R}$.

Suppose now that there exist x and y such that $f(x+y) = -f(x) - f(y)$. Then

$$f(2x+y) = f(x + (x+y)) = \varepsilon_1 (f(x) + f(x+y)) = -\varepsilon_1 f(y)$$

and

$$f(2x+y) = \varepsilon_2 (f(2x) + f(y)) = \varepsilon_2 (2f(x) + f(y))$$

where $\varepsilon_1, \varepsilon_2 \in \{-1, 1\}$. Hence $2f(x) + f(y) = \varepsilon_3 f(y)$ and analogously

$$2f(y) + f(x) = \varepsilon_4 f(x).$$

If $\varepsilon_3 = -1$ then $f(x) + f(y) = 0$ and therefore

$$f(x+y) = -f(x) - f(y) = f(x) + f(y).$$

If $\varepsilon_3 = \varepsilon_4 = 1$ then $f(x) = f(y) = 0$ and again $f(x + y) = f(x) + f(y)$.

Hence the function f is additive. $\square$

Problem 12.4. (German TST 2009) Find all functions $f : \mathbb{R} \mapsto \mathbb{R}$ such that if $x^3 + f(y) \cdot x + f(z) = 0$, then $f(x)^3 + y \cdot f(x) + z = 0$.

Solution. Taking $y = z = 0$ and x_0 such that $x_0^3 + f(0)x_0 + f(0) = 0$, we deduce that $f(x_0) = 0$. Now, take $x = 0, y = 0, z = x_0$ and then $x = 0, y = 1, z = x_0$ to obtain $f(0) = 0$ (by subtracting the two relations obtained from the functional equation). Now, choose $y = 0$ and $x = -\sqrt[3]{f(z)}$ for arbitrary z . We obtain $f(-\sqrt[3]{f(z)}) = -\sqrt[3]{z}$, showing that f is bijective.

Now, fix x, y and choose $z = f^{-1}(-x^3 - xf(y))$ to obtain

$$-(f(x)^3 + yf(x)) = f^{-1}(-x^3 - xf(y)).$$

Then

$$f(-f(x)^3 - yf(x)) = -x^3 - xf(y).$$

Choose $y = f^{-1}(-x^2)$, so that $x^3 + xf(y) = 0$. Using injectivity and the last relation we get $f(x)(y + f(x)^2) = 0$. So for all $x \neq 0$ we have $f(x)^2 + f^{-1}(-x^2) = 0$, so $f(-f(x)^2) = -x^2$ and this clearly holds for $x = 0$ too. By changing x and $-x$ and using the injectivity of f we deduce that f is odd. Therefore $f(f(x)^3 + yf(x)) = x^3 + xf(y)$ and $f(f(x)^2) = x^2$. Put $f(x)^2$ instead of x in the last relation to deduce that $f(x^4) = f(x)^4$. Using this for $x = 1$ we obtain $f(1) = 1$. By taking $x = 1$ in the identity above we also obtain $f(y + 1) = f(y) + 1$.

Now, we will find all bijective odd functions f that satisfy $f(x^4) = f(x)^4$ and $f(x+1) = f(x)+1$. Note that $f(x) \geq 0$ for $x \geq 0$ by the first relation. Also, $f(n) = n$ for all integers n due to the second relation. Therefore, $f(x) - \lfloor x \rfloor = f(x - \lfloor x \rfloor) \geq 0$ and $f(x) - (\lfloor x \rfloor + 1) = -f(\lfloor x \rfloor + 1 - x) \leq 0$, which shows that $x - 1 \leq f(x) \leq x + 1$. Hence $1 - \frac{1}{x} \leq \frac{f(x)}{x}$ for $x > 0$ and $\frac{f(x)}{x}$ tends to 1 for $x \rightarrow \infty$. Combining this with $(\frac{f(x)}{x})^{\frac{x}{4^n}} = \frac{f(x^{4^n})}{x^{4^n}}$ we deduce that $f(x) = x$ for $x > 1$. (Indeed, if $x > 1$ is fixed, then $(f(x)/x)^{\frac{x}{4^n}}$ tends to 0 if $f(x) < x$ and to ∞ if $f(x) > x$.) This together with $f(x+1) = f(x)+1$ shows that $f(x) = x$ for all $x \in \mathbb{R}$. Therefore the only solution of the equation is the identity function.

Problem 12.5. (CTSJ Competition, Romania 2008) Let O be a fixed point in the plane $\mathbb{R}^2$ and $f : \mathbb{R}^2 \setminus \{O\} \rightarrow \mathbb{R}$ be a function such that

$$f(A) - f(B) + f(C) - f(D) = 0$$

for all four distinct points $A, B, C, D \in \mathbb{R}^2 \setminus \{O\}$ with $\triangle AOB \sim \triangle COD$. Prove that f is a constant.

Solution. Let $X, Y \in \mathbb{R}^2 \setminus \{O\}$ be arbitrary points. Denote by A the point on the line OY such that $OA = OX$ and O is between A and Y , and by B the point on the line OX such that $OB = OY$ and O is between B and X . Then $\triangle AOX \sim \triangle BOY$ and $\triangle AOX \sim \triangle YOB$, so $f(A) - f(X) + f(B) - f(Y) = 0$ and $f(A) - f(X) + f(Y) - f(B) = 0$. Adding up these identities gives $f(A) = f(X)$ and $f(B) = f(Y)$. On the other hand $\triangle AOB \cong \triangle XOY$, so $f(A) - f(B) + f(X) - f(Y) = 0$, whence $f(X) = f(Y)$.

Problem 12.6. Let f be a real function on the plane such that

$$f(A_1) + f(A_2) + \cdots + f(A_n) = 0$$

for all regular n -gons $A_1 A_2 \dots A_n$. Prove that f is identically zero.

First Solution. Let P be a point in the plane and let us consider the regular n -gon $PA_1 A_2 \dots A_{n-1}$. After a rotation centered in P through the angle $2k\pi/n$, $k = 0, 1, \dots, n-1$, of the given n -gon, we obtain a regular n -gon $A_{k0} A_{k1} \dots A_{k(n-1)}$, where $A_{k0} = P$ and A_{ki} is the point obtained by rotating the point A_i , for all $i = 1, 2, \dots, n-1$. Taking into account the hypothesis for each regular n -gon before obtained, we get:

$$\sum_{k=0}^{n-1} \sum_{i=0}^{n-1} f(A_{ki}) = 0.$$

In this sum, the number $f(P)$ appears n times and then

$$nf(P) + \sum_{k=0}^{n-1} \sum_{i=1}^{n-1} f(A_{ki}) = 0.$$

But after a small analysis of the sum it is obvious that

$$\sum_{k=0}^{n-1} \sum_{i=1}^{n-1} f(A_{ki}) = \sum_{i=1}^{n-1} \sum_{k=0}^{n-1} f(A_{ki}) = 0$$

because $A_{0i}A_{1i}\dots A_{(n-1)i}$ are all regular n -gons. Thus, from the previous two relations, we get that $f(P) = 0$, and so $f \equiv 0$. $\square$

Second Solution. We shall use complex numbers. Set $\omega = \cos \frac{2\pi}{n} + i \sin \frac{2\pi}{n}$. Then the condition is that for any $z \in \mathbb{C}$ and any positive real t we have

$$\sum_{j=1}^n f(z + t\omega^j) = 0.$$

In particular, for each $k = 1, \dots, n$, we have

$$\sum_{j=1}^n f(z - \omega^k + \omega^j) = 0.$$

Summing over k we get

$$\sum_{m=1}^n \sum_{k=1}^n f(z - (1 - \omega^m)\omega^k) = 0.$$

For $m = n$ the inner sum is equal to $nf(z)$ since $\omega^n = 1$. For other m the inner sum runs over a regular polygon, hence vanishes. Thus $f(z) = 0$ for all $z \in \mathbb{C}$.

Problem 12.7. (Romania TST 1996) Let $n \geq 3$ be an integer and $p \geq 2n - 3$ be a prime. Let M be a set of n points in the plane no three of which are collinear and $f : M \rightarrow \{0, 1, \dots, p-1\}$ be a function such that:

- (i) f is zero for only one point of M ;
- (ii) if $A, B, C \in M$ are distinct points and (ABC) is the circumcircle of triangle ABC , then

$$\sum_{P \in M \cap (ABC)} f(P) \equiv 0 \pmod{p}.$$

Prove that all points of M lie on the same circle.

Solution. Let $X \in M$ be the point for which $f(X) = 0$. Note first that if every circle through X and two other points of M contains one more point of M then all the points of M lie on the same circle. This can be seen by applying an inversion with pole X which reduces the above problem to the well-known analogous problem for lines: If a finite set of points N has the property that every line containing two points of N contains also a third point of N , then all points of N are collinear. Suppose now that not all points of M lie on the same circle. Then there exists a circle through X which passes through only two other points A and B of M . Let $f(A) = i$. Then $f(B) = p - i$ since $f(A) + f(B) + f(X) = 0$. Let a be the number of the circles which pass through X, A and other points of M , b be the analogous number determined by X and B , and let S be the sum of the values of f at the points of M . Considering the circles passing through X and A we get

$$S + (a - 1)i \equiv 0 \pmod{p}.$$

Analogously $S + (b - 1)(p - i) \equiv 0 \pmod{p}$. Hence $(a + b - 2)i \equiv 0 \pmod{p}$ which implies $a + b \equiv 2 \pmod{p}$. But $1 \leq a, b \leq n - 2$ and therefore $2 \leq a + b \leq 2n - 4 < p$. Hence $a + b = 2$, i.e. $a = b = 1$. This shows that all the points of M lie on the circle (XAB) , a contradiction.

Problem 12.8. (Romania TST 1997) Let S be a set of $n \geq 4$ points in the plane, not all lying on a circle, and no three of them collinear. Let $f : S \rightarrow \mathbb{R}$ be a function such that for each circle C containing three or more points from S we have

$$\sum_{P \in C \cap S} f(P) = 0.$$

Prove that f is identically 0.

Solution. Let $A_1, A_2, \dots, A_n, n \geq 4$, be the points in the set S and let $a_1, a_2, \dots, a_n$ be the values of f at these points. For every two points A and B in S denote by $n(A, B)$ the number of circles that pass through A, B and other points in S . Then $n(A, B) \geq 2$ and adding the values of f over all of these circles we get 0. On the other hand in this sum we have the sum

$a_1 + a_2 + \cdots + a_n$ of the values of f at all points and $(n(A, B) - 1)(a + b)$ as the values of f at A and B have been counted $n(A, B)$ times. Hence

$$a_1 + a_2 + \cdots + a_n = (1 - n(A, B))(a + b).$$

But $1 - n(A, B) < 0$, so $a + b$ and $a_1 + a_2 + \cdots + a_n$ either have opposite signs or both vanish. Applying this for the pairs $(A_1, A_2), (A_2, A_3), \dots, (A_n, A_1)$ and adding up we see that either $2(a_1 + a_2 + \cdots + a_n)$ and $a_1 + a_2 + \cdots + a_n$ have opposite signs or they vanish. Since the first option is not possible we conclude that $a_1 + a_2 = 0, a_2 + a_3 = 0, \dots, a_n + a_1 = 0$ which clearly implies that $a_1 = a_2 = \cdots = a_n = 0$.

Problem 12.9. Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be a map such that if $|XY| = 1$, then $|f(X)f(Y)| = 1$. Prove that f is an isometry, i.e. $|f(X)f(Y)| = |XY|$ for all $X, Y \in \mathbb{R}^2$.

Solution. We say that a positive number d has property P (and write $P(d)$) if whenever $|XY| = d$ we have $|f(X)f(Y)| = d$. Note first that $P(1) \Rightarrow P(\sqrt{3})$, as we can see by gluing together two equilateral triangles of side 1. It is also easy to see that $P(d) \Rightarrow P(nd)$, $n \in \mathbb{N}$. Pentagons $ABCDE$ with $BC = CD = DE = EB = \sqrt{3}, AB = AD = 1$ are isometrically invariant under f , and since $AC = \sqrt{2}$ we have $P(\sqrt{2})$. Hence squares of side 1 are also isometrically invariant under f . It is not hard to prove now that a pentagon $ABCDE$ with $ABCD$ a square of side 1 and CDE an equilateral triangle outside $ABCD$ is isometrically invariant (it requires some work to prove that $f(E)$ is outside $f(ABCD)$, and we have $P(|AE|)$). Set $d = |AE|$ and note that $d > \sqrt{3}$. Considering two squares $ABCD$ and $ABEF$ of side d , and G, H on different sides of AB such that $AH = HB = BG = GA$, we see that $P(|GH|)$ with $t = |GH| < 1$. It follows by induction (scaling all the constructions above) that we have $P(t^n)$, so $P(mt^n)$ for all $m, n \in \mathbb{N}$. Thus we have a dense subset $\{a_n\}$ of $(0, \infty)$ with $P(a_n)$. Suppose now that $|f(X)f(Y)| > |XY|$ for some $X, Y \in \mathbb{R}^2$. We can find n such that $|XY| < 2a_n < |f(X)f(Y)|$ and a point A such that $|AX| = |AY| = a_n$. Then $|f(A)f(X)| = |f(A)f(Y)| = a_n$ and by triangle inequality we get $|f(X)f(Y)| \leq 2a_n$, a contradiction. Thus $|f(X)f(Y)| \leq |XY|$ for all $X, Y \in \mathbb{R}^2$ and we conclude easily that $|f(X)f(Y)| = |XY|$, i.e. f is an isometry.

19.13 Approximating by Linear Functions

Problem 13.1. (IMO 1993) Determine whether there exists a strictly increasing function $f : \mathbb{N} \rightarrow \mathbb{N}$ such that $f(1) = 2$ and $f(f(n)) = f(n) + n$ for all $n \in \mathbb{N}$.

Solution. If we try to set $f(x) \sim cx$, then we get $f(f(n)) \sim c^2n$. Thus $c^2n \sim cn + n$, so $c^2 = c + 1$ and hence $c = \frac{\sqrt{5}+1}{2}$. We shall show that

$$f(n) = \left\lfloor cn + \frac{1}{2} \right\rfloor$$

(the closest integer to cn) satisfies the requirements. Observe that f is strictly increasing and $f(1) = 2$. By the definition of f ,

$$|f(n) - cn| \leq \frac{1}{2}$$

and $f(f(n)) - f(n) - n$ is an integer. On the other hand,

$$\begin{aligned} |f(f(n)) - f(n) - n| &= |f(f(n)) - f(n) - c^2n + cn| \\ &= |(c-1)(f(n) - cn) + (f(f(n)) - cf(n))| \\ &\leq (c-1)|f(n) - cn| + |f(f(n)) - cf(n)| \\ &\leq \frac{c-1}{2} + \frac{1}{2} = \frac{c}{2} < 1. \end{aligned}$$

This shows that $f(f(n)) - f(n) - n = 0$.

Problem 13.2. (IMO 1979) Find all increasing functions $f : \mathbb{N} \rightarrow \mathbb{N}$ with the property that all natural numbers which are not in the image of f are those of form $f(f(n)) + 1, n \in \mathbb{N}$.

Solution. We shall prove that f is unique. If $f(x) \sim cx$, then we conclude that $m \sim c^2(m - n)$, where $m = f(n) \sim cn$. So $c = c^2(c - 1)$ or $c^2 - c - 1 = 0$, i.e. $c = \frac{1 + \sqrt{5}}{2}$ is the positive root of the equation. So we try to set $f(x) = \lfloor cx + d \rfloor$ for some constant d . Now we compute $f(1) = 1$, $f(2) = 3$,

$f(3) = 4, f(4) = 6, f(5) = 8$, and we can try to put $d = 0$, so $f(n) = \lfloor cn \rfloor$. Let's prove that it satisfies the hypothesis. If $f(n) = m$ then $m < cn < m + 1$ so $\frac{m}{c} < n < \frac{m+1}{c}$. As $\frac{1}{c} = c - 1$ we get $(c-1)m < n < (c-1)(m+1)$ and so m is in $\text{Im}(f)$ if and only if the interval $(cm, cm + c - 1)$ contains an integer which is equivalent to the fact that $\{cm\} > 2 - c$. And if $f(f(n)) + 1 = m$ then $\lfloor c\lfloor cn \rfloor \rfloor = m - 1$ so $\lfloor cn \rfloor \in ((m-1)(c-1), m(c-1))$ so

$$n \in ((m-1)(c-1)^2, m(c-1)^2 + (c-1)) = ((2-c)m + c - 2, (2-c)m + c - 1),$$

so $n = \lfloor (2-c)m + c - 1 \rfloor = 2m - \lfloor c(m-1) \rfloor - 2$. Therefore $m = f(f(n)) + 1$ if and only if the number $n = 2m - \lfloor c(m-1) \rfloor - 2$ satisfies the condition $f(f(n)) + 1 = m$. Set $u = \{c(m-1)\}$. Then $n = (2-c)m + c - 2 + u$ so

$$\begin{aligned} f(n) &= \lfloor c(2-c)m + cu - 2c + c^2 \rfloor = \lfloor (c-1)m + cu - c + 1 \rfloor = \lfloor (c-1)(m-1) + cu \rfloor \\ &= \lfloor c(m-1) - m + 1 + cu \rfloor = c(m-1) - m + 1 + cu - \{u(c+1)\}. \end{aligned}$$

Set $s = \{u(c+1)\}$. Then

$$f(f(n)) = \lfloor c(c-1)(m-1) + c^2u - cs \rfloor = \lfloor m - 1 + (c+1)u - cs \rfloor.$$

So $f(f(n)) + 1 = m$ if and only if $0 < (c+1)u - cs < 1$. If $t = u(c+1) \in (0, 1+c)$ this is equivalent to $t - c\{t\} \in (0, 1)$. When $t < 1$ this is false as the requested value is negative. When $1 < t < 2$ we have

$$t - c\{t\} = t - c(t-1) = c - (c-1)t \in (0, 1).$$

When $t > 2$ we have

$$t - c\{t\} = t - c(t-2) = 2c - (c-1)t > 2c - (c-1)(c+1) = 2c - c^2 + 1 = c > 1.$$

So our condition is equivalent to $t \in (1, 2)$ or

$$u \in \left(\frac{1}{c+1}, \frac{2}{c+1} \right) = (2-c, 4-2c)$$

so $\{cm - c\} \in (2-c, 4-2c)$ or $\{cm\} \in \{0, 2-c\}$. This condition is equivalent to $\{cm\} < 2 - c$.

Thus we see that the condition $m = f(n)$ is equivalent to $\{cm\} > 2 - c$ and the condition $m = f(f(n)) + 1$ is equivalent to $\{cm\} < 2 - c$. So these two conditions are complementary and the proof is finished.

Problem 13.3. Find all increasing functions $f: \mathbb{N} \rightarrow \mathbb{N}$ such that the only natural numbers which are not in the image of f are those of form $2n + f(n)$, $n \in \mathbb{N}$.

Solution. Like in the previous problem we prove that f is unique. It remains now to find it. Again setting $f(x) \sim cx$ we get $m \sim 2(m - n) + c(m - n)$, where $m = f(n)$. Hence $c = 2(c - 1) + c(c - 1)$, so $c^2 = 2$. Therefore $f(n) \sim n\sqrt{2}$ and we can suppose that $f(n) = \lfloor n\sqrt{2} + a \rfloor$. By computing some values of f we can even infer $a = 0$, so $f(n) = \lfloor n\sqrt{2} \rfloor$. To prove that this function satisfies the condition we must prove that the sets $\{\lfloor n\sqrt{2} \rfloor | n \in \mathbb{N}\}$ and $\{\lfloor (2 + \sqrt{2})n \rfloor | n \in \mathbb{N}\}$ partition $\mathbb{N}$. This follows from the fact that

$$\frac{1}{\sqrt{2}} + \frac{1}{2 + \sqrt{2}} = \frac{1}{\sqrt{2}} + \frac{2 - \sqrt{2}}{2} = 1$$

and the following well-known theorem which we prove for completeness:

Theorem (Beatty). If $\alpha, \beta \in \mathbb{R} \setminus \mathbb{Q}$ and $\frac{1}{\alpha} + \frac{1}{\beta} = 1$, then the sets

$$A = \{\lfloor n\alpha \rfloor | n \in \mathbb{N}\} \text{ and } B = \{\lfloor n\beta \rfloor | n \in \mathbb{N}\}$$

partition $\mathbb{N}$.

Proof. Note that

$$|A \cap \{1, 2, \dots, n\}| = \left\lfloor \frac{n+1}{\alpha} \right\rfloor$$

(the number of m 's with $m\alpha < n + 1$) and

$$|B \cap \{1, 2, \dots, n\}| = \left\lfloor \frac{n+1}{\beta} \right\rfloor.$$

Therefore

$$|A \cap \{1, 2, \dots, n\}| + |B \cap \{1, 2, \dots, n\}| = \left\lfloor \frac{n+1}{\alpha} \right\rfloor + \left\lfloor \frac{n+1}{\beta} \right\rfloor.$$

On the other hand

$$\left\lfloor \frac{n+1}{\alpha} \right\rfloor \in \left(\frac{n+1}{\alpha} - 1, \frac{n+1}{\alpha} \right)$$

and

$$\left\lfloor \frac{n+1}{\beta} \right\rfloor \in \left(\frac{n+1}{\beta} - 1, \frac{n+1}{\beta} \right)$$

thus

$$\left\lfloor \frac{n+1}{\alpha} \right\rfloor + \left\lfloor \frac{n+1}{\beta} \right\rfloor \in \left(\frac{n+1}{\alpha} + \frac{n+1}{\beta} - 2, \frac{n+1}{\alpha} + \frac{n+1}{\beta} \right) = (n-1, n+1).$$

As this number is an integer, we finally conclude that

$$|A \cap \{1, 2, \dots, n\}| + |B \cap \{1, 2, \dots, n\}| = n.$$

By writing the condition for n and for $n+1$ and subtracting them we deduce that

$$|A \cap \{n\}| + |B \cap \{n\}| = 1$$

which implies that A and B partition $\mathbb{N}$. □

Problem 13.4. Find all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ that satisfy

$$f(1) = 1, \quad f(n+1) = f(n) + 2 \text{ if } f(f(n) - n + 1) = n$$

and $f(n+1) = f(n) + 1$ otherwise.

Solution. Let us put $f(n) \sim cn$. Then $1 < c < 2$. Also $f(f(n) - n + 1)$ must equal n an infinite number of times. So $c(cn - n + 1) \sim n$ or $c(c-1) = 1$ and hence $c = \frac{1+\sqrt{5}}{2}$. Next we compute $f(2) = 3, f(3) = 4$. So we suppose that $f(n) = \lfloor cn \rfloor$. First, we prove that $\lfloor cn \rfloor$ satisfies the condition. Indeed, let $t = \{cn\}$. Then $\lfloor c(n+1) \rfloor - \lfloor cn \rfloor = \lfloor cn + c \rfloor - \lfloor cn \rfloor$ is 2 when $t \geq 2 - c$ and 1 otherwise. However

$$\lfloor c(\lfloor cn \rfloor - n + 1) \rfloor = \lfloor c(cn - t - n + 1) \rfloor = \lfloor (c^2 - c)n + c - ct \rfloor = \lfloor n + c(1 - t) \rfloor.$$

It equals n when $1 - t < \frac{1}{c}$ or $t > 1 - \frac{1}{c} = 1 - (c - 1) = 2 - c$ and $n + 1$ otherwise. Hence we see that both conditions

$$\lfloor c(n+1) \rfloor - \lfloor cn \rfloor = 2 \text{ and } \lfloor c(\lfloor cn \rfloor - n + 1) \rfloor = n$$

are equivalent to the same condition $\{cn\} > 2 - c$ and thus $\lfloor cn \rfloor$ satisfies the condition.

Now we prove $f(n) = \lfloor cn \rfloor$ by induction on n . It is true for $n \leq 3$. Next assume it holds for all $n \leq k$. We prove it for $k + 1$. We have $f(k + 1) = f(k) + 2 = \lfloor ck \rfloor + 2$ if $f(f(k) - k + 1) = k$ and $f(k + 1) = f(k) + 1 = \lfloor ck \rfloor + 1$ otherwise. Note that $f(k) - k + 1 < ck - k + 1 = k + 1 - (2 - c)k < k + 1$. Hence $f(f(k) - k + 1) = \lfloor c(f(k) - k + 1) \rfloor = \lfloor c(\lfloor ck \rfloor - k + 1) \rfloor$ using the induction step. However we have proven that $\lfloor c(n + 1) \rfloor = \lfloor cn \rfloor + 2$ if $\lfloor c(\lfloor ck \rfloor - k + 1) \rfloor = k$ and $\lfloor c(n + 1) \rfloor = \lfloor cn \rfloor + 1$ otherwise. Therefore $f(k + 1) = \lfloor c(k + 1) \rfloor$ and the induction is finished.

Problem 13.5. (China TST 2006) Suppose that a function f defined on nonnegative integers satisfies $f(0) = 0$, $f(n) = n - f(f(n - 1))$ for $n \geq 1$. Find all real polynomials g such that $f(n) = \lfloor g(n) \rfloor$, $n = 0, 1, 2, \dots$

Solution. (by A. Frimu) Setting $f(n) \sim \frac{n}{\alpha}$ we get from the given condition for f that α is the positive root of $x^2 + x - 1 = 0$. We shall prove by induction on n that $f(n) = \left\lfloor \frac{n+1}{\alpha} \right\rfloor$. The first few values can be easily verified. Suppose that it is true for any $k \leq n - 1$. Then $f(n - 1) = \left\lfloor \frac{n}{\alpha} \right\rfloor$. Let $k = \left\lfloor \frac{n}{\alpha} \right\rfloor$. We have to prove that

$$\left\lfloor \frac{n+1}{\alpha} \right\rfloor + \left\lfloor \frac{f(n-1)+1}{\alpha} \right\rfloor = n,$$

or

$$\left\lfloor \frac{n+1}{\alpha} \right\rfloor + \left\lfloor \frac{k+1}{\alpha} \right\rfloor = n.$$

First case. $k < \frac{n}{\alpha} < \frac{n+1}{\alpha} < k + 1$. Then we have to prove that

$$E = k + \left\lfloor \frac{k+1}{\alpha} \right\rfloor = n.$$

It is clear that E is an integer and it is sufficient to prove that $E > n - 1$ and $E < n + 1$. Indeed, we have

$$k + \left\lfloor \frac{k+1}{\alpha} \right\rfloor > k + \frac{k+1}{\alpha} - 1 = k\alpha + \frac{1}{\alpha} - 1.$$

Then $k\alpha + \frac{1}{\alpha} - 1 > n - 1$ becomes $k\alpha + \frac{1}{\alpha} > n$ or $k + \frac{1}{\alpha^2} > \frac{n}{\alpha}$. Adding $\frac{1}{\alpha}$ to both sides we get $k + 1 > \frac{n+1}{\alpha}$, which is true by our assumption. For $E < n + 1$ we use $E < k + \frac{k+1}{\alpha} = k\alpha + \frac{1}{\alpha} < n + 1$ since each term of the LHS is smaller than the respective term of the RHS.

Second case. $k < \frac{n}{\alpha} < k + 1 < \frac{n+1}{\alpha}$. We have to show that

$$E = k + 1 + \frac{k+1}{\alpha} = n.$$

As in the first case, E is an integer and we prove that $n - 1 < E < n + 1$. Indeed $E < k + 1 + \frac{k+1}{\alpha} = (k+1)\alpha < n + 1$, or $k + 1 < \frac{n+1}{\alpha}$, which is true from our assumption. Also

$$E > k + 1 + \frac{k+1}{\alpha} - 1 = (k+1)\alpha - 1 > \frac{n}{\alpha} \cdot \alpha - 1 = n - 1,$$

also from our assumption.

Now, clearly g has degree 1. Let $g = ux + v$. Since

$$f(n) = \left\lfloor \frac{n+1}{\alpha} \right\rfloor = \lfloor g(n) \rfloor,$$

we get $\left| n \left(\frac{1}{\alpha} - u \right) + \frac{1}{\alpha} - v \right| < 1$ for $n = 0, 1, 2, \dots$. Clearly this implies $u = \frac{1}{\alpha}$ and then using the density of $\left\{ \frac{n}{\alpha} \right\}$ in $(0, 1)$ we see that $v = \frac{1}{\alpha}$. So

$$g(n) = \frac{n+1}{\alpha}.$$

19.14 Extremal Element Method

Problem 14.1. Find all functions $f: \mathbb{N} \rightarrow \mathbb{N}$ such that

$$f(f(f(n))) + f(f(n)) + f(n) = 3n$$

for all $n \in \mathbb{N}$.

Solution. We shall show by induction that $f(n) = n$ for all $n \in \mathbb{N}$. We have

$$f(f(f(1))) + f(f(1)) + f(1) = 3$$

and therefore

$$f(f(f(1))) = f(f(1)) = f(1) = 1.$$

Suppose that $f(k) = k$ for all $k \leq n$. It follows by the given condition that the function f is injective. This implies that $f(m) > n$ for $m > n$. Hence

$$f(n+1) \geq n+1, f(f(n+1)) \geq n+1$$

and

$$f(f(f(n+1))) \geq n+1.$$

Summing up gives

$$f(f(f(n+1))) + f(f(n+1)) + f(n+1) \geq 3(n+1).$$

On the other hand we have that

$$f(f(f(n+1))) + f(f(n+1)) + f(n+1) = 3(n+1)$$

and therefore

$$f(f(f(n+1))) = f(f(n+1)) = f(n+1) = n+1.$$

Hence it follows by induction that

$$f(n) = n$$

for all $n \in \mathbb{N}$.

Problem 14.2. Find all bijections $f, g, h: \mathbb{N} \rightarrow \mathbb{N}$ for which

$$f(n)^3 + g(n)^3 + h(n)^3 = 3ng(n)h(n)$$

whenever $n \in \mathbb{N}$.

Solution. The functions $f(n) = g(n) = h(n) = n$ satisfy the condition. Next, by AM-GM inequality we have $f(n)^3 + g(n)^3 + h(n)^3 \geq 3f(n)g(n)h(n)$ with equality if and only if $f(n) = g(n) = h(n)$. Therefore $f(n) \leq n$ with equality if and only if $f(n) = g(n) = h(n) = n$. The problem now follows clearly from the fact that f is a bijection. If $f(1) = a$, then $1 \geq a$ so $a = 1$ and then equality holds so $f(1) = g(1) = h(1) = 1$. We then proceed by induction: if we have shown that $f(k) = g(k) = h(k) = k$ for $k \leq n$ and set $f(n+1) = m$, then $n+1 \leq m$ (all numbers less than $n+1$ are already occupied). On the other hand, $n+1 \geq m$, hence and we have $f(n+1) = g(n+1) = h(n+1) = n+1$, as desired.

Problem 14.3. (IMO 1977) Let $f: \mathbb{N} \rightarrow \mathbb{N}$ be a function such that

$$f(n+1) > f(f(n))$$

for all $n \in \mathbb{N}$. Show that $f(n) = n$ for all $n \in \mathbb{N}$.

Solution. First note that if $f(n) \geq k$ for some $k \in \mathbb{N}$ and all $n \geq k$, then

$$f(n+1) > f(f(n)) \geq f(k) \geq k.$$

Hence it follows by induction on k that $f(n) \geq k$ for all $n \geq k$. In particular, $f(k) \geq k$ for all $k \in \mathbb{N}$. Suppose that $f(k) > k$ for some $k \in \mathbb{N}$. Denote by $f(m)$ the least element of the set $A = \{f(n) : n \geq k\}$. If $m-1 > k$ then $f(m-1) \geq m-1 > k$ and if $m-1 = k$ then $f(m-1) = f(k) > k$. Hence $f(m-1) \in A$. On the other hand $f(m) > f(f(m-1))$ which is a contradiction. Thus $f(k) = k$ for all $k \in \mathbb{N}$.

Problem 14.4. Find all injective functions $f: \mathbb{N} \rightarrow \mathbb{N}$ such that $f(1) = 2$, $f(2) = 4$ and

$$f(f(m) + f(n)) = f(f(m)) + f(n)$$

for all $m, n \in \mathbb{N}$.

Solution. Setting respectively $n = 1$ and $m = 1$ we get that $f(f(n)) = f(n) + 2$ for all $n \in \mathbb{N}$. Since $f(2) = 4$ we get by induction that $f(2n) = n + 2$ for all $n \in \mathbb{N}$. So, every even integer is the image of 1 or an even integer and having in mind that the function f is injective we conclude that $f(2n + 1)$ is odd for all $n \in \mathbb{N}$. Consider the set A of all odd integers $x > 1$ such that $f(x) = x + 2$. Since $f(f(n)) = f(n) + 2$ it follows that the set A is not empty and we denote by a its minimal element. If $a > 3$ we know that $f(x) \in A$ for all odd $x \in [3, a)$ and therefore $f(x) \geq a$. But if $f(x) \geq a + 2$, then $f(x) - 2 \in A$ and $f(f(x) - 2) = f(x)$, a contradiction. So $f(x) = a$ and since f is injective it follows that $a = 5$ (else the interval $[3, a)$ would contain at least two odd integers both having the same image a). But $a = 5$ implies $f(3) = 5$, a contradiction. Thus $a = 3$ which shows that the only solution is the function defined by $f(1) = 2$ and $f(n) = n + 2$ for all $n > 1$.

Problem 14.5. (BMO 2002) Find all functions $f: \mathbb{N} \rightarrow \mathbb{N}$ such that

$$2n + 2001 \leq f(f(n)) + f(n) \leq 2n + 2002$$

for all $n \in \mathbb{N}$.

Solution. First, we shall show that $f(n) > n$ for all $n \in \mathbb{N}$. Assume the contrary and let $m \in \mathbb{N}$ be such that $f(m) \leq m$ and $k = f(m)$ be the least possible. Then for $l = f(k)$ we have

$$k + l \geq 2m + 2001 \text{ and } f(l) + l \leq 2k + 2002.$$

Hence

$$2k + 2002 \geq f(l) + 2m + 2001 - k$$

and we get

$$f(l) \leq 3k - 2m + 1 < k < m < l,$$

a contradiction. Hence the function $g(n) = f(n) - n$ is positive. Let $g(p)$ be its least value and set $q = f(p)$. Then it follows by the given condition that

$$2g(p) + g(q) \geq 2001 \text{ and } 2g(q) + g(f(q)) \leq 2002.$$

These inequalities imply that

$$4g(p) \geq 4002 - 2g(q) \geq 2000 + g(f(q)) \geq 2000 + g(p),$$

i.e. $g(p) \geq 667$. Now it follows by the inequality

$$2g(n) + g(f(n)) \leq 2002$$

that $g(n) = 667$, i.e. $f(n) = n + 667$ for all $n \in \mathbb{N}$. Conversely, it is easy to check that this function satisfies the given conditions.

Problem 14.6. (IMO 1972) Let f and g be real-valued functions defined for all real values of x and y , and satisfying the equation

$$f(x+y) + f(x-y) = 2f(x)g(y)$$

for all x, y . Prove that if f is not identically zero, and if $|f(x)| \leq 1$ for all x then $|g(y)| \leq 1$ for all y .

First Solution. Let $M = \sup |f(x)| \leq 1$ and $\{x_k\}_{n=1}^\infty$, a sequence such that $|f(x_k)| \rightarrow M$, $k \rightarrow \infty$. Then the triangle inequality implies

$$|g(y)| = \frac{|f(x_k+y) + f(x_k-y)|}{2|f(x_k)|} \leq \frac{2M}{2|f(x_k)|}$$

and letting $k \rightarrow \infty$ we get $|g(y)| \leq 1$.

Second Solution. Suppose that $f(x_0) \neq 0$ and for a given y define a sequence $\{x_k\}$ by

$$x_{k+1} = \begin{cases} x_k + y & \text{if } |f(x_k+y)| \geq |f(x_k-y)| \\ x_k - y, & \text{otherwise} \end{cases}$$

It follows from the given equation that $|f(x_{k+1})| \geq |g(y)||f(x_k)|$ and we get $|f(x_{k+1})| \geq |g(y)|^k |f(x_0)|$ by induction on k . Since $|f(x_k)| \leq 1$ for all k we obtain $|g(y)| \leq 1$.

Problem 14.7. (Bulgaria 2008) Find all real numbers a for which there exists a function $f : (0, \infty) \rightarrow (0, \infty)$ such that

$$3f(x)^2 = 2f(f(x)) + ax^4$$

for all $x > 0$.

Solution. Let $m = \inf\{f(x) \mid x^2 : x > 0\}$. We have that $m \geq 0$ and

$$3f(x)^2 \geq 2mf(x)^2 + ax^4 \geq (2m^3 + a)x^4.$$

Hence $3m^2 \geq 2m^3 + a$. Then the inequality $2m^3 - 3m^2 + 1 = (m-1)^2(2m+1)$ implies that $a \leq g(m) = 3m^2 - 2m^3 \leq 1$. On the other hand, since $g(1) = 1$, $\lim_{m \rightarrow +\infty} g(m) = -\infty$ and g is a continuous function, it follows that for any $a \leq 1$ the equation $g(c) = a$ has a solution $c_a \geq 1$. Then the function $f(x) = c_a x^2$ satisfies the given condition.

Remark. We show two more ways to prove that $a \leq 1$.

It follows by induction on n that $f(x) > a_n x^2$, where (a_n) is the sequence defined by

$$a_0 = 0, a_n = \sqrt{\frac{2a_{n-1}^3 + a}{3}}, n \in \mathbb{N}.$$

Hence $a \leq 1$, since otherwise the sequence (a_n) increases and tends to ∞ , a contradiction.

A third way to show that $a \leq 1$ is to set $b_n = f^{(n)}(x)$, $c_{n+1} = b_{n+1}/b_n^2$ and to note that $c_{n+1} = \frac{3 - a/c_n^2}{2} > 0$. If $a > 1$, then $c_{n+1} < c_n$ and hence this sequences converges. For its limit c we have $2c^3 - 3c^2 + a = 0$. This equation has one real root and it is smaller than $-1/2$, a contradiction.

Problem 14.8 Is there a bounded function $f: \mathbb{R} \rightarrow \mathbb{R}$ such that $f(1) = 1$ and

$$f\left(x + \frac{1}{x^2}\right) = f(x) + f\left(\frac{1}{x}\right)^2$$

for all $x \neq 0$.

First Solution. We will show that the answer is no. So assume for sake of contradiction that there is such a function f . Then, $f(2) = 2$ as it can be seen by letting $x = 1$. Hence $\sup |f(x)| \geq 2$ and set $\sup |f(x)| = 2 + a$, where $a \geq 0$. (We know that $\sup |f(x)|$ is finite since the function f is bounded.) Assume that $a > 0$, and then pick x such that $|f(x)| = 2 + a - \epsilon$, where $\epsilon < \frac{a^2}{2(2+a)}$.

Then plugging $\frac{1}{x}$ in the relation gives us

$$f\left(x^2 + \frac{1}{x}\right) \geq -2 - a + (2 + a - \epsilon)^2 = 2 + 3a + a^2 - 2\epsilon(2 + a) + \epsilon^2 > 2 + a,$$

a contradiction. So $\sup |f(x)| = 2$. Now letting $x = \frac{1}{2}$ yields

$$f\left(\frac{9}{2}\right) \geq f\left(\frac{1}{2}\right) + 4 \geq 2$$

and since $f\left(\frac{9}{2}\right) \leq 2$ the above can only happen when $f\left(\frac{1}{2}\right) = -2$. But then letting $x = 2$ gives us

$$f\left(\frac{9}{4}\right) = 6,$$

which is impossible. Hence, we infer that there are no functions f with the requirements from the hypothesis.

Second Solution. Suppose f is a function satisfying the conditions of the problem. Denote by c the smallest integer multiple of $1/4$ such that $f(x) \leq c$ for all x . Since $f(2) = 2$ we get that $c \geq 2$. Also, by the definition of c there exists a real number x such that $f(x) \geq c - 1/4$. Hence

$$c \geq f\left(x + \frac{1}{x^2}\right) = f(x) + f\left(\frac{1}{x}\right)^2 \geq c - \frac{1}{4} + f\left(\frac{1}{x}\right)^2$$

and therefore $f\left(\frac{1}{x}\right) \geq -1/2$. Now

$$c \geq f\left(\frac{1}{x} + x^2\right) = f\left(\frac{1}{x}\right) + f(x)^2 \geq -\frac{1}{2} + \left(c - \frac{1}{4}\right)^2$$

which contradicts $c \geq 2$.

Problem 14.9. (G. Dospinescu) Find all integers k for which there is a function $f: \mathbb{N} \rightarrow \mathbb{N}$ such that

$$f(f(f(n))) = f(n+1) + k$$

for all positive integers n .

Solution. (Solution by Bojan Basic, simplified by G. Dospinescu) Note that for $k = 0$ there are constant solutions and for $k \geq -1$ we have the solution

$$f(n) = n + \frac{k+1}{2}.$$

We will prove that these are the only possibilities.

Step 1. We will prove here that no integer $k \leq -2$ works. Suppose we have a function f such that $f(n+1) \geq 2 + f(f(f(n)))$ for all n . We will prove by induction on j that $f(n) \geq j$ for all $n \geq j$. This is clear for $j = 1$, so assume it holds for j and take $n \geq j+1$. Then $n-1 \geq j$, so $f(n-1) \geq j$ and we can repeat to get $f(f(f(n-1))) \geq j$. Therefore $f(n) \geq j+1$ and we are done. Thus, if $n \geq j$ we have $f(n) \geq j$. Take $j = n$ to get $f(n) \geq n$. Thus, $f(n+1) \geq f(n) + 2$ and f is increasing. But since $f(f(f(n))) < f(n+1)$, we deduce that $f(f(n)) \leq n$. Therefore $f(n) = n$ for all n , which is clearly impossible.

Step 2. We prove here that no $k \geq 2$ can work. Suppose there exists a solution f for such a k . First, we will prove that f is injective. Indeed, if $f(a) = f(b)$ we clearly have $f(a+1) = f(b+1)$ and so $f(a+n) = f(b+n)$, which implies that f takes only a finite number of values. We will prove that this is not the case, but this is pretty hard, so we break it into small pieces.

Step 3. We shall find $\text{Im}(f)$. Suppose that f has a finite image and note that if $x \in \text{Im}(f)$ and $x \neq f(1)$, then $x+k \in \text{Im}(f)$. Thus, by considering x to be the largest element of $\text{Im}(f)$ we find that $x = f(1)$ and moreover that $f(n) \leq f(1) - k$ for all $n \geq 2$. We claim that there is n with $f(n) = 1$. Assuming the contrary, then we have $f(f(n)) \neq 1$ and so $f(f(f(n))) \leq f(1) - k$, thus $f(n) \leq f(1) - 2k$ for $n \geq 2$. By induction we get $f(n) \leq f(1) - jk$ for all j and all $n \geq 2$, a contradiction.

Choose now n with $f(n) = 1$ and define $x_0 = n, x_j = f(f(x_{j-1} - 1))$ if x_{j-1} is defined and $x_{j-1} > 1$. Otherwise, we stop the sequence. Using the functional equation we obtain $f(x_j) = 1 + kj$ when x_j is defined. Definitely, x_1 is defined, as we saw that $f(1) \geq 1 + k > 1$. Now, since $\text{Im}(f)$ is bounded, the sequence x_j cannot be defined for all j and there is a maximal j for which x_j is defined. Then $x_j = 1$ (since x_{j+1} is not defined) and $f(1) = 1 + kj$, but also $f(x_1) = 1 + k, \dots, f(x_{j-1}) = 1 + k(j-1)$. Therefore, $\text{Im}(f)$ contains all numbers $1, 1+k, \dots, 1+jk$.

Finally, we claim that $\text{Im}(f) = \{1, 1+k, \dots, 1+jk\}$. Suppose there is r not congruent to $1 \pmod k$ such that $f(n) = r$ for some n . Define $y_0 = n, y_i = f(f(y_{i-1} - 1))$ as above and note again that there is i with $y_i = 1$ and then $1 + jk = f(1) = f(y_i) = r + ki$, a contradiction.

Consider now some n such that $f(n) = 1 + k(j - 1)$ (we saw that $j \geq 1$ and that such n exists). Then $f(f(f(n - 1))) = f(1)$ and we saw that this forces $f(f(n - 1)) = 1$. In particular, $f(n - 1) \neq 1$ and so $f(n - 1) \in \{1 + k, \dots, 1 + jk\}$. Hence there is m with $f(m) = f(n - 1) - k$. Clearly, $m > 1$ and since $f(f(f(m - 1))) = f(n - 1)$ and $f(f(n - 1)) = 1$ we have

$$f(1 + f(m - 1)) + k = f(f(f(f(m - 1)))) = f(f(n - 1)) = 1,$$

obviously impossible. This finally finishes the proof of the fact that f cannot be periodic and so f is injective.

Step 4. Define $A = \mathbb{N} \setminus f(\mathbb{N})$, $B = f(A)$ and $C = f(B)$. These sets are pairwise disjoint as f is injective. Clearly,

$$B = f(\mathbb{N} \setminus f(f(\mathbb{N}))), \quad C = f(f(\mathbb{N})) \setminus f(f(f(\mathbb{N})))$$

and

$$B \cup C = f(\mathbb{N}) \setminus f(f(f(\mathbb{N}))).$$

Notice that from the given functional equation also follows

$$s \in f(f(f(\mathbb{N}))) \iff s - k \in f(\mathbb{N}) \setminus \{f(1)\}.$$

For $0 \leq i \leq k - 1$ let us define $r_i = \min\{f(x) : f(x) \equiv i \pmod{k}\}$. Denote also $r_k = \min\{f(1) + jk : f(1) + jk \in f(\mathbb{N}), j \in \mathbb{N}\}$. It could happen that an r_i doesn't exist, in that case we simply ignore it onward. We claim that $B \cup C = \{r_i : 0 \leq i \leq k\}$. For a starting point, it is obvious that $r_i \in f(\mathbb{N})$ for each $0 \leq i \leq k$. On the other hand, if for an i held $r_i \in f(f(f(\mathbb{N})))$ then we would have $r_i - k \in f(\mathbb{N}) \setminus \{f(1)\}$, which is in contradiction with minimality of r_i . Let us now check whether an element different from every r_i belongs to the considered union. Write it in form $x = r_k + jk$ if that is possible, otherwise if that is not possible in form $x = r_i + jk$ for an $0 \leq i \leq k - 1$, where $j \in \mathbb{N}$. Then it holds $x - k \in f(\mathbb{N}) \setminus \{f(1)\}$, and it follows $x \in f(f(f(\mathbb{N})))$, hence $x \notin B \cup C$.

Therefore, $B \cup C$ has finitely many elements, and hence it holds for A too, i. e. all but finitely many natural numbers are contained in the set of values of the function f . We conclude that every r_i exists, and hence $B \cup C$ has precisely $k + 1$ elements. However, since B and C are disjoint and with the same number of elements, their union must contain even number of elements, while $k + 1$ is odd. Contradiction!

19.15 Fixed Points

Problem 15.1. Prove that every increasing function $f : [0, 1] \rightarrow [0, 1]$ has a fixed point.

Solution. Suppose f has no fixed point. Then $f(0) > 0$ and set

$$\alpha = \sup\{x \in [0, 1] : f(x) > x\}.$$

Note that $f(\alpha) \geq \alpha$. Indeed, take a sequence $\{x_n\}_{n=1}^\infty$ tending to α and such that $f(x_n) > x_n$ for all n . Since $\alpha > x_n$ we get $f(\alpha) > f(x_n) > x_n$ which implies $f(\alpha) \geq \alpha$. If $f(\alpha) > \alpha$ then $f(f(\alpha)) > f(\alpha)$ which implies $f(\alpha) < \alpha$, a contradiction. Hence α is a fixed point of f .

Problem 15.2. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function such that

$$f(f(x)) = x^3 + \frac{3}{4}x$$

for all $x \in \mathbb{R}$. Prove that there exist three distinct real numbers a, b, c such that $f(a) + f(b) + f(c) = 0$.

Solution. Set $g(x) = x^3 + \frac{3}{4}x$. Then $f(f(x)) = g(x)$ and it follows that $f(g(x)) = f(f(f(x))) = g(f(x))$. Note also that the function $g(x)$ is injective hence so is $f(x)$.

Let x_0 be a fixed point of $g(x)$, i.e. $g(x_0) = x_0$. Then

$$f(x_0) = f(g(x_0)) = g(f(x_0))$$

and $f(x_0)$ is also a fixed point of $g(x)$. Since the fixed points of $g(x)$ are $-\frac{1}{2}, 0, \frac{1}{2}$ it follows that f is a bijection of the set $\left\{-\frac{1}{2}, 0, \frac{1}{2}\right\}$. Thus

$$f\left(-\frac{1}{2}\right) + f(0) + f\left(\frac{1}{2}\right) = -\frac{1}{2} + 0 + \frac{1}{2} = 0.$$

Problem 15.3. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function such that for every $x \in \mathbb{R}$ there is a positive integer $n = n(x)$ for which $f^{(n)}(x) = 1$. Show that f has a fixed point.

Solution. We shall show that 1 is a fixed point of f . Suppose that $f(1) \neq 1$. Then f has no fixed points, because if $a \neq 1$ is a fixed point then $f^{(n)}(a) = a \neq 1$ for all $n \in \mathbb{N}$. Now, by the Intermediate value theorem, either $f(x) < x$ for all $x \in \mathbb{R}$, or $f(x) > x$ for all $x \in \mathbb{R}$. In the former case the sequence $2, f(2), f^{(2)}(2), \dots$ is increasing and $f^{(n)}(2) > 1$ for all $n \in \mathbb{N}$, a contradiction. In the latter case the sequence $0, f(0), f^{(2)}(0), \dots$ is decreasing and $f^{(n)}(0) < 1$ for all $n \in \mathbb{N}$, again a contradiction. Hence $f(1) = 1$.

Problem 15.4. (M. Tetiva) Find all continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(f(x)) = 3f(x) - 2x$$

for all $x \in \mathbb{R}$.

Solution. First of all, it is clear that f is injective, so by continuity it is monotonic. But then $f \circ f$ is increasing and since $f(x) = (2x + f(f(x)))/3$, f itself is increasing. Then f has limits at ∞ and $-\infty$ and from the functional equation we get that they are ∞ and $-\infty$ respectively. So f is surjective and f is an increasing bijection. Therefore, for any real number a we can define $x_n = f^{(n)}(a)$ for all integers n and from the functional equation we get $x_{n+2} = 3x_{n+1} - 2x_n$. Now using a double induction we obtain $x_n = 2a - f(a) + 2^n(f(a) - a)$ for all integers n . Since $f(x_n) = x_{n+1}$, by taking $n \rightarrow -\infty$ we obtain $f(2a - f(a)) = 2a - f(a)$.

Let us consider now the set A of fixed points of f . By the previous result, A is not empty and we will prove that A is an interval of a very special type. Take two elements $a < b$ of A . But for all $c \in (a, b)$ the function $g(x) = f(x) - (2x - c)$ takes values of opposite signs at a and b and so vanishes at some x between a and b . Thus $c = 2x - f(x)$ and by the previous paragraph $c \in A$. Thus if $a, b \in A$ we also have $[a, b] \subset A$. Now, if $A = \mathbb{R}$, we obtain $f(x) = x$ for all x , which works.

Suppose that $A = (-\infty, a]$ for some a . Then since $2x - f(x) \in A$ for all x , we have $2x - f(x) \leq a$ for all x . We claim that if $x > a$ then we actually have $2x - f(x) = a$. Choose $x > 0$ and define as usual $x_n = f^{(n)}(x)$, then we saw that $x_n = 2x - f(x) + 2^n(f(x) - x)$. If $2x - f(x) < a$, there is n such that $x_n \leq a$ and then $f(x_n) = x_{n+1}$, so $x_n = x_{n+1}$, which forces $f(x) = x$ and $x \in A$, a contradiction as $x > a$. So $f(x) = x$ for $x \leq a$, $f(x) = 2x - a$ for $x \geq a$ and one easily checks that this works.

We have two more cases with similar treatments. If $A = [a, \infty)$, then $f(x) = x$ for $x \geq a$ and $f(x) = 2x - a$ for $x \leq a$; if $A = [a, b]$ then $f(x) = 2x - a$ for $x \leq a$, $f(x) = x$ for $x \in [a, b]$ and $f(x) = 2x - b$ for $x \geq b$. This gives a whole family of solutions and we proved these are all solutions.

19.16 Functional Equations for Polynomials

Problem 16.1. Let n be a positive integer. Find all polynomials P with real coefficients such that

$$P\left(x + \frac{1}{n}\right) + P\left(x - \frac{1}{n}\right) = 2P(x)$$

for all $x \in \mathbb{R}$.

Solution. Set $Q(x) = P\left(x + \frac{1}{n}\right) - P(x)$. Then $Q(x) = Q\left(x - \frac{1}{n}\right)$ and by induction $Q(x) = Q\left(x - \frac{k}{n}\right)$ for all $k \in \mathbb{N}$. Let x_0 be a (complex) zero of $Q(x)$. Then $x_0 - \frac{k}{n}$ is a zero of $Q(x)$ for any $k \in \mathbb{N}$. Hence $Q(x) = c$ for some $c \in \mathbb{R}$ and we get $P\left(x + \frac{1}{n}\right) - P(x) = c$. If we set $R(x) = P(x) - cnx$, the latter condition reads as $R(x) = R\left(x - \frac{1}{n}\right)$ and we show as above that R is a constant. Therefore $P(x) = ax + b$ for some constants $a, b \in \mathbb{R}$. It is easy to check that all linear polynomials satisfy the problem condition.

Problem 16.2. (MR 2008, O. Furdui) Find all monic polynomials P and Q with real coefficients such that

$$P(1) + P(2) + \cdots + P(n) = Q(1 + 2 + \cdots + n)$$

for all $n \geq 1$.

Solution. Since

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}$$

it follows that

$$P(x) = Q\left(\frac{x(x+1)}{2}\right) - Q\left(\frac{x(x-1)}{2}\right) \quad (1)$$

for all $x \in \mathbb{N}$, so this identity holds for all $x \in \mathbb{R}$. Let $\deg(Q) = n$. Then it is easy to see that the leading term of the right hand side of (1) is $\frac{nx^{2n-1}}{2^{n-1}}$ and we get that $n = 2^{n-1}$. Hence $n \in \{1, 2\}$ since it follows by induction that $2^{n-1} > n$ for $n \geq 3$. On the other hand we have from the given condition that $P(1) = Q(1)$, while (1) gives $P(1) = Q(1) - Q(0)$. Hence $Q(0) = 0$ and therefore either $Q(x) = x$ or $Q(x) = x^2 + bx$. Now it follows from (1) that $P(x) = x$ or $P(x) = x^3 + bx$, respectively. It is easy to check that all these polynomials satisfy the condition of the problem.

Problem 16.3. Find all polynomials $F, G \in \mathbb{R}[X]$ such that

$$F(G(x)) = F(x)G(x)$$

for all $x \in \mathbb{R}$.

Solution. If $G \equiv 0$, then F is a polynomial with $F(0) = 0$. If $F \equiv 0$, then the equation holds for all $G \in \mathbb{R}[X]$. Suppose now that $F, G \not\equiv 0$. Then $\deg F \cdot \deg G = \deg F + \deg G$ which is possible only if $\deg F = \deg G = 0$ or $\deg F = \deg G = 2$. In the first case F and G are constant polynomials and we have $F \equiv c$ and $G \equiv 1$. Let now F and G be quadratic polynomials. If x_0 is a complex zero of G , then

$$F(0) = F(G(x_0)) = F(x_0)G(x_0) = 0$$

and therefore $F(x) = ax(x+c)$ for some real constants $a \neq 0$ and c . Hence

$$aG(x)(G(x)+c) = ax(x+c)G(x).$$

Since $a \neq 0$ and $G \not\equiv 0$, we get $G(x) = x^2 + cx - c$.

Problem 16.4. (Putnam 2010) Find all pairs of polynomials $P(x)$ and $Q(x)$ with real coefficients for which

$$P(x)Q(x+1) - P(x+1)Q(x) = 1$$

for all $x \in \mathbb{R}$.

Solution. The pairs (P, Q) satisfying the given equation are those of the form $P(x) = ax + b, Q(x) = cx + d$ for $a, b, c, d \in \mathbb{R}$ such that $bc - ad = 1$.

Suppose P and Q satisfy the given equation; note that neither P nor Q can be identically zero. By subtracting the equations

$$P(x)Q(x+1) - P(x+1)Q(x) = P(x-1)Q(x) - P(x)Q(x-1) = 1,$$

we obtain the equation

$$P(x)(Q(x+1) + Q(x-1)) = Q(x)(P(x+1) + P(x-1)).$$

The original equation implies that $P(x)$ and $Q(x)$ have no common nonconstant factor, so $P(x)$ divides $P(x+1) + P(x-1)$. Since each of $P(x+1)$ and $P(x-1)$ has the same degree and leading coefficient as P , we must have $P(x+1) + P(x-1) = 2P(x)$. If we define the polynomials $R(x) = P(x+1) - P(x)$, $S(x) = Q(x+1) - Q(x)$, we have $R(x+1) = R(x)$, and similarly $S(x+1) = S(x)$. Put $a = R(0), b = P(0), c = S(0), d = Q(0)$. Then $R(x) = a, S(x) = c$ for all $x \in \mathbb{Z}$, and hence identically; consequently, $P(x) = ax + b, Q(x) = cx + d$ for all $x \in \mathbb{Z}$, and hence identically. For P and Q of this form,

$$P(x)Q(x+1) - P(x+1)Q(x) = bc - ad,$$

so we get a solution if and only if $bc - ad = 1$, as claimed.

Problem 16.5. (Bulgaria 2001) Find all polynomials $P \in \mathbb{R}[X]$ such that

$$P(x)P(x+1) = P(x^2)$$

for all $x \in \mathbb{R}$.

Solution. If $P \equiv \text{const}$ then $P \equiv 0$ or $P \equiv 1$. Suppose now that $P \not\equiv \text{const}$. Note that the given identity is satisfied for all $x \in \mathbb{C}$. If $\alpha \in \mathbb{C}$ is a zero of P , then $P(\alpha^2) = 0$ and it follows by induction on n that $P(\alpha^{2^n}) = 0$. Hence either $\alpha = 0$ or $|\alpha| = 1$ since otherwise P would have infinitely many zeros.

On the other hand, it follows from the given identity that $(\alpha - 1)^2$ is a zero of P and therefore $\beta = ((\alpha - 1)^2 - 1)^2$ is also a zero of P . Then either

$\beta = 0$ or $|\beta| = 1$, i.e. $\alpha = 0, \alpha = 2$ or $|\alpha(\alpha - 2)| = 2$. If $\alpha \neq 0$ then $|\alpha| = 1$ and $|\alpha - 2| = 1$, i.e. $\alpha = 1$.

Hence all zeros of the polynomial $P(x)$ are equal to 0 or 1. It follows that $P(x) = cx^n(x-1)^m$ and the given condition implies that $P(x) = x^n(x-1)^n$, where $n \in \mathbb{N}$.

Problem 16.6. (IMO Shortlist 1979) Find all polynomials $P \in \mathbb{R}[X]$ such that

$$P(x)P(2x^2) = P(2x^3 + x)$$

for all $x \in \mathbb{R}$.

First Solution. If $P \equiv \text{const}$ then $P \equiv 0$ or $P \equiv 1$. Now set

$$P(x) = \sum_{k=0}^n a_k x^{n-k},$$

where $n = \deg P \geq 1$ and $a_0 \neq 0$. Comparing the coefficients of x^{3n} on both sides of the given identity we get $a_0^2 = a_0$, i.e. $a_0 = 1$. Let $P(x) = x^k P_1(x)$, where $k \geq 0$ and $P_1(0) \neq 0$. Then the given identity can be rewritten as

$$2^k x^{2k} P_1(x) P_1(2x^2) = (2x^2 + 1)^k P_1(2x^3 + x).$$

It follows that $k = 0$ since otherwise $P_1(0) = 0$, a contradiction. Now the above identity for $x = 0$ gives $a_n = P(0) = 1$. Hence by Vietta's theorem we see that the product of all zeros of P is equal to 1.

Now let $\alpha \in \mathbb{C}$ be the zero of P whose modulus is a maximum. Then $P(2\alpha^3 + \alpha) = 0$ and therefore $|\alpha| \leq 1$ since otherwise

$$|2\alpha^3 + \alpha| \geq |\alpha|(2|\alpha|^2 - 1) > |\alpha|,$$

a contradiction. On the other hand, the product of the zeros of P is equal 1, so it follows that $|\alpha| \geq 1$. Hence $|\alpha| = 1$ and $|2\alpha^2 + 1| = 1$. Set

$$\alpha = \cos \varphi + i \sin \varphi.$$

Then $2\alpha^2 + 1 = (2 \cos 2\varphi + 1) + i \cdot 2 \sin 2\varphi$, i.e. $(2 \cos 2\varphi + 1)^2 + (2 \sin 2\varphi)^2 = 1$. It follows that $\cos 2\varphi = -1$. Hence $\alpha = \pm i$ and since the coefficients of $P(x)$ are real we conclude that i and $-i$ are zeros of $P(x)$. Set

$$P(x) = (x^2 + 1)^m Q(x),$$

where $m \geq 1$ and $Q(i)Q(-i) \neq 0$. Then using the identity

$$(x^2 + 1)((2x^2)^2 + 1) = ((2x^3 + x)^2 + 1)$$

we see that the polynomial $Q(x)$ satisfies the given conditions. Now the same arguments as above show that $Q \equiv 0$ or $Q \equiv 1$ (recall that $Q(i)Q(-i) \neq 0$). Proceeding in the same way as above, we conclude that the solutions of the problem are the polynomials $Q \equiv 0$, $Q \equiv 1$ and $Q(x) = (x^2 + 1)^n$, where $n \in \mathbb{N}$.

Second Solution. If u is a zero of $P \not\equiv \text{const}$, then by setting $x = u$ we get that $2u^3 + u$ is also a zero of P . Let r be a zero with maximal absolute value. Then $|2r^3 + r| \leq |r|$ hence $2|r|^3 - |r| \leq |r|$ so $|r| \leq 1$. So all zeros of P have absolute value at most 1. Now replace x by $-x$ to get

$$P(-x)P(2x^2) = P(-2x^3 - x).$$

Comparing it with the condition we get

$$\frac{P(x)}{P(-x)} = \frac{P(2x^3 + x)}{P(-2x^3 - x)}.$$

Hence if we set $P(x) = Q(x^2)R(x)$, where R has no two zeros that add up to zero (thus $R(x)$ and $R(-x)$ have no common zero) then the above identity reads as

$$\frac{R(x)}{R(-x)} = \frac{R(2x^3 + x)}{R(-2x^3 - x)}.$$

Now $R(2x^3 + x)$ and $R(-2x^3 - x)$ also have no common zero: if w is a common zero of $R(2x^3 + x)$ and $R(-2x^3 - x)$ then $2w^3 + w, -2w^3 - w$ are two zeros of $R(x)$ which add up to zero. Hence we get $R(2x^3 + x)R(-x) = R(-2x^3 - x)R(x)$ and thus $R(2x^3 + x) \mid R(x)$. This is possible only when R is a constant because otherwise the degree of $R(2x^3 + x)$ is bigger than the degree of R . So $P(x) = Q(x^2)$ and we get $Q(x^2)Q(4x^4) = Q(4x^6 + 4x^4 + x^2)$. Now if we replace x^2 by x we get $Q(x)Q(4x^2) = Q(4x^3 + 4x + 1)$. All zeros of Q have absolute value at most 1, as we have shown. Now let

$$Q(x) = a \prod_{i=1}^n (x - r_i).$$

Then

$$Q(4x^2) = 4^n a \prod_{i=1}^n \left(x - \sqrt{\frac{r_i}{4}} \right) \left(x + \sqrt{\frac{r_i}{4}} \right)$$

and

$$Q(4x^3 + 4x^2 + x) = 4^n a \prod_{i=1}^n \left(x^3 + x^2 + \frac{x}{4} - r_i \right).$$

Consider now the sum of zeros of $Q(x)Q(4x^2)$ and the sum of zeros of $Q(4x^3 + 4x^2 + 1)$. The sum of zeros of $Q(x)$ is $\sum_{i=1}^n r_i$, the sum of zeros of $Q(4x^2)$ is zero, and the sum of zeros of $Q(4x^3 + 4x^2 + 1)$ is $-n$ because we have n factors $x^3 + x^2 + \frac{x}{4} - r_i$ the sum of whose zeros is -1 . So $\sum_{i=1}^n r_i = -n$ and as $|r_i| \leq 1$ this is possible only when $r_i = -1$. Thus $Q(x) = a(x+1)^n$. Now if $Q \neq 0$ then we get $a^2 = a$ if looking at the leading coefficient of

$$Q(x)Q(4x^2) = Q(4x^3 + 4x + x)$$

so $a = 1$. Finally as $(x+1)(4x^2+1) = (4x^3+4x^2+x+1)$ we deduce that $Q(x) = (x+1)^n$ satisfies the conditions.

Since among the constant polynomials only 0 and 1 are solutions we see that all solutions of the problem are $P \equiv 0$, $P \equiv 1$ and $P(x) = (x^2+1)^n$, where $n \in \mathbb{N}$.

Problem 16.7. (Romania 1990) Find all polynomials $P \in \mathbb{R}[X]$ such that

$$2P(2x^2 - 1) = P(x)^2 - 2$$

for all $x \in \mathbb{R}$.

Solution. Suppose that $P(x) \not\equiv P(1)$ and set $P(x) = (x-1)^n Q(x) + P(1)$, where $n \in \mathbb{N}$ and $Q(1) \neq 0$. Then

$$\begin{aligned} & 2^{n+1}(x-1)^n(x+1)^n Q(2x^2-1) + 2P(1) \\ &= (x-1)^{2n} Q(x)^2 + 2(x-1)^n Q(x)P(1) + P(1)^2 - 2. \end{aligned}$$

Hence $2P(1) = P(1)^2 - 2$ and

$$2^{n+1}(x+1)^n Q(2x^2-1) = (x-1)^n Q(x) + 2Q(x)P(1).$$

This identity for $x = 1$ gives $Q(1)(2^{2n+1} - P(1)) = 0$. Taking into account that $P(1) = 1 \pm \sqrt{3} \notin \mathbb{Q}$ we get $Q(1) = 0$, a contradiction. Thus $P(x) \equiv P(1)$ and the only solutions are the constant polynomials $P(x) \equiv 1 + \sqrt{3}$ and $P(x) \equiv 1 - \sqrt{3}$.

Problem 16.8. Let $k, l \in \mathbb{N}$. Find all nonconstant polynomials $P \in \mathbb{R}[X]$ such that

$$xP(x - k) = (x - l)P(x)$$

for all $x \in \mathbb{R}$.

Solution. If $\deg P = n \geq 1$ and $P(x) = ax^n + bx^{n-1} + \dots$ then

$$P(x - k) = ax^n + (b - nak)x^{n-1} + \dots$$

Hence $xP(x - k) = ax^{n+1} + (b - nak)x^n + \dots$ and

$$(x - l)P(x) = ax^{n+1} + (b - la)x^n + \dots$$

So $l = nk$. Thus $xP(x - k) = (x - nk)P(x)$. Then $x|P(x)$, k is a zero of $P(x)$ and hence $2k$ is a zero of $P(x - k)$. As $P(x - k)|(x - nk)P(x)$ we conclude that if $n > 2$ then $2k$ is a zero of x and so on, obtaining

$$x(x - k) \dots (x - (n - 1)k) | P(x).$$

If $P(x) = x(x - k) \dots (x - (n - 1)k)Q(x)$ the condition transforms to $Q(x - k) = Q(x)$ which is possible only if Q is a constant. Hence there is a polynomial P with the given property if and only if $l = nk$ for some $n \in \mathbb{N}$ and $P(x) = cx(x - k) \dots (x - (n - 1)k)$, $c \neq 0$, which is clearly a solution.

Problem 16.9. (H. Shapiro) Find all polynomials $P \in \mathbb{R}[X]$ that satisfy the identity

$$P(x)P(x + 1) = P(x^2 + x + 1)$$

for all $x \in \mathbb{R}$.

Solution. Let w be a zero of $P \not\equiv \text{const}$ with maximal absolute value. Set $x = w$ to conclude that $w_1 = w^2 + w + 1$ is a zero of P and $x = w - 1$ to conclude that $w_2 = w^2 - w + 1$ is a zero of P . Then $|w_1 - w_2| = 2|w|$ but $|w_1 - w_2| \leq |w_1| + |w_2| = 2|w|$. The equality can hold only if $w_1 + w_2 = 0$. So

$w^2 + 1 = 0$ and hence $w = \pm i$. In this case $(w_1, w_2) = (i, -i)$ and $x^2 + 1 | P(x)$. However $Q(x) = x^2 + 1$ satisfies $Q(x)Q(x+1) = Q(x^2 + x + 1)$ and therefore $\frac{P}{Q}$ satisfies the same condition. We can repeat the same operation until we reach a constant polynomial. Then it is easy to see that all solutions are $P \equiv 0$ and $P(x) = (x^2 + 1)^n$, $n \in \mathbb{N}_0$.

Problem 16.10. Find all polynomials $P \in \mathbb{C}[X]$ that satisfy the identity

$$P(x)P(-x) = P(x^2)$$

for all $x \in \mathbb{C}$.

Solution. If $P \equiv c$ is a constant, then $c^2 = c$ and so $c = 0, 1$. Assume now P is not constant. If $r \neq 0$ is a zero of P then r^2 is also a zero of P . Then r^{2^k} is also a zero of P by induction on k . As P has finitely many zeros, $r^{2^k} = r^{2^m}$ for some k, m hence $|r| = 1$. We have found thus a zero w for which $w^{2^n} = w$ thus $w^{2^n-1} = 1$. If n is minimal with this property, then $w, w^2, \dots, w^{2^n-1}$ are different zeros of P hence $Q(x) = (x - w)(x - w^2) \dots (x - w^{2^n-1})$ divides P . Now

$$\begin{aligned} Q(x)Q(-x) &= (x - w)(-x - w) \dots (x - w^{2^n-1})(-x - w^{2^n-1}) \\ &= (-1)^n (x^2 - w^2) \dots (x^2 - w^{2^n}) = (-1)^n Q(x^2) \end{aligned}$$

because $w^{2^n} = w$. Thus $(-1)^n Q$ satisfies the original condition, so we can divide P by Q and the condition holds for $(-1)^n \frac{P}{Q}$. Now repeating this argument as many times as necessary we shall reach a polynomial with no non-zero zeros. Then if $P(x) = cx^n$ we have $cx^n cx^n (-1)^n = cx^{2n}$ which gives $c = (-1)^n$.

Concluding, all solutions of the problem are given by

$$P(x) = (-x)^n \prod Q_i(x),$$

where each

$$Q_i(x) = (w - x)(w^2 - x)(w^4 - x) \dots (w^{2^{m-1}} - x)$$

for a complex number w with $w^{2^m} = w$.

Problem 16.11. (Putnam 2003) Do there exist polynomials $A(x)$, $B(x)$, $C(y)$, $D(y)$ such that

$$1 + xy + x^2y^2 = A(x)C(y) + B(x)D(y)$$

holds identically?

Solution. No, there do not. Suppose the contrary. By setting $y = -1, 0, 1$ in succession, we see that the polynomials $1 - x + x^2, 1, 1 + x + x^2$ are linear combinations of $a(x)$ and $b(x)$. But these three polynomials are linearly independent, so cannot all be written as linear combinations of two other polynomials, contradiction.

Problem 16.12. Find all polynomials $P \in \mathbb{C}[X]$ such that

$$P(x^2 - y^2) = P(x - y)P(x + y)$$

for all $x, y \in \mathbb{C}$.

Solution. Suppose w is a zero of P . Then $x - y = w$ implies

$$x^2 - y^2 = w(x + y) = w(2x - w)$$

is also a zero of P . Now if $w \neq 0$ then $w(2x - w)$ can take any value being a non-constant linear function in x . So $P \equiv 0$. If $P = cx^n$, we get

$$c(x^2 - y^2)^n = c(x - y)^n c(x + y)^n = c^2(x^2 - y^2)^n$$

and so $c = 1$. Hence all solutions are $P \equiv 0$, $P \equiv 1$ and $P(x) = x^n$, $n \in \mathbb{N}$.

Problem 16.13. Let $a, b \in \mathbb{R}$ and $(a, b) \neq (0, 0)$. Find all polynomials $P(x, y) \in \mathbb{R}[x, y]$ that satisfy

$$P(x + a, y + b) = P(x, y)$$

for all $x, y \in \mathbb{R}$.

Solution. Assume that $b \neq 0$. Consider the polynomial $R \in \mathbb{R}[x, y]$ defined by

$$R(x, y) = P\left(x + \frac{a}{b}y, y\right).$$

Then

$$P(x, y) = R\left(x - \frac{a}{b}y, y\right).$$

So $P(x + a, y + b) = P(x, y)$ can be rewritten in terms of R as

$$R\left((x + a) - \frac{a}{b}(y + b), y + b\right) = R\left(x - \frac{a}{b}y, y\right)$$

or

$$R\left(x - \frac{a}{b}y, y + b\right) = R\left(x - \frac{a}{b}y, y\right).$$

If we replace x by $x - \frac{a}{b}y$ we get $R(x, y) = R(x, y + b)$. Then it follows by induction on n that $R(x, y) = R(x, y + nb)$ for all $n \in \mathbb{N}$. For a fixed $x \in \mathbb{R}$ set $Q_x(y) = R(x, y)$. Then $Q_x(y) = Q_x(y + nb)$ for all $n \in \mathbb{N}$ and therefore Q_x is constant. So $Q_x(y) = Q_x(0) = R(x, 0)$. This shows that $R(x, y)$ is a polynomial in x , so $R(x, y) = Q(x)$ for some polynomial $Q \in \mathbb{R}[x]$. Then

$$P(x, y) = R\left(x - \frac{a}{b}y, y\right) = Q\left(x - \frac{a}{b}y\right) = S(bx - ay).$$

Note that every polynomial of this form satisfies the condition, as

$$P(x + a, y + b) = S(b(x + a) - a(y + b)) = S(bx - ay) = P(x, y).$$

If $b = 0$, then $a \neq 0$ and we repeat the reasoning by replacing b with a and y with x to get again $P(x, y) = S(bx - ay)$.

Problem 16.14. (IMO 1975) The function $f(x, y)$ is a homogeneous polynomial of degree n in x and y . If $f(1, 0) = 1$ and for all a, b, c ,

$$f(a + b, c) + f(b + c, a) + f(c + a, b) = 0,$$

prove that $f(x, y) = (x - 2y)(x + y)^{n-1}$.

Solution. We shall prove that $f(x, y) = x - 2y$ for all x, y with $x + y = 1$. Indeed, in this case $f(x, y) = f(x, 1 - x)$ can be regarded as a polynomial in $z = x - 2y = 3x - 2$, say $f(x, 1 - x) = F(z)$. Putting in the given relation $a = b = x/2, c = 1 - x$, we obtain $f(x, 1 - x) + 2f(1 - x/2, x/2) = 0$; hence $F(z) + 2F(-z/2) = 0$. Since $F(1) = f(1, 0) = 1$ we get that $F((-2)^k) = (-2)^k$ for all $k \in \mathbb{N}_0$. Thus $F(z) = z$ for infinitely many values of z , so $F(z) \equiv z$.

Consequently $f(x, y) = x - 2y$ if $x + y = 1$. Hence for all x, y with $x + y \neq 0$ we have

$$\begin{aligned} f(x, y) &= (x + y)^n f\left(\frac{x}{x + y}, \frac{y}{x + y}\right) \\ &= (x + y)^n \left(\frac{x}{x + y} - 2\frac{y}{x + y}\right) = (x + y)^{n-1}(x - 2y). \end{aligned}$$

The same is true for $x + y = 0$, because f is a polynomial.

Remark. Using the problem above one can describe all polynomials f satisfying the given conditions. To do this use the well-known fact that every polynomial can be written in a unique way as a sum of homogeneous polynomials. Thus every polynomial with the given properties has the form $f(x, y) = (x - 2y)p(x + y)$, where $p(t)$ is a polynomial with $p(1) = 1$.

Problem 16.15. (MR 6, 2008, M. Becheanu, T. Dumitrescu) Let $P \in \mathbb{Z}[X]$ be a polynomial such that $P(1) = P(-1)$. Prove that there is a polynomial $Q(x, y)$ with integer coefficients such that

$$P(t) = Q(t^2 - 1, t^3 - t)$$

for all $t \in \mathbb{R}$.

Hint. Use induction on degree of P . To prove the inductive step denote by a the leading coefficient of P and consider the polynomial

$$R(x) = P(x) - a(x^2 - 1)^n \text{ when } \deg P = 2n \text{ is even}$$

and

$$R(x) = P(x) - a(x^2 - 1)^{n-1}(x^3 - x) \text{ when } \deg P = 2n + 1 \text{ is odd.}$$

Problem 16.16. Let $P \in \mathbb{R}[X]$ be a polynomial such that $P(\sin t) = P(\cos t)$ for all real t . Prove that there exists a polynomial $Q \in \mathbb{R}[X]$ such that

$$P(x) = Q(x^4 - x^2) \text{ for all } x \in \mathbb{R}.$$

Solution. Since $P(\sin t) = P(\cos t) = P(\cos(-t)) = P(\sin(-t)) = P(-\sin t)$, we conclude that $P(u) = P(-u)$ for all $u \in [-1, 1]$. Hence $P(x) = P(-x)$ for all $x \in \mathbb{R}$. Comparing coefficients in this identity shows that those with odd indexes vanish, so $P(x) = T(x^2)$, where $T \in \mathbb{R}[X]$. Thus we get

$$T(\sin^2 t) = T(\cos^2 t) = T(1 - \sin^2 t)$$

which implies that $T(x) = T(1 - x)$. Set

$$R(y) = T\left(\frac{1}{2} + y\right).$$

Then for $x = \frac{1}{2} + y$ we get $R(y) = R(-y)$ and we conclude as above that $R(y) = S(y^2)$. Hence

$$T(x) = R\left(x - \frac{1}{2}\right) = S\left(x^2 - x + \frac{1}{4}\right).$$

Now we set $Q(x) = S\left(x + \frac{1}{4}\right)$ and get that

$$T(x) = Q(x^2 - x) \text{ and } P(x) = T(x^2) = Q(x^4 - x^2).$$

Note that these polynomials satisfy the condition as

$$\sin^4 x - \sin^2 x = \cos^4 x - \cos^2 x = -\sin^2 x \cos^2 x.$$

Problem 16.17. (G. Dospinescu) Find all polynomials $f(x, y, z)$ with real coefficients such that

$$f\left(a + \frac{1}{a}, b + \frac{1}{b}, c + \frac{1}{c}\right) = 0$$

whenever $abc = 1$.

Solution. It is natural to set $x = a + \frac{1}{a}$, $y = b + \frac{1}{b}$, $z = c + \frac{1}{c}$. The key observation now is that the condition $abc = 1$ is equivalent to

$$x^2 + y^2 + z^2 - xyz - 4 = 0, \quad |x|, |y|, |z| \geq 2.$$

Hence it is clear that the polynomials $f(x, y, z)$ divisible by $x^2 + y^2 + z^2 - xyz - 4$ are solutions of the problem. We shall prove that these are all solutions. Using the long division theorem for polynomials we see that there are polynomials $g(x, y, z), h(y, z), k(y, z)$ with real coefficients such that

$$f(x, y, z) = (x^2 + y^2 + z^2 - xyz - 4)g(x, y, z) + xh(y, z) + k(y, z).$$

For arbitrary y, z with $\min(|y|, |z|) > 2$ set

$$x + \frac{yz + \sqrt{(y^2 - 4)(z^2 - 4)}}{2}.$$

Then $x^2 + y^2 + z^2 - xyz - 4 = 0$ and we get

$$(yz + \sqrt{(y^2 - 4)(z^2 - 4)})h(y, z) + k(y, z) = 0$$

whenever $\min(|y|, |z|) > 2$. Note now that the function $\sqrt{x^2 - 4}$ is not rational (i.e. not the ratio of two polynomials). Indeed, if

$$\sqrt{x^2 - 4} = \frac{p(x)}{q(x)}$$

for some relatively prime polynomials $p(x)$ and $q(x)$ then it follows that 2 is a double zero of $x^2 - 4$, a contradiction. Consequently the above identity implies that for each z with $|z| > 2$ we have $h(y, z) = k(y, z) = 0$ for all y with $|y| > 2$. Hence the Identity principle implies that $h(y, z) = k(y, z) = 0$ for all y, z and therefore $f(x, y, z)$ is divisible by $x^2 + y^2 + z^2 - xyz - 4$.

Problem 16.18. Prove that the equation $f^4 + g^4 = h^2$ has only trivial solutions in $\mathbb{C}[X]$.

Proof. Use Theorem 16.3.

Problem 16.19. Prove that there are no nonconstant polynomials $f, g \in \mathbb{C}[X]$ such that $f^3 - g^2$ is a nonzero constant.

First Solution. Let $h = f^3 - g^2$ be a nonzero constant. Then the polynomials f and g are relatively prime and from Davenport's theorem (Theorem 16.2) we get $\deg f \leq 2 \deg h - 2 = -2$, a contradiction.

Second Solution. Note first that there do not exist nonconstant polynomials $p, q \in \mathbb{C}[X]$ and a nonzero constant $\alpha \in \mathbb{C}[X]$ such that $p(x)^2 = \alpha + q(x)^2$. Indeed, if

$$\alpha = p(x)^2 - q(x)^2 = (p(x) - q(x))(p(x) + q(x)),$$

then the polynomials $p(x) - q(x)$ and $p(x) + q(x)$ are constant, so $p(x)$ and $q(x)$ are constant, a contradiction. Returning to the problem let $a = f^3 - g^2$ be a nonzero constant. Then

$$g(x)^2 = (f(x) - b)(f(x)^2 + bf(x) + b^2),$$

where $b = a^{1/3} \neq 0$. The polynomials $f(x) - b$ and $f(x)^2 + bf(x) + b^2$ are relatively prime. Indeed, if they share a common root x_0 , then

$$f(x_0) = b, f(x_0)^2 + bf(x_0) + b^2 = 3b^2 = 0,$$

a contradiction. Therefore each of these polynomials is a square of a polynomial in $\mathbb{C}[X]$. Hence $f(x) = h(x)^2 + b$, thus

$$\begin{aligned} (h(x)^2 + b)^2 + b(h(x)^2 + b) + b^2 &= h(x)^4 + 3bh(x)^2 + 3b^2 \\ &= (h(x)^2 + c)(h(x)^2 + d), \end{aligned}$$

where

$$c = \frac{-3b + ib\sqrt{3}}{2}, d = \frac{-3b - ib\sqrt{3}}{2}.$$

By the same argument as above the polynomials $h(x)^2 + c$ and $h(x)^2 + d$ are relatively prime. Thus $h(x)^2 + c = p(x)^2$ and the remark at the beginning leads to $c = 0$, a contradiction.

Problem 16.20. Prove that the equation $f^2 = g^3 + g$ has only trivial solutions in rational functions with complex coefficients.

Solution. Write $f = \frac{n}{N}, g = \frac{m}{M}$, for polynomials n, m, N, M with complex coefficients such that n, N and m, M are relatively prime. Clearing denominators we get $n^2M^3 = m^3N^2 + mM^2N^2$. This implies that $N^2|M^3$ and $M^2|N^2$. Hence $N = PM, P \in \mathbb{C}[X]$ and $P^2|M$. Thus $M = QP^2$ and $N = QP^3$, $P, Q \in \mathbb{C}[X]$. Plugging in the identity above we see that $Q|m^3$ and since m, M

are relatively prime it follows that Q is a constant. Hence setting $Q = c$ for some $c \in \mathbb{C}$ we get $M^3 = cN^2$. Writing N for $\sqrt{c}N$ we may assume that $M^3 = N^2$. Now the unique factorization of complex polynomials shows that $M = e^2$ and $N = e^3$ for some $e \in \mathbb{C}[X]$. Thus $n^2e^6 = m^3e^6 + me^{10}$ and therefore $n^2 = m(m^2 + e^4)$. Since the polynomials m and $m^2 + e^4$ are relatively prime we conclude that $m = k^2$ is a square and we have $n^2 = k^2(k^4 + e^4)$. This in turn implies $k^4 + e^4 = z^2$ for some $z \in \mathbb{C}[X]$, which contradicts Problem 16.18.

19.17 Functional Inequalities

Problem 17.1. (Romania 2011) Find all functions $f : [0, 1] \rightarrow \mathbb{R}$ such that

$$|x - y|^2 \leq |f(x) - f(y)| \leq |x - y|$$

for all $x, y \in [0, 1]$.

Solution. Let $P(x, y)$ be the assertion $|x - y|^2 \leq |f(x) - f(y)| \leq |x - y|$. Setting $y \rightarrow x$ in $P(x, y)$, we conclude that $f(x)$ is continuous. If $f(a) = f(b)$, then $P(a, b) \implies (a - b)^2 \leq 0$ and so $a = b$ and $f(x)$ is injective. Now $f(x)$ continuous and injective implies monotonous. Note also that if $f(x)$ is a solution then $f(x) + c$ and $c - f(x)$ are solutions too. So we may assume that $f(0) = 0$ and $f(x)$ is increasing. Then $P(1, 0) \implies f(1) = 1$ and so $f(x) \in [0, 1]$. Now $P(x, 0) \implies f(x) \leq x$ and $P(x, 1) \implies 1 - f(x) \leq 1 - x$. Thus $f(x) = x$ which indeed is a solution. Hence all solutions of the problem are the functions $f(x) = x + a$ and $f(x) = a - x$, where a is a real number.

Problem 17.2. (Romania TST 2007). Prove that every function $f : \mathbb{Q} \rightarrow \mathbb{R}$ such that

$$|f(x) - f(y)| \leq (x - y)^2$$

for all $x, y \in \mathbb{Q}$ is constant.

First Solution. We have

$$\left| f(x) - f\left(\frac{x+y}{2}\right) \right| \leq \frac{|x-y|^2}{4}$$

and

$$\left| f(y) - f\left(\frac{x+y}{2}\right) \right| \leq \frac{|x-y|^2}{4}.$$

So the triangle inequality implies that

$$|f(x) - f(y)| \leq \frac{|x-y|^2}{2}.$$

Now it follows by induction on n that

$$|f(x) - f(y)| \leq \frac{|x-y|^2}{2^n},$$

for all $n \in \mathbb{N}$ which implies that $f(x) = f(y)$. Thus $f(x) = \text{const.}$

Second Solution. Take two rational numbers $x < y$. Partition the interval $[x, y]$ in n subintervals by $x = x_0 < x_1 < \cdots < x_n = y$, with $x_{i+1} - x_i = \frac{y-x}{n}$,

for $0 \leq i \leq n-1$. Then $|f(x_{i+1}) - f(x_i)| \leq (x_{i+1} - x_i)^2 = \frac{(y-x)^2}{n^2}$.

But then, by telescoping and triangle inequality,

$$|f(y) - f(x)| \leq \sum_{i=0}^{n-1} |f(x_{i+1}) - f(x_i)| \leq \sum_{i=0}^{n-1} (x_{i+1} - x_i)^2 = \frac{(y-x)^2}{n}.$$

Since n can be taken arbitrarily large, this implies $f(y) = f(x)$, thus the only such functions are the constant ones.

Problem 17.3. (Bulgaria 2008) Let $f : (0, \infty) \rightarrow (0, \infty)$ be a function such that

$$2f(x^2) \geq xf(x) + x$$

for all $x > 0$. Prove that $f(x^3) \geq x^2$ for all $x > 0$.

Solution. Obviously $f(x^2) > x/2$. Assume that we have proved

$$f(x) > x^{a_n}/2^{1/2^n}$$

for all $x > 0$ ($a_0 = 1/2$). Since $f(x^2) \geq x\sqrt{f(x)}$, then

$$f(x) \geq \sqrt{x}\sqrt{f(\sqrt{x})} > x^{a_{n+1}}/2^{1/2^{n+1}},$$

where $a_{n+1} = 1/2 + a_n/4$. Now the equality $a_{n+1} - 2/3 = (a_n - 2/3)/4$ implies that $a_n \rightarrow 2/3$. Having in mind that $2^{1/2^n} \rightarrow 1$, we get that $f(x) \geq x^{2/3}$ for $x > 0$.

Problem 17.4. (China 1990) Let $f : [0, \infty) \rightarrow \mathbb{R}$ be a function such that

$$f(x)f(y) \leq y^2 f\left(\frac{x}{2}\right) + x^2 f\left(\frac{y}{2}\right)$$

for all $x, y \geq 0$ and $|f(x)| \leq M$ for $0 \leq x \leq 1$, where M is a fixed constant. Prove that $f(x) \leq x^2$ for all $x \geq 0$.

Solution. We shall prove the stronger inequality $f(x) \leq \frac{x^2}{2}$. To do this set $x = y$ in the given inequality to obtain

$$f(x)^2 \leq 2x^2 f\left(\frac{x}{2}\right).$$

In particular, $f(x) \geq 0$ for all $x \geq 0$ and $f(0) = 0$. So, we have to prove the desired inequality for $x > 0$. Set $g(x) = \frac{2f(x)}{x^2}$ for $x > 0$. Then $g^2(x) \leq g\left(\frac{x}{2}\right)$ and it follows by induction that

$$g^{2^n}(x) \leq g\left(\frac{x}{2^n}\right)$$

for all $x > 0$ and $n \in \mathbb{N}$. Note that $g(x) \geq 0$. Hence

$$g(x) \leq \sqrt[2^n]{g\left(\frac{x}{2^n}\right)} \leq \sqrt[2^n]{M \frac{2^{2n+1}}{x^2}}$$

if $\frac{x}{2^n} \in (0, 1)$. Now letting $n \rightarrow \infty$ and using the fact that $\lim_{n \rightarrow \infty} \frac{n}{2^n} = 0$ we get $g(x) \leq 1$. Thus $f(x) \leq \frac{x^2}{2}$ for all $x \geq 0$.

Problem 17.5. (G. Dospinescu) Find all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ such that

$$f(n+1) > \frac{f(n) + f(f(n))}{2}$$

for all $n \in \mathbb{N}$.

Solution. We will prove first of all that $f(n) \geq n$. For this, it is enough to prove that $f(n) \geq j$ if $n \geq j$. We prove this by induction on j . For $j = 1$ this is clear. If it holds for j , then take $n > j$. We have $n - 1 \geq j$, so $f(n - 1) \geq j$, $f(f(n - 1)) \geq j$ and from the functional equation we get $f(n) > j$, thus $f(n) \geq j + 1$.

Next, we prove that $f(n) - n \leq 1$ for all n . Suppose that for some n_0 we have $f(n_0) - n_0 \geq 2$. Then $f(f(n_0)) \geq f(n_0 + 2) \geq n_0 + 2$ and from the functional equation we get $f(n_0 + 1) \geq n_0 + 3$. By induction we get that $f(n) \geq n + 2$ for all $n \geq n_0$. Therefore $2f(n + 1) > f(n) + f(n + 2)$ for $n \geq n_0$, which shows that the sequence $f(n + 1) - f(n)$ becomes decreasing from n_0 on, obviously impossible.

Finally, the identity function is a solution, so we assume now that n_0 is the smallest integer for which $f(n_0) = n_0 + 1$. Using the previous step and the functional equation we obtain that $f(n) = n + 1$ for all $n \geq n_0$. One can easily check that all these functions are solutions.

Problem 17.6. (Romania 2001) Prove that there is no function $f : (0, \infty) \rightarrow (0, \infty)$ such that

$$f(x + y) \geq f(x) + yf(f(x))$$

for all $x, y \in (0, \infty)$.

Hint. Use the same arguments as in the second part of the solution to Problem 17.6.

Problem 17.7. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x) \geq x + 1 \text{ and } f(x + y) \geq f(x)f(y)$$

for all $x, y \in \mathbb{R}$.

Solution. The second condition implies that

$$f(x_1 + x_2 + \cdots + x_n) \geq f(x_1)f(x_2) \cdots f(x_n)$$

for all $x_i \in \mathbb{R}$, $1 \leq i \leq n$. In particular $f(x) \geq f\left(\frac{x}{n}\right)^n$ for all $x \in \mathbb{R}$ and $n \in \mathbb{N}$. Hence

$$f(x) \geq f\left(\frac{x}{n}\right)^n \geq \left(1 + \frac{x}{n}\right)^n$$

and letting $n \rightarrow \infty$ gives $f(x) \geq e^x$. In particular, $f(0) \geq 1$. On the other hand, $f(0) \geq (f(0))^2$ and therefore $f(0) = 1$. Now

$$1 = f(0) \geq f(x)f(-x) \geq e^x \cdot e^{-x} = 1$$

which shows that $f(x) = e^x$. It is easy to check that this function satisfies the conditions of the problem.

Problem 17.8. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x+y) \geq (y+1)f(x)$$

for all $x, y \in \mathbb{R}$.

Solution. As in the solution to Problem 17.6 we get that $f \geq 0$,

$$f(x) \geq \left(1 + \frac{x}{n}\right)^n \cdot f(0) \text{ and } f(0) \geq \left(1 - \frac{x}{n}\right)^n \cdot f(x).$$

Since $\left(1 \pm \frac{x}{n}\right)^n \rightarrow e^{\pm x}$, it follows that $f(x) = f(0)e^x$. Conversely, for every $c \geq 0$ the function $f(x) = ce^x$ satisfies the given condition since $e^y \geq 1 + y$ for all $y \in \mathbb{R}$ (prove!).

Problem 17.9. (Hungary 2008) Determine all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfying for all real numbers x and y the relation $f(xy) \leq xf(y)$.

Solution. Set $y = 0$ to get $(1-x)f(0) \leq 0$ for all $x \in \mathbb{R}$ which implies $f(0) = 0$. On the other hand for $x, y > 0$ we have

$$xf(y) = xf\left(\frac{y}{x}x\right) \leq x\frac{y}{x}f(x) = yf(x).$$

Similarly, $yf(x) \leq xf(y)$, hence $xf(y) = yf(x)$, or $\frac{f(x)}{x} = \frac{f(y)}{y}$. Hence there is a constant a such that $f(x) = ax$ for all $x \geq 0$.

We also have

$$xf(-y) = xf\left(\left(\frac{y}{x}\right)(-x)\right) \leq x\frac{y}{x}f(-x) = yf(-x).$$

Similarly, $yf(-x) \leq xf(-y)$. Hence $xf(-y) = yf(-x)$ which shows that there exists a constant b such that $f(x) = bx$ for all $x < 0$.

But then $a = f(1) = f((-1)(-1)) \leq -f(-1) = -(-b) = b$, so we need $a \leq b$. Conversely, one can easily check that, for all $a \leq b$, the function

$$f(x) = \begin{cases} ax, & \text{for } x \geq 0 \\ bx, & \text{for } x \leq 0 \end{cases}$$

satisfies the given inequality.

Problem 17.10. (Bulgaria 1998) Prove that there is no function $f : (0, \infty) \rightarrow (0, \infty)$ such that

$$f(x)^2 \geq f(x+y)(f(x)+y)$$

for all $x, y \in (0, \infty)$.

Solution. Suppose that there is a function f with the given property. Then

$$f(x) - f(x+y) \geq \frac{f(x)y}{f(x)+y} \quad (1)$$

which shows that f is a strictly decreasing function. Given an $x \in (0, \infty)$ we choose an $n \in \mathbb{N}$ such that $nf(x+1) \geq 1$. Then

$$f\left(x + \frac{k}{n}\right) - f\left(x + \frac{k+1}{n}\right) \geq \frac{f\left(x + \frac{k}{n}\right) \cdot \frac{1}{n}}{f\left(x + \frac{k}{n}\right) + \frac{1}{n}} > \frac{1}{2n}$$

for any $k \in \mathbb{N}, k < n$. (Note that $nf\left(x + \frac{k}{n}\right) > nf(x+1) > 1$.) Adding up these inequalities for $k = 0, 1, \dots, n-1$ we get

$$f(x) - f(x+1) > \frac{1}{2}.$$

Now take an $m \in \mathbb{N}$ such that $m \geq 2f(x)$. Then

$$f(x) - f(x+m) = (f(x) - f(x+1)) + \dots + (f(x+m-1) - f(x+m)) > \frac{m}{2} \geq f(x).$$

Hence $f(x+m) < 0$, a contradiction.

Problem 17.11. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ that are continuous at 0 and

$$f(2x) - 3x^2 - x \leq f(x) \leq f(3x) - 8x^2 - 2x$$

for all $x \in \mathbb{R}$.

Solution. For all $t \in \mathbb{R}$ and $k \in \mathbb{N}$ we have

$$f(t) - f\left(\frac{t}{2^n}\right) = \sum_{k=1}^n \left(f\left(\frac{t}{2^{k-1}}\right) - f\left(\frac{t}{2^k}\right) \right) \leq \sum_{k=1}^n \left(\frac{3t^2}{4^k} + \frac{t}{2^k} \right).$$

Letting $n \rightarrow \infty$, we get $f(t) - f(0) \leq t^2 + t$. Proceeding in the same way by using the inequality $f(3x) - f(x) \geq 8x^2 + 2x$, we get $f(t) - f(0) \geq t^2 + t$. Hence all functions with the required properties have the form $f(x) = x^2 + x + a$, $a \in \mathbb{R}$.

Problem 17.12. (ZIMO 2009) Find all real numbers a such that there exists a function $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfying the inequality

$$x + af(y) \leq y + f(f(x))$$

for all $x, y \in \mathbb{R}$.

Solution. Let first $a \geq 0$. Setting $c_1 = af(0)$ and $c_2 = f(f(0))$, it follows that

$$x + c_1 \leq f(f(x)), \quad af(y) \leq y + c_2.$$

Hence

$$a^2(x + c_1) \leq a^2 f(f(x)) \leq a(f(x) + c_2) \leq x + c_2(a + 1)$$

and then $(a^2 - 1)x \leq c_2(a + 1) - c_1a^2$ for all x . This is possible only if $a = 1$.

In this case the function $f(x) = x$ obviously satisfies the desired condition.

If $a < 0$, then, for example, the function $f(x) = c|x|$, where $c = \max\{1, -1/a\}$ also satisfies the given inequality.

Problem 17.13. (IMO 2009) Find all functions $f: \mathbb{N} \rightarrow \mathbb{N}$ such that $x, f(y)$ and $f(y + f(x) - 1)$ are the sides of a triangle for all $x, y \in \mathbb{N}$.

Solution. Put $x = 1$, then the condition tells us that $f(y) = f(y + f(1) - 1)$. We claim that $f(1) = 1$. Otherwise $f(1) - 1 > 0$, which means that f repeats itself every $f(1) - 1$ numbers. This means that f can take finitely many values, so if we take x sufficiently large, $x, f(y), f(y + f(x) - 1)$ cannot be the sides of a triangle. The condition for $y = 1$ implies that $f(f(x)) = x$ and we conclude that f is bijective.

Now we shall prove by induction on n that $f(n) = (n-1)f(2) - (n-2)$ for all $n \in \mathbb{N}$. Since f is bijective it follows that $x, y, f(f(x) + f(y) - 1)$ are sides of a triangle. This means that $f(f(x) + f(y) - 1) < x + y$. Take $x = y = 2$, then $f(2f(2) - 1) < 4$, i.e. it can either be 1, 2 or 3. But the first two cases are impossible since in both of them we get $f(2) = 1$, contradicting to bijectivity of f . In the third case we get $2f(2) - 1 = f(3)$, proving the hypothesis for $n = 3$. The induction step involves exactly the same argument, by taking $a = 2, b = n$, and argue that $f(f(2) + f(n) - 1) = n + 1$.

The formula for f tells us that f is strictly increasing. Suppose that $f(i) = i$ for $i = 1, 2, \dots, k-1$. If $f(k) > k$, then as f is increasing nothing can map back to k , contradicting to the bijectivity of f . So $f(k) = k$. By induction $f(n) = n$ for all n .

Problem 17.14. (IMO 2011) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be such a function that

$$f(x+y) \leq yf(x) + f(f(x))$$

for all $x, y \in \mathbb{R}$. Prove that $f(x) = 0$ for all $x \leq 0$.

Solution. Substituting $y = t - x$, we may rewrite the given condition as

$$f(t) \leq tf(x) - xf(x) + f(f(x)). \quad (1)$$

Now setting $t = f(a)$, $x = b$ and $t = f(b)$, $x = a$, we get

$$f(f(a)) - f(f(b)) \leq f(a)f(b) - bf(b), \quad f(f(b)) - f(f(a)) \leq f(a)f(b) - af(a)$$

and hence $af(a) + f(b) \leq 2f(a)f(b)$. If $b = f(a)$, it follows that $af(a) \leq 0$ for any a . In particular, $f(a) \geq 0$ for $a < 0$. Suppose now that $f(x) > 0$ for some x . Then (1) implies that $f(t) < 0$ if $t < x - f(f(x))/f(x)$, a contradiction. Hence $f(x) \leq 0$ for any x , and $f(x) = 0$ for $x < 0$. Finally, setting $t = x < 0$ in (1), we obtain $f(x) \leq f(f(x))$. Since $f(x) = 0$, this means $0 \leq f(0)$. On the other hand, $f(0) \leq 0$ and so $f(x) = 0$ for any $x \leq 0$.

Remark. The above solution implies that the functions satisfying (1) have the form $f(x) = 0$ for $x \leq 0$ and $f(x) = -g(x)$ for $x > 0$, where $g : (0, \infty) \rightarrow [0, \infty)$ is an arbitrary function with $g(x+y) \leq yg(x)$, $x, y > 0$. For example, we may take $g(x) = ce^x - d$, where $c \geq \max(d, 0)$ (use that $e^x \geq 1 + x$).

Problem 17.15. (IMAR 2009) Show that for each function $f : (0, \infty) \rightarrow (0, \infty)$ there exist $x, y \in (0, \infty)$ such that $f(x + y) < yf(f(x))$.

First Solution. Assume $f(x + y) \geq yf(f(x))$ for all $x, y > 0$. Take $a > 1$ and $t = f(f(a)) > 0$. Then for $b \geq a \left(1 + \frac{1}{t} + \frac{1}{t^2}\right) > a$ we have

$$f(b) = f(a + (b - a)) \geq (b - a)f(f(a)) = (b - a)t \geq a \left(1 + \frac{1}{t}\right) > a$$

and we get

$$f(f(b)) = f(a + (f(b) - a)) \geq (f(b) - a)t \geq a.$$

Furthermore, if we take $x \geq \frac{ab + 2}{a - 1} > b$, it results

$$f(x) = f(b + (x - b)) \geq (x - b)f(f(b)) \geq (x - b)a \geq x + 2,$$

hence $f(x) > x + 1$. But now

$$f(f(x)) = f(x + (f(x) - x)) \geq (f(x) - x)f(f(x))$$

implies $1 \geq f(x) - x$, i.e. $f(x) \leq x + 1$, a contradiction.

Second Solution. Suppose, if possible, that $f(x + y) \geq yf(f(x))$, whatever $x > 0$ and $y > 0$. Fix a positive real number a , write $\alpha = f(f(a))$ to get $f(t) = f(a + (t - a)) \geq \alpha(t - a)$ for $t > a$, and deduce that $f(t) > a$ for all large enough t . For such t , write

$$f(f(t)) = f(a + (f(t) - a)) \geq \alpha(f(t) - a) \geq \alpha(\alpha(t - a) - a),$$

to infer that $\beta = f(f(b)) > 1$, for some large enough b .

Hence for $t > b + (b - 1)/(\beta - 1)$ we get

$$f(t) = f(b + (t - b)) \geq \beta(t - b) > t + 1. \quad (1)$$

But then

$$f(f(t)) = f(t + (f(t) - t)) \geq (f(t) - t)f(f(t))$$

which implies $f(t) \leq t + 1$, a contradiction to (1).

Problem 17.16. (IMO Shortlist 2009) Prove that for each function $f : \mathbb{R} \rightarrow \mathbb{R}$ there exist $x, y \in \mathbb{R}$ such that $f(x - f(y)) > yf(x) + x$.

Solution. Assume that

$$f(x - f(y)) \leq yf(x) + x \quad (1)$$

for all real x, y . Let $a = f(0)$. Setting $y = 0$ in (1) gives $f(x - a) \leq x$ for all real x , or equivalently

$$f(y) \leq y + a \quad (2)$$

for all real y . Setting $x = f(y)$ in (1) yields in view of (2)

$$a = f(0) \leq yf(f(y)) + f(y) \leq yf(f(y)) + y + a.$$

This implies $0 \leq y(f(f(y)) + 1)$ and thus

$$f(f(y)) \geq -1 \quad (3)$$

for all $y > 0$. From (2) and (3) we obtain $-1 \leq f(f(y)) \leq f(y) + a$ for all $y > 0$, so

$$f(y) \geq -a - 1 \quad (4)$$

for all $y > 0$. Now we show that $f(x) \leq 0$ for all real x . Assume the contrary, i.e. $f(x) > 0$ for some x . Take any y such that $y < x - a$ and $y < \frac{-a - x - 1}{f(x)}$. Then in view of (2), $x - f(y) \geq x - (y + a) > 0$ and by (1) and (4) we obtain

$$yf(x) + x \geq f(x - f(y)) \geq -a - 1,$$

whence

$$y \geq \frac{-a - x - 1}{f(x)}$$

contrary to our choice of y . Thus $f(x) \leq 0$ for all real x . In particular, $a = f(0) \leq 0$ and (2) then yields $f(x) \leq x$ for all real x . Now choose y such that $y > 0$ and $y > -f(-1) - 1$, and set $x = f(y) - 1$. From (1), (4) and the above inequality we obtain

$$\begin{aligned} f(-1) = f(x - f(y)) &\leq yf(x) + x = yf(f(y) - 1) + f(y) - 1 \\ &\leq y(f(y) - 1) - 1 \leq -y - 1, \end{aligned}$$

i.e. $y \leq -f(-1) - 1$, a contradiction to the choice of y .

19.18 Miscellaneous Problems

19.18.1 Inductive Arguments

Problem 18.1. Find all functions $f: \mathbb{N} \rightarrow \mathbb{R}$ such that $f(1) \neq 0$ and

$$f(1)^2 + f(2)^2 + \dots + f(n)^2 = f(n)f(n+1)$$

for all $n \in \mathbb{N}$.

Solution. It is clear that $f(n) \neq 0$ as the LHS of the condition is positive. Now by subtracting the condition for n from that for $n+1$ we get

$$f(n+1)^2 = f(n+1)(f(n+2) - f(n)).$$

Thus reducing by $f(n+1) \neq 0$ we get the Fibonacci-type relation

$$f(n+2) = f(n+1) + f(n).$$

If we set $a = f(1)$ then taking $n = 1$ we get $f(2) = a$. Thus $f(n) = aF_n$ by induction on n , using the recurrent relation. It is clear that this function satisfies the condition by induction, as this is clear for $n = 1$ and the induction step was proven to be equivalent to the true relation

$$f(n+2) = f(n+1) + f(n).$$

Problem 18.2. Find all functions $f: \mathbb{N} \rightarrow \mathbb{N}$ such that

$$f(1)^3 + f(2)^3 + \dots + f(n)^3 = (f(1) + f(2) + \dots + f(n))^2$$

for all $n \in \mathbb{N}$.

Solution. The function $f(x) = x$ comes to mind, in view of the famous identity

$$1^3 + 2^3 + \dots + n^3 = \left(\frac{n(n+1)}{2} \right)^2 = (1 + 2 + \dots + n)^2.$$

We shall prove this is the only solution, proving in the meantime the identity, too. By setting $n = 1$ we get $f(1) = 1$. If we subtract the identity for n from the identity for $n + 1$ we get

$$f(n+1)^3 = f(n+1)(2f(1) + 2f(2) + \dots + 2f(n) + f(n+1))$$

so we get an identity of a smaller degree:

$$f(n+1)^2 = 2f(1) + 2f(2) + \dots + 2f(n) + f(n+1).$$

By setting $n = 1$ we get $f(2) = 2$ and doing the same procedure with the identity above we get

$$f(n+2)^2 - f(n+1)^2 = f(n+2) + f(n+1),$$

i.e. $f(n+2) - f(n+1) = 1$. It is thus clear by induction on n that $f(n) = n$. The verification is clear by the same induction, as we actually worked by equivalence.

Problem 18.3.(IMO 1981) Assume that $f(x, y)$ is defined for all nonnegative integers x and y , and that the following three equations are satisfied:

- i) $f(0, y) = y + 1$;
- ii) $f(x + 1, 0) = f(x, 1)$;
- iii) $f(x + 1, y + 1) = f(x, f(x + 1, y))$.

Determine $f(4, 1981)$.

Solution. We have $f(1, 0) = f(0, 1) = 2$. Then we get

$$f(1, y + 1) = f(0, f(1, y)) = f(1, y) + 1;$$

hence $f(1, y) = y + 2$ for $y \geq 0$. Next we find that $f(2, 0) = f(1, 1) = 3$ and $f(2, y + 1) = f(1, f(2, y)) = f(2, y) + 2$, from which $f(2, y) = 2y + 3$. In particular, $f(2, 2) = 7$. Further, $f(3, 0) = f(2, 1) = 5$ and $f(3, y + 1) = f(2, f(3, y)) = 2f(3, y) + 3$. This gives by induction $f(3, y) = 2^{(y+3)} - 3$. In particular, $f(3, 3) = 61$. Finally, from $f(4, 0) = f(3, 1) = 13$ and $f(4, y + 1) = f(3, f(4, y)) = 2^{f(4, y)+3} - 3$, we conclude that

$$f(4, y) = 2^{2^{\dots^2}} - 3,$$

where we have $y + 3$ two's.

Problem 18.4. (Putnam 1992) Prove that $f(n) = 1 - n$ is the only integer-valued function defined on the integers that satisfies the following conditions:

- (i) $f(f(n)) = n$ for all integers n ;
- (ii) $f(f(n+2) + 2) = n$ for all integers n ;
- (iii) $f(0) = 1$.

Solution. We show that $f(k) = 1 - k$ and $f(1 - k) = k$ for every $k \geq 0$. For $k = 0$ and 1 we have $f(0) = 1$ and $f(1) = f(f(0)) = 0$. For $k \geq 2$ by induction we have $f(k-2) = 3 - k$ and $f(3 - k) = k - 2$. Note that

$$f(n+2) + 2 = f(f(f(n+2) + 2)) = f(n)$$

So $f(k) = f(k-2) - 2 = 1 - k$ and $f(1 - k) = f(3 - k) + 2 = k$ as desired. As k and $1 - k$ for $k \geq 1$ cover the integers, we are done.

Problem 18.5. Find all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ such that

$$f^{(19)}(n) + 97f(n) = 98n + 232$$

for all $n \in \mathbb{N}$.

Solution. We have

$$f(n) \leq \frac{98n + 232}{97} = n + 2 + \frac{n + 38}{97}.$$

So $f(n) \leq n + 2$ when $n \leq 59$. Now it is not hard to show that $f^{(19)}(n) \leq n + 38$ for all $n \leq 21$. Then for $n \leq 21$ we have

$$f^{(19)}(n) + 97f(n) = 98n + 232 = 97(n + 2) + n + 38f(n) \geq f^{(19)}(n) + 97f(n)$$

which shows that $f(n) = n + 2$, $f^{(19)}(n) = n + 38$. Actually $f^{(19)}(20) = 20 + 38$ and $f^{(19)}(21) = 21 + 38$ imply $f(n) = n + 2$ for all $n \leq 57$.

Now we will prove by induction that $f(n) = n + 2$ for all n . Suppose $f(n) = n + 2$ for all $n < m$. Then we have

$$\begin{aligned} f^{(19)}(m - 36) &= f(m) = 98(m - 36) + 232 - 97f(m - 36) \\ &= 98(m - 36) + 232 - 97(m - 34) = m + 2. \end{aligned}$$

Hence the only solution of the equation is $f(n) = n + 2$.

Problem 18.6. (IMO Shortlist 2011) Determine all pairs (f, g) of functions from the set of positive integers to itself that satisfy

$$f^{(g(n)+1)}(n) + g^{(f(n))}(n) = f(n+1) - g(n+1) + 1$$

for every positive integer n .

Solution. The given identity implies $f(n+1) \geq g(n+1) + 1 \geq 2$ for all $n \geq 1$ and so $f(n) \geq 2$ for all $n \geq 2$. Hence we get $f(n+1) \geq g(n+1) + 2 \geq 3$ for all $n \geq 2$ and so $f(n) \geq 3$ for all $n \geq 3$. Using the same reasoning we easily get by induction that $f(n) \geq n$ for all $n \geq 1$. So

$$f(n+1) = f^{(g(n))}(f(n)) + g^{(f(n))}(n) + g(n+1) - 1 \geq f(n) + 1$$

and $f(n)$ is increasing.

If $f(n) \geq n+1$ for some n , then $f(f(n)) \geq f(n+1)$ and so $f^{(g(n)+1)}(n) \geq f(n+1)$ and then $LHS \geq f(n+1) + 1$ while $RHS \leq f(n+1)$, a contradiction.

So $f(n) = n$ for all n and the equation takes the form $g^{(n)}(n) = 2 - g(n+1)$. So $g(n) = 1$ for all $n \geq 2$. Setting $n = 1$ gives $g(1) = 2 - g(2) = 1$. Hence $f(n) = n$ and $g(n) = 1$ for all $n \in \mathbb{N}$, and obviously these functions satisfy the given equation.

Problem 18.7. (Canada 2002) Find all functions $f : \mathbb{N}_0 \rightarrow \mathbb{N}_0$ such that

$$mf(n) + nf(m) = (m+n)f(m^2 + n^2)$$

for all $m, n \in \mathbb{N}_0$.

First Solution. We shall prove that the only solutions are the constant functions. To do this we first plug in $m = 0$ and $n = 1$ to get $f(0) = f(1)$. Now assume that there exist $a, b \in \mathbb{N}_0$ neither equal to 0 and such that $f(a) > f(b)$. Then $af(b) + bf(a) = (a+b)f(a^2 + b^2)$ and therefore $(a+b)f(b) < (a+b)f(a^2 + b^2) < (a+b)f(a)$, i.e. $f(b) < f(a^2 + b^2) < f(a)$. Proceeding in the same way with a and $a^2 + b^2$ we get another positive integer $c = (a^2 + b^2)^2 + a^2$ such that $f(a^2 + b^2) < f(c) < f(a)$ and it follows by induction that there are infinitely many positive integers in the interval $(f(b), f(a))$, a contradiction. So, the only solutions are the constant functions.

Second Solution. We have $0f(n) + nf(0) = nf(n^2)$ and so $f(n^2) = f(0)$ for all n . Now the identity

$$3n^2f(4n^2) + 4n^2f(3n^2) = 7n^2f(25n^4)$$

implies

$$3n^2f(0) + 4n^2f(3n^2) = 7n^2f(0),$$

so $f(3n^2) = f(0)$ for all n .

On the other hand

$$12n^2f(5n^2) + 5n^2f(12n^2) = 17n^2f(169n^4)$$

and using the above identities we get

$$12n^2f(5n^2) + 5n^2f(0) = 17n^2f(0),$$

so $f(5n^2) = f(0)$ for all n .

Then using the identity

$$2n^2f(n^2) + n^2f(2n^2) = 3n^2f(5n^4)$$

we get

$$2n^2f(0) + n^2f(2n^2) = 3n^2f(0),$$

so $f(2n^2) = f(0)$ for all n .

Finally, we have

$$2nf(n) = nf(n) + nf(n) = 2nf(2n^2) = 2nf(0)$$

which shows that $f(n) = f(0)$ for all n .

Problem 18.8. Find all functions $f: \mathbb{Z} \rightarrow \mathbb{Z}$ that obey

$$f(m+n) + f(mn-1) = f(m)f(n) + 2$$

for all $m, n \in \mathbb{Z}$.

Solution. If $f = c$ is a constant function, we get $2c = c^2 + 2$ and so $(c-1)^2 + 1 = 0$ which is impossible. Now set $m = 0$ to get $f(n) + f(-1) = f(0)f(n) + 2$ so

$f(n)(1 - f(0)) = 2 - f(-1)$. As f is not constant we get $f(0) = 1, f(-1) = 2$. Next set $m = -1$ to get $f(n-1) + f(-n-1) = 2f(n) + 2$. If we replace n by $-n$, the left-hand side does not change therefore the right-hand side does not change too, so f is even. Hence $f(n-1) + f(n+1) = 2f(n) + 2$. Now we can easily prove by induction on $n \geq 0$ that $f(n) = n^2 + 1$ and the evenness of f implies $f(n) = n^2 + 1$ for all n . It is easy to check that this function satisfies the given identity.

Problem 18.9. (Nordic Contest 1999) A function $f: \mathbb{N} \rightarrow \mathbb{R}$ satisfies for some positive integer m the conditions

$$f(m) = f(1995), \quad f(m+1) = 1996, \quad f(m+2) = 1997$$

and

$$f(n+m) = \frac{f(n) - 1}{f(n) + 1}$$

for all $n \in \mathbb{N}$. Prove that $f(n+4m) = f(n)$ and find the least m for which such a function f exists.

Solution. If $h(x) = \frac{x-1}{x+1}$ then $f(n+m) = h(f(n))$ so $f(n+4m) = h^{(4)}(f(n))$.

We need to check that $h^{(4)}(x) = x$. Indeed

$$h^{(2)}(x) = \frac{\frac{x-1}{x+1} - 1}{\frac{x-1}{x+1} + 1} = \frac{-1}{x}$$

and therefore $h^{(4)}(x) = h^{(2)}(h^{(2)}(x)) = x$. We have solved the first part of the problem. If the least possible value of m is 1, then $f(n+4) = f(n)$, so $f(1997) = f(5) = f(1)$. But we know that

$$f(1997) = f(3) = h^{(2)}(f(1)) = \frac{-1}{f(1)}.$$

Thus $m = 1$ gives us $f(1)^2 = -1$ which is impossible. Similarly $m = 2$ gives

$$f(1995) = f(3) = f(2), \quad f(1996) = f(4) = f(3), \quad f(1997) = f(5) = f(4)$$

thus $f(2) = f(3) = f(4) = f(5)$. Then $h(f(2)) = f(2)$. But the equation $\frac{x-1}{x+1} = x$ gives us $x-1 = x^2+x$, so again $x^2 = -1$ which is impossible. Finally if $m = 3$ then for any value of $f(1), f(2), f(3)$ we can compute f inductively. Because 12|1992, we get $f(1995) = f(3) = f(m)$ and so on, so $m = 3$ is the answer.

Problem 18.10. Suppose $f: \mathbb{Q} \rightarrow \{0, 1\}$ is such that $f(1) = 1, f(0) = 0$ and if $f(x) = f(y)$, then $f\left(\frac{x+y}{2}\right) = f(x) = f(y)$. Prove that $f(x) = 1$ whenever $x \geq 1$.

Solution. Let $A = \{x | f(x) = 1\}$ and $B = \{x | f(x) = 0\}$. We have that if x, y belong to one of the sets A and B , then $\frac{x+y}{2}$ also belongs to the same set.

If $2 \in B$, then $1 = \frac{2+0}{2} \in B$, a contradiction. So $2 \in A$. Next we prove by induction on n that $n \in A$. If $n = 2k$, then as $0 \in B, 2k \in B$ would imply $\frac{2k+0}{2} = k \in B$, a contradiction to the induction step. If $n = 2k+1$, then

we prove like above that $2k, 2k+2 \in A$ and so $2k+1 = \frac{2k+2k+2}{2} \in A$.

Now assume that $f(1+a) = 0$ for $a > 0$. We prove by induction on n that $1+na \in B$ like above: if $n = 2k$ then $1+2ka \in A$ together with $1 \in A$ would imply $1+ka \in A$ contradicting the induction hypothesis, and if $n = 2k+1$, then we show that $1+2ka, 1+(2k+2)a \in B$; hence their mean $1+(2k+1)a$ is also in B . Finally, if n is such that $na \in B$, then $1+na \in B$ which contradicts our conclusion above that all positive integers are in A .

Problem 18.11. (Nordic Contest 1998) Find all functions $f: \mathbb{Q} \rightarrow \mathbb{Q}$ such that

$$f(x+y) + f(x-y) = 2(f(x) + f(y))$$

for all $x, y \in \mathbb{Q}$.

Proof. Set $x = y = 0$ to obtain that $f(0) = 0$; then set $y = -x$ and deduce $f(x) = f(-x)$ for any $x \in \mathbb{Q}$. By setting $y = \frac{n}{t}, x = \frac{1}{y}$, we prove by induction on n that

$$f\left(\frac{n}{t}\right) = n^2 f\left(\frac{1}{t}\right)$$

for any $n, t \in \mathbb{N}$. The solutions are $f(x) = cx^2$, where c is a rational number.

Problem 18.12. (Iran 1995) Find all functions $f: \mathbb{Z} \setminus \{0\} \rightarrow \mathbb{Q}$ such that

$$f\left(\frac{x+y}{3}\right) = \frac{f(x) + f(y)}{2}$$

for all $x, y \in \mathbb{Z} \setminus \{0\}$.

Solution. We will first show by induction that $f(x) = c$ for $x \in \mathbb{N}$. Set $f(1) = c$. Then by taking $x = 1, y = 2$ we get $f(2) = f(1) = c$. Now plugging $x = y = 3$ we obtain $f(3) = c$. Assume that for every $1 \leq x \leq n$ where $n \geq 3$ we have $f(x) = c$. Let t be an integer in the interval $[1, 3]$ such that $(n+1) + t$ is a multiple of 3. Then

$$f\left(\frac{n+1+t}{3}\right) = \frac{f(n+1) + f(t)}{2}$$

and it follows from $t \leq 3, \frac{n+1+t}{3} \leq n$ and the inductive assumption that $f(n+1) = c$. Hence $f(x) = c$ for every $x \geq 1$.

Now let k be a negative integer. There is an integer $l > 0$ such that $l+k > 0$ and $l+k$ is a multiple of 3. Then plugging $x = l, y = k$ in the given equation leads us to $f(k) = c$. Thus $f(x) = c$ for a constant $c \in \mathbb{Q}$.

Problem 18.13. (IMO 1982) The function $f(n)$ is defined for all positive integers n and takes on nonnegative integer values. Also, for all m, n ,

$$f(m+n) - f(m) - f(n) = 0$$

or 1; $f(2) = 0, f(3) > 0$ and $f(9999) = f(3333)$. Determine $f(1982)$.

Solution. From $f(1) + f(1) \leq f(2) = 0$ we obtain $f(1) = 0$. Since $0 < f(3) \leq f(1) + f(2) + 1$, it follows that $f(3) = 1$. Note that if $f(3n) \geq n$, then $f(3n+3) \geq f(3n) + f(3) \geq n+1$. Hence by induction $f(3n) \geq n$ holds for all $n \in \mathbb{N}$. Moreover, if the inequality is strict for some n , then it is so for all integers greater than n as well. Since $f(9999) = f(3333)$, we deduce that $f(3n) = n$ for all $n \leq 3333$. By the given condition, we have $3f(n) \leq f(3n) \leq 3f(n) + 2$. Therefore $f(n) = \lfloor f(3n)/3 \rfloor = \lfloor n/3 \rfloor$ for $n \leq 3333$. In particular, $f(1982) = \lfloor 1982/3 \rfloor = 660$.

Problem 18.14. (Romania TST 2004) Find all injective functions $f : \mathbb{N} \rightarrow \mathbb{N}$ such that

$$f(f(n)) \leq \frac{n + f(n)}{2}$$

for all $n \in \mathbb{N}$.

Solution. We shall prove that $f(n) \leq n$ for all non-negative n which together with the injectivity of f implies by induction that $f(n) = n$.

Suppose, by way of contradiction, that $f(n) > n$ for some non-negative n and denote by $f^{(k)}$ the k -th iterate of f . We easily obtain by induction on k that $f^{(k)}(n) < f(n)$ for all $k = 2, 3, \dots$. Indeed, if $f^{(m)}(n) < f(n)$ for $m = 2, \dots, k$ then

$$f^{(m+1)}(n) \leq \frac{f^{(m)}(n) + f^{(m-1)}(n)}{2} < f(n).$$

Thus $\{f^{(k)}(n)\}_{k=1}^{\infty}$ is a bounded sequence of non-negative integers and so $f^{(p)}(n) = f^{(q)}(n)$ for some $p < q$. Since f is injective it follows that $f^{(q-p)}(n) = n$, therefore $f^{(q-p+1)}(n) = f(n)$, a contradiction with the inequality proved above. Hence the only solution to the problem is the identity function.

Remark. The problem can be solved also by using the well-known fact that if $\{a_k\}_{k=1}^{\infty}$ is a sub-additive sequence, i.e. $a_{k+1} \leq \frac{a_k + a_{k-1}}{2}$, then it has a finite limit. In our case, for every $n \in \mathbb{N}$ the sequence $\{f^{(k)}(n)\}_{k=1}^{\infty}$ is sub-additive and being a convergent sequence of integers is constant beginning from some term. This contradicts the injectivity of f if it is not the identity function.

Problem 18.15. (Romania 1986) Find all surjective functions $f : \mathbb{N} \rightarrow \mathbb{N}$ such that

$$f(n) \geq n + (-1)^n$$

for all $n \in \mathbb{N}$.

Solution. Let S_n be the set of positive integers x such that $x + (-1)^x \leq n$. Obviously, this is the set of all even numbers $\leq n - 1$ and all odd numbers $\leq n + 1$. So $S_{2k} = \{1, 2, 3, \dots, 2k - 1, 2k + 1\}$ and $S_{2k+1} = \{1, 2, 3, \dots, 2k + 1\}$. In particular, $S_1 = \{1\}$ and $f(1) = 1$.

We clearly have

$$f^{-1}([1, n]) \subseteq \cup_{k \in [1, n]} S_k.$$

Hence

$$f^{-1}([1, 2k]) \subseteq \{1, 2, 3, \dots, 2k-1, 2k+1\}$$

and

$$f^{-1}([1, 2k+1]) \subseteq \{1, 2, 3, \dots, 2k+1\}.$$

So $|f^{-1}([1, n])| = n$ and this implies that

$$f^{-1}(\{n\}) = f^{-1}([1, n]) \setminus f^{-1}([1, n-1]).$$

Thus we have a unique solution f defined by $f(1) = 1$, $f(2k) = 2k+1$ and $f(2k+1) = 2k$ for all $k \geq 1$.

Problem 18.16. Find all functions $f: \mathbb{N} \rightarrow \mathbb{N}$ such that $f(1) = 1$ and

$$f(n+1) = \left\lfloor f(n) + \sqrt{f(n)} + \frac{1}{2} \right\rfloor$$

for all $n \in \mathbb{N}$.

Solution. Note that $f(n+1)$ depends on $\left\lfloor \sqrt{f(n)} + \frac{1}{2} \right\rfloor$. So suppose that

$$\left\lfloor \sqrt{f(n)} + \frac{1}{2} \right\rfloor = m,$$

thus

$$\left(m - \frac{1}{2}\right)^2 \leq f(n) < \left(m + \frac{1}{2}\right)^2$$

or $m^2 - m \leq f(n) \leq m^2 + m$. Since $f(n+1) = f(n) + m$ so

$$m^2 \leq f(n+1) \leq m^2 + 2m < (m+1)(m+2).$$

Hence

$$\left\lfloor \sqrt{f(n+1)} + \frac{1}{2} \right\rfloor$$

is either m or $m+1$ and $f(n+2)$ is either $f(n)+2m$ or $f(n)+2m+1$, so $m^2+m \leq f(n+2) \leq m^2+3m+1$, i.e. $m(m+1) \leq f(n+2) \leq (m+1)(m+2)$. Thus if we denote

$$g(x) = \lfloor \sqrt{x} + \frac{1}{2} \rfloor$$

then

$$g(f(n+2)) = g(f(n)) + 1.$$

As $g(f(1)) = 1$ and $g(f(2)) = 1$, we deduce by induction that

$$g(f(n)) = \left\lfloor \frac{n+1}{2} \right\rfloor.$$

Hence

$$f(n+1) = f(n) + \left\lfloor \frac{n+1}{2} \right\rfloor$$

and it is easy to prove by induction on n that

$$f(n) = \left\lfloor \frac{n}{2} \right\rfloor \left\lfloor \frac{n+1}{2} \right\rfloor + 1.$$

Problem 18.17. (IMO Shortlist 2001) Find all functions $f: \mathbb{N}_0^3 \rightarrow \mathbb{R}$ that satisfy $f(p, q, r) = 0$ if $pqr = 0$ and

$$\begin{aligned} f(p, q, r) = 1 + \frac{1}{6} & (f(p+1, q-1, r) + f(p-1, q+1, r) + f(p-1, q, r+1) \\ & + f(p+1, q, r-1) + f(p, q+1, r-1) + f(p, q-1, r+1)) \end{aligned}$$

otherwise.

Solution. It is clear that the second most important condition computes $f(p, q, r)$ in terms of $f(p+1, q-1, r)$ etc. and the sum of coordinates remains the same: if $p+q+r = s$ then $(p+1)+(q-1)+r = s$ etc. This implies that it suffices to compute f on each of the planes $p+q+r = s$, as the conditions take place within this planes. Also the fact that $f(p, q, r) = 0$ when $pqr = 0$ may suggest that $f(p, q, r) = kpqr$. So let us try to set $f(p, q, r) = kpqr$. Then

$$\frac{1}{6} (f(p+1, q-1, r) + f(p-1, q+1, r) + f(p-1, q, r+1) + f(p+1, q, r-1)$$

$$\begin{aligned}
& +f(p, q+1, r-1) + f(p, q-1, r+1)) - f(p, q, r) \\
& = \frac{k}{6}((p+1)(q-1)r + (p-1)(q+1)r + (p+1)q(r-1) \\
& + (p-1)q(r+1) + p(q-1)(r+1) + p(q+1)(r-1) - 6pqr) \\
& = \frac{k}{6}(6pqr - 2p - 2q - 2r - 6pqr) = -\frac{k}{3}(p+q+r)
\end{aligned}$$

and

$$k = \frac{3}{p+q+r} \text{ for } p+q+r \neq 0.$$

So we have found a solution

$$f(p, q, r) = \frac{3pqr}{p+q+r}$$

for $p+q+r \neq 0$. We try to prove now that it is unique. As it satisfies the condition, it suffices to show that the values of f may be deduced inductively on each (x, y, z) s.t. $x+y+z=s$. Note that we already know the values of the function on the boundary $xyz=0$. Therefore we might consider starting from the boundary and compute other values of the function one by one using the condition of the problem. Indeed, we can perform an induction on

$$s^2 - x^2 - y^2 - z^2 = (x+y+z)^2 - x^2 - y^2 - z^2 = 2xy + 2yz + 2zx.$$

When it is zero, the condition follows as two of the numbers must be zero. Now assume we have proven the claim when $s^2 - x^2 - y^2 - z^2 \leq k$ and let us prove it when $s^2 - x^2 - y^2 - z^2 = k+1$. Without loss of generality $x \leq y \leq z$. If $x=0$, we are done; otherwise set $p=x-1$, $q=y$, $r=z+1$. We see

$$(p^2 + q^2 + r^2) - (x^2 + y^2 + z^2) = 2(z-x+1) > 0$$

and so the induction assumption applies to (p, q, r) . Moreover it also applies for $(p-1, q+1, r)$, $(p-1, q, r+1)$, $(p, q-1, r+1)$, $(p, q+1, r-1)$, $(p+1, q+1, r)$ as it's easy to check, because the sum of squares of coordinates of each of these numbers equals $p^2 + q^2 + r^2$ plus one of $2(p-q+1)$, $2(q-p+1)$, $2(q-r+1)$, $2(r-q+1)$, $2(p-r+1)$, whereas the sum of squares of coordinates of x, y, z

is $p^2 + q^2 + r^2 + 2(r - p + 1)$ and is clearly less than all the others, because $p < q < r$. So the induction assumption allows us to say f is computed on all points (p, q, r) , $(p - 1, q, r + 1)$, $(p, q + 1, r - 1)$, $(p, q - 1, r + 1)$, $(p - 1, q, r + 1)$, $(p + 1, q - 1, r)$ and writing the condition for (p, q, r) allows us to compute the value of $f(p + 1, q, r - 1) = f(x, y, z)$. Therefore f is unique and it equals $\frac{3pqr}{p + q + r}$ when $(p, q, r) \neq (0, 0, 0)$ and 0 when $p = q = r = 0$.

Problem 18.18. Find all functions $f: \mathbb{N} \rightarrow \mathbb{N}$ such that

$$f(f(m) + f(n)) = m + n$$

for all $m, n \in \mathbb{N}$.

Solution. If $f(m_1) = f(m_2)$ then by setting $m = m_1, m_2$ we deduce $m_1 + n = m_2 + n$ so $m_1 = m_2$. Thus f is injective. Hence if $m + n = k + l$ then $f(f(m) + f(n)) = f(f(k) + f(l))$ so $f(m) + f(n) = f(k) + f(l)$. Therefore $f(m + n - 1) + f(1) = f(m) + f(n)$. Hence by setting $m = 2$ we get $f(n + 1) = f(n) + f(2) - f(1)$. If we set $f(2) - f(1) = a$, $f(1) = b$ we deduce by induction on n that $f(n) = a(n - 1) + b = an + b - a$. Hence

$$\begin{aligned} f(f(m) + f(n)) &= f(am + b - a + an + b - a) = f(a(m + n) + 2(b - a)) \\ &= a(a(m + n) + 2(b - a)) + (b - a) = a^2(m + n) + (2a + 1)(b - a). \end{aligned}$$

As $f(f(m) + f(n)) = m + n$ we get $a^2 = 1$. Hence $a = 1$ ($a = -1$ yields $f(n) < 0$ for sufficiently large n) and $(2a + 1)(b - a) = 3(b - a) = 0$. So $b = a = 1$ and f is the identity function. It satisfies our condition.

Problem 18.19. Find all functions $f: \mathbb{N} \rightarrow \mathbb{N}$ that satisfy

$$f(m^2 + f(n)) = f(m)^2 + n$$

for all $m, n \in \mathbb{N}$.

Solution. Set $a = f(1)$. Then $f(1 + f(n)) = a^2 + n$ and hence

$$f(1 + a^2 + n) = f(1 + f(1 + f(n))) = a^2 + 1 + f(n),$$

i.e. $f(n + b) = f(n) + b$, where $b = a^2 + 1$. It follows by induction on k that $f(n + kb) = f(n) + kb$ for any k . Then

$$f(m)^2 + n + (2m + b)b = f(m^2 + f(n)) + (2m + b)b = f((m + b)^2 + f(n))$$

$$= f(m+b)^2 + n = (f(m) + b)^2 + n = f(m)^2 + n + (2f(m) + b)b.$$

Comparing the first and the last term above, we get that $f(m) = m$ for all $m \in \mathbb{N}$ and this function satisfies the given condition.

Problem 18.20. Find all functions $f: \mathbb{N}_0 \rightarrow \mathbb{N}_0$ that satisfy

$$f(f(m)^2 + f(n)^2) = m^2 + n^2$$

for all $m, n \in \mathbb{N}_0$.

Solution. If $f(n_1) = f(n_2)$, then setting $n = n_1, n = n_2$ we conclude that $n_1 = n_2$. So f is injective.

First, we shall find $f(i)$, $0 \leq i \leq 6$. If $f(0) = a$, then $f(2a^2) = 0$. Hence setting $m = n = 2a^2$ we get $f(0) = 8a^4$. So $a = 8a^4$ and thus $a = 0$. Now, observe that if $m^2 + n^2 = x^2 + y^2$, then

$$f(f(m)^2 + f(n)^2) = m^2 + n^2 = x^2 + y^2 = f(f(x)^2 + f(y)^2).$$

So the injectivity of f implies that $f(m)^2 + f(n)^2 = f(x)^2 + f(y)^2$ if and only if $m^2 + n^2 = x^2 + y^2$. If $f(1) = b$, then set $m = 0, n = 1$ and $m = 0, n = b^2$ to get $f(b^2) = 1$ and $f(1) = b^4$. Hence $b^4 = b$, i.e. $f(1) = 1$ or $f(1) = 0$. As $f(0) = 0$ we conclude that $f(1) = 1$. Set $m = n = 1$, $m = 2, n = 0$, $m = 2, n = 1$ and $m = n = 2$ to get $f(2) = 2$, $f(4) = 4$, $f(5) = 5$ and $f(8) = 8$. As $3^2 + 4^2 = 5^2 + 0^2$ we see that $f(3) = 3$. Set $m = 1, n = 3$ to get $f(10) = 10$ and then as $6^2 + 8^2 = 10^2 + 0^2$ we obtain that $f(6) = 6$.

Now, the identities

$$(4k+i)^2 = (4k-i)^2 + (2k+2i)^2 - (2k-2i)^2,$$

$$(4k+4) = (4k)^2 + (2k+5)^2 - (2k-3)^2$$

show that for every $n \geq 7$ there exist positive integers x, y, m smaller than n and such that $n^2 = x^2 + y^2 - m^2$. Then the equality

$$f(n)^2 = f(x)^2 + f(y)^2 - f(m)^2$$

allows us to conclude by induction that $f(n) = n$ for all $n \in \mathbb{N}_0$.

Problem 18.21. (BMO 2009) Find all functions $f : \mathbb{N} \rightarrow \mathbb{N}$ such that

$$f(f(m)^2 + 2f(n)^2) = m^2 + 2n^2$$

for all $m, n \in \mathbb{N}$.

First Solution. Note that f is injective and hence

$$f(m)^2 + 2f(n)^2 = f(p)^2 + 2f(q)^2 \Leftrightarrow m^2 + 2n^2 = p^2 + 2q^2. \quad (1)$$

Now setting $f(1) = a$ we get $f(3a^2) = 3$ and $f(5a^2)^2 + 2f(a^2)^2 = 3f(3a^2)^2 = 27$. Since the solutions of the equation $x^2 + 2y^2 = 27$ in positive integers are $(x, y) = (3, 3)$ and $(x, y) = (5, 1)$, it follows that $f(a^2) = 1$ and $f(5a^2) = 5$. By (1), $2f(4a^2)^2 - 2f(2a^2)^2 = f(5a^2)^2 - f(a^2)^2 = 24$ and so $f(2a^2) = 2$ and $f(4a^2) = 4$. Using (1) again, we get

$$f((k+4)a^2)^2 = 2f((k+3)a^2)^2 - 2f((k+1)a^2)^2 + f(ka^2)^2$$

and therefore $f(ka^2) = k$ by induction on k . Then $f(a^3) = a = f(1)$ and thus $a = 1$. It is clear that the function $f(k) = k$ satisfies the given condition.

Second Solution. Set $a_n = f(n)^2$. We know from the first solution that

$$a_{n+4} = 2a_{n+3} - 2a_{n+1} + a_n.$$

The characteristic equation of this recurrence relation is

$$x^4 - 2x^3 + 2x - 1 = (x-1)^3(x+1) = 0$$

and hence

$$a_n = c_1(-1)^n + c_2 + c_3n + c_4n^2.$$

Then using that $f(3f(n)^2)^2 = 9n^4$, it is easy to show that $c_1 = c_2 = c_3 = 0$ and $c_4 = 1$, i.e. $f(n) = n$.

Problem 18.22. (Korea 1998) Find all functions $f : \mathbb{N}_0 \rightarrow \mathbb{N}_0$ that satisfy

$$2f(m^2 + n^2) = f(m)^2 + f(n)^2$$

for all $m, n \in \mathbb{N}_0$.

Solution. The key is not the condition itself but an observation that follows directly from it: if $m^2 + n^2 = x^2 + y^2$, then $f(m)^2 + f(n)^2 = f(x)^2 + f(y)^2$. Using this we can compute $f(n)$ from $f(0)$ and $f(1)$: set $m = n = 1$ to compute $f(2)$, then $m = 0, n = 2$ to compute $f(4)$, $m = 1, n = 2$ to compute $f(5)$, then $m = 3, n = 4$ to compute $f(3)$, $m = n = 2$ to find $f(8)$, $m = 1, n = 3$ to get $f(10)$, then $m = 6, n = 8$ to compute $f(6)$. If $n > 6$ we can compute $f(n)$ inductively on n as follows: we only need to find $x, y, z < n$ with $x^2 + n^2 = y^2 + z^2$ or equivalently $n^2 - y^2 = z^2 - x^2$ or $(n + y)(n - y) = (z - x)(z + x)$. Setting $y = n - 2k$ then $(n + y)(n - y) = (2n - 2k)2k$. So if we have $z + x = n - k, z - x = 4k$, we satisfy the condition. This implies

$$z = \frac{n - k + 4k}{2} = \frac{n + 3k}{2}, x = \frac{n - 5k}{2}$$

which is possible if $n - k$ is even, $n \geq 5k$. Now take $k = 0, 1$ such that

$$k \equiv n \pmod{2}.$$

Then $\frac{n + 3k}{2} \leq \frac{n + 3}{2} < n$ and we are done.

It remains therefore to investigate $f(0)$ and $f(1)$. If we set $m = n = 0$, we get $2f(0)^2 = 2f(0)$, so $f(0) = 1$ or $f(0) = 0$. If $f(0) = 1$, set $m = 1, n = 0$ to get $2f(1) = f(1)^2 + 1$ which implies $f(1) = 1$. Now $f(x) = 1$ satisfies the conditions. As f can be uniquely computed from $f(0)$ and $f(1)$ we conclude that $f \equiv 1$ is the only solution in this case. Now if $f(0) = 0$, then set $m = 1, n = 0$ to get $2f(1) = f(1)^2$. So $f(1) = 0$ or $f(1) = 2$. Note that $f \equiv 0$ satisfies the condition. So it is the only solution with $f(0) = f(1) = 0$. The function $f(x) = 2x$ also satisfies the condition and it is the only solution with $f(0) = 0, f(1) = 2$. Therefore the solutions to our equation are the functions $f(x) = 2x, f(x) = 1$ and $f(x) = 0$.

Problem 18.23. Find all functions $f: \mathbb{Z} \rightarrow \mathbb{R}$ satisfying

$$f(a^3 + b^3 + c^3) = f(a)^3 + f(b)^3 + f(c)^3$$

whenever $a, b, c \in \mathbb{Z}$.

Solution. Set $f(x) = ax$ to get $a = 0, 1, -1$ and set $f(x) = c$ to get $c = 0, c = \pm \frac{1}{\sqrt{3}}$. We shall prove that these are the only solutions. First, we shall find

a way of computing $f(n)$ by induction on $|n|$. Since $f(0) = f(n^3 + (-n)^3 + 0^3) = f(n)^3 + f(-n)^3 + f(0)^3$, we compute $f(-n)$ by $f(n)$. Further, note that if $a^3 + b^3 + c^3 = m^3 + n^3 + p^3$, then $f(a)^3 + f(b)^3 + f(c)^3 = f(m)^3 + f(n)^3 + f(p)^3$. So if we can write n^3 as a sum of at most five cubes of numbers having absolute value less than n , then we are done. Let us find such representations. Set $n = 2^m(2a + 1)$. Note that $5^3 = 4^3 + 4^3 - 1^3 - 1^3 - 1^3$, $6^3 = 3^3 + 4^3 + 5^3$ and $7^3 = 6^3 + 4^3 + 4^3 - 1^3$. If $a = 1$ and $m \geq 1$, $a = 2$ or $a = 3$, then n is a multiple of 6, 5 or 7 and we are done. For $a \geq 4$, we get the result by the identity $(2a + 1)^3 = (2a - 1)^3 + (a + 4)^3 - (a - 4)^3 - 5^3 - 1^3$. Finally, $2^{3m} = 2^{3(m-2)}(3^3 + 3^3 + 2^3 + 1^3 + 1^3)$ and therefore this also holds if $a = 0$ and $m \geq 2$. So f is uniquely determined by $f(0)$ and $f(1)$, since then we compute $f(2) = f(1^3 + 1^3)$ and $f(3) = f(1^3 + 1^3 + 1^3)$. Now, let us look at $f(0)$ and $f(1)$. By setting $a = b = c = 0$ we get $f(0) = 3f(0)^3$. Thus $f(0) = 0$ or $f(0) = \pm \frac{1}{\sqrt{3}}$. If $f(0) = 0$, then setting $a = 1, b = c = 0$ we get $f(1) = f(1)^3$. Hence $f(1) = 0, -1, 1$ and we obtain the solutions $f(x) = 0$, $f(x) = -x$ and $f(x) = x$. Now assume that $f(0) \neq 0$. Without loss of generality let $f(0) = \frac{1}{\sqrt{3}}$ (the second case is analogous since we could look at $-f$ instead of f). Then by setting $a = 1, b = c = 0$ we deduce $f(1) = f(1)^3 + 2f(0)^3$ which is a polynomial equation w.r.t. $f(1)$ with solutions $\frac{1}{\sqrt{3}}, \frac{-2}{\sqrt{3}}$. If $f(1) = \frac{1}{\sqrt{3}}$, then we get the solution $f(x) = \frac{1}{\sqrt{3}}$. We are left with the case $f(1) = -\frac{2}{\sqrt{3}}$. We have $f(0) = f(-x^3 + x^3 + 0^3) = f(-x)^3 + f(x)^3 + f(0)^3$ and therefore $f(-x)^3 = -f(x)^3 + \frac{2}{3\sqrt{3}}$. Thus $f(-1)^3 = \frac{2}{\sqrt{3}} + \frac{2}{3\sqrt{3}} = \frac{8}{3\sqrt{3}}$ and so $f(-1) = \frac{2}{\sqrt{3}} = -f(1)$. Then $f(x^3) = f(x^3 + 1^3 + (-1)^3) = f(x)^3 + f(1)^3 + f(-1)^3 = f(x)^3$. On the other hand, $f(x^3) = f(x^3 + 0^3 + 0^3) = f(x)^3 + f(0)^3 + f(0)^3 = f(x)^3 + \frac{2}{3\sqrt{3}}$, a contradiction. All the cases are now investigated.

Problem 18.24. Find all functions $f : \mathbb{Q}^+ \rightarrow \mathbb{Q}^+$ such that

$$f(m^{2010} + f(n)) = m^{10} f(m^{50})^{40} + n$$

for all $m, n \in \mathbb{Q}^+$.

Solution. Let $a, b, c \in \mathbb{N}$ and $d \in \mathbb{N}_0$ be such that $a = bc + d$. We shall prove that if

$$f(m^a + f(n)) = m^d f(m^c)^b + n$$

for all $m, n \in \mathbb{Q}^+$ then $f(r) = r$ for all $r \in \mathbb{Q}^+$.

Set $\alpha = f(1), \beta = \alpha^b + 1$. Then $f(1 + f(n)) = \alpha^b + n$ and we get

$$f(1 + \alpha^b + n) = f(1 + f(1 + f(n))) = \alpha^b + 1 + f(n),$$

i.e. $f(n + \beta) = f(n) + \beta$. Now it follows by induction on $k \in \mathbb{N}_0$ that $f(n + k\beta) = f(n) + k\beta$. Using the binomial formula it is easy to see that for each $m \in \mathbb{Q}^+$ there exists $e \in \mathbb{N}$ such that $\frac{(m+e)^a - m^a}{\beta}$ and $\frac{(m+e)^c - m^c}{\beta}$ are integers. Then

$$\begin{aligned} m^d f(m^c)^b + 1 + (m+e)^a - m^a &= f(m^a + \alpha) + (m+e)^a - m^a \\ &= f((m+e)^a + \alpha) = (m+e)^d f^b((m+e)^c) + 1 \\ &= (m+e)^d [f(m^c) + (m+e)^c - m^c]^b + 1, \end{aligned}$$

i.e.

$$(m+e)^a - m^a = (m+e)^d [f(m^c) + (m+e)^c - m^c]^b - m^d f(m^c)^b.$$

Setting $f(m^c) = x$ we see that the right hand side of the above identity is a nonconstant polynomial P of x whose coefficients are positive integers. Hence it is an increasing function for $x \geq 0$. Since $P(m^c) = (m+e)^a - m^a$ we get $f(m^c) = m^c$. Thus $f(m^a + f(n)) = m^a + n$ and it follows by induction on k that $f(km^a + f(n)) = km^a + n$ for all $k \in \mathbb{N}$ and $m, n \in \mathbb{Q}^+$. In particular, $f(p/q + 1) = f(pq^{a-1}/q^a + 1) = p/q + 1$ for all $p, q \in \mathbb{N}$, i.e. $f(r) = r$ for all $1 < r \in \mathbb{Q}$. Hence for every $r \in \mathbb{Q}^+$ we have $1 + f(r) = f(1 + f(r)) = 1 + r$ and therefore $f(r) = r$.

Remark. We leave as an exercise to the reader to show that the statement of the above problem holds also true when b or c is 0, or if we replace $\mathbb{Q}^+$ by $\mathbb{N}, \mathbb{Z}, \mathbb{R}$ or $(0, \infty)$.

Problem 18.25. (IMO Shortlist 2010) Determine all functions $f : \mathbb{Q}^+ \rightarrow \mathbb{Q}^+$ which satisfy the equation

$$f(f(x)^2y) = x^3f(xy)$$

for all $x, y \in \mathbb{Q}^+$.

Solution. The only solution is the function $f(x) = \frac{1}{x}$. Indeed, by substituting $y = 1$, we get

$$f(f(x)^2) = x^3f(x). \quad (1)$$

Hence whenever $f(x) = f(y)$ we have

$$x^3 = \frac{f(f(x)^2)}{f(x)} = \frac{f(f(y)^2)}{f(y)} = y^3$$

which implies $x = y$. So the function f is injective. Now replace x by xy in (1), and apply the given equation twice, second time to $(y, f(x)^2)$ instead of (x, y) :

$$f(f(xy)^2) = (xy)^3f(xy) = y^3f(f(x)^2y) = f(f(x)^2f(y)^2).$$

Since f is injective, we get $f(xy) = f(x)f(y)$. This implies $f(1) = 1$ and $f(x^n) = f(x)^n$ for all integers n . Then the given functional equation can be rewritten as

$$f(f(x))^2f(y) = x^3f(x)f(y)$$

which gives

$$f(f(x)) = \sqrt{x^3f(x)}. \quad (2)$$

Let $g(x) = xf(x)$. Then by (2) we have

$$\begin{aligned} g(g(x)) &= g(xf(x)) = xf(x) \cdot f(xf(x)) = xf^2(x)f(f(x)) \\ &= xf(x)^2\sqrt{x^3f(x)} = (xf(x))^{5/2} = (g(x))^{5/2}, \end{aligned}$$

and by induction

$$g^{(n)}(x) = (g(x))^{(5/2)^n} \quad (3)$$

for all positive integers n , where $g^{(n)}(x)$ is the n -th iterate of g . Consider (3) for a fixed x . The left-hand side is always rational, so $(g(x))^{(5/2)^n}$ must be rational for every n . Suppose that $g(x) \neq 1$, and let the prime factorization of $g(x)$ be $g(x) = p_1^{\alpha_1} \cdots p_k^{\alpha_k}$, where $p_1, \dots, p_k$ are distinct primes and $\alpha_1, \dots, \alpha_k$ are nonzero integers. Then the unique prime factorization of (3) is

$$g^{(n)}(x) = (g(x))^{(5/2)^n} = p_1^{(5/2)^n \alpha_1} \cdots p_k^{(5/2)^n \alpha_k},$$

where the exponents should be integers. But this is not true for large values of n , for example when $2^n \nmid \alpha_1$. Thus $g(x) = 1$ which shows that $f(x) = \frac{1}{x}$. It is easy to check that this function satisfies the problem conditions.

Remark. Note that if we consider the above functional equation for functions $f : (0, \infty) \rightarrow (0, \infty)$ then $f(x) = \frac{1}{x}$ is not the only solution. Another one is, for example, $g(x) = x^{3/2}$ and one can show that in this case there are in fact infinitely many solutions.

19.18.2 Basic Properties of Functions

Problem 18.26. (Iran 1998) Let $f : (0, \infty) \rightarrow (0, \infty)$ be a decreasing function that satisfies

$$f(x+y) + f(f(x) + f(y)) = f(f(x + f(y)) + f(y + f(x)))$$

for all $x, y \in (0, \infty)$. Show that $f(f(x)) = x$.

Solution. Set $y = x$ to get $E = f(2x) + f(2a) = f(2f(a+x))$, where $a = f(x)$. Now replace x by $f(x)$ to get $F = f(2b) + f(2a) = f(2f(a+b))$, where $b = f(f(x))$. If $b < x$ then $f(a+b) > f(a+x)$, so $f(2f(a+b)) < f(2f(a+x))$ as f is decreasing. Also $f(2b) > f(2x)$ and we get $f(2x) + f(2a) < f(2b) + f(2a)$. Hence $F > E$ from the first relation and $F < E$ from the second one, a contradiction. If $b > x$ then we get $F < E$ and $F > E$, again a contradiction. So $b = x$.

Problem 18.27. (Bulgaria 2011) Determine all functions $f : \mathbb{Q} \rightarrow \mathbb{R}$ such that

$$(f(x) + f(y) - 2f(xy)) \cdot (f(x) + f(z) - 2f(xz)) \geq 0$$

for all $x, y, z \in \mathbb{Q}$.

Solution. Setting $y = 0, z = 1$ gives $(f(x) - f(0))(f(x) - f(1)) \leq 0$, i.e. either $f(0) \leq f(x) \leq f(1)$ or $f(1) \leq f(x) \leq f(0)$ for all x .

Suppose that the function f is not constant. If $f(0) \leq f(x) < f(1)$ or $f(1) < f(x) \leq f(0)$ for some $x \neq 0$, then setting $y = 1/x, z = 1$ gives $f(1) + f(1) > f(a) + f(1/a) \geq 2f(1)$ or $f(1) + f(1) < f(a) + f(1/a) \leq 2f(1)$, a contradiction. Hence $f(x) = f(1)$ for $x \neq 0$ and $f(0)$ is an arbitrary real number.

Conversely, a direct check shows that every function defined as above satisfies the given inequality.

Problem 18.28. Find all functions $f : \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}$ such that

$$f(x^2)(f(x)^2 + f\left(\frac{1}{y^2}\right)) = 1 + f\left(\frac{1}{xy}\right)$$

for all $x, y \in \mathbb{R} \setminus \{0\}$.

Solution. Set $a = f(1)$. Then setting $x = 1, y = \frac{1}{x}$ we get

$$f(x) = a^3 + af(x^2) - 1$$

and so $f(x)$ is even. Now setting $y = \frac{1}{x}$ gives

$$f(x^2)(f(x)^2 + f(x^2)) = 1 + a.$$

Plugging $f(x) = a^3 + af(x^2) - 1$ in this identity, we get

$$f(x^2)((a^3 + af(x^2) - 1)^2 + f(x^2)) = 1 + a$$

and so $f(x^2)$ is a root of the equation

$$a^2X^3 + (2a^4 - 2a + 1)X^2 + (a^3 - 1)^2X - (a + 1) = 0.$$

Setting $x = y = 1$ in the given equation we get $(a + 1)(a^2 - 1) = 0$ and so $a = \pm 1$. If $a = -1$, the cubic is $X(X + 1)(X + 4) = 0$. The equation $f(x) = a^3 + af(x^2) - 1$ becomes $f(x) = -f(x^2) - 2$. If $f(x^2) = 0$ for some

x , then $f(x) = -2$, impossible. If $f(x^2) = -4$ for some x , then $f(x) = 2$, impossible. So $f(x) = -1$ for all $x \neq 0$ which is a solution.

If $a = 1$, the cubic is $(X - 1)(X^2 + 2X + 2) = 0$ and so $f(x^2) = 1$ for all $x > 0$. Since f is an even function we get $f(x) = 1$ for all $x \neq 0$ which is a solution.

Problem 18.29. (Bulgaria 2011) Prove that for every nonconstant function $f : \mathbb{Q}^+ \rightarrow \mathbb{R}$ there exist numbers $x, y, z \in \mathbb{Q}^+$, such that

$$(f(x) + f(y) - 2f(xy)) \cdot (f(x) + f(z) - 2f(xz)) < 0.$$

Solution. Assume the contrary. Then setting $x = 1$ gives

$$(f(1) - f(y))(f(1) - f(z)) \geq 0$$

for all $y, z \in \mathbb{Q}^+$. Thus either $f(x) \geq f(1)$ for all $x \in \mathbb{Q}^+$ or $f(x) \leq f(1)$ for all $x \in \mathbb{Q}^+$. We shall consider the first case only since the second one is analogous. Our goal is to prove that $f(x) \leq f(1)$ for all $x \in \mathbb{Q}^+$ which implies $f(x) = 1$, a contradiction.

Suppose first that $f(x) + f(y) \geq 2f(xy)$ for all $x, y \in \mathbb{Q}^+$. Then setting $y = 1$ gives $f(x) \leq f(1)$. If there is $x \in \mathbb{Q}^+$ such that

$$f(x) + f(y) \leq 2f(xy) \tag{1}$$

for all $y \in \mathbb{Q}^+$ setting $y = 1/x$ gives $f(1) + f(1) \leq f(x) + f(1/x) \leq 2f(1)$. Therefore $f(x) = f(1)$ for all x satisfying (1). Suppose that there is $x \in \mathbb{Q}^+$ such that

$$f(x) + f(y) \geq 2f(xy) \tag{2}$$

for all $y \in \mathbb{Q}^+$. Then setting $y = 1$ gives $f(x) \leq f(1)$ for every x satisfying (2) and the problem is solved.

Problem 18.30. Is there a function $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(xy) = \max\{f(x), y\} + \min\{f(y), x\}$$

for all $x, y \in \mathbb{R}$?

Solution. Suppose that f satisfies the given equation. First we shall prove that $f(f(x)) = x$. Setting $y = f(x)$ gives

$$f(xf(x)) = \max\{f(x), f(x)\} + \min\{f(f(x)), x\} = \min\{f(f(x)), x\} + f(x).$$

Also

$$f(f(x)x) = \max\{f(f(x)), x\} + \min\{f(x), f(x)\} = \max\{f(f(x)), x\} + f(x).$$

Hence $\max\{f(f(x)), x\} = \min\{f(f(x)), x\}$ which means that $f(f(x)) = x$.

The next step is to prove that $f(1) = 0$. Setting $x = 1, y = f(x)$ we get $f(f(x)) = \max\{f(f(x)), 1\} + \min\{f(1), f(x)\}$ which shows that $x = \max\{x, 1\} + \min\{f(1), f(x)\}$. Also, $x = \min\{x, 1\} + \max\{f(1), f(x)\}$. Therefore $1 = \max\{1, 1\} + \min\{f(1), f(1)\}$ and we conclude that $f(1) = 0$.

Now using the above two identities for $x \geq 1$ we get respectively that $x = \min\{x, 1\} + \max\{f(1), f(x)\} = 1 + \max\{0, f(x)\}$ and $x = \max\{x, 1\} + \min\{0, f(x)\}$. Hence $f(x) = x - 1$.

For $x \leq 1$ we have $x = \min\{x, 1\} + \max\{f(1), f(x)\} = x + \max\{0, f(x)\}$ and $x = \max\{x, 1\} + \min\{0, f(x)\} = 1 + \min\{0, f(x)\}$ which imply that $f(x) = x - 1$. Thus $f(x) = x - 1$ for all $x \in \mathbb{R}$. However this solution does not work, so there is no function with the desired property.

Problem 18.31. Prove that there is no function $f : (0, \infty) \rightarrow (0, \infty)$ satisfying the following conditions:

- (i) $f(x + f(y)) = f(x)f(y)$ for all $x, y > 0$;
- (ii) there are at most finitely many x with $f(x) = 1$.

Solution. Suppose that there is a function satisfying the given conditions. First we will prove that $f(x) > 1$ and $f(x) \geq x$ for all $x > 0$. If $f(u) = 1$, then $f(x + 1) = f(x)$ and so $f(u + n) = 1$ for all positive integers n which contradicts (ii). So $f(x) \neq 1$ for all $x \in (0, \infty)$. If $f(u) < u$, then setting $x = u - f(u), y = u$ in the given identity we get that $f(u - f(u)) = 1$ which is in contradiction with the above result. So $f(x) \geq x$ for all $x \in (0, \infty)$.

Now set $y = x$ in the initial equation to get $f(x + f(x)) = f(x)^2$. Then $f(x + 2f(x)) = f(x)^3$ and we get by induction on n that $f(x + (n - 1)f(x)) = f(x)^n$ for all $n \in \mathbb{N}$. Hence $f(x)^n \geq x + (n - 1)f(x) > x$ and so $f(x) > x^{\frac{1}{n}}$. Now letting $n \rightarrow +\infty$, we get $f(x) > 1$ for all $x \in (0, \infty)$.

Let $A = f(\mathbb{R})$. The given identity implies that $f(f(x) + f(1)) = f(f(x))f(1)$ and $f(f(x) + f(1)) = f(f(1))f(x)$. So, $f(f(x)) = cf(x)$ for some constant $c = \frac{f(f(1))}{f(1)} > 1$. Hence $f(x) = cx$ for all $x \in A$ and using the initial identity we see that if $x, y \in A$, then $xy \in A$. Thus $x \in A$ implies $x^2 \in A$ and so $f(x^2) = cx^2$.

Since $f(x)^2 > f(x)$ setting $x = f(x)^2 - f(x)$, $y = x$ in the given identity we get $f(f(x)^2) = f(f(x)^2 - f(x))f(x)$. On the other hand $f(f(x)^2) = cf(x)^2$ and therefore $cf(x)^2 = f(f(x)^2 - f(x))f(x)$. Hence $f(f(x)^2 - f(x)) = cf(x) > f(x)^2 - f(x)$ and so $f(x) < c + 1$. Then for every $n \in \mathbb{N}$ we have $c + 1 > f(f(x)^{2n}) = cf(x)^{2n}$ which is a contradiction since $f(x) > 1$. So, there is no function with the desired properties.

Problem 18.32. (IMO Shortlist 1997) Prove that if the function $f: \mathbb{R} \rightarrow \mathbb{R}$ is such that $|f(x)| \leq 1$ and

$$f(x) + f\left(x + \frac{13}{42}\right) = f\left(x + \frac{1}{6}\right) + f\left(x + \frac{1}{7}\right)$$

for all $x \in \mathbb{R}$, then it is periodic.

Solution. We have

$$f\left(x + \frac{1}{6} + \frac{1}{7}\right) - f\left(x + \frac{1}{7}\right) = f\left(x + \frac{1}{6}\right) - f(x)$$

which implies

$$f\left(x + \frac{k}{6} + \frac{1}{7}\right) - f\left(x + \frac{k-1}{6} + \frac{1}{7}\right) = f\left(x + \frac{k}{6}\right) - f\left(x + \frac{k-1}{6}\right)$$

for $1 \leq k \leq 6$. Summing up these inequalities gives

$$f\left(x + 1 + \frac{1}{7}\right) - f\left(x + \frac{1}{7}\right) = f(x + 1) - f(x).$$

Set $g(x) = f(x + 1) - f(x)$. Then $g\left(x + \frac{1}{7}\right) = g(x)$ which implies

$$g(x) = g\left(x + \frac{1}{7}\right) = g\left(x + \frac{2}{7}\right) = \cdots = g(x + 1).$$

Hence $g(x) = g(x + n)$ for all $n \in \mathbb{N}$. Then

$$\begin{aligned} f(x + n) - f(x) &= (f(x + n) - f(x + n - 1)) + \cdots + (f(x + 1) - f(x)) \\ &= g(x + n - 1) + \cdots + g(x) = ng(x), \end{aligned}$$

i.e. $f(x + n) - f(x) = ng(x)$ for all $x \in \mathbb{R}$ and $n \in \mathbb{N}$. Hence

$$n|g(x)| = |f(x + n) - f(x)| \leq |f(x + n)| + |f(x)| \leq 2,$$

i.e. $n|g(x)| \leq 2$ for all $x \in \mathbb{R}$ and $n \in \mathbb{N}$. This shows that $g(x) = 0$ for all $x \in \mathbb{R}$, i.e. $f(x + 1) = f(x)$.

Problem 18.33. (Romania TST 2007) Let $f \in \mathbb{R}[X]$ be a polynomial of degree n . Prove that:

- (i) f can not be written as a sum of at most n periodic functions;
- (ii) if $n = 1$, then f can be written as a sum of two real periodic functions;
- (iii) if $n = 1$ and f is the sum of two periodic functions, then they are unbounded on every interval;
- (iv) f can be written as a sum of $n + 1$ periodic functions;
- (v) there are real functions which can not be written as a finite sum of periodic functions.

Solution. (i) Let f be a polynomial that can be written as the sum of n periodic functions $f_1, f_2, \dots, f_n$ with periods $t_1, t_2, \dots, t_n$. Then

$$f(x + t_1) - f(x) = f_2(x + t_1) - f_2(x) + \cdots + f_n(x + t_1) - f_n(x).$$

Note that $f(x + t_1) - f(x)$ is a polynomial of degree at most $n - 1$, and it is written as sum of $n - 1$ periodic functions. So, we "go down" until a polynomial of degree 1 is written as sum of only one periodic function, a contradiction.

(ii) Let $A = \{n + m\sqrt{2} \mid n, m \in \mathbb{Z}\}$ and for every $a \in \mathbb{R}$ denote

$$[a] = \{b \in \mathbb{R} \mid a - b \in A\}.$$

By the Axiom of Choice there exists a choice function $c : \{[a] \mid a \in \mathbb{R}\} \rightarrow \mathbb{R}$, i.e. a function that chooses a number $c([a])$ in each class $[a]$. Using the fact

that the number $\sqrt{2}$ is irrational it follows that for each $x \in \mathbb{R}$ there exist unique integers n, m such that $x = c([x]) + n + m\sqrt{2}$. Then we define

$$f_1(x) = f_1(c([x]) + n + m\sqrt{2}) = c([x]) + m\sqrt{2}$$

and

$$f_2(x) = f_2(c([x]) + n + m\sqrt{2}) = n.$$

The functions f_1 and f_2 are periodic since

$$f_1(x+1) = f_1(c([x]) + (n+1) + m\sqrt{2}) = c([x]) + m\sqrt{2} = f_1(x)$$

and analogously

$$f_2(x + \sqrt{2}) = f_2(x).$$

We have also

$$f_1(x) + f_2(x) = c([x]) + m\sqrt{2} + n = x$$

and therefore if $f(x) = ax + b$ then $f = (af_1 + b) + af_2$ is a sum of two periodic functions.

(iii) Since f is periodic if and only if $af + b$ is, with a, b nonzero constants and the same for boundedness, we can suppose that that $f(x) = x$. Let now $f_1(x) + f_2(x) = x$ for all $x \in \mathbb{R}$, and the periods of f_1 and f_2 be t_1 and t_2 , respectively. Without loss of generality we may assume that $f_1(0) = 0$. Then $f_1(nt_1) = 0$ and $f_2(nt_1 + mt_2) = nt_1$ for all integers n, m . If the ratio t_1/t_2 is a rational number, then $x = f_1(x) + f_2(x)$ is a periodic function, a contradiction. Hence t_1/t_2 is irrational, and as it is well-known (Kronecker's theorem) the numbers of the form $nt_1 + mt_2$, where $n, m \in \mathbb{Z}$ are dense in $\mathbb{R}$. Hence for each $\epsilon > 0$, there are integers n, m such that $0 < nt_1 + mt_2 < \epsilon$, and we can make nt_1 very large by going first nearer to 0, and then multiplying n and m by a large factor - that is, choose $0 < nt_1 + mt_2 < \frac{\epsilon}{N}$ then we get $0 < (nN)t_1 + (mN)t_2 < \epsilon$ and $(nN)t_1$ is large (larger than Nt_1 for example). Then we can get close to every real with a very large nt_1 , to which corresponds a very large value of f_2 . This shows that f_2 can not be bounded on an interval, and the same for f_1 .

(iv) This can be proved by induction, using (ii). More generally, we prove that we can write f as a sum of $n+1$ functions of given periods $t_1, t_2, \dots, t_{n+1}$

as long as all ratios t_i/t_j are irrational. The base case $n = 1$ was done in part ii): the method for $(1, \sqrt{2})$ adapts to any (t_1, t_2) with $t_1/t_2 \notin \mathbb{Q}$.

To proceed further, we need a little lemma:

Lemma. Assume that s, t are non-zero reals with $s/t \notin \mathbb{Q}$. If $g: \mathbb{R} \rightarrow \mathbb{R}$ is a function of period s , then g can be written as $g(x) = g'(x) - g'(x - t)$, where g' is also periodic of period s .

Proof of lemma. As in part ii), define the class $[a] = \{a + ms + nt \mid m, n \in \mathbb{Z}\}$ and define a choice function $c: \{[a] \mid a \in \mathbb{R}\} \rightarrow \mathbb{R}$. It suffices to define g' on every class $[a]$ because the condition only uses number from a class. Set $g'(c([a])) = 0$. Then this forces $g'(c([a]) + kt)$ to equal

$$g(c([a]) + kt) + g(c([a]) + (k-1)t) + \dots + g(c([a]) + t)$$

for $k > 0$ and

$$g(c([a])) + g(c([a]) - t) + \dots + g(c([a]) + (k+1)t)$$

for $k < 0$. We extend this to the entire class by making g' periodic of period s . It is straightforward to see that this satisfies the condition. The lemma is proved. $\square$

Now let's return to the induction statement. Suppose the statement is true for all polynomials of degree less than n and let f be a polynomial of degree n . Then $f(x) - f(x - t_{n+1})$ is a polynomial of degree $n-1$ and there exist periodic functions $g_k(x)$ of periods t_k , $1 \leq k \leq n$, such that

$$f(x) - f(x - t_{n+1}) = g_1(x) + g_2(x) + \dots + g_n(x).$$

Using the lemma we write $g_k(x) = f_k(x) - f_k(x - t_{n+1})$, where f_k still has period t_k . Therefore

$$f(x) - f(x - t_{n+1}) = f_1(x) - f_1(x - t_{n+1}) + \dots + f_n(x) - f_n(x - t_{n+1})$$

and this shows that

$$f_{n+1}(x) = f(x) - f_1(x) - \dots - f_n(x)$$

is periodic of period t_{n+1} , as desired.

(v) Take $f(x) = 2^x$ and suppose that

$$2^x = \sum_{i=1}^n f_i(x)$$

with $f_i(x + t_i) = f_i(x)$ and $t_i > 0$. Then it is easy to check that

$$2^x = \sum_{i=1}^{n-1} h_i(x),$$

where

$$h_i(x) = \frac{f_i(x + t_n) - f_i(x)}{2^{t_n} - 1}$$

and $h_i(x + t_i) = h_i(x)$, $1 \leq i \leq n-1$. Thus if 2^x is sum of n periodic functions, then it is sum of $n-1$ periodic functions. This implies that 2^x is a periodic function, a contradiction.

Remark. A good intuition for introducing classes as in parts ii) and iv) above is as follows. A class $[a] = \{a + ms + nt\}$ is in bijection with $\mathbb{Z} \times \mathbb{Z}$. In this language, a periodic function with periods s, t simply becomes a function $f: \mathbb{Z}^2 \rightarrow \mathbb{R}$ that only depends on the second, respectively first coordinate. Thus if we restrict to $[a]$ and view $[a]$ as $\mathbb{Z}^2$, part ii) asks to represent the function $f(m, n) = a + ms + nt$ as a sum $g(m) + h(n)$ of functions that only depend on one of the coordinates. Respectively, part iv) asks to represent $f(m)$ as $g(m) - g(m-1)$. From this point of view, the required functions are very easy to construct, and that is what we did above.

The Axiom of Choice was needed to perform this simultaneously for all classes. The interpretation of a class as $\mathbb{Z}^2$ requires the choice of an "origin" $c([a])$ that corresponds to $(0, 0)$. That is easy to do for any one single class in particular, but to do it for all of them at once we need to employ the Axiom of Choice. Writing a class as $[a]$ is counter-intuitive, because then one has the natural choice $c(a) = a$, however writing all classes as $[a]$ (for some a) simultaneously is exactly a choice function.

Also note that part (v) can also be solved using the above idea. If we consider a map $\mathbb{Z}^n \rightarrow \mathbb{R}$ given by $(c_1, c_2, \dots, c_n) \rightarrow x + c_1 t_1 + \dots + c_n t_n$ then

a function of period t_i becomes a function that does not depend on the i -th component in this language. In particular by cancelation

$$\sum_{\epsilon_i \in \{0,1\}} (-1)^{\epsilon_1 + \dots + \epsilon_n} f(\epsilon_1, \epsilon_2, \dots, \epsilon_n) = 0$$

for any such function f , hence also for f a sum of such functions. However this is not true for the function $f(x) = 2^x$ and the above sum will simply be

$$2^x \sum_{\epsilon_i \in \{0,1\}} (-1)^{\epsilon_1 + \dots + \epsilon_n} 2^{\epsilon_1 + \dots + \epsilon_n} = 2^x \cdot \prod_{i=1}^n (1 - 2) = (-1)^n \cdot 2^x \neq 0.$$

This is an alternative solution to part (v) and this idea was used in the solution of Problem 9.2.

Problem 18.34. (Kiev 2007) Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x^2 - f(y)^2) = xf(x) - y^2$$

for all $x, y \in \mathbb{R}$.

Solution. Suppose that $f(y_0) = 0$. Then setting $x = 1, y = y_0$ in the given equation we get that $y_0 = 0$. Now setting $x = y = 0$ we obtain $f(f(0)^2) = 0$, hence $f(0) = 0$. Thus for $y = 0$ we get $f(x^2) = xf(x)$ which together with $f(0) = 0$ imply that f is an odd function. Now set $x = 0$ to get $f(-f(y)^2) = -y^2$, i.e. $f(f(y)^2) = y^2$ for all $y \in \mathbb{R}$. Then

$$f(y^4) = f(f(f(y)^2)^2) = f(y)^4$$

which shows that $f(t) > 0$ for $t > 0$. Setting $y = x > 0$ gives

$$f(x^2 - f(x)^2) = xf(x) - x^2.$$

In particular, if $x > f(x)$ then the LHS is positive, whereas the RHS is negative, a contradiction. Analogously we see that the case $x < f(x)$ is also impossible and we conclude that the only solution is the function $f(x) = x$.

Problem 18.35. (IMO Shortlist 2005) Determine all functions $f : (0, \infty) \rightarrow (0, \infty)$ such that

$$f(x)f(y) = 2f(x + yf(x))$$

for all positive real numbers x and y .

Solution. The answer is the constant function $f(x) = 2$ which obviously satisfies the equation.

First, we show that a function f satisfying the equation is nondecreasing. Indeed, suppose that $f(x) < f(z)$ for some positive real numbers $x > z$. Set $y = (x - z)/(f(z) - f(x)) > 0$, so that $x + yf(x) = z + yf(z)$. The equation now implies

$$f(x)f(y) = 2f(x + yf(x)) = 2f(z + yf(z)) = f(z)f(y),$$

therefore $f(x) = f(z)$, a contradiction. Thus f is nondecreasing. Assume now that f is not strictly increasing, i.e. $f(x) = f(z)$ holds for some positive numbers $x > z$. If y belongs to the interval $(0, (x - z)/f(x)]$ then $z < z + yf(z) \leq x$. Since f is nondecreasing, we obtain

$$f(z) \leq f(z + yf(z)) \leq f(x) = f(z),$$

leading to $f(z + yf(z)) = f(x)$. Thus,

$$f(z)f(y) = 2f(z + yf(z)) = 2f(x) = 2f(z).$$

Hence $f(y) = 2$ for all y in the above interval. But if $f(y_0) = 2$ for some y_0 then

$$2 \cdot 2 = f(y_0)f(y_0) = 2f(y_0 + y_0f(y_0)) = 2f(3y_0).$$

Therefore $f(3y_0) = 2$ and we get by induction that $f(3^k y_0) = 2$ for all positive integers k . Hence the above observation leads to the conclusion that $f(x) = 2$ for all $x \in \mathbb{R}$.

Assume now that f is a strictly increasing function. The equation implies that

$$2f(x + 1 \cdot f(x)) = f(x)f(1) = f(1)f(x) = 2f(1 + xf(1))$$

for all $x > 0$ and since f is injective we get that $x + f(x) = 1 + xf(1)$, i.e. $f(x) = cx + 1$, where $c = f(1) - 1$. But now a direct check shows that f is not a solution. Thus the only solution to the given equation is the constant function $f(x) = 2$.

Problem 18.36. (India 2008) Find all functions $f : (0, \infty) \rightarrow (0, \infty)$ such that $f(x + f(y)) = yf(xy + 1)$ for all $x, y \in (0, \infty)$.

Solution. First, we shall prove that $f(y) \leq 1$ for all $y > 1$. Assume the contrary, i.e. $f(y) > 1$ for some $y > 1$. Setting $x = \frac{f(y) - 1}{y - 1} > 0$ gives $x + f(y) = xy + 1$ and hence $y = \frac{f(x + f(y))}{f(xy + 1)} = 1$, a contradiction. In the same way we get that $f(y) \geq 1$ for all $y < 1$.

Next, we shall prove that $f(y) = \frac{1}{y}$ for all $y > 1$. Setting $x = 1 - \frac{1}{y} > 0$, then $f(1 + f(y) - 1/y) = yf(y)$. If $f(y) > \frac{1}{y}$ for some $y > 1$, then $f(1 + f(y) - 1/y) < 1$, a contradiction. Analogously, $f(y) < \frac{1}{y}$ is impossible for $y > 1$ which implies that $f(y) = \frac{1}{y}$. Then $f(x + 1/y) = f(x + f(y)) = yf(xy + 1) = \frac{1}{x + 1/y}$, $x > 0, y > 1$. Since for $0 < x' < x \leq 1$ and $y = \frac{1}{x - x'} > 1$ we have that $x = x' + 1/y$, it follows that $f(x) = \frac{1}{x}$ for all $x > 0$. This function obviously satisfies the given condition.

Problem 18.37. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x + y)f(x - y) = f(x^2) - f(y^2)$$

for all $x, y \in \mathbb{R}$.

Solution. Clearly, $f(0) = 0$ and taking $y = 0$ we get $f(x^2) = f(x)^2$, which shows that $f(x) \geq 0$ if $x \geq 0$. Take $x = 0$ to obtain $f(y)(f(y) + f(-y)) = 0$. Suppose that $f(y) = 0$ for some y . Then $f(-y)^2 = f(y^2) = f(y)^2 = 0$ and $f(y) + f(-y) = 0$. But this also holds when $f(y) \neq 0$ and therefore f is an odd function. Hence $f(x) \leq 0$ for $x \leq 0$. As f respects nonnegative and negative numbers, we conclude from the given condition that the function f is nondecreasing. We have also $f(1)^2 = f(1)$, i.e. $f(1) = 0$ or 1 . Suppose that $f(1) = 0$. As f is nondecreasing we obtain that f is identically zero on $[0, 1]$. But we also know that $f(y + 1)^2 = f(y)^2$ for all y which follows by setting $x = y + 1$ in the given identity. So by induction f is identically zero on $[0, n]$

for all $n \in \mathbb{N}$. Thus f is identically zero on $[0, \infty]$ and hence on $\mathbb{R}$ since f is odd.

Suppose now that $f(1) = 1$. Then $f(n-1)f(n+1) = f(n)^2 - 1$ and we claim that $f(n) = n$ for all integers n . By the previous relation, it is enough to show that $f(2) = 2$. But if $f(2) = x$, then $f(3) = x^2 - 1$ and $f(4) = x(x^2 - 2)$. On the other hand $f(4) = f(2)^2 = x^2$ and we immediately infer that $x = 2$ (we exclude the case $x = -1$ since f is nondecreasing). Therefore $f(n) = n$ for all integers n . But then $[x] + 1 \geq f([x] + 1) \geq f(x) \geq f([x]) = [x]$ and so $f(x)/x$ tends to 1 as $x \rightarrow \infty$. This together with $(f(x)/x)^{2^n} = f(x^{2^n})/x^{2^n}$ implies that $f(x) = x$ for $x > 1$. Now, fix x arbitrary and take y very large. Then $f(x-y) = -f(y-x) = -(y-x) = x-y$, $f(x+y) = x+y$ and $f(y) = y$. Plugging these back in the functional equation we obtain $f(x)^2 = x^2$ for all $x \in \mathbb{R}$. Hence $f(x) = x$ since $f(x)$ respects the sign of x . Thus only the identity function satisfies the original functional equation.

Problem 18.38. Find all functions $f : (0, \infty) \rightarrow (0, \infty)$ such that

$$f(x)f(yf(x)) = f(x+y)$$

for all $x, y \in (0, \infty)$.

Solution. The first key point is to note that if $x + y = yf(x)$, then $f(x) = 1$. But we can take such y if $f(x) > 1$ and this argument shows that $f(x) \leq 1$ for all x . Thus $f(x+y) \leq f(x)$ and so f is non-increasing.

We shall prove now that if $f(x) = 1$ for some $x > 0$ then f is constant. This is easy, as the functional equation gives $f(y+x) = f(y)$ and so $f(nx) = 1$ for all $n \in \mathbb{N}$. Take now any $y > 0$ and n such that $y < nx$. Then $f(y) \geq f(nx) = 1$ and by the first observation we get $f(y) = 1$. Thus f is constant.

Now, suppose that $f(x) < 1$ for all $x > 0$. Then f is strictly decreasing. Replace y by $y/f(x)$ to deduce that

$$f(x)f(y) = f\left(x + \frac{y}{f(x)}\right).$$

This identity together with the injectivity of f imply that

$$x + \frac{y}{f(x)} = y + \frac{x}{f(y)}$$

for all $x, y \in (0, \infty)$. Now setting $y = 1$ in this equation and solving it for $f(x)$ we obtain $f(x) = \frac{1}{1+ax}$ for some constant $a > 0$. One can easily check that all functions of this form are indeed solutions.

Problem 18.39. (G. Dospinescu) Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(xf(y)) + f(yf(x)) = 2xy$$

for all $x, y \in \mathbb{R}$.

Solution. Note that $f(0) = 0$ and $f(xf(x)) = x^2$. So $f(f(1)) = 1$ and if we take $x = f(1)$ in the previous relation we find that $f(1)^2 = 1$.

We prove now that f is injective. First, let us suppose that $f(1) = 1$. Taking $y = 1$, we find that $f(x) + f(f(x)) = 2x$ and the injectivity follows. Next, suppose that $f(1) = -1$. Then using $f(xf(x)) = x^2$ we find that $f(-f(-1)) = 1$. Put $x = -f(-1)$ in $f(xf(x)) = x^2$ to obtain $f(-1)^2 = 1$. Since $f(-f(-1)) = 1$ and $f(1) \neq 1$, we have $f(-1) = 1$. Next, take $y = -1$ to get $f(x) + f(-f(x)) = -2x$, from where the injectivity follows.

Another step is to prove that $f(1/x) = 1/f(x)$. This is the trickier one. Take $y = f(1/x) \cdot 1/x$. Using $f(xf(x)) = x^2$ again, we find that

$$f(1/x \cdot f(1/x) \cdot f(x)) = f(1/x)$$

and using injectivity we obtain that $f(1/x) = 1/f(x)$.

Now the problem is solved. If $f(1) = 1$, we saw that $f(x) + f(f(x)) = 2x$. Then substituting x by $1/x$ and using the last step we find that

$$1/f(x) + 1/f(f(x)) = 2/x.$$

From here $1/f(x) + 1/(2x - f(x)) = 2/x$ yielding $f(x) = x$ for all x . If $f(1) = -1$ we use the same arguments to find that $f(x) = -x$ for all x . Both functions verify the problem condition.

Problem 18.40. (Japan 2009) Find all functions $f : [0, \infty) \rightarrow [0, \infty)$ such that

$$f(x^2) + f(y) = f(x^2 + y + xf(4y))$$

for all $x, y \in [0, \infty)$.

Solution. We shall use the fact that for each constant k the quadratic function $g(x) = x^2 + kx$ takes all nonnegative values. We first prove that f is a nondecreasing function. Indeed, let $z = y + a$, $a > 0$. Then there is $x > 0$ such that $a = g(x) = x^2 + f(4y)x$ and we have

$$f(z) = f(y + a) = f(y + x^2 + xf(4y)) = f(x^2) + f(y) \geq f(y).$$

Suppose now that $f(z) = 0$ for some $z > 0$. Then $f(x) = 0$ for all $x \in [0, z]$. Replacing x by $\sqrt{x}$ and y by $\frac{z}{4}$ in the given equation we get

$$f(x) = f(x) + f\left(\frac{z}{4}\right) = f\left(x + \frac{z}{4}\right).$$

This shows that the function f is periodic with period $\frac{z}{4}$ and therefore $f = 0$.

Thus we may assume that $f(x) > 0$ for all $x > 0$. Then the function f is strictly increasing, i.e. injective and the identity

$$f(x + y + \sqrt{x}f(4y)) = f(x + y) = f(y + x + \sqrt{y}f(4x))$$

implies that $\sqrt{x}f(4y) = \sqrt{y}f(4x)$. Hence

$$\frac{f(x)}{\sqrt{x}} = \text{const}$$

and therefore $f(x) = c\sqrt{x}$. Now setting $x = y = 1$ in the given equation we obtain that $c = 1$, i.e. $f(x) = \sqrt{x}$.

Problem 18.41. (G. Dospinescu) Find all functions $f : (0, \infty) \rightarrow (0, \infty)$ such that

$$f(1 + xf(y)) = yf(x + y)$$

for all $x, y \in (0, \infty)$.

Solution. The answer is $f(x) = 1/x$, which can be easily checked to be a solution. Let us prove that it is the unique solution. First of all, we will show that $f(x) \leq 1$ for $x > 1$ and $f(x) \geq 1$ for $x < 1$. Suppose for example that for some $y > 1$ we have $f(y) > 1$. Then we set $x = (y - 1)/(f(y) - 1)$ to obtain $f(x + y) = yf(x + y)$, a contradiction.

Thus, $f(x+y) = f(1+xf(y))/y \leq 1/y$ for all $x, y \in (0, \infty)$, from where $f(x) \leq 1/x$. Indeed, if $f(a) > 1/a$ for some $a \in \mathbb{R}$ then for some $b \in (0, a)$ we have $f(a) > 1/(a-b) > 1/a$. Hence for $x = b$ and $y = a-b$ we get $f(x+y) > 1/y$, a contradiction. Next, note that $f(1+f(z)) = zf(1+z)$, so

$$yf(x+y) = f(1+xf(y)) = f(1+f(xf(y)))/xf(y)$$

and $xyf(y)f(x+y) = f(1+f(xf(y)))$. Taking $x = 1/f(y)$ we find that $f(x) \geq (1-k)/(kx)$ for all x , where $k = 1 - f(1+f(1)) > 0$.

We prove now that f is injective. If $f(a) = f(b)$, then for all x we have $af(x+a) = bf(x+b)$ and so $a^n f(na) = b^n f(bn)$ for all $n \geq 1$. We may assume that $a < b$, in which case for n large enough we have $f(nb) < (a/b)^n$, which contradicts the fact that $f(nb) \geq (1-k)/(knb)$ for all n .

Finally, let $y > 1$ and $x > 0$. Then

$$\begin{aligned} yf(x+y) &= f(1+xf(y)) = f((1+xf(y)-1/y)+1/y) \\ &= yf(1+f(1/y)(1+xf(y)-1/y)) \end{aligned}$$

and from injectivity of f we deduce that $x+y = 1 + f(1/y)(1+xf(y)-1/y)$. Hence $f(1/y)f(y) = 1$ and $y = 1 + f(1/y) - 1/yf(1/y)$. From here we find that for $y > 1$ we have $f(y) = 1/y$ and since $f(1/x) = 1/f(x)$ we get $f(x) = 1/x$ for $x \neq 1$. But then it is easy to see that $f(1) = 1$ and therefore $f(x) = 1/x$ for all $x \in (0, \infty)$.

Problem 18.42. (Ukraine TST 2007) Find all functions $f : \mathbb{Q} \rightarrow \mathbb{Q}$ such that

$$f(x^2 + y + f(xy)) = 3 + (x + f(y) - 2)f(x)$$

for all $x, y \in \mathbb{Q}$.

Solution. We will show that $f(x) = x + 1$ is the only solution. The hardest part of the proof is to compute $f(0)$. First of all, by taking $x = 0$ we obtain $f(x + f(0)) = 3 + f(0)(f(x) - 2)$. Taking $y = 0$ gives

$$f(x^2 + f(0)) = 3 + (x + f(0) - 2)f(x).$$

By setting $x = 1$ in these two relations and eliminating the term $f(1 + f(0))$ we obtain that $f(1) = 2f(0)$. Also, by taking $x = -1$ we obtain

$$3 + (f(0) - 3)f(-1) = f(1 + f(0)) = 3 + (f(0) - 1)f(1),$$

which implies $f(-1) = \frac{2f(0)^2 - 2f(0)}{f(0) - 3}$. Therefore, setting $x = y = -1$ in the original functional equation and using the identity

$$f(x + f(0)) = 3 + f(0)(f(x) - 2)$$

we deduce that

$$\begin{aligned} f(2f(0)) &= f(f(1)) = 3 + (f(-1) - 3)f(-1) \\ &= 3 + f(0)(f(f(0)) - 2) = 3 + f(0)[3 + f(0)(f(0) - 2) - 2]. \end{aligned}$$

Thus if $x = f(0)$ we have $f(-1) = \frac{2x^2 - 2x}{x - 3}$ as well as

$$f(-1)^2 - 3f(-1) + 3 = x^3 - 2x^2 + x + 3.$$

Replacing $f(-1)$ as a function of x in the last relation we obtain a fifth degree equation with obvious roots 0, 1 and no other rational roots. Suppose that $f(0) = 0$. Then by setting $x = y = 0$ in the initial relation we obtain a contradiction.

So we finally obtained that $f(0) = 1$ and since

$$f(x + f(0)) = 3 + f(0)(f(x) - 2),$$

we get $f(x + 1) = f(x) + 1$. So $f(x + n) = f(x) + n$ and $f(n) = n + 1$ for all integers. We also have $f(x^2) = 2 + (x - 1)f(x)$. Thus, if we take $x = q + \frac{p}{q}$ we obtain

$$f\left(\frac{p^2}{q^2}\right) = 2 - p - q + \left(q - 1 + \frac{p}{q}\right)f\left(\frac{p}{q}\right).$$

On the other hand

$$f\left(\frac{p^2}{q^2}\right) = 2 + \left(\frac{p}{q} - 1\right)f\left(\frac{p}{q}\right)$$

and solving this system gives $f(x) = x + 1$ for all rational numbers x .

Problem 18.43. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfy the equations $f(x^2) = f(x)^2$ and $f(x + 1) = f(x) + 1$. Prove that $f(x) = x$ for all $x \in \mathbb{R}$.

Solution. The equations imply that $f(x) \geq 0$ for $x \geq 0$ and

$$f(x+n) = f(x) + n$$

for all integers n . Then

$$(f(x) - 1)^2 = f(x-1)^2 = f((x-1)^2) = f(1-x)^2 = (1 + f(-x))^2.$$

Expanding both sides and noting that $f(x)^2 = f(-x)^2$ we get $f(-x) = -f(x)$ for all $x \in \mathbb{R}$. In particular $f(0) = 0$ and therefore $f(n) = n$ for all integers n . Now we show two ways to prove that $f(x) = x$. The first one is to proceed as follows. We have $f(x) - [x] = f(x - [x]) \geq 0$ and

$$f(x) - ([x] + 1) = -f([x] + 1 - x) \leq 0,$$

which show that $x - 1 \leq f(x) \leq x + 1$. Hence $1 - \frac{1}{x} \leq \frac{f(x)}{x} \leq 1 + \frac{1}{x}$ for $x > 0$ and $\frac{f(x)}{x}$ tends to 1 for $x \rightarrow \infty$. Combining this with

$$(f(x)/x)^{2^n} = f(x^{2^n})/x^{2^n}$$

we deduce that $f(x) = x$ for $x > 1$. (Indeed, if $x > 1$ is fixed, then $(f(x)/x)^{2^n}$ tends to 0 if $f(x) < x$ and to ∞ if $f(x) > x$.) This together with

$$f(x+1) = f(x) + 1$$

shows that $f(x) = x$ for all $x \in \mathbb{R}$.

The second way is the following. We show first that if $x \in [n, n+1]$ then $f(x) \in [n, n+1]$. Indeed, if $x - n \geq 0$ then $f(x) - n = f(x - n) \geq 0$. Similarly if $(n+1) - x \geq 0$ then $(n+1) - f(x) = n+1 + f(-x) = f(n+1-x) \geq 0$. Hence we have proved that $|f(x) - x| \leq 1$ for all x . Now set $k = 2^n$, where n is a positive integer. For all $x \geq 0$ we have $f(x) \geq 0$ and

$$\begin{aligned} 1 &\geq |x^k - f(x^k)| = |x^k - f(x)^k| \\ &= |x - f(x)| \cdot |x^{k-1} + x^{k-2}f(x) + \cdots + f(x)^{k-1}| \geq |x - f(x)| x^{k-1}. \end{aligned}$$

Therefore $|x - f(x)| \leq x^{1-k}$. If $x > 1$ then letting $k \rightarrow \infty$ we have $x^{1-k} \rightarrow 0$, so $f(x) = x$ when $x > 1$. Now using the identity $f(x+n) = f(x) + n$ we conclude that $f(x) = x$ for all $x \in \mathbb{R}$.

Problem 18.44. Determine all monotonic functions $f : [0, \infty) \rightarrow \mathbb{R}$ such that

$$f(x+y) - f(x) - f(y) = f(xy+1) - f(xy) - f(1),$$

for all $x, y \geq 0$ and $f(3) + 3f(1) = 3f(2) + f(0)$.

Solution. If $f(x)$ is a solution, then so is $f(x) + a$ and we may assume that $f(1) = 1$.

Let $P(x, y)$ be the assertion $f(x+y) - f(x) - f(y) = f(xy+1) - f(xy) - 1$.

Let $m, n, p \in \mathbb{N}$ and set $g(x) = f\left(\frac{x}{p}\right)$. Comparing $P\left(\frac{2m}{p}, \frac{n}{p}\right)$ and

$P\left(\frac{2n}{p}, \frac{m}{p}\right)$ we get $g(2m+n) - g(2m) - g(n) = g(2n+m) - g(2n) - g(m)$.

This suggests to look for all solutions of the following problem: "Find all functions $g : \mathbb{N} \rightarrow \mathbb{R}$ such that $g(2x+y) - g(2x) - g(y) = g(2y+x) - g(2y) - g(x)$ for all $x, y \in \mathbb{N}$ ".

The set $\mathbb{S}$ of solutions is a real vector space. Setting $y = 1$, we get

$$g(2x+1) = g(2x) + g(1) + g(x+2) - g(2) - g(x).$$

Setting $y = 2$, we get

$$g(2x+2) = g(2x) + g(2) + g(x+4) - g(4) - g(x).$$

From these two equations, we see that the values $g(1)$, $g(2)$, $g(3)$, $g(4)$ and $g(6)$ of g determine uniquely $g(x) \forall x \in \mathbb{N}$ and so the dimension of $\mathbb{S}$ is at most 5. But the five functions below are independent solutions: $g_1(x) = 1$, $g_2(x) = x$, $g_3(x) = x^2$, $g_4(x) = 1$ if $x \equiv 0 \pmod{2}$ and $g_4(x) = 0$ if $x \not\equiv 0 \pmod{2}$ and $g_5(x) = 1$ if $x \equiv 0 \pmod{3}$ and $g_5(x) = 0$ if $x \not\equiv 0 \pmod{3}$. So the general solution has the form

$$g(x) = a \cdot x^2 + b \cdot x + c + d \cdot g_4(x) + e \cdot g_5(x).$$

Now

$$f\left(\frac{x}{p}\right) = a_p x^2 + b_p x + c_p + d_p g_4(x) + e_p g_5(x)$$

for all $x \in \mathbb{N}$ and choosing $x = kp$ we get

$$f(k) = a_p k^2 p^2 + b_p kp + c_p + d_p g_4(kp) + e_p g_5(kp).$$

Thus $a_p p^2 = a$ and $b_p p = b$ for some real a, b . Choosing $x = 2kp$, $x = 3kp$ and $x = 6kp$, we get $c_p = c$ and $d_p = e_p = 0$. Hence

$$f\left(\frac{x}{p}\right) = a \frac{x^2}{p^2} + b \frac{x}{p} + c$$

for all $x \in \mathbb{N}$ and therefore $f(x) = ax^2 + bx + c$ for all $x \in \mathbb{Q}^+$. But $f(x)$ monotonous implies that $a = 0$ or $\frac{b}{a} \geq 0$ and now it is easy to conclude that $f(x) = ax^2 + bx + c$ for all $x \in (0, \infty)$. Finally, the condition

$$f(3) + 3f(1) = 3f(2) + f(0)$$

shows that $f(x) = ax^2 + bx + c$ for all $x \in [0, \infty)$ and it is easy to check back that this mandatory form indeed is a solution.

Problem 18.45. (IMO Shortlist 2009) Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(xf(x+y)) = f(yf(x)) + x^2$$

for all $x, y \in \mathbb{R}$.

Solution. It is easy to check that the functions $f(x) = x$ and $f(x) = -x$ are solutions. We shall prove that there are no other solutions.

Let f be a function satisfying the given equation. It is clear that f is not a constant. We first show that $f(0) = 0$. Suppose that $f(0) \neq 0$. For any real t substituting $(x, y) = (0, t/f(0))$ into the given equation we have $f(0) = f(t)$, a contradiction to the fact that f is not a constant. Therefore $f(0) = 0$. Now substituting $(x, y) = (t, 0), (t, -t)$ into the given equation we get $f(tf(t)) = f(0) + t^2 = t^2$ and $f(tf(0)) = f(-t(f(t)) + t^2$, respectively. Hence

$$f(tf(t)) = t^2, \quad f(-t(f(t))) = -t^2 \quad (1)$$

for every real t . Consequently, f is surjective. We also see that if $f(t) = 0$ then $0 = f(tf(t)) = t^2$, i.e. $t = 0$. We next show that f is an odd function,

i.e. $f(-s) = -f(s)$ for all $s \in \mathbb{R}$. This is clear if $f(s) = 0$. If $f(s) < 0$ then there is $t \neq 0$ with $f(s) = -t^2$. We can also find a real number a such that $af(t) = s$. Substituting $(x, y) = (t, a)$ into the given equation, we get

$$f(tf(t+a)) = f(af(t)) + t^2 = f(s) + t^2 = 0,$$

and therefore $tf(t+a) = 0$ which implies $t+a = 0$. Hence $s = -tf(t)$ and using (1) gives $f(-s) = f(tf(t)) = t^2 = -f(s)$. Now suppose that $f(s) > 0$. Then there is a $t \neq 0$ with $f(s) = t^2$ and we can choose a such that $tf(a) = s$. Substituting $(x, y) = (t, a-t)$ into the given equation and proceeding as above we get $s = tf(t)$. Hence $f(-s) = f(-tf(t)) = -t^2 = -f(s)$ holds also in this case, finishing the proof that the function f is odd.

Now substituting $(x, y) = (s, t), (t, -s-t)$ and $(-s-t, s)$ into the given equation and using that f is an odd function we obtain the identities

$$f(tf(s)) - f(sf(s+t)) = -s^2,$$

$$f(tf(s)) - f((s+t)f(t)) = -t^2,$$

$$f((s+t)f(t)) + f(sf(s+t)) = (s+t)^2.$$

Adding up these three equations yields $f(tf(s)) = ts$. By fixing s so that $f(s) = 1$ we obtain $f(x) = sx$ for all real x . Substituting into the given equation shows that $s^2 = 1$, i.e. $s = \pm 1$.

Problem 18.46. (IMO Shortlist 2008) Let $f : \mathbb{R} \rightarrow \mathbb{N}$ be a function such that

$$f\left(x + \frac{1}{f(y)}\right) = f\left(y + \frac{1}{f(x)}\right)$$

for all $x, y \in \mathbb{R}$. Prove that the function f is not surjective.

Solution. Suppose $f(\mathbb{R}) = \mathbb{N}$. We prove several properties of the function f in order to reach a contradiction. To start with, observe that one can assume that $f(0) = 1$. Indeed, let $f(a) = 1$. Consider the function $g(x) = f(x+a)$. By substituting $x+a$ and $y+a$ for x and y in the given identity we have

$$g(x+1/g(y)) = f(x+a+1/f(y+a)) = f(y+a+1/f(x+a)) = g(y+1/g(x)).$$

So g satisfies the given functional equation with the additional property

$$g(0) = 1.$$

Also, $g(\mathbb{R}) = f(\mathbb{R}) = \mathbb{N}$ and we can assume that $f(0) = 1$.

Claim 1. For every $c \in \mathbb{R}$ we have $\{f(c + 1/n) \mid n \in \mathbb{N}\} = \mathbb{N}$.

Proof. The given identity implies

$$\begin{aligned} f(\mathbb{R}) &= \{f(x + 1/f(c)) \mid x \in \mathbb{R}\} = \{f(c + 1/f(x)) \mid x \in \mathbb{R}\} \\ &\subseteq \{f(c + 1/n) \mid n \in \mathbb{N}\} \subseteq f(\mathbb{R}) \end{aligned}$$

and the claim follows. $\square$

We shall use *Claim 1* in the special cases $c = 0$ and $c = 1/3$:

$$\{f(1/n) \mid n \in \mathbb{N}\} = \{f(1/3 + 1/n) \mid n \in \mathbb{N}\} = \mathbb{N}.$$

Claim 2. If $f(u) = f(v)$ for some $u, v \in \mathbb{R}$ then $f(u+q) = f(v+q)$ for all non-negative rational numbers q . Furthermore, if $f(q) = 1$ for some nonnegative rational number q then $f(kq) = 1$ for all $k \in \mathbb{N}$.

Proof. We have

$$f(u + 1/f(x)) = f(x + 1/f(u)) = f(x + 1/f(v)) = f(v + 1/f(x)).$$

Since $f(x)$ attains all positive integers this yields $f(u + 1/n) = f(v + 1/n)$ for all $n \in \mathbb{N}$. Let $q = k/n$ be a positive rational number. Then k repetitions of the last step yield

$$f(u + q) = f(u + k/n) = f(v + k/n) = f(v + q).$$

Now let $f(q) = 1$ for some nonnegative rational q , and let $k \in \mathbb{N}$. As $f(0) = 1$, the previous conclusion yields successively

$$f(q) = f(2q), f(2q) = f(3q), \dots, f((k-1)q) = f(kq),$$

as needed. $\square$

Claim 3. The equality $f(q) = f(q+1)$ holds for all nonnegative real q .

Proof. As noted above there is a positive integer m such that $f(1/m) = 1$. Applying the second statement of *Claim 2* with $q = 1/m$ and $k = m$ yields $f(1) = 1$. Since $f(0) = f(1) = 1$ the first statement of *Claim 2* implies $f(q) = f(q+1)$ for all nonnegative real q . $\square$

Claim 4. The equality $f(1/n) = n$ holds for every $n \in \mathbb{N}$.

Proof. For a nonnegative rational q we set $x = q, y = 0$ in the given identity and use *Claim 3* to obtain

$$f(1/f(q)) = f(q + 1/f(0)) = f(q + 1) = f(q).$$

As we noted above, for each $n \in \mathbb{N}$ there exists $k \in \mathbb{N}$ such that $f(1/k) = n$. Then the last equation with $q = 1/k$ gives

$$n = f(1/k) = f(1/f(1/k)) = f(1/n). \quad \square$$

Now we are ready to obtain a contradiction. Let $n \in \mathbb{N}$ be such that $f(1/3 + 1/n) = 1$. Write $1/3 + 1/n$ as s/t , where s and t are co-prime. Observe that $t > 1$ since $1/3 + 1/n$ is not an integer. By Bezout's lemma there are $k, l \in \mathbb{N}$ such that $ks - lt = 1$. Because $f(0) = f(s/t) = 1$, *Claim 2* implies $f(ks/t) = 1$. Now $(f(ks/t) = f(1/t + l))$. On the other hand $f(1/t + l) = f(1/t)$ by l successive applications of *Claim 3*. Finally, $f(1/t) = t$ by *Claim 4*, leading to $t = 1$, a contradiction.

19.18.3 Continuous Functions

Problem 18.47. Find all continuous functions $f : (0, \infty) \rightarrow (0, \infty)$ such that

$$f(x) + \frac{1}{f(x)} = x + \frac{1}{x}$$

for all $x \in (0, \infty)$.

Solution. Solving with respect to $f(x)$ we get $f(x) = x$ or $f(x) = \frac{1}{x}$ for every $x > 0$. Note that the functions x and $\frac{1}{x}$ coincide only in $x = 1$, so $f(1) = 1$. Now, we claim that $f(x) = x$ for all $x \in (0, 1)$ or $f(x) = \frac{1}{x}$ for all $x \in (0, 1)$.

Indeed, assume that $f(u) = u, f(v) = \frac{1}{v}$ for $0 < u, v < 1$. Then $f(u) < 1$ and $f(v) > 1$ and by continuity there exists r between u and v such that $f(r) = 1$. However $r \in (0; 1)$ thus either $f(r) = r$ or $f(r) = \frac{1}{r}$, a contradiction. Similarly, $f(x) = x$ for all $x \in (1, \infty)$ or $f(x) = \frac{1}{x}$ for all $x \in (1, \infty)$. Therefore we have four solutions, according to whether $f(x)$ is x or $\frac{1}{x}$ on $(0, 1)$ and whether $f(x)$ is x or $\frac{1}{x}$ on $(1, \infty)$. All these four functions are continuous and satisfy the given identity.

Problem 18.48. (Bulgaria 2006) Let $f : (0, \infty) \rightarrow (0, \infty)$ be a function such that

$$f(x+y) - f(x-y) = 4\sqrt{f(x)f(y)}$$

for all $x > y > 0$.

- a) Prove that $f(2x) = 4f(x)$ for all $x \in (0, \infty)$.
- b) Find all such functions f .

Solution. a) Since $f(x+y) - f(x-y) > 0$, then f is a (strictly) increasing function. Hence $f(x)$ has a limit $l \geq 0$ as $x \rightarrow 0, x > 0$. Letting $x, y \rightarrow 0, x > y > 0$ gives $l - l = 4\sqrt{l^2}$, i.e. $l = 0$. For a fixed x letting $y \rightarrow 0, y > 0$ implies that $f(x+y) - f(x-y) \rightarrow 0$. Since f is monotonic we conclude that it is continuous at x . Finally, letting $y \rightarrow x, y < x$, we get that $f(2x) = 4f(x)$.

- b) Set $x = ny > 0$ and $n \geq 2$. Then

$$f((n+1)y) = f((n-1)y) + 4\sqrt{f(ny)f(y)}.$$

Using that $f(2y) = 4f(y)$, we conclude by induction on n that $f(ny) = n^2f(y)$. Set $f(1) = c > 0$. Then $f(n) = cn^2$. For $p, q \in \mathbb{N}$ one has that

$$cp^2 = f(q \cdot p/q) = q^2f(p/q),$$

i.e. $f(p/q) = c(p/q)^2$. Since f is a continuous function and $\mathbb{Q}^+$ is a dense subset of $(0, \infty)$, it follows that $f(x) = cx^2$ for all $x > 0$. Conversely, the functions of this form satisfy the given condition.

Problem 18.49. Find all continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfying

$$f(f(x)y - xf(y)) = yf(f(x)) - f(x)f(y)$$

for all $x, y \in \mathbb{R}$.

Solution. (by Batominovski) Take $x = y = 0$ to get $f(0) = -f(0)^2$.

Case 1. $f(0) = -1$.

Let $x = 0$, then $f(-y) = yf(-1) + f(y)$ and by taking $y = 1$ we obtain $f(1) = 0$. Now, take $x = 1$ to get $f(-f(y)) = -y$. Next, take $x = y$ to get $xf(f(x)) = f(x)^2 - 1$. Substitute $-f(x)$ instead of x in the last relation, to obtain $f(x)f(-x) = 1 - x^2$. Finally, we saw that $f(-x) = xf(-1) + f(x)$. Let us write $f(-1) = -2k$, then by eliminating $f(-x)$ we obtain that

$$f(x) = kx \pm \sqrt{(k^2 - 1)x^2 + 1}$$

and this for all x . As $f(0) = -1$ and f is continuous, we deduce that

$$f(x) = kx - \sqrt{(k^2 - 1)x^2 + 1}$$

for all x and so we must have $k^2 \geq 1$. As $f(1) = 0$, we obtain $k \geq 0$ and so $k \geq 1$. It is a painful task to check that all these functions are indeed solutions!

Case 2. $f(0) = 0$.

This case is extremely hard, so we cut it into many steps. By taking $x = y$ we get $xf(f(x)) = f(x)^2$.

Subcase 1. There is $t \neq 0$ such that $f(t) = 0$.

We shall use the following

Lemma. We have $f(1) = 0$, $xf(f(x)) = f(x)^2$, $f(-f(y)) = 0$, and $f(f(x)f(y)) = 0$ for all x and y .

Proof of lemma. We already saw that $xf(f(x)) = f(x)^2$. Now, take $x = t$, then $y = t$ and then $x = -tf(y)$ in the original functional equation to obtain that

$$f(tf(x)) = tf(f(x)), \quad f(-tf(x)) = 0, \quad f(tf(x)^2) = 0.$$

Therefore

$$f(tf(f(1))) = tf(f(f(1)))$$

and $f(tf(1)^2) = 0$. But $f(1)^2 = f(f(1))$ and so $f(f(f(1))) = 0$. Hence from $xf(f(x)) = f(x)^2$ we easily deduce that $f(1) = 0$. Now take $x = 1$ in the functional equation to get the third identity. But then the fourth identity is also clear and the lemma is proved. $\square$

Let $S = f^{-1}(\mathbb{R} \setminus \{0\})$. If $S = \emptyset$, then f is the zero function. Suppose S is nonempty and let F be the restriction of f on S . By the second identity of the lemma, F is injective.

We shall prove now that $f(a) \leq 0$ for all a . Suppose that there exists $a \neq 0$ such that $f(a) > 0$. Thus, $f(f(a)) = \frac{f(a)^2}{a} \neq 0$, so there is y such that $yf(f(a)) \gg \max\{f(a), 1\}$. (Here the symbol " $A \gg B$ " means that the difference $A - B$ can be made arbitrarily large.) Since

$$yf(f(a)) = f(f(a)y - af(y)) + f(a)f(y)$$

we have either $f(f(a)y - af(y)) \gg 1$ or $f(y) \gg 1$. In any case, using the continuity of f , there must be a number $u \neq 0$ such that $f(u) = 1$. Consequently,

$$0 = uf(1) = uf(f(u)) = f(u)^2 = 1,$$

a contradiction.

Now, if $f(p) < 0$ and $f(q) < 0$ for some $p > 0$ and $q < 0$, then the injectivity of F is violated (argued by continuity of f). Thus, either (i) $f(x) = 0$ for all $x \geq 0$ and $f(x) \leq 0$ for $x < 0$, or (ii) $f(x) = 0$ for all $x \leq 0$ and $f(x) \leq 0$ for $x > 0$. The latter case cannot happen; otherwise, there would exist $s > 0$ such that $f(s) < 0$ and $f(f(s)) = \frac{f(s)^2}{s}$ would be positive.

Hence, $f(x) = 0$ if $x \geq 0$ and $f(x) \leq 0$ for $x < 0$. Now, assume that $f(r) < 0$ for some $r < 0$. Thus, for all $y \geq 0$,

$$f(f(r)y) = f(f(r)y - rf(y)) = yf(f(r)) - f(r)f(y) = y\frac{f(r)^2}{r},$$

or $f(f(r)y) = k(yf(r))$, where $k = \frac{f(r)}{r} > 0$. Since y is arbitrary, we then prove that $f(x) = kx$ for all negative x .

We can easily check that both the zero function and the functions in the form discussed above satisfy the condition. Therefore, for any $k \geq 0$, $f(x) = 0$ if $x \geq 0$ and $f(x) = kx$ for $x < 0$ is a solution to the functional equation.

Subcase 2. $f(t) = 0$ iff $t = 0$.

Note first that f is injective and

$$xf(f(x)y - xf(y)) = f(x)(f(x)y - xf(y))$$

for all x and y . For each x , define

$$S_x = \{f(x)y - xf(y) \mid y \in \mathbb{R}\}$$

(of course, $0 \in S_x$ for each x) and for $x \neq 0$ set $m_x = \frac{f(x)}{x}$. Also, let

$$T = \{x \in \mathbb{R} \mid S_x \neq \{0\}\}.$$

Thus, for all $x \in \mathbb{R}$ and $y \in S_x$ we have $f(y) = m_x y$. If $T = \emptyset$, then there exists $k \in \mathbb{R}$ such that $f(x) = kx$ for all x .

Assume that $T \neq \emptyset$ (but keep in mind that $0 \notin T$). Then, whenever a and b are in T , S_a and S_b contain a common nonzero element or else, one of them contains only nonnegative elements and the other contains only non-positive elements. Thus, $T = T^+ \cup T^-$, where $T^+ = \{x \in T \mid S_x \subseteq [0, +\infty)\}$ and $T^- = \{x \in T \mid S_x \subseteq (-\infty, 0]\}$. Hence, m_x are equal for all $x \in T^+$ and we denote their common value by m_+ (if $T^+ = \emptyset$, we just set $m_+ = 0$). The number m_- is defined similarly.

Now, if $x \in \mathbb{R} \setminus T$ and $y \in T$, we have $S_x = \emptyset$, or $f(x)y - xf(y) = 0$.

Since $f(y) = m_{\pm y}$, we then have $f(x) = m_{\pm x}$ as well. This shows that $f(x) = m_{\pm x}$ for all $x \in \mathbb{R}$.

By continuity of f it follows that $f(x) = k_+ x$ for all $x \geq 0$ and some $k_+ \in \{m_{\pm}\}$. Similarly, for all $x \leq 0$, $f(x) = k_- x$ for some $k_- \in \{m_{\pm}\}$.

(i) If $k_+ > 0$ and $k_- > 0$, then for $x > 0$ and $y < 0$,

$$f((k_+ - k_-)xy) = f(f(x)y - f(x)y) = yf(f(x)) - f(x)f(y).$$

This shows that $f((k_+ - k_-)xy) = k_+((k_+ - k_-)xy)$ or $0 < k_+ < k_-$.

(ii) If $k_+ > 0$ and $k_- < 0$, then for $x > 0$ and $y < 0$,

$$f((k_+ - k_-)xy) = f(f(x)y - f(x)y) = yf(f(x)) - f(x)f(y).$$

Hence,

$$f((k_+ - k_-)xy) = k_+((k_+ - k_-)xy).$$

However, since $(k_+ - k_-)xy < 0$,

$$f((k_+ - k_-)xy) = k_-((k_+ - k_-)xy).$$

Therefore, $k_+ = k_-$, which is absurd.

(iii) If $k_+ < 0$, then for $x > 0$ and $y < 0$,

$$f((k_+ - k_-)xy) = f(f(x)y - f(x)y) = yf(f(x)) - f(x)f(y)$$

implies $f((k_+ - k_-)xy) = 0$. Thus $k_+ = k_-$ which contradicts the fact that T is nonempty.

Combining the results in all cases considered above we conclude that the solutions to the given functional equation are the following functions:

$$f(x) = kx,$$

where k is an arbitrary constant, $f(x) = k_+x$ if $x \geq 0$ and $f(x) = k_-x$ if $x \leq 0$, where $0 \leq k_+ < k_-$, and $f(x) = kx - \sqrt{(k^2 - 1)x^2 + 1}$, where $k \geq 1$.

Problem 18.50. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function satisfying the equation

$$f(x + y) = f(x + f(y))$$

for all real x and y . Prove that:

- (a) if f is continuous then it is either constant or the identity function;
- (b) there are noncontinuous solutions to the given functional equation.

Solution. (a) Letting $x = 0$ we get $f(y) = f(f(y))$. So $f(x) = x$ on the range of f . Let $a = \inf f$ and $b = \sup f$. As f is continuous, the Intermediate value theorem tells us that the interval (a, b) lies in the range of f . So f is the identity on (a, b) . If $a = b$ then f is constant. If $a = -\infty$ and $b = +\infty$, then $f(x) = x$. Suppose that a is finite and $b > a$. The continuity of f implies that $f(a) = \lim_{x \rightarrow a+} f(x) = a$. So f must be the identity on $[a, b)$. Hence $f(x) = x$ on some interval $[a, a + 2d]$ for $d > 0$. Let $0 < t < d$. Then $f(a - t) = a + s$ for some $s > d$. (If $s \leq d$ then $a = f(t + a - t) = f(t + a + s) = a + t + s$, which is a contradiction.) However, the statements $f(a - t) > a + d$ for all $t \in (0, a)$ and $f(a) = a$ contradict the Intermediate Value theorem. The case where b is finite and $a < b$ is similar.

(b). Prove that $f(x) = x - [x]$ gives the desired example.

Problem 18.51. Prove that there is no continuous function $f : [0, 1] \rightarrow \mathbb{R}$ such that

$$f(x) + f(x^2) = x$$

for all $x \in [0, 1]$.

Solution. Assume that there is f with the given property. Then

$$\begin{aligned} f(x) &= x - f(x^2) = x - (x^2 - f(x^4)) \\ &= x - x^2 + x^4 - \cdots + (-1)^n (x^{2^n} - f(x^{2^{n+1}})) \end{aligned}$$

for $x \in [0, 1]$ and $n \geq 0$. Since $f(0) = 0$ and $\lim_{n \rightarrow \infty} x^{2^{n+1}} = 0$ for all $x \in (0, 1)$, it follows by continuity of f that $\lim_{n \rightarrow \infty} f(x^{2^{n+1}}) = 0$. Hence

$$f(x) = x - x^2 + x^4 - \cdots + (-1)^n x^{2^n} + \cdots$$

for all $x \in (0, 1)$. Fix $x_0 \in (0, 1)$ and define $x_n = (x_{n-1})^{1/2}$, $n \geq 1$. Then $(x_n)_{n \geq 0}$ is an increasing sequence converging to 1 and

$$\lim_{n \rightarrow \infty} f(x_n) = f(1) = 1/2.$$

On the other hand f is not constant and there exists $x_0 \in (0, 1)$ such that $f(x_0) \neq f(1) = 1/2$. Assume that $f(x_0) = \alpha + 1/2$, with $\alpha > 0$. Using the identity $f(x_n) = x_n - f(x_{n-1})$ we get that

$$f(x_{2n+1}) = x_1 - x_2 + x_3 - x_4 + \cdots + x_{2n-1} - x_{2n} + x_{2n+1} - f(x_0).$$

But $x_{2k-1} - x_{2k} < 0$ for $1 \leq k \leq n$ and $x_{2n+1} < 1$. Therefore

$$f(x_{2n+1}) < 1 - f(x_0) = 1/2 - \alpha,$$

a contradiction. The case $\alpha < 0$ is analogous, and we deduce that there is no function with the given property.

Problem 18.52. Determine all continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x + f(x)) = f(x)$$

for all $x \in \mathbb{R}$.

Solution. We shall prove that the only solutions are the constant functions. Suppose that f is a nonconstant solution and take $a < d$ such that $f(a) \neq f(d)$. We may assume that $f(d) > 0$. Let $b_1 = \max\{x \in [a, d] \mid f(x) = f(d)\}$. Since $f(b_1) > 0$ and $f(a) \neq f(d)$, then $b_1 > a$ and there is b such that $a \leq b < b_1$ and $f(b) > 0$. It is clear that $f(b) \neq f(d)$. Let $c = \max\{x \in [b, d] \mid f(x) = f(b)\}$. Since $f(c) = f(b) \neq f(d)$ it follows that $c < d$ and (*) $f(x) \neq f(b)$ for all $x \in (c, d]$. We may assume that $f(c) < f(d)$. Take $n \in \mathbb{N}$ such that

$$d + nf(d) \geq c + nf(c) + f(b)$$

and let m be the largest positive integer such that

$$b + mf(b) \leq d + nf(d).$$

Then

$$b + (m+1)f(b) > d + nf(d) \geq c + nf(c) + f(b)$$

which is equivalent to

$$b + mf(b) > c + nf(c).$$

Now the Intermediate value theorem implies that

$$(c + nf(c), d + nf(d)] \subset \{x + nf(x) \mid x \in (c, d]\}.$$

Hence there exists $x_0 \in (c, d]$ such that $x_0 + nf(x_0) = b + mf(b)$. The given identity implies by induction on k that $f(x + kf(x)) = f(x)$ for all $k \in \mathbb{N}$ and $x \in \mathbb{R}$. Thus $f(x_0) = f(b)$, a contradiction to (*). $\square$

Problem 18.53. (Tuymaada 2003) Find all continuous functions $f: (0, \infty) \rightarrow \mathbb{R}$ that satisfy

$$f\left(x + \frac{1}{x}\right) + f\left(y + \frac{1}{y}\right) = f\left(x + \frac{1}{y}\right) + f\left(y + \frac{1}{x}\right)$$

for all $x, y \in (0, \infty)$.

Solution. Linear functions satisfy the condition, and we prove that only these do.

If we replace y by $\frac{1}{y}$ we can enhance the condition:

$$f\left(x + \frac{1}{x}\right) + f\left(y + \frac{1}{y}\right) = f\left(x + \frac{1}{y}\right) + f\left(y + \frac{1}{x}\right) = f(x+y) + f\left(\frac{1}{x} + \frac{1}{y}\right).$$

Then

$$f(x+y) - f\left(x + \frac{1}{y}\right) = f\left(\frac{1}{x} + y\right) - f\left(\frac{1}{x} + \frac{1}{y}\right).$$

Now pick up a fixed $1 < y < C$ for some $C > 1$ and let x be sufficiently large. Now as f is continuous then f is uniformly continuous on the segment $\left[\frac{1}{C}, 2C\right]$. Hence for every $\epsilon > 0$ there exists $a > 0$ such that $|f(x) - f(y)| < \epsilon$ whenever $x, y \in \left[\frac{1}{C}, 2C\right]$, $|x - y| < a$. Thus taking $x > \max\left\{C, \frac{1}{a}\right\}$ we deduce that

$$\left|f\left(\frac{1}{x} + \frac{1}{y}\right) - f\left(\frac{1}{y}\right)\right|, \left|f\left(\frac{1}{x} + y\right) - f(y)\right| < \epsilon.$$

This means

$$\left(f\left(\frac{1}{x} + y\right) - f\left(\frac{1}{x} + \frac{1}{y}\right)\right) - \left(f(y) - f\left(\frac{1}{y}\right)\right) < 2\epsilon$$

and we have proven that

$$\lim_{x \rightarrow \infty} \left(f(x+y) - f\left(x + \frac{1}{y}\right)\right) = f(y) - f\left(\frac{1}{y}\right)$$

and the convergence is uniform on any interval $[1, C]$ for y . Now every $b > 0$ can be written as $y - \frac{1}{y}$ in a unique way for $y > 1$. Set $g(b) = f(y) - f\left(\frac{1}{y}\right)$. Then we see that $g(b) = \lim_{x \rightarrow \infty} (f(x+b) - f(x))$. It is pretty clear from here that $g(a+b) = g(a) + g(b)$ and since g is continuous we find that $g(x) = cx$. Now we can suppose that $c = 0$ otherwise take $f(x) - cx$ instead of f . So we

then have $f(x) = f\left(\frac{1}{x}\right)$ and also $\lim_{x \rightarrow \infty} (f(x+a) - f(x)) = 0$ uniformly for $a \in [0, C]$. Now take y be fixed and let $x \rightarrow \infty$. The condition

$$f\left(x + \frac{1}{x}\right) + f\left(y + \frac{1}{y}\right) = f\left(x + \frac{1}{y}\right) + f\left(y + \frac{1}{x}\right)$$

is rewritten as

$$f\left(x + \frac{1}{x}\right) - f\left(x + \frac{1}{y}\right) = f\left(y + \frac{1}{x}\right) - f\left(y + \frac{1}{y}\right).$$

The right-hand side now tends to zero according to the limit result obtained just before, and the left-hand side tends to $f\left(x + \frac{1}{x}\right) - f(x)$. Thus

$$f\left(x + \frac{1}{x}\right) = f(x).$$

This is enough to prove that f is constant. Indeed, let $1 \leq a < b$. Consider the sequences defined by $x_0 = a, y_0 = b, x_{i+1} = x_i + \frac{1}{x_i}, y_{i+1} = y_i + \frac{1}{y_i}$. As $x + \frac{1}{x}$ is increasing for $x \geq 1$ and also $\left(x + \frac{1}{x}\right)^2 \geq x^2 + 2$ we have $x_i < y_i$ and x_i, y_i grow arbitrarily large. Finally we can note that

$$\left|x + \frac{1}{x} - y - \frac{1}{y}\right| < |x - y|$$

for $x, y \geq 1$, therefore $|x_i - y_i| \leq |a - b|$. Hence $\lim_{n \rightarrow \infty} (f(x_n) - f(y_n)) = 0$. But $f(x_n) = f(a), f(y_n) = f(b)$ and we conclude that $f(a) = f(b)$. This finishes the proof. $\square$

Problem 18.54. Find all continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(xf(y)) + f(yf(x)) = \frac{1}{2}f(2x)f(2y)$$

for all real values x, y .

Solution. In this proof we show that when f is not constant, it is bijective on the separate domains $(-\infty, 0]$ and $[0, \infty)$, (not necessarily on $\mathbb{R}$) and we first find all solutions on those domains. Then we get all functions f , by joining any two functions from the separate domains and checking they work.

Assume f is not constant and let $P(x, y)$ be the assertion

$$f(xf(y)) + f(yf(x)) = \frac{1}{2}f(2x)f(2y).$$

Then

$$P(0, 0) : 4f(0) = f(0)^2 \implies f(0) = 0, 4.$$

Injectivity. As $f(x) = |x|$ is a solution, we cannot prove that f is injective on $\mathbb{R}$, instead we show it is injective on the domains $(-\infty, 0]$ and $[0, \infty)$. So suppose there are two real numbers $a \neq b$ such that $f(a) = f(b)$. Then we have

$$\frac{1}{4}f(2a)^2 + \frac{1}{4}f(2b)^2 = f(af(a)) + f(bf(b)) = f(af(b)) + f(bf(a)) = \frac{1}{2}f(2a)f(2b)$$

which implies

$$\frac{1}{4}[f(2a) - f(2b)]^2 = 0 \implies f(2a) = f(2b).$$

Moreover,

$$f(af(x)) + f(xf(a)) = \frac{1}{2}f(2a)f(2x) = \frac{1}{2}f(2b)f(2x) = f(bf(x)) + f(xf(b)).$$

This then implies

$$f(af(x)) = f(bf(x)) \tag{1}$$

for all $x \in \mathbb{R}$

Case 1. $f(0) = 0$.

First we will show that f is injective on $[0, \infty)$. So for the sake of contradiction assume there existed $a > b > 0$ such that $f(a) = f(b)$. Since $f(x)$ is continuous and not constant when $x > 0$, there is some interval $[0, c_1]$ or

$[-c_1, 0]$ such that f is surjective onto that interval. We may assume that interval is $[0, c_1]$. So, motivated by (1) we define a strictly decreasing sequence $u_0 \in [0, c_1]$, $u_{n+1} = \frac{b}{a}u_n$. We find that $u_n \in [0, c_1]$ for all n and therefore

$$f(au_0) = f(bu_0) = f(au_1) = \cdots = f(au_n).$$

Now $\lim_{n \rightarrow \infty} u_n \rightarrow 0$, so by the continuity of f we have

$$\lim_{n \rightarrow \infty} f(au_n) = f\left(\lim_{n \rightarrow \infty} au_n\right) = f(0) = 0.$$

This implies that $f(au_0) = 0$ for all $u_0 \in [0, c_1]$, and therefore $f(x) = 0$ when $x \in [0, ac_1]$. But then for all $x \in [0, ac_1]$ we have

$$P(x, x) : 0 = f(xf(x)) = \frac{1}{4}f(2x)^2,$$

hence $f(2x) = 0$. Inductively we find that $f(x) = 0$ for all $x \in [0, \infty)$, contradicting the assumption that f is not constant on that domain. Hence f is injective on the domain $[0, \infty)$.

As for the domain $(-\infty, 0]$, simply alter the original assumption to $a < b < 0$ such that $f(a) = f(b)$ and the same proof applies. Hence f is injective on $(-\infty, 0]$ and $[0, \infty)$.

Case 2. $f(0) = 4$.

Again we will consider the case $x \in [0, \infty)$. Assume there exist $a > b > 0$ such that $f(a) = f(b)$. Then

$$P\left(\frac{x}{2}, 0\right) : f(2x) + 4 = 2f(x) \iff f(2x) - 4 = 2[f(x) - 4],$$

and inductively $f(2^n x) - 4 = 2^n[f(x) - 4]$. So assuming there exists at least one value such that $f(x) - 4 \neq 0$, we will have $f(2^n) \rightarrow \pm\infty$. And since f is continuous, f will also be surjective onto at least one of $[4, \infty)$ or $(-\infty, 4]$. We may assume it for $[4, \infty)$.

Similar to the previous case we define the increasing sequence $u_0 \in \left[4, \frac{4a}{b}\right]$

and $u_{n+1} = \frac{au_n}{b}$. Again $u_n \in [4, \infty)$ and therefore

$$f(bu_0) = f(au_0) = f(bu_1) = \cdots = f(bu_n).$$

Now for any $y \in [4, \infty)$ there exists $u_0 \in \left[4, \frac{4a}{b}\right]$, such that

$$y = bu_n = b \frac{a^n u_0}{b^n}$$

for some n . Hence for each value v in the range of f , there exists some $x \in [4b, 4a]$ such that $f(x) = v$. But f is continuous on the domain $[4b, 4a]$ therefore it attains a (finite) maximum. This contradicts the fact that f is surjective on $[4, \infty)$, hence our assumption is false and $f(x)$ is injective on the domain $[0, \infty)$.

We handle the negative domain $(-\infty, 0]$ by changing the assumption to $a < b < 0$ and $f(a) = f(b)$. Therefore $f(x)$ is injective on both domains $x \in (-\infty, 0]$ and $[0, \infty)$.

Surjectivity. We already know that $f(x)$ is surjective on either $(-\infty, 4]$ or $[4, \infty)$ when $f(0) = 4$, so consider, $f(0) = 0$. We know that there exists some interval $[-c_1, 0]$ or $[0, c_1]$ such that f is surjective onto that range and f is monotonic increasing/decreasing (following from f being injective and continuous), so we consider two cases.

Case 1. f is surjective on $[0, c_1]$.

Suppose f is bounded above and let

$$\lim_{x \rightarrow \infty} f(x) \rightarrow L_1.$$

Then when $f(y) > 0$ we have

$$P(\infty, y) : L_1 + f(L_1 y) = \frac{L_1}{2} f(2y).$$

So let $y = u_0 > 0$, and $u_{n+1} = \frac{u_n}{L}$, and as we send $n \rightarrow \infty$, by the continuity of f we have $L_1 + f(0) = \frac{L_1}{2} f(0) \implies L_1 = 0$.

But this implies f is constant, and contradicts that f is surjective on $[0, c_1]$, hence f is not bounded above, and must be surjective onto $[0, \infty)$.

Case 2. f is surjective on $[-c_1, 0]$.

Suppose f is bounded below and let

$$\lim_{n \rightarrow \infty} f(x) \rightarrow L_2,$$

then when $f(y) < 0$ we have

$$P(\infty, y) : L_2 + f(L_1 y) = \frac{L_1}{2} f(2y).$$

By a similar argument to Case 1, we find $L_2 = 0$, contradicting that f is not constant. Hence $f(x)$ has no lower bound and must be surjective onto $(-\infty, 0]$.

So, if $f(0) = 0$, we know that there exists $2c \in \mathbb{R}$ such that $f(2c) = 4$. Hence

$$f(cf(c)) = \frac{1}{4} f(2c)^2 = 4 = f(2c)$$

and the fact that f is injective implies that $f(c) = 2$. Now

$$P(x, c) : f(2x) + f(cf(x)) = \frac{1}{2} f(2c) f(2x) = 2f(2x) \implies f(cf(x)) = f(2x)$$

and the injectivity of f shows that $f(x) = kx$, where $k = 2/c$.

If $f(0) = 4$ the above does not work because $c = 0$. But we do know that $f(2^n x) = 4 + 2^n[f(x) - 4]$ and letting $f(x) = g(x) + 4$ we get

$$g(2^n x) = 2^n g(x). \quad (2)$$

On the other hand

$$P(x, x) : f(xf(x)) = \frac{1}{4} f(2x)^2 = (f(x) - 2)^2 \Leftrightarrow g(xg(x) + 4x) = g(x)^2 + 4g(x)$$

and applying (2) gives $g(2^n xg(x) + x) = 2^n g(x)^2 + g(x)$, which holds for all $n \in \mathbb{Z}, x \in (0, \infty)$.

Now there exists $c \in \mathbb{R}$ such that $f(c) = 1$, so, letting $x = c$ gives

$$g(2^n c + c) = 2^n + 1.$$

Hence applying (2) we get

$$f(2^{n+m} c + 2^m) = 2^{n+m} + 2^m \quad (3)$$

which also holds for all $n, m \in \mathbb{Z}$ and $x \in \mathbb{R}$.

Now for each $a \in (0, \infty)$ we will define a sequence converging to a and show that $g(ac) = a$. Then it will follow that $g(cx) = x$ for all $x \in (0, \infty)$.

So pick two integers $k, \ell \in \mathbb{Z}$ such that $2^k + 2^\ell < a$, and let $u_0 = 2^k + 2^\ell$. The next terms of the sequence are defined inductively by $u_{n+1} = 2^{k_{n+1}} u_n^2 + u_n$, where k_{n+1} is the largest possible integer such that $u_{n+1} < a$. Then the limit of this sequence as $n \rightarrow \infty$ is a .

But from (3) we have $g(cu_n) = u_n$ for all $n \in \mathbb{N}$, so the continuity of g implies $\lim_{n \rightarrow \infty} g(cu_n) = g(\lim_{n \rightarrow \infty} cu_n) = g(ca) = a$. This is true for all $a \in (0, \infty)$, so we have $g(x) = \frac{x}{c}$ or $f(x) = kx + 4$, for $k = \frac{1}{c}$.

Thus all solutions of the problem are the following functions: $f(x) = kx$, $k \in \mathbb{R}$; $f(x) = kx + 4$, $k \in \mathbb{R}$;

$$f(x) = \begin{cases} k_1 x & x < 0 \\ k_2 x & x \geq 0 \end{cases}$$

and

$$f(x) = \begin{cases} k_1 x + 4 & x < 0 \\ k_2 x + 4 & x \geq 0 \end{cases}$$

where $k_1 \leq 0$, $k_2 \geq 0$.

Problem 18.55. (IMC 2008). Find all continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f(x) - f(y)$ is rational for all reals x and y such that $x - y$ is rational.

Solution. We prove that $f(x) = ax + b$, where $a \in \mathbb{Q}$ and $b \in \mathbb{R}$. These functions obviously satisfy the conditions.

Suppose that a function $f(x)$ fulfills the required properties. For an arbitrary rational q , consider the function $g_q(x) = f(x + q) - f(x)$. This is a continuous function which attains only rational values, therefore g_q is constant (prove it!).

Set $a = f(1) - f(0)$ and $b = f(0)$. Let n be an arbitrary positive integer and let $r = f(1/n) - f(0)$. Since $f(x + 1/n) - f(x) = f(1/n) - f(0) = r$ for all x , we have

$$\begin{aligned} f(k/n) - f(0) &= (f(1/n) - f(0)) + (f(2/n) - f(1/n)) + \dots \\ &\quad + (f(k/n) - f((k-1)/n)) = kr \end{aligned}$$

and similarly $f(-k/n) - f(0) = -kr$ for $k \geq 1$. In the case $k = n$ we get $a = f(1) - f(0) = nr$, so $r = a/n$. Hence, $f(k/n) - f(0) = kr = ak/n$ and then $f(k/n) = a \cdot k/n + b$ for all integers k and $n > 0$.

So, we have $f(x) = ax + b$ for all rational x . Since the function f is continuous and the rational numbers form a dense subset of $\mathbb{R}$, the same holds for all real x .

19.18.4 The Odd and Even Parts of Functions

Problem 18.56. Find all continuous functions $f: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x+y) + f(x)f(y) = f(xy+1)$$

for all $x, y \in \mathbb{R}$.

Solution. If we replace x, y by $-x, -y$ and compare with the initial condition we get

$$f(x+y) + f(x)f(y) = f(-x-y) + f(-x)f(-y).$$

Now write $f = g + h$, where $g(x) = \frac{f(x) + f(-x)}{2}$ and $h(x) = \frac{f(x) - f(-x)}{2}$ are the even and odd parts of f . Then

$$\begin{aligned} g(x+y) + h(x+y) + (g(x) + h(x))(g(y) + h(y)) \\ = g(x+y) - h(x+y) + (g(x) - h(x))(g(y) - h(y)) \end{aligned}$$

and we get

$$2h(x+y) + 2g(x)h(y) + 2h(x)g(y) = 0.$$

Next we replace y by $-y$ to get

$$2h(x-y) - 2g(x)h(y) + 2h(x)g(y) = 0$$

and from here

$$h(x+y) + h(x-y) = -2h(x)g(y).$$

We have solved this equation in Problem 8.3. So h is either $c \sin(ax) + d \cos(ax)$ or $c \sinh(ax) + d \cosh(ax)$ and $g = \cos(ax)$ which do not satisfy the original

condition, or $h(x)$ is a linear function and $g(x) = -1$, or $h(x) = 0$ and g is arbitrary. If $h = a + bx$ is linear, then $a = 0$ as h is odd. As $f(x) = g(x) + h(x)$, we get $f(x) = -1 + bx$ and substituting into the original condition we see that only $f(x) = x - 1$ satisfies the condition. If $h(x) = 0$, then f is an even function and if we replace y by $-y$, we get

$$f(x - y) + f(x)f(y) = f(xy - 1).$$

So

$$f(x + y) - f(x - y) = f(xy + 1) - f(xy - 1) = k(4xy),$$

where $k(x) = f\left(\frac{x}{4} + 1\right) - f\left(\frac{x}{4} - 1\right)$. Then we set $r(t) = f(\sqrt{t})$ for $t > 0$ and conclude that $r(x) - r(y) = k(x - y)$ for all $x, y \geq 0$ due to the identity

$$(x + y)^2 - (x - y)^2 = 4xy.$$

Hence $r(x) - r(y) = r(x - y) - r(0)$ for all $x \geq y \geq 0$ which shows that $r(x) + r(0)$ is an additive continuous function on $(0, \infty)$. Hence it follows from Theorem 1.5 that r is a linear function on $(0, \infty)$ and since f is an even function we conclude that $f(x) = a + bx^2$. Then replacing into the original condition we get $f(x) = x^2 - 1$ or $f(x) = 0$. Hence there are three solutions: $f(x) = x - 1$, $f(x) = x^2 - 1$ and $f(x) = 0$.

Problem 18.57. Find all continuous functions $f, g, h: \mathbb{R} \rightarrow \mathbb{R}$ that obey

$$f(x + y) + g(xy) = h(x)h(y) + 1$$

for all $x, y \in \mathbb{R}$.

Solution. If $y = 0$, then $f(x) = h(x)h(0) + 1 - g(0)$. We can assume $g(0) = 0$ otherwise replace f by $f + g(0)$ and g by $g - g(0)$ to get the same condition. We get $f(x) = h(x)h(0) + 1$ so

$$h(x + y)h(0) + g(xy) = h(x)h(y).$$

If $h(0) = 0$, we get

$$h(x)h(y) = g(xy) = h(xy)h(1)$$

and so it follows from Theorem 1.11 that $h(x) = ax^k$ for some k , $g(x) = a^2x^k$. If $h(0) \neq 0$, we can assume $h(0) = 1$; otherwise replace h by $\frac{h}{h(0)}$ and g by $\frac{g}{g_0}$ to maintain the condition. So we have $h(x+y) + g(xy) = h(x)h(y)$. If we set $y = 1$, we get

$$g(x) = ah(x) - h(x+1),$$

where $a = h(1)$. So we get

$$h(x+y) + ah(xy) = h(x)h(y) + h(xy+1).$$

Now let u, v be the even and odd parts of h . We have

$$\begin{aligned} & u(x+y) + v(x+y) + au(xy) + av(xy) \\ &= (u(x) + v(x))(u(y) + v(y)) + u(xy+1) + v(xy+1). \end{aligned}$$

Now if we replace x, y by $-x, -y$ we get

$$\begin{aligned} & u(x+y) - v(x+y) + au(xy) + av(xy) \\ &= (u(x) - v(x))(u(y) - v(y)) + u(xy+1) + v(xy+1). \end{aligned}$$

Subtracting these two relations we get

$$2v(x+y) = 2v(x)u(y) + 2u(x)v(y)$$

and replacing y by $-y$ the relation turns into

$$v(x+y) - v(x-y) = 2u(x)v(y).$$

Now using Problem 8.3) it is easy to check that the only solutions u, v with u even and v odd are $v = c \sin(ax), u = \cos(ax)$ or $v = c \sinh(ax), u = \cosh(ax)$ or $v = cx, u = 1$ or $v = 0$, where u any even function. We can eliminate the first two solutions immediately, as an expression of h as sine-cosine surely does not satisfy the condition (just look that $h(xy+1) - ah(xy)$ has nothing to share with $h(x+y)$ or $h(x)h(y)$). If $v = cx, u = 1$, then $h(x) = 1+cx, a = 1+c$ and so we get

$$1 + c(x+y) + (1+c)(1+cxy) = (1+cx)(1+cy) + 1 + c(xy+1)$$

which satisfies the condition, and

$$g(x) = (1 + c)(1 + cx) - c(x + 1) = c^2x + 1.$$

If $v = 0$, then h is even. Then writing the condition on h for x, y and $x, -y$ then subtracting them we get

$$h(x + y) - h(x - y) = h(1 + xy) - h(xy - 1).$$

This functional equation was solved in the solution to the previous problem with solutions $h(x) = ax^2 + b$. Since $h(0) = 1$ we get $b = 1$ and hence $h(x) = ax^2 + 1$, $f(x) = ax^2 + 2$ and $g(x) = a^2x^2 - 2ax$.

19.18.5 Constructive Problems

Problem 18.58. (Romania Shortlist 2001) (a) Let $g, h : \mathbb{Z} \rightarrow \mathbb{Z}$ be one-to-one functions. Show that the function $f : \mathbb{Z} \rightarrow \mathbb{Z}$ defined by $f(x) = g(x)h(x)$ for all $x \in \mathbb{Z}$ is not surjective.

(b) Let $f : \mathbb{Z} \rightarrow \mathbb{Z}$ be a surjective function. Show that there exist surjective functions $g, h : \mathbb{Z} \rightarrow \mathbb{Z}$ such that $f(x) = g(x)h(x)$ for all $x \in \mathbb{Z}$.

Solution. (a) Suppose f is a surjective function and let $a, b \in \mathbb{Z}$ be such that $f(a) = 1$ and $f(b) = -1$. Obviously $a \neq b$ and the identities

$$g(a)h(a) = 1 \text{ and } g(b)h(b) = -1$$

imply that $f(a), g(a), f(b), g(b) \in \{1, -1\}$. If $f(a) \neq f(b)$ and $g(a) \neq g(b)$ then $f(a)f(b) = g(a)g(b) = -1$ which contradicts $f(a)g(a)f(b)g(b) = -1$. Therefore either $f(a) = f(b)$ or $g(a) = g(b)$, contradicting the injectivity of f or g .

(b) Let $a_0 \in \mathbb{Z}$ be such that $f(a_0) = 0$. We define $g(a_0) = h(a_0) = 0$. For every positive integer n , let a_n and b_n be integers such that $f(a_n) = n^2$ and $f(b_n) = -n^2$. Then we define $g(a_n) = n, h(a_n) = n, g(b_n) = -n, h(b_n) = n$. Thus all integers are in the range of g and all nonnegative integers are in the range of h . For any positive integer n let c_n be an integer with $f(c_n) = n(n+1)$. Since $a_m \neq c_n$ and $b_m \neq c_n$ for every m ($n(n+1)$ is not a square) we can

define $g(c_n) = -(n+1)$ and $h(c_n) = -n$. Thus the range of h covers $\mathbb{Z}$. In the case when the set

$$A = \mathbb{Z} \cup_{n \in \mathbb{N}} \{a_n, b_n, c_n\}$$

is nonempty we define $g(k) = f(k)$ and $h(k) = 1$ for all $k \in A$.

Problem 18.59. (G. Dospinescu) Find all functions f defined on the non-negative reals with values in the same set such that

$$f(\lfloor f(x) \rfloor) + \{f(x)\} = x$$

for all $x \geq 0$. Here $\lfloor a \rfloor$ and $\{a\}$ are the integer and fractional parts of a real number a .

Solution. Put $x = 0$ to get $f(0)$ is an integer and $f(f(0)) = 0$. By taking $x = n$ an integer and then $x = \lfloor f(n) \rfloor$ and by comparing the two relations we obtain

$$f(\lfloor n - \{f(n)\} \rfloor) + \{n - \{f(n)\}\} = \lfloor f(n) \rfloor$$

for all integers n . This will be used to prove that $f(n)$ is an integer for all integers n . Assuming the contrary, consider the smallest nonnegative integer n for which $f(n)$ is not an integer. By the first observation $n \geq 1$ and clearly $\lfloor n - \{f(n)\} \rfloor = n - 1$. The above relation shows that $\{n - \{f(n)\}\}$ is an integer and so $f(n)$ is an integer itself, a contradiction.

Using the fact that $f(n)$ is an integer, we deduce from $x = n$ that $f(f(n)) = n$. Also, the relation $f(\lfloor f(x) \rfloor) + \{f(x)\} = x$ shows that $\{f(x)\} - x$ is an integer and so $f(x) - x$ is an integer for all x .

Now, consider $x > 0$ and put $n = \lfloor f(x) \rfloor$, so $f(n) + \{f(x)\} = x$. By taking the integer part we obtain $f(n) = \lfloor x \rfloor$ and so $f(\lfloor f(x) \rfloor) = \lfloor x \rfloor$. Since we saw that $f(f(n)) = n$, we obtain

$$f(\lfloor x \rfloor) = f(f(\lfloor f(x) \rfloor)) = \lfloor f(x) \rfloor,$$

so $f(\lfloor x \rfloor) = \lfloor f(x) \rfloor$. This and the relation $\{f(x)\} = \{x\}$ show that f is determined uniquely by its restriction to the set of nonnegative integers, which is a function $g : \{0, 1, \dots\} \rightarrow \{0, 1, \dots\}$ such that $g(g(n)) = n$. Conversely, it is easy to check that all these functions are solutions of the problem.

Problem 18.60. (IMO Shortlist 1997) Do there exist functions $f, g : \mathbb{R} \rightarrow \mathbb{R}$ such that:

- a) $f(g(x)) = x^2$ and $g(f(x)) = x^4$;
 b) $f(g(x)) = x^2$ and $g(f(x)) = x^3$
 for all $x \in \mathbb{R}$?

Solution. a) We shall show that there are functions f and g satisfying the given conditions. To do this we first construct f and g on the restricted domain $[1, \infty]$. From the two equations we get

$$f(x^4) = f(g(f(x))) = f(x)^2$$

and

$$g(x^2) = g(f(x))^4.$$

These identities suggest considering the following functions

$$f(x) = 2^{(\log_2 x)^{1/2}}, g(x) = 16^{(\log_2 x)^2}, x \geq 1,$$

and to extend them to $\mathbb{R}$. On the interval $(0, 1)$ define f and g to be $1/f(1/x)$ and $1/g(1/x)$, respectively. Next set $f(0) = g(0) = 0$ and

$$f(x) = f(-x), \quad g(x) = g(-x)$$

for $x \in (-\infty, 0)$. It is easy to check that the functions f and g defined in this way on $\mathbb{R}$ satisfy the required equations.

b) No! Suppose f and g satisfy the given equations. As the function $h(x) = x^3$ is one-to-one it follows that f is also a one-to-one function. Thus $f(1), f(-1)$ and $f(0)$ are three distinct numbers. On the other hand

$$f(x^3) = f(g(f(x))) = f(x)^2$$

and therefore the above numbers are roots of the equation $t = t^2$, a contradiction.

Problem 18.61. Find a solution to the equation:

- a) $f(x^2) = 1 + f(x)$ for all $x \in (0, \infty)$;
 b) $f(x+1) = f(x)^2$ for all $x \in \mathbb{R}$.

Solution. We shall find solutions to both equations by reducing them to linear ones.

a) Set $F(x) = f(a^x)$, where $a > 0$. Then $F(2x) = 1 + F(x)$ which is satisfied by the function $F(x) = \log_2 x$. Hence a solution to the given equation is given by the function $f(x) = \log_2 \log_a x$, where $a > 0$ is an arbitrary constant.

b) We can make the given equation linear by letting $F(x) = \log_a f(x)$, where $a > 0$. Then the equation becomes $F(x+1) = 2F(x)$ with an obvious solution $f(x) = 2^x$. Hence a solution to the original equation is $f(x) = a^{2^x}$.

Problem 18.62. (IMC 2003) (a) Show that for each function $f : \mathbb{Q} \times \mathbb{Q} \rightarrow \mathbb{R}$ there exists a function $g : \mathbb{Q} \rightarrow \mathbb{R}$ such that

$$f(x, y) \leq g(x) + g(y)$$

for all $x, y \in \mathbb{Q}$.

(b) Find a function $f : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ for which there is no function $g : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x, y) \leq g(x) + g(y)$$

for all $x, y \in \mathbb{R}$.

Solution.(a) Let $\phi : \mathbb{Q} \rightarrow \mathbb{N}$ be a bijection. Define

$$g(x) = \max |f(s, t)| : s, t \in \mathbb{Q}, \phi(s) \leq \phi(x), \phi(t) \leq \phi(x).$$

We have

$$f(x, y) \leq \max\{g(x), g(y)\} \leq g(x) + g(y).$$

(b) We give two solutions to this part.

First Solution. We shall show that the function defined by $f(x, y) = \frac{1}{|x - y|}$ for $x \neq y$ and $f(x, x) = 0$ satisfies the problem. If, by contradiction there exists a function g as above, it results that

$$g(y) \geq \frac{1}{|x - y|} - f(x)$$

for all $x, y \in \mathbb{R}, x \neq y$. Hence for each $x \in \mathbb{R}$, $\lim_{y \rightarrow x} g(y) = \infty$. We now show that there are no functions g having an infinite limit at each point of a bounded and closed interval $[a, b]$.

For each $k \in \mathbb{N}$ we set $A_k = \{x \in [a, b] \mid |g(x)| \leq k\}$. We obviously have that

$$[a, b] = \bigcup_{k=1}^{\infty} A_k.$$

The set $[a, b]$ is uncountable, so at least one of the sets A_k is infinite (in fact uncountable). For this set, say A_k , there exists a sequence in A_k having distinct terms. This sequence will contain a convergent subsequence $(x_n)_{n \in \mathbb{N}}$ that converges to a point $x \in [a, b]$. But $\lim_{y \rightarrow x} g(y) = \infty$ implies that $g(x_n) \rightarrow \infty$, which is a contradiction because $|g(x_n)| \leq k$, for all $n \in \mathbb{N}$.

Second Solution. Let S be the set of all sequences of real numbers. The cardinality of S is

$$|S| = |\mathbb{R}|^{\aleph_0} = 2^{\aleph_0^2} = 2^{\aleph_0} = |\mathbb{R}|.$$

Thus, there exists a bijection $h : \mathbb{R} \rightarrow S$.

Now, define the function f in the following way. For any real x and positive integer n , let $f(x, n)$ be the n th element of sequence $h(x)$. If y is not a positive integer then let $f(x, y) = 0$. We prove that this function has the required property.

Let g be an arbitrary $\mathbb{R} \rightarrow \mathbb{R}$ function. We show that there exist real numbers x, y such that $f(x, y) > g(x) + g(y)$. Consider the sequence

$$(n + g(n))_{n \geq 1}.$$

This sequence is an element of S , thus $(n + g(n))_{n \geq 1} = h(x)$ for a certain real x . Then for an arbitrary positive integer n , $f(x, n)$ is the n th element, $f(x, n) = n + g(n)$. Choosing n such that $n > g(x)$, we obtain

$$f(x, n) = n + g(n) > g(x) + g(n).$$

Problem 18.63. (IMO Shortlist 2008) Let $S \subseteq \mathbb{R}$ be a set of real numbers. We say that a pair (f, g) of functions from S into S is a Spanish Couple on S , if they satisfy the following conditions:

(i) Both functions are strictly increasing, i.e. $f(x) < f(y)$ and $g(x) < g(y)$ for all $x, y \in S$ with $x < y$;

(ii) The inequality $f(g(g(x))) < g(f(x))$ holds for all $x \in S$.

Decide whether there exists a Spanish Couple

(a) on the set $S = \mathbb{N}$ of positive integers;

(b) on the set $S = (a - \frac{1}{b} \mid a, b \in \mathbb{N})$.

Solution. We show that the answer is NO for part (a) and YES for part (b).

(a) Let $g^{(k)}(x)$ denote the k -th iterate of $g(x)$ and $g^{(0)}(x) = x$. Suppose that there exists a Spanish Couple (f, g) on the set $\mathbb{N}$. From property (i) we have $f(x) \geq x$ and $g(x) \geq x$ for all $x \in \mathbb{N}$. We claim that $g^{(k)}(x) \leq f(x)$ for all $k \geq 0$. The proof is by induction on k . We already have the base case $k = 0$ since $x \leq f(x)$. For the inductive step from k to $k + 1$ we have from (ii) that

$$g(g^{(k+1)}(x)) = g^{(k)}(g^{(2)}(x)) \leq f(g^{(2)}(x)) < g(f(x)).$$

Since g is increasing it follows that $g^{(k+1)}(x) < f(x)$ and the claim is proved.

If $g(x) = x$ for all $x \in \mathbb{N}$ then $f(g(g(x))) = f(x) = g(f(x))$, a contradiction with (ii). Hence there is $x_0 \in \mathbb{N}$ for which $x_0 < g(x_0)$. Now consider the sequence $x_0, x_1, \dots$, where $x_k = g^{(k)}(x_0)$. This sequence is increasing since $x_0 < g(x_0) = x_1$ and $x_k < x_{k+1}$ implies $x_{k+1} = g(x_k) < g(x_{k+1} = x_{k+2})$. Hence we obtain a strictly increasing sequence $x_0 < x_1 < \dots$ of positive integers which on the other hand has an upper bound, namely $f(x_0)$. This cannot happen in $\mathbb{N}$, thus no Spanish Couple exists on $\mathbb{N}$.

(b) We present a Spanish Couple on the set $S = (a - \frac{1}{b} \mid a, b \in \mathbb{N})$. Let

$$f(a - 1/b) = a + 1 - 1/b,$$

$$g(a - 1/b) = a - 1/(b + 3^a).$$

These functions are clearly increasing. Condition (ii) holds, since

$$f(g(g(a - 1/b))) = (a + 1) - 1/(b + 2 \cdot 3^a)$$

$$< (a + 1) - 1/(b + 3^{a+1}) = g(f(a - 1/b)).$$

19.18.6 Functional Equations Using Special Groups

Problem 18.64. (IMO Shortlist 2001) Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfying

$$f(xy)(f(x) - f(y)) = (x - y)f(x)f(y)$$

for all $x, y \in \mathbb{R}$.

Solution. Let (*) denote the given functional equation. Substituting $y = 1$ we get $f(x)^2 = xf(x)f(1)$. If $f(1) = 0$, then $f(x) = 0$ for all x , which is the trivial solution. Suppose $f(1) = a \neq 0$ and let $G = \{x \in \mathbb{R} \mid f(x) \neq 0\}$. Then

$$f(x) = \begin{cases} ax & \text{if } x \in G \\ 0 & \text{otherwise.} \end{cases}$$

We shall prove now that G is a multiplicative subgroup of $\mathbb{R}$.

- (1) Clearly $1 \in G$, because $f(1) \neq 0$.
- (2) If $x \in G, y \notin G$, then by (*) it follows that $f(xy)f(x) = 0$, so $xy \notin G$.
- (3) If $x, y \in G$, then $x/y \in G$ (otherwise by (2), $y(xy) = x \notin G$).
- (4) If $x, y \in G$, then by (3) we have $x^{-1} \in G$, so $xy = y/x^{-1} \in G$.

Conversely, it is easy to verify that given $a \neq 0$ every multiplicative subgroup G of $\mathbb{R}$ gives a function f as above satisfying (*). An obvious example of such a group is the set of nonzero real numbers. Note however, that there are many other choices for G , including the nonzero rational numbers.

Problem 18.65. Find all functions $f : \mathbb{Q} \rightarrow \mathbb{Q}$ such that

$$f(x + y + f(x)) = x + f(x) + f(y)$$

for all $x, y \in \mathbb{Q}$.

Solution. Let A be any additive subgroup of $\mathbb{Q}$, i.e. if $x, y \in A$ then $x - y \in A$. Denote by $\sim$ the equivalence relation defined by $x \sim y \iff x - y \in A$ and let $r(x)$ be any choice function which associates at each rational number an element of its equivalence class (unique per class). Let $m : \mathbb{Q} \rightarrow A$ be an arbitrary function. First we shall show that the function

$$f(x) = m(r(x)) + x - 2r(x)$$

is a solution. We have

$$x + f(x) = m(r(x)) + 2(x - r(x)) \in A,$$

so

$$y + x + f(x) \sim y \text{ and } r(y + x + f(x)) = r(y).$$

Hence

$$\begin{aligned} f(x + y + f(x)) &= m(r(x + y + f(x))) + x + y + f(x) - 2r(x + y + f(x)) \\ &= m(r(y)) + x + y + f(x) - 2r(y) = x + f(x) + f(y). \end{aligned}$$

We shall prove now that every solution f to the given equation has the above form. Set $A = \{x \in \mathbb{Q} \mid f(x + y) = f(y) + x, \forall y \in \mathbb{Q}\}$. We shall show that A is an additive subgroup of $\mathbb{Q}$. It is clear that $0 \in A$. If $x \in A$, we have $f(x + y) = f(y) + x$ for all $y \in \mathbb{Q}$. Set $y = y_1 - x$. Then

$$f(x + y_1 - x) = f(y_1 - x) + x$$

and so $f(y_1) = f(y_1) - x$ for all $y_1 \in \mathbb{Q}$. Hence $-x \in A$. If $x \in A$ and $x_1 \in A$ then

$$f(x + x_1 + y) = f(x + (x_1 + y)) = f(x_1 + y) + x.$$

But $f(x_1 + y) = f(y) + x_1$ since $x_1 \in A$, and so $f(x + x_1 + y) = f(y) + x + x_1$ implying $x + x_1 \in A$.

Note that $f(x) + x \in A$ for all $x \in \mathbb{Q}$ since

$$f(y + (x + f(x))) = f(y) + (x + f(x)).$$

Now the fact that A is an additive group implies that $x \sim y \iff x - y \in A$ is an equivalence relation. Let $r(x)$ be any choice function which associates at each rational number an element of its equivalence class (unique per class). Set $m(x) = f(x) + x$. Since $f(x) + x \in A$ it follows that $m(x)$ is a function from $\mathbb{Q}$ to $\mathbb{Q}$. Then $x = r(x) + (x - r(x))$ and since $x - r(x) \in A$ (x and $r(x)$ are in the same equivalence class), we have

$$f(x) = f(r(x) + (x - r(x))) = f(r(x)) + x - r(x) = m(r(x)) + x - 2r(x).$$

Here are some examples of functions satisfying the given relation:

- 1). Let $A = \mathbb{Q}$, $r(x) = 0$ and $m(x) = p$. Then $f(x) = x + p$.
- 2). Let $A = \{0\}$, $r(x) = x$ and $m(x) = 0$. Then $f(x) = -x$.
- 3). Let $A = \mathbb{Z}$, $r(x) = x - \lfloor x \rfloor$ and $m(x) = n$. Then $f(x) = 2\lfloor x \rfloor - x + n$.
(Here $\lfloor x \rfloor$ is the integral part of a real number x .)
- 4). Let $A = \mathbb{Z}$, $r(x) = x - \lfloor x \rfloor$ and $m(x) = \lfloor 5 \sin(\pi x) \rfloor$. Then

$$f(x) = \lfloor 5 \sin(\pi(x - \lfloor x \rfloor)) \rfloor + 2\lfloor x \rfloor - x.$$

19.18.7 Density

Problem 18.66. (IMO Shortlist 2003) Find all functions $f : (0, \infty) \rightarrow (0, \infty)$, which are increasing on the segment $[1, \infty)$ and such that

$$f(xyz) + f(x) + f(y) + f(z) = f(\sqrt{xy})f(\sqrt{yz})f(\sqrt{zx})$$

for all $x, y, z \in (0, \infty)$.

Solution. Replacing x, y and z by $\frac{x}{y}, \frac{y}{x}$ and xy , respectively, we get that

$$2f(xy) + f\left(\frac{x}{y}\right) + f\left(\frac{y}{x}\right) = f(1)f(x)f(y).$$

For $y = 1$ it follows that $3f(x) + f\left(\frac{1}{x}\right) = f(1)^2 f(x)$. In particular,

$$4f(1) = f(1)^3.$$

Since $f(1) > 0$, then $f(1) = 2$. Thus, $f(x) = f\left(\frac{1}{x}\right)$ and the above identity can be written as

$$f(xy) = f(x)f(y) - f\left(\frac{x}{y}\right).$$

Further, since $e > 1$, then $f(e) \geq f(1) = 2$ and hence $f(e) = e^\alpha + e^{-\alpha}$ for some $\alpha \geq 0$. Using that $f(x^2) = f(x)^2 - 2$, it follows by induction that

$$f(e^{2^{-n}}) = e^{\alpha 2^{-n}} + e^{-\alpha 2^{-n}}$$

for all $n \in \mathbb{N}_0$. Having in mind the equality

$$f(e^{(m+1)2^{-n}}) = f(e^{2^{-n}})f(e^{m2^{-n}}) - f(e^{(m-1)2^{-n}}),$$

we get again by induction that

$$f(e^{m2^{-n}}) = e^{\alpha m 2^{-n}} + e^{-\alpha m 2^{-n}}$$

for all $m, n \in \mathbb{N}_0$.

Since the set of numbers of the form $m2^{-n}$, $m, n \in \mathbb{N}_0$, is dense in $(0, \infty)$ (use the binary representation of the positive real numbers) and f is a monotonic function in the segment $[1, \infty)$, we conclude that $f(x) = x^\alpha + x^{-\alpha}$ in this segment. The same is true in the segment $(0, 1)$ in virtue of the equality

$$f(x) = f\left(\frac{1}{x}\right).$$

Conversely, it is easy to see that any function of this form satisfies the condition of the problem.

Problem 18.67. Show that there exists a continuous function $f: \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(x) + f(2x) + f(3x) = 0$$

for all $x \in \mathbb{R}$.

Solution. The key idea here is to consider a complex valued function and take the real part. Consider the function $f(x) = |x|^\alpha$. It will satisfy the conditions of the problem if $1^\alpha + 2^\alpha + 3^\alpha = 0$. Of course, this will not happen for real α , but may happen for complex α . Namely, set $g(t) = e^{\alpha |\ln(t)|}$, then

$$g(x) + g(2x) + g(3x) = (1 + e^{\alpha \ln 2} + e^{\alpha \ln 3})g(x).$$

If we write $\alpha = a + bi$, where $a, b \in \mathbb{R}$, then we need $1 + e^{\alpha \ln 2} + e^{\alpha \ln 3} = 0$, that is

$$1 + 2^a(\cos(b \ln 2) + i \sin(b \ln 2)) + 3^a(\cos(b \ln 3) + i \sin(b \ln 3)) = 0.$$

We thus seek a solution to the system

$$1 + 2^a \cos(b \ln 2) + 3^a \cos(b \ln 3) = 0, 2^a \sin(b \ln 2) + 3^a \sin(b \ln 3) = 0.$$

This is equivalent to the identity

$$(1 + 2^a \cos(b \ln 2) + 3^a \cos(b \ln 3))^2 + (2^a \sin(b \ln 2) + 3^a \sin(b \ln 3))^2 = 0,$$

which can be rewritten as

$$\begin{aligned} 1 + 4^a + 9^a + 2 \cdot 6^a (\sin(b \ln 2) \sin(b \ln 3) + \cos(b \ln 2) \cos(b \ln 3)) \\ + 2(2^a \cos(b \ln 2) + 3^a \cos(b \ln 3)) = 0. \end{aligned}$$

Now set $a = 0$ and let $x = b \ln 2, y = b \ln 3$. We then seek

$$\begin{aligned} f(b) &= 3 + 2(\sin x \sin y + \cos x \cos y) + 2(\cos x + \cos y) \\ &= 3 + 2 \cos(x - y) + 2(\cos x + \cos y) = 0. \end{aligned}$$

If we let $b = 0$ then $x = 0, y = 0$ and $f(0) = 9 > 0$. From the other side, if $x = \frac{3\pi}{4}$ and $y = -\frac{3\pi}{4}$ then

$$f(b) = 3 + 2 \cos \frac{3\pi}{2} - 4 \cos \frac{\pi}{4} = 3 - 2 - 2\sqrt{2} < 0.$$

Therefore we would be done if we would have a b such that $x = \frac{3\pi}{4}, y = -\frac{3\pi}{4}$, because then $f(0) > 0, f(b) < 0$ and then f will have a zero by continuity. Unfortunately, such a b does not exist, but we can relax the condition to

$$x = 2m\pi + \frac{3\pi}{4}, y = 2l\pi - \frac{3\pi}{4}$$

because $\sin$ and $\cos$ are periodic functions. If we let $b = \ln 2 \left(2m\pi + \frac{3\pi}{4} \right)$ then

$$x = 2m\pi + \frac{3\pi}{4}, y = \frac{\ln 3}{\ln 2} \cdot 2m\pi + \frac{3\pi \ln 3}{4 \ln 2}.$$

We want $y = 2l\pi - \frac{3\pi}{4}$ for $l \in \mathbb{Z}$, that is

$$m \frac{\ln 3}{\ln 2} - l = -\frac{3}{8} - \frac{3 \ln 3}{8 \ln 2}.$$

Again, this is not achievable, but from Kronecker's theorem we can find integers m, l such that $m \frac{\ln 3}{\ln 2} - l$ is as close to $-\frac{3}{8} - \frac{3 \ln 3}{8 \ln 2}$ as we need. Then for such b we will have $f(b) < 0$ and we are done.

Problem 18.68. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$f(xy(x+y)) = f(x)f(y)(f(x) + f(y))$$

for all $x, y \in \mathbb{R}$.

Solution. We have the obvious constant solutions $f(x) = 0$ and $f(x) = \pm \frac{1}{\sqrt{2}}$ as well as $f(x) = x$ and $f(x) = -x$. We shall show these are the only solutions. Suppose that f is not constant in what follows.

Step 1. $f(0) = 0$, $f(-x) = -f(x)$ and f is injective.

Suppose that $f(0) \neq 0$. Then $f(x)(f(x) + f(0)) = 1$. On the other hand setting $y = x$ we get $f(2x^3) = 2f(x)^3$, so $2f(0)^2 = 1$. This implies that $f(x) = f(0)$ or $f(x) = -2f(0)$ for all x . Combining this with $f(2x^3) = 2f(x)^3$, we easily see that f is constant, a contradiction. So $f(0) = 0$. Next, take $y = -x$ to get

$$f(x)f(-x)(f(x) + f(-x)) = 0. \quad (1)$$

We will show that $f(x) = 0$ implies $x = 0$. Note that if $x \neq 0$ and $f(x) = 0$, then $f(xy(x+y)) = 0$ for all y , that is $f(z) = 0$ if $z/x \geq -x^2/4$. This immediately implies that f is identically 0, a contradiction. So $f(x) = 0$ implies $x = 0$ and by (1) we get $f(x) = -f(-x)$. Finally, if $f(x_1) = f(x_2)$, put $y = -x_2$ and use what we found to get $x_1 = x_2$, which shows injectivity.

Step 2. $f(nx) = nf(x)$ for all $n \in \mathbb{Z}$ and $x \in \mathbb{R}$.

Put $y = -2x$ in order to have $xy(x+y) = 2x^3$. Combining the obtained relation from the functional equation with $f(2x^3) = 2f(x)^3$ and $f(-2x) = -f(2x)$ (by Step 1) we get $f(2x) = 2f(x)$ or $f(2x) = -f(x) = f(-x)$. By injectivity we get $f(2x) = 2f(x)$ for all x . So $2f(x)^3 = f(2x^3) = 2f(x^3)$ and so $f(x^3) = f(x)^3$. Put now $y = nx$ with n integer to get $f(n(n+1)x^3) = f(x)f(nx)(f(x) + f(nx))$. We prove by induction on n that $f(nx) = nf(x)$ for all $n \in \mathbb{N}$ and $x \in \mathbb{R}$. For $n = 1$ this is clear and if it is true for $n \in \mathbb{N}$, then it follows from $f(n(n+1)x^3) = nf((n+1)x^3)$ that $f((n+1)x^3) = (n+1)x^3$ for all x . Hence $f((n+1)x) = (n+1)f(x)$ for all x and the induction is finished.

Since f is an odd function (Step 1) we conclude that $f(nx) = nf(x)$ for all $n \in \mathbb{Z}$ and $x \in \mathbb{R}$.

Now we are ready to solve the problem. It follows by Step 2 that $f(x) = xf(1)$ for all rational x . On the other hand $f(1)^3 = f(1)$ by the proof of Step 2 and $f(1) \neq 0$, so $f(1) = 1$ or $f(1) = -1$. By changing f into $-f$ we may assume that $f(1) = 1$ and therefore $f(x) = x$ for all rational x . Note that if $b^4 + 4ab \geq 0$, then $f(b)^4 + 4f(a)f(b) \geq 0$, as the equation $xb(x+b) = a$ has real solutions if and only if $b^4 + 4ab \geq 0$. Therefore, by taking large rational values for a we deduce that for all $b > 0$ we have $f(b) > 0$. Taking $a > 0$ and a rational number b , we obtain that $a \leq -b^3/4$ implies that $f(a) \leq -b^3/4$ and by density we deduce that $f(a) \leq a$. Playing the same game (take $b > 0$ and a rational such that $-b^3/4 \leq a$, which implies $a \geq -f(b)^3/4$) we deduce that $f(b) \leq b$ for all $b > 0$. Thus $f(b) = b$ for all $b > 0$ and since f is odd we conclude that f is the identity function.

Problem 18.69. (Tuymaada 2006) Find all functions $f : (0, \infty) \rightarrow (0, \infty)$ with the following two properties:

$$(i) \ f(x+1) = f(x) + 1;$$

$$(ii) \ f\left(\frac{1}{f(x)}\right) = \frac{1}{x}$$

for all $x \in (0, \infty)$.

Solution. It is clear that $f(x) = x$ works and we will show that this is the only solution.

Let f have the above properties. Then by (ii) it follows that f is injective and surjective, i.e. bijective. By induction over k we will show that

$$f((n-1)/k, n/k) = ((n-1)/k, n/k) \text{ and } f(n/k) = n/k$$

for all positive integers n . Then we are done as this implies $f(x) \in (a, b)$ for all positive rational numbers a, b such that $x \in (a, b)$. Indeed, let $f(x) \neq x$ for some $x \in (0, \infty)$. Since $\mathbb{Q}$ is a dense subset of $(0, \infty)$ it follows that there are $a, b \in \mathbb{Q}$ such that $x \in (a, b)$ but $f(x) \notin (a, b)$, a contradiction.

For the induction we first do the case $k = 1$. Using (i) we see that

$$A_n = f((n, n+1]) = f((0, 1]) + n$$

and the bijectivity of f implies that these sets are disjoint. Hence their union is $(0, \infty)$ and $A_0 = (0, 1]$. (Indeed for $x \notin A_0$ we have $x \in A_{n+1}$ for some $n \geq 0$, so $x - 1 \in A_n$ by (i), so $x > 1$. And for $x > 1$ we have $x - 1 \in A_n$ for some $n \geq 0$, so by (i) we have $x \in A_{n+1}$, so $x \notin A_0$.) Now it suffices to show $f(1) = 1$. Then the rest of the case $k = 1$ follows from (i). Note that $f(1) \leq 1$ follows from the above and $f(1) \geq 1$ follows from (ii) since if $a = f(1)$, then $f(1/a) = 1$, and thus by the above $1/a \leq 1$, so $a = f(1) \geq 1$.

For the inductive step we assume the assertion is true for $1, \dots, k - 1$. By (i) it suffices to show that $f((n - 1)/k, n/k) = ((n - 1)/k, n/k)$ for all $1 \leq n \leq k$ and $f(n/k) = n/k$ for all integers $1 \leq n < k$. The second assertion follows from $f(n/k) = f(1/(k/n)) = f(1/f(k/n)) = 1/(k/n) = n/k$, where the second step holds by the inductive hypothesis as $n < k$, and the third step holds by (ii). Likewise the first assertion follows from

$$\begin{aligned} f((n - 1)/k, n/k) &= \{f(x) \mid (n - 1)/k < x < n/k\} \\ &= \{f(1/y) \mid k/n < y < k/(n - 1)\} = \{f(1/f(y)) \mid k/n < y < k/(n - 1)\} \\ &= \{1/y \mid k/n < y < k/(n - 1)\} = ((n - 1)/k, n/k), \end{aligned}$$

where in the first step we substituted $y = 1/x$, in the second step we have used the inductive hypothesis as $n, n - 1 < k$, so

$$f((k/n, (k + 1)/n)) = (k/n, (k + 1)/n)$$

and

$$f((k - 1)/(n - 1), k/(n - 1)) = ((k - 1)/(n - 1), k/(n - 1)),$$

which implies $f((k/n, k/(n - 1))) = (k/n, k/(n - 1))$, and in the third step we have used (ii).

19.18.8 Iterations

Problem 18.70. Find all functions $f: \mathbb{R} \setminus \{0; 1\} \rightarrow \mathbb{R}$ satisfying

$$f(x) + f\left(\frac{1}{1 - x}\right) = \frac{2(1 - 2x)}{x(1 - x)}$$

for all x in the domain of f .

Solution. The condition links $f(x)$ to $f\left(\frac{1}{1-x}\right)$ and only. Set

$$g(x) = \frac{1}{1-x}, h(x) = g^{-1}(x) = 1 - \frac{1}{x}.$$

Using the condition we can establish a dependence only between

$$f(x), f(g(x)), f(h(x)), f(g(g(x))), f(h(h(x))), \dots$$

So we have to look at properties of g or h . We see that

$$g(g(x)) = \frac{x-1}{x} = 1 - \frac{1}{x} = h(x)$$

so $g(g(g(x))) = x$ and we know

$$f(x) + f(g(x)), f(g(x)) + f(g(g(x))), f(g(g(x))) + f(x).$$

From here we can find $f(x)$ by solving the linear system. We can do this manually by substituting into the condition, and we get

$$f(x) = \frac{x+1}{x-1}.$$

It satisfies the conditions because we worked by equivalency.

Problem 18.71. (China 1988) For any integer $n \geq 3$, let $f(n)$ denote the least positive integer which is not a divisor of n . For each integer $n \geq 3$ find the positive integer k such that $f^{(k)}(n) = 2$.

Solution. Clearly, $f(n) = 2$ if n is odd, and we have $k = 1$. We now consider even integers $n \geq 4$. Suppose $f(n) = ab$, where a and b are relatively prime integers greater than 1. Then $a < f(n)$ and $b < f(n)$, hence both a and b divide n . Since they are relatively prime, ab also divides n , which contradicts the definition of $f(n)$. It follows that $f(n) = p^l$ for some prime p and some positive integer l . If $p = 2$, then $f(2^l) = 3$ and $f(3) = 2$, so that $k = 3$. If $p \neq 2$, then $f(p^l) = 2$, so that $k = 2$.

Problem 18.72. For a given positive integer m , find all positive integers a such that there exist functions f and g from $\mathbb{N}$ to itself, with g bijective and

$$f^{(m)}(x) = g(x) + a$$

for all $x \in \mathbb{N}$.

Solution. The answer is all positive integers a divisible by m . First, note that if $a = km$ then we can take the functions $g(x) = x, f(x) = x + k$. Now, suppose we have two such functions and define a family of sets A_k by the rule: for $0 \leq k < m$, A_k is the set of those numbers from $\mathbb{N}$ which can be written as $f^{(k)}(x)$ but not as $f^{(k+1)}(y)$ for some x, y . Here $f^{(k)}$ is k -th iterate of f and $f^{(0)}$ is the identity. As g is injective, f is also injective and A_{k+1} is the image of A_k through the function f . Also, it is trivial to show that the union of $A_0, \dots, A_{m-1}$ is $\mathbb{N}$ minus the set $f^{(m)}(\mathbb{N})$ which is exactly the set $\{1, 2, \dots, a\}$ (as g is surjective). Thus, all these sets are finite and since f is injective they have the same number of elements. Moreover, any two are disjoint. Thus, we have found a partition of the set $\{1, 2, \dots, a\}$ into m sets with the same cardinality. This means that $m|a$.

Remark. This is a vast generalization of several competition problems, among which the most recent is an IMO Training Test China 2009: There is a function f on positive integers such that $f^{(p)}(n) = n + k$ iff p divides k .

Problem 18.73. (Bulgaria 1996) Find all strictly monotonic functions $f : (0, \infty) \rightarrow (0, \infty)$ such that

$$f\left(\frac{x^2}{f(x)}\right) = x$$

for all $x \in (0, \infty)$.

Solution. We shall show that the function $g(x) = \frac{f(x)}{x}$ is a constant. We have

$$g\left(\frac{x}{g(x)}\right) = g(x)$$

and it follows by induction that

$$g\left(\frac{x}{g(x)^n}\right) = g(x),$$

i.e.

$$f\left(\frac{x}{g(x)^n}\right) = \frac{x}{g(x)^{n-1}}$$

for all $n \in \mathbb{N}$. On the other hand, the given condition implies that

$$f\left(\frac{f(x)^2}{f(f(x))}\right) = f(x).$$

Since the function f is injective we get $\frac{f(x)^2}{f(f(x))} = x$, i.e. $g(xg(x)) = g(x)$.

Now it follows by induction that $g(xg(x)^n) = g(x)$, i.e. $f(xg(x)^n) = xg(x)^{n+1}$ for all $n \in \mathbb{N}$. Denote by $f^{(m)}(x)$ the m -th iterate of $f(x)$. Then

$$f^{(m)}(xg(x)^{-k}) = xg(x)^{m-k} \quad (1)$$

for all $k, m \in \mathbb{N}$.

Now suppose that the function $g(x)$ is not constant. Then $g(x_1) < g(x_2)$ for some $x_1 \neq x_2$. Now choose k such that $\left(\frac{g(x_2)}{g(x_1)}\right)^k \geq \frac{x_2}{x_1}$. Since the function f is monotonic, it follows that $f^{(2m)}$ is a strictly increasing function and (1) implies that

$$\left(\frac{g(x_1)}{g(x_2)}\right)^{2m-k} \geq \frac{x_2}{x_1}$$

for all $m \in \mathbb{N}$. On the other hand, for m large enough the converse inequality holds, a contradiction. Thus the function $g(x)$ is a constant and therefore $f(x) = ax$, where $a > 0$. It is easy to check that these functions satisfy the given conditions.

Problem 18.74. Let $a, b \in (0, 1/2)$ and $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function such that

$$f(f(x)) = af(x) + bx$$

for all $x \in \mathbb{R}$. Prove that there exists a real constant c such that $f(x) = cx$ for all $x \in \mathbb{R}$.

Solution. Note first that the function f is injective and its continuity implies that f is strictly monotonic. Moreover, f is not bounded since the function

bx is not bounded and therefore f is onto. For arbitrary $x_0 \in \mathbb{R}$ define $x_{n+1} = f(x_n)$ for $n > 0$ and $x_{n-1} = f^{-1}(x_n)$ for $n \leq 0$. Then it follows from the given condition for f that $x_{n+2} = ax_{n+1} + bx_n$ for all $n \in \mathbb{Z}$. Let t_1 and t_2 be the roots of the characteristic equation $x^2 - ax - b = 0$. Then $t_1 > 0 > t_2$, $1 > |t_1| > |t_2|$ and there exist $c_1, c_2 \in \mathbb{R}$ such that $x_n = c_1 t_1^n + c_2 t_2^n$ for all $n \in \mathbb{Z}$. Assume that f is increasing. If $c_2 > 0$, then $0 < x_n < x_{n+2}$ for $n < 0$ small enough, a contradiction since $f(x_n) > f(x_{n+2})$. Similarly, the case $c_2 < 0$ is impossible and we conclude that $c_2 = 0$. So, $x_0 = c_1$ and $x_1 = c_1 t_1 = t_1 x_0$. Thus $f(x) = t_1 x$ for all $x \in \mathbb{R}$. The same arguments show that if f is decreasing, then $t_1 = 0$ and $f(x) = t_2 x$ for all $x \in \mathbb{R}$.

Problem 18.75. Find all continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that the function $f(f(x)) - 2f(x) + x$ is constant.

Solution. We show that all functions f with the given property have the form $f(x) = x + a$, $a \in \mathbb{R}$.

Suppose that

$$f(f(x)) - 2f(x) + x = c$$

for all $x \in \mathbb{R}$, where c is a real constant. Then f is injective and by continuity it follows that f is strictly monotonic. If f is strictly decreasing then the function $f(f(x))$ and hence $f(f(x)) - 2f(x) + x$ is strictly increasing, a contradiction. Hence f is strictly increasing and we shall show that it is onto. If $\lim_{x \rightarrow \infty} f(x) = a < \infty$, then $\lim_{x \rightarrow \infty} (f(f(x)) - 2f(x) + x) = \infty$, a contradiction. Hence $\lim_{x \rightarrow \infty} f(x) = \infty$ and analogously $\lim_{x \rightarrow -\infty} f(x) = -\infty$. Thus $f(\mathbb{R}) = \mathbb{R}$ and $f : \mathbb{R} \rightarrow \mathbb{R}$ is a bijection. Now we shall prove that $c = 0$. If $f(x_0) = x_0$ for some $x_0 \in \mathbb{R}$, then

$$x_0 = f(x_0) = f(f(x_0)) = 2f(x_0) - x_0 + c,$$

i.e. $c = 0$. If $f(x_0) \neq x_0$, we define $x_n = f^{(n)}(x_0)$ for all $n \in \mathbb{Z}$. Here f^{-1} is the inverse of f . Since $x_1 \neq x_0$, we have $x_1 = x_0 + r$ with some nonzero r . Therefore the above sequence is strictly increasing if $r > 0$ and strictly decreasing if $r < 0$. Moreover, for each integer n , $x_{n+2} = 2x_{n+1} - x_n + c$, or equivalently $x_{n+2} - x_{n+1} = x_{n+1} - x_n + c$. By induction, we get

$$x_n = x_0 + n(r + \frac{(n-1)}{2}c), n = 0, \pm 1, \dots$$

If $c \neq 0$, then for sufficiently large $|n|$ the terms x_n are either greater than x_0 or less than x_0 , which contradicts the strict monotonicity of the above sequence. Thus we have to consider only the case $c = 0$. Now $f(x) = x$ satisfies the condition and we may assume that $f(x_0) \neq x_0$ for some $x_0 \in \mathbb{R}$. Then for the sequence defined above we have $x_{n+1} = x_n + r$ and therefore $x_n = x_0 + nr$, where $r = x_1 - x_0 = f(x_0) - x_0$. Choose an arbitrary y_0 from the open interval with end points x_0 and x_1 , and consider the sequence $y_n = f^{(n)}(y_0)$, $n = 0, \pm 1, \dots$. Then $y_n = y_0 + ns$, where $s = y_1 - y_0$. Since f is strictly increasing, we see that each y_n is between x_n and x_{n+1} and therefore

$$x_0 + nr \leq y_0 + ns \leq x_0 + (n+1)r.$$

Now dividing by n the above inequalities and letting $n \rightarrow \infty$ we get $r = s$. This shows that the function $f(x) - x$ is a constant on the open interval with end points x_0 and x_1 . Using the same arguments and the continuity of f we conclude that the function $f(x) - x$ is constant, i.e. $f(x) = x + c$ for all $x \in \mathbb{R}$.

Problem 18.76. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function such that $f^{(n)}(x) = x$ for some integer $n \geq 2$ and all $x \in \mathbb{R}$. Prove that:

- (i) if n is odd, then $f(x) = x$ for all $x \in \mathbb{R}$;
- (ii) if n is even, then $f^{(2)}(x) = x$ for all $x \in \mathbb{R}$.

Solution. The function f is injective and by continuity it follows that it is strictly monotonic.

(i) If n is odd then the function f is increasing since otherwise f and hence $f^{(n)}$ is decreasing, a contradiction. Suppose that $f(x_0) > x_0$ for some $x_0 \in \mathbb{R}$. Then $f^{(n)}(x_0) > x_0$, a contradiction. Similarly, the case $f(x_0) < x_0$ is not possible and therefore $f(x) = x$ for all $x \in \mathbb{R}$.

(ii) Let $n = 2k$ be even and set $g(x) = f^{(2)}(x)$. Then $g^{(k)}(x) = x$ for all $x \in \mathbb{R}$. Since f is monotonic it follows that the function g is increasing and the same reasoning as in (i) shows that $g(x) = x$ for all $x \in \mathbb{R}$.

Problem 18.77. (Romanian TST 2011) Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous, decreasing function, such that $g(\mathbb{R}) = (-\infty, 0)$. Prove that there are no continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that $f^{(k)} = g$ for an integer $k \geq 2$.

Solution. Suppose that such a function f exists. Injectivity of g implies the injectivity of the continuous function f , which in turn is strictly monotonic.

As g is increasing, we conclude that f is increasing and k is an odd number. Moreover, f is not surjective. Because $f(\mathbb{R})$ is an interval with

$$(-\infty, 0) = g(\mathbb{R}) = f(f^{(k-1)}(\mathbb{R})) \subset f(\mathbb{R}),$$

we deduce that f is bounded from above. Let m be a real such that $f(x) < m$, for all reals x . Then, $f^{(k-1)}(x) < m$, for all $x \in \mathbb{R}$, so

$$g(x) = f(f^{(k-1)}(x)) > f(m),$$

for all reals x , in contradiction with $g(\mathbb{R}) = (-\infty, 0)$.

Problem 18.78. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function such that

$$f^{(2)}(x) = x$$

for all $x \in \mathbb{R}$. Prove that either $f(x) = x$ for all $x \in \mathbb{R}$, or f has the form $f(x) = f_0(x - p) + p$, where $p \in \mathbb{R}$ and

$$f_0(x) = \begin{cases} g(x) & \text{if } x \geq 0 \\ g^{-1}(x) & \text{if } x < 0 \end{cases}$$

for some continuous function $g : \mathbb{R} \rightarrow \mathbb{R}$ such that $g(0) = 0$, $g(x) < 0$ for $x > 0$, $g(x)$ is decreasing for $x > 0$ and $g^{-1}(x)$ is the inverse function to $g(x)$.

Solution. Since the function f is injective and continuous we conclude that it is strictly monotonic. In what follows we shall always assume that $f(x) \neq x$. Then the function f is decreasing. Indeed, suppose that f is increasing and take x_0 such $f(x_0) \neq x_0$. If $f(x_0) > x_0$, then $x_0 = f(f(x_0)) > f(x_0)$, a contradiction. If $x_0 < f(x_0)$, then $f(x_0) < f(f(x_0)) = x_0$, a contradiction again. Hence, the function f is decreasing. Consider the function $g(x) = f(x) - x$. It is continuous, decreasing and $\lim_{x \rightarrow \pm\infty} f(x) = \mp\infty$. Hence there exists a unique point p such that $g(p) = 0$, i.e. f has a unique fixed point p . Set $f_0(x) = f(x + p) - p$. Then $f(x) = f_0(x - p) + p$, $f_0(0) = 0$ and $f_0(f_0(x)) = x$ which implies that $f_0^{-1}(x) = f_0(x)$. Hence the function $f_0(x)$ can be written in the given form. Conversely, suppose that f has the given form. Since $g(x) \leq 0$ for $x \geq 0$ we have $f_0(f_0(x)) = g^{-1}(g(x)) = x$. For $x < 0$, we put $u = g^{-1}(x)$. Then $g(u) = x < 0$, which implies that $g^{-1}(x) = u > 0$.

Hence $f_0(f_0(x)) = g(g^{-1}(x)) = x$ for $x < 0$. Thus $f_0(f_0(x)) = x$ for all $x \in \mathbb{R}$ and therefore

$$f(f(x)) = f(f_0(x - p) + p) = f_0(f_0(x - p)) + p = (x - p) + p = x$$

for all $x \in \mathbb{R}$.

Remark. The functional equation $f(f(x)) = x$ and more generally $f^{(n)}(x) = x$ is called the *Babbage equation* [23] since it was investigated first by the British mathematician Charles Babbage [5] who is known as a founder of the modern computing.

Problem 18.79. (IMO Shortlist 2009) Let $P(x)$ be a non-constant polynomial with integer coefficients. Prove that there is no function $T : \mathbb{Z} \rightarrow \mathbb{Z}$ such that the number of integers x with $T^{(n)}(x) = x$ is equal to $P(n)$ for every $n \geq 1$, where $T^{(n)}$ is the n -th iterate of T .

First Solution. Assume there is a polynomial P of degree at least 1 with the desired property for a given function T . Let $A(n)$ denote the set of all $x \in \mathbb{Z}$ such that $T^{(n)}(x) = x$ and let $B(n)$ denote the set of all $x \in \mathbb{Z}$ for which $T^{(n)}(x) = x$ and $T^{(k)}(x) \neq x$ for all $1 \leq k < n$. Both sets are finite under the assumption made. For each $x \in A(n)$ there is a smallest $k \geq 1$ such that $T^{(k)}(x) = x$, i.e. $x \in B(k)$. Let $d = \gcd(k, n)$. There are positive integers r, s such that $rk - sn = d$ and hence

$$x = T^{(rk)}(x) = T^{(sn+d)}(x) = T^{(d)}(T^{(sn)}(x)) = T^{(d)}(x).$$

The minimality of k implies $d = k$, i.e. $k \mid n$. On the other hand one clearly has $B(k) \subseteq A(n)$ if $k \mid n$. Thus we have

$$A(n) = \bigcup_{d \mid n} B(d)$$

as a disjoint union and hence

$$|A(n)| = \sum_{d \mid n} |B(d)|.$$

Furthermore, for every $x \in B(n)$ the elements

$$x, T^{(1)}(x), T^{(2)}(x), \dots, T^{(n-1)}(x)$$

are n distinct elements of $B(n)$. The fact that they are in $A(n)$ is obvious. If for some $k < n$ and some $0 \leq i < n$ we had $T^{(k)}(T^{(i)}(x)) = T^{(i)}(x)$, i.e. $T^{(k+i)}(x) = T^{(i)}(x)$, that would imply

$$x = T^{(n)}(x) = T^{(n-i)}(T^{(i)}(x)) = T^{(n-i)}(T^{(k+i)}(x)) = T^{(k)}(T^{(n)}(x)) = T^{(k)}(x)$$

contradicting the minimality of n . Thus $T^{(i)}(x) \in B(n)$ and $T^{(i)}(x) \neq T^{(j)}(x)$ for $0 \leq i < j \leq n-1$.

So indeed, T permutes the elements of $B(n)$ in (disjoint) cycles of length n and in particular one has $n \mid |B(n)|$. Now let

$$P(x) = \sum_{i=0}^k a_i x^i, a_i \in \mathbb{Z}, k \geq 1, a_k \neq 0$$

and suppose that $|A(n)| = P(n)$ for all $n \geq 1$. Let p be any prime. Then

$$p^2 \mid |B(p^2)| = |A(p^2)| - |A(p)| = a_1(p^2 - p) + a_2(p^4 - p^2) + \dots$$

Hence $p \mid a_1$ and since this is true for all primes we have $a_1 = 0$. Now consider any two different primes p and q . Since $a_1 = 0$ we have that

$$|A(p^2q)| - |A(pq)| = a_2(p^4q^2 - p^2q^2) + a_3(p^6q^3 - p^3q^3) + \dots$$

is a multiple of p^2q . But we also have

$$p^2q \mid |B(p^2q)| = |A(p^2q)| - |A(pq)| - |B(p^2)|.$$

This implies

$$p^2q \mid |B(p^2)| = |A(p^2)| - |A(p)| = a_2(p^4 - p^2) + a_3(p^6 - p^3) \dots + a_k(p^{2k} - p^k).$$

Since this is true for every prime $q \neq p$ we see that the left hand side of the above identity must vanish for every prime p . But it is a polynomial of p of degree $2k \geq 2$ which is a contradiction.

Second Solution. As in the first solution define $A(n)$ and $B(n)$ and assume that a polynomial P with the required property exists. This again implies that $|A(n)|$ and $|B(n)|$ are finite for all positive integers n , and that

$$P(n) = |A(n)| = \sum_{d \mid n} |B(d)|$$

and $n \mid |B(n)|$. Now for all distinct primes p and q , we have

$$\begin{aligned} P(0) \equiv P(pq) &\equiv |B(1)| + |B(p)| + |B(q)| + |B(pq)| \\ &\equiv |B(1)| + |B(p)| \pmod{q}. \end{aligned}$$

Thus, for a fixed p , the expression $P(0) - |B(1)| - |B(p)|$ is divisible by arbitrarily large primes q which means that $P(0) = |B(1)| + |B(p)| = P(p)$ for all primes p . This implies that the polynomial P is constant, a contradiction.

19.18.9 Discrete Subharmonic Functions

Problem 18.80. Let $f : \mathbb{Z} \rightarrow \mathbb{R}$ be a discrete subharmonic function, i.e.

$$f(k) \leq \frac{f(k-1) + f(k+1)}{2}$$

for all $k \in \mathbb{Z}$. Prove that if f is bounded above, then it is constant.

Hint. For a function $h : \mathbb{Z} \rightarrow \mathbb{R}$ set

$$(\Delta h)(x) = h(x+1) + h(x-1) - 2h(x).$$

Then use the same reasoning as in the solution to Problem 18.82 by replacing the function $\ln(|z|^2 - 1)$ by $|x|$.

Problem 18.81. Find all functions $f : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$4f(x) = 2f(x+1)^2 + 2f(x-1)^2 + 1$$

for all $x \in \mathbb{R}$.

Solution. If $g(x) = f(x) - 1/2$, then $g(x) \geq -1/4$ and

$$2g(x) = g(x+1) + g(x-1) + g(x+1)^2 + g(x-1)^2.$$

Fix $x \in \mathbb{R}$ and define $h : \mathbb{Z} \rightarrow \mathbb{R}$ by $h(n) = g(x+n)$. Then

$$2h(n) = h(n+1) + h(n-1) + h(n+1)^2 + h(n-1)^2 \geq h(n+1) + h(n-1)$$

and therefore $-h(n)$ is a bounded from above discrete subharmonic function. Hence it is constant (Problem 18.80), i.e. $f(x) = f(x+n)$ for all $n \in \mathbb{Z}$. Now

the given condition on f implies that $4f(x) = 4f(x)^2 + 1$ which shows that $f(x) = 1/2$ for all $x \in \mathbb{R}$.

Problem 18.82. Let $f : \mathbb{Z}^2 \rightarrow \mathbb{R}$ be a discrete subharmonic function, i.e.

$$f(x, y) \leq \frac{f(x+1, y) + f(x, y+1) + f(x-1, y) + f(x, y-1)}{4}$$

for all $x, y \in \mathbb{Z}$. Prove that if f is bounded above, then it is constant.

Solution. For a function $h : \mathbb{Z}^2 \rightarrow \mathbb{R}$ set

$$(\Delta h)(z) = h(x+1, y) + h(x, y+1) + h(x-1, y) + h(x, y-1) - 4h(x, y),$$

where $z = (x, y)$. Consider the function $g(z) = \ln(|z|^2 - 1)$, where

$$|z|^2 = x^2 + y^2.$$

It is easy to check that $(\Delta g)(z) < 0$ for $|z| > 2$. Note that $g(z)$ can be defined on the whole $\mathbb{Z}^2$ such that $(\Delta g)(z) < 0$ for any $z \neq 0$. (Set, for example, $g(0) = \ln a$ and $g(z) = \ln b$ for $|z| = 1$, where $0 < 3a < b^4$ and $0 < 4b < 1$).

For every $\varepsilon > 0$ set $f_\varepsilon = f - \varepsilon g$. Since $\lim_{z \rightarrow \infty} f_\varepsilon(z) = -\infty$ it follows that f_ε has a maximum which we denote by M_ε . If $z \neq 0$, then $(\Delta f_\varepsilon)(z) > 0$ implies that $f_\varepsilon(z) < M_\varepsilon$. Hence $f_\varepsilon(0) = M_\varepsilon$ and we get $f(0) = \lim_{\varepsilon \rightarrow 0} M_\varepsilon = M := \sup f$. Then it follows from $(\Delta f)(z) \geq 0$ that $f(z) = M$ for each of the four neighboring points z of 0. Continuing in the same way we see that $f(z) = M$ for every z .

Remark. The same reasoning as above shows that the statement of Problem 18.82 remains true under the weaker condition

$$\limsup_{(x,y) \rightarrow \infty} \frac{f(x, y)}{\ln(x^2 + y^2)} \leq 0.$$

It is also interesting to note that for every $n \geq 3$ there exist nonconstant bounded subharmonic functions on $\mathbb{Z}^n$.

Problem 18.83. Find all positive constants c such that

$$g(x, y) = \ln(x^2 + y^2 + c)$$

is a discrete subharmonic function on $\mathbb{Z}^2$.

Answer. $c \in \left[\frac{1}{2}, \infty\right)$.

Hint. Show that the subharmonicity of $g(x, y)$ is equivalent to the inequality

$$\left(c + \frac{1}{2}\right)(x^2 + y^2 + c)^2 + \left(c + \frac{1}{2}\right)^2 \geq (x^2 - y^2)^2$$

for all $x, y \in \mathbb{Z}$.

Remark. Problem 18.83 shows that the boundness condition mentioned in the Remark above is the optimal one ensuring the constancy of a discrete subharmonic function on $\mathbb{Z}^2$.

Problem 18.84. Every lattice point in the space is assigned a real number from the open interval $(0, 1)$ so that each of these numbers is equal to the arithmetic mean of the numbers assigned to the six neighboring lattice points. Prove that all numbers are equal.

Solution. Let $f(x, y, z)$ be the number assigned to the point with coordinates (x, y, z) . Then

$$\begin{aligned} 6f(x, y, z) &= f(x+1, y, z) + f(x-1, y, z) + f(x, y+1, z) \\ &\quad + f(x, y-1, z) + f(x, y, z+1) + f(x, y, z-1). \end{aligned}$$

Suppose that not all numbers are equal. Then there exist two points at distance 1, whose assigned numbers are different and after a rotation we may assume that $f(x_0+1, y_0, z_0) > f(x_0, y_0, z_0)$ for some $x_0, y_0, z_0 \in \mathbb{Z}$. Set

$$g(x, y, z) = f(x+1, y, z) - f(x, y, z).$$

Then $M = \sup_{x \in \mathbb{Z}} g(x) \in (0, 1]$ and

$$\begin{aligned} 6g(x, y, z) &= g(x+1, y, z) + g(x-1, y, z) + g(x, y+1, z) \\ &\quad + g(x, y-1, z) + g(x, y, z+1) + g(x, y, z-1). \end{aligned}$$

In particular, if $g(a, b, c) \geq M - \varepsilon$, then

$$g(a+1, b, c) = 6g(a, b, c) - g(a-1, b, c) - g(a, b+1, c)$$

$$-g(a, b-1, c) - g(a, b, c+1) - g(a, b, c-1) \geq 6(M - \varepsilon) - 5M = M - 6\varepsilon.$$

Now it follows by induction that $g(a+n, b, c) \geq M - 6^n \varepsilon$ for all $n \in \mathbb{N}$.

By choosing first $n \geq \frac{2}{M}$, then $\varepsilon \in \left(0, \frac{M}{2 \cdot 6^n}\right]$ and finally $a, b, c \in \mathbb{Z}$, such that $g(a, b, c) \geq M - \varepsilon$, we get

$$1 > f(a+n, b, c) > f(a+n, b, c) - f(a, b, c) = \sum_{k=0}^{n-1} g(a+k, b, c) \geq n \frac{M}{2} \geq 1,$$

a contradiction.

Chapter 20

Notation and Abbreviations

20.1 Notation

We assume familiarity with the standard mathematical terminology. Some of the essential notations are listed below.

- $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ - the sets of positive integers (natural numbers), integers, rational numbers, real numbers, complex numbers respectively.

- For $F = \mathbb{Z}, \mathbb{Q}, \mathbb{R}$, the set $F^+, F_+, F^{>0}$ or $F_{>0}$ contains the positive elements of F . Similarly, $F^-, F_-, F^{<0}$ or $F_{<0}$ denotes the negative elements of F , and $F^{\geq 0}, F_{\geq 0}$ represents the set of nonnegative elements of F whereas $F^{\leq 0}, F_{\leq 0}$ is the set of non-positive elements of F . For F that contains 0, F^* denotes $F \setminus \{0\}$, the set of non-zero elements of F . $\mathbb{N}_0 = \mathbb{N} \cup \{0\}$ is the set of nonnegative integers.

- For n a positive integers, $\mathbb{Z}/n\mathbb{Z}$ denotes the the residues modulo n , whereas $(\mathbb{Z}/n\mathbb{Z})^*$ denotes the residues modulo n that are coprime to n .

- For $F = \mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ or $\mathbb{Z}/n\mathbb{Z}$ we denote by $F[X]$ the set of polynomials in the variable X with coefficients in F , and $F(X)$ the set of rational functions in X with coefficients in F . A rational function is the ratio of two polynomials.

- For x a real number, $\lfloor x \rfloor$ is the greatest integer smaller than or equal to x .

- For A, B points in the plane or space, AB denotes the line through A and B , the segment through A and B , or the distance between A and B ,

depending on the context.

- For S a plane figure, $[S]$ denotes the area of S . If S is a solid, $[S]$ denotes the volume of S .
- For S a finite set, $|S|$ denotes the number of elements in S . If S is infinite, one can write $|S| = \infty$.

20.2 Abbreviations

We indicated the sources of the problems where possible. The meaning of the abbreviations is explained below.

- AMM - American Mathematical Monthly
- AoPS - Art of Problem Solving
- BMO - Balkan Mathematical Olympiad
- IMAR - Institute of Mathematics of the Romanian Academy
- IMC - International Mathematics Competition
- IMO - International Mathematical Olympiad
- MOSP - Mathematical Olympiad Summer Program
- MR - Mathematical Reflections
- USAMO - United States of America Mathematical Olympiad
- TST - Team Selection Test
- ZIMO - Zhautykov International Mathematical Olympiad

The following abbreviations were also used in the text.

- gcd, lcm - the greatest common divisor, respectively the smallest common multiple of a set of integers.
- e.g. - for example.
- i.e. - in other words.
- RHS, LHS - the right-hand side, respectively left-hand side of a given equality of inequality.

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